Integrated course material from local sources

Econometrics 2 Unified Course Flow

Capra-style unified course flow. May 2026.

Study edition

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How to Use These Notes

Study legend

Definitions introduce the precise objects and assumptions to memorize.

Propositions collect results and properties that should be used in exercises and proofs.

Examples show how a formula or model is applied.

Notes / Review boxes contain interpretation, reminders, and links between concepts.

Revision checks applied

This edition includes a 10-pass boundary/unboxed audit. This edition was revised for study use: mathematical statements that appeared as raw inline OCR blocks were promoted where possible, subsection starts were made visible, Definition/Proposition/Theorem/Example/Note boxes were visually separated, early-closed statement boxes were extended where continuation text was detected, and all bundled figures were compiled from the accompanying image folder.

Source Map

The course flow below records the source files used to build the integrated material. Figures are inserted near the corresponding text block from which they were extracted.

No.	Role	Source
6	Core material	Corematerial/Timeseries_Set1_Univariatemodels(ARMAprocesses-part1).pdf
22	Figures / slide material	Figurematerial/Timeseries_Set1_slides(part1).pdf
23	Figures / slide material	Figurematerial/Timeseries_Set1_slides(part2).pdf
7	Core material	Corematerial/Timeseries_Set2_Univariatemodels(ARMAprocesses-part2).pdf
24	Figures / slide material	Figurematerial/Timeseries_Set2_slides.pdf
8	Core material	Corematerial/Timeseries_Set3_Univariatemodels(Non-stationaryseries).pdf
25	Figures / slide material	Figurematerial/Timeseries_Set3_slides.pdf
9	Core material	Corematerial/Timeseries_Set4_Univariatemodels(ARCHprocesses).pdf
26	Figures / slide material	Figurematerial/Timeseries_Set4_slides.pdf
10	Core material	Corematerial/Timeseries_Set5_Multivariatemodels(Dynamicmodelsforstationaryseries).pdf
27	Figures / slide material	Figurematerial/Timeseries_Set5_slides.pdf
11	Core material	Corematerial/Timeseries_Set6_Multivariatemodels(Dynamicmodelsfornon-stationaryseries).pdf
28	Figures / slide material	Figurematerial/Timeseries_Set6_slides.pdf
12	Core material	Corematerial/Timeseries_Set7_Multivariatemodels(VectorAutoRegressive-VAR-models).pdf
29	Figures / slide material	Figurematerial/Timeseries_Set7_slides.pdf
13	Core material	Corematerial/Timeseries_Set8_Multivariatemodels(CointegrationinVARmodels).pdf
30	Figures / slide material	Figurematerial/Timeseries_Set8_slides.pdf
14	Exam / exercises	Corematerial/Timeseries_Solutionstoexercises_1.pdf
15	Exam / exercises	Corematerial/Timeseries_Solutionstoexercises_2.pdf

No.	Role	Source
16	Exam / exercises	Output/TMP/extracted/Sampleexampapers-20260505/Ectx2_Timeseries_Examsample1.pdf
17	Exam / exercises	Output/TMP/extracted/Sampleexampapers-20260505/Ectx2_Timeseries_Examsample2.pdf
18	Exam / exercises	Output/TMP/extracted/Sampleexampapers-20260505/Ectx2_Timeseries_Examsample3.pdf
19	Exam / exercises	Output/TMP/extracted/Sampleexampapers-20260505/Ectx2_Timeseries_Examsample4.pdf
20	Exam / exercises	Output/TMP/extracted/Sampleexampapers-20260505/Ectx2_Timeseries_Examsample5.pdf
21	Exam / exercises	Output/TMP/extracted/Sampleexampapers-20260505/Ectx2_Timeseries_Examsample6.pdf
1	Optional readings	Optionalreadings/BlanchardQuah_AER89.pdf
2	Optional readings	Optionalreadings/Cochrane_QJE94.pdf
3	Optional readings	Optionalreadings/Engle_AER04.pdf
4	Optional readings	Optionalreadings/Engle_JEP01.pdf
5	Optional readings	Optionalreadings/StockWatson_JEP01.pdf

Chapter 1

Integrated Course Flow

1.1 Set 1 – Stochastic Processes and ARMA Models

Time series econometrics Univariate time series models F. C. Bagliano February 2024 Many applied econometric studies, particularly in the fields of macroeconomics and finance, use data organized as time series, where the variables evolve over time and are sampled at a regular frequency (annual, quarterly, monthly, daily, etc.).

This part of the course will deal with the properties and the basic econometric techniques used in time series analysis. First, we will focus on a single variable, introducing univariate models; then, we will analyse the dynamic interrelationships among several variables by means of basic versions of multivariate time series models.

From a statistical perspective, economic time series are viewed as generated by stochastic processes with precise properties that allow for the definition and use of appropriate estimators. Therefore, we begin by reviewing the general properties of stochastic processes of greatest interest for time series analysis.

DEFINITION 1.1– Stochastic process

Stochastic processes: general properties A stochastic process $\{y_t\}$ is a sequence of random variables ordered with respect to time. Measuring time at discrete dates we have:

$\{y_t\}$ for $t = ..., 0, 1, 2, ...$ A complete description of the properties of a stochastic process would require the specification of the joint cumulative probability distribution function $F(.)$ of any possible set of random variables y_t:

$F(y_{t_1}, y_{t_2}, \ldots, y_{t_n})$ for any set of n time dates $t_1, t_2, \ldots, t_n$ In practice, a stochastic process is characterized by only some quantities, such as ex- pected values $E(.)$, variances $Var(.)$, (auto)covariances $Cov(.)$ and (auto)correlations $Corr(.)$. Therefore the focus is on the first and second moments of the process.

In general, we have:

$$E(y_t) = \mu_t$$
$$Var(y_t) = E[(y_t - \mu_t)^2] = E(y_t^2) - \mu_t^2$$
$$Cov(y_t, y_{t-k}) = E\left[(y_t - \mu_t)(y_{t-k} - \mu_{t-k})\right] = E(y_t y_{t-k}) - \mu_t \mu_{t-k}$$

$$\equiv \gamma_{k,t} \quad \text{for} \quad k = 0, \pm 1, \ldots$$
$$Corr(y_t, y_{t-k}) = \frac{Cov(y_t, y_{t-k})}{\sqrt{Var(y_t) Var(y_{t-k})}} \equiv \rho_{k,t} \quad \text{for} \quad k = 0, \pm 1, \ldots$$

Notes

1. In the general formulation above, all quantities are indexed by time t and therefore are potentially time-varying.

2. For $k = 0$, $Cov(y_t, y_{t-k}) = Var(y_t) \equiv \gamma_{0,t}$ and $Corr(y_t, y_{t-k}) \equiv \rho_{0,t} = 1$ Problem: Ideally, we would like to have all necessary information to infer the probability law generating process y_t for each date t. However, in practice only a unique realization of the stochastic process is available (an observed economic data

series, such as the price level or the GDP growth rate, in a given time period).

Only under some conditions on the probabilistic structure of the process, it is possible to reduce the number of unknown parameters and proceed to inference on the general features of the process on the basis of the unique realization available.

Such conditions are restrictions on two features of a stochastic process:

(a) time heterogeneity: restrictions are aimed to ensure that the probability law is time-invariant, leading to the property of stationarity (b) "memory" of the process: restrictions are introduced to limit the degree of temporal dependence among the observations, leading to the property of ergodicity **2**

1.1 Stationarity

For a process to be "homogeneous" with respect to time (i.e. with the same probability law for each date t), it must be stationary. Two concepts of stationarity can be defined: "strong" and "weak" stationarity.

(i) strong stationarity (also called strict stationarity): for all sets of time dates $t_1, t_2, ..., t_n$ and for all $h > 0$ $F(y_{t_1}, y_{t_2}, \ldots, y_{t_n}) = F(y_{t_1+h}, y_{t_2+h}, \ldots, y_{t_n+h})$ The probability distribution of the process is therefore time-invariant: all

moments of the distribution do not change over time. In particular, the probability distributions of each individual observation are the same:

$F(y_t) = F(y_{t+h})$ (ii) weak stationarity (also called covariance stationarity): expected values, variances and covariances (first and second moments of the distributions) are finite and do not depend on time. For all t and k we have:

$$E(y_t) = \mu < \infty$$
$$Var(y_t) = \gamma_0 < \infty$$
$$Cov(y_t, y_{t-k}) = \gamma_k < \infty$$
$$Corr(y_t, y_{t-k}) = \rho_k = \frac{\gamma_k}{\gamma_0}$$

Under weak stationarity what matters is only the relative position of the observations along time (i.e. the distance k), and not their absolute position (i.e.

t and $t - k$). No restrictions are imposed on higher moments of the distributions. The behavior of weakly stationary processes is fully characterized by means of the autocorrelation function (ACF), whereby the autocorrelations ρ_k are expressed as a function of the lag k between observations (the plot of this function is called correlogram).

Notes

1. If the expected value and the variance of a process are finite, strong stationarity implies weak stationarity, whereas the latter does not imply the former (since it does not impose restrictions on moments higher than the second).

2. If the elements of the process are jointly normally distributed, weak stationarity implies strong stationarity (since the normal distribution is completely described by its first two moments).

3. Stationary processes can be effectively used to describe economic quantities that approach over time their long-run ("steady-state") equilibrium value, but are continuously hit by random disturbances.

EXAMPLE – Illustration

Example. The white noise process $\{\varepsilon_t\}$ with the following properties:

$$E(\varepsilon_t) = 0$$
$$\gamma_k = E(\varepsilon_t \varepsilon_{t-k}) = \begin{cases} \sigma^2 & \text{for } k = 0 \\ 0 & \text{for } k > 0 \end{cases}$$

is the simplest example of a weakly stationary process: $\varepsilon \sim WN(0, \sigma^2)$. This process does not require the stochastic independence of its elements (only their uncorrelatedness) and is more general than the process defined by a sequence of independently and identically distributed (IID) random variables, with zero expected value and variance σ^2: $\varepsilon \sim IID(0, \sigma^2)$. This latter process is strongly stationary (and also weakly stationary if σ^2 is finite).

Note

all weakly stationary processes $\{y_t\}$ can be represented as the sum of a deterministic (i.e. non-stochastic) component and a stochastic component, the latter being a linear combination of current and past values of a white noise process $\{\varepsilon_t\}$ as follows:

$y_t = \mu + \sum_{i=0}^{\infty} \varphi_j \varepsilon_{t-j}$ where $\varepsilon \sim WN(0, \sigma^2)$, $\varphi_0 = 1$, $\sum_{j=0}^{\infty} \varphi_j^2 < \infty$ (to guarantee a finite variance) and μ is the expected value of y_t: $E(y_t) = \mu$. This expected value

is the deterministic (and therefore perfectly forecastable) component of the process, and $\sum_{j=0}^{\infty} \varphi_j \varepsilon_{t-j}$ is the stochast component, which is not completely forecastable by means of a linear function of past values of $\{y_t\}$. According to this representation, ε_t is the forecasting error made when a linear pre-

diction of y_t is constructed on the basis of past realizations of the process:

$\varepsilon_t = y_t - E(y_t \mid y_{t-1}, y_{t-2}, ...)$. This result is a simple formulation of the Wold decomposition theorem.

1.2 Ergodicity Even if a process is stationary, further restrictions are needed on its "memory" (i.e.

on the dependence of the elements of the process at different dates) in order to describe its general properties with a limited number of parameters, and to be able to make statistical inferences on the basis of the unique realization available. For a weakly stationary process the ergodicity conditions ensure that the moments of the process can be consistently estimated by means of the corresponding sample moments.

More precisely, a stationary process $\{y_t\}$ with $t = 1, 2, ...T$ is:

- ergodic for the mean if plim $\frac{1}{T}\sum_{t=1}^{T} y_t = \mu$

- ergodic for the variance if plim $\frac{1}{T}\sum_{t=1}^{T}(y_t - \mu)^2 = Var(y_t)$

- ergodic for the covariance if plim $\frac{1}{T}\sum_{t=k+1}^{T}(y_t - \mu)(y_{t-k} - \mu) = \gamma_k$ Under the above conditions, as T goes to infinity (i.e. as the length of the sam- ple increases), sample moments converge in probability to the corresponding theoretical moments, and can therefore been relied upon as consistent estima- tors of the mean, variance and covariances of the process.

Notes

1. Similar definitions of ergodicity can be applied also to moments higher than the second.

2. If a process is not ergodic for a given moment (e.g. the mean), then it is not possible to obtain an estimate of that moment from a unique realization of the process, even of an infinite length.

3. A stochastic process can be weakly stationary but not ergodic. As an example of a weakly stationary but not ergodic process consider the following:

$y_t = \varepsilon_t + z_1$ where $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$ and $z_1 \sim N(0, \sigma_z^2)$ independent of ε_t for $t = 1, ..., T$.

This process is weakly stationary since:

$$E(y_t) = E(\varepsilon_t) + E(z_1) = 0$$
$$Var(y_t) = E(\varepsilon_t^2) + E(z_1^2) = \sigma_\varepsilon^2 + \sigma_z^2$$
$$Cov(y_t, y_{t-k}) = \sigma_z^2 \quad \forall k \geq 1$$

If the mean of the process μ (here equal to 0) is estimated by the sample mean $\bar{y}$ we get:

$\bar{y} \equiv \frac{1}{T}\sum_{t=1}^{T} y_t = z_1 + \frac{1}{T}\sum_{t=1}^{T}\varepsilon_t$ with plim $\bar{y} = z_1 \neq 0$

$\Rightarrow$ the process $\{y_t\}$ is not ergodic for the mean.

2.1 Definition and basic properties of the lag operator

DEFINITION 1.2– Lag operator

The lag operator L is defined as:

$Ly_t = y_{t-1}$ with the following properties:

(i) $L^2 y_t = L(Ly_t) = y_{t-2}$. In general:

$L^k y_t = y_{t-k}$ with $L^0 y_t = y_t$.

(ii) The property above applies also to negative values of k:

$L^{-k} y_t = y_{t+k}$ In this case the operator becomes a forward or lead operator.

(iii) For a constant μ, $L\mu = \mu$.

DEFINITION 1.3– Lag polynomials

2.2 Lag polynomials **Definition.**

$a(L) = 1 + a_1 L + a_2 L^2 + \ldots + a_k L^k = \sum_{j=0}^{k} a_j L^j$ (where $a_0 = 1$).

Application:

$a(L)y_t = y_t + a_1 y_{t-1} + a_2 y_{t-2} + \ldots + a_k y_{t-k} = \sum_{k=0}^{k} a_j y_{t-j}$ The polynomial $a(L)$ linearly "filters" the process $\{y_t\}$: a new series is obtained by

means of a linear combinations of the elements of the original series.

EXAMPLE – Illustration

Example. given the series $u_t = a_1 u_{t-1} + \varepsilon_t$ where $\varepsilon \sim WN(0, \sigma^2)$, the application of the "filter" $a(L) = 1 - a_1 L$ transforms u_t into a white noise series:

$$a(L)u_t = (1 - a_1 L)u_t = u_t - a_1 u_{t-1} = \varepsilon_t$$

DEFINITION 1.4– Invertibility of lag polynomials

Invertibility of lag polynomials **2.3 Definition.** under certain conditions, a lag polynomial $a(L)$ is invertible; in this case there exists a lag polynomial, denoted by $a(L)^{-1}$, such that:

EXAMPLE – Illustration

$a(L)^{-1}a(L) = 1$ **Example.** given $a(L) = 1 - aL$ with $|a| < 1$, the polynomial $1 + aL + a^2L^2 + \ldots = \sum_{i=1}^{\infty} a^j L^j = \frac{1}{1-aL}$ can be defined as $a(L)^{-1}$ since

$$\left(\sum_{i=1}^{\infty} a^j L^j\right)(1 - aL) = \sum_{i=1}^{\infty} a^j L^j - \sum_{i=1}^{\infty} a^j L^j = 1$$

Generalization: a polynomial of degree k $a(L) = 1 + a_1 L + a_2 L^2 + \ldots + a_k L^k$ can be "factorized" as follows $a(L) = \left(1 - \frac{1}{z_1}L\right)\left(1 - \frac{1}{z_2}L\right)\ldots\left(1 - \frac{1}{z_h}L\right)$

where $z_1, z_2, ..., z_k$ are the solutions, or "roots", (maybe complex) of the "character- istic equation" associated with the polynomial:

$a(z) = 1 + a_1 z + a_2 z^2 + \ldots + a_k z^k = 0$ All the factors $\left(1 - \frac{1}{z_i} L\right)$ are invertible only if $\left|\frac{1}{z_1}\right|, \left|\frac{1}{z_2}\right|, \ldots, \left|\frac{1}{z_h}\right| < 1$

$\Rightarrow |z_1|, |z_2|, \ldots |z_k| > 1$ The polynomial $a(L)$ is therefore invertible only if all the solutions ("roots") of the characteristic equation $a(z) = 0$ are (in modulus) larger than one (or "outside the unit circle").

Note

invertbility can be checked in an algebraically different but substantially equivalent method (used by some econometric software packages, e.g. EViews) by solving the equation $m^k + a_1 m^{k-1} + \ldots + a_{k-1} m + a_k = 0$

In this case the solutions ("roots") $m_1, m_2, ...m_k$ correspond to $\frac{1}{z_1}, \frac{1}{z_2}, ...\frac{1}{z_k}$ in the formulation above and are therefore often defined as "inverse roots". The invertibility condition now becomes $|m_1|, |m_2|, \ldots |m_k| < 1$ Exercise: check the invertibility of the following lag polynomials applied to the

process $\{y_t\}$:

$$a(L)y_t = y_t - 1.5y_{t-1} + 0.5y_{t-2}$$
$$b(L)y_t = y_t - 1.3y_{t-1} + 0.4y_{t-2}$$

AutoRegressive Moving Average (ARMA) processes The class of $ARMA$ models (Box-Jenkins 1970, 1976) is particularly relevant in the analysis of univariate time series, since:

any weakly stationary and ergodic stochastic process can be approximated (in terms of its autocorrelation function ACF) by an ARMA process The specification of ARMA models is based on the following idea:

the time series $\{y_t\}$ for $t = ...0, 1, ...T, ...$, is expressed as a linear combination of its past values and of a white noise stochastic component $\{\varepsilon_t\}$ The latter component ε_t has the following properties:

$E(\varepsilon_t) = 0, \qquad E(\varepsilon_t^2) = \sigma_\varepsilon^2, \qquad E(\varepsilon_t \varepsilon_{t-k}) = 0 \quad \forall k \geq 1$ We first analyze purely moving average (MA) models, than purely autoregressive (AR) models, and finally the general class of $ARMA$ models that feature both

components.

DEFINITION 1.5– Moving-average process MA(q)

3.1 Moving Average (MA) models **Definition.** a moving average process of order q, $MA(q)$, is defined as:

$y_t = \varepsilon_t + \alpha_1 \varepsilon_{t-1} + \alpha_2 \varepsilon_{t-2} + \ldots + \alpha_a \varepsilon_{t-a}$ where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$, $\alpha_0 = 1$ and y_t is measured as deviation from its mean

$(y_t = Y_t - \mu)$. Using the lag operator we can express y_t as:

$y_t = \alpha(L)\varepsilon_t$ where $\alpha(L) = 1 + \alpha_1 L + \alpha_2 L^2 + \ldots + \alpha_q L^q = \sum_{i=0}^{q} \alpha_i L^i$ Properties: an $MA(q)$ process is weakly stationary for all values of the parameters

$$\alpha_i$$

$$E(y_t) = 0$$

$$Var(y_t) = \sigma_\varepsilon^2(1 + \alpha_1^2 + \alpha_2^2 + \cdots + \alpha_q^2) = \sigma_\varepsilon^2 \sum_{i=0}^{q} \alpha_j^2$$

$$Cov(y_t, y_{t-k}) \equiv \gamma_k = \sigma_\varepsilon^2(\alpha_k + \alpha_{k+1}\alpha_1 + \alpha_{k+2}\alpha_2 + \cdots + \alpha_q\alpha_{q-k})$$

$= \sigma_\varepsilon^2 \sum_{k=1}^{q-\kappa} \alpha_{j+k}\alpha_j$ for $1 \leq k \leq q$ $Cov(y_t, y_{t-k}) \equiv \gamma_k = 0$

for $k > q$

Therefore, an $MA(q)$ process displays non-zero autocorrelations up to lag q and not beyond. The associated ACF has the following form:

$$\rho_k = \begin{cases} 1 & \text{for } \kappa = 0 \\ \frac{\sum_{j=0}^{q-k} \alpha_{j+k}\alpha_j}{1+\alpha_1^2+\alpha_2^2+\cdots+\alpha_q^2} & \text{for } 1 \leq k \leq q \end{cases}$$

EXAMPLE – Exercise

Exercise. derive the ACF of an $MA(1)$ and an $MA(2)$ process.

General case: $MA(\infty)$. A moving average process of an infinite order is defined as:

$$y_t = \varepsilon_t + \alpha_1\varepsilon_{t-1} + \alpha_2\varepsilon_{t-2} + \cdots + \alpha_q\varepsilon_{t-q} + \cdots = \sum_{i=0}^{\infty} \alpha_j\varepsilon_{t-j}$$

where an "absolute summability" condition is imposed on the α_i coefficients to ensure convergence of the series:

Note

$\sum_{j=1}^{\infty} |\alpha_j| < \infty$ **Note.** absolute summability implies square summability of the α_i's
$\sum_{i=0}^{\infty} |\alpha_j| < \infty \implies \sum_{i=0}^{\infty} \alpha_j^2 < \infty$

yielding a finite value for the variance of the process:

DEFINITION 1.6– Autoregressive process AR(p)

$Var(y_t) = \sigma_\varepsilon^2 \sum_{i=1}^{\infty} \alpha_j^2$ 3.2 Autoregressive models (AR) **Definition.** an autoregressive process of order p, $AR(p)$, is defined as:

$y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \ldots + \theta_n y_{t-n} + \varepsilon_t$ where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$. Using the lag operator we can express this process as:

$\theta(L) y_t = \varepsilon_t$ where $\theta(L) = 1 - \theta_1 L - \theta_2 L^2 - \ldots - \theta_p L^p = 1 - \sum_{i=1}^{p} \theta_j L^j$

with $\theta_0 = 1$.

PROPOSITION 1.1– Stationarity of AR(p)

an $AR(p)$ process is weakly stationary only under certain conditions on the θ_j parameters. Under stationarity, an $AR(p)$ process has an infinite moving average $MA(\infty)$ representation:

$$\theta(L) y_t = \varepsilon_t \Rightarrow y_t = \theta(L)^{-1} \varepsilon_t \equiv \varphi(L) \varepsilon_t$$
$$y_t = (1 + \varphi_1 L + \varphi_2 L^2 + ...) \varepsilon_t = \sum_{i=0}^{\infty} \varphi_j \varepsilon_{t-j}$$

where the φ_j parameters are functions of the θ_j parameters in the autoregressive representation. The stationarity conditions coincide with the invertibility conditions for the lag polynomial $\theta(L)$:

all solutions of the characteristic equation associated with $\theta(L)$ $\theta(z) = 0$ must lie outside the unit circle.

$AR(1): \quad y_t = \theta_1 y_{t-1} + \varepsilon_t \quad \rightarrow \quad \theta(L) y_t = \varepsilon_t \quad$ with $\theta(L) = 1 - \theta_1 L$. Example.

The stationarity condition on the solution of the characteristic equation is:

$\theta(z) = 0 \Rightarrow 1 - \theta_1 z = 0 \Rightarrow z = \frac{1}{\theta_1} \Rightarrow |z| > 1$ if $|\theta_1| < 1$ Under stationarity, the process for y_t has the following representation:

$$y_t = \theta(L)^{-1} \varepsilon_t = \frac{1}{1 - \theta_1 L} \varepsilon_t = \sum_{i=0}^{\infty} \theta_1^j L^j \varepsilon_t$$

$= \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_1^2 \varepsilon_{t-2} + \ldots$ If an $AR(p)$ process is stationary, the autocovariances γ_k that form the ACF can be computed using the so-called Yule-Walker equations, obtained from the covariance between y_t and a generic lag y_{t-k} (multiplying y_t by y_{t-k} and taking the expected

value of both sides of the equation):

$$Cov(y_t, y_{t-k}) = \theta_1 Cov(y_{t-1}, y_{t-k}) + \theta_2 Cov(y_{t-2}, y_{t-k}) + \cdots + \theta_p Cov(y_{t-p}, y_{t-k}) + Cov(\varepsilon_t, y_{t-k})$$

which can be rewritten as:

$$\gamma_k = \theta_1\gamma_{k-1} + \theta_2\gamma_{k-2} + \cdots + \theta_p\gamma_{k-p} + Cov(\varepsilon_t, y_{t-k})$$

Considering successive values of k from 0 up to p we get (noting that $\gamma_{-j} = \gamma_j$):

$$\begin{aligned} \gamma_0 &= \theta_1\gamma_1 + \theta_2\gamma_2 + \ldots + \theta_p\gamma_p + \sigma_\varepsilon^2 \qquad \text{for } \; k = 0 \\ \gamma_1 &= \theta_1\gamma_0 + \theta_2\gamma_1 + \ldots + \theta_p\gamma_{p-1} \qquad \text{for } \; k = 1 \end{aligned}$$

$\gamma_p = \theta_1\gamma_{p-1} + \theta_2\gamma_{p-2} + \ldots + \theta_p\gamma_0$ for $k = p$ This system of $p+1$ linear equations can be solved for the first $p+1$ autocovariances $(\gamma_0, \gamma_1, ..., \gamma_p)$ as functions of the parameters of the process $(\theta_1, \theta_2, ..., \theta_p$ and $\sigma_\varepsilon^2)$. The

autocovariances of higher order can then be obtained recursively from the relation:

$\gamma_k = \theta_1\gamma_{k-1} + \theta_2\gamma_{k-2} + \ldots + \theta_p\gamma_{k-p} \qquad$ for $k > p$

DEFINITION 1.7– Autoregressive-moving-average process ARMA(p,q)

Autoregressive moving average (ARMA) models 3.3 Definition: an autoregressive moving average process of order (p, q), called $ARMA(p, q)$,

is defined as:

$$y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \cdots + \theta_p y_{t-p} + \varepsilon_t + \alpha_1\varepsilon_{t-1} + \alpha_2\varepsilon_{t-2} + \cdots + \alpha_q\varepsilon_{t-q}$$

where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$. Using the lag operator we can express this process as:

$\theta(L)\, y_t = \alpha(L)\, \varepsilon_t$ where $\theta(L) = 1 - \theta_1 L - \theta_2 L^2 - \ldots - \theta_n L^p$ $\alpha(L) = 1 + \alpha_1 L + \alpha_2 L^2 + \ldots + \alpha_a L^q$

with $\theta_0 = \alpha_0 = 1$.

PROPOSITION 1.2– Stationarity and invertibility of ARMA(p,q)

ARMA processes may possess two relevant properties: "stationar- ity" and "invertibility", leading to two different representations of the process with exactly the same autocorrelation properties.

(a) stationarity: an $ARMA(p, q)$ process is stationary if all the roots of the char- acteristic equation associated with the lag polynomial of the autoregressive component $\theta(L)$ $\theta(z) = 0$ lie outside the unit circle. Under stationarity the process can be represented as a $MA(\infty)$:

$$\theta(L)y_t = \alpha(L)\varepsilon_t \quad \Rightarrow \quad y_t = \varphi(L)\varepsilon_t = \sum_{i=0}^{\infty} \varphi_i \varepsilon_{t-i}$$

where $\varphi(L) = \frac{1+\alpha_1 L+\alpha_2 L^2+\ldots+\alpha_q L^q}{1-\theta_1 L-\theta_2 L^2-\ldots-\theta_r L^p}$ satisfies the convergence condition $\sum_{i=0}^{\infty} |\varphi_j| < \infty$

and the coefficients φ_i are functions of the coefficients in the autoregressive and moving

average components of the process.

EXAMPLE – Illustration

Example. the following simple $ARMA(1,1)$ process $y_t = \theta_1 y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$ with $|\theta_1| < 1$ can be written, using polynomials in the lag operator, as $\underbrace{(1-\theta_1 L)}_{\theta(L)} y_t = \underbrace{(1+\alpha_1 L)}_{\alpha(L)} \varepsilon_t$

which, under stationarity (since $|\theta_1| < 1$), can be represented as the following $MA(\infty)$ process:

$$\begin{aligned} y_t &= \frac{1+\alpha_1 L}{1-\theta_1 L}\varepsilon_t \\ &= (1+\alpha_1 L)(1+\theta_1 L+\theta_1^2 L^2+...)\varepsilon_t \\ &= \underbrace{\left[1+(\alpha_1+\theta_1)L+\theta_1(\alpha_1+\theta_1)L^2+\theta_1^2(\alpha_1+\theta_1)L^3+\ldots\right]}_{\varphi(L)}\varepsilon_t \end{aligned}$$

$$= \varepsilon_t + (\alpha_1+\theta_1)\varepsilon_{t-1} + \theta_1(\alpha_1+\theta_1)\varepsilon_{t-2} + \theta_1^2(\alpha_1+\theta_1)\varepsilon_{t-3} + \ldots$$

$= \varphi(L)\varepsilon_t$ The convergence condition is satisfied since1 $\sum_{j=1}^{\infty} |\varphi_j| = 1 + |(\alpha_1+\theta_1)| + |\theta_1(\alpha_1 + \theta_1)| + |\theta_1^2(\alpha_1+\theta_1)| + \ldots$

$= \left|\frac{1+\alpha_1}{1-\theta_1}\right| < \infty$ 1Note that this infinite sum can be easily obtained just by setting $L = 1$ in the $MA(\infty)$ expression of the process.

PROPOSITION 1.3– Unboxed proposition

(b) invertibility: an $ARMA(p,q)$ process is invertible if it admits an autoregres- sive representation of infinite order, $AR(\infty)$:

$\theta(L)y_t = \alpha(L)\varepsilon_t \implies \varepsilon_t = \pi(L)y_t = \sum_{j=1}^{\infty}(-\pi_j)y_{t-j}$ with $\pi_0 = -1$ and $\sum_{j=0}^{\infty}|\pi_j| < \infty$

$\Rightarrow y_t = \sum_{i=1}^{\infty}\pi_j y_{t-j} + \varepsilon_t$ where $\pi(L) = \frac{1-\theta_1 L-\theta_2 L^2-\ldots-\theta_p L^p}{1+\alpha_1 L+\alpha_2 L^2+\ldots+\alpha_q L^q}$

and the coefficients π_j are functions of the coefficients in the autoregressive and moving average components of the process. The invertibility condition pertains to the moving average component of the process: the ARMA process is invertible if all the roots of the characteristic equation associated with the lag polynomial of the MA component $\alpha(L)$ $\alpha(L) = 0$ lie outside the unit circle.

EXAMPLE – Illustration

Example. consider again the $ARMA(1,1)$ process $y_t = \theta_1 y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$ with $|\theta_1| < 1$ written, using polynomials in the lag operator, as $\underbrace{(1-\theta_1 L)}_{\theta(L)} y_t = \underbrace{(1+\alpha_1 L)}_{\alpha(L)} \varepsilon_t$

Under invertibility (assuming $|\alpha_1| < 1$), $\{y_t\}$ can be represented as the fol- low-

ing $AR(\infty)$ process:2 $\frac{1-\theta_1 L}{1+\alpha_1 L} y_t = \varepsilon_t \underbrace{(1-\theta_1 L)(1-\alpha_1 L+\alpha_1^2 L^2-\alpha_1^3 L^3+...)}_{\pi(L)} y_t = \varepsilon_t$

$\underbrace{[1-(\alpha_1+\theta_1)L+\alpha_1(\alpha_1+\theta_1)L^2-\alpha_1^2(\alpha_1+\theta_1)L^3+...]}_{\pi(L)} y_t = \varepsilon_t$

2The convergence condition is satisfied, since $\sum_{j=0}^{\infty}|\pi_j| = \left|\frac{1-\theta_1}{1+\alpha_1}\right| < \infty$

$$\Rightarrow y_t = (\alpha_1+\theta_1)y_{t-1} - \alpha_1(\alpha_1+\theta_1)y_{t-2} + \alpha_1^2(\alpha_1+\theta_1)y_{t-3} + \cdots + \varepsilon_t$$

Notes

(i) invertibility allows to express y_t as a linear combination of its own past values and of a contemporaneous (i.e. dated t) white noise stochastic term: it is then possible to forecast future values y_{t+h} on the basis of observed past realizations of the process;

(ii) invertibility (defined above for an ARMA process) may obviously be referred also to a pure $MA(q)$ process;

(iii) the same autocovariance function is compatible with two ARMA processes (with different parameters), only one of which is invertible. For instance, take the following two $MA(1)$ processes (a) $y_t = \varepsilon_t + \alpha_1\varepsilon_{t-1}$, $\varepsilon_t \sim WN(0,\sigma_\varepsilon^2)$ with $|\alpha_1| < 1$

(b) $y_t = \eta_t + \beta_1\eta_{t-1}$, $\eta_t \sim WN(0,\sigma_\eta^2)$ with $\beta_1 = \frac{1}{\alpha_t}$ and $\sigma_\eta^2 = \alpha_1^2\sigma_\varepsilon^2$

with autocovariance functions:

(a) $\gamma_0 = (1+\alpha_1^2)\sigma_\varepsilon^2$, $\gamma_1 = \alpha_1\sigma_\varepsilon^2$, $\gamma_k = 0$ for $k \geq 2$ (b) $\gamma_0 = (1+\beta_1^2)\,\sigma_\eta^2 = \left[1+\left(\frac{1}{\alpha_1}\right)^2\right]\alpha_1^2\sigma_\varepsilon^2$ $= (1+\alpha_1^2)\,\sigma_\varepsilon^2$,

$$\gamma_1 = \beta_1\sigma_\eta^2 = \left(\frac{1}{\alpha_1}\right)\alpha_1^2\sigma_\varepsilon^2 = \alpha_1\sigma_\varepsilon^2,\ \gamma_k = 0 \text{ for } k \geq 2$$

The two processes display the same autocovariance function but only (a) is invertible.

(*iv*) The autocovariances of an ARMA process can be obtained by means of the Yule-Walker equations, as shown above in the case of an AR process, taking into account here also the moving average component. For example, given the following $ARMA(1,1)$ process $y_t = \theta_1 y_{t-1} + \varepsilon_t + \alpha_1\varepsilon_{t-1}$ from the covariance between y_t and a generic lag y_{t-k}:

$Cov(y_t, y_{t-k}) = \theta_1 Cov(y_{t-1}, y_{t-k}) + Cov(\varepsilon_t, y_{t-k}) + \alpha_1 Cov(\varepsilon_{t-1}, y_{t-k})$ rewritten as (to simplify notation):

$$\gamma_k = \theta_1\gamma_{k-1} + Cov(\varepsilon_t, y_{t-k}) + \alpha_1 Cov(\varepsilon_{t-1}, y_{t-k})$$

setting $k = 0, 1$, we get (making use of $Cov(\varepsilon_t, y_t) = Cov(\varepsilon_{t-1}, y_{t-1}) = \sigma_\varepsilon^2$, $Cov(\varepsilon_{t-1}, y_t)$ $= (\alpha_1+\theta_1)\sigma_\varepsilon^2$):

$\gamma_0 = \theta_1\gamma_1 + [1+\alpha_1(\alpha_1+\theta_1)]\sigma_\varepsilon^2$ for $k = 0$ $\gamma_1 = \theta_1\gamma_0 + \alpha_1\sigma_\varepsilon^2$

for $k = 1$ This two-equation system is solved for γ_0 and γ_1 as functions of the process parameters θ_1, α_1 and σ_ε^2:

$$\gamma_0 = \frac{1+\alpha_1^2+2\theta_1\alpha_1}{1-\theta_1^2}\sigma_\varepsilon^2$$
$$\gamma_1 = \frac{(1+\theta_1\alpha_1)(\alpha_1+\theta_1)}{1-\theta_1^2}\sigma_\varepsilon^2$$

Higher-order autocovariances are then obtained by the recursive relation:

$\gamma_k = \theta_1\gamma_{k-1}$ for $k \geq 2$ Finally, autocorrelations can be derived as:

$$\rho_1 \equiv \frac{\gamma_1}{\gamma_0} = \frac{(1+\theta_1\alpha_1)(\alpha_1+\theta_1)}{1+\alpha_1^2+2\theta_1\alpha_2}$$

and $\rho_k \equiv \frac{\gamma_k}{\gamma_2} = \frac{\theta_1\gamma_{k-1}}{\gamma_2} = \theta_1\rho_{k-1}$ for $k \geq 2$

(v) If the polynomials $\theta(L)$ and $\alpha(L)$ of an $ARMA(p,q)$ process have an identical root (i.e. they display a common factor), there exists an $ARMA(p-1,q-1)$ process with the same autocovariance structure of the (higher-order) original process. For instance, if the polynomials of the $ARMA(2,2)$ process $y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t + \alpha_1\varepsilon_{t-1} + \alpha_2\varepsilon_{t-2}$

can be factorized as $(1-\varphi_1 L)(1-\varphi_2 L)y_t = (1-\psi_1 L)(1-\psi_2 L)\varepsilon_t$ with $\varphi_1 = \psi_1$, then the original process reduces to the following $ARMA(1,1)$:

$$(1-\varphi_2 L)y_t = (1-\psi_2 L)\varepsilon_t$$
$$\Rightarrow y_t = \varphi_2 y_{t-1} + \varepsilon_t - \psi_2\varepsilon_{t-1}$$

EXAMPLE – Exercise

Exercise. Suppose that the stochastic process $\{y_t\}$ is generated by two unobserved components $\{x_t\}$ and $\{z_t\}$, defined as:

$x_t = v_t + v_{t-1}$ with $v_t \sim WN(0,1)$ $z_t \sim WN(0,1)$ independent of v_t $y_t = x_t + z_t$ (*)
1. Prove that the variance and the autocovariances of $\{y_t\}$ in (*) have the fea- tures typical of an $MA(1)$ process.

2. Now, suppose that you want to represent $\{y_t\}$ by means of a specific $MA(1)$ process of the following form:

$y_t = \varepsilon_t + \alpha\varepsilon_{t-1}$ with $\varepsilon_t \sim WN(0,\sigma_\varepsilon^2)$ (**) Find α and σ_ε^2 such that the variance and the autocovariances of $\{y_t\}$ obtained

from $(*)$ are equal to those obtained from $(**)$.

EXAMPLE – Exercise

Exercise. ($AR(1)$ *plus noise process*). Let the process $\{y_t\}$ be the sum of two components, $\{x_t\}$ and $\{v_t\}$:

$y_t = x_t + v_t$ $(*)$ with $\{v_t\}$ being a white noise process, $v_t \sim WN(0,\sigma_v^2)$, and $\{x_t\}$ being a stationary $AR(1)$ process $x_t = \theta\, x_{t-1} + u_t$ $|\theta| < 1$

(**) where $\{u_t\}$ is a white noise process, $u_t \sim WN(0,\sigma_u^2)$. The two white noise pro-

cesses $\{v_t\}$ and $\{u_t\}$ are uncorrelated for all t.

1. Check that the process $\{y_t\}$ is (weakly) stationary.

2. Find the autocorrelation function (ACF) of $\{y_t\}$ and evaluate the effect of $\{v_t\}$ on the autocorrelation pattern of the process.

Univariate time series models Stochastic processes: general properties Stochastic process $\{y_t\}$: sequence of random variables ordered with respect to time. In discrete time:

$\{v_t\}$ for $t = ..., 0, 1, 2, ...$ A complete description of the properties of a stochastic process would require the specification of the joint cumulative probability distribution function $F(.)$ of any possible set of random variables y_t $F(y_{t_1}, y_{t_2}, \ldots, y_{t_n})$ for any set of n time dates $t_1, t_2, \ldots, t_n$

In practice, a stochastic process is characterized by expected values, variances, (auto)covariances and (auto)correlations In general:

$$E(v_t) = \mu_t$$
$$Var(y_t) = E[(y_t - \mu_t)^2] = E(y_t^2) - \mu_t^2$$
$$Cov(y_t, y_{t-k}) = E[(y_t - \mu_t)(y_{t-k} - \mu_{t-k})] = E(y_t y_{t-k}) - \mu_t \mu_{t-k}$$

$\equiv \gamma_{k,t}$ for $k = 0, \pm 1, \ldots$ $Corr(y_t, y_{t-k}) = \frac{Cov(y_t, y_{t-k})}{\sqrt{Var(v_t)Var(v_{t-k})}} \equiv \rho_{k,t}$ for $k = 0, \pm 1, \ldots$

In general:

$$E(v_t) = \mu_t$$
$$Var(y_t) = E[(y_t - \mu_t)^2] = E(y_t^2) - \mu_t^2$$
$$Cov(v_t, v_{t-k}) = E[(v_t - \mu_t)(v_{t-k} - \mu_{t-k})] = E(v_t v_{t-k}) - \mu_t \mu_{t-k}$$

$\equiv \gamma_{k,t}$ for $k = 0, \pm 1, \ldots$ $Corr(y_t, y_{t-k}) = \frac{Cov(y_t, y_{t-k})}{\sqrt{Var(y_t)Var(y_{t-k})}} \equiv \rho_{k,t}$ for $k = 0, \pm 1, \ldots$

Note

1. All quantities are indexed by t 2. For $k = 0$, $Cov(y_t, y_{t-k}) = Var(y_t) \equiv \gamma_{0,t}$ and $Corr(v_t, v_{t-k}) \equiv \rho_{0,t} = 1$

yt tn t2 t1

ALGORITHM – Problem

Problem. Ideally, we would like to have all necessary information to infer the probability law generating process y_t for each date t.

However, in practice only a unique realization of the stochastic process is available. Only under some conditions on the probabilistic structure of the process it is possible to reduce the number of unknown parameters and proceed to inference on the general features of the process on the basis of the only available realization.

Such conditions are restrictions on two features of a stochastic process:

(a) time heterogeneity (to ensure that the probability law is time-invariant) ⇒ stationarity (b) "memory" of the process (to limit the degree of temporal dependence) ⇒ ergodicity

Stationarity For a process to be "homogeneous" with respect to time, it must be stationary. Two concepts of stationarity:

(i) strong stationarity (or strict stationarity): for all sets of time dates $t_1, t_2, \ldots, t_n$ and for all $h > 0$ $F(y_{t_1}, y_{t_2}, \ldots, y_{t_n}) = F(y_{t_1+h}, y_{t_2+h}, \ldots, y_{t_n+h})$ The probability distribution of the process is time-invariant:

all moments do not change over time. In particular, the probability distributions of the individual observations are the same:

$$F(y_t) = F(y_{t+h})$$

(ii) weak stationarity (or covariance stationarity): expected values, variances and covariances are finite and do not depend on time. For all t and k:

$$E(v_t) = \mu < \infty$$
$$Var(y_t) = \gamma_0 < \infty$$
$$Cov(y_t, y_{t-k}) = \gamma_k < \infty$$
$$Corr(y_t, y_{t-k}) = \rho_k = \frac{\gamma_k}{\gamma_0}$$

No restrictions are imposed on higher moments. Weakly stationary processes are characterized by the autocorrelation function (ACF): autocorrelations ρ_k are expressed as a function of the lag k between observations (the plot is the correlogram)

yt tn t_2 t1

-71 >

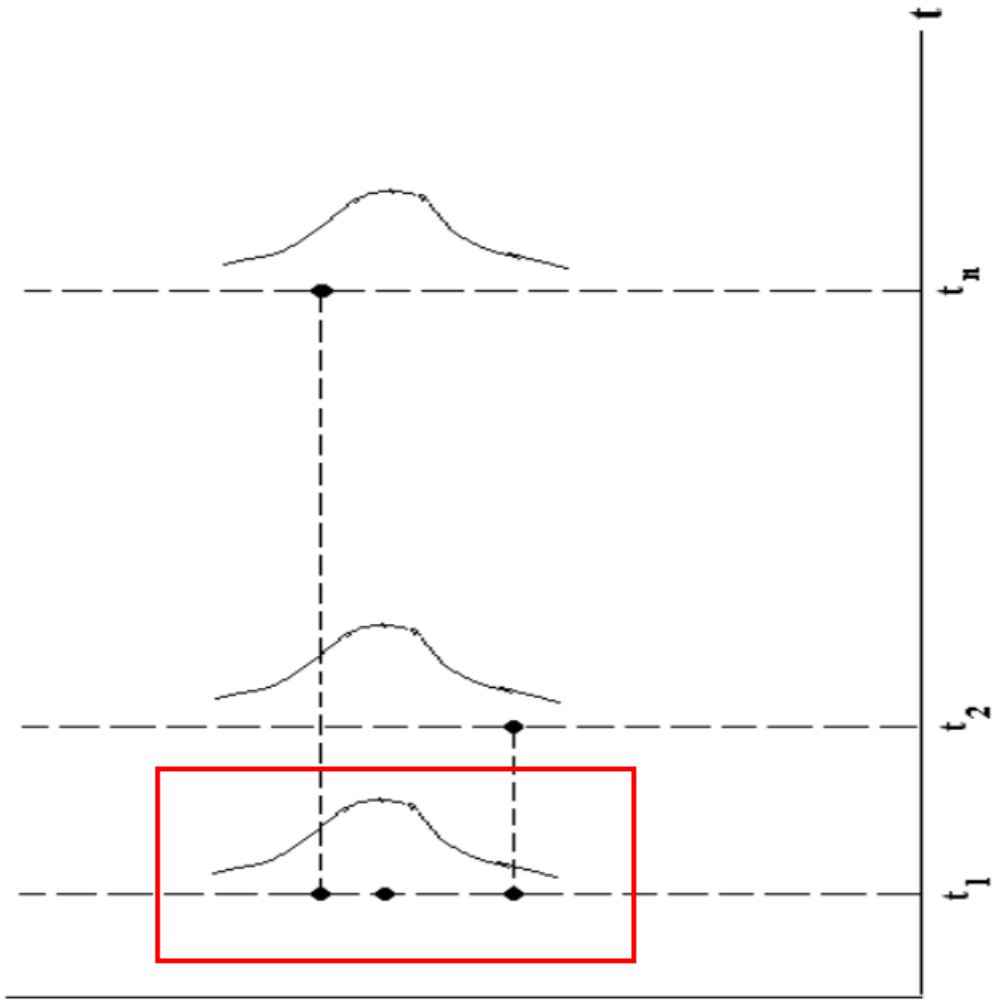


Figure 1.1: Course figure, source page 9, crop 1.

Notes

1. If the expected value and the variance of a process are finite, strong stationarity implies weak stationarity, whereas the latter does not imply the former (since it does not impose restrictions on moments higher than the second).

2. If the elements of the process are jointly normally distributed, weak stationarity implies strong stationarity (since the normal distribution is completely described by its first two moments)

3. Stationary processes can be used to describe economic quantities that approach long-run ("steady-state") equilibrium values, but are continuously hit by random disturbances.

EXAMPLE – Illustration

Example. The white noise process $\{\varepsilon_t\}$ with:

$$E(\varepsilon_t) = 0$$
$$\gamma_k = E(\varepsilon_t \varepsilon_{t-k}) = \begin{cases} \sigma^2 & \text{for } k = 0 \\ 0 & \text{for } k > 0 \end{cases}$$

is a weakly stationary process: $\varepsilon \sim WN(0, \sigma^2)$. It does not require the stochastic independence of its elements (only their uncorrelatedness) and is more general than the process defined by a sequence of independently and identically distributed (IID) random variables: $\varepsilon \sim IID(0, \sigma^2)$. This latter process is strongly stationary (and also weakly stationary if σ^2 is finite)

Note

all weakly stationary processes $\{y_t\}$ can be represented in terms of a white noise process $\{\varepsilon_t\}$:

$y_t = \mu + \sum_{j=0} \varphi_j \varepsilon_{t-j}$ where $\varepsilon \sim WN(0, \sigma^2)$, $\varphi_0 = 1$, $\sum_{j=0}^{\infty} \varphi_j^2 < \infty$ and μ is the expected value of y_t: $E(y_t) = \mu$.

ε_t is the forecasting error made when a linear prediction of y_t is constructed on the basis of past realizations of the process:

$\varepsilon_t = y_t - E(y_t \mid y_{t-1}, y_{t-2}, \dots)$. This result is a simple formulation of the Wold decomposition theorem

Ergodicity Even if a process is stationary, further restrictions are needed on its "memory" (the dependence of the elements of the process at different dates) in order to describe its general properties with a limited number of parameters, and to be able to make statistical inferences on the basis of the unique realization available.

For a weakly stationary process the ergodicity conditions ensure that the moments of the process can be consistently estimated by means of the corresponding sample moments

A stationary process $\{y_t\}$ with $t = 1, 2, ..., T$ is:

- ergodic for the mean if plim $\frac{1}{T} \sum_{i=1}^{t} y_t = \mu$ ergodic for the variance if plim $\frac{1}{T} \sum (y_t - \mu)^2 = \text{Var}(y_t)$

ergodic for the covariance if plim $\frac{1}{T} \sum_{k=1}^{\infty} (y_t - \mu)(y_{t-k} - \mu) = \gamma_k$ $t = k + 1$

Note

1. Similar definitions of ergodicity can be applied also to moments higher than the second.

2. If a process is not ergodic for a given moment (e.g. the mean), then it is not possible

to obtain an estimate of that moment from a unique realization of the process, even of an infinite length.

yt tn t_2 t1

C1 $\geq$

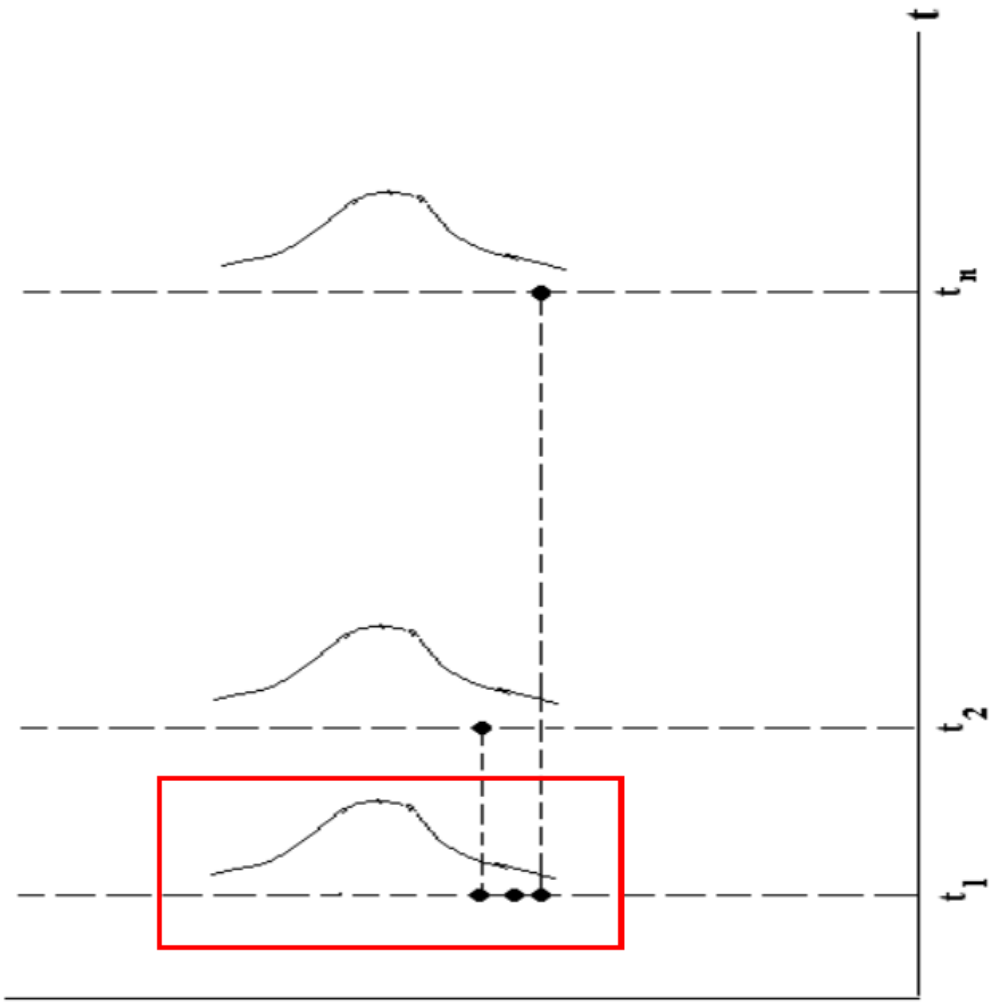


Figure 1.2: Course figure, source page 18, crop 1.

Note

1. Similar definitions of ergodicity can be applied also to moments higher than the second.

2. If a process is not ergodic for a given moment (e.g. the mean), then it is not possible to obtain an estimate of that moment from a unique realization of the process, even of an infinite length.

3. A stochastic process can be weakly stationary but not ergodic. As an example consider the following:

$y_t = \varepsilon_t + z_1$ where $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$ and $z_1 \sim N(0, \sigma_\varepsilon^2)$ independent of ε_t for $t = 1, \dots T$.

This process is weakly stationary since:

$$E(v_t) = E(\varepsilon_t) + E(z_1) = 0$$
$$Var(v_t) = E(\varepsilon_t^2) + E(z_1^2) = \sigma_\varepsilon^2 + \sigma_\tau^2$$
$$Cov(v_t, v_{t-k}) = \sigma_z^2 \qquad \forall k \geq 1$$

If the mean of the process μ (here 0) is estimated by the sample mean $\bar{y}$ we get:

$\bar{y} \equiv \frac{1}{T}\sum y_t = z_1 + \frac{1}{T}\sum \varepsilon_t$ with plim $\bar{v} = z_1 \neq 0 \Rightarrow$ the process $\{y_t\}$ is not ergodic for the mean.

Definition and basic properties of the lag operator

DEFINITION 1.8– Lag operator

The lag operator L is defined as:

$Ly_t = y_{t-1}$ with the following properties:

(i) $L^2 y_t = L(Ly_t) = y_{t-2}$. In general:

$L^k y_t = y_{t-k}$ with $L^0 y_t = y_t$.

(ii) The property above applies also to negative values of k:

$L^{-k} y_t = y_{t+k}$ In this case the operator becomes a forward or lead operator.

(iii) For a constant μ, $L\mu = \mu$

DEFINITION 1.9– Lag polynomials

Lag polynomials **Definition.**

$a(L) = 1 + a_1 L + a_2 L^2 + \ldots + a_k L^k = \sum_{j=0}^{n} a_j L^j$ (where $a_0 = 1$).

Application:

$a(L)y_t = y_t + a_1 y_{t-1} + a_2 y_{t-2} + \ldots + a_k y_{t-k} = \sum_{i=0}^{k} a_i y_{t-i}$ The polynomial $a(L)$ linearly "filters" the process $\{y_t\}$: a new series is obtained by means of a linear combinations of the

elements of the original series

EXAMPLE – Illustration

Example. given the series $u_t = a_1 u_{t-1} + \varepsilon_t$ where $\varepsilon \sim WN(0, \sigma^2)$, the application of the "filter" $a(L) = 1 - a_1 L$ transforms u_t into a white noise series:

$$a(L)u_t = (1 - a_1 L)u_t = u_t - a_1 u_{t-1} = \varepsilon_t$$

DEFINITION 1.10– Invertibility of lag polynomials

Invertibility of lag polynomials **Definition.** if a lag polynomial $a(L)$ is invertible, there exists a lag polynomial, $a(L)^{-1}$, such that:

$$a(L)^{-1}a(L) = 1$$

EXAMPLE – Illustration

Example. given $a(L) = 1 - aL$ with $|a| < 1$, the polynomial $1 + aL + a^2L^2 + \ldots = \sum a^j L^j = \frac{1}{1-aL}$ can be defined as $a(L)^{-1}$ since $\left(\sum_{i=0}^{\infty} a^j L^j\right)(1 - aL) = 1$

Generalization: a polynomial of degree k $a(L) = 1 + a_1L + a_2L^2 + \ldots + a_kL^k$ can be "factorized" as follows $a(L) = \left(1 - \frac{1}{Z_1}L\right)\left(1 - \frac{1}{Z_2}L\right)\ldots\left(1 - \frac{1}{Z_k}L\right)$

where $z_1, z_2, ..., z_k$ are the solutions (maybe complex) of the "characteristic equation" associated with the polynomial:

$$a(z) = 1 + a_1z + a_2z^2 + \ldots + a_kz^k = 0$$

All the factors $(1 - \frac{1}{z_i}L)$ are invertible only if $\left|\frac{1}{Z_1}\right|, \left|\frac{1}{Z_2}\right|, \ldots \left|\frac{1}{Z_k}\right| < 1$

$\Rightarrow |z_1|, |z_2|, \ldots |z_k| > 1$ The polynomial $a(L)$ is therefore invertible only if all the solutions ("roots") of the characteristic equation $a(z) = 0$ are (in modulus) larger than one (or "outside the unit circle")

Note

invertbility can be checked by solving the equation $m^k + a_1m^{k-1} + \ldots + a_{k-1}m + a_k = 0$ In this case the solutions $m_1, m_2, \ldots m_k$ correspond to $\frac{1}{z_1}, \frac{1}{z_2},$

$\frac{1}{z_k}$ in the formulation above, and the invertibility condition becomes $|m_1|, |m_2|,$ $\ldots |m_k| < 1$ "inverse roots"

EXAMPLE – Exercise

Exercise. check the invertibility of the following lag polynomials applied to the process $\{y_t\}$:

$$a(L)y_t = y_t - 1.5y_{t-1} + 0.5y_{t-2}$$
$$b(L)y_t = y_t - 1.3y_{t-1} + 0.4y_{t-2}$$

Polynomial: $a(L) * y(t) = y(t) - 1.3 * y(t-1) + 0.4 * y(t-2)$ Inverse Roots of AR/MA Polynomial(s) Modulus Root(s) Cycle 0.800000 0.500000 No root lies outside the unit circle.

Polynomial is invertible

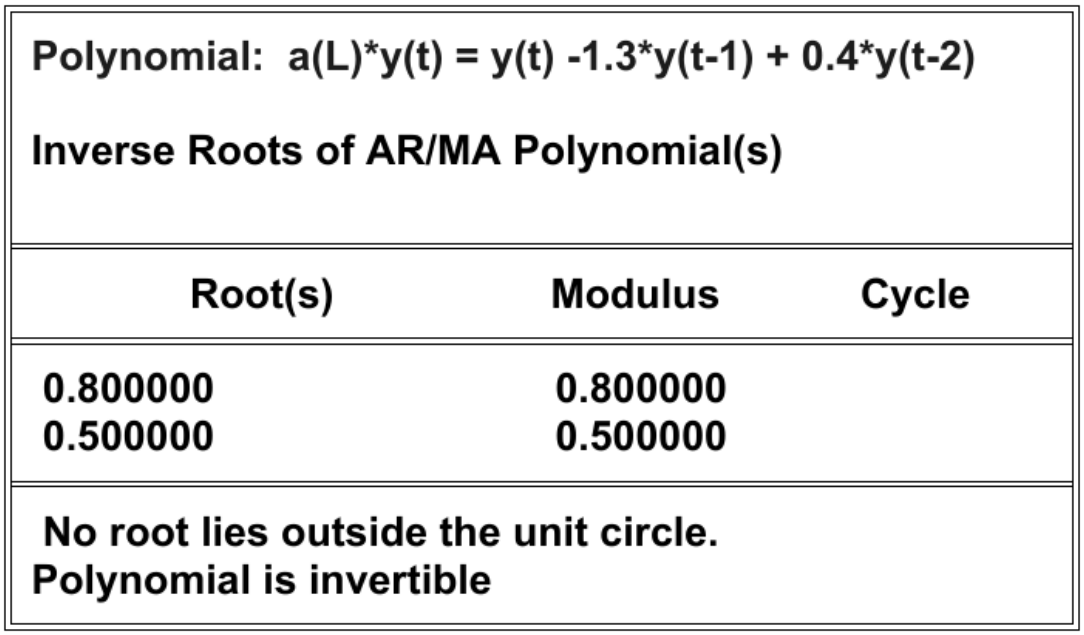


Figure 1.3: Course figure, source page 32, crop 1.

Inverse Roots of Polynomial 1.5 1.0 0.5 0.0 Roots

-0.5

-1.0

-1.5

-0.5 0.5 1.0

-1.5

-1.0 0.0 1.5

ARMA processes Importance of ARMA models (Box-Jenkins 1970, 1976):

any weakly stationary and ergodic stochastic process can be approximated (in terms of its autocorrelation function ACF) by an ARMA process Basic idea:

the time series $\{y_t\}$ for $t = \dots 0, 1, \dots, T, \dots$, is expressed as a linear combination of its past values and of a white noise stochastic component $\{\varepsilon_t\}$ The component ε_t has the following properties:

$$E(\varepsilon_t) = 0, \quad E(\varepsilon_t^2) = \sigma_\varepsilon^2, \quad E(\varepsilon_t \varepsilon_{t-k}) = 0 \quad \forall k \geq 1$$

DEFINITION 1.11– Moving-average process MA(q)

Moving Average (MA) models **Definition.** a moving average process of order q, $MA(q)$, is defined as:

$v_t = \varepsilon_t + \alpha_1\varepsilon_{t-1} + \alpha_2\varepsilon_{t-2} + \ldots + \alpha_a\varepsilon_{t-a}$ where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$, $\alpha_0 = 1$ and y_t is measured as deviation

from its mean ($v_t = Y_t - \mu$).

Using the lag operator:

$y_t = \alpha(L)\varepsilon_t$ where $\alpha(L) = 1 + \alpha_1 L + \alpha_2 L^2 + \ldots + \alpha_q L^q = \sum_{j=1}^{n} \alpha_j L^j$

PROPOSITION 1.4– Weak stationarity and ACF of MA(q)

an $MA(q)$ process is weakly stationary for all values of the parameters α_i:

$$E(v_t) = 0$$

$$Var(y_t) = \sigma_\varepsilon^2(1 + \alpha_1^2 + \alpha_2^2 + \ldots + \alpha_q^2) = \sigma_\varepsilon^2 \sum_{i=0}^{\infty} \alpha_j^2$$

$$Cov(y_t, y_{t-k}) \equiv \gamma_k = \sigma_\varepsilon^2(\alpha_k + \alpha_{k+1}\alpha_1 + \alpha_{k+2}\alpha_2 + \ldots + \alpha_q\alpha_{q-k})$$

$= \sigma_\varepsilon^2 \sum_{j=0}^{q-k} \alpha_{j+k}\alpha_j$ for $1 \leq k \leq q$ $Cov(y_t, y_{t-k}) \equiv \gamma_k = 0$ for $k > q$

The associated ACF has the following form:

$$\rho_k = \begin{cases} 1 & \text{for } k = 0 \\ \frac{\sum_{j=0}^{q-k} \alpha_{j+k}\alpha_j}{1+\alpha_1^2+\alpha_2^2+\cdots+\alpha_q^2} & \text{for } 1 \leq k \leq q \\ 0 & \text{for } k > q \end{cases}$$

EXAMPLE – Exercise

Exercise. derive the ACF of an $MA(1)$ and an $MA(2)$ process.

General case: $MA(\infty)$. A moving average process of an infinite order is defined as:

$$y_t = \varepsilon_t + \alpha_1\varepsilon_{t-1} + \alpha_2\varepsilon_{t-2} + \ldots + \alpha_q\varepsilon_{t-q} + \ldots = \sum_{j=0}^{\infty} \alpha_j\varepsilon_{t-j}$$

where an "absolute summability" conditions is imposed on the α_i coefficients to ensure convergence of the series:

$$\sum_{j=0} |\alpha_j| < \infty$$

Note

absolute summability implies square summability of the α_i'S $\sum_{j=0}^{\infty} |\alpha_j| < \infty \quad \Rightarrow \sum_{j=0}^{\infty} \alpha_j^2 < \infty$

$$\Rightarrow \quad Var(y_t) = \sigma_\varepsilon^2 \sum_{j=0}^{\infty} \alpha_j^2$$

DEFINITION 1.12– Infinite moving-average process

Autoregressive models (AR) **Definition.** an autoregressive process of order p, $AR(p)$, is defined as:

$v_t = \theta_1 v_{t-1} + \theta_2 v_{t-2} + \ldots + \theta_p v_{t-p} + \varepsilon_t$ where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.

Using the lag operator:

$\theta(L) y_t = \varepsilon_t$ where (with $\theta_0 = 1$) $\theta(L) = 1 - \theta_1 L - \theta_2 L^2 - \ldots - \theta_p L^p = 1 - \sum_{j=1}^{n} \theta_j L^j$

PROPOSITION 1.5– Stationarity of AR(p)

an $AR(p)$ process is weakly stationary only under certain conditions on the θ_i parameters. Under stationarity, an $AR(p)$ process has an infinite moving average $MA(\infty)$ representation:

$$\theta(L) y_t = \varepsilon_t \implies y_t = \theta(L)^{-1} \varepsilon_t \equiv \varphi(L) \varepsilon_t$$
$$y_t = (1 + \varphi_1 L + \varphi_2 L^2 + \ldots) \varepsilon_t = \sum_{j=0}^{\infty} \varphi_j \varepsilon_{t-j}$$

The stationarity conditions coincide with the invertibility conditions for the lag polynomial $\theta(L)$:

all solutions of the characteristic equation associated with $\theta(L)$ $\theta(z) = 0$ must lie outside the unit circle

EXAMPLE – Illustration

Example. $AR(1)$: $y_t = \theta_1 y_{t-1} + \varepsilon_t \to \theta(L) y_t = \varepsilon_t$ with $\theta(L) = 1 - \theta_1 L$. The stationarity condition on the solution of the characteristic equation is:

$\theta(z) = 0 \implies 1 - \theta_1 z = 0 \implies z = \frac{1}{\theta_1} \Rightarrow |z| > 1$ if $|\theta_1| < 1$ Under stationarity, the process for y_t has the following representation:

$$y_t = \frac{1}{1 - \theta_1 L} \varepsilon_t = \sum_{i=0}^{\infty} \theta_1^j L^j \varepsilon_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_1^2 \varepsilon_{t-2} + \ldots$$

If an $AR(p)$ process is stationary, the autocovariances γ_k (that form the ACF) can be computed using the Yule-Walker equations obtained from the covariance between y_t and a generic lag y_{t-k}:

$$Cov(y_t, y_{t-k}) = \theta_1 Cov(y_{t-1}, y_{t-k}) + \theta_2 Cov(y_{t-2}, y_{t-k}) + \cdots + \theta_p Cov(y_{t-p}, y_{t-k}) + Cov(\varepsilon_t, v_{t-k})$$

which can be rewritten as:

$$\gamma_k = \theta_1 \gamma_{k-1} + \theta_2 \gamma_{k-2} + \ldots + \theta_p \gamma_{k-p} + Cov(\varepsilon_t, y_{t-k})$$

Considering successive values of k from 0 up to p we get (noting that $\gamma_{-i} = \gamma_i$):

for $k = 0$ $\gamma_0 = \theta_1 \gamma_1 + \theta_2 \gamma_2 + \ldots + \theta_p \gamma_p + \sigma_\epsilon^2$ $\gamma_1 = \theta_1 \gamma_0 + \theta_2 \gamma_1 + \ldots + \theta_p \gamma_{p-1}$

for $k = 1$ . . .

$\gamma_p = \theta_1 \gamma_{p-1} + \theta_2 \gamma_{p-2} + \ldots + \theta_p \gamma_0$ for $k = p$ This system of $p+1$ linear equations can be solved for the first $p+1$ autocovariances $(\gamma_0, \gamma_1, \ldots, \gamma_p)$ as functions of the

parameters $\theta_1, \theta_2, \ldots, \theta_p$ and σ_ϵ^2

The autocovariances of higher order can then be obtained recursively from the relation:

$\gamma_k = \theta_1 \gamma_{k-1} + \theta_2 \gamma_{k-2} + \ldots + \theta_p \gamma_{k-p}$ for $k > p$

DEFINITION 1.13– Autoregressive-moving-average process ARMA(p,q)

Autoregressive moving average (ARMA) models **Definition.** an autoregressive moving average process of order (p, q), called $ARMA(p, q)$, is defined as:

$$y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \ldots + \theta_p y_{t-p} + \varepsilon_t + \alpha_1 \varepsilon_{t-1} + \alpha_2 \varepsilon_{t-2} + \ldots + \alpha_q \varepsilon_{t-q}$$

where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.

Using the lag operator we can express this process as:

$\theta(L) v_t = \alpha(L) \varepsilon_t$ where (with $\theta_0 = \alpha_0 = 1$) $\theta(L) = 1 - \theta_1 L - \theta_2 L^2 - \ldots - \theta_p L^p$ $\alpha(L) = 1 + \alpha_1 L + \alpha_2 L^2 + \ldots + \alpha_q L^q$

PROPOSITION 1.6– Stationarity of ARMA(p,q)

(a) stationarity: an $ARMA(p, q)$ process is stationary if all the roots of the characteristic equation associated with the lag polynomial of the autoregressive component $\theta(L)$ $\theta(z) = 0$ lie outside the unit circle. Under stationarity the process can be represented as a $MA(\infty)$:

$\theta(L) y_t = \alpha(L) \varepsilon_t \implies y_t = \varphi(L) \varepsilon_t = \sum_{i=0}^{\infty} \varphi_i \varepsilon_{t-i}$ where $\varphi(L) = \frac{1 + \alpha_1 L + \alpha_2 L^2 + \ldots + \alpha_q L^q}{1 - \theta_1 L - \theta_2 L^2 - \ldots - \theta_p L^p}$ with $\sum_{i=0} |\varphi_i| < \infty$

EXAMPLE – Illustration

Example. stationary $ARMA(1,1)$ process $y_t = \theta_1 y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$ with $|\theta_1| < 1$ written as $\underbrace{(1-\theta_1 L)}_{t} y_t = \underbrace{(1+\alpha_1 L)}_{\varepsilon_t} \varepsilon_t$

$$\theta(L)$$
$$\alpha(L)$$

with $MA(\infty)$ representation:

$$\begin{aligned} y_t &= \frac{1+\alpha_1 L}{1-\theta_1 L}\varepsilon_t \\ &= (1+\alpha_1 L)(1+\theta_1 L+\theta_1^2 L^2+\dots)\varepsilon_t \\ &= \underbrace{\left[1+(\alpha_1+\theta_1)L+\theta_1(\alpha_1+\theta_1)L^2+\theta_1^2(\alpha_1+\theta_1)L^3+\dots\right]}_{\varepsilon_t}\varepsilon_t \end{aligned}$$

$$\varphi(L)$$
$$= \varepsilon_t + (\alpha_1+\theta_1)\varepsilon_{t-1} + \theta_1(\alpha_1+\theta_1)\varepsilon_{t-2} + \theta_1^2(\alpha_1+\theta_1)\varepsilon_{t-3} + \dots$$

$$= \varphi(L)\varepsilon_t$$

and convergence condition satisfied $\infty \sum |\varphi_j| = 1 + |(\alpha_1+\theta_1)| + |\theta_1(\alpha_1+\theta_1)| + |\theta_1^2(\alpha_1+\theta_1)| + \dots$ j=0 $= \left|\frac{1+\alpha_1}{1-\theta_1}\right| < \infty$

PROPOSITION 1.7– Unboxed proposition

(b) invertibility: an $ARMA(p,q)$ process is invertible if it admits an autoregressive representation of infinite order, $AR(\infty)$:

$\theta(L)v_t = \alpha(L)\varepsilon_t \implies \varepsilon_t = \pi(L)v_t = \sum_{j=0}^{\infty}(-\pi_j)y_{t-j}$ with $\pi_0 = -1$ and $\sum_{j=0}^{\infty}|\pi_j| < \infty$

$\Rightarrow y_t = \sum \pi_j y_{t-j} + \varepsilon_t$ where $\pi(L) = \frac{1-\theta_1 L-\theta_2 L^2-\dots-\theta_p L^p}{1+\alpha_1 L+\alpha_2 L^2+\dots+\alpha_q L^q}$

The invertibility condition pertains to the moving average component of the process: the ARMA process is invertible if all the roots of the characteristic equation associated with the lag polynomial of the MA component $\alpha(L)$ $\alpha(L) = 0$ lie outside the unit circle.

EXAMPLE – Illustration

Example. stationary $ARMA(1,1)$ process $y_t = \theta_1 y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$ with $|\theta_1| < 1$ written as $\underbrace{(1-\theta_1 L)}_{t} y_t = \underbrace{(1+\alpha_1 L)}_{\varepsilon_t} \varepsilon_t$

$$\theta(L)$$
$$\alpha(L)$$

Under invertibility (assuming $|\alpha_1| < 1$), $\{y_t\}$ can be represented as the following $AR(\infty)$ process:

$$\frac{1-\theta_1 L}{1+\alpha_1 L} y_t = \varepsilon_t$$
$$(1-\theta_1 L)(1-\alpha_1 L+\alpha_1^2 L^2-\alpha_1^3 L^3+\dots)y_t = \varepsilon_t$$

$$[1-(\alpha_1+\theta_1)L+\alpha_1(\alpha_1+\theta_1)L^2-\alpha_1^2(\alpha_1+\theta_1)L^3+\dots]y_t = \varepsilon_t$$
$$\pi(L)$$
$$\Rightarrow y_t = (\alpha_1+\theta_1)y_{t-1}-\alpha_1(\alpha_1+\theta_1)y_{t-2}+\alpha_1^2(\alpha_1+\theta_1)y_{t-3}+\dots+\varepsilon_t$$

Notes

(i) invertibility allows to express y_t as a linear combination of its own past values and of a contemporaneous (i.e. dated t) white noise stochastic term: it is then possible to forecast future values v_{t+h} on the basis of observed past realization of the process;

(ii) invertibility (defined above for an ARMA process) may obviously be referred also to a pure $MA(q)$ process;

(iii) the same ACF is compatible with two ARMA processes (with different parameters), only one of which is invertible.

(iii) For instance, take the two $MA(1)$ processes (a) $v_t = \varepsilon_t + \alpha_1\varepsilon_{t-1}$, $\varepsilon_t \sim WN(0,\sigma_\varepsilon^2)$ with $|\alpha_1| < 1$ (b) $y_t = \eta_t + \beta_1\eta_{t-1}$, $\eta_t \sim WN(0,\sigma_\eta^2)$ with $\beta_1 = \frac{1}{\alpha_1}$ and $\sigma_\eta^2 = \alpha_1^2\sigma_\varepsilon^2$

with ACF :

(a) $\gamma_0 = (1+\alpha_1^2)\sigma_\varepsilon^2$, $\gamma_1 = \alpha_1\sigma_\varepsilon^2$, $\gamma_k = 0$ for $k \geq 2$ (b) $\gamma_0 = (1+\beta_1^2)\sigma_\eta^2 = \left[1+\left(\frac{1}{\alpha_1}\right)^2\right]\alpha_1^2\sigma_\varepsilon^2$ $= (1+\alpha_1^2)\sigma_\varepsilon^2$,

$\gamma_1 = \beta_1\sigma_\eta^2 = \left(\frac{1}{\alpha_1}\right)\alpha_1^2\sigma_\varepsilon^2 = \alpha_1\sigma_\varepsilon^2$, $\gamma_k = 0$ for $k \geq 2$

The two processes display the same ACF but only (a) is invertible.

(iv) The autocovariances of an ARMA process can be obtained by means of the Yule-Walker equations.

For example, given the following $ARMA(1,1)$ process $y_t = \theta_1 y_{t-1} + \varepsilon_t + \alpha_1\varepsilon_{t-1}$ from the covariance between y_t and a generic lag y_{t-k}:

$Cov(y_t, y_{t-k}) = \theta_1 Cov(y_{t-1}, y_{t-k}) + Cov(\varepsilon_t, y_{t-k}) + \alpha_1 Cov(\varepsilon_{t-1}, y_{t-k})$ rewritten as:

$$\gamma_k = \theta_1\gamma_{k-1} + Cov(\varepsilon_t, v_{t-k}) + \alpha_1 Cov(\varepsilon_{t-1}, v_{t-k})$$

Setting $k = 0, 1$, we get (making use of $Cov(\varepsilon_t, y_t) = Cov(\varepsilon_{t-1}, y_{t-1}) = \sigma_\varepsilon^2$, $Cov(\varepsilon_{t-1}, y_t)$ $= (\alpha_1+\theta_1)\sigma_\varepsilon^2$:

$\gamma_0 = \theta_1\gamma_1 + [1+\alpha_1(\alpha_1+\theta_1)]\sigma_\varepsilon^2$ for $k = 0$ $\gamma_1 = \theta_1\gamma_0 + \alpha_1\sigma_\varepsilon^2$ for $k = 1$

This two-equation system is solved for γ_0 and γ_1 as functions of the process parameters θ_1, α_1 and σ_ε^2:

$$\gamma_0 = \frac{1 + \alpha_1^2 + 2\theta_1\alpha_1}{1 - \theta_1^2}\sigma_\varepsilon^2$$
$$\gamma_1 = \frac{(1 + \theta_1\alpha_1)(\alpha_1 + \theta_1)}{1 - \theta_1^2}\sigma_\varepsilon^2$$

The autocovariances of higher order are then obtained by the recursive relation:

$\gamma_k = \theta_1\gamma_{k-1}$ for $k \geq 2$

Autocorrelations can be derived as:

$$\rho_1 \equiv \frac{\gamma_1}{\gamma_0} = \frac{(1 + \theta_1\alpha_1)(\alpha_1 + \theta_1)}{1 + \alpha_1^2 + 2\theta_1\alpha_1}$$

and $\rho_k \equiv \frac{\gamma_k}{\gamma_0} = \frac{\theta_1\gamma_{k-1}}{\gamma_0} = \theta_1\rho_{k-1} \quad$ for $k \geq 2$

(v) If the polynomials $\theta(L)$ and $\alpha(L)$ of an $ARMA(p, q)$ have an identical root (i.e. they display a common factor), there exists an $ARMA(p-1, q-1)$ process with the same autocovariance structure of the (higher-order) original process.

For instance, if the polynomials of the $ARMA(2, 2)$ process $y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t + \alpha_1\varepsilon_{t-1} + \alpha_2\varepsilon_{t-2}$ can be factorized as $(1 - \varphi_1 L)(1 - \varphi_2 L)v_t = (1 - \psi_1 L)(1 - \psi_2 L)\varepsilon_t$

with $\varphi_1 = \psi_1$, then the original process reduces to the following $ARMA(1, 1)$:

$$(1 - \varphi_2 L)y_t = (1 - \psi_2 L)\varepsilon_t$$
$$\Rightarrow y_t = \varphi_2 y_{t-1} + \varepsilon_t - \psi_2\varepsilon_{t-1}$$

1.1.1 Additional Figures

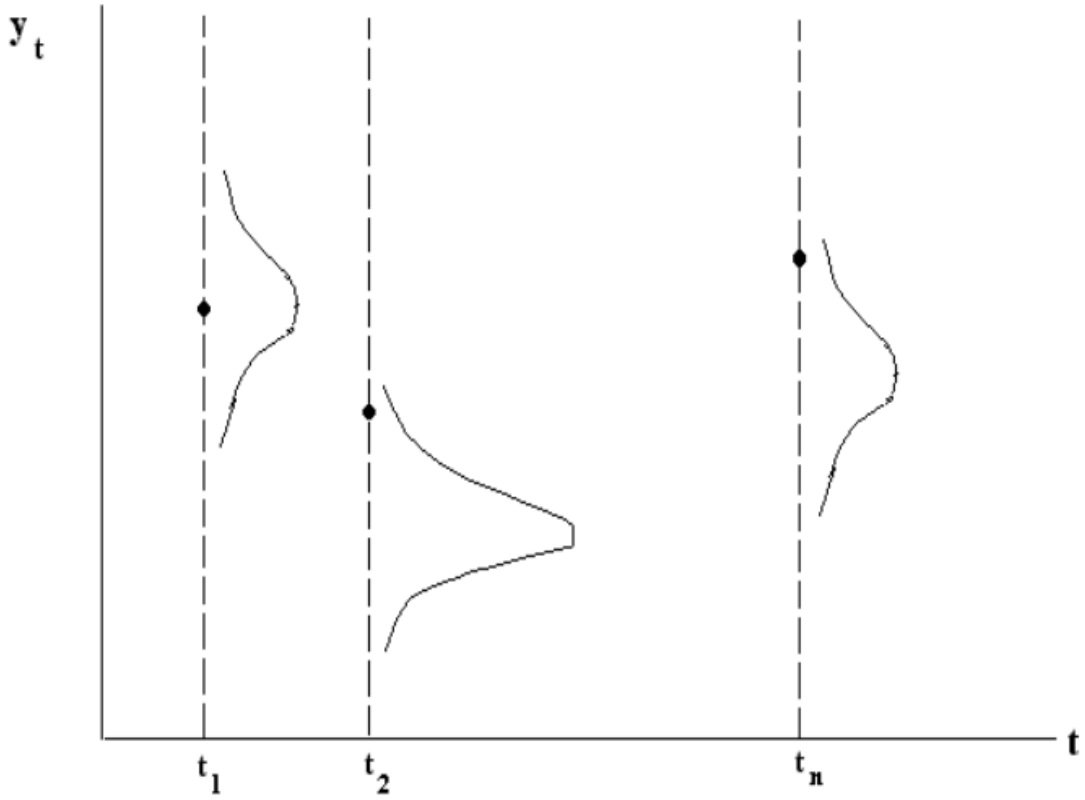


Figure 1.4: Additional figure from the original slide image set 1, image 1.

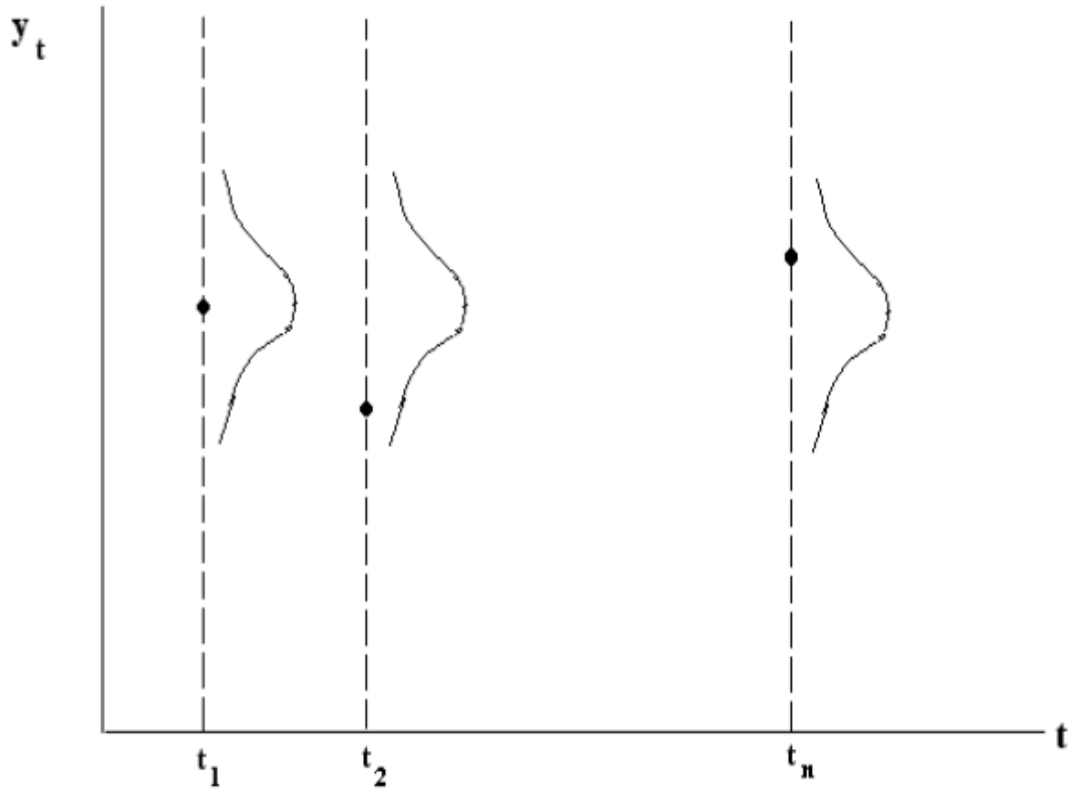


Figure 1.5: Additional figure from the original slide image set 1, image 2.

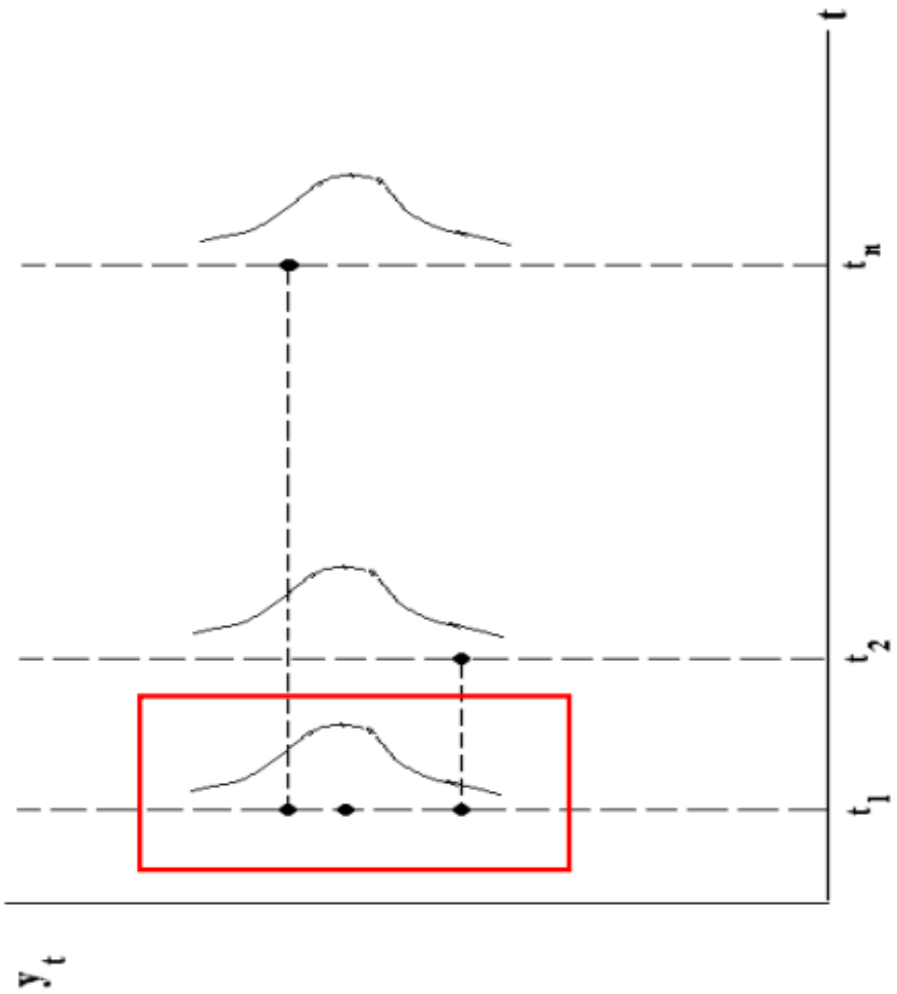


Figure 1.6: Additional figure from the original slide image set 1, image 3.

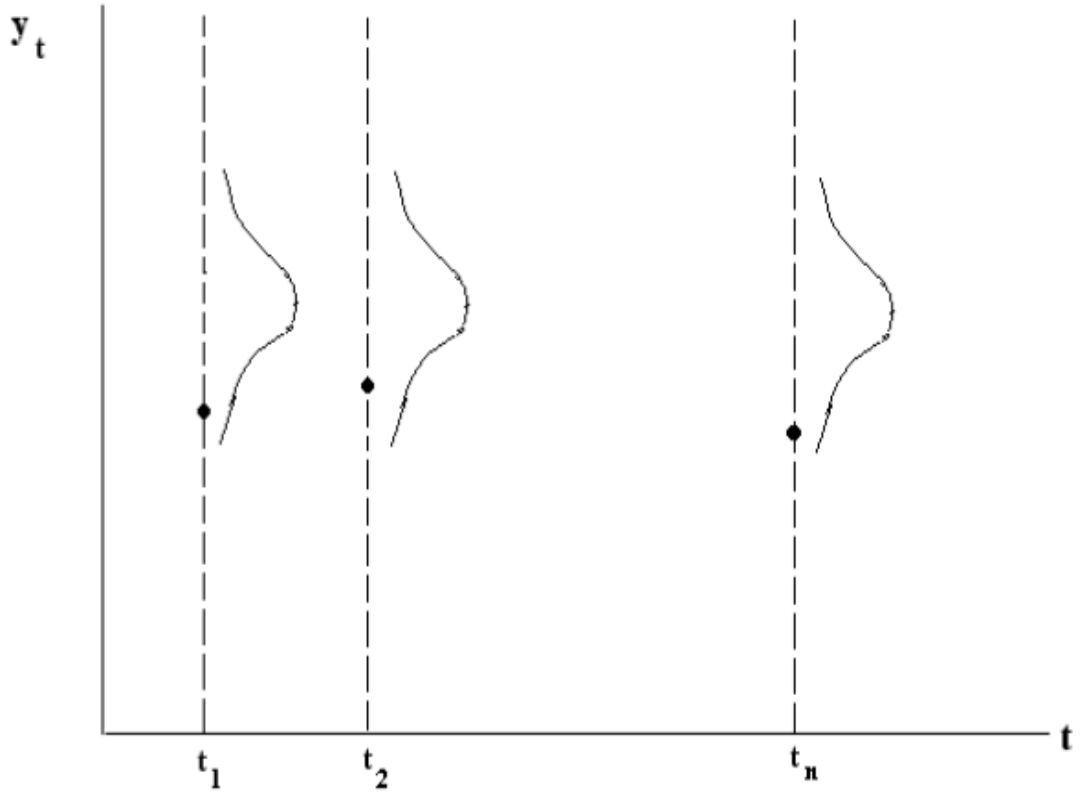


Figure 1.7: Additional figure from the original slide image set 1, image 4.

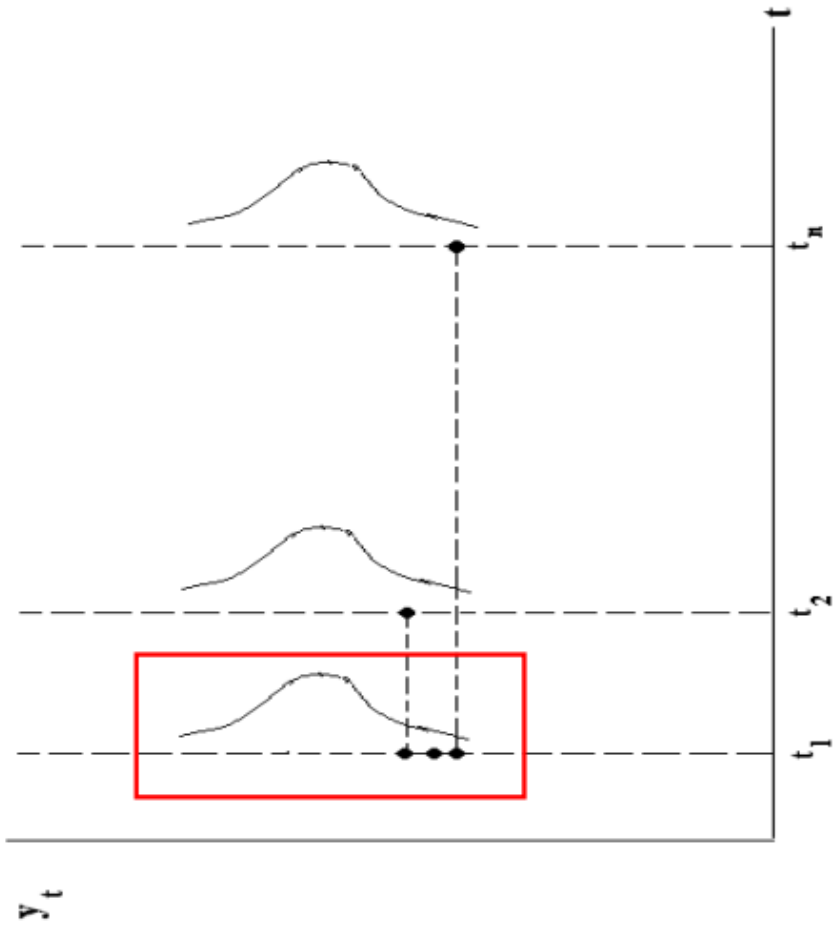


Figure 1.8: Additional figure from the original slide image set 1, image 5.

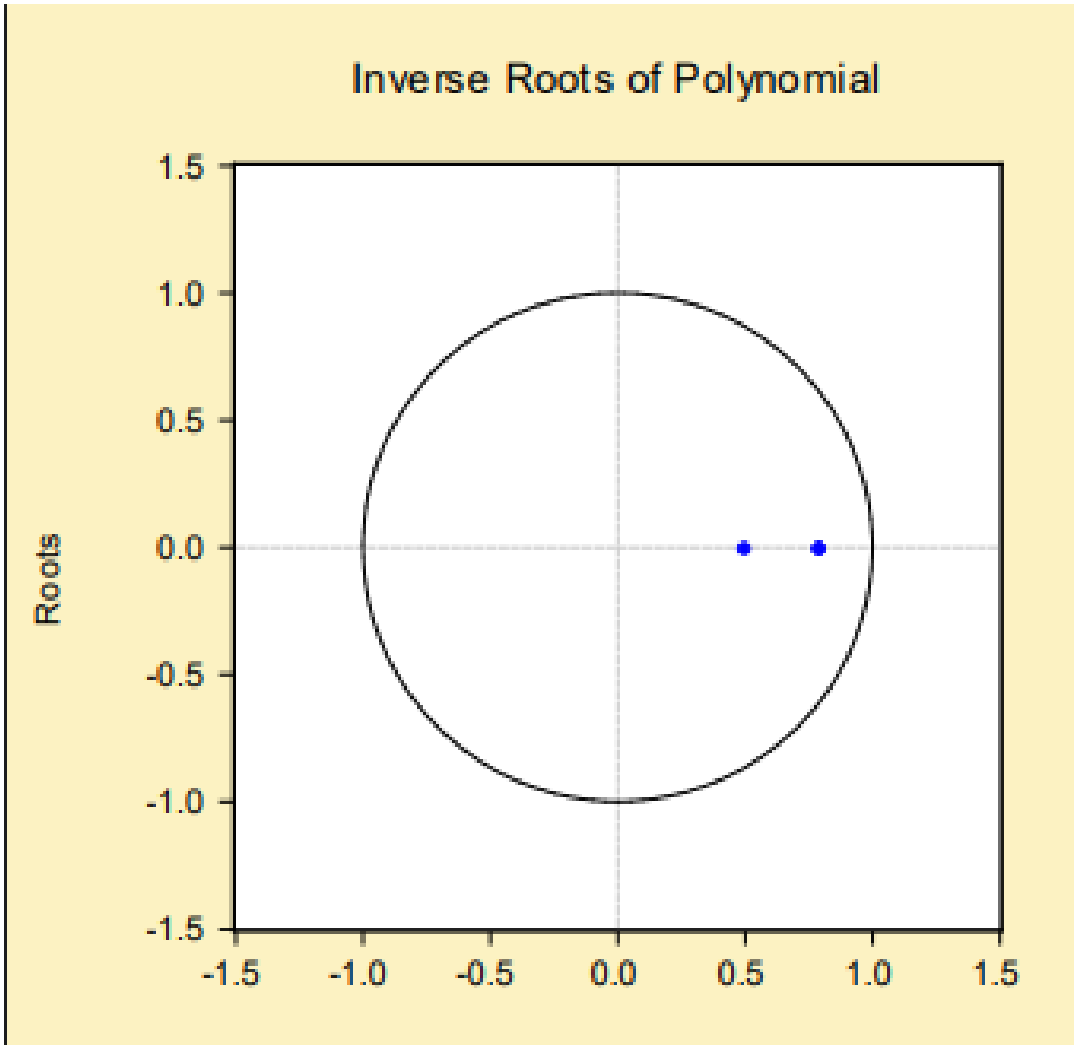


Figure 1.9: Additional figure from the original slide image set 1, image 6.

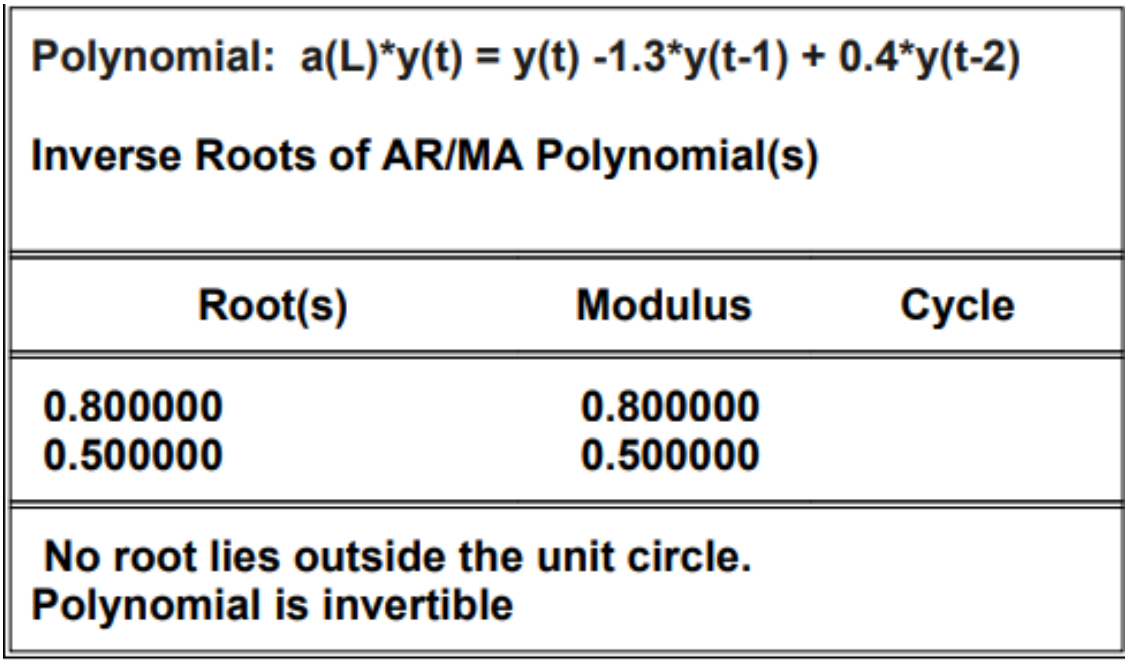


Figure 1.10: Additional figure from the original slide image set 1, image 7.

1.2 Set 2 – Choice of an ARMA Model

Choice of an ARMA model Given an observed stationary time series $\{y_t\}$, the choice of the specific ARMA model to estimate can be carried out by analyzing the features of the series in terms of its:

• autocorrelation function (ACF) • partial autocorrelation function (PACF) 4.1 The autocorrelation function The ACF describes the correlation between y_t and y_{t-k} as a function of the lag k (the correlogram is its graphical representation). For stationary processes we have $\rho_k = \frac{Cov(y_t, y_{t-k})}{Var(y_t)} \equiv \frac{\gamma_k}{\gamma_0}$ for $k = 0, 1, \ldots$

From a sample of T observations of the stochastic process $\{y_t\}$, the autocorrelations can be estimated by means of the sample correlation coefficients $\hat{\rho}_k$, computed as:

$$\hat{\rho}_k = \frac{\frac{1}{T-k}\sum_{t=k+1}^{T}[(y_t - \bar{y})(y_{t-k} - \bar{y})]}{\frac{1}{\pi}\sum_{t=1}^{T}(y_t - \bar{y})^2}$$

where $\bar{y}$ is the sample average. The estimated $\hat{\rho}_k$ can also be obtained as the OLS estimator β from a bivariate linear regressions:

$y_t = \delta + \beta\, y_{t-k} + \varepsilon_t$ (here, differently from the sample correlation coefficient, the estimator of ρ_k is computed with the same number of observations in the numerator and denominator, $T - k$).

If the process is stationary and ergodic (for mean, variance and covariances), then $\hat{\rho}_k$ is a consistent estimator of ρ_k (see below for Monte Carlo simulation results):

plim $\hat{\rho}_k = \rho_k$ Moreover, the following asymptotic property holds: if $\rho_1, \rho_2, \ldots, \rho_{j-1} \neq 0$ and $\rho_j = 0$, then $\hat{\rho}_i$ is asymptotically normally distributed

$\sqrt{T}\,(\hat{\rho}_j - \rho_j) \xrightarrow{d} N\,(0, v_j)$ with $v_i = 1 + 2\rho_1^2 + 2\rho_2^2 + \ldots + 2\rho_{i-1}^2$

It is therefore possible to test the hypothesis $\rho_i = 0$ using the estimated $\hat{\rho}_i$ and the critical values

of the standard normal distribution, since:

$$\sqrt{T}\frac{\hat{\rho}_j}{\sqrt{1+2\hat{\rho}_1^2+\ldots+2\hat{\rho}_{j-1}^2}} \underset{d}{\rightarrow} N(0,1)$$

Given the typical form of the autocorrelation function of $MA(q)$ processes (with $\rho_j \neq 0$ for $j \leq q$ and $\rho_j = 0$ for $j > q$, this test is particularly appropriate for choosing between an $MA(q)$ and an $MA(q-1)$ models by testing the null hypothesis $\rho_q = 0$ (i.e. the process is $MA(q-1)$) against the alternative $\rho_q \neq 0$ (favoring a

$MA(q)$ process).

AR and general $ARMA$ processes have more complicated theoretical autocorrelation functions (i.e. they do not go sharply to zero beyond a certain lag, as in the case of MA processes). In order to choose the appropriate order for a process with an autoregressive component, the analysis of the partial autocorrelation function provides necessary complementary information.

Note

the asymptotic distribution of $\hat{\rho}_i$ under the null hypothesis of no autocorre- lation at all lags between 1 and $j-1$ ($\rho_1 = \ldots = \rho_{j-1} = 0$) is $N(\rho_j, \frac{1}{T})$.

Consistency of $\hat{\rho}_k$: Monte Carlo simulation As an illustration of the consistency property of the sample autocorrelation coeffi- cients $\hat{\rho}_k$, we use the following $AR(1)$ data generating process:

$y_t = 0.8y_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim WN(0,1)$ for $t = 1, ..., T$ and generate 1000 time series $\{y_t\}$ for four different sample lengths $T = 50, 100, 200$ and 1000 using independent random drawings from the white noise, (standard) normally distributed process $\{\varepsilon_t\}$. For each simulated realization of the stochastic

process $\{y_t\}$, we compute the sample autocorrrelation coefficients $\hat{\rho}_k$ for each sample length T for $k = 1, ..., 20$.

The resulting plots of the ACF (correlograms) are displayed in Figure 1, together with the theoretical ACF (i.e. the population values given by $\rho_k = 0.8^k$). For each sample length ($T = 50, 100, 200, 1000$) the average $\hat{\rho}_k$ over the 1000 simulations are shown: as T increases, the estimated (average) $\hat{\rho}_k$ converge to the corresponding true values for all lags k.

Focusing on the first-order autocorrelation coefficient, Figure 2 shows the empirical distribution of the 1000 estimated $\hat{\rho}_1$ for each sample length T: as T increases, the distribution shrinks around the true value $\rho_1 = 0.8$ (vertical line).

1.0

—— ACF for T = 50 Theoretical ACF

- ACF for T = 100 - ACF for T = 1000

— ACF for T = 200 0.8 0.6 0.4 0.2 0.0

-0.2 Figure 1. Theoretical ACF and estimated ACF from $AR(1)$ process for $T = 50, 100, 200, 1000$. T = 50 T = 100 T = 1000 T = 200 0.3 0.4 0.5 0.6 0.9 0.7 0.2 0.8 1.0 Figure 2 Empirical distributions of $\hat{\rho}_1$ across 1000 simulations for $T = 50, 100, 200, 1000$.

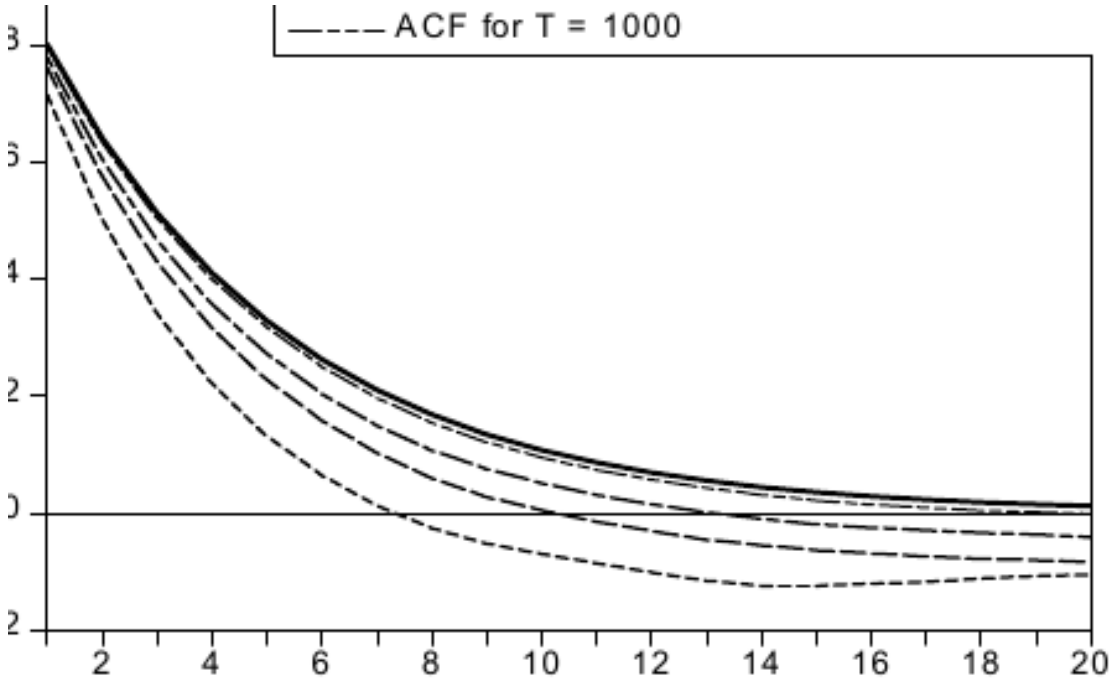


Figure 1.11: Course figure, source page 3, crop 1.

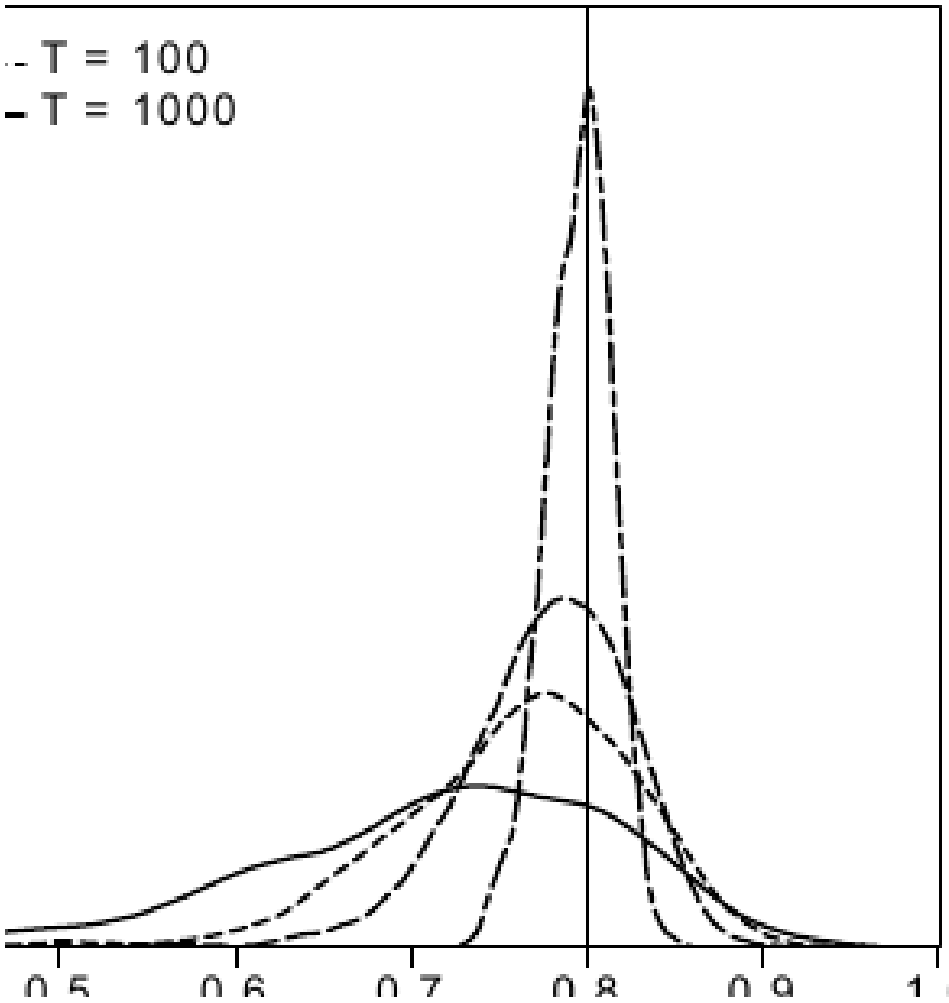


Figure 1.12: Course figure, source page 3, crop 2.

4.2 The partial autocorrelation function

The PACF provides a measure of the autocorrelation between y_t and y_{t-k} as a function of the lag k after having eliminated the effect of all intermediate lags $y_{t-1}, y_{t-2}, \ldots, y_{t-k+1}$. The partial autocorrelation coefficients of order k, denoted ρ_k^P, for a generic process $\{y_t\}$ can be estimated by applying OLS to (purely) autore-

gressive regression models of increasing order:

$$\begin{array}{lll} AR(1) & : & y_t = \delta + \beta_1 y_{t-1} + \varepsilon_t \qquad \Rightarrow \hat{\beta}_1 \text{ is an estimate of } \rho_1^P \\ AR(2) & : & y_t = \delta + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \varepsilon_t \quad \Rightarrow \hat{\beta}_2 \text{ is an estimate of } \rho_2^P \end{array}$$

$AR(k) : y_t = \delta + \beta_1 y_{t-1} + \ldots + \beta_k y_{t-k} + \varepsilon_t \Rightarrow \hat{\beta}_k$ is an estimate of ρ_k^P Note that the purpose of the above (autoregressive) regressions is not that of un-

covering the form of the "true" data generating process, but only that of collecting information on the autocorrelation structure of $\{y_t\}$ at different lags controlling for intermediate past levels of the series. Each regression delivers valuable information (partial autocorrelation coefficients) on the last lag in the estimated equation; in each regression the controlling variables are the intermediate past values of $\{y_t\}$.

If the true process generating the observed $\{y_t\}$ is $AR(p)$, then the partial autocorre- lation coefficients are equal to zero for lags $k > p$, and the corresponding estimated coefficients β_k yield consistent estimates of the ρ_k^P:

plim $\hat{\beta}_k = 0$ for $k > p$ and $\sqrt{T}\left(\hat{\beta}_k - 0\right) \xrightarrow{d} N(0,1)$ The estimate of the coefficients of partial autocorrelation can then provide useful information for choosing the order of the AR model.

In general, the joint analysis of the ACF and $PACF$ is useful for the selection of the appropriate model, taking into account that:

• for an $AR(p)$ process: the ACF decreases gradually (maybe with oscillations), whereas the PACF is zero for $k > p$;

• for an $MA(q)$ process: the ACF is zero for $k > q$, whereas the PACF decreases gradually (maybe with oscillations).

4.3 Specification (diagnostic) tests

After having chosen and estimated a specific $ARMA(p,q)$ model, the appropriate- ness of the model choice can be evaluated by analyzing the estimated residuals: if the model is correctly specified, it should deliver white noise residuals. The so-called Ljung-Box "portmanteau" test, Q_K, tests the hypothesis of absence of autocorrelation in the estimated residual series up to order K:

$Q_K = T(T+2)\sum_{k=1}^{K} \frac{1}{T-k} r_k^2$ where r_k is the k-th order autocorrelation coefficient of the estimated residuals, and T is the number of sample observations. Under the assumption of correct specification of an $ARMA(p,q)$ process we have $Q_K \sim \chi^2(K-p-q-1)$

4.4 Model selection criteria

General criteria for model selection can also be used to choose the most appropriate specification among various estimated ARMA models, characterized by different values of p and q. Two of the most widely used criteria are the following:

• Akaike's information criterion:

$AIC = \log \hat{\sigma}^2 + 2\frac{p+q+1}{T}$ • Bayesian information criterion:

$BIC = \log \hat{\sigma}^2 + \frac{p+q+1}{T} \log T$ where $\hat{\sigma}^2$ is the estimated residual variance, and $p+q+1$ is the number of esti- mated parameters (including a constant term). In comparing alternative models, preference goes to the one displaying lower values for the AIC and BIC criteria.

Both criteria penalize specifications with a high number of parameters ($p \in q$), but, since $\log T > 2$ for reasonable sample sizes, the BIC criterion favors more strongly "parsimonious" models, and is the one used more often in practice.

Examples of AR and MA processes. The following figures show some exam- ples of stationary $AR(1)$, $AR(2)$ and MA processes. For each process an time series of 100 observations is displayed, followed by the theoretical (solid line) and estimated (white bars) autocorrelation and partial autocorrelation functions (ACF and $PACF$) for the first 20 lags.

ACF and PACF of stationary $AR(1)$ models 2.5 ACF-Theoretical ACF-Estimated PACF-Theoretical PACF-Estimated 0.0

-2.5 5.0 ACF-Theoretical CF-Estimated PACF-Theoretical PACF-Estimated $= 0.80 \cdot y_{t-1} + \varepsilon_t$ 2.5 0.0

-2.5 5.0 ACF-Theoretical ACF-Estimated PACF-Theoretical PACF-Estimated $= -0.80 \cdot y_1 + \varepsilon$ _1 2.5 0.0

-2.5 ACF and PACF of stationary AR(2) models 5.0 CF-Theoretical CF Estimated PACF-Theoretical PACF-Estimated $y_t = 0.50y_t$ 40.40y_1 2.5 0.0

-2.5 ACF-Theoretical PACF-Theoretical PACF-Estimated $y_t = -0.80y_{t-2} + \varepsilon_t$ ACE Estimated

-5 = 1.30v $\frac{1}{1}$ -0.80 y_{t-2} $+\varepsilon_t$ ACF-Theoretical PACF-Theoretical y_t ACF-Estimated PACF-Estimated

-5

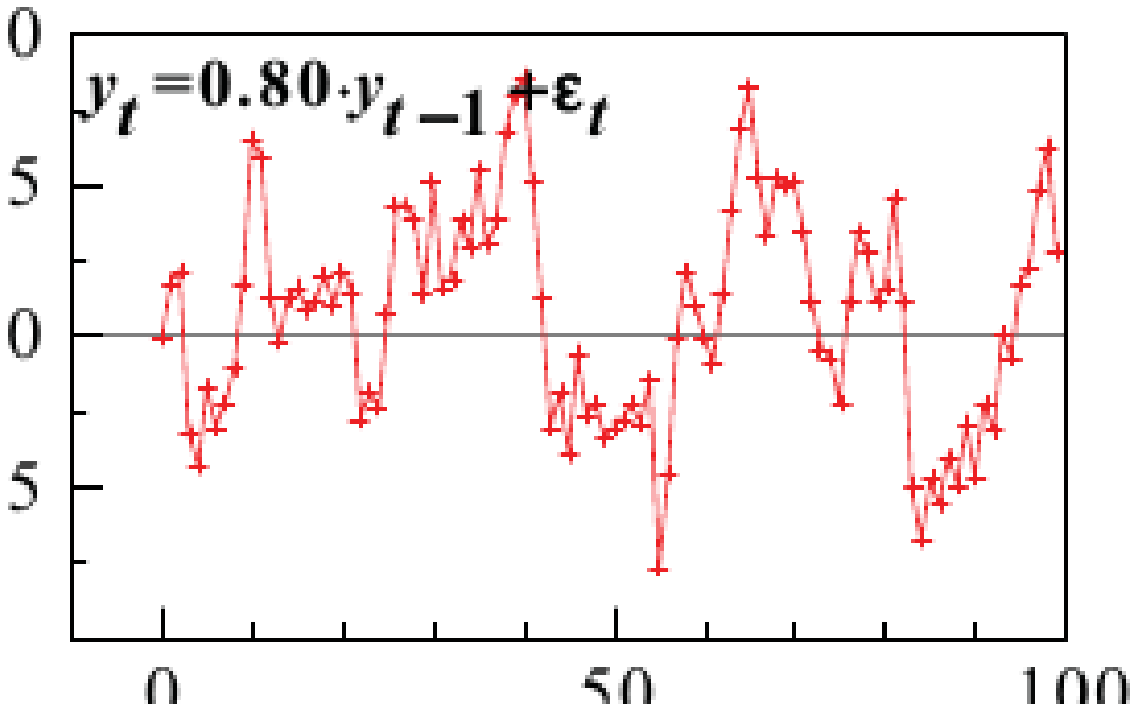


Figure 1.13: Course figure, source page 6, crop 1.

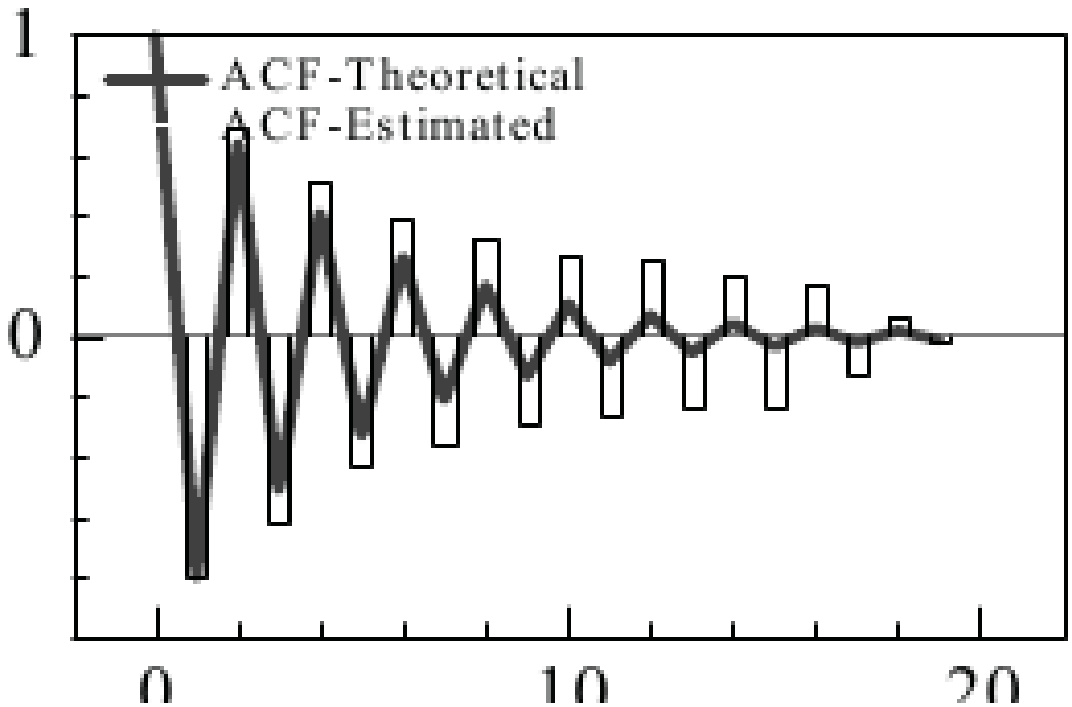


Figure 1.14: Course figure, source page 6, crop 2.

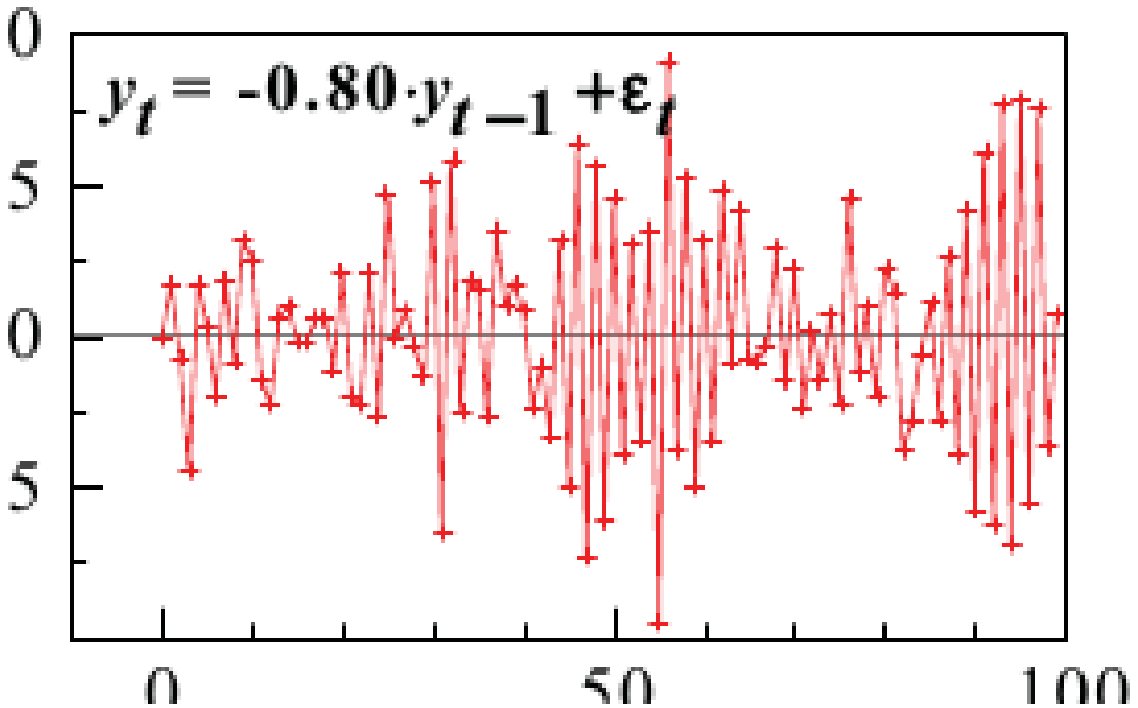


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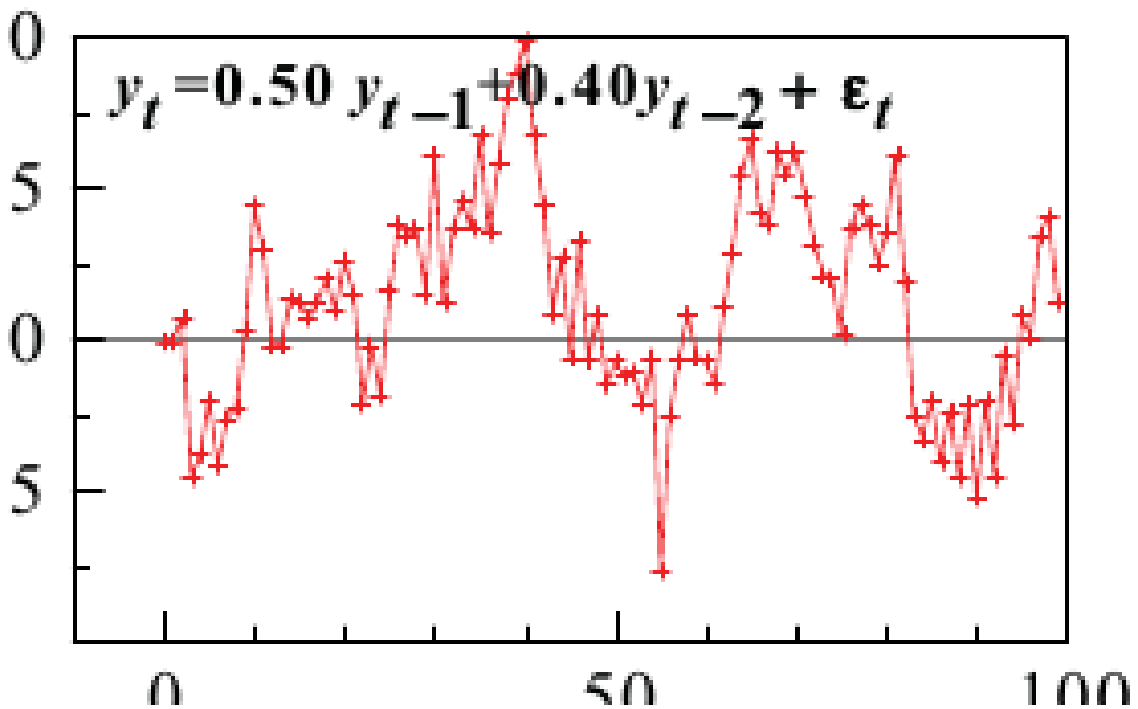


Figure 1.16: Course figure, source page 6, crop 4.

ACF and PACF of MA models 5.0 • PACF-Theoretical PACF-Estimated ACF-Theoretical ACF-Estimated $y_t = \varepsilon_t - 0.90 \cdot \varepsilon_{t-1}$ 2.5 ÷ H U 0.0

-2.5 $y_t = \varepsilon_t + 0.90 \cdot \varepsilon_{t-1}$ • ACF-Theoretical ACF-Estimated • PACF-Theoretical PACF-Estimated 2.5 0.0 MI

-2.5 $-1^{\pm 0.50 \cdot \varepsilon_t} - 2^{\pm 0.50 \cdot \varepsilon_t} - 3 = \epsilon_f - 0.60 \cdot \epsilon_f$ • ACF-Theoretical ACF-Estimated PACF-Theoretical PACF-Estimated 2.5 0.0 M

-2.5

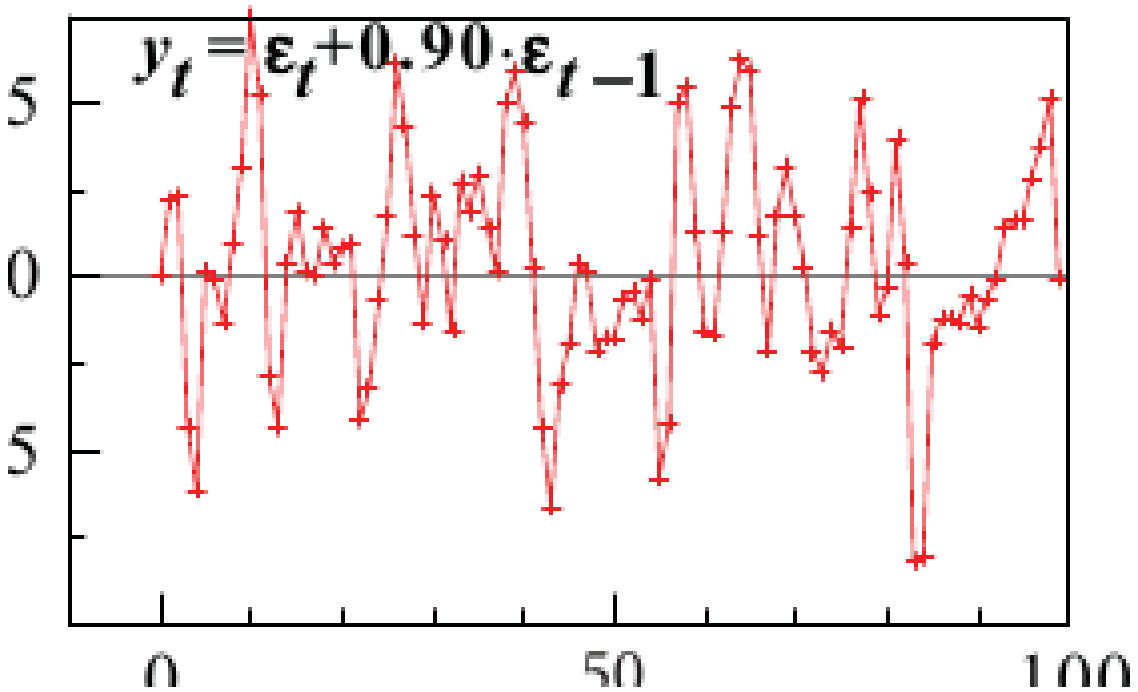


Figure 1.17: Course figure, source page 7, crop 1.

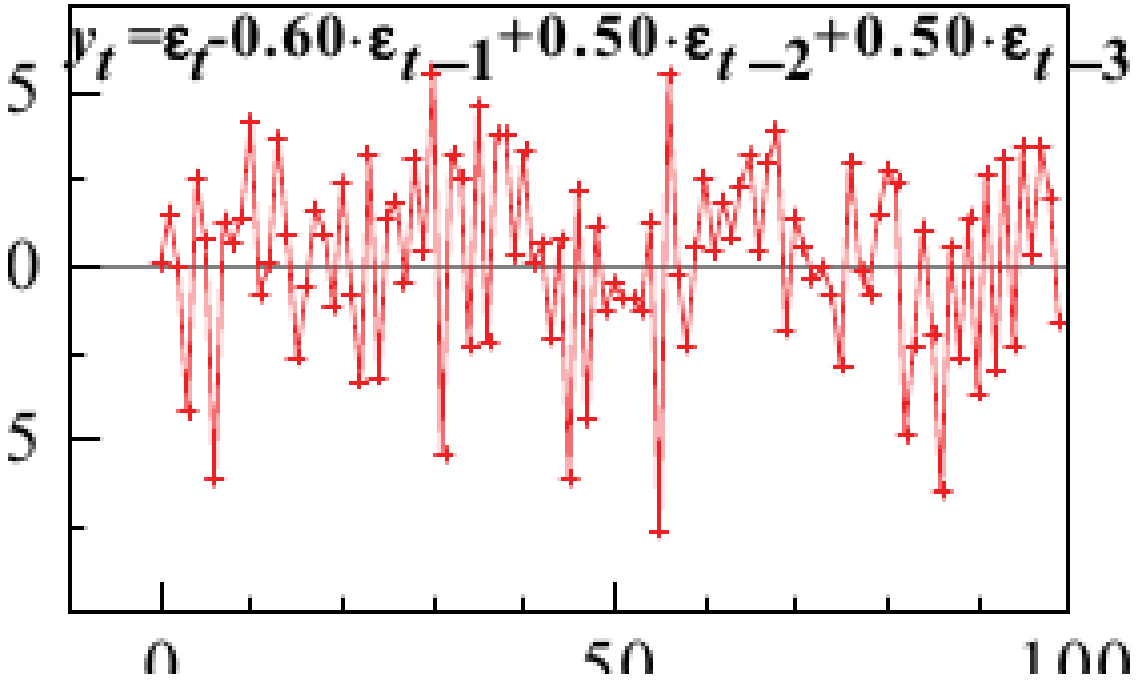


Figure 1.18: Course figure, source page 7, crop 2.

Estimation of ARMA models (outline) Several methods can be used to estimate the parameters of $ARMA$ models. We will start with some special cases before presenting the general method, based on the maximum likelihood principle.

$AR(p)$ process 5.1 In the case of the following process $y_t = \delta + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \ldots + \theta_n y_{t-n} + \varepsilon_t$ with $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$ we have

$E(y_{t-i}\varepsilon_t) = 0$ for $j = 1, \ldots p$ Interpreting the process as a regression model, there is no correlation between the error term ε_t and the regressors $y_{t-1}, y_{t-2}, \ldots, y_{t-p}$. However, each ε_t is correlated with the realization of the process at the same time date y_t (i.e. $E(y_t\varepsilon_t) = \sigma_\varepsilon^2$) and,

through the autoregressive nature of the process, with all future realizations y_{t+1}, y_{t+2},... This violates one of the basic assumptions of the classical linear regression model (CLRM), i.e. strict exogeneity of the regressors (stating that regressors should be orthogonal to the error term for all observations). Therefore, estimation of the AR process by ordinary least squares yields biased estimates of the population parameters in finite samples. However, the orthogonality of the error term with the set of regressors for each date t is enough to guarantee that OLS estimation delivers consistent and asymptotically normal estimators of the process parameters.

EXAMPLE – Illustration

Example. Take the following simple stationary $AR(1)$ form for the data generating process of $\{y_t\}$:

$y_t = \theta y_{t-1} + \varepsilon_t$ with $|\theta| < 1$ and $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$ Given the white noise properties of ε_t, we have $E(y_{t-1}\varepsilon_t) = 0$ but

$$E(y_t\,\varepsilon_t) = \sigma_\varepsilon^2$$
$$E(y_{t+1}\varepsilon_t) = E\left[(\theta y_t + \varepsilon_{t+1})\,\varepsilon_t\right] = \theta E(y_t\varepsilon_t) = \theta\sigma_c^2$$

In general $E(y_{t+i}\varepsilon_t) = \theta^i\sigma_\varepsilon^2 \neq 0$ for all $i > 0$ so that each ε_t is correlated with all present and future realizations of the process y_{t+i} (for $i \geq 0$). If the following regression is estimated by OLS on the realized

values of $\{y_t\}$ over a sample of T observations:

$$y_t = \delta + \beta\, y_{t-1} + u_t$$

the correlation between the error term and future values of the regressor will deliver biased estimates of θ in finite samples. The bias disappears as the sample size $T \to \infty$, so that the OLS estimator is consistent. To evaluate the magnitude of the bias we perform a Monte Carlo exercise, running the above OLS regression on 1000 generated series for $\{y_t\}$ over different sample sizes $T = 10, \ldots 200$ and with different values of the population parameter $\theta = 0.2, 0.5, 0.9$. The average estimated β

across replications is plotted in the figure below for all the sample sizes and the true values of θ considered.

Average OLS estimated AR(1) coefficients 1.0 theta=0.9 0.8 0.6 theta=0.5 0.4 theta=0.2 0.2 0.0 Sample size T Figure 3. Average OLS estimate of $AR(1)$ parameter θ for sample size $T = 10, \ldots 200$. It is apparent from Figure 3 that in small samples there is a sizable downward bias that tends to disappear as the sample size increases, becoming relatively negligible when the number of available observations reaches 200 (the largest sample size in the simulation). It can be shown analytically that the expected value of the OLS estimate of the $AR(1)$ parameter is $E(\hat{\beta}_{OLS}) = \theta\left(1 - \frac{2}{T}\right)$

and the bias is $-\frac{2\theta}{T}$. Therefore, in finite samples the estimated persistence of the process (captured by the magnitude of θ) is biased towards 0.

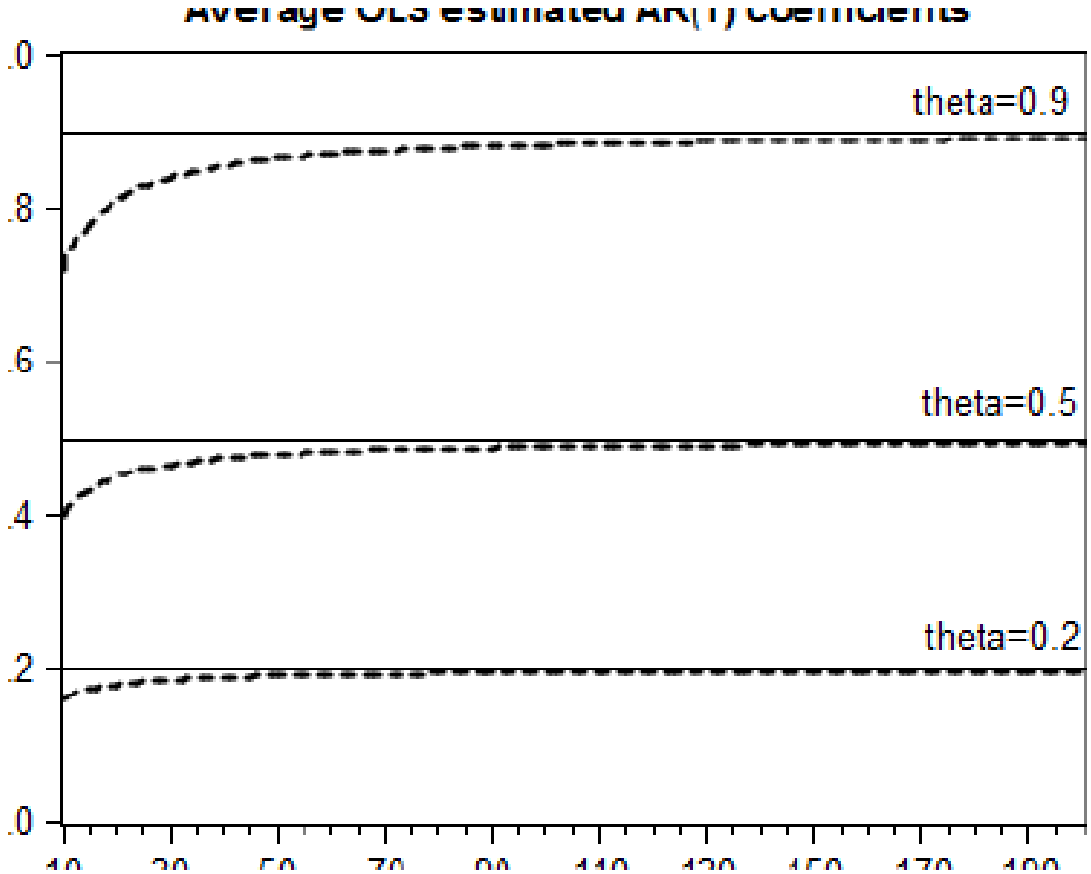


Figure 1.19: Course figure, source page 9, crop 1.

5.2 $MA(1)$ process

In the case of $y_t = \mu + \varepsilon_t + \alpha_1\varepsilon_{t-1}$ for $t = 1, 2, ...T$ with $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$ it is not possible to immediately apply a regression method since the process $\{\varepsilon_t\}$ is not observable. However, if the $MA(1)$ process is invertible

(and therefore ε_{t-1} can be expressed as an infinite linear combination of past ob- served values $y_{t-1}, y_{t-2}, ...$, it is nevertheless possible to obtain consistent estimators of μ and α_1.

Starting from the (theoretical) residual sum of squares (RSS) to be minimized:

$S(\mu, \alpha_1) = \sum^T (y_t - \mu - \alpha_1\varepsilon_{t-1})^2$ and expressing (under invertibility, ensured by $|\alpha_1| < 1$) each term ε_{t-1} as $\varepsilon_{t-1} = \frac{1}{1+\alpha_t L}(y_{t-1} - \mu) = (1 - \alpha_1 L + \alpha_1^2 L^2 - ...)(y_{t-1} - \mu)$

$= \sum_{j=1}^{\infty}(-\alpha_1)^j(y_{t-1-j} - \mu)$ $S(\mu, \alpha_1)$ can be rewritten as follows:

$$S(\mu, \alpha_1) = \sum_{i=1}^{T}\left(y_t - \mu - \alpha_1\left(\sum_{i=1}^{\infty}(-\alpha_1)^j(y_{t-1-j} - \mu)\right)\right)^2$$

Operationally, since an infinite series of past observations on y is not available, $S(.)$ is approximated by $S(\mu, \alpha_1) = \sum_{i=1}^{T}\left(y_t - \mu - \alpha_1\left(\sum_{i=1}^{t-2}(-\alpha_1)^j(y_{t-1-j} - \mu)\right)\right)^2$

In principle, minimization of $S(.)$ with respect to μ and α_1 yields the first-order conditions necessary to obtain consistent estimators $\hat{\mu}$ and $\hat{\alpha}_1$. Being $S(.)$ a (highly) non-linear function of the parameters, non linear least squares (NLS) techniques are required. Such method is applied by means of numerical optimization procedures to minimize $S(.)$. The resulting NLS estimators are consistent and asymptotically

normal.

5.3 Maximum likelihood estimation

An alternative estimation procedure, that can be applied to general ARMA processes, is based on the maximum likelihood (ML) principle, here adapted to the case of

observations that are not independent. In order to apply ML estimation, an as- sumption on the form of the distribution of ε is required: in the following such distribution is assumed to be normal.

Principle: given T observations of $\{y_t\}$, if the observations were independent, the likelihood function $L(.)$ could be written as $L(\boldsymbol{\vartheta}) = f(y_1, y_2, ..., y_T; \boldsymbol{\vartheta}) = \prod_{t=1}^{T} f(y_t; \boldsymbol{\vartheta})$

where $f(y_1, y_2, ..., y_T; \vartheta)$ is the joint density function of the T observations, $f(y_t; \vartheta)$ is the density of the t-th observation and vector $\boldsymbol{\vartheta}$ collects all parameters (to be estimated) defining the process $\{y_t\}$.

Since the observations of the process $\{y_t\}$ are not independent, the likelihood function must be obtained using a sequence of conditional density functions:

$L(\boldsymbol{\vartheta}) = \prod_{t=0}^{T} f(y_t \mid y_{t-1}, ..., y_1; \boldsymbol{\vartheta}) \cdot f(y_1; \boldsymbol{\vartheta})$ where $f(y_t \mid y_{t-1}, ..., y_1; \vartheta)$ is the density of y_t conditional on past realizations of the

process, and $f(y_1; \vartheta)$ is the (unconditional) density of the first observation in the sample. The corresponding loglikelihood function is then:

$\log L(\boldsymbol{\vartheta}) = \sum_{t=0}^{T} \log f(y_t \mid y_{t-1}, ..., y_1; \boldsymbol{\vartheta}) + \log f(y_1; \boldsymbol{\vartheta})$ The maximum likelihood estimates of parameters in ϑ are obtained by maximiz-

ing the loglikelihood function $\log L(\boldsymbol{\vartheta})$ with respect to each element of vector $\boldsymbol{\vartheta}$, ignoring the last term. Under stationarity and with a sufficiently large sample, the unconditional density of the first observation can be dropped from the maximization procedure.

The $ARMA(1,1)$ case. Consider the process $y_t = \theta_1 y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$ with ε normally distributed white noise, $N(0, \sigma_\varepsilon^2)$. Conditioning on the initial ob- servation y_1 (that we consider as fixed), the likelihood function to be maximized

is:

$$L(\theta_1, \alpha_1, \sigma_\varepsilon^2) = \prod_{i=1}^{T} f(y_t \mid y_{t-1}, \ldots, y_1; \theta_1, \alpha_1, \sigma_\varepsilon^2)$$

$$= \prod_{t=2}^{T} \left(\frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \right) e^{-\frac{\varepsilon_t^2}{2\sigma_\varepsilon^2}} = \left(\frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \right)^{T-1} e^{-\frac{\sum_{t=2}^{T} \varepsilon_t^2}{2\sigma_\varepsilon^2}}$$

with $\varepsilon_t = y_t - \theta_1 y_{t-1} - \alpha_1 \varepsilon_{t-1}$ from which we get $\log L(\theta_1, \alpha_1, \sigma_\varepsilon^2) = -\frac{T-1}{2} \log\left(2\pi\sigma_\varepsilon^2\right) - \frac{1}{2} \frac{\sum_{t=2}^{T} \varepsilon_t^2}{\sigma^2}$

To implement the maximization of $\log L(\theta_1, \alpha_1, \sigma_\varepsilon^2)$ it is necessary to recursively generate the

error terms ε_t starting from an assumption on the value of ε_1, usually set equal to its expected value: $\varepsilon_1 = 0$. After substituting the terms ε_t as func-

tions of observed y_t and parameters into the loglikelihood function, it is possible to proceed to the maximization making use of numerical techniques.

Special cases:

(i) $AR(1)$ process: here $\alpha_1 = 0$ and $\varepsilon_t = y_t - \theta_1 y_{t-1}$. The maximum likelihood estimator of θ_1 coincides with the OLS estimator.

(ii) $MA(1)$ process: here $\theta_1 = 0$ and it is necessary to generate recursively the ε's, starting from $\varepsilon_1 = 0$:

$$\varepsilon_2 = y_2 - \alpha_1 \varepsilon_1 = y_2$$
$$\varepsilon_3 = y_3 - \alpha_1 \varepsilon_2 = y_3 - \alpha_1 y_2$$
$$\varepsilon_4 = y_4 - \alpha_1 \varepsilon_3 = y_4 - \alpha_1 y_3 + \alpha_1^2 y_2$$

$\varepsilon_T = y_T - \alpha_1 \varepsilon_{T-1} = \sum_{t=0}^{T-2} (-\alpha_1)^t y_{T-t}$ After substitution of the ε series into the loglikelihood function, maximization is performed by numerical techniques; the estimator of α_1 in this case has no

closed-form expression.

Exercise. Suppose that we want to estimate the following dynamic model $y_t = \theta y_{t-1} + u_t$ with $u_t = \varepsilon_t + \alpha \varepsilon_{t-1}$, $|\theta| < 1$, $\varepsilon \sim WN(0, \sigma^2)$ being interested only in estimating θ.

1. What are the properties of the OLS estimator of θ?

2. Is it possible to use y_{t-2} as an instrument to get a consistent estimator of θ by means of instrumental variable (IV) estimation?

Forecasting with ARMA models Optimal predictor 6.1 Given a stochastic process $\{y_t\}$, it is possible to construct predictions (or "fore- casts") over different horizons spanning h periods, using the observations on the process up to time T. Using the information set $I_T = \{y_{-\infty}, \ldots, y_{T-1}, y_T\}$ we can predict y_{T+h} (the value of y after h periods). Let the prediction on a h-

period horizon be denoted by $\hat{y}_{T+h|T}$. A predictor will in general be a function of the elements in the information set I_T:

$\hat{y}_{T+h|T} = q(y_T, y_{T-1}, ...)$ The criterion to choose the function $q(.)$ is the minimization of the mean square forecast error (MSE):

$\min MSE = E\left(\left(y_{T+h} - \hat{y}_{T+h|T}\right)^2 \mid I_T\right)$ It is easy to prove that the optimal predictor of y_{T+h} is the expected value of y_{T+h} conditional onto the information available in T:

$\hat{y}_{T+h|T} = E\left(y_{T+h} \mid I_T\right) = E\left(y_{T+h} \mid y_T, y_{T-1}, \ldots\right)$ Proof. The stationary $ARMA(p, q)$ process $\theta(L) y_t = \alpha(L) \varepsilon_t$

can be written as an $MA(\infty)$:

$y_t = \theta(L)^{-1} \alpha(L) \varepsilon_t = \pi(L) \varepsilon_t = \sum_{i=0}^{\infty} \pi_j \varepsilon_{t-j}$ Therefore:

$$y_{T+h} = \varepsilon_{T+h} + \pi_1 \varepsilon_{T+h-1} + \cdots + \pi_{h-1} \varepsilon_{T+1} + \underbrace{\pi_h \varepsilon_T + \pi_{h+1} \varepsilon_{T-1} + \ldots}_{\text{information up to } T}$$

$$= \varepsilon_{T+h} + \pi_1\varepsilon_{T+h-1} + \cdots + \pi_{h-1}\varepsilon_{T+1} + \sum_{i=0}^{\infty} \pi_{h+j}\varepsilon_{T-j}$$

A predictor of y_{T+h} can be written as a linear combination of the ε's with coefficients π^* to be determined:

$$\hat{y}_{T+h|T} = \sum_{i=0}^{\infty} \pi^*_{h+j}\varepsilon_{T-j}$$

The π^* coefficients are chosen to minimize the mean square forecast error. Given the forecast error $y_{T+h} - \hat{y}_{T+h|T} = \varepsilon_{T+h} + \pi_1\varepsilon_{T+h-1} + \ldots + \pi_{h-1}\varepsilon_{T+1} + \sum_{i=0}^{\infty}\left(\pi_{h+j} - \pi^*_{h+j}\right)\varepsilon_{T-j}$

since $E(\varepsilon_t\varepsilon_s) = 0$ for all $t \neq s$, the MSE is $E\left(\left(y_{T+h} - \hat{y}_{T+h|T}\right)^2 \mid I_T\right) = \sigma^2_\varepsilon + \pi^2_1\sigma^2_\varepsilon + \ldots + \pi^2_{h-1}\sigma^2_\varepsilon$ $+ \sum_{l=1}^{\infty}\left(\pi_{h+j} - \pi^*_{h+j}\right)^2\sigma^2_\varepsilon$

To minimize MSE we must set $\pi^*_{h+i} = \pi_{h+i}$ from which $\hat{y}_{T+h|T} = \sum_{i=0}^{\infty}\pi_{h+j}\varepsilon_{T-j} = E(y_{T+h} \mid y_T,$ $y_{T-1}, ...)$

$AR(1)$ process 6.1.1 For the process (expressed as deviations from the mean μ and with $|\theta| < 1$) $y_t = \theta y_{t-1} + \varepsilon_t$ the optimal predictors at various horizons are $\hat{y}_{T+1|T} = E(y_{T+1} \mid y_T, y_{T-1}, ...) =$ $\theta y_T + E(\varepsilon_{T+1} \mid y_T, ...) = \theta y_T$

$$\hat{y}_{T+2|T} = E(y_{T+2} \mid y_T, y_{T-1}, ...) = \theta\,\hat{y}_{T+1|T} + E(\varepsilon_{T+2} \mid y_T, ...) = \theta^2\,y_T$$

$$\hat{y}_{T+h|T} = E(y_{T+h} \mid y_T, y_{T-1}, ...) = \theta\,\hat{y}_{T+h-1|T} + E(\varepsilon_{T+h} \mid y_T, ...) = \theta^h\,y_T$$

For $h \to \infty$, $\hat{y}_{T+h|T} \to 0$ and the predicted level of the process tends to the uncon- ditional mean.

6.1.2 $MA(1)$ process

In the case of an invertible moving average ($|\alpha| < 1$) $y_t = \varepsilon_t + \alpha\,\varepsilon_{t-1}$ we can express the ε's in terms of the observed y's, and obtain the following pre- dictions:

$$\hat{y}_{T+1|T} = E(y_{T+1} \mid y_T, y_{T-1}, ...) = E(\varepsilon_{T+1} \mid y_T, ...) + \alpha E(\varepsilon_T \mid y_T, ...) = \alpha\varepsilon_T$$
$$= \alpha\left(\sum_{j=1}^{\infty}(-\alpha)^j y_{T-j}\right)$$

$\hat{y}_{T+h|T} = E(y_{T+h} \mid y_T, y_{T-1}, ...) = 0$ for $h \geq 2$

6.1.3 $ARMA(p, q)$ process

The same procedure used to construct predictions for $AR(1)$ and $MA(1)$ processes can be extended to stationary and invertible $ARMA(p, q)$ processes. For example, in the $ARMA(2, 1)$ case $y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t + \alpha_1\varepsilon_{t-1}$ we get the following one-step ahead forecast:

$\hat{y}_{T+1|T} = \theta_1 y_T + \theta_2 y_{T-1} + \alpha_1\varepsilon_T$ In order to express $\hat{y}_{T+1|T}$ in terms of observed values of $\{y_t\}$ we exploit the invert-

ibility property of the process, that we now write as $y_t - \theta_1 y_{t-1} - \theta_2 y_{t-2} = (1 + \alpha_1 L)\varepsilon_t$ from which, inverting the moving average component, we get ε_t in terms of current and past values of $\{y_t\}$:

$$\begin{aligned}\varepsilon_t &= (1 + \alpha_1 L)^{-1}(y_t - \theta_1 y_{t-1} - \theta_2 y_{t-2}) \\ &= \sum_{i=0}^{\infty}(-\alpha_1)^j L^j (y_t - \theta_1 y_{t-1} - \theta_2 y_{t-2})\end{aligned}$$

Using that expression, ε_T can be expressed in terms of observed y_{T-j} $(j = 0, ...\infty)$ in the one-step ahead forecast above:

$$\begin{aligned}\hat{y}_{T+1|T} &= \theta_1 y_T + \theta_2 y_{T-1} + \alpha_1 \varepsilon_T \\ &= \theta_1 y_T + \theta_2 y_{T-1} + \alpha_1 \sum_{i=0}^{\infty}(-\alpha_1)^j L^j (y_T - \theta_1 y_{T-1} - \theta_2 y_{T-2})\end{aligned}$$

$$= (\theta_1 + \alpha_1) y_T + (\theta_2 - \alpha_1\theta_1 - \alpha_1^2) y_{T-1} - \alpha_1(\theta_2 - \alpha_1\theta_1 - \alpha_1^2) y_{T-2} + \dots$$

Then, the two-period ahead forecast is obtained as $\hat{y}_{T+2|T} = \theta_1 \hat{y}_{T+1|T} + \theta_2 y_T = (\theta_1^2 + \theta_1\alpha_1 + \theta_2) y_T + \theta_1(\theta_2 - \alpha_1\theta_1 - \alpha_1^2) y_{T-1}$

$-\theta_1\alpha_1 \left(\theta_2 - \alpha_1\theta_1 - \alpha_1^2\right) y_{T-2} + \dots$ and for longer horizons $(h \geq 3)$ forecast are obtained recursively as:

$$\hat{y}_{T+h|T} = \theta_1 \hat{y}_{T+h-1|T} + \theta_2 \hat{y}_{T+h-2|T}$$

6.2 Prediction accuracy

The forecasting power of an $ARMA$ model can be evaluated by estimating the model on a sample shorter than the whole set of observations available (estimation) sample), and then constructing the predictors for the out-of-sample observations until the last date available (forecast sample). To evaluate the prediction accuracy, the mean square (or variance) forecast error is constructed:

$$C_h = E\left(\left(y_{T+h} - \underbrace{\hat{y}_{T+h|T}}_{F(x)\to I_T}\right)^2\right) = Var\left(y_{T+h} \mid I_T\right)$$

In the general stationary $ARMA(p, q)$ case, with the following $MA(\infty)$ representa- tion (with $\alpha_0 = 1$) $y_t = \varepsilon_t + \alpha_1\varepsilon_{t-1} + \alpha_2\varepsilon_{t-2} + \ldots + \ldots = \sum_{i=1}^{\infty} \alpha_j \varepsilon_{t-j}$

the h-period ahead prediction is (in terms of the stochastic elements $\varepsilon_T, \varepsilon_{T-1}, ...$) $\hat{y}_{T+h|T} = E(\varepsilon_{T+h} + \alpha_1\varepsilon_{T+h-1} + \alpha_2\varepsilon_{T+h-2} + \ldots + \alpha_h\varepsilon_T + \alpha_{h+1}\varepsilon_{T-1} + \ldots | \varepsilon_T, \varepsilon_{T-1}, \ldots)$

$= \alpha_h\varepsilon_T + \alpha_{h+1}\varepsilon_{T-1} + \ldots = \sum_{i=1}^{\infty} \alpha_j \varepsilon_{T+h-j}$ with the following prediction error $y_{T+h} - \hat{y}_{T+h|T} = \varepsilon_{T+h} + \alpha_1\varepsilon_{T+h-1} + \alpha_2\varepsilon_{T+h-2} + \ldots + \alpha_{h-1}\varepsilon_{T+1}$

$= \sum_{j=1}^{n-1} \alpha_j \varepsilon_{T+h-j}$ with variance $E\left(\left(y_{T+h} - \hat{y}_{T+h|T}\right)^2\right) = \sigma_\varepsilon^2 + \alpha_1^2\sigma_\varepsilon^2 + \alpha_2^2\sigma_\varepsilon^2 + \ldots + \alpha_{h-1}^2\sigma_\varepsilon^2$

$= \sigma_\varepsilon^2 \sum_{i=1}^{h-1} \alpha_j^2$ In the case of stationary $ARMA$ processes, the variance of the forecasting error tends to increase with the forecasting horizon, reaching the unconditional variance of the process

for $h \to \infty$. For each ARMA model there is an horizon of intermediate length over which the predictions are more efficient than the unconditional mean

(e.g. for an $MA(1)$ process this applies only to the one-period ahead prediction; for an $AR(1)$ process this happens over all horizons, with efficiency gradually decreasing as h extends into the future).

Choice of an ARMA model The choice of the specific ARMA model to estimate can be carried out by analyzing the features of the series in terms of its:

autocorrelation function (ACF) • partial autocorrelation function (PACF)

Autocorrelation function ACF: correlation between y_t and y_{t-k} as a function of lag k (graphical representation: correlogram). For stationary processes:

$\rho_k = \frac{Cov(y_t, y_{t-k})}{Var(y_t)} \equiv \frac{\gamma_k}{\gamma_0}$ for $k = 0, 1, \ldots$ From a sample of T observations of $\{y_t\}$, autocorrelations can

be estimated by means of the correlation coefficients $\hat{\rho}_k$, computed as:

$$\hat{\rho}_k = \frac{\frac{1}{T-k}\sum_{t=k+1}^{T}[(y_t - \bar{y})(y_{t-k} - \bar{y})]}{\frac{1}{T}\sum_{t=k+1}^{T}(y_t - \bar{y})^2}$$

where $\bar{y}$ is the sample average.

The estimated $\hat{\rho}_k$ can also be obtained from bivariate linear regressions:

$v_t = \delta + \beta v_{t-k} + \varepsilon_t$ (here the estimator of ρ_k is computed with the same number of observations in the numerator and denominator).

If the process is stationary and ergodic (for mean, variance and covariances), then $\hat{\rho}_k$ is a consistent estimator of ρ_k:

plim $\hat{\rho}_k = \rho_k$

Moreover, the following asymptotic property holds: if $\rho_1, \rho_2, \ldots, \rho_{i-1} \neq 0$ and $\rho_i = 0$, then $\hat{\rho}_i$ is asymptotically normally distributed $\sqrt{T}(\hat{\rho}_j - \rho_j) \xrightarrow{d} N(0, v_j)$

with $v_i = 1 + 2\rho_1^2 + 2\rho_2^2 + \ldots + 2\rho_{i-1}^2$ It is therefore possible to test the hypothesis $\rho_i = 0$ using the estimated $\hat{\rho}_i$ and the critical values of the standard normal distribution, since:

$$\sqrt{T} \xrightarrow{p_j} \to N(0, 1)$$
$$\sqrt{1 + 2\hat{\rho}_1^2 + \ldots + 2\hat{\rho}_{i-1}^2} \xrightarrow{d} N(0, 1)$$

Given the typical form of the autocorrelation function of $MA(q)$ processes (with $\rho_j \neq 0$ for $j \leq q$ and $\rho_j = 0$ for $j > q$), this test is particularly appropriate for choosing between an $MA(q)$ and an $MA(q-1)$ models.

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- X Run Print Save SaveAs Cut Copy Paste InsertTxt Find Replace Wrap+/- LineNum+/- Encrypt Evaluating consistency of sample (auto)correlation statistics by Monte Carlo simulation:

sample sizes: T = 50, 100, 200, 1000 number of replications (for each sample size): 1000 "true" data generating process for y(t):

(1) AR(1) $y(t) = 0.8y(t-1) + \epsilon(t)$ with epsilon ~ N (0, 1) for each sample size:

generate 1000 series y(t) drawing randomly epsilon(t) from its distribution compute rho(k) for k=1, ... 20 for each replication plot average rho(k) against k : 'correlogram' compare with "theoretical" ACF of the true process Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05

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-8 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom 1.0 Theoretical ACF — ACF for $T = 50$

- ACF for $T = 100$ — ACF for $T = 200$ ACF for $T = 1000$ 0.8 0.6 av. $\hat{\rho}_k$ 0.4 0.2 0.0

-0.2 Lag (k) Path = c:\bagliano\papers\di martino DB = none WF = acf_plots xR 14:06 29/03/2022 **N** S

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-23 EViews - [Graph: RHO1_DISTR Workfile: ACF_PLOTS::Untitled\] File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom − − − − $T = 50$ − − − − − $T = 100$ — T = 200 — T = 1000 Density 0.5 ρ0.6 0.3 0.4 0.8 0.7 0.9 0.2 1.0 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = acf_plots xR V 12:26 08/03/2022 ....

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AR, and general ARMA processes have more complicated theoretical autocorrelation functions (i.e. they do not go sharply to zero beyond a certain lag, as in the case of MA processes). In order to choose the appropriate order for a process with an autoregressive component, the analysis of the partial autocorrelation function is necessary.

Partial autocorrelation function PACF: autocorrelation between y_t and y_{t-k} as a function of lag k after having eliminated the effect of all intermediate lags $y_{t-1}, y_{t-2}, \ldots, y_{t-k+1}$.

The partial autocorrelation coefficients of order k, denoted ρ_k^P, for $\{y_t\}$ can be estimated by applying OLS to (purely) autoregressive models of increasing order:

$\Rightarrow \hat{\beta}_1$ is an estimate of ρ_1^P $AR(1)$: $y_t = \delta + \beta_1 y_{t-1} + \varepsilon_t$ $AR(2)$: $y_t = \delta + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \varepsilon_t \Rightarrow \hat{\beta}_2$ is an estimate of ρ_2^P

$AR(k)$: $y_t = \delta + \beta_1 y_{t-1} + \ldots + \beta_k y_{t-k} + \varepsilon_t \Rightarrow \hat{\beta}_k$ is an estimate of ρ_k^P

If the true process is $AR(p)$, the partial autocorrelation coefficients are equal to zero for lags $k > p$, and the estimated $\hat{\beta}_k$ yields a consistent estimate of ρ_k^P:

plim $\hat{\beta}_k = 0$ for $k > p$ and $\sqrt{T}\left(\hat{\beta}_k - 0\right) \underset{d}{\to} N(0,1)$

The estimate of the coefficients of partial autocorrelation can then provide useful information for choosing the order of the AR model.

In general, the joint analysis of the ACF and PACF is useful for the selection of the appropriate model, taking into account that:

for an $AR(p)$ process: the ACF decreases gradually (maybe with oscillations), whereas the PACF is zero for $k > p$;

• for an $MA(q)$ process: the ACF is zero for $k > q$, whereas the PACF decreases gradually (maybe with oscillations).

Examples of Stationary $AR(1)$ Models 2.5 ACF-Theoretical PACF-Theoretical PACF-Estimated 0.0

-2.5 5.0 ACF-Theoretical PACF-Theoretical $y_t = 0.80 \cdot y_{t-1} + \varepsilon$ CF-Estimated PACF-Estimated 2.5

- 0.0

-2.5 5.0 $= -0.80 \cdot y_{t-1} + \varepsilon_t$ CF-Theoretical CF-Estimated PACF-Estimated 2.5 0.0

-2.5

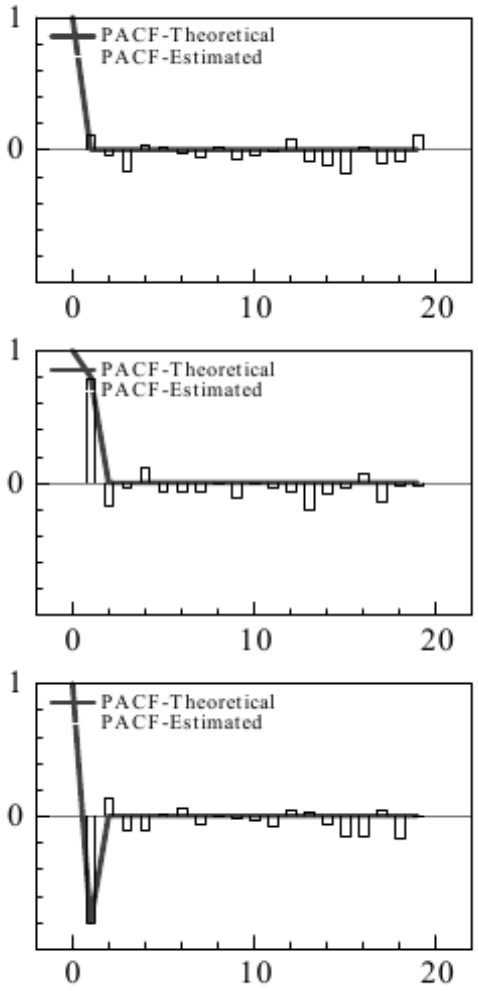


Figure 1.20: Course figure, source page 15, crop 1.

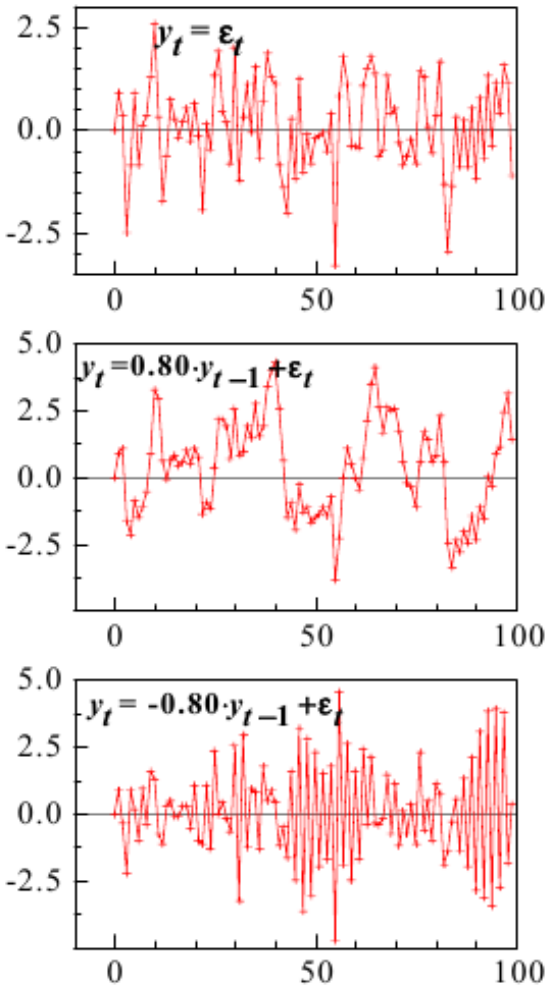


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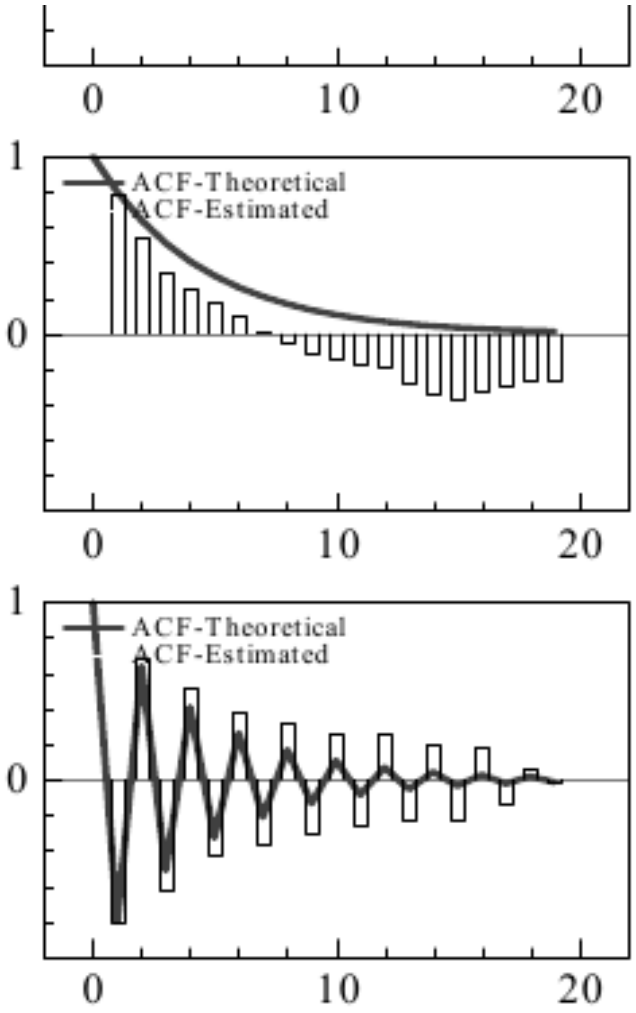


Figure 1.22: Course figure, source page 15, crop 3.

Examples of $AR(2)$ Models 5.0 CF-Theoretical CF Estimated PACF-Theoretical $v_\star = 0.50v_\star + 0.40v_\star$ PACF-Estimated 2.5 0.0 , Jusonniilli

-2.5 ACF-Theoretical ACE-Estimated PACF-Theoretical PACF-Estimated $\overline{y_t} = -0.80y_{t-2} + \varepsilon_t$

-5 Ω .30 v 1-0.80 v $y_t = 1 + \varepsilon$.

ACF-Theoretical ACF-Estimated PACF-Estimated UTT

-5

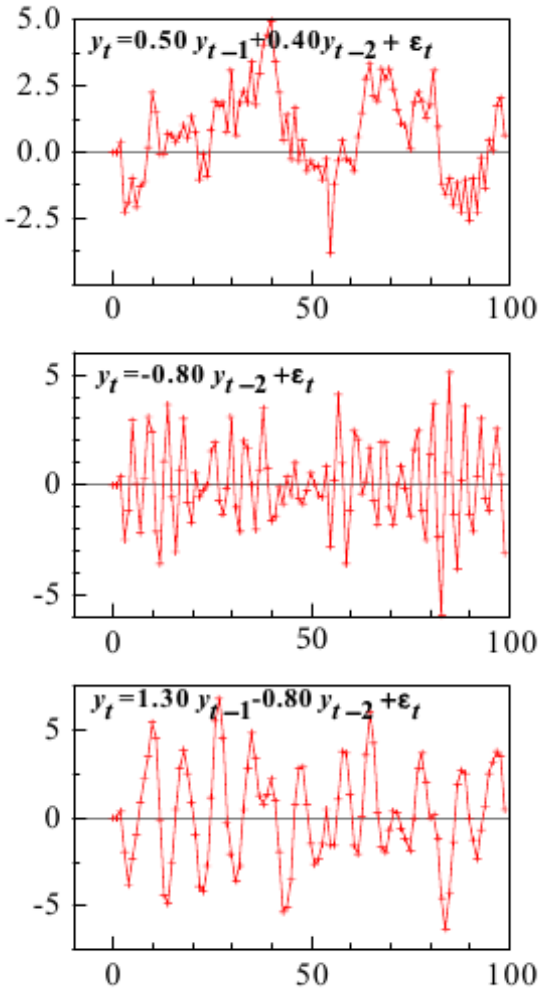


Figure 1.23: Course figure, source page 16, crop 1.

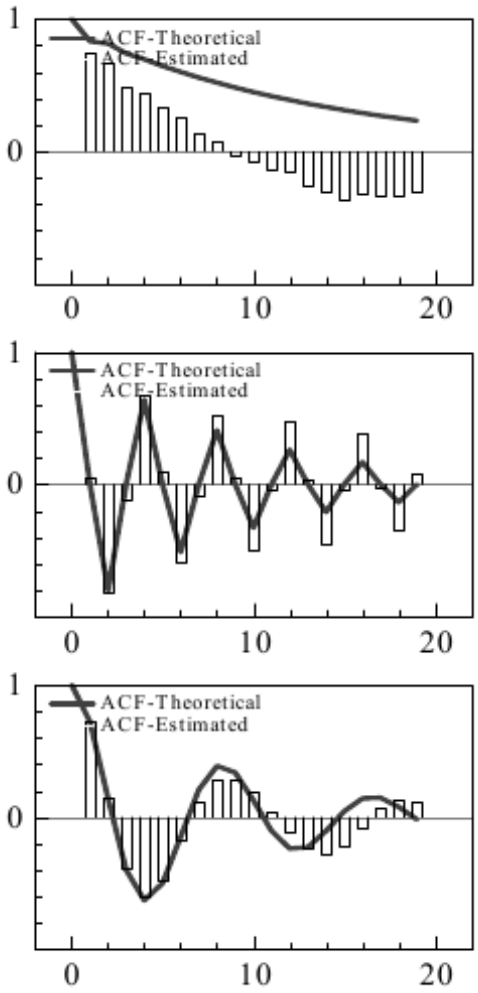


Figure 1.24: Course figure, source page 16, crop 2.

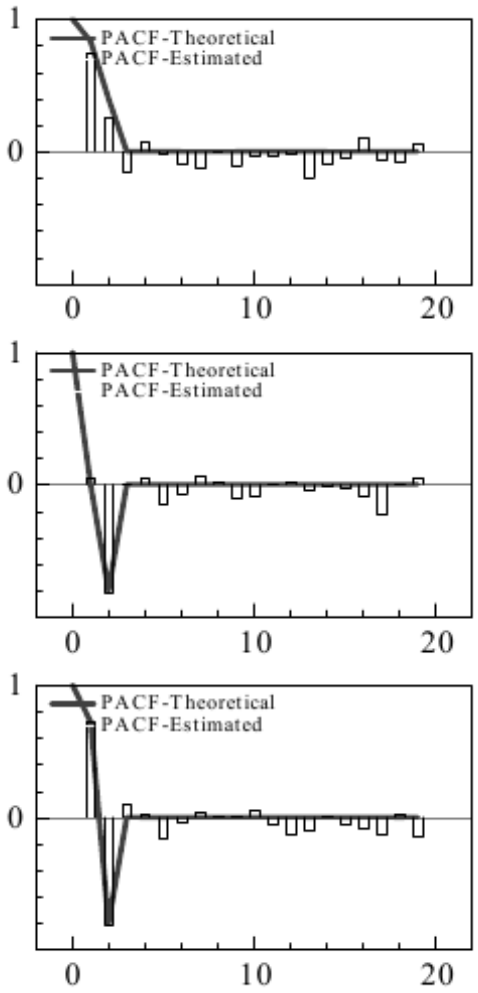


Figure 1.25: Course figure, source page 16, crop 3.

Examples of MA Models 5.0 CF-Theoretical PACE-Theoretical $y_t = \varepsilon_t - 0.90 \cdot \varepsilon_{t-1}$ PACF-Estimated ACF-Estimated 2.5 0.0

-2.5 ACF-Theoretical PACF-Theoretical 2.5 ACF-Estimated PACF-Estimated 0.0

-2.5 $= \varepsilon_t - 0.60 \cdot \varepsilon_{t-1} + 0.50 \cdot \varepsilon_{t-2} + 0.50 \cdot \varepsilon_t$ ACF-Theoretical PACF-Theoretical 2.5 ACF-Estimated PACF-Estimated 0.0

-2.5

$$
\begin{aligned}
\gamma_1 &= E[y_t y_{t-1}] \\
&= E[(\epsilon_t + \alpha\epsilon_{t-1})(\epsilon_{t-1} + \alpha\epsilon_{t-2})] \\
&= E[\epsilon_t\epsilon_{t-1} + \alpha\epsilon_t\epsilon_{t-2} + \alpha\epsilon_{t-1}^2 + \alpha^2\epsilon_{t-1}\epsilon_{t-2}] = \alpha\sigma^2 \\
\gamma_2 &= E[y_t y_{t-2}] \\
&= E[(\epsilon_t + \alpha\epsilon_{t-1})(\epsilon_{t-2} + \alpha\epsilon_{t-3})] \\
&= E[\epsilon_t\epsilon_{t-2} + \alpha\epsilon_t\epsilon_{t-3} + \alpha\epsilon_{t-1}\epsilon_{t-2} + \alpha^2\epsilon_{t-1}\epsilon_{t-3}] = 0 \\
&\vdots
\end{aligned}
$$

Figure 1.26: Course figure, source page 17, crop 1.

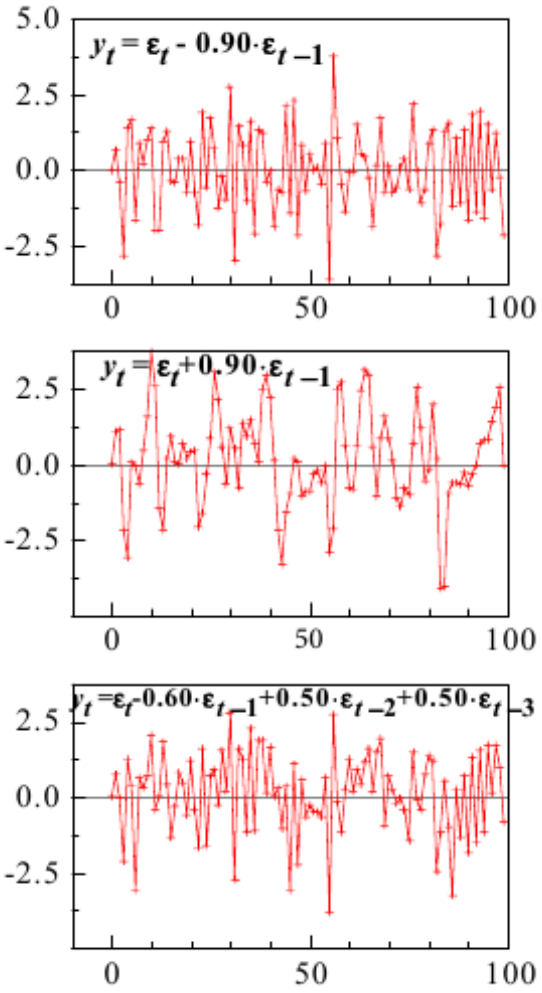


Figure 1.27: Course figure, source page 17, crop 2.

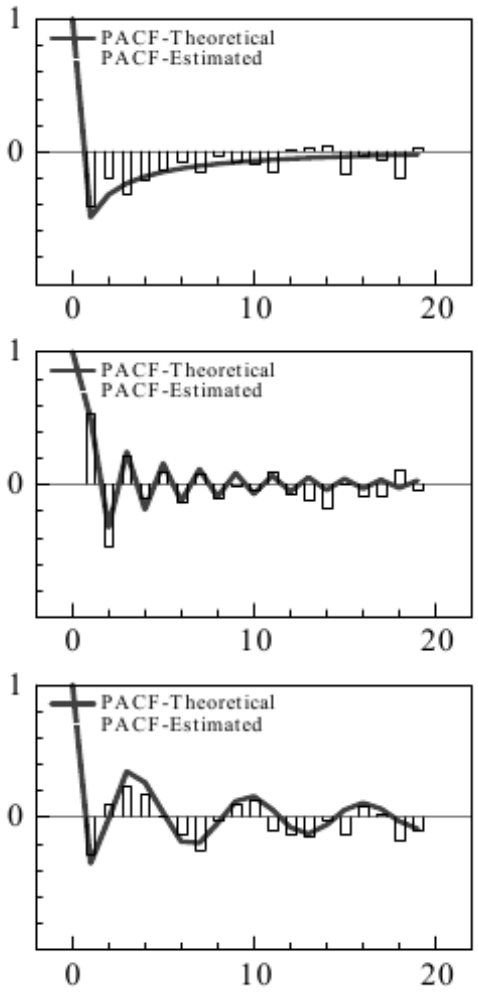


Figure 1.28: Course figure, source page 17, crop 3.

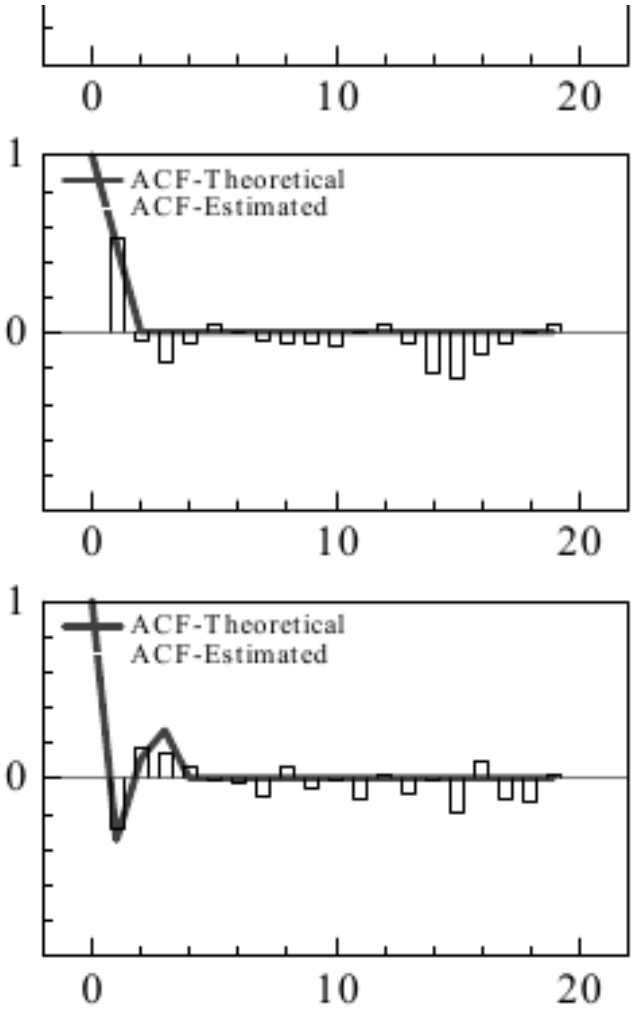


Figure 1.29: Course figure, source page 17, crop 4.

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-8 II File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom ACF and PACF for ARMA(2,1) process 1.0 $y(t) = 0.6y(t-1) - 0.4y(t-2) + e(t) + 0.4e(t-1)$ Autocorrelation 0.5 0.0

-0.5 1.0 Partial autocorrelation 0.5 0.0

-0.5 Lag Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = acf_teor_plots xR ^ 12:40 08/03/2022 e ~ i P Fz r ~ S x w PE (1)

Diagnostic (specification) tests After having estimated a specific ARMA model, the appropriateness of the model choice can be evaluated by analyzing the estimated residuals: if the model is correctly specified, it should deliver white noise residuals.

Ljung-Box "portmanteau" test, Q_K: tests the hypothesis of absence of autocorrelation in the estimated residual series up to order K:

$Q_K = T(T+2)\sum_{k=1}^{K} \frac{1}{T-k} r_k^2$ where r_k is the k-th order autocorrelation coefficient of the estimated residuals.

Under the assumption of correct specification of an $ARMA(p,q)$ process we have $Q_K \sim \chi^2(K - p - q - 1)$

Model selection criteria General criteria for model selection:

- Akaike's information criterion:

$AIC = \log \hat{\sigma}^2 + 2\frac{p+q+1}{T}$ Bayesian information criterion:

$BIC = \log \hat{\sigma}^2 + \frac{p+q+1}{T} \log T$ where $\hat{\sigma}^2$ is the estimated residual variance, and $p + q + 1$ is the number of estimated parameters (including a constant).

Preference goes to the model displaying lower values for the AIC and BIC criteria. Both criteria penalize specifications with a high number of parameters ($p \in q$), but BIC is more often used.

Estimation of ARMA models (outline) $AR(p)$ process $v_t = \delta + \theta_1 v_{t-1} + \theta_2 v_{t-2} + \ldots + \theta_p v_{t-p} + \varepsilon_t$ with $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$. Since

$E(y_{t-j}\varepsilon_t) = 0$ for $j = 1, \ldots p$ there is no correlation between the error term ε_t and the regressors $y_{t-1}, y_{t-2}, \ldots, y_{t-p}$ but:

$E(y_t\varepsilon_t) = \sigma_\varepsilon^2 > 0$ and $E(y_{t+i}\varepsilon_t) \neq 0$ for all future observations y_{t+i} with $i > 0$

Consequence:

due to the violation of the assumption of "strict exogeneity" of regressors in the CLRM $\Rightarrow OLS$ estimates biased in finite samples But, due to orthogonality of the error term with the set of regressors for each date t $\Rightarrow OLS$ estimates consistent and asymptotically normal

EXAMPLE – Illustration

Example. $AR(1)$ data generating process of $\{y_t\}$:

$y_t = \theta y_{t-1} + \varepsilon_t$ with $|\theta| < 1$ and $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$ We have $E(y_{t-1}\varepsilon_t) = 0$ but $E(v_t\varepsilon_t) = \sigma_\varepsilon^2$

and $E(y_{t+1}\varepsilon_t) = E[(\theta y_t + \varepsilon_{t+1})\varepsilon_t] = \theta E(y_t\varepsilon_t) = \theta\sigma_\varepsilon^2$ In general $E(v_{t+i}\varepsilon_t) = \theta^i\sigma_\varepsilon^2 \neq 0$ for all $i > 0$

If the following regression is estimated by OLS over a sample of T observations:

$y_t = \delta + \beta y_{t-1} + u_t$ the OLS estimate of θ is biased in finite samples:

$E(\hat{\beta}_{OLS}) = \theta\left(1 - \frac{2}{T}\right)$ The bias $-\frac{2\theta}{T}$ disappears as the sample size $T \to \infty$, so that the OLS estimator is consistent

Monte Carlo experiment for $T = 10, \ldots 200$ and $\theta = 0.2, 0.5, 0.9$ (1000 replications for each sample size)

Average OLS estimated AR(1) coefficients 1.0 theta=0.9 0.8 0.6 theta=0.5 0.4 theta=0.2 0.2 0.0 Sample size T

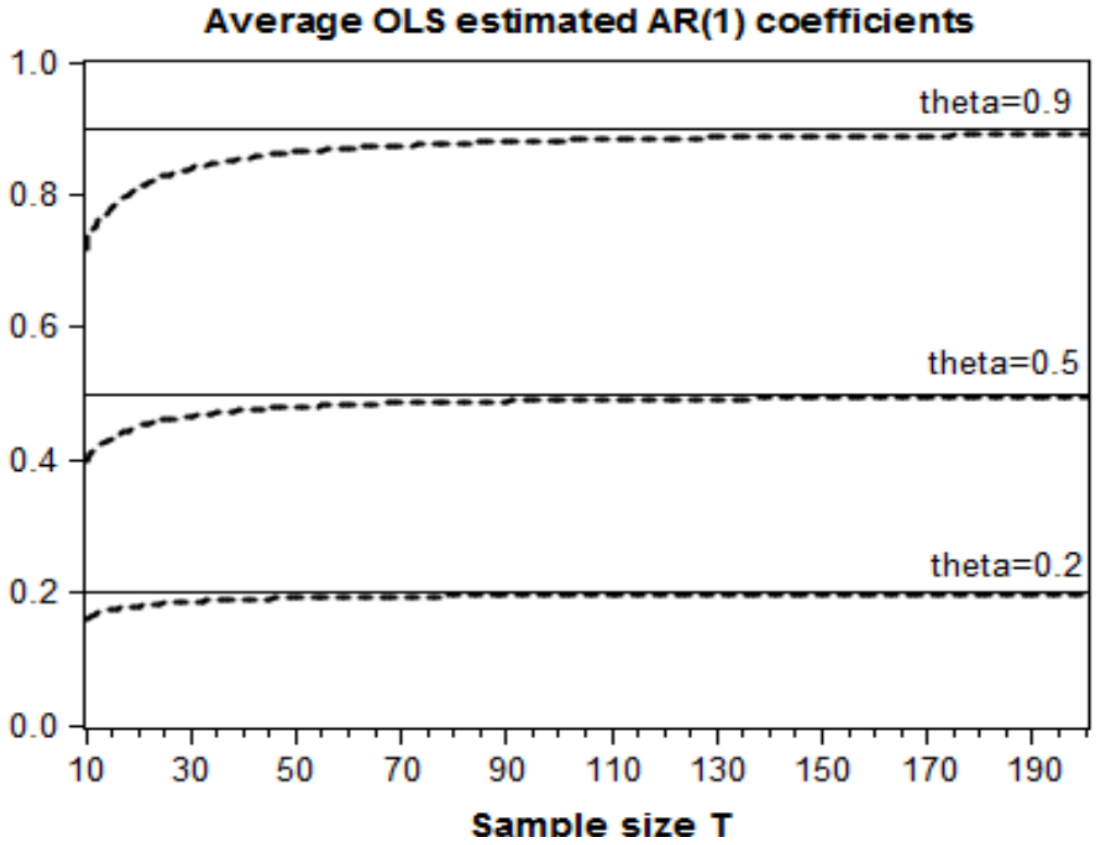


Figure 1.30: Course figure, source page 27, crop 1.

$MA(1)$ process $v_t = \mu + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$ for $t = 1, 2, ..., T$ with $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$. No regression method is applicable since $\{\varepsilon_t\}$ is unobserved.

However, if the $MA(1)$ process is invertible (ε_{t-1} can be expressed as an infinite linear combination of past observed values $y_{t-1}, y_{t-2}, \dots$), it is possible to obtain consistent estimators of μ and α_1.

Residual sum of squares (RSS) to be minimized:

The residual sum of squares can be written as

$$S(\mu, \alpha_1) = \sum_i (y_t - \mu - \alpha_1 \varepsilon_{t-1})^2.$$

Under invertibility each term ε_{t-1} can be expressed as

$$\varepsilon_{t-1} = \frac{1}{1 + \alpha_1 L}(y_{t-1} - \mu) = (1 - \alpha_1 L + \alpha_1^2 L^2 - \cdots)(y_{t-1} - \mu).$$

$$= \sum_{j=0}^{j} (-\alpha_1)^j (y_{t-1-j} - \mu)$$

$S(\mu, \alpha_1)$ can be rewritten as follows:

$$S(\mu, \alpha_1) = \sum_{t=2}^{T} \left(y_t - \mu - \alpha_1 \left(\sum_{i=0}^{\infty} (-\alpha_1)^j (y_{t-1-j} - \mu) \right) \right)^2$$

Operationally, $S(.)$ is approximated by

$$S(\mu, \alpha_1) = \sum_{t=2}^{T} \left(y_t - \mu - \alpha_1 \left[\sum_{i=0}^{t-2} (-\alpha_1)^j (y_{t-1-j} - \mu) \right] \right)^{-1}.$$

Minimization of S(.) with respect to μ and α_1 yields the first-order conditions necessary to obtain consistent estimators $\hat{\mu}$ and $\hat{\alpha}_1$.

Non linear least squares (NLS) techniques are required (numerical optimization procedures). The NLS estimators are consistent and asymptotically normal.

Maximum likelihood estimation Alternative estimation procedure: maximum likelihood (ML), adapted to the case of not independent observations.

In order to apply ML estimation, an assumption on the form of the distribution of ε is required: normality.

Principle: given T observations of $\{y_t\}$, the likelihood function L(.) for independent observations is $L(\boldsymbol{\vartheta}) = f(y_1, y_2, \ldots, y_T; \boldsymbol{\vartheta}) = \prod f(y_t; \boldsymbol{\vartheta})$ where $f(y_1, y_2, \ldots, y_T; \vartheta)$ is the joint density function of the T

observations, $f(y_t; \vartheta)$ is the density of the t-th observation and vector ϑ collects all parameters (to be estimated) defining the process $\{y_t\}$.

Since the observations of $\{y_t\}$ are not independent, the likelihood function is obtained by a sequence of conditional density functions:

$L(\boldsymbol{\vartheta}) = \prod f(y_t \mid y_{t-1}, \ldots, y_1; \boldsymbol{\vartheta}) \cdot f(y_1; \boldsymbol{\vartheta})$ where $f(y_t \mid y_{t-1}, \ldots, y_1; \vartheta)$ is the density of y_t conditional on past

realizations of the process, and $f(y_1; \vartheta)$ is the (unconditional) density of the first observation in the sample.

The corresponding loglikelihood function is:

$$\log L(\boldsymbol{\vartheta}) = \sum_{t=1}^{T} \log f(y_t \mid y_{t-1}, \ldots, y_1; \boldsymbol{\vartheta}) + \log f(y_1; \boldsymbol{\vartheta})$$

The maximum likelihood estimates of parameters in ϑ are obtained by maximizing the loglikelihood function $\log L(\vartheta)$ with respect to each element of vector ϑ.

The $ARMA(1,1)$ case.

$y_t = \theta_1 y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1}$ with ε normally distributed white noise, $N(0, \sigma_\varepsilon^2)$.

Conditioning on y_1 (treated as fixed), the likelihood function is:

$$L(\theta_1, \alpha_1, \sigma_\varepsilon^2) = \prod_{t=2} f(y_t \mid y_{t-1}, \ldots, y_1; \theta_1, \alpha_1, \sigma_\varepsilon^2)$$

$$= \prod_{t=2}^{T} \left(\frac{1}{\sqrt{2\pi\sigma_\epsilon^2}} \right) e^{-\frac{\varepsilon_t^2}{2\sigma_\epsilon^2}} = \left(\frac{1}{\sqrt{2\pi\sigma_\epsilon^2}} \right)^{T-1} e^{-\frac{\sum_{t=2}^{T} \varepsilon_t^2}{2\sigma_\epsilon^2}}$$

with $\varepsilon_t = v_t - \theta_1 v_{t-1} - \alpha_1 \varepsilon_{t-1}$

from which we get $\log L(\theta_1, \alpha_1, \sigma_\varepsilon^2) = -\frac{T-1}{2} \log\left(2\pi\sigma_\varepsilon^2\right) - \frac{1}{2} \frac{\sum_{t=2}^{T} \varepsilon_t^2}{\sigma_\varepsilon^2}$

To maximize $\log L(\theta_1, \alpha_1, \sigma_\epsilon^2)$ it is necessary to recursively generate the error terms ε_t starting from an assumption on ε_1, usually $\varepsilon_1 = 0$.

After substituting the terms ε_t as functions of observed y_t and parameters into the loglikelihood function, it is possible to proceed to the maximization making use of numerical techniques.

Special cases:

(i) AR(1) process: $\alpha_1 = 0$ and $\varepsilon_t = y_t - \theta_1 y_{t-1}$. The maximum likelihood estimator of θ_1 coincides with the OLS estimator.

(ii) $MA(1)$ process: $\theta_1 = 0$ and it is necessary to generate recursively the ε's, starting from $\varepsilon_1 = 0$:

$$\varepsilon_2 = v_2 - \alpha_1 \varepsilon_1 = v_2$$
$$\varepsilon_3 = y_3 - \alpha_1 \varepsilon_2 = y_3 - \alpha_1 y_2$$
$$\varepsilon_4 = v_4 - \alpha_1 \varepsilon_3 = v_4 - \alpha_1 v_3 + \alpha_1^2 v_2$$

$T - 2$ $\varepsilon_T = y_T - \alpha_1 \varepsilon_{T-1} = \sum(-\alpha_1)^t y_{T-t}$ After substitution of the ε's into the loglikelihood function, maximization is performed by numerical techniques; the estimator of α_1 has no closed-form expression

Forecasting with ARMA models Optimal predictor Given a stochastic process $\{y_t\}$, it is possible to construct predictions (or "forecasts") over different horizons h using the observations on the process up to time T.

Using the information set $I_T = \{y_{-\infty}, \dots, y_{T-1}, y_T\}$ we can predict y_{T+h} (the value of y after h periods). Let the prediction on a h-period horizon be denoted by $\hat{y}_{T+h|T}$.

A predictor will in general be a function of the elements in the information set I_T:

$\hat{v}_{T+h|T} = g(v_T, v_{T-1}, \dots)$ The criterion to choose the function $g(.)$ is the minimization of the mean square forecast error (MSE):

$\min MSE = E\left(\left(y_{T+h} - \hat{y}_{T+h|T}\right)^2 \mid I_T\right)$ It can be shown that the optimal predictor of y_{T+h} is the expected value of y_{T+h} conditional onto the information available in T:

$$\hat{v}_{T+h|T} = E(v_{T+h} \mid I_T) = E(v_{T+h} \mid v_T, v_{T-1}, \dots)$$

Proof. The stationary $ARMA(p, q)$ process $\theta(L)y_t = \alpha(L)\varepsilon_t$ can be written as an $MA(\infty)$:

∞ $y_t = \theta(L)^{-1}\alpha(L)\varepsilon_t = \pi(L)\varepsilon_t = \sum \pi_j \varepsilon_{t-j}$ $i = 0$ Therefore:

$$y_{T+h} = \varepsilon_{T+h} + \pi_1 \varepsilon_{T+h-1} + \dots + \pi_{h-1}\varepsilon_{T+1} + \underbrace{\pi_h \varepsilon_T + \pi_{h+1}\varepsilon_{T-1} + \dots}_{\gamma}$$

information up to T $\infty = \varepsilon_{T+h} + \pi_1 \varepsilon_{T+h-1} + \dots + \pi_{h-1}\varepsilon_{T+1} + \sum_n \pi_{h+j}\varepsilon_{T-j}$

$$i = 0$$

A predictor of y_{T+h} can be written as a linear combination of the ε's with coefficients π^* to be determined:

$\hat{y}_{T+h|T} = \sum_{j=0} \pi^*_{h+j}\varepsilon_{T-j}$ The π^* coefficients are chosen to minimize the mean square forecast error. Given the forecast error ∞ $y_{T+h} - \hat{y}_{T+h|T} = \varepsilon_{T+h} + \pi_1\varepsilon_{T+h-1} + \ldots + \pi_{h-1}\varepsilon_{T+1} + \sum(\pi_{h+j} - \pi^*_{h+j})\varepsilon_{T-j}$

since $E(\varepsilon_t\varepsilon_s) = 0$ for all $t \neq s$, the MSE is $E\left(\left(y_{T+h} - \hat{y}_{T+h|T}\right)^2 \mid I_T\right) = \sigma^2_\varepsilon + \pi^2_1\sigma^2_\varepsilon + \ldots + \pi^2_{h-1}\sigma^2_\varepsilon + \sum\left(\pi_{h+j} - \pi^*_{h+j}\right)^2\sigma^2_\varepsilon$

To minimize MSE we must set $\pi^*_{h+j} = \pi_{h+j}$ from which ∞ $\hat{y}_{T+h|T} = \sum \pi_{h+j}\varepsilon_{T-j} = E(y_{T+h} \mid y_T, y_{T-1}, ...)$

j=0

$AR(1)$ process For the process $y_t = \theta y_{t-1} + \varepsilon_t$ the optimal predictors at various horizons are $\hat{y}_{T+1|T} = E(y_{T+1} \mid y_T, y_{T-1}, \dots) = \theta y_T + E(\varepsilon_{T+1} \mid y_T, \dots) = \theta y_T$

$$\hat{y}_{T+2|T} = E(y_{T+2} \mid y_T, y_{T-1}, \dots) = \theta\hat{y}_{T+1|T} + E(\varepsilon_{T+2} \mid y_T, \dots) = \theta^2 y_T$$

. . .

$$\hat{y}_{T+h|T} = E(y_{T+h} \mid y_T, y_{T-1}, \dots) = \theta\hat{y}_{T+h-1|T} + E(\varepsilon_{T+h} \mid y_T, \dots) = \theta^h y_T$$

For $h \to \infty$, $\hat{y}_{T+h|T} \to 0$ and the predicted level of the process tends to the unconditional mean μ.

$MA(1)$ process In the case of an invertible moving average $v_t = \varepsilon_t + \alpha\varepsilon_{t-1}$ we can express the ε's in terms of the observed y's, and obtain the following predictions:

$$\begin{aligned}\hat{y}_{T+1|T} = E(y_{T+1} \mid y_T, y_{T-1}, \dots) &= E(\varepsilon_{T+1} \mid y_T, \dots) + \alpha E(\varepsilon_T \mid y_T, \dots) \\ &= \alpha\varepsilon_T\end{aligned}$$

$= \alpha\left(\sum_{i=0}^{\infty}(-\alpha)^j y_{T-j}\right)$ for $h \geq 2$ $\hat{y}_{T+h|T} = E(y_{T+h} \mid y_T, y_{T-1}, \dots) = 0$

$ARMA(p, q)$ process The same procedure used to construct predictions for $AR(1)$ and $MA(1)$ processes can be extended to stationary and invertible $ARMA(p, q)$ processes.

In the $ARMA(2, 1)$ case $y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t + \alpha_1\varepsilon_{t-1}$

we get the following one-step ahead forecast:

$\hat{y}_{T+1|T} = \theta_1 y_T + \theta_2 y_{T-1} + \alpha_1\varepsilon_T$ To express $\hat{y}_{T+1|T}$ in terms of observed values of $\{y_t\}$ we exploit invertibility. From

$y_t - \theta_1 y_{t-1} - \theta_2 y_{t-2} = (1 + \alpha_1 L)\varepsilon_t$ inverting the MA component, we get:

$$\begin{aligned}\varepsilon_t &= (1 + \alpha_1 L)^{-1}(y_t - \theta_1 y_{t-1} - \theta_2 y_{t-2}) \\ &= \sum(-\alpha_1)^j L^j(y_t - \theta_1 y_{t-1} - \theta_2 y_{t-2})\end{aligned}$$

Using that expression for ε_T the one-step ahead forecast becomes:

$$\begin{aligned}\hat{y}_{T+1|T} &= \theta_1 y_T + \theta_2 y_{T-1} + \alpha_1\varepsilon_T \\ &\infty \\ &= \theta_1 y_T + \theta_2 y_{T-1} + \alpha_1\sum_{j=0}^{\infty}(-\alpha_1)^j L^j(y_T - \theta_1 y_{T-1} - \theta_2 y_{T-2})\end{aligned}$$

$$= (\theta_1 + \alpha_1)y_T + (\theta_2 - \alpha_1\theta_1 - \alpha_1^2)y_{T-1} - \alpha_1(\theta_2 - \alpha_1\theta_1 - \alpha_1^2)y_{T-2} + \dots$$

The two-period ahead forecast is obtained as $\hat{v}_{T+2|T} = \theta_1\hat{v}_{T+1|T} + \theta_2 v_T = (\theta_1^2 + \theta_1\alpha_1 + \theta_2)y_T + \theta_1(\theta_2 - \alpha_1\theta_1 - \alpha_1^2)y_{T-1}$

$-\theta_1\alpha_1(\theta_2 - \alpha_1\theta_1 - \alpha_1^2)y_{T-2} + \dots$ and for longer horizons ($h \geq 3$) forecast are obtained recursively as:

$$\hat{y}_{T+h|T} = \theta_1\hat{y}_{T+h-1|T} + \theta_2\hat{y}_{T+h-2|T}$$

Prediction accuracy The forecasting power of an ARMA model can be evaluated by:

estimating the model on a sample shorter than the whole set of observations available (estimation sample) constructing the predictors for the out-of-sample observations until the last date available (forecast sample)

To evaluate the prediction accuracy: mean square (or variance) forecast error:

$$C_h = E\left(\left(y_{T+h} - \hat{y}_{T+h|T}\right)^2\right) = Var(y_{T+h} \mid I_T)$$

For a stationary $ARMA(p, q)$, with the $MA(\infty)$ representation ($\alpha_0 = 1$) $y_t = \varepsilon_t + \alpha_1\varepsilon_{t-1} + \alpha_2\varepsilon_{t-2} + \dots + \dots = \sum_{j=0}^{\infty}\alpha_j\varepsilon_{t-j}$

the h-period ahead prediction is (in terms of $\varepsilon_T, \varepsilon_{T-1}, \dots$) $\hat{y}_{T+h|T} = E(\varepsilon_{T+h} + \alpha_1\varepsilon_{T+h-1} + \dots + \alpha_h\varepsilon_T + \alpha_{h+1}\varepsilon_{T-1} + \dots | \varepsilon_T, \varepsilon_{T-1}, \dots)$

$\infty = \alpha_h\varepsilon_T + \alpha_{h+1}\varepsilon_{T-1} + \dots = \sum \alpha_j\varepsilon_{T+h-j}$ i=h

with prediction error $y_{T+h} - \hat{y}_{T+h|T} = \varepsilon_{T+h} + \alpha_1\varepsilon_{T+h-1} + \alpha_2\varepsilon_{T+h-2} + \dots + \alpha_{h-1}\varepsilon_{T+1}$

$h - 1 = \sum_{j=0}\alpha_j\varepsilon_{T+h-j}$ with variance $E((y_{T+h} - \hat{y}_{T+h|T})^2) = \sigma_\varepsilon^2 + \alpha_1^2\sigma_\varepsilon^2 + \alpha_2^2\sigma_\varepsilon^2 + \dots + \alpha_{h-1}^2\sigma_\varepsilon^2$

$h - 1 = \sigma_\varepsilon^2\sum\alpha_j^2$ The variance of the forecasting error tends to increase with h, reaching the unconditional variance of the process for $h \to \infty$.

× EViews

-0 File Edit Object View Proc Quick Options Add-ins Window Help Workfile: ARMA_FOR_EX1 - (c:\bagl... - View Proc Object Save Freeze Details+/- Show Program: ACF_TEXT - (c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016\acf_text.prg)

- X Range: 1 2000 - 2000 obs Filter: * Run Print Save SaveAs Cut Copy Paste InsertTxt Find Replace Wrap+/- LineNum+/- Encrypt Sample: 1981 2000 - 20 obs Order: Name alpha1 alpha2 arma22 c delta epsilon for20a for20b for20b for_st_err mean mi resid theta1 theta2 y y___acf y_plot yf yf_l yf_se yf_u yf_u 'Forecasting with ARMA models' "Estimation sample":

= 1, ..., 1980 "Forecast sample" (h = 1, ... 20): T+1, ... T+h = 1981, ..., 2000 Untitled / New Page Workfile: ARMA_FOR_EX2 - (c:\bag... _ $y(t) = 10 + 0.9y(t-1) + \epsilon(t)$

—-> E(y) = 100 View Proc Object Save Freeze Details+/- Show AR(1) Range: 1 2000 - 2000 obs Filter: * Sample: 1981 2000 - 20 obs Order: Name # alpha1 $y(t) = 10 + 0.9y(t-1) - 0.4y(t-2) + \epsilon(t)$ - - - - > $E(y) = 20$ AR(2) # alpha2 arma22 arma22 C delta epsilon for20a for20b for_sr_err mean Mi resid theta1 theta2 Y y y_____y_acf y___plot yf_l yf_se yf_u with epsilon $\sim N(0, 1)$ Untitled / New Page / Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex2 Line: 1 xR 09:55 e Fz r ∼ N S D

- G w O 3)

× EViews - [Graph: Y_PLOT Workfile: ARMA_FOR_EX1::Untitled\]

-0 II File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Y 11111111 1750 1000 1250 1500 2000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex1 xR 10:02 .

D S G

×

-23 EViews - [Graph: Y_ACF Workfile: ARMA_FOR_EX1::Untitled\] File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom 1.0 Autocorrelation 0.8 0.6 0.4 0.2 0.0 Lag (k) Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex1 7 | 0.98 xR 10:04 i S D O e G Fz r S w

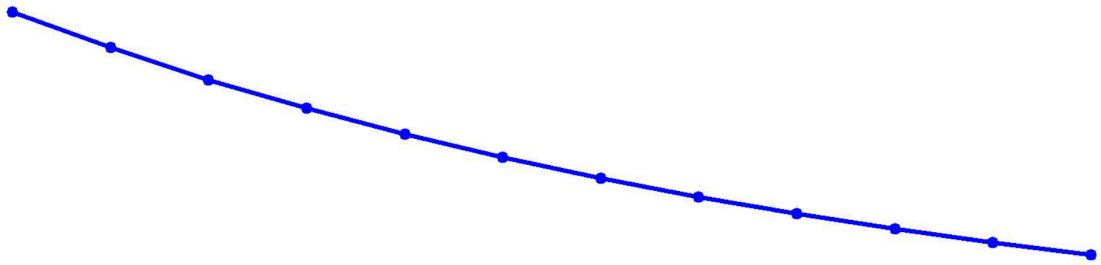


Figure 1.31: Course figure, source page 56, crop 1.

× EViews - [Graph: FOR20A Workfile: ARMA_FOR_EX1::Untitled\]

-3 III File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom + 2 st. dev. of forecast error 104 - forecast E(y)

- 2 st. dev. of forecast error 1975 1970 1980 1985 1990 1995 2000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex1 xR 09/03/2022 ~ O e .....

Fz r S w I P≥

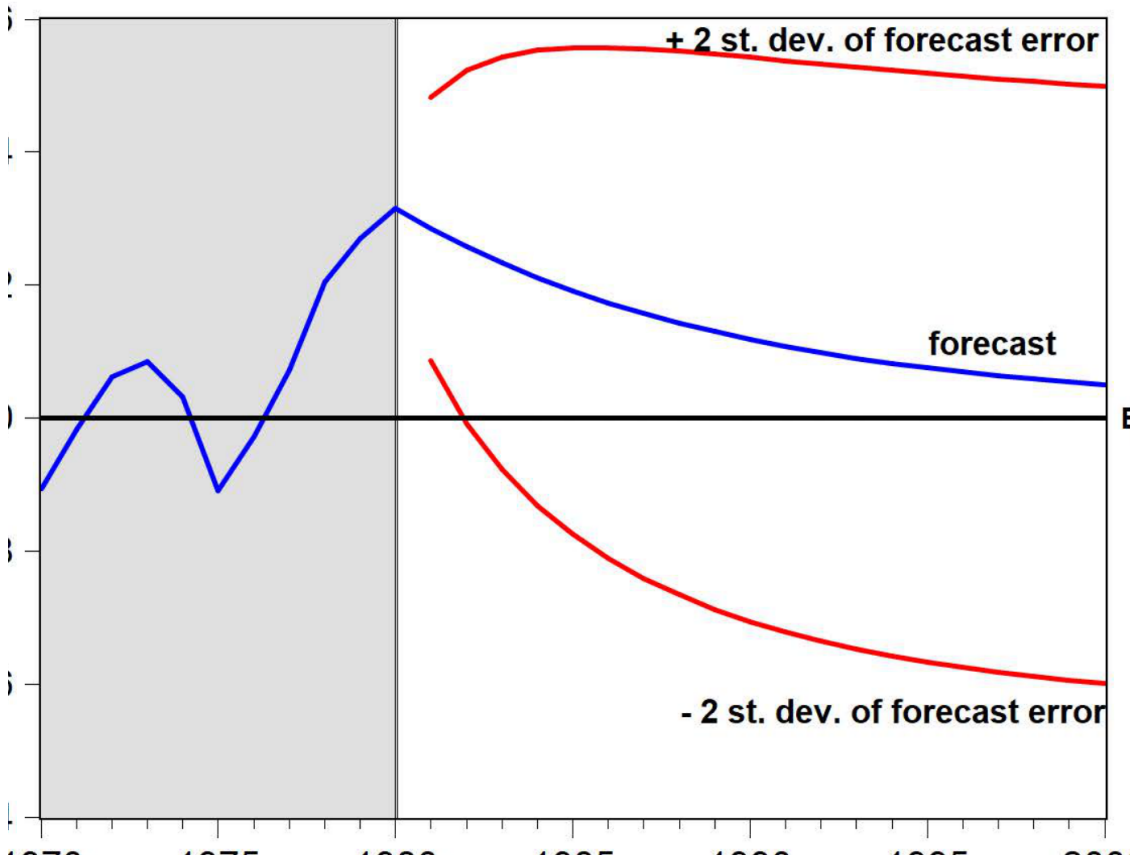


Figure 1.32: Course figure, source page 57, crop 1.

X

- EViews - [Graph: FOR20B Workfile: ARMA_FOR_EX1::Untitled\] II File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom realized forecast 1995 1975 1980 1985 1990 1970 2000 1976 | 99.3 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex1 xR 10:16 09/03/2022 ~ Sa i, = e Fz r Z S w I PB

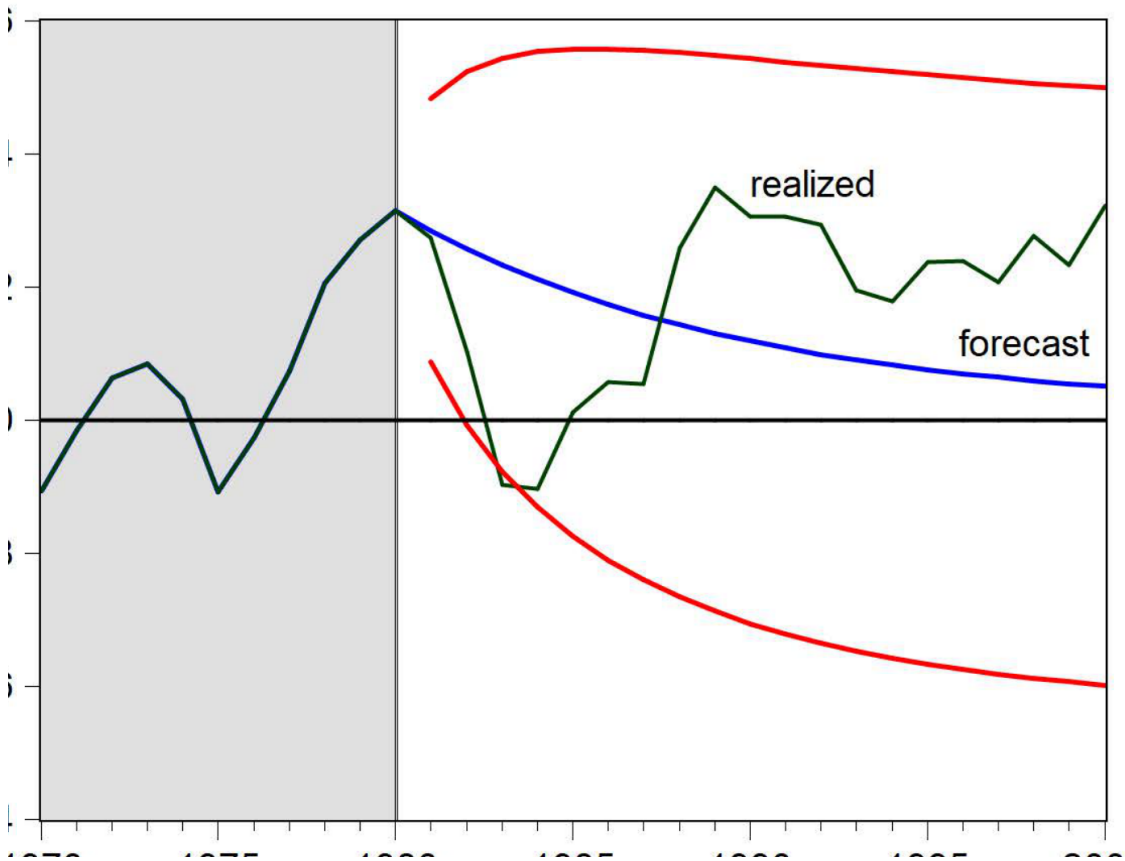


Figure 1.33: Course figure, source page 58, crop 1.

× EViews

-0 File Edit Object View Proc Quick Options Add-ins Window Help Workfile: ARMA_FOR_EX1 - (c:\bagl... - View Proc Object Save Freeze Details+/- Show Program: ACF_TEXT - (c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016\acf_text.prg)

- X Range: 1 2000 - 2000 obs Filter: * Run Print Save SaveAs Cut Copy Paste InsertTxt Find Replace Wrap+/- LineNum+/- Encrypt Sample: 1981 2000 - 20 obs Order: Name alpha1 alpha2 arma22 c delta epsilon for20a for20b for20b for_st_err mean mi resid theta1 theta2 y y___acf y_plot yf yf_l yf_se yf_u yf_u 'Forecasting with ARMA models' "Estimation sample":

= 1, ..., 1980 "Forecast sample" (h = 1, ... 20): T+1, ... T+h = 1981, ..., 2000 Untitled / New Page Workfile: ARMA_FOR_EX2 - (c:\bag... _ $y(t) = 10 + 0.9y(t-1) + \epsilon(t)$

—-> E(y) = 100 View Proc Object Save Freeze Details+/- Show AR(1) Range: 1 2000 - 2000 obs Filter: * Sample: 1981 2000 - 20 obs Order: Name # alpha1 $y(t) = 10 + 0.9y(t-1) - 0.4y(t-2) + \epsilon(t)$ − − − − > $E(y) = 20$ AR(2) # alpha2 arma22 arma22 C delta epsilon for20a for20b for_sr_err mean Mi resid theta1 theta2 Y y y_____y_acf y___plot yf_l yf_se yf_u with epsilon $\sim N(0,1)$ Untitled / New Page / Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex2 Line: 1 xR 09:55 e Fz r ∼ N S D

- G w O 3)

× EViews - [Graph: Y_PLOT Workfile: ARMA_FOR_EX2::Untitled\]

-0 II File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom

Y 2000 1000 1250 1750 1500 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex2 _____ D S RR G

× 3—33 EViews - [Graph: Y_ACF Workfile: ARMA_FOR_EX2::Untitled\] File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom .8 Autocorrelation .6 .4 .2 .0

-.2 Lag (k) Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex2 xR ^ 10:24 .

< Sa e r x D Fz S w PB O *

× EViews - [Graph: FOR20A Workfile: ARMA_FOR_EX2::Untitled\]

-8 III File Edit Object View Proc Quick Options Add-ins Window Help _ X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom + 2 st. dev. of forecast error E(y) forecast

- 2 st. dev. of forecast error 1975 1980 1985 1995 1970 1990 2000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex2 xR ^ 10:26 09/03/2022 O ∼ .....

r Fz S w x

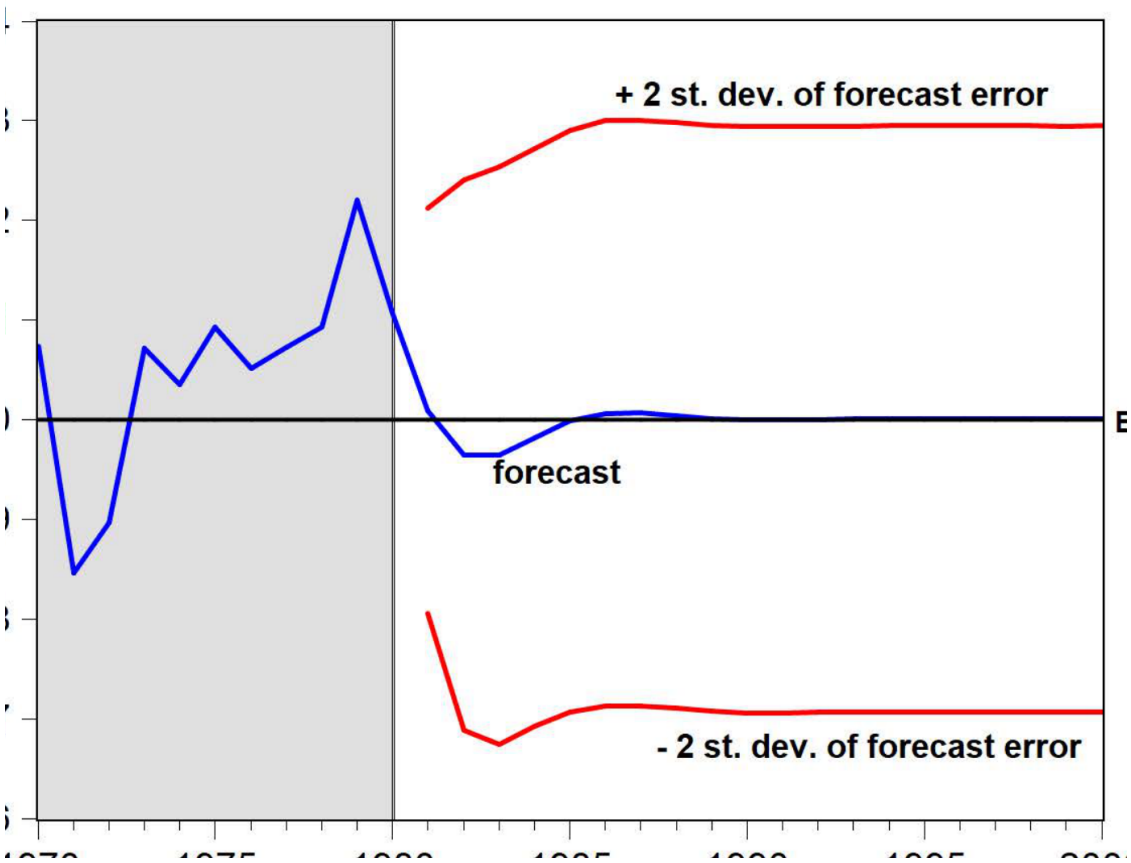


Figure 1.34: Course figure, source page 62, crop 1.

– X EViews - [Graph: FOR20B Workfile: ARMA_FOR_EX2::Untitled\] II File Edit Object View

Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom realized forecast 1970 1975 1985 1980 1990 1995 2000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arma_for_ex2 xR ^ 10:29 09/03/2022 O ∼

- S Fz w x

1.2.1 Additional Figures

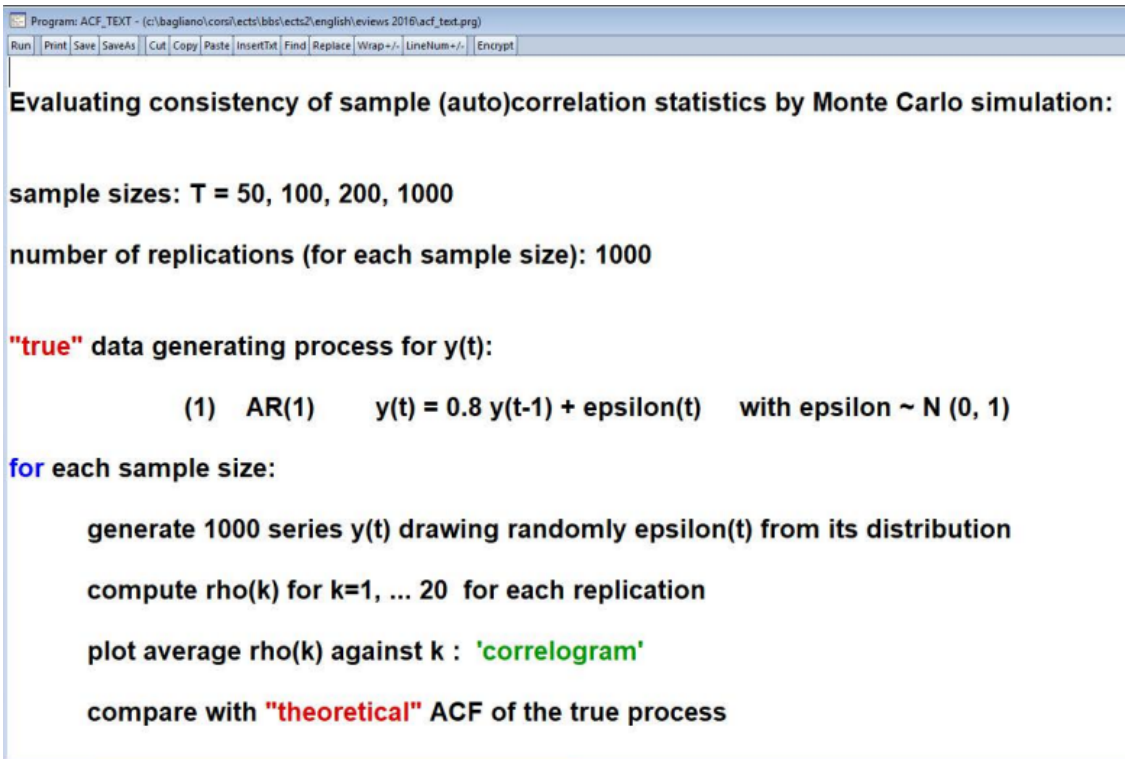


Figure 1.35: Additional figure from the original slide image set 2, image 1.

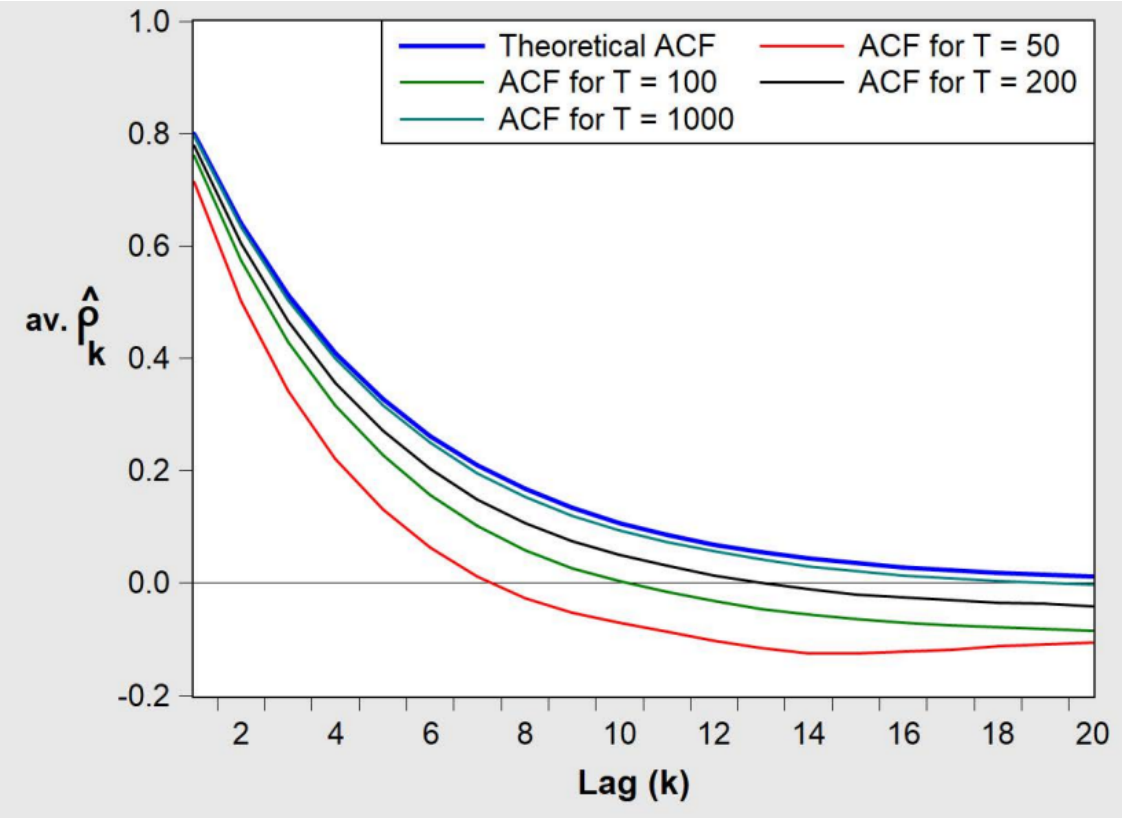


Figure 1.36: Additional figure from the original slide image set 2, image 2.

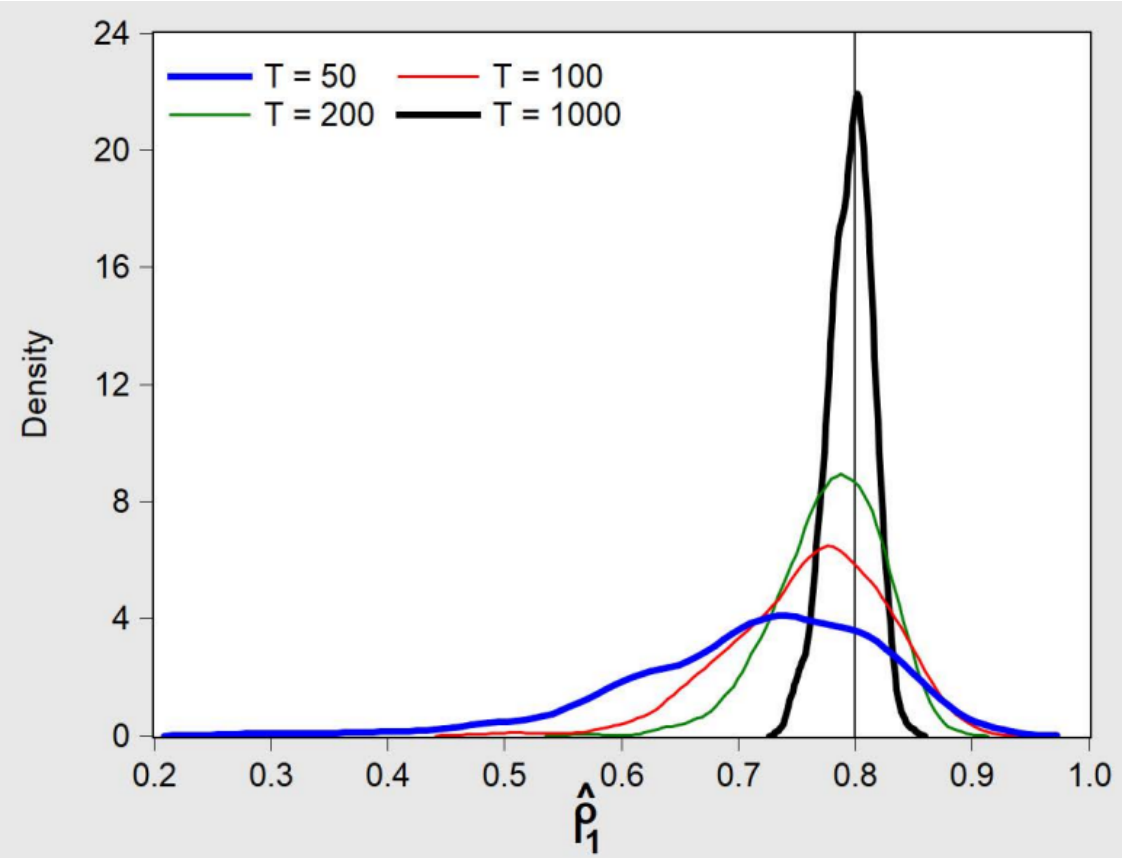


Figure 1.37: Additional figure from the original slide image set 2, image 3.

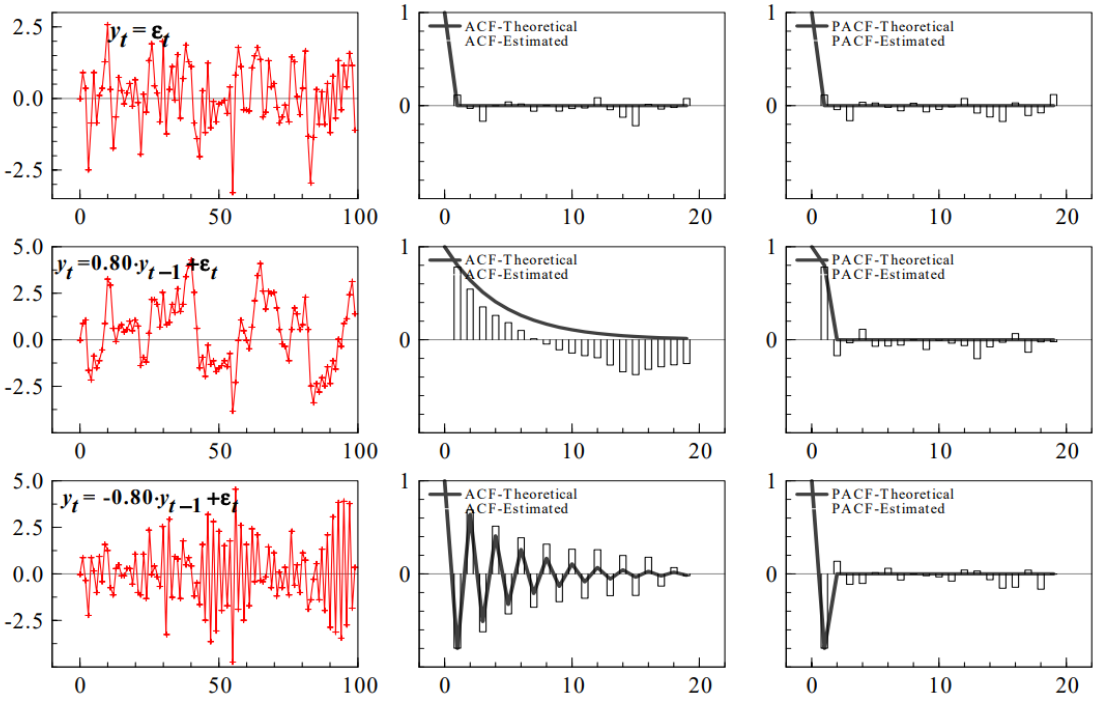


Figure 1.38: Additional figure from the original slide image set 2, image 4.

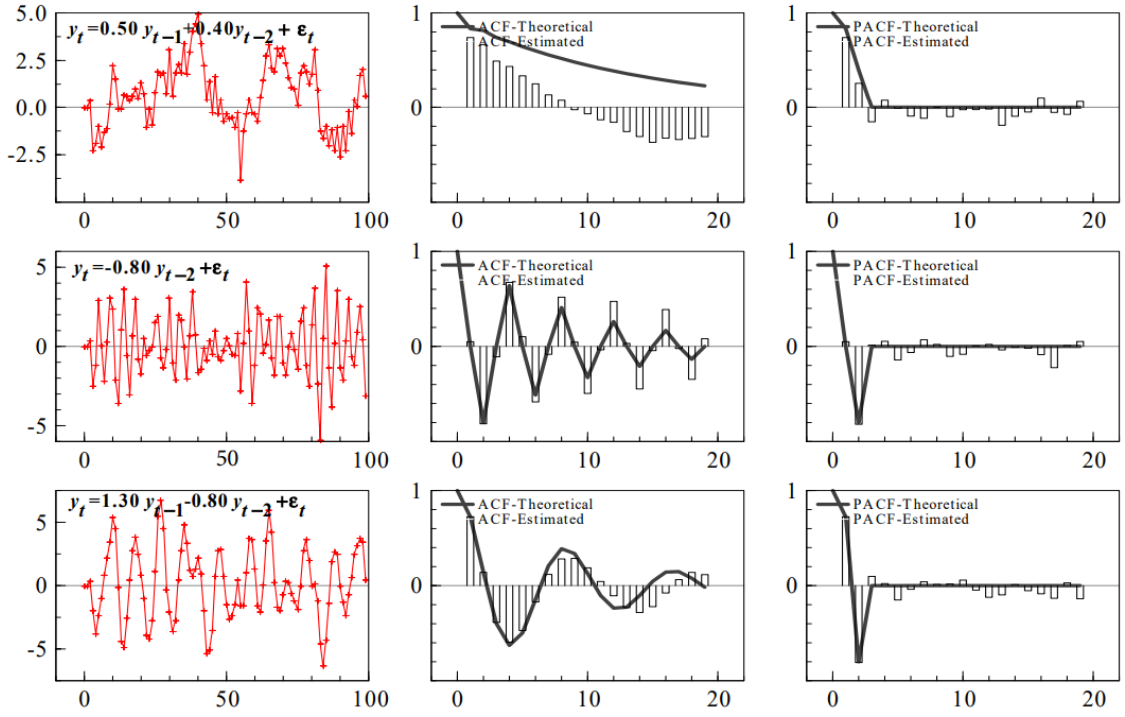


Figure 1.39: Additional figure from the original slide image set 2, image 5.

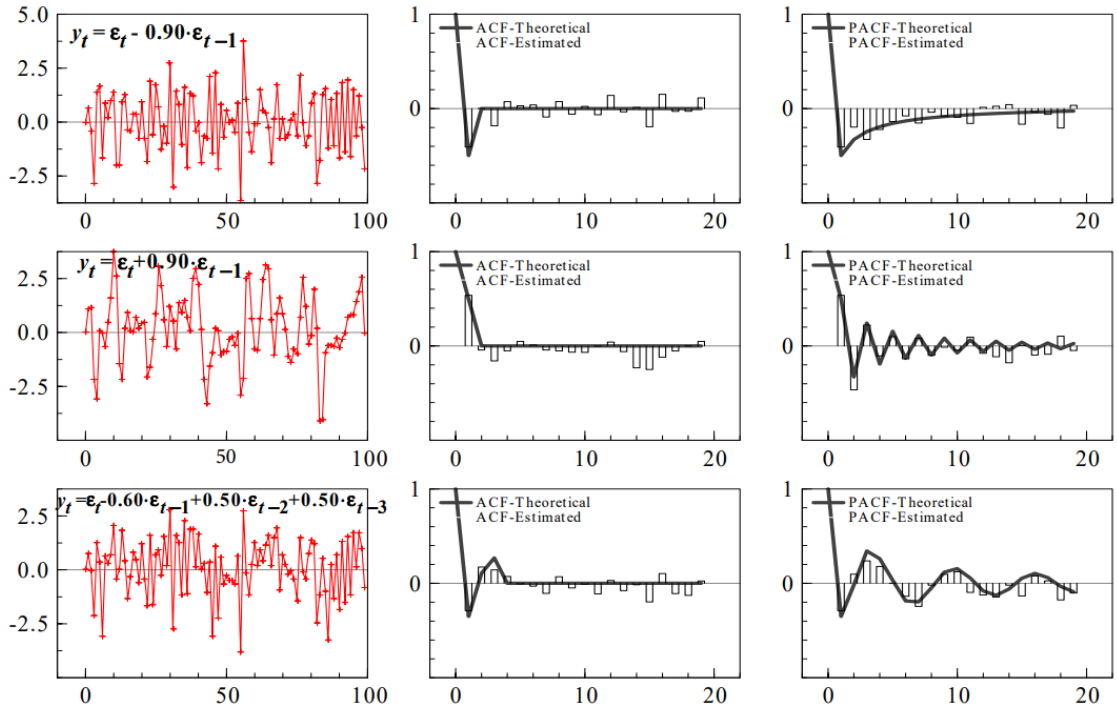


Figure 1.40: Additional figure from the original slide image set 2, image 6.

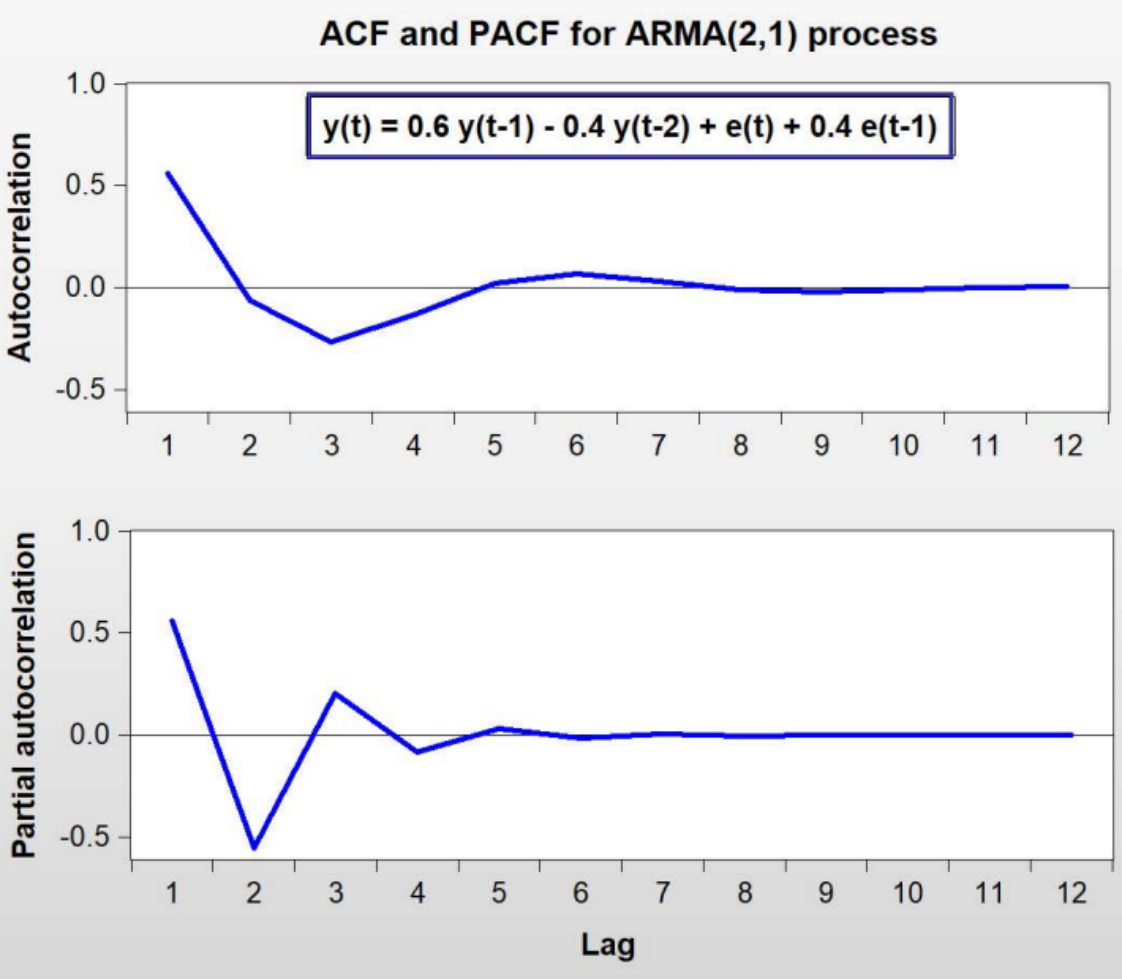


Figure 1.41: Additional figure from the original slide image set 2, image 7.

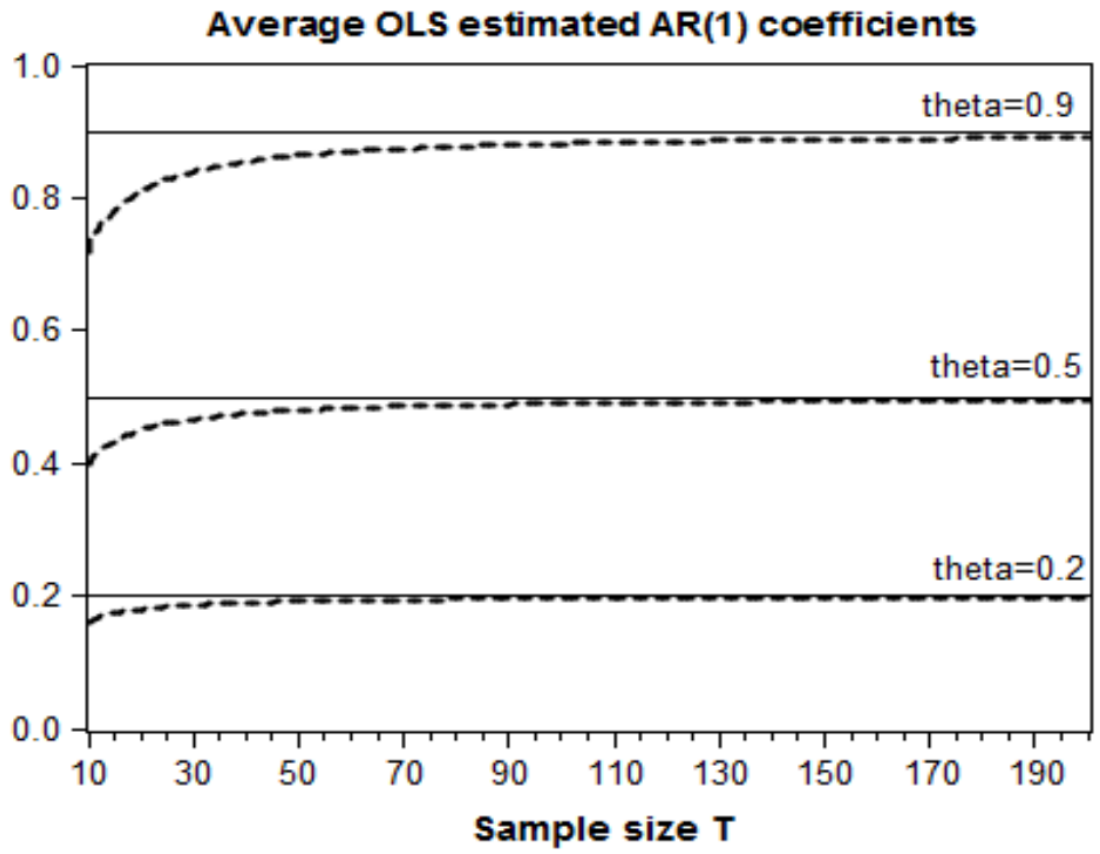


Figure 1.42: Additional figure from the original slide image set 2, image 8.

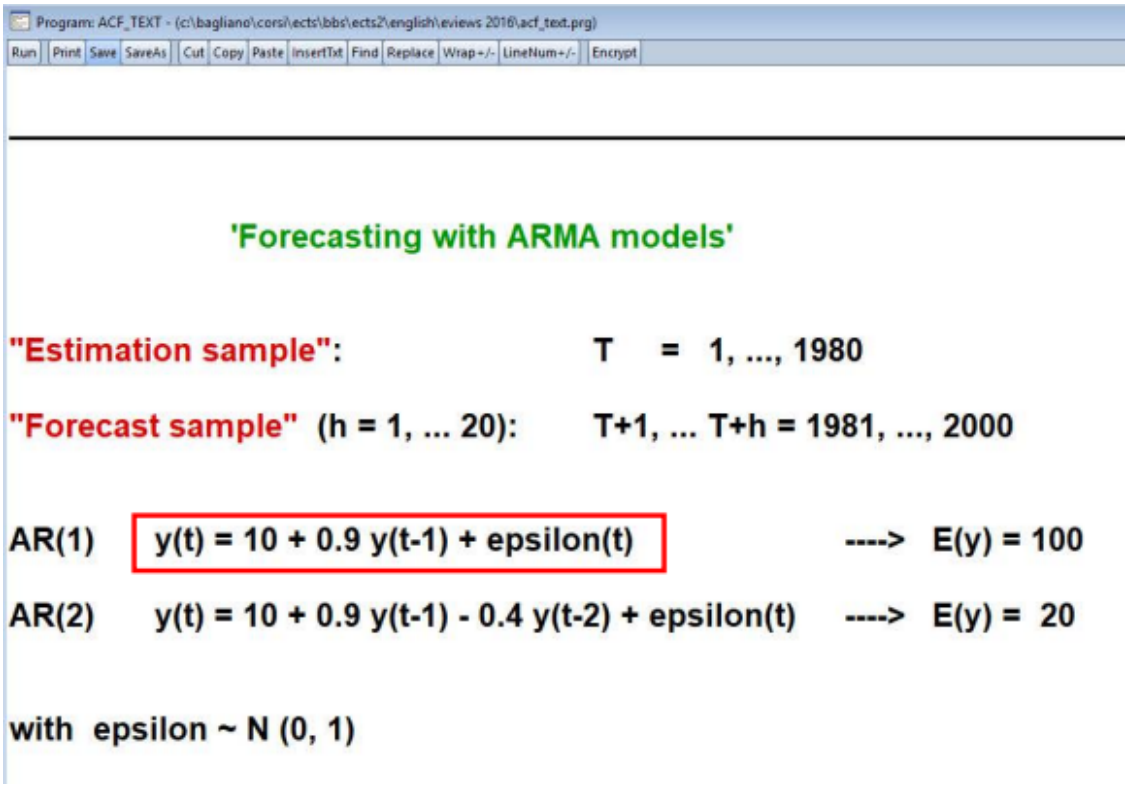


Figure 1.43: Additional figure from the original slide image set 2, image 9.

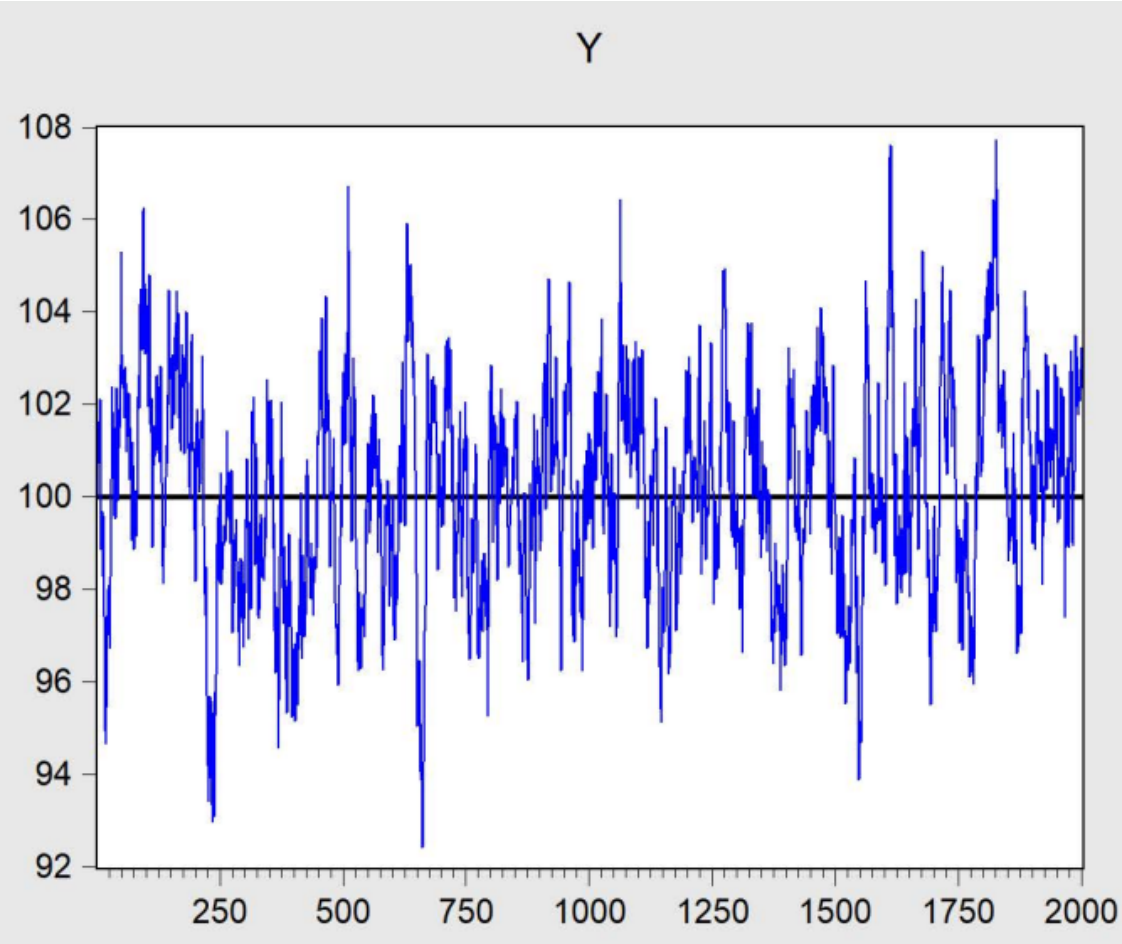


Figure 1.44: Additional figure from the original slide image set 2, image 10.

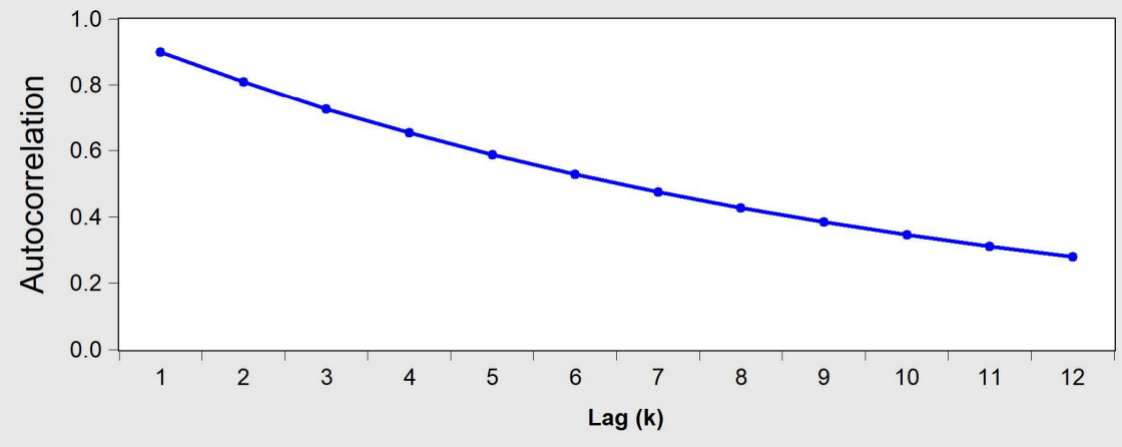


Figure 1.45: Additional figure from the original slide image set 2, image 11.

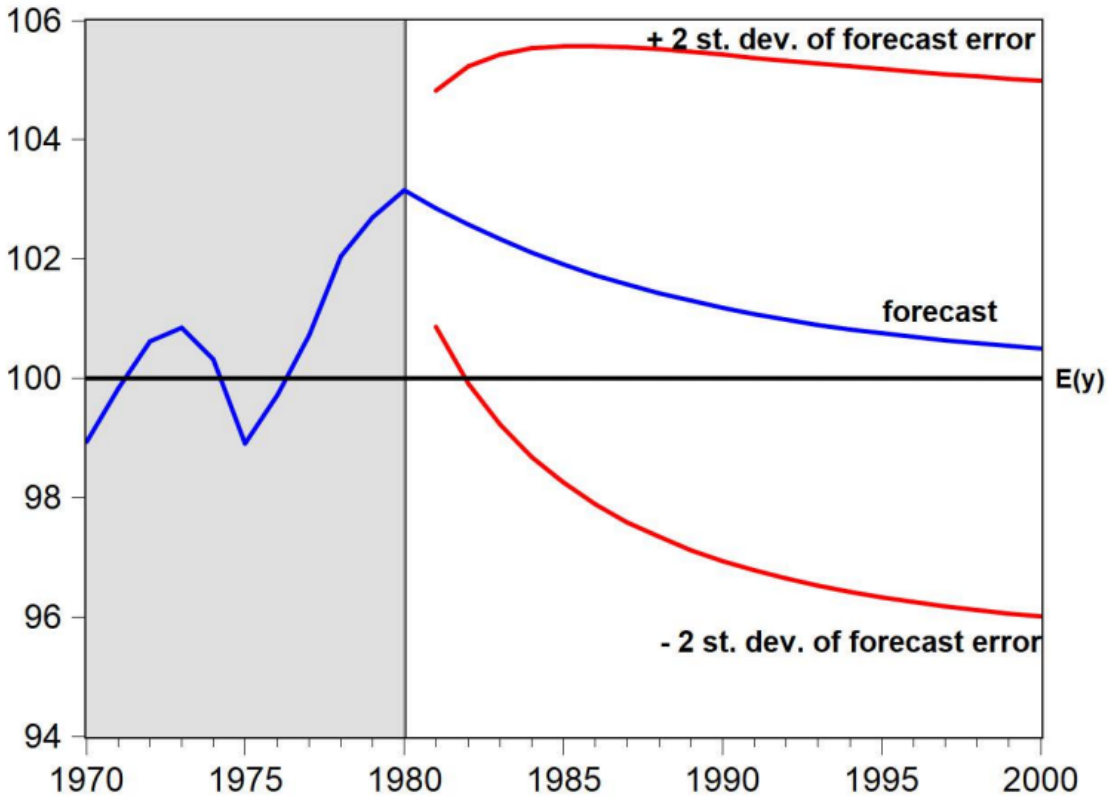


Figure 1.46: Additional figure from the original slide image set 2, image 12.

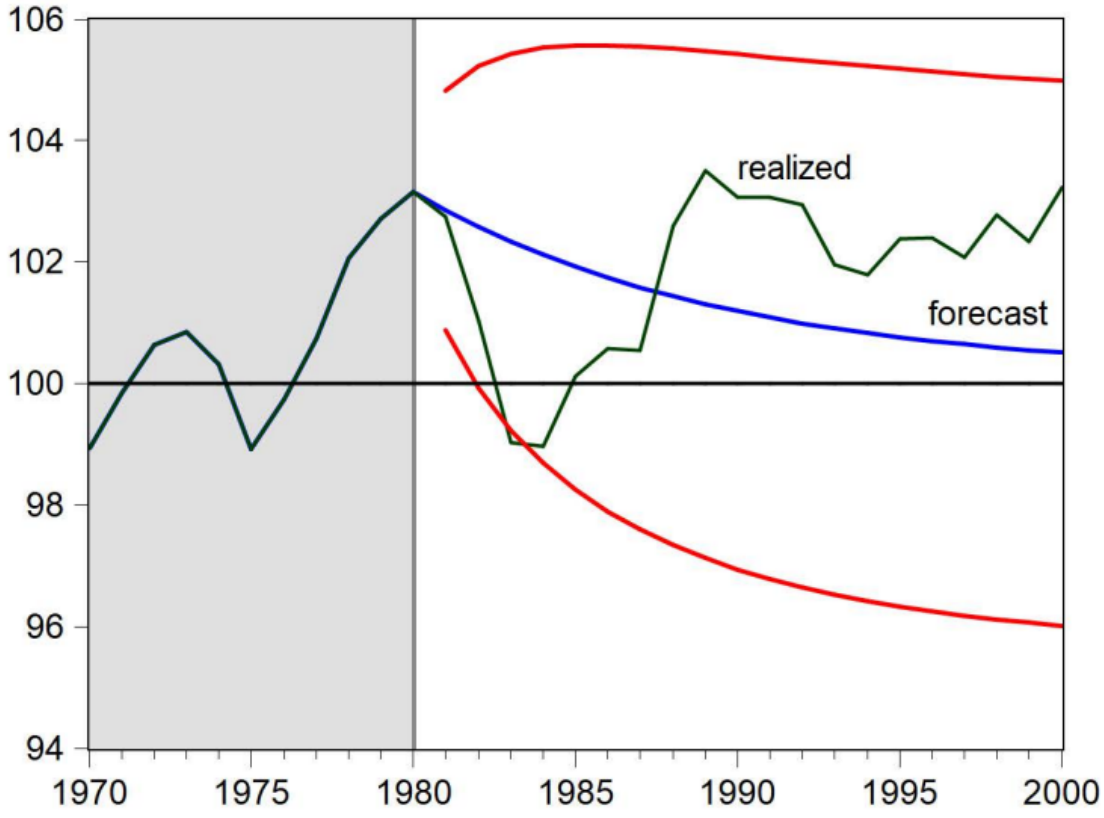


Figure 1.47: Additional figure from the original slide image set 2, image 13.

'Forecasting with ARMA models'

"Estimation sample": T = 1, ..., 1980

"Forecast sample" (h = 1, ... 20): T+1, ... T+h = 1981, ..., 2000

AR(1) y(t) = 10 + 0.9 y(t-1) + epsilon(t) ----> E(y) = 100

AR(2) y(t) = 10 + 0.9 y(t-1) - 0.4 y(t-2) + epsilon(t) ----> E(y) = 20

with epsilon ~ N (0, 1)

Figure 1.48: Additional figure from the original slide image set 2, image 14.

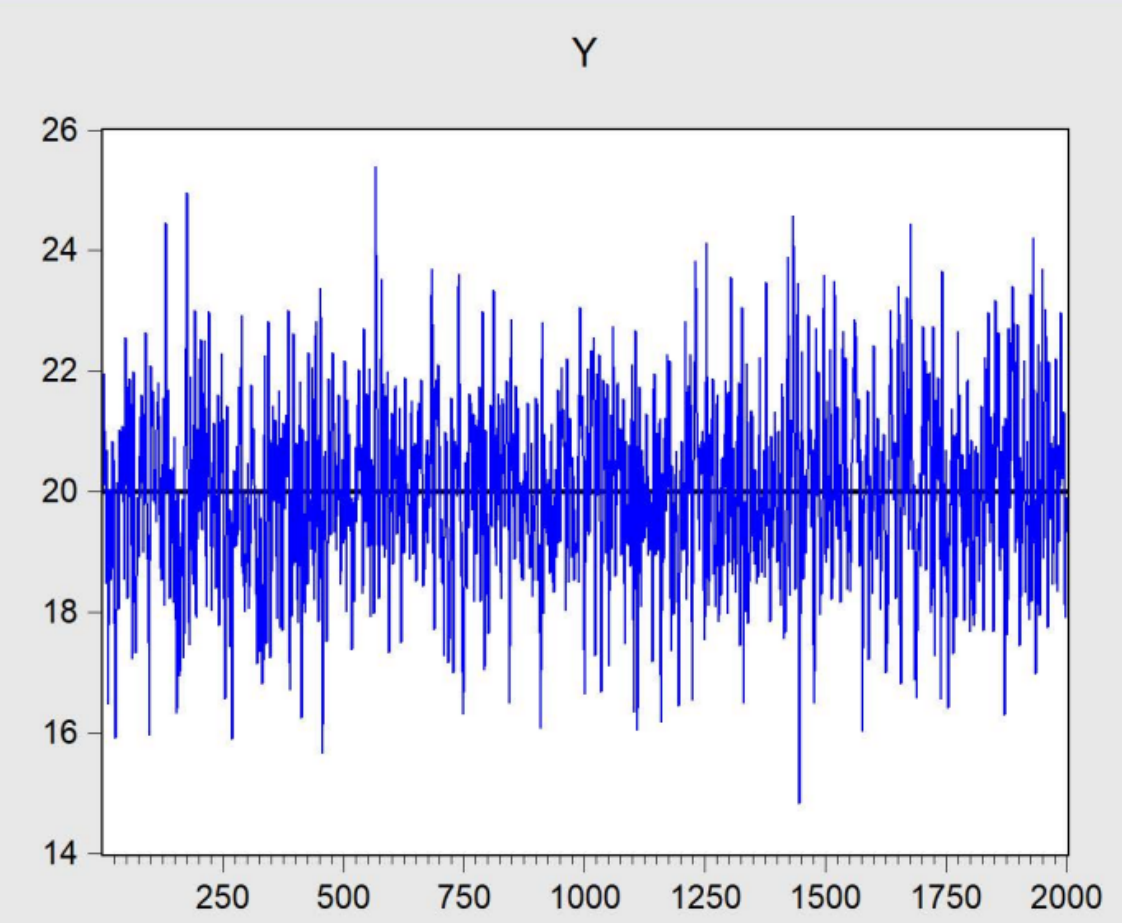


Figure 1.49: Additional figure from the original slide image set 2, image 15.

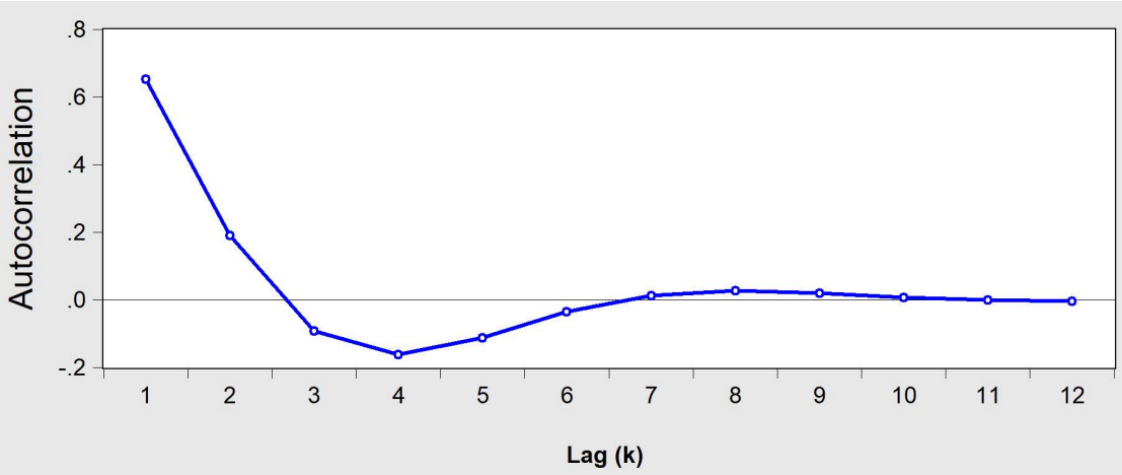


Figure 1.50: Additional figure from the original slide image set 2, image 16.

1.3 Set 3 – Non-stationary Time Series

Non-stationary time series Many economic time series do not satisfy the stationarity assumption, displaying a behavior over time that contrasts with the assumption of time-invariance of (at least the first two moments of) the probability distribution of the individual observations.

The non-stationary nature of the series entails some relevant problems:

1. Traditional asymptotic theory, based on the convergence of sample moments to their theoretical counterparts, can be applied to stationary series. This result does not apply to non-stationary series, and inference cannot be based on standard distributions.

2. If the series are non-stationary, regression models can yield evidence of "spu- rious" (statistically significant) or "nonsense" relationships among truly in- dependent variables.

Therefore, appropriate techniques are needed to detect non-stationarity and to ob- tain non-spurious relationships among non-stationary variables, when they exist.

Non-stationarity of a series can be due to several causes, the most important of which are:

1. the existence of structural breaks in the process generating the series, with changes in the properties of the probability distributions of the observations;

2. the presence of a persistent, long-run trend component driving the evolution of the series.

In the following we will deal with the latter cause of non-stationarity, starting with the distinction between two kinds of trend.

7.1 Deterministic trend vs. stochastic trend Two models are mostly used to describe the non-stationarity of a series due to long-run trends.

7.1.1 Deterministic trend

Let us assume that the process $\{y_t\}$ is described by $y_t = q(t) + u_t$ where • $g(t)$ is a deterministic function of time t: the deterministic trend. Examples are:

$$g(t) = \delta + \gamma t$$
$$g(t) = \delta + \gamma_1 t + \gamma_2 t^2$$

- u_t is a stationary and ergodic stochastic process of the following form:

$u_t = \sum_{i=0}^{\infty} \theta_j \varepsilon_{t-j}$ where $\varepsilon \sim IID(0, \sigma_\varepsilon^2), \sum_{j=0}^{\infty} |\theta_j| < \infty$.

Therefore, the process $\{y_t\}$ can be expressed as $y_t = g(t) + \sum_{i=0}^{\infty} \theta_j \varepsilon_{t-j}$ from which $y_t - g(t) = \sum_{i=0}^{\infty} \theta_i \varepsilon_{t-j}$

Deviations of y_t from the deterministic component follow a mean zero, stationary and ergodic process. In this case the series y_t is said to be trend stationary.

Main implication: there is a tendency of the series to return toward the deterministic trend $g(t)$ after a deviation due to a shock ε (this behavior is called mean reversion).

The effect of ε on y is therefore only temporary.

7.1.2 Stochastic trend

Let now be the following stochastic process (known as random walk):

$y_t = y_{t-1} + \varepsilon_t$ with $\varepsilon \sim IID(0, \sigma_\varepsilon^2)$. Substituting backward the past observations of y we get $y_t = \varepsilon_t + \varepsilon_{t-1} + \ldots + \varepsilon_1 + y_0$

$= \sum_{i=1}^{t} \varepsilon_i + y_0$ The value of y at time t is the sum of the initial value of the series y_0 and of all past realizations of the stationary process ε with unit coefficients. The term $\sum_{i=1}^{t} \varepsilon_i$ is the stochastic trend of the series.

Main implication: in this case there is no mean reversion (each ε_t has permanent effects on y).

PROPOSITION 1.8– Conditional moments of a random walk

(unconditional moments of a non-stationary process do not exixt; there- fore we can only define moments conditional on the initial value y_0; therefore means, variances and covariances defined below must all be interpreted as conditional on y_0):

(i) $E(y_t) = E(y_0 + \sum_{i=1}^{t} \varepsilon_i) = y_0$ (ii) $Var(y_t) = Var(y_0 + \sum_{i=1}^{t} \varepsilon_i) = \sum_{i=1}^{t} Var(\varepsilon_i) = t\sigma_\varepsilon^2$. The variance is not

constant over time and tends to ∞ as $t \to \infty$ (non-stationary process) (iii) the autocovariance function is :

$$\gamma_{kt} = E\left(\sum_{i=1}^{t} \varepsilon_i \sum_{j=1}^{t-k} \varepsilon_j\right) = \sum_{i=1}^{t-k} E(\varepsilon_j^2) = (t-k)\,\sigma_\varepsilon^2$$

Also the autocovariances tend to ∞ as $t \to \infty$ (non-stationary process).

(iv) the autocorrelation function (ACF) is then $\rho_{kt} = \frac{Cov(y_t, y_{t-k})}{\sqrt{Var(y_t)Var(y_{t-k})}} \equiv \frac{(t-k)\sigma_\varepsilon^2}{\sqrt{t\sigma^2(t-k)\sigma^2}} = \sqrt{\frac{t-k}{t}}$

yielding two implications:

– for any given t, $\rho_{k\,t}$ decreases slowly as k increases;

$-\rho_{kt} \to 1$ as $t \to \infty$ (for all k): the realizations of y are asymptotically perfectly correlated, therefore the process has an infinite "memory".

The random walk is an example of a process integrated of order 1:

$y_t \sim I(1)$ The first difference of this process, $\Delta y_t \equiv y_t - y_{t-1} = \varepsilon_t$ is a stationary (and ergodic) process with a zero expected value, which is defined as a process integrated of order zero:

$\Delta u_t \sim I(0)$ Therefore, the original non-stationary process $\{y_t\}$ can be made stationary by taking first differences: this process is called difference stationary.

DEFINITION 1.14– Integrated process and ARIMA(p,d,q)

General definition: a stochastic process $\{y_t\}$ is integrated of order $d, y_t \sim I(d)$, if its d-th difference, $\Delta^d y_t \equiv (1-L)^d y_t$, is stationary. For example:

$$
\begin{aligned}
& I(0): && y_t = \varepsilon_t && \\
& I(1): && y_t = y_{t-1} + \varepsilon_t && \Rightarrow \quad \Delta y_t = \varepsilon_t \\
& I(2): && y_t = 2y_{t-1} - y_{t-2} + \varepsilon_t && \Rightarrow \quad \Delta y_t = \Delta y_{t-1} + \varepsilon_t \\
& && \Rightarrow \quad \Delta^2 y_t = \varepsilon_t &&
\end{aligned}
$$

If $\Delta^d y_t$ is described by a stationary $ARMA(p,q)$ process, the original $\{y_t\}$ process is defined as an autoregressive integrated moving average $ARIMA(p,d,q)$.

Notes

1. The property of non-stationarity can be formulated in terms of the roots of the characteristic equation of the autoregressive component.

EXAMPLE – Illustration

Example. if the $AR(2)$ process $y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t$ $(1 - \theta_1 L - \theta_2 L^2) y_t = \varepsilon_t$ has the following roots of the associated characteristic equation: $|z_1| > 1$ and $|z_2| = 1$, then y_t can be expressed as

$$
\begin{aligned}
& (1 - \varphi_1 L)(1 - L) y_t = \varepsilon_t \quad \text{with} \quad |\varphi_1| < 1 \\
\Rightarrow \quad & (1 - \varphi_1 L)\, \Delta y_t = \varepsilon_t \\
\Rightarrow \quad & \Delta y_t = \varphi_1 \Delta y_{t-1} + \varepsilon_t
\end{aligned}
$$

Then Δy_t is described by a stationary $AR(1)$ process and y_t has a unit root, being generated by a non-stationary $ARIMA(1,1,0)$ process.

2. A random walk with drift process is:

$y_t = \delta + y_{t-1} + \varepsilon_t$ Given an initial observation y_0, we now have $y_t = \underbrace{\delta t}_{\text{deterministic trend}} + \underbrace{\sum_{i=1}^{t} \varepsilon_i}_{i=1} + y_0$

The presence of a drift term δ does not affect the difference stationarity of the process, since $\Delta y_t = \delta + \varepsilon_t$ is stationary. However, now also the expected value (conditional on the initial y_0) tends to ∞ as $t \to \infty$:

$E(y_t) = y_0 + \delta t$ 3. In general, we can summarize the main features of $I(0)$ and $I(1)$ series as follows I(0) $\frac{I(1)}{y_0 \text{ or function of } t}$ constant mean constant $|\to \infty$ as $t \to \infty$ variance $\to 0$

ACF $\to 1$ effect of shocks || transitory | permanent

7.2 Non-stationarity and estimation of an $AR(1)$ model

The asymptotic theory used to derive the properties of estimators in models with stationary variables cannot be applied if the series involved are non-stationary.

An example is the OLS estimator from estimation of an $AR(1)$ model when the underlying process is a random walk. In this case y_t is generated by $y_t = y_{t-1} + \varepsilon_t$ (1) where the ε's are distributed independently from a normal distribution with mean 0 and variance σ_ε^2, and with $y_0 = 0$. OLS estimation of the AR(1) model

$y_t = \beta\, y_{t-1} + v_t$ (2) yields the following estimator:

$$\hat{\beta}_{OLS} = \frac{\sum_{t=1}^{T} y_t y_{t-1}}{\sum_{t=1}^{T} y_{t-1}^2} = 1 + \frac{\sum_{t=1}^{T} y_{t-1}\varepsilon_t}{\sum_{t=1}^{T} y_{t-1}^2}$$

Asymptotic theory for non-stationary variables yields a number of important results, that are stated below and illustrated by means of a simple Monte Carlo simulation.

Using the random walk in (1) as the data generating process, with the process $\{\varepsilon_t\}$ generated as independent random drawings from a standard normal distribution (mean 0 and unit variance), the $AR(1)$ model in (2) is estimated by OLS over a relatively small ($T_1 = 100$) and a relatively large sample size ($T_2 = 500$). The experiment is replicated 1000 times: for each replication (for both sample sizes), the estimated coefficients ($\hat{\beta}_1$ and $\hat{\beta}_2$, respectively) and the associated t-statistics

(t_1 and t_2) are recorded.

The main results are the following:

1. the estimator $\hat{\beta}_{OLS}$ is biased in finite samples ($E(\hat{\beta}_{OLS}) \neq 1$) but converges in probability to 1 as the sample size goes to ∞ ($T \to \infty$):

$p\lim \hat{\beta}_{OLS} = 1$ and the "velocity of convergence" of β_{OLS} is proportional to $\frac{1}{T}$, and is greater than that of the OLS estimator for a stationary and ergodic process (that has a velocity proportional to $\frac{1}{\sqrt{T}}$). Therefore

$\hat{\beta}_{OLS}$ is a superconsistent estimator of the true $\beta = 1$ (Figure 4).

$T_2 = 500$ T1= 100 Au 12 Deu 2 1.025 1.050 0.850 0.875 0.900 0.850 0.875 0.900 0.925 0.950 0.975 1.000 0.925 0.950 1.025 0.975 1.000 1.050 Estimated beta1 coefficient Estimated beta2 coefficient Figure 4. Empirical distributions of $\hat{\beta}_{OLS}$ coefficients across 1000 replications for $T_1 = 100$ and $T_2 = 500$.

2. The asymptotic distribution of $T(\hat{\beta}_{OLS} - 1)$ is not standard normal:

$T\left(\hat{\beta}_{OLS} - 1\right) \not\to N(0,1)$ Note that, given the superconsistency of the estimator, it is necessary to multiply the deviation of $\hat{\beta}_{OLS}$ from 1 by T and not by $\sqrt{T}$ (as in the case of consistent

estimators) to obtain a variable with an asymptotic distribution, even though not normal in this case (Figure 5).

3. The ML estimator of the residual variance is a consistent estimator of σ_ε^2:

$p\lim s^2 = p\lim \frac{\sum_{t=1}^{T}(y_t - \hat{\beta}_{OLS} y_{t-1})^2}{T} = \sigma_\varepsilon^2$ 4. The t-statistic for the test of the null hypothesis $H_0 : \beta = 1$

$t_{\hat{\beta}} = \frac{\hat{\beta}_{OLS} - 1}{s_{\hat{\beta}_{OLS}}}$ converges in distribution (under H_0) to a non standard random variable (Figure 6).

5. The superconsistency property of $\hat{\beta}_{OLS}$ obtains also when ε_t is autocorrelated (whereas in the stationary case the OLS estimator is inconsistent).

.24 Distribution of T(beta - 1) for T=100 Distribution of T(beta - 1) for T=500 .20 .16 .12 .08 .04 .00

-20

-16

-12

-8

-24

-4 Figure 5. Empirical distributions of $T(\hat{\beta}_{OLS} - 1)$ for $T_1 = 100$ and $T_2 = 500$.

.5 Distribution of t-stat for T=100 Distribution of t-stat for T=500 .4 .3 .2 .1 .0

-2

-1

-3

-4 Figure 6. Empirical distributions of t-statistics for $T_1 = 100$ and $T_2 = 500$ compared with t distribution.

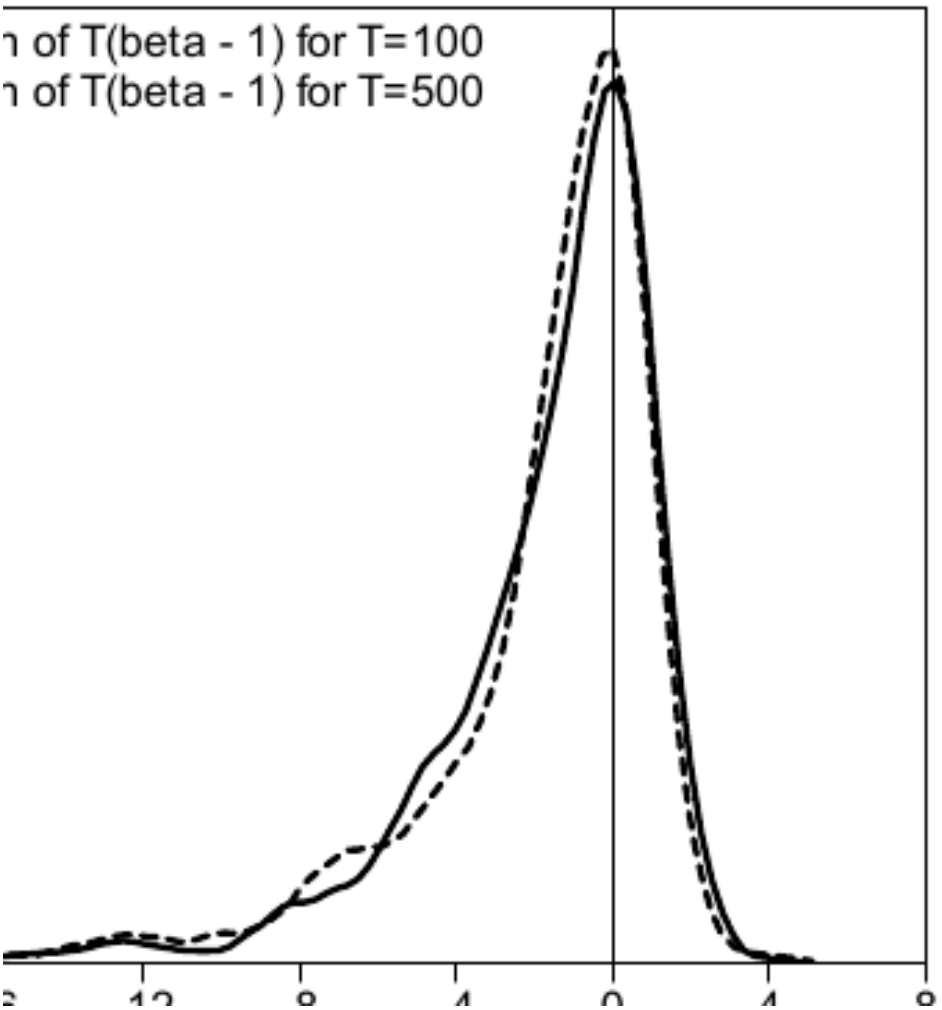


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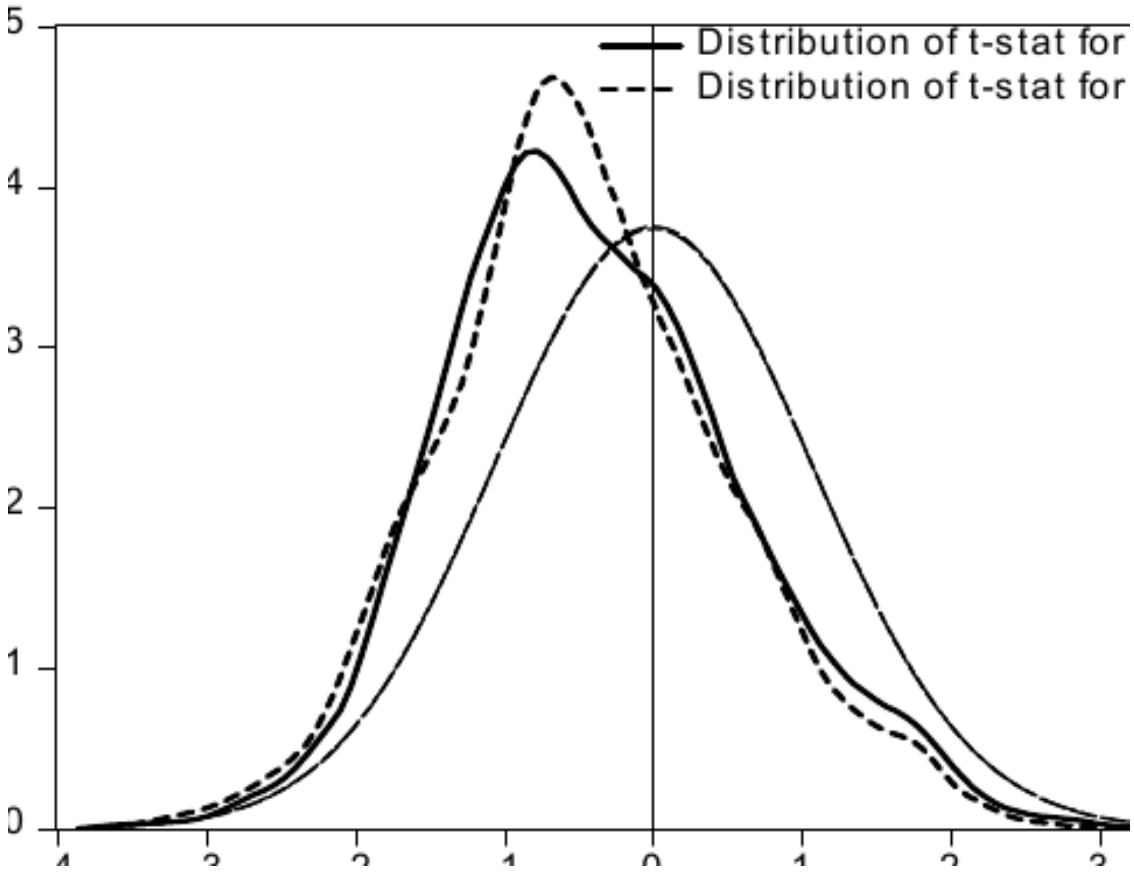


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Exercise. Consider the following random walk plus noise stochastic process $y_t = y_0 + \sum_{i=1}^{t} \varepsilon_i + \eta_t$ where $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$ and $\eta \sim WN(0, \sigma_\eta^2)$ are not correlated: $E(\varepsilon_t \eta_{t-s}) = 0$ for all

$$t$$

$$s$$

(a) Show that y_t follows an $ARIMA(p, d, q)$ process and find the values of p, d and q.

(b) Derive (conditional onto the initial observation y_0) the following properties of y: $E(.), Var(.)$, γ_k and ρ_k. Compare the ACF of this process with that of a random walk (without noise).

(c) Check the stationarity of Δy_t, find its ACF and compare it with that of an $MA(1)$ process.

7.3 Unit root tests In order to apply correctly the results of asymptotic theory, and to proceed to forecasting, it is extremely important to distinguish between deterministic and sto- chastic trends.

The most common tests for non-stationarity differ as to the null hypothesis tested and are in general very sensitive to the specification of the de- terministic component of the model. The kind of test most widely used has been proposed by Dickey and Fuller (1979, 1981).

7.3.1 Dickey-Fuller (DF) tests Specification DF tests are based on the following specification of $AR(1)$ models:

(i) $y_t = \delta + \theta y_{t-1} + \varepsilon_t$ (ii) $y_t = \delta + \gamma t + \theta y_{t-1} + \varepsilon_t$ with $\varepsilon \sim IID(0, \sigma_\varepsilon^2)$. The null and alternative

hypotheses are:

$H_0 : \theta = 1 \Rightarrow y \sim I(1)$ $H_1 : |\theta| < 1 \Rightarrow y \sim I(0)$ In the two cases above we have:

(i) under H_0 the process has a unit root and is a random walk with drift $y_t = \delta + y_{t-1} + \varepsilon_t \Rightarrow \Delta u_t = \delta + \varepsilon_t$ whereas under H_1 the process is a stationary $AR(1)$ with a non-zero uncon- ditional mean;

(ii) under H_0 the process has a stochastic trend (unit root) and a linear deter- ministic trend in its first difference Δy_t $y_t = \delta + \gamma t + y_{t-1} + \varepsilon_t \Rightarrow \Delta y_t = \delta + \gamma t + \varepsilon_t$ whereas under H_1 the process is trend stationary (i.e. stationary around a

linear deterministic trend).

Test procedure The test is implemented by estimating with OLS the following models:

(i) $\Delta y_t = \delta + (\theta - 1) y_{t-1} + \varepsilon_t$ (ii) $\Delta u_t = \delta + \gamma t + (\theta - 1) u_{t-1} + \varepsilon_t$

and testing the estimated coefficient on y_{t-1}: under H_0 (presence of a unit root, yielding non-stationarity) this coefficient is 0, whereas under H_1 it is < 0. From the estimate, the following statistic is calculated:

$DF = \frac{\hat{\theta}-1}{st.err.(\hat{\theta})}$ whose distribution is non-standard and has the following critical values:

model (i) model (ii) 5% 1% 1% 5% $T = 50$

-3.58 -2.93

-3.50 -4.14 $T = 100$ -3.51 -2.89 -4.04 -3.45 $T = \infty$ -3.43 -2.86 -3.96 -3.41 As an illustration of the features of the distribution of the Dickey-Fuller unit root test statistics, we set up a Monte Carlo simulation, generating 10000 random walks processes as in model (i) above, using independent drawings of ε_t from a standard normal distribution, for a sample of length $T = 200$. For each replication an OLS regression of the form $\Delta y_t = \beta_0 + \beta_1 y_{t-1} + v_t$ is run and the empirical

distributions of the estimated coefficient on y_{t-1} and the corresponding DF-statistic (i.e. $\beta_1 / st.err.(\beta_1)$) are shown in Figure 7. In the left panel, it can be seen that the distribution of β_1 is not centered around the true value of the corresponding parameter under the null hypothesis ($\theta - 1 = 0$) and is remarkably skewed to the left.

In the right panel of Figure 7, the distribution of the DF test statistic is displayed:

the 5% quantile of the distribution is -2.87, close to the asymptotic critical value reported in the table above (-2.86). Moreover, the DF distribution is compared with a t distribution (centered on the true value of zero under the null hypothesis) in order to show that, using the 5% critical value of -1.65 from the t distribution to test for a unit root would result in a rejection of the (true) null hypothesis of non-stationarity around 46% of the times (i.e. the "size" of the test would be 46% instead of 5%): the correct distribution to employ is the DF distribution, featuring lower critical values.

Empirical distribution of DF statistic vs. t distribution Empirical distribution of coefficient on y(t-1) .6 5% crit, value 5% crit. value DF statistic DF distribution t distribution

— t (-2.87) (-1.65) .5 Density .3 .2 .1 .0

-.10

-.08

-.06

-.04 -.02 .00 .02

-.16

-.14

-.12 .04

-6

-5

-3

-2

-.18

-4

-1 DF statistic Estimated coefficient on y(t-1) Figure 7. Empirical distributions of the coefficient on y_{t-1} and of the DF statistic for a sample $T = 200$.

DF tests suffer from two main problems:

Problems • in the presence of autocorrelation of the error term ε_t, its distribution depends on the autocovariances of ε_t; a possible solution is the use of the augmented $DF(ADF)$ test, illustrated in subsection 7.3.2 • it has a low power against alternative stationary processes with roots close (though less than) unity; in this case it could be useful to combine the infor- mation provided by DF tests with that obtained by other tests that adopt stationarity (i.e. the absence of a unit root) as a null hypothesis, e.g. the KPSS test (subsection 7.3.3). Whenever possible, increasing the span of the data used in the analysis (i.e. the length of the observation period) would

lead to more powerful unit root tests. Note that what matters more for power is not the number of observations per se (T), but the extension of the time period analyzed: for a given T, power is enhanced if the time span is longer;

on the other hand, for a given span, increasing the number of observations by sampling the data at a higher frequency yields only a marginal increase in power. For example, a series of annual data over an extended time period will often lead to unit root tests with higher power than those obtained using monthly data over a shorter time span.

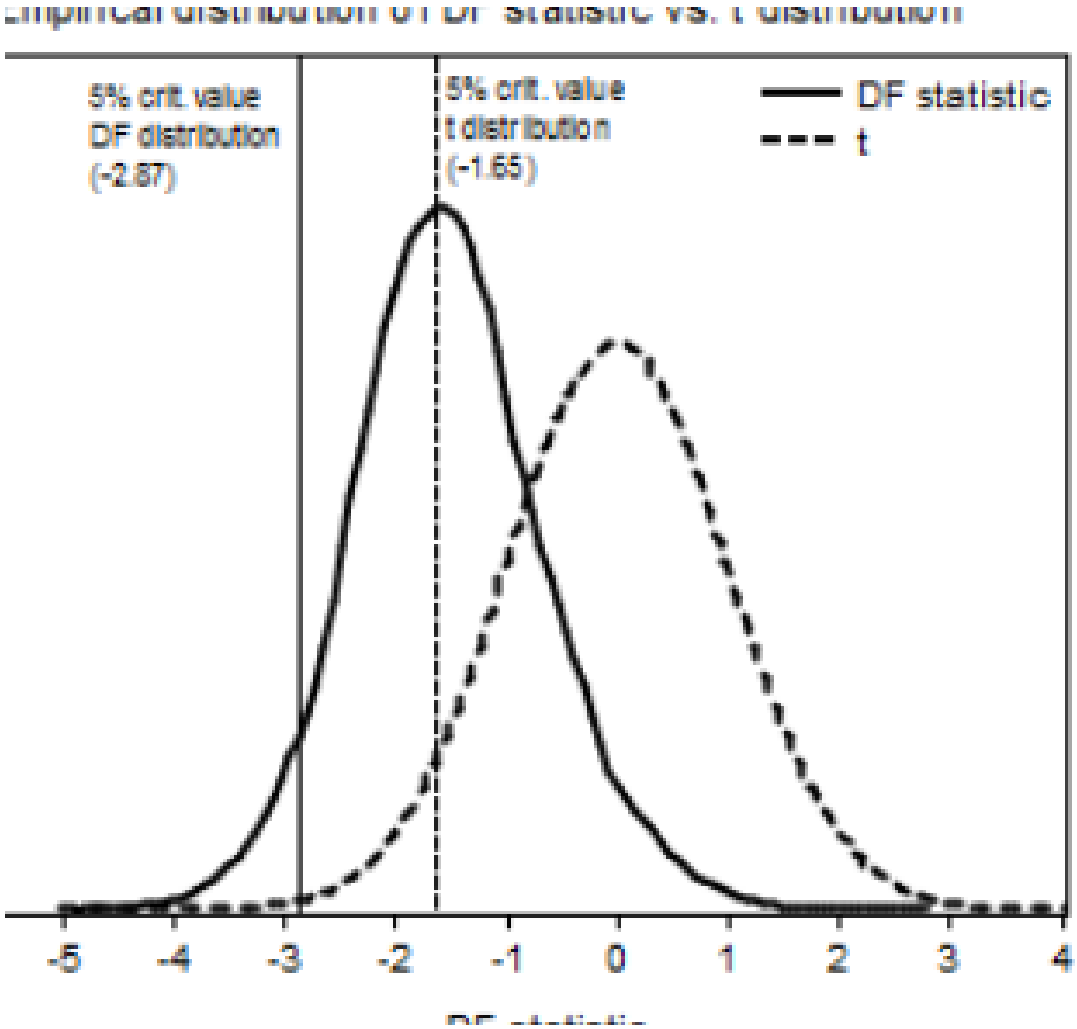


Figure 1.53: Course figure, source page 11, crop 1.

The ADF test 7.3.2 If the $AR(1)$ hypothesis of the DF test is substituted with a more general $AR(p)$ hypothesis, in the case of more general deterministic component we have $y_t = \delta + \gamma t + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \ldots + \theta_n y_{t-n} + \varepsilon_t$ that can be rewritten (by means of a simple reparameterization) as

$$\Delta y_t = \delta + \gamma t + \pi y_{t-1} + c_1 \Delta y_{t-1} + c_2 \Delta y_{t-2} + \cdots + c_{p-1} \Delta y_{t-p+1} + \varepsilon_t$$

with $\pi = \theta_1 + \theta_2 + \ldots + \theta_n - 1$ $c_1 = -(\theta_2 + \theta_3 + \ldots + \theta_n)$ $c_2 = -(\theta_3 + \theta_4 + \ldots + \theta_n)$ ...

$c_{p-1} = -\theta_p$ The null hypothesis of non-stationarity $H_0 : \pi = 0$ is equivalent to the existence of a unit root $(z = 1)$ for the associated characteristic equation $\theta(z) = 0$:

$\theta(1) = 0 \Rightarrow 1 - \theta_1 - \theta_2 - \ldots - \theta_n = 0 \Rightarrow \pi = 0$ The test is implemented on the OLS estimate of the π coefficient:

$ADF = \frac{\hat{\pi}}{st.err(\hat{\pi})} \sim DF$ The ADF statistic has the same distribution as the DF statistic.

The KPSS test 7.3.3 Differently from the DF and ADF tests, the $KPSS$ (Kwiatkowski, Phillips, Schmidt e Shin, 1992) test has a null hypothesis of stationarity of the process and is based on the OLS regression of y_t on a deterministic trend:

$y_t = \delta + \gamma t + \varepsilon_t$ From the estimated residuals $e_t = y_t - \hat{\delta} - \hat{\gamma} t$ an estimate of the variance is

obtained $s^2 = \frac{1}{T}\sum_{t=1}^{T} e_t^2$

and, for each t, the partial sum of the residuals is computed as $S_t = \sum_{i=1}^{t} e_i$ from which the following statistic is constructed $KPSS = \frac{\frac{1}{T^2}\sum_{t=1}^{T} S_t^2}{s^2}$

The asymptotic distribution of this statistic has the following critical values: 0.146 (5%) e 0.216 (1%). If the computed value for KPSS is larger than the chosen critical value, then the null hypothesis of stationarity is rejected.

Non-stationary time series The stationarity property does not apply to many series The non-stationary nature of the series entails some relevant problems:

1. Traditional asymptotic theory, based on the convergence of sample moments to their theoretical counterparts, cannot apply to non-stationary series, and inference cannot be based on standard distributions 2. If the series are non-stationary, regression models can yield evidence of "spurious" (statistically significant) or "nonsense" relationships among truly independent variables

Non-stationarity can be due to several causes, the most important of which are:

1. the existence of structural breaks in the process generating the series, with changes in the properties of the probability distributions of the observations;

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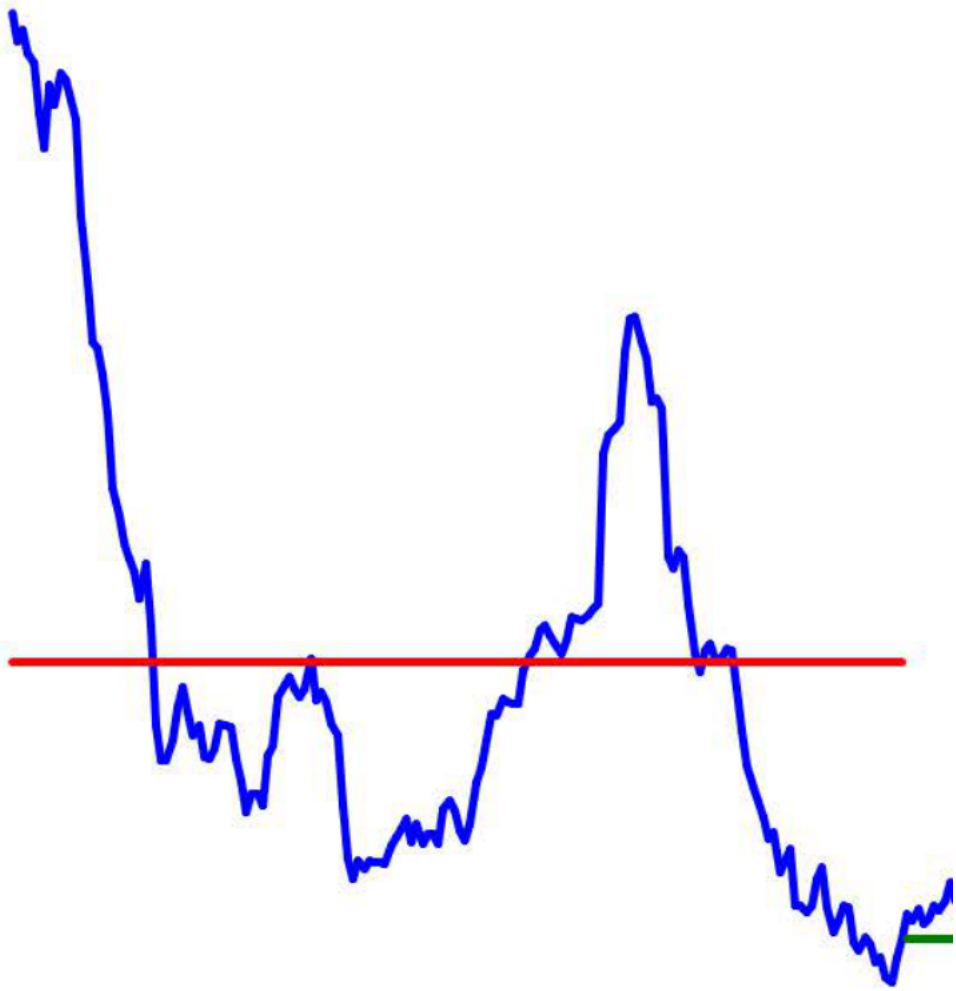


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Non-stationarity can be due to several causes, the most important of which are:

1. the existence of structural breaks in the process generating the series, with changes in the properties of the probability distributions of the observations;

2. the presence of a persistent, long-run trend component driving the evolution of the series.

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Deterministic vs. stochastic trend Deterministic trend Let us assume that $\{y_t\}$ is described by $y_t = g(t) + u_t$ where • $g(t)$ is a deterministic function of time t: the deterministic trend.

Examples:

$$g(t) = \delta + \gamma t$$
$$g(t) = \delta + \gamma_1 t + \gamma_2 t^2$$

• u_t is a stationary and ergodic stochastic process of the following form:

$u_t = \sum_{j=0} \theta_j \varepsilon_{t-j}$ where $\varepsilon \sim IID(0, \sigma_\varepsilon^2), \sum_{j=0}^{\infty} |\theta_j| < \infty$.

Process $\{y_t\}$ can be expressed as $y_t = g(t) + \sum_{j=0} \theta_j \varepsilon_{t-j}$ from which $y_t - g(t) = \sum_{j=0}^{\infty} \theta_j \varepsilon_{t-j}$

Deviations of y_t from the deterministic component follow a mean zero, stationary and ergodic process. In this case the series y_t is said to be trend stationary

Main implication: there is a tendency of the series to return toward the deterministic trend $g(t)$ after a deviation due to a shock ε (this behavior is called mean reversion) ⇒ the effect of ε on y is only temporary

Stochastic trend Let the following stochastic process (random walk):

$y_t = y_{t-1} + \varepsilon_t$ with $\varepsilon \sim IID(0, \sigma_\varepsilon^2)$

Substituting backward the past observations of y we get $y_t = \varepsilon_t + \varepsilon_{t-1} + \ldots + \varepsilon_1 + y_0 = \sum_{i=1}^{i} \varepsilon_i$ $+ y_0$ stochastic trend The value of y at time t is the sum of the initial value y_0 and of all

past realizations of the stationary process ε with unit coefficients.

The term $\sum_{i=1}^{t} \varepsilon_i$ is the stochastic trend of the series.

Main implication: in this case there is no mean reversion ⇒ each ε_t has permanent effects on y

Properties (conditional on y_0):

(i) $E(y_t) = E(y_0 + \sum_{i=1}^{t} \varepsilon_i) = y_0$ (ii) $Var(y_t) = Var(y_0 + \sum_{i=1}^{t} \varepsilon_i) = \sum_{i=1}^{t} Var(\varepsilon_i) = t\sigma_\varepsilon^2$. The

variance is not constant over time and tends to ∞ as $t \to \infty$ (non-stationarity)

(iii) the autocovariance function is :

$$\gamma_{kt} = E\left(\sum_{i=1}^{t} \varepsilon_i \sum_{i=1}^{t-k} \varepsilon_j\right) = \sum_{i=1}^{t-k} E(\varepsilon_j^2) = (t-k)\sigma_\varepsilon^2$$

autocovariances tend to ∞ as $t \to \infty$ (non-stationarity)

(iv) the autocorrelation function (ACF) is then $\rho_{kt} = \frac{Cov(y_t, y_{t-k})}{\sqrt{Var(v_t)Var(v_{t-k})}} \equiv \frac{(t-k)\sigma_\varepsilon^2}{\sqrt{t\sigma_\varepsilon^2 (t-k)\sigma_\varepsilon^2}} = \sqrt{\frac{t-k}{t}}$

with two implications:

- for any given t, ρ_{kt} decreases slowly as k increases;

- $\rho_{kt} \to 1$ as $t \to \infty$ (for all k): the realizations of y are asymptotically perfectly correlated: the process has an infinite "memory"

The random walk is an example of a process:

integrated of order 1 or $y_t \sim I(1)$ The first difference of this process, $\Delta y_t \equiv y_t - y_{t-1} = \varepsilon_t$ is a stationary (and ergodic) process with a zero expected value:

integrated of order 0 or $\Delta y_t \sim I(0)$ The original non-stationary process $\{y_t\}$ can be made stationary by taking first differences: this process is called difference stationary

DEFINITION 1.15– Integrated process and ARIMA(p,d,q)

General definition: a stochastic process $\{y_t\}$ is integrated of order d, $y_t \sim I(d)$, if its d-th difference, $\Delta^d y_t \equiv (1-L)^d y_t$, is stationary. For example:

$$I(0)$$
$$y_t = \varepsilon_t$$
$$I(1) : y_t = y_{t-1} + \varepsilon_t \implies \Delta y_t = \varepsilon_t$$
$$I(2) : y_t = 2y_{t-1} - y_{t-2} + \varepsilon_t \implies \Delta y_t = \Delta y_{t-1} + \varepsilon_t$$

$\Rightarrow \Delta^2 y_t = \varepsilon_t$ If $\Delta^d y_t$ is a stationary $ARMA(p,q)$ process, the original $\{y_t\}$ process is defined as autoregressive integrated moving average $ARIMA(p,d,q)$

× EViews - [Graph: GRAF_USCPI Workfile: USUK05::Untitled\]

- File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom 6.0 .016 US cpi D (US cpi) 5.8 .012 5.6 .008 5.4 .004 5.2 .000 5.0

-.004 4.8

-.008 2005 1995 1985 1990 2000 1980 1985 1990 2000 2005 1995 .012 D D (UScpi) .008 .004 .000

-.004

-.008

-.012 1985 1990 1995 2000 1980 2005 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 x8 11:38 09/03/2022

Notes

1. The property of non-stationarity can be formulated in terms of the roots of the characteristic equation of the autoregressive component.

EXAMPLE – Illustration

Example. if the $AR(2)$ process $y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t$ $(1 - \theta_1 L - \theta_2 L^2)v_t = \varepsilon_t$ has the following roots of the associated characteristic equation: $|z_1| > 1$ and $|z_2| = 1$, then y_t can be expressed as

$(1 - \varphi_1 L)(1 - L)y_t = \varepsilon_t$ with $|\varphi_1| < 1 \Rightarrow (1 - \varphi_1 L)\Delta y_t = \varepsilon_t \Rightarrow \Delta v_t = \varphi_1 \Delta v_{t-1} + \varepsilon_t$

Then Δy_t is described by a stationary $AR(1)$ process and y_t has a unit root, being generated by a non-stationary $ARIMA(1, 1, 0)$ process

2. A random walk with drift process is:

$y_t = \delta + y_{t-1} + \varepsilon_t$ Given y_0, we have $y_t = \underbrace{\delta t}_{\text{deterministic trend}} + \sum_{i=1}^{i} \varepsilon_i + y_0$ stochastic trend

The drift δ does not affect the difference stationarity of y_t, since $\Delta v_t = \delta + \varepsilon_t$ is stationary. Now also the expected value (conditional on y_0) tends to ∞ as $t \to \infty$:

$$E(y_t) = y_0 + \delta t$$

3. In general, we can summarize the main features of $I(0)$ and $I(1)$ series as follows I(0) I(1) y_0 or function of t constant mean constant variance $\to \infty$ as $t \to \infty$ ACF $\to 0 \to 1$ effect of shocks transitory permanent

mean	constant	y_0 or function of t
variance	constant	$\to \infty$ as $t \to \infty$
ACF	$\to 0$	$\to 1$
effect of shocks	transitory	permanent

Figure 1.55: Course figure, source page 23, crop 1.

d X EViews

-01 File Edit Object View Proc Quick Options Add-ins Window Help Workfile: RWALK_FOR_PLOTS... - Workfile: RWALK PLOTS - (c:\bagliano\cor... - - X Program: ACF_TEXT - (c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016\acf_text.prg)

- X View Proc Object Save Freeze Details+/- View Proc Object Save Freeze Details+/- Show Fetch S Run Print Save SaveAs Cut Copy Paste InsertTxt Find Replace Wrap+/- LineNum+/- Encrypt Range: 1 2000 - 2000 obs Filter: * Range: 1 1100 - 1100 obs Filter: * ^ Sample: 1 200 - 200 obs Order: Name Sample: 1 500 - 500 obs Order: Name B c delta S epsilon art1 'Deterministic versus stochastic trends: implications for forecasting' art2 beta1 eq_det eq_rw BC for20_det delta distr1 "Estimation sample":

T = 1, ..., 1980 for20_det_st_err for20_rw distr1_norm for20_rw_st_err M distr2 gamma gamma resid theta distr2_norm "Forecast sample" (h = 1, ... 20): T+1, ... T+h = 1981, ..., 2000

epsilon grafb1 time y_det y_det_200 grafb2 grafbeta1 grafbeta2 y_dt_rw_200 grafici1 ∑ y_rw y_rw_200 grafici2 Compare forecast from graft1 yf_det yf_det_l yf_det_se yf_det_u yf_det_u yf_rw yf_rw_l yf_rw_se yf_rw_u graft2 m DI plot_beta plot_beta_distr TS model: y(t) = 1.0 + 0.5 t + epsilon(t) –> deterministic (linear) trend plot t distr M resid # t1 # t1emp025 DS model: $y(t) = 0.5 + y(t-1) + \epsilon(t)$ –> random walk with drift # t1emp975 # t2 # t2emp025 t2emp975 tstat1 tstat2 with epsilon ~ normal WN (0, 1) W y_rw_500 J y_rw_acf \ Untitled / New Page <» Untitled / New Page

Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_for_plots xR V V 10:30 x e **N** P Fz S o ×∃ w P (3) G S D

× EViews - [Graph: Y_DT_RW_200 Workfile: RWALK_FOR_PLOTS::Untitled\]

-8 II File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Y_Linear time trend Y_Random walk with drift

-20 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_for_plots xR ⌢ 10/03/2022 **N** S D G

- Fz A w

d X EViews

-01 File Edit Object View Proc Quick Options Add-ins Window Help Workfile: RWALK_FOR_PLOTS... - Workfile: RWALK PLOTS - (c:\bagliano\cor... - - X Program: ACF_TEXT - (c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016\acf_text.prg)

- X View Proc Object Save Freeze Details+/- View Proc Object Save Freeze Details+/- Show Fetch S Run Print Save SaveAs Cut Copy Paste InsertTxt Find Replace Wrap+/- LineNum+/- Encrypt Range: 1 2000 - 2000 obs Filter: * Range: 1 1100 - 1100 obs Filter: * ⌢ Sample: 1 200 - 200 obs Order: Name Sample: 1 500 - 500 obs Order: Name B c delta S epsilon art1 'Deterministic versus stochastic trends: implications for forecasting' art2 beta1 eq_det eq_rw BC for20_det delta distr1 "Estimation sample":

T = 1, ..., 1980 for20_det_st_err for20_rw distr1_norm for20_rw_st_err M distr2 gamma gamma resid theta distr2_norm "Forecast sample" (h = 1, ... 20): T+1, ... T+h = 1981, ..., 2000 epsilon grafb1 time y_det y_det_200 grafb2 grafbeta1 grafbeta2 y_dt_rw_200 grafici1 ∑ y_rw y_rw_200 grafici2 Compare forecast from graft1 yf_det yf_det_l yf_det_se yf_det_u yf_det_u yf_rw yf_rw_l yf_rw_se yf_rw_u graft2 m DI plot_beta plot_beta_distr TS model: y(t) = 1.0 + 0.5 t + epsilon(t) –> deterministic (linear) trend plot t distr M resid # t1 # t1emp025 DS model: $y(t) = 0.5 + y(t-1) + \epsilon(t)$ –> random walk with drift # t1emp975 # t2 # t2emp025 t2emp975 tstat1 tstat2 with epsilon ~ normal WN (0, 1) W y_rw_500 J y_rw_acf \ Untitled / New Page <» Untitled / New Page

Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_for_plots xR V V 10:30 x e **N** P Fz S o ×∃ w P (3) G S D

× EViews - [Graph: FOR20_DET Workfile: RWALK_FOR_PLOTS::Untitled\]

-3 III File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Y with linear deterministic trend 1,004 1,000

- Y realized

— Y forecast 1975 1985 1995 1980 1990 2000 1970 Path =

c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_for_plots xR ⌢ 10:42 **N** S D

- Fz w

d X EViews

-01 File Edit Object View Proc Quick Options Add-ins Window Help Workfile: RWALK_FOR_PLOTS... - Workfile: RWALK PLOTS - (c:\bagliano\cor... - - X Program: ACF_TEXT - (c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016\acf_text.prg)

- X View Proc Object Save Freeze Details+/- View Proc Object Save Freeze Details+/- Show Fetch S Run Print Save SaveAs Cut Copy Paste InsertTxt Find Replace Wrap+/- LineNum+/- Encrypt Range: 1 2000 - 2000 obs Filter: * Range: 1 1100 - 1100 obs Filter: * ⌢ Sample: 1 200 - 200 obs Order: Name Sample: 1 500 - 500 obs Order: Name B c delta S epsilon art1 'Deterministic versus stochastic trends: implications for forecasting' art2 beta1 eq_det eq_rw BC for20_det delta distr1 "Estimation sample":

T = 1, ..., 1980 for20_det_st_err for20_rw distr1_norm for20_rw_st_err M distr2 gamma gamma resid theta distr2_norm "Forecast sample" (h = 1, ... 20): T+1, ... T+h = 1981, ..., 2000 epsilon grafb1 time y_det y_det_200 grafb2 grafbeta1 grafbeta2 y_dt_rw_200 grafici1 ∑ y_rw y_rw_200 grafici2 Compare forecast from graft1 yf_det yf_det_l yf_det_se yf_det_u yf_det_u yf_rw yf_rw_l yf_rw_se yf_rw_u graft2 m DI plot_beta plot_beta_distr TS model: y(t) = 1.0 + 0.5 t + epsilon(t) –> deterministic (linear) trend plot t distr M resid # t1 # t1emp025 DS model: $y(t) = 0.5 + y(t-1) + \epsilon(t)$ –> random walk with drift # t1emp975 # t2 # t2emp025 t2emp975 tstat1 tstat2 with epsilon ~ normal WN (0, 1) W y_rw_500 J y_rw_acf \ Untitled / New Page <» Untitled / New Page

Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_for_plots xR V V 10:30 x e **N** P Fz S o ×∃ w P (3) G S D

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- EViews - [Graph: FOR20_RW Workfile: RWALK_FOR_PLOTS::Untitled\] III File Edit Object View Proc Quick Options Add-ins Window Help _ X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Y generated by random walk with drift 1,030 Y realized Y forecast 1,025 1,020 1,015 1,010 1,005 1975 1980 1985 1990 1995 1970 2000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_for_plots xR ⌢ 10:46 O **N** D e

- Fz w S

Non-stationarity and estimation of an $AR(1)$ model The asymptotic theory used to derive the properties of estimators in models with stationary variables cannot be applied if the series involved are non-stationary.

EXAMPLE – Illustration

Example.

OLS estimator from estimation of an $AR(1)$ model when the process yt is a random walk $v_t = v_{t-1} + \varepsilon_t$ where the ε's are distributed independently from $N(0, \sigma_\varepsilon^2)$ and $y_0 = 0$. OLS estimation of the $AR(1)$ model $v_t = \beta v_{t-1} + v_t$

yields the following estimator:

$$\hat{\beta}_{OLS} = \frac{\sum_{t=1}^{T} y_t y_{t-1}}{\sum_{t=1}^{T} y_{t-1}^2} = 1 + \frac{\sum_{t=1}^{T} y_{t-1}\varepsilon_t}{\sum_{t=1}^{T} y_{t-1}^2}$$

Results from asymptotic theory for non-stationary variables:

1. the estimator $\hat{\beta}_{OLS}$ is biased in finite samples ($E(\hat{\beta}_{OLS}) \neq 1$) but converges in probability to 1 as the sample size goes to ∞ ($T \to \infty$):

$p\lim \hat{\beta}_{OLS} = 1$ and the "velocity of convergence" of $\hat{\beta}_{OLS}$ is proportional to $\frac{1}{T}$, greater than that of the OLS estimator for stationary processes (velocity proportional to $\frac{1}{\sqrt{T}}$)

$\hat{\beta}_{OLS}$ is a superconsistent estimator of the true $\beta = 1$.

× EViews - [Graph: Y_RW_500 Workfile: RWALK_PLOTS::Untitled\]

-3 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Y

-4 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_plots xR ⌃ 11:05 10/03/2022 O S D **N** G

- S w

× EViews - [Graph: Y_RW_ACF Workfile: RWALK_PLOTS::Untitled\]

-2 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Autocorrelation function (ACF) 1.0 0.8 Autocorrelation 0.6 0.4 0.2 0.0 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_plots x8 11:07 10/03/2022 i e **N** D Fz Sec.

S w

× EViews - [Graph: PLOT_BETA Workfile: RWALK_PLOTS::Untitled\]

-8 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Estimated Beta1 (T=100) Estimated Beta2 (T=500) Density 0.88 0.92 0.84 0.96 0.76 0.80 1.00 1.04 1.08 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_plots xR 11:09 10/03/2022 S ∼ O e r D Fz

- W S w x P 3)

2. The asymptotic distribution of $T(\hat{\beta}_{OLS} - 1)$ is not standard normal:

$T(\hat{\beta}_{OLS} - 1) \not\to N(0, 1)$ Note that, given the superconsistency of the estimator, it is necessary to multiply the deviation of $\hat{\beta}_{OLS}$ from 1 by T and not by $\sqrt{T}$ (as in the case of consistent estimators) to obtain a variable with an asymptotic (non-normal) distribution.

× EViews - [Graph: PLOT_BETA_DISTR Workfile: RWALK_PLOTS::Untitled\]

-8 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom .24 Distribution of T(beta - 1) for T=100 Distribution of T(beta - 1) for T=500 .20 .16 Density .12 .08 .04 .00

-20

-8

-24

-12

-16

-4 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_plots x8 ^ 11:13 10/03/2022 S e ∼ D

- Fz r S w *** P≥ 3)

3. The ML estimator of the residual variance is a consistent estimator of σ_ε^2:

$$p\lim s^2 = p\lim \frac{\sum_{t=1}^{T}(y_t - \hat{\beta}_{OLS} y_{t-1})^2}{T} = \sigma_\varepsilon^2$$

4. The t-statistic for the test of the null hypothesis $H_0 : \beta = 1$ $t_{\hat{\beta}} = \frac{\hat{\beta}_{OLS}-1}{s_{\hat{\beta}_{OLS}}}$

converges in distribution (under H_0) t o a non standard random variable

× EViews - [Graph: PLOT_T_DISTR Workfile: RWALK_PLOTS::Untitled\]

-0 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom .5 Distribution of t-stat for T=100 Distribution of t-stat for T=500 .4 .3 Density .2 .1 .0

-2

-3

-1

-4 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = rwalk_plots xR 11:16 10/03/2022 .

e ∼ S D

- G Fz w ×∃ S P * 3)

5. The superconsistency property of $\hat{\beta}_{OLS}$ obtains also when ε_t is autocorrelated (whereas in the stationary case the OLS estimator is inconsistent).

Unit root tests In order to apply correctly the results of asymptotic theory, and to proceed to forecasting, it is extremely important to distinguish between deterministic and stochastic trends.

Widely used non-stationarity tests:

1. differ for the null hypothesis tested 2. are in general sensitive to the specification of the deterministic component of the model

Dickey-Fuller (DF) tests Specification DF tests are based on the following specification of $AR(1)$ models:

(i) $v_t = \delta + \theta v_{t-1} + \varepsilon_t$ (ii) $y_t = \delta + \gamma t + \theta y_{t-1} + \varepsilon_t$ with $\varepsilon \sim IID(0, \sigma_\varepsilon^2)$. The null and alternative hypotheses are:

$$H_0 : \theta = 1 \implies y \sim I(1)$$
$$H_1 : |\theta| < 1 \implies y \sim I(0)$$

In the two cases above we have:

(i) under H_0 the process has a unit root and is a random walk with drift $y_t = \delta + y_{t-1} + \varepsilon_t \Rightarrow \Delta y_t = \delta + \varepsilon_t$ whereas under H_1 it is a stationary $AR(1)$;

(ii) under H_0 the process has a stochastic trend (unit root) and a linear deterministic trend in its first difference $y_t = \delta + \gamma t + y_{t-1} + \varepsilon_t \Rightarrow \Delta y_t = \delta + \gamma t + \varepsilon_t$ whereas under H_1 process is trend stationary (stationary around a linear deterministic trend)

Test procedure The test is implemented by estimating with OLS the following models:

(i) $\Delta y_t = \delta + (\theta - 1) y_{t-1} + \varepsilon_t$ (ii) $\Delta y_t = \delta + \gamma t + (\theta - 1) y_{t-1} + \varepsilon_t$ and testing the estimated coefficient on y_{t-1}:

under H_0 (presence of a unit root, yielding non-stationarity) the coefficient is 0 under H_1 the coefficient is < 0

From the estimate, the following statistic is calculated:

$DF = \frac{\hat{\theta} - 1}{st.err.(\hat{\theta})}$ whose distribution is non-standard with critical values:

model (i) model (ii) 1% 5% 1% 5% $T = 50$ | -3.58 -2.93 | -4.14 -3.50 $T = 100$ -3.51 -2.89 -4.04 -3.45 $T = \infty$ -3.43 -2.86 -3.96 -3.41

× EViews - [Graph: PLOT_DF_T_DISTR Workfile: DF::Untitled\] III File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Empirical distribution of DF statistic vs. t distribution Empirical distribution of coefficient on y(t-1) .6 5% crit. value

—- DF statistic 5% crit. value t distribution DF distribution (-1.65) (-2.87) .5 .4 15 - Density .3 .2 .1 .0

-.18 -.16 -.14 -.12 -.10 -.08 -.06 -.04 -.02 .00

-2

-5

-3 .02

-6 .04

-4

-1

Δ DF statistic Estimated coefficient on y(t-1) Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = df xR 11:36 10/03/2022 e D ∼ Fz r

- x∃ S w PE a

Problems DF tests suffer from two main problems:

in the presence of autocorrelation of the error term ε_t, its distribution depends on the autocovariances of ε_t; a possible solution is the use of the augmented DF (ADF) test • it has a low power

against alternative stationary processes with roots close (though less than) unity; in this case it could be useful to combine the information provided by DF tests with that obtained by other tests that adopt stationarity (i.e. the absence of a unit root) as a null hypothesis, e.g. the KPSS test.

The ADF test If the $AR(1)$ hypothesis of the DF test is substituted with a more general $AR(p)$ hypothesis, in the case of more general deterministic component we have $y_t = \delta + \gamma t + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \ldots + \theta_p y_{t-p} + \varepsilon_t$

$y_t = \delta + \gamma t + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \ldots + \theta_p y_{t-p} + \varepsilon_t$ can be rewritten (by reparameterization) as $\Delta y_t = \delta + \gamma t + \pi y_{t-1} + c_1 \Delta y_{t-1} + c_2 \Delta y_{t-2} + \ldots + c_{p-1} \Delta y_{t-p+1} + \varepsilon_t$

with $\pi = \theta_1 + \theta_2 + \ldots + \theta_p - 1$ $c_1 = -(\theta_2 + \theta_3 + \ldots + \theta_p)$ $c_2 = -(\theta_3 + \theta_4 + \ldots + \theta_n)$ • • • • • • • • •

$$c_{p-1} = -\theta_p$$

The null hypothesis of non-stationarity $H_0 : \pi = 0$ is equivalent to the existence of a unit root ($z = 1$) for the associated characteristic equation $\theta(z) = 0$:

$\theta(1) = 0 \implies 1 - \theta_1 - \theta_2 - \ldots - \theta_p = 0 \implies \pi = 0$ The test is implemented on the OLS estimate of the π coefficient:

$ADF = \frac{\hat{\pi}}{st.\,err.\,(\hat{\pi})} \sim DF$ The ADF statistic has the same distribution as the DF statistic

The KPSS test The KPSS test has a null hypothesis of stationarity and is based on the OLS regression of y_t on a deterministic trend:

$$y_t = \delta + \gamma t + \varepsilon_t$$

From the estimated residuals $e_t = y_t - \hat{\delta} - \hat{\gamma} t$ an estimate of the variance is obtained $s^2 = \frac{1}{T} \sum_{t=1}^{T} e_t^2$ and, for each t, the partial sum of the residuals is computed as $S_t = \sum e_i$

from which the following statistic is constructed $KPSS = \frac{\frac{1}{T^2} \sum_{t=1}^{T} S_t^2}{S_t^2}$ The asymptotic distribution of this statistic has the following critical values: $0.146(5\%) = 0.216(1\%)$

× EViews - [Graph: PLOT_Y_TS Workfile: KPSS_EXAMPLE::Untitled\]

-2 III File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options Update AddText Line/Shade Remove Template Zoom y(t) = d + g t + epsilon(t)

-40 1250 1500 2000 1000 1750 955 | 170 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = kpss_example xR 11:59 i **N** G S D

- w A

× EViews - [Graph: PLOT_EQ_TS Workfile: KPSS_EXAMPLE::Untitled\]

-2 III File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Estimated:

y(t)=d+gt+epsilon(t) Data generated by:

y(t)=d+gt+epsilon(t) Residual (left scale) Actual Fitted

- 150

- 100

- 50

- 0

-50 0 -

-2

-4 1000 1250 1500 1750 2000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = kpss_example x8 12:08 S D i C .

× EViews - [Graph: PLOT_EQ_DS Workfile: KPSS_EXAMPLE::Untitled\]

- II File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Estimated:

y(t)=d+gt+epsilon(t) Data generated by: y(t)=g+y(t-1)+epsilon(t) Residual (left scale) Actual Fitted

-50

-10

-20

-30 1250 2000 1000 1500 1750 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = kpss_example xR 12:11 10/03/2022 D i S G

× EViews - [Graph: GRAF_E_DS_TS Workfile: KPSS_EXAMPLE::Untitled\] III File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom

-10

-20 Residual, true model: Random walk Residual, true model: Linear deterministic trend

-30 2000 1750 1000 1250 1500 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = kpss_example xR 12:16 10/03/2022 D S S.

1.4 Set 4 – ARCH Processes

Modelling variance: ARCH processes Many (especially financial) time series, such as stock price indices, exchange rates, price levels, display periods of lower volatility followed by periods of higher volatility.

If we want to appropriately describe those variables by means of time series models, the classical assumption of homoskedasticity (i.e. constant variance over time) of the error term cannot apply. Therefore, in addition to a model for the (conditional) mean of the series, it is necessary to specify also a model capturing the evolution of the variance over time. To this aim, an useful class of models is based on the concept of autoregressive conditional heteroskedasticity (ARCH), originally developed by R.

Engle (1982), who showed how to estimate jointly the mean and the conditional variance of a time series process.

As an example of the typical appearance of $ARCH$ effects in financial series, we look at the behavior of daily stock returns on the UK stock market from January 1970 to August 2005. Panel (a) of Figure 8 shows the level of the $FTSE$ stock market index over this period (around 9300 daily observations), which displays the features of an integrated series with an upward trend and wide and long-lasting fluctuations.

In panel (b) of the Figure, the daily stock market returns are shown (denoting the index level by y_t, daily returns are computed as $\Delta \log y_t$): the series looks stationary, but its variability does not appear to be constant over the whole sample. In fact, there are periods of relatively high volatility (e.g. in the mid-1970s, at the end of the 1990s and again in 2002-2003) and periods of low volatility (e.g. during most of the 1980s and 1990s, and in the final years of the sample). To see more clearly the existence of periods characterized by different volatility, panel (c) shows the moving standard deviation of stock returns computed using a rolling window of 120 market days: wide fluctuations are evident throughout the sample, with

persistent deviations from the full-sample standard deviation of daily returns (the horizontal line at 0.9%). Finally, panel (d) of Figure 8 shows the squared residuals from an $AR(1)$ model fitted to daily returns (i.e. the residuals from the following regression: $\Delta \log y_t = \mu + \theta_1 \Delta \log y_{t-1} + \varepsilon_t$, that should capture the variability of the innovations to the stock return process through time. Here, clusters of large residuals are evident in some periods (mid-1970s, end of the 1990s, first half of the

2000s), with other periods featuring smaller shocks: again, this purely descriptive evidence suggests that stock return volatility is changing through time.

(a) FTSE index (daily) (b) Daily stock returns 3,500 .12 3,000 .08 2,500 .04 2,000 .00 1,500

-.04 1,000

-.08

-.12

-.16 1975 2005 1970 1980 1985 1990 1995 2000 2005 1975 1980 1985 1995 2000 1970 1990 (d) Squared residuals from AR(1) model of stock returns (c) Moving standard deviation of daily stock returns .030 .012 .025 .010 .020 .008 .015 .006 .010 .004 .005 .002 .000 1995 1970 1975 1980 1985 1990 1995 2000 2005 1970 1975 1980 1985 1990 2000 2005 Figure 8. UK FTSE 100 stock market index (daily, January 1970-August 2005):

(a) levels; (b) daily returns; (c) moving standard deviation of returns; (d) squared residuals from $AR(1)$ model of returns.

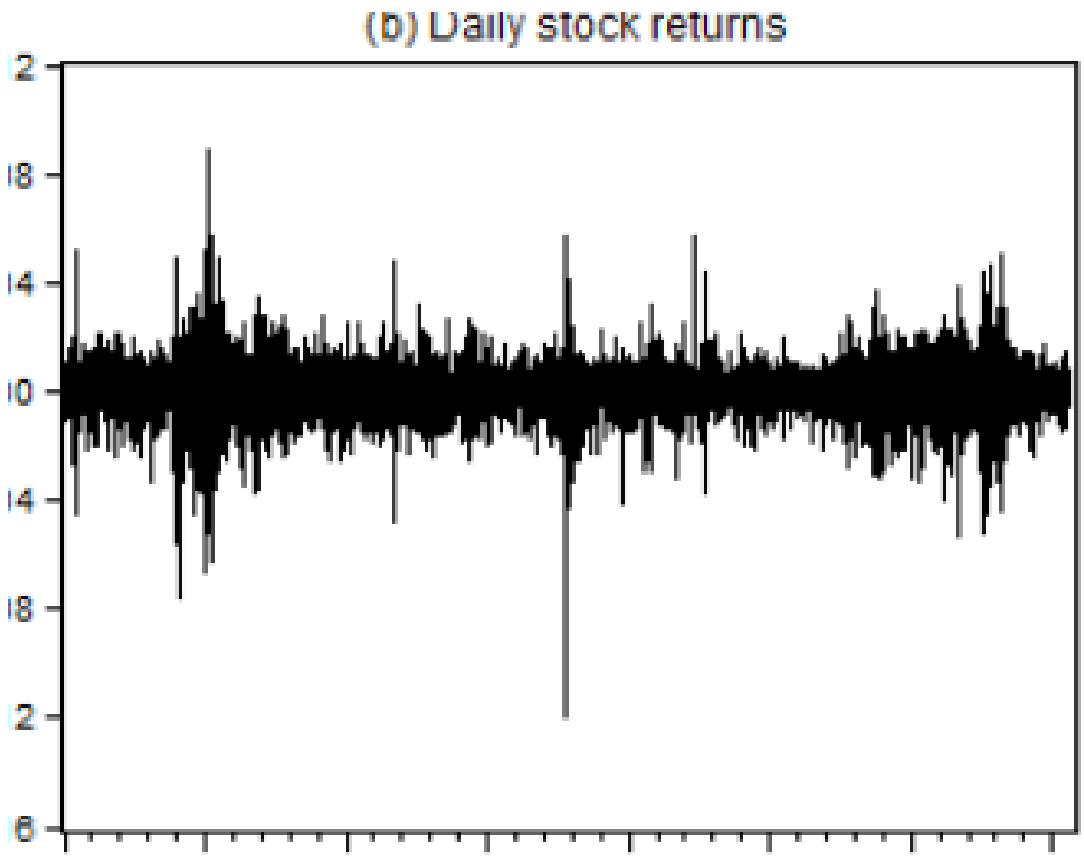


Figure 1.56: Course figure, source page 2, crop 1.

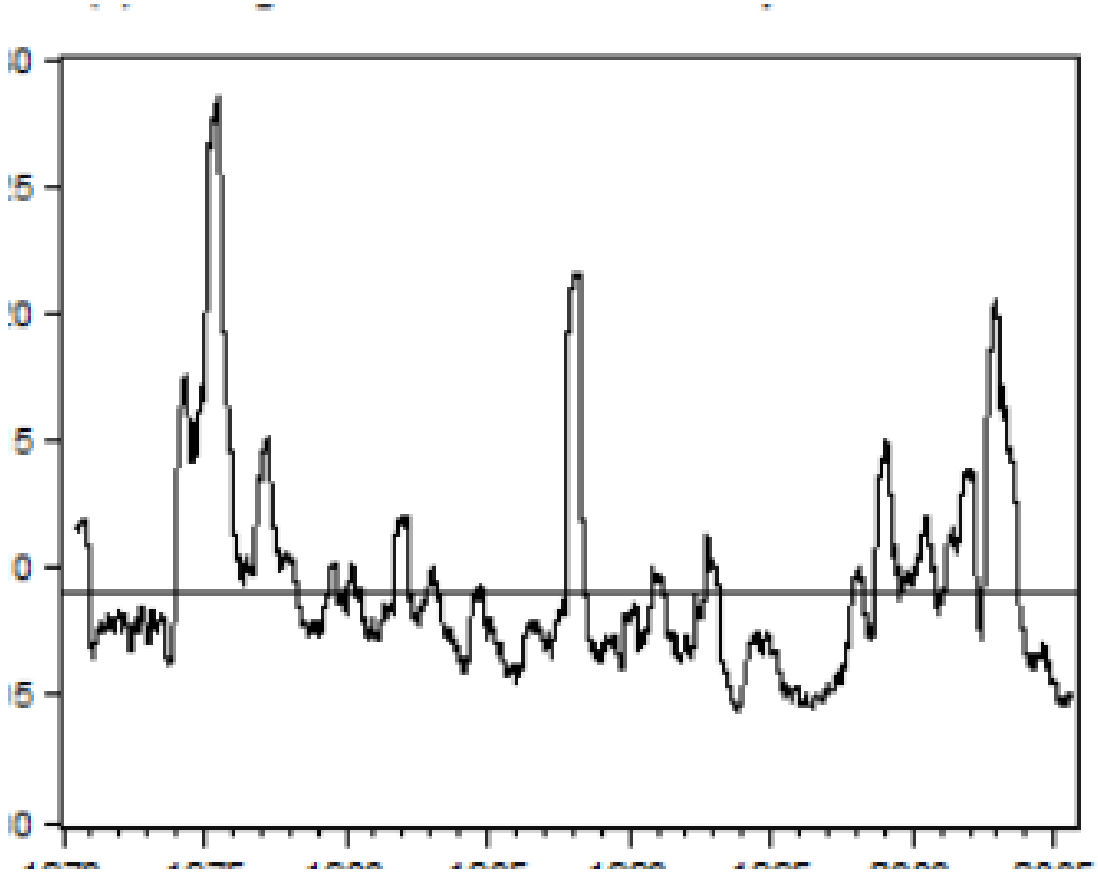


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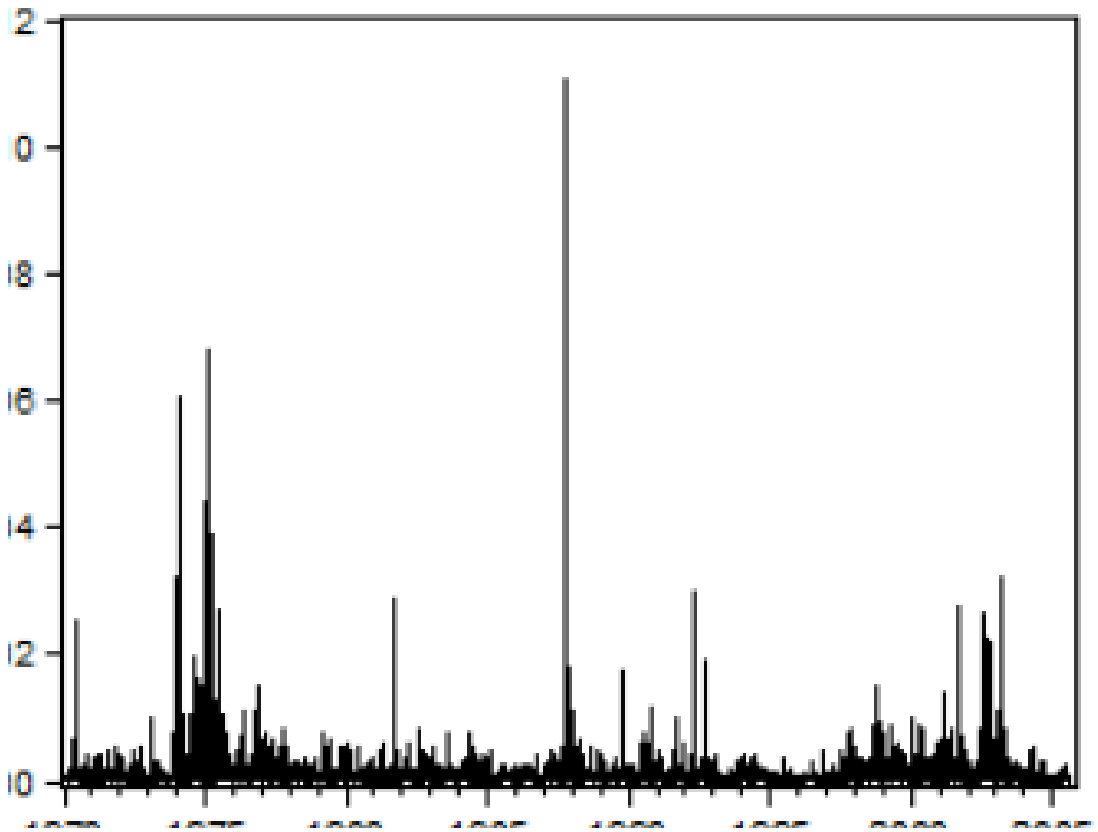


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8.1 The simple case: $ARCH(1)$

Given a process $\{y_t\}$ and the information set $I_{t-1} = \{y_{t-1}, y_{t-2}, ...\}$, the mean of y_t conditional on I_{t-1} is defined as $E(y_t \mid I_{t-1})$, and the error term of the process (or

"innovation") ε_t is:

$\varepsilon_t = y_t - E(y_t \mid I_{t-1})$ The variance of ε_t conditional on I_{t-1} might change over time, being influenced by the past realization ε_{t-1}. Such dependence can be formalized as follows $E(\varepsilon_t^2 \mid I_{t-1}) \equiv \sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2$

where $\omega \geq 0$ and $0 \leq \alpha < 1$ (the non-negativity of the coefficients ensures a non- negative variance). A large (positive or negative) value of ε_{t-1} causes an increase of the variance of ε in the following period (therefore increasing the probability that a relatively large value of ε -of either sign- occurs also in period t): ε_t^2 and ε_{t-1}^2 are

then correlated. This process is called $ARCH(1)$ since the correlation is limited to the first lag.

To describe in detail the properties of an $ARCH(1)$ process, let us start from the definition of the innovation ε:

$\varepsilon_t = v_t\sqrt{\omega + \alpha\,\varepsilon_{t-1}^2}$ where v_t is a white noise process with $\sigma_v^2 = 1$, independent of ε_{t-1}. The process $\{\varepsilon_t\}$ has zero mean, is not autocorrelated, and, due to the assumptions on ω and α

above, is stationary. In fact, the unconditional properties of the process are:

- unconditional mean:

$$E(\varepsilon_t) = E\left(v_t\sqrt{\omega + \alpha\,\varepsilon_{t-1}^2}\right)$$
$$= \underbrace{E(v_t)}\,E\left(\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right) = 0$$

where use has been made of the assumption of independence between v_t and ε_{t-1}.

- unconditional variance:

$$E(\varepsilon_t^2) = E\left|\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right)^2\right| = E\left[v_t^2\left(\omega + \alpha\varepsilon_{t-1}^2\right)\right]$$

$= \underbrace{E(v_t^2)}_{t}\,E\left(\omega + \alpha\varepsilon_{t-1}^2\right) = \omega + \alpha E(\varepsilon_{t-1}^2)$ yielding the stationary solution (letting $E(\varepsilon_t^2) = E(\varepsilon_{t-1}^2)$), ensured by the assump-

tion $\alpha < 1$:

$$E(\varepsilon_t^2) \equiv \sigma^2 = \frac{\omega}{1-\omega}$$

- unconditional autocovariances:

$$E(\varepsilon_t\varepsilon_{t-1}) = E\left[\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right)\left(v_{t-1}\sqrt{\omega + \alpha\varepsilon_{t-2}^2}\right)\right]$$

$$= \underbrace{E(v_t)}_{E}\,E\left(\sqrt{\omega + \alpha\varepsilon_{t-1}^2}v_{t-1}\sqrt{\omega + \alpha\varepsilon_{t-2}^2}\right) = 0$$

and, more generally $E(\varepsilon_t\varepsilon_{t-i}) = 0 \qquad \forall i \neq 0$ As for the conditional mean and variance we have:

- conditional mean:

$$E(\varepsilon_t \mid I_{t-1}) = E\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2} \mid I_{t-1}\right)$$
$$= \underbrace{E(v_t)}_{t}\left(\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right) = 0$$

- conditional variance:

$$E(\varepsilon_t^2 \mid I_{t-1}) = E\left[\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right)^2 \mid I_{t-1}\right]$$
$$= \underbrace{E(v_t^2)}_{t-1}\left(\omega + \alpha\,\varepsilon_{t-1}^2\right) = \omega + \alpha\,\varepsilon_{t-1}^2$$

The conditional variance of ε_t is not constant over time but depends on the square of the error occurred in the previous period, ε_{t-1}^2: a large realization of ε^2 at time $t-1$ causes an increase in the error variance in the following period t. Although the errors ε are not autocorrelated ($E(\varepsilon_t\varepsilon_{t-i}) = 0$), they are not independent. Autocorrelation

measures the linear dependence between the errors at different time dates, whereas the ARCH process specified above captures a relation between the error variances, which is a non linear form of dependence.

Note

1. The variability over time of σ_t^2 around its mean does not affect the time- invariance of the unconditional variance σ^2; however, such variability has an effect on the higher order moments of the unconditional distribution of ε_t. In particular, the latter distribution has "fatter" tails than a normal distribution (i.e. the kurtosis -when the fourth moment exists- is larger than 3).

2. The $ARCH(1)$ process for the variance σ_t^2 can be expressed as an $AR(1)$ for ε_t^2. Using the relation $\sigma_t^2 = \varepsilon_t^2 - (\varepsilon_t^2 - \sigma_t^2)$

and the definition of the $ARCH(1)$ process for σ_t^2 we get $\varepsilon_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + (\varepsilon_t^2 - \sigma_t^2)$ which is a stationary (since $\alpha < 1$) $AR(1)$ process for ε_t^2, where the innovation

term $\varepsilon_t^2 - \sigma_t^2$ has zero mean and is heterosked astic.

Generalizations: $ARCH(p)$ and $GARCH(p,q)$ 8.1.1 (i) The $ARCH(1)$ process can be generalized by assuming the dependence of σ_t^2 on several past realizations of ε_t:

$$E(\varepsilon_t^2 \mid I_{t-1}) \equiv \sigma_t^2 = \omega + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \cdots + \alpha_p\varepsilon_{t-n}^2$$

$= \omega + \alpha(L)\varepsilon_{t-1}^2$ with $\alpha_1, \alpha_2, ..., \alpha_p \geq 0$ and (as a necessary and sufficient stationarity condi- tion) $\alpha(1) = \alpha_1 + \alpha_2 + \ldots + \alpha_p < 1$. This process is called ARCH(p). Its

unconditional mean is:

$E(\varepsilon_t^2) \equiv \sigma^2 = \frac{\omega}{1-\alpha_1-\alpha_2-\ldots-\alpha_n}$ Also in this more general case it is possible to express the process as an $AR(p)$ for ε_t^2, with a mean zero, heteroskedastic error term:

$$\varepsilon_t^2 = \omega + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \cdots + \alpha_n\varepsilon_{t-n}^2 + (\varepsilon_t^2 - \sigma_t^2)$$

Note on testing for ARCH effects. The above expression suggests a way of testing for the existence of $ARCH(p)$ effects in the series of residuals obtained from estimation of a regression model (e.g. an ARMA model) for a process $\{y_t\}$, based on the Lagrange Multiplier principle. Once a regression model for y_t has been estimated and the residual series $\hat{\varepsilon}_t$ obtained, the following auxiliary regression of $\hat{\varepsilon}_t^2$ onto $\hat{\varepsilon}_{t-1}^2$, $\hat{\varepsilon}_{t-2}^2$, ... $\hat{\varepsilon}_{t-p}^2$ can be run:

$$\hat{\varepsilon}_t^2 = \omega + \alpha_1\,\hat{\varepsilon}_{t-1}^2 + \alpha_2\,\hat{\varepsilon}_{t-2}^2 + \cdots + \alpha_p\,\hat{\varepsilon}_{t-n}^2 + \nu_t$$

Under the null hypothesis of no $ARCH(p)$ effects (i.e. $\alpha_1 = \alpha_2 = \ldots = \alpha_p = 0$), the test statistic $T \cdot R^2$ (where T is the number of observations and R^2 is the coefficient of determination of the auxiliary regression) has a $\chi^2(p)$ distribution and allows to perform a simple test for $ARCH(p)$ effects.

(ii) A further generalization has generated the class of $GARCH(p,q)$ (general- ized ARCH) models. In this case the conditional variance follows the process:

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_j \varepsilon_{t-j}^2 + \sum_{i=1}^{q} \beta_j \sigma_{t-j}^2$$
$$= \omega + \alpha(L)\varepsilon_{t-1}^2 + \beta(L)\sigma_{t-1}^2$$

with $\alpha_1, \alpha_2, ..., \alpha_p \geq 0,\ \beta_1, \beta_2, ..., \beta_q \geq 0$ and (as a necessary and sufficient stationarity condition) $\alpha(1) + \beta(1) < 1$. The nature and properties of this process are similar to those of an $ARMA$ process. In particular, consider the case (widely used in applied work) of a $GARCH(1,1)$ process:

$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$ This process, though having only three parameters, is able to capture a long- lasting persistence of volatility over time. In fact, by repeated substitution of past values of σ_{t-1}^2, we get:

$\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{i=1}^{\infty} \beta^{j-1} \varepsilon_{t-j}^2$ Now σ_t^2 depends on the infinite series of past realizations of ε_{t-j}^2 with exponentially decreasing weights. As for ARCH processes, also in this case $\{\varepsilon_t\}$

has a zero mean and is not autocorrelated; its unconditional variance is $E(\varepsilon_t^2) = \omega + \alpha E(\varepsilon_{t-1}^2) + \beta E(\sigma_{t-1}^2)$ which yields (noting that, under stationarity, $E(\varepsilon_t^2) = E(\varepsilon_{t-1}^2)$ and $E(\varepsilon_{t-1}^2) = E(\varepsilon_{t-1}^2)$

$$E(\sigma_{t-1}^2)$$

$E(\varepsilon_t^2) \equiv \sigma^2 = \frac{\omega}{1-\alpha-\beta}$ The $GARCH(1,1)$ process for σ_t^2 can be rewritten as a stationary $ARMA(1,$ 1) process for ε_t^2:

$$\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + (\varepsilon_t^2 - \sigma_t^2) - \beta(\varepsilon_{t-1}^2 - \sigma_{t-1}^2)$$

Finally, the $GARCH(1,1)$ model can be intuitively interpreted as a predictor of the variance of the innovation ε_t based on information up to time $t-1$, constructed as a weighted average of three components. Using the definition of the unconditional variance σ^2 derived above, the model can be written as:

$\sigma_t^2 = (1 - \alpha - \beta)\,\sigma^2 + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2$ Here σ_t^2 combines three different variance forecasts: (i) a long-run forecast,

captured by the unconditional variance σ^2, with weight $(1 - \alpha - \beta)$; (ii) the forecast of the innovation variance made in the previous period $\sigma_{t-1}^2 \equiv E(\varepsilon_{t-1}^2)\ I_{t-2})$, with weight β; and (iii) the new information on the innvation process

obtained in period $t-1$ and not available earlier, captured by ε_{t-1}^2 (and interpreted as a variance forecast based only on the information in period $t-1$), with weight α. In the average, the weights determine how fast the conditional variance changes with new information and how fast it reverts to its long-run mean.

Comparison of ARCH and GARCH models: an example. To illustrate the ability of $GARCH(1,$ 1) processes to yield high volatility persistence, we compare the conditional variances generated from the two following processes:

$$\sigma_{1t}^2 = \omega + \alpha_1\,\varepsilon_{1t-1}^2$$
$$\sigma_{2t}^2 = \omega + \alpha_2\,\varepsilon_{2t-1}^2 + \beta_2\,\sigma_{2t-1}^2$$

ARCH(1) GARCH(1,1) with $\omega = 1$, $\alpha_1 = 0.9$, $\alpha_2 = 0.1$ and $\beta_2 = 0.8$. The two processes share the same unconditional variance (since $\alpha_1 = \alpha_2 + \beta_2 = 0.9$), but yield a different degree of volatility persistence. Figure 9 shows the realizations of σ_{1t}^2 and σ_{2t}^2 for a sample

of 250 observations obtained using the same drawings for the white noise process v_t necessary to generate the series for ε_{1t} and ε_{2t}. The effect of large realizations of v_t on the conditional variance is sharply different: in the $ARCH(1)$ case the variance increases remarkably and reverts to smaller values relatively quickly, whereas in the $GARCH(1,1)$ case the impact effect is smaller but is followed by much longer periods of relatively large volatility.

ARCH(1)

– GARCH(1,1) Figure 9. Conditional variances generated by an $ARCH(1)$ and a $GARCH(1,1)$ processes.

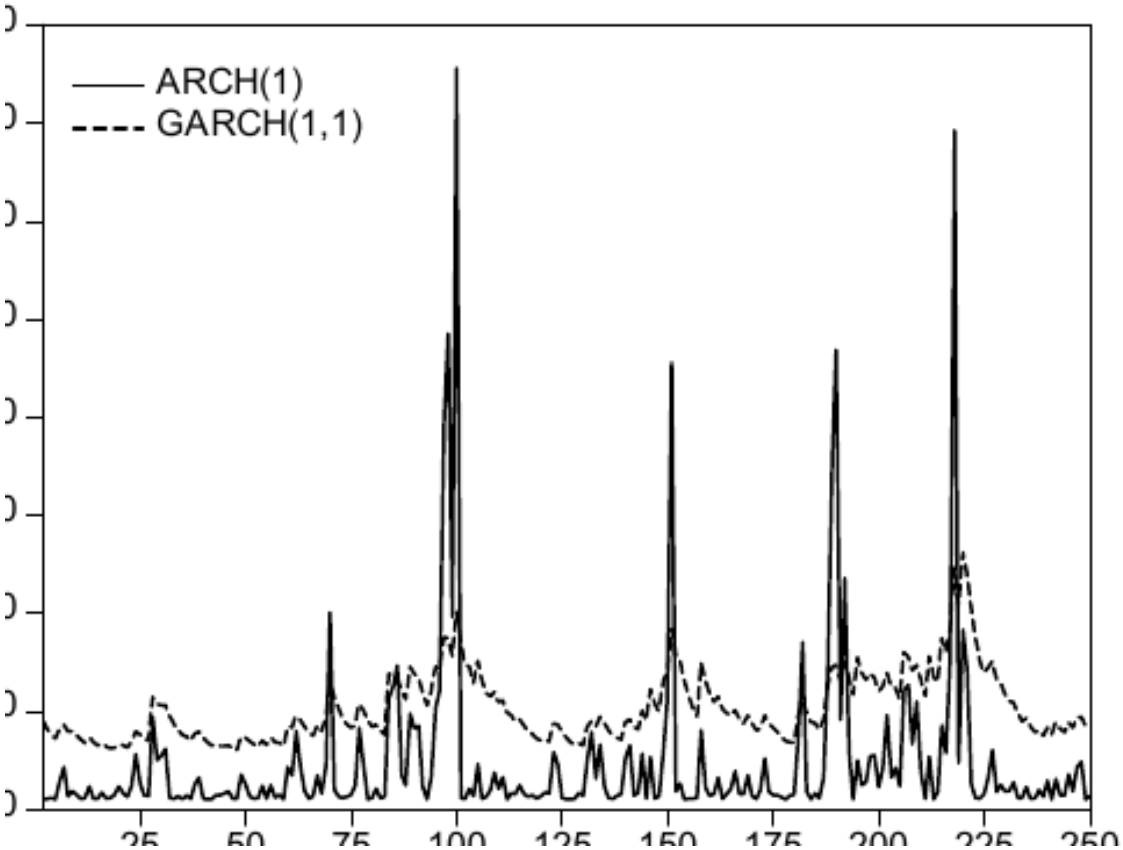


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8.1.2 Some extensions GARCH models have been extended in several ways in order to analyze particular patterns of time-varying volatility of economic time series.

We present here two such extensions of the basic model suggested both by the theory of finance and by the empirical evidence on the behavior of financial series.

The $ARCH - M$ (ARCH in mean) process The $ARCH - M$ model allows for a link between the conditional second moments (variances) and the first moments (means) of the variables of interest. Such a relation is suggested by basic financial principles, showing that the ex-ante riskiness of financial assets is an important de- terminant of expected returns. It is possible to formalize this relation for empirical purposes by means of an extended $ARCH$ model, where the shocks to the condi- tional variance not only persist over time, but also affect the conditional mean of the process.

Consider the case of a risky asset, with excess return (i.e. the return in excess of a riskfree rate) described by:

$y_t = \underbrace{\mu_t}_{\text{risk premium}} + \underbrace{\varepsilon_t}_{\text{unpredictable shock}}$ The conditional mean of the excess return (expected excess return) is:

$E(y_t \mid I_{t-1}) = \mu_t$ and, in addition of being driven by several economic determinants collected in vector $\mathbf{x}_t$, is an increasing function of the conditional variance of ε_t (σ_t^2):

$\mu_t = \mathbf{x}_t'\boldsymbol{\theta} + \delta\,\sigma_t^2$ The parameter $\delta \geq 0$ measures the trade-off between risk and return for this asset.

The conditional variance of ε_t can be modelled by means of an $ARCH(p)$ process:

$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_j\,\varepsilon_{t-j}^2$ Past realizations of ε not only change the variance of the shock distribution, but also affect the risk premium of the asset. Now we have $E(y_t \mid I_{t-1}) = \mathbf{x}_t'\boldsymbol{\theta} + \delta\left(\omega + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2\right)$

Asymmetric volatility models: the exponential GARCH (EGARCH) The empirical evidence of a negative correlation between current returns and future volatility in the behavior of many financial assets (especially stocks) motivated the extension of GARCH models to allow for asymmetries in the effect of ε_t on the future conditional variance of the process: positive innovations ("good news", $\varepsilon_t > 0$ have a smaller effect on volatility than negative shocks (“bad news”, $\varepsilon_t < 0$).

The EGARCH process captures this asymmetry (often called "leverage effect") by assuming the following relation between volatility and realizations of the shocks:

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \alpha \frac{|\varepsilon_{t-1}|}{\sigma_{t-1}}$$

Notes

1. the logarithmic form ensures that positive variances are obtained without imposing non-negativity constraints on the model’s parameters;

2. the asymmetry is present if $\gamma \neq 0$. In particular, if $\varepsilon_{t-1} > 0$ the effect on σ_t^2 is $\alpha + \gamma$ if $\varepsilon_{t-1} < 0$ the effect on σ_t^2 is $\alpha - \gamma$

If $\gamma < 0$ good news increase volatility less than bad news;

3. the relation between σ_t^2 and ε_{t-1} can be represented by the news impact curve, which is symmetric in basic GARCH models, and asymmetric in EGARCH models, as shown in Figure 10.

σ_t^2 GARCH EGARCH e_{t-1} Figure 10. News impact curves for GARCH and EGARCH models.

8.2 Estimation and forecasting of ARCH models (outline) 8.2.1 Estimation Estimation of models with ARCH and GARCH stochastic components is implemented by applying the maximum likelihood (ML) principle. Differently from models with a constant variance, that yield linear first order conditions, in the presence of autoregressive conditional heteroskedasticity non-linearities arise in the equations that should provide the expressions for the ML estimators. Analytical solutions are not available and numerical optimization procedures are needed.

We illustrate the main steps of the ML procedure in the simple $ARCH(1)$ case.

Consider the regression model with $ARCH(1)$ errors:

$y_t = \mathbf{x}_t'\boldsymbol{\theta} + \varepsilon_t$ $\varepsilon_t = v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2} = v_t\sigma_t$ with v_t independently distributed as a standard normal; then, the conditional dis-

tribution of ε_t is normal as well:

$\varepsilon_t \mid I_{t-1} \sim N(0, \sigma_t^2)$ Under normality, the conditional density function of y_t is $f(y_t \mid \mathbf{x}_t, I_{t-1}; \boldsymbol{\theta}, \omega,$ $\alpha) = \frac{1}{\sqrt{2\pi\sigma_t^2}} e^{-\frac{1}{2}\left(\frac{\varepsilon_t}{\sigma_t}\right)^2}$

yielding the following likelihood function $L(\boldsymbol{\theta}, \omega, \alpha) = \prod^T \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) e^{-\frac{1}{2}\left(\frac{\varepsilon_t}{\sigma_t}\right)^2}$

and log-likelihood function $\log L(\boldsymbol{\theta}, \omega, \alpha) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^T \log\sigma_t^2 - \frac{1}{2}\sum_{t=1}^T \left(\frac{\varepsilon_t}{\sigma_t}\right)^2$

Substituting for σ_t^2 and for ε_t and ε_{t-1}, the log-likelihood function can be written in terms of the parameters to be estimated (losing the initial observation of the sample):

$$\log L(\boldsymbol{\theta}, \omega, \alpha) = -\frac{T-1}{2}\log(2\pi) - \frac{1}{2}\sum_{l=1}^{T} \log\left(\omega + \alpha\left(y_{t-1} - \mathbf{x}_{t-1}'\boldsymbol{\theta}\right)^2\right)$$

$$-\frac{1}{2}\sum_{t=2}^{T} \frac{(y_t - \mathbf{x}_t'\boldsymbol{\theta})^2}{\left(\omega + \alpha\left(y_{t-1} - \mathbf{x}_{t-1}'\boldsymbol{\theta}\right)^2\right)}$$

This function can be maximized (with numerical optimization procedures) with respect to $\boldsymbol{\theta}, \boldsymbol{\omega}$, $\boldsymbol{\alpha}$, obtaining maximum likelihood estimates (the estimates of $\boldsymbol{\theta}$ are asymptotically independent from those of the ARCH process, ω and α).

8.2.2 Forecasting Forecasts of the conditional variance are obtained following a procedure similar to that used for $ARMA$ processes. Looking again at the $GARCH(1,1)$ case, for the future date $t+h$ we have $\sigma_{t+b}^2 = \omega + \alpha\,\varepsilon_{t+b-1}^2 + \beta\,\sigma_{t+b-1}^2$

which yields the following prediction conditional on time t:

$E(\sigma_{t+h}^2|I_t) = \omega + \alpha E(\varepsilon_{t+h-1}^2|I_t) + \beta E(\sigma_{t+h-1}^2|I_t)$ Using the fact that $E(\varepsilon_{t+j}^2|I_t) = E(\sigma_{t+j}^2|I_t)$ for all j, we get

$E(\sigma_{t+h}^2|I_t) = \omega + (\alpha + \beta)E(\sigma_{t+h-1}^2|I_t)$ Using this relation recursively, we can write the prediction at time t of the variance h periods ahead as $\sigma_{t+h|t}^2 \equiv E(\sigma_{t+h}^2|I_t) = \omega\left[1 + (\alpha+\beta) + (\alpha+\beta)^2 + \ldots + (\alpha+\beta)^{h-1}\right] + (\alpha+\beta)^h\sigma_t^2$

$$= \omega \frac{1-(\alpha+\beta)^h}{1-(\alpha+\beta)} + (\alpha+\beta)^h \sigma_t^2$$
$$= \frac{\omega}{1-\alpha-\beta} + (\alpha+\beta)^h \left(\sigma_t^2 - \left(\frac{\omega}{1-\alpha-\beta}\right)\right)$$

where the last expression tells us that, as the forecasting horizon $h \to \infty$, the prediction of the variance converges (since $\alpha + \beta < 1$) to the unconditional variance $\sigma^2 = \frac{\omega}{1-\alpha-\beta}$, with a rate of convergence given by $\alpha + \beta$.

Modelling variance: ARCH processes Many time series display periods of lower volatility followed by periods of higher volatility ⇒ the classical assumption of homoskedasticity cannot apply.

In addition to a model for the (conditional) mean of the series, it is necessary to specify also a model capturing the evolution over time of the variance.

An useful class of models is based on the concept of autoregressive conditional heteroskedasticity (ARCH, Engle 1982)

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1 1 1975 1980 1985 1990 2000 1975 1970 1995 2005 1970 1980 1985 1990 1995 2000 2005 (d) Squared residuals from AR(1) model of stock returns (c) Moving standard deviation of daily stock returns .030 .012 .025 - .010 .020 - .008 .015 - .006 .010 - .004 man .005 - .002 .000 .000 - 1. 12 2000 2005 1975 1980 1990 2000 2005 1970 1995 1970 1985 1995 1975 1980 1985 1990 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arch_ftse_plot x8 12:25 i S D G

- .

The simple case: ARCH(1) Given $\{y_t\}$ and the information set $I_{t-1} = \{y_{t-1}, y_{t-2}, \dots\}$, let us define:

the mean of y_t conditional on I_{t-1} : $E(y_t|I_{t-1})$ the associated error term (or "innovation") ε_t:

$\varepsilon_t = v_t - E(v_t \mid I_{t-1})$ The variance of ε_t conditional on I_{t-1} can change over time, depending on the past realization ε_{t-1}:

$E(\varepsilon_t^2 \mid I_{t-1}) \equiv \sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2$ with $\omega > 0$ and $0 < \alpha < 1$.

ε_t^2 and ε_{t-1}^2 are correlated: ARCH(1) process

Properties of the ARCH(1) process Definition of the innovation ε_t:

$\varepsilon_t = v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}$ where $v_t \sim WN(0,1)$ is independent of ε_{t-1} $\{\varepsilon_t\}$ has zero mean, is not autocorrelated, and is stationary

Unconditional properties:

- unconditional mean:

$$E(\varepsilon_t) = E\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right)$$
$$= E(v_t)E\left(\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right) = 0$$

where use has been made of the independence between v_t and $\mathcal{E}_{t-1}$

- unconditional variance:

$$E(\varepsilon_t^2) = E\left[\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right)^2\right] = E[v_t^2(\omega + \alpha\varepsilon_{t-1}^2)]$$

$= \underbrace{E(v_t^2)}_{t} E(\omega + \alpha\varepsilon_{t-1}^2) = \omega + \alpha E(\varepsilon_{t-1}^2)$ yielding the stationary solution (letting $E(\varepsilon_t^2) = E(\varepsilon_{t-1}^2)$), ensured

by $\alpha < 1$:

$$E(\varepsilon_t^2) \equiv \sigma^2 = \frac{\omega}{1-\alpha}$$

- unconditional autocovariances:

$$E(\varepsilon_t\varepsilon_{t-1}) = E\left[\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right)\left(v_{t-1}\sqrt{\omega + \alpha\varepsilon_{t-2}^2}\right)\right]$$

$$= \underbrace{E(v_t)}_{t} E\left(\sqrt{\omega + \alpha\varepsilon_{t-1}^2}v_{t-1}\sqrt{\omega + \alpha\varepsilon_{t-2}^2}\right) = 0$$

and, more generally $E(\varepsilon_t\varepsilon_{t-i}) = 0 \qquad \forall i \neq 0$

Conditional properties:

- conditional mean:

$$E(\varepsilon_t|I_{t-1}) = E\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}|I_{t-1}\right)$$
$$= \underbrace{E(v_t)}_{t}\left(\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right) = 0$$

- conditional variance:

$$E(\varepsilon_t^2|I_{t-1}) = E\left[\left(v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2}\right)^2|I_{t-1}\right]$$
$$= \underbrace{E(v_t^2)}_{\bullet}(\omega + \alpha\varepsilon_{t-1}^2) = \omega + \alpha\varepsilon_{t-1}^2$$

The conditional variance of ε_t is not constant over time but depends on ε_{t-1}^2 Though not autocorrelated ($E(\varepsilon_t\varepsilon_{t-i}) = 0$), the errors ε are not independent

Note

1. The variability over time of σ_t^2 around its mean does not affect the time-invariance of the unconditional variance σ^2;

however, it affects the higher-order moments of the unconditional distribution of ε_t. In particular, the latter distribution has "fatter" tails than a normal distribution (i.e.

the kurtosis is larger than 3)

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-8.20e-21 Mean 1,600 Median 1.34e-05 0.082376 Maximum 1,200 −0.105138 Minimum 0.009988 Std. Dev.

-0.255762 Skewness 10.76652 Kurtosis Jarque-Bera 23482.52 Probability 0.000000

-0.075

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-0.025 0.000 0.025 0.050 0.075

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2. The ARCH(1) process for σ_t^2 can be expressed as an AR(1) for ε_t^2. Using the relation $\sigma_t^2 = \varepsilon_t^2 - (\varepsilon_t^2 - \sigma_t^2)$ and the definition of the ARCH(1) process for σ_t^2 we get

$\varepsilon_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + (\varepsilon_t^2 - \sigma_t^2)$ which is a stationary ($\alpha < 1$) AR(1) process for ε_t^2, where the innovation term $\varepsilon_t^2 - \sigma_t^2$ has zero mean and is

heteroskedastic

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Generalizations: (i) ARCH(p) The ARCH(1) process can be generalized by assuming that σ_t^2 depends on several past realizations of ε_t:

$$E(\varepsilon_t^2 \mid I_{t-1}) \equiv \sigma_t^2 = \omega + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \ldots + \alpha_p\varepsilon_{t-p}^2$$

$= \omega + \alpha(L)\varepsilon_{t-1}^2$ with $\alpha_1, \alpha_2, \ldots, \alpha_p \geq 0$ and (as a necessary and sufficient stationarity condition) $\alpha(1) = \alpha_1 + \alpha_2 + \ldots + \alpha_p < 1$. This process is

called ARCH(p).

Its unconditional mean is:

$E(\varepsilon_t^2) \equiv \sigma^2 = \frac{\omega}{1-\alpha_1-\alpha_2-\ldots-\alpha_n}$ It is possible to express the process as an $AR(p)$ for ε_t^2, with a mean zero, heteroskedastic error term:

$$\varepsilon_t^2 = \omega + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \ldots + \alpha_p\varepsilon_{t-p}^2 + (\varepsilon_t^2 - \sigma_t^2)$$

Testing for ARCH effects in the residuals of a regression

- estimation of a regression model (e.g. ARMA) for $\{y_t\} \Rightarrow \hat{\varepsilon}_t$
- run auxiliary regression $\hat{\varepsilon}_t^2 = \omega + \alpha_1\hat{\varepsilon}_{t-1}^2 + \alpha_2\hat{\varepsilon}_{t-2}^2 + \ldots + \alpha_p\hat{\varepsilon}_{t-p}^2 + v_t$
- under H_0: $\alpha_1 = \alpha_2 = ... = \alpha_p = 0$ (no ARCH(p) effects):

$$T \cdot R^2 \sim \gamma^2(p)$$

Generalizations: (ii) $GARCH(p, q)$ In this case the conditional variance follows the process:

$$\sigma_t^2 = \omega + \sum_{r=1}^{r} \alpha_j\varepsilon_{t-j}^2 + \sum_{r=1}^{q} \beta_j\sigma_{t-j}^2$$

i=1 j=1 $= \omega + \alpha(L)\varepsilon_{t-1}^2 + \beta(L)\sigma_{t-1}^2$ with $\alpha_1, \alpha_2, \ldots, \alpha_p \geq 0$, $\beta_1, \beta_2, \ldots, \beta_q \geq 0$ and (as a necessary and

sufficient stationarity condition) $\alpha(1) + \beta(1) < 1$ The nature and properties of this process are similar to those of an ARMA

GARCH(1,1) process:

$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$ This process is able to capture a long-lasting persistence of volatility over time.

By repeated substitution of past values of σ_{t-1}^2, we get:

$\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha\sum_{j=1}^{j-1}\beta^{j-1}\varepsilon_{t-j}^2$ Now σ_t^2 depends on the infinite series of past realizations of ε_{t-j}^2

with exponentially decreasing weights

As for ARCH processes, also in this case $\{\varepsilon_t\}$ has a zero mean and is not autocorrelated; its unconditional variance is $E(\varepsilon_t^2) = \omega + \alpha E(\varepsilon_{t-1}^2) + \beta E(\sigma_{t-1}^2)$ which yields (noting that, under stationarity, $E(\varepsilon_t^2) = E(\varepsilon_{t-1}^2)$ and

$$E(\varepsilon_{t-1}^2) = E(\sigma_{t-1}^2)$$

$E(\varepsilon_t^2) \equiv \sigma^2 = \frac{\omega}{1-\alpha-\beta}$ The GARCH(1,1) process for σ_t^2 can be rewritten as a stationary ARMA(1,1) process for ε_t^2:

$$\varepsilon_t^2 = \omega + (\alpha + \beta)\varepsilon_{t-1}^2 + (\varepsilon_t^2 - \sigma_t^2) - \beta(\varepsilon_{t-1}^2 - \sigma_{t-1}^2)$$

Interpretation of $GARCH(1,1)$ model:

$\sigma_t^2 = (1 - \alpha - \beta)\sigma^2 + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$ as predictor of variance of ε_t based on information available up to $t-1$, obtained as a weighted average of three elements

(i) long-run forecast: unconditional variance σ^2, weight $1-\alpha-\beta$ (ii) previous conditional variance σ_{t-1}^2, weight β (iii) information on variance obtained at $t-1$, ε_{t-1}^2, weight α

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- - x View Proc Object Save Freeze Details+/- Show Fetch View Proc Object Save Freeze Details+/- Show Run Print Save SaveAs Cut Copy Paste InsertTxt Find Replace Wrap+/- LineNum+/- Encrypt Range: 1/01/1970 8/31/2005 - 9305 obs Filter: * Range: 1 250 – 250 obs Filter: * Sample: 1/01/1970 8/31/2005 - 9305 obs Name Sample: 2 250 - 249 obs Order: Name acf_e acf_e2 acf_resid ar1 arch6 c corr corr_e2 correl_e2 data diftse e e2 ftse gra1</l # alpha1 # alpha2 # beta1 'Comparison of ARCH(1) and GARCH(1,1) processes' # beta2 BC epsilon1 Consider two processes with the same v(t) ~ WN(0,1) :

epsilon2 M epsilon2_sq garchconfr omega resid theta ARCH(1) with coefficients:

(1) V Var1 $alpha1 = 0.9$ omega = 1 var1t var2t var2t var2t y1 (2) GARCH(1,1) with coefficients:

Note

y2 y2garch $alpha2 = 0.1$ $beta2 = 0.8$ omega = 1 alpha1 = (alpha2 + beta2) = 0.9 **Note.**

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Extensions: (i) ARCH – M (ARCH in mean) Allows for a link between the conditional second moments (variances) and the first moments (means) of the variables of interest (suggested by basic financial principles: ex-ante riskiness of financial assets is an important determinant of expected returns) It is possible to formalize this relation for

empirical purposes by means of an extended ARCH model, where the shocks to the conditional variance not only persist over time, but also affect the conditional mean of the process.

Consider the case of a risky asset, with excess return (i.e. the return in excess of a riskfree rate) described by:

$y_t = \underbrace{\mu_t}_{\leftarrow} + \underbrace{\varepsilon_t}_{\leftarrow}$ risk premium unpredictable shock The conditional mean of the excess return (expected excess) return) is:

$E(y_t|I_{t-1}) = \mu_t$ and, in addition of being driven by several economic determinants collected in vector $\mathbf{x}_t$, is an increasing function of the conditional variance of ε_t (σ_t^2):

$\mu_t = \mathbf{x}_t'\theta + \delta\sigma_t^2$ $\delta > 0$ measures the trade-off between risk and return for the asset

The conditional variance of ε_t can be modelled by means of an $ARCH(p)$ process:

$\sigma_t^2 = \omega + \sum_{i=1}^{P} \alpha_i \varepsilon_{t-i}^2$ Past realizations of ε not only change the variance of the shock distribution, but also affect the risk premium of the asset:

$$E(y_t \mid I_{t-1}) = \mathbf{x}_t'\theta + \delta\left(\omega + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2\right)$$

Extensions: (ii) EGARCH (exponential GARCH) The empirical evidence of a negative correlation between current returns and future volatility motivated GARCH models with asymmetric effects of ε_t on the future conditional variance of the process:

positive innovations ("good news", $\varepsilon_t > 0$) have a smaller effect on volatility than negative shocks ("bad news", $\varepsilon_t < 0$) The EGARCH process captures this asymmetry ("leverage" effect") by assuming the following relation between volatility and realizations of the shocks:

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \alpha \frac{|\varepsilon_{t-1}|}{\sigma_{t-1}}$$

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \alpha \frac{|\varepsilon_{t-1}|}{\sigma_{t-1}}$$

Note

1. the logarithmic form ensures that positive variances are obtained without imposing non-negativity constraints on the model's parameters;

2. the asymmetry is present if $\gamma \neq 0$. In particular, if $\varepsilon_{t-1} > 0$ the effect on σ_t^2 is $\alpha + \gamma$ if $\varepsilon_{t-1} < 0$ the effect on σ_t^2 is $\alpha - \gamma$

If $\gamma < 0$ good news increase volatility less than bad news;

3. the relation between σ_t^2 and ε_{t-1} can be represented by the news impact curve, which is symmetric in basic GARCH models, and asymmetric in EGARCH models:

σ_t^2 GARCH EGARCH $\mathbf{e}_{t-1}$ News Impact Curve

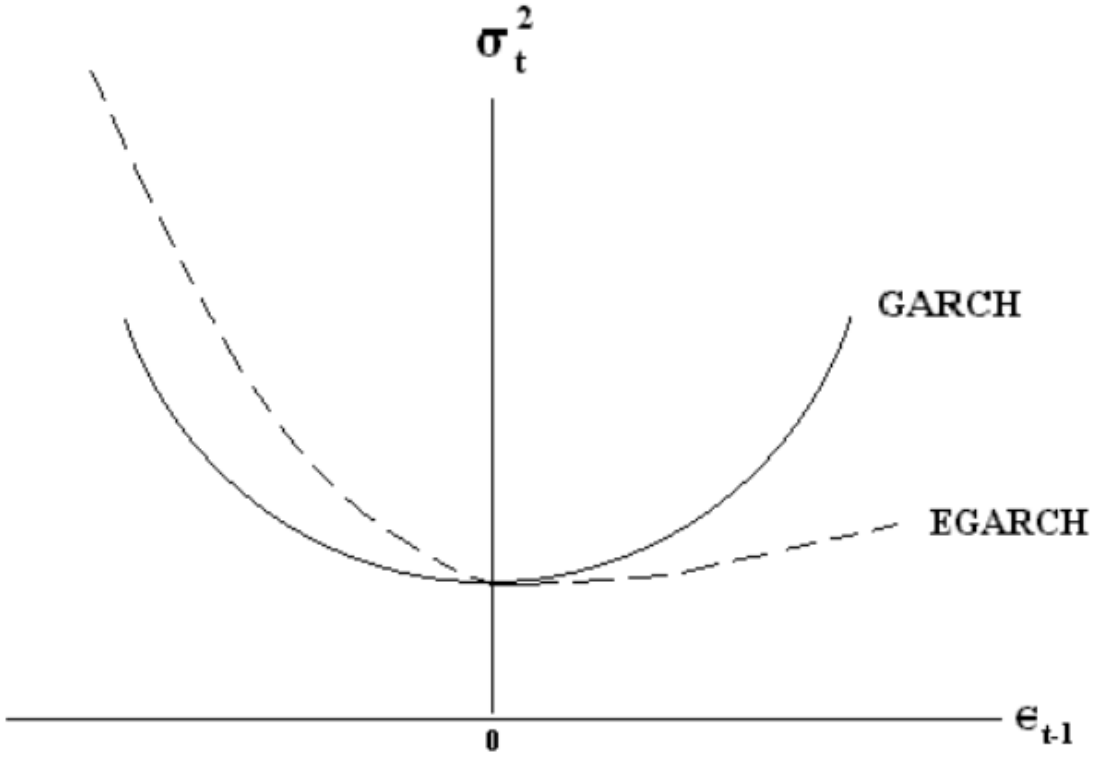


Figure 1.60: Course figure, source page 29, crop 1.

Estimation of ARCH models Estimation of models with ARCH and GARCH stochastic components is implemented by applying the maximum likelihood (ML) principle.

In ARCH models non-linearities arise in the equations that should provide the expressions for the ML estimators. Analytical solutions are not available and numerical optimization procedures are needed. We illustrate the main steps of the ML procedure in the simple ARCH(1) case

Main steps of the ML procedure for a $ARCH(1)$ model:

- regression model with $ARCH(1)$ errors:

$v_t = \mathbf{x}_t'\theta + \varepsilon_t$ $\varepsilon_t = v_t\sqrt{\omega + \alpha\varepsilon_{t-1}^2} = v_t\sigma_t$ with v_t independently distributed as a standard normal; then, the

conditional distribution of ε_t is normal as well:

$$\varepsilon_t \mid I_{t-1} \sim N(0, \sigma_t^2)$$

- conditional density function of y_t $f(y_t \mid \mathbf{x}_t, I_{t-1}; \boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\alpha}) = \frac{1}{\sqrt{2\pi\sigma_t^2}}\mathbf{e}^{-\frac{1}{2}\left(\frac{\varepsilon_t}{\sigma_t}\right)^2}$

yielding the following likelihood function $L(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\alpha}) = \prod_{t=1}^{T}\left(\frac{1}{\sqrt{2\pi\sigma_t^2}}\right)\mathbf{e}^{-\frac{1}{2}\left(\frac{\varepsilon_t}{\sigma_t}\right)^2}$

and log-likelihood function $\log L(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\alpha}) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\sigma_t^2 - \frac{1}{2}\sum_{t=1}^{T}\left(\frac{\varepsilon_t}{\sigma_t}\right)^2$

Substituting for σ_t^2 and for ε_t and ε_{t-1}, the log-likelihood function can be written in terms of the parameters to be estimated:

$$\log L(\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\alpha}) = -\frac{T-1}{2}\log(2\pi) - \frac{1}{2}\sum_{i=1}^{r}\log\left(\boldsymbol{\omega} + \boldsymbol{\alpha}\left(y_{t-1} - \mathbf{x}'_{t-1}\boldsymbol{\theta}\right)^2\right)$$

$$-\frac{1}{2}\sum_{t=2}^{T}\frac{(y_t - \mathbf{x}'_t\boldsymbol{\theta})^2}{(\omega + \alpha(y_{t-1} - \mathbf{x}'_{t-1}\boldsymbol{\theta})^2)}$$

This function can be maximized (with numerical optimization procedures) with respect to θ, ω, α, obtaining ML estimates (the estimates of θ are asymptotically independent from those of the ARCH process, ω and α)

Forecasting Forecasts of the conditional variance are obtained following a procedure similar to that used for ARMA processes From a $GARCH(1,1)$ model, for the future date $t+h$ we have $\sigma^2_{t+h} = \omega + \alpha\varepsilon^2_{t+h-1} + \beta\sigma^2_{t+h-1}$ which yields the following prediction conditional on time t:

$E(\sigma^2_{t+h}|I_t) = \omega + \alpha E(\varepsilon^2_{t+h-1}|I_t) + \beta E(\sigma^2_{t+h-1}|I_t)$ Using the fact that $E(\varepsilon^2_{t+i}|I_t) = E(\sigma^2_{t+i}|I_t)$ for all j, we get

$$E(\sigma^2_{t+h}|I_t) = \omega + (\alpha+\beta)E(\sigma^2_{t+h-1}|I_t)$$

Using this relation recursively, we can write the prediction at time t of the variance h periods ahead as $\sigma^2_{t+h|t} \equiv E(\sigma^2_{t+h}|I_t) = \omega\left[1 + (\alpha+\beta) + (\alpha+\beta)^2 + \ldots + (\alpha+\beta)^{h-1}\right]$

$$+(\alpha+\beta)^h\sigma^2_t$$
$$= \omega\frac{1-(\alpha+\beta)^h}{1-(\alpha+\beta)} + (\alpha+\beta)^h\sigma^2_t$$

$$= \frac{\omega}{1-\alpha-\beta} + (\alpha+\beta)^h\left(\sigma^2_t - \left(\frac{\omega}{1-\alpha-\beta}\right)\right)$$

$\Rightarrow$ as the forecasting horizon $h \to \infty$, the prediction of the variance converges (since $\alpha+\beta<1$) to the unconditional variance $\sigma^2 = \frac{\omega}{1-\alpha-\beta}$, with a rate of convergence given by $\alpha+\beta$

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time arch_condvar arch_garch_condvar arch_garch_forecast archgarch_forecast_e2 Modelling UK stock market returns (FTSE100) with AR(1) model BC G corr Corr_e Corr_e2 correl_e2 (i) estimation sample: 1/01/1970 - 10/20/1987 data data_forecast (note: stock market crashed on October 19 and 20 1987: - 25% overall) diftse diftsef e e2 e2_1988 (ii) (conditional variance) forecast sample: 10/21/1987 - 3/31/1988 M forecast_arch forecast_arch1 M forecast_garch forecast_garch11 M ftse In fise forecast garch11 garch_cond_var garch_condvar gra1 gra2 <» Untitled / New Page / Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = arch_ftse_forecast Group: CORR Modifed: 5/6/12 11:44 xR ^ 13:29 ∼

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—— ARCH(1) Forecast

—— GARCH(1,1) Forecast

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1.5 Set 5 – Dynamic Models for Stationary Multivariate Series

Time series econometrics Multivariate time series models F. C. Bagliano March 2024 In economics and finance, we are often interested in studying the dynamic links among several variables. To this aim, univariate (e.g. $ARMA$) models (which are useful to describe the dynamic properties of one individual time series and obtain forecast thereof) are not appropriate. What is necessary is a model capable to capture the joint evolution of the variables over time, describing the dynamic reaction of one series to a change in other variables. This set of notes presents the basics of some simple dynamic models for stationary variables (Section 1) and for non-stationary series (Section 2). Section 3 introduces the basics of vector autoregressive (VAR) models, widely used to study the joint dynamics of a set of

economic series, starting with the case of stationary variables and extending the analysis to non-stationary (but cointegrated) series in the final Section 4.

Dynamic models for stationary series A class of single-equation models often used to describe the dynamic response of a stationary series to changes in its determinants (stationary as well) is that of the autoregressive distributed lag (ADL) models; in what follows, we study a simple example

of an ADL model.

Given two stationary variables y_t and x_t, assume that the following model de- scribes the dynamic behavior of y as a function of its own lags (the "autoregressive") component) and of the contemporaneous and one-lagged values of x (the "distrib- uted lag" component):

$y_t = \delta + \theta y_{t-1} + \phi_0 x_t + \phi_1 x_{t-1} + \varepsilon_t$ where $\varepsilon \sim WN(0, \sigma^2)$, $E(x_t \varepsilon_t) = 0$ and $|\theta| < 1$. With only one lag of y and x the

model is an $ADL(1, 1)$.

PROPOSITION 1.9– Dynamic multipliers in an ADL model

1. The contemporaneous effect (or impact multiplier) on y of a change in x is simply measured by:

$$\frac{\partial y_t}{\partial x_t} = \phi_0$$

2. In order to find the effect on y of the change in x in subsequent periods (i.e.

after t), we rewrite the model using the lag operator L:

$(1 - \theta L)y_t = \delta + (\phi_0 + \phi_1 L)x_t + \varepsilon_t$ getting $y_t = \frac{\delta}{1-\theta} + \frac{\phi_0 + \phi_1 L}{1-\theta L} x_t + \frac{1}{1-\theta L}\varepsilon_t$

(1) Unfolding the polynomials:

$$\begin{aligned} y_t &= \frac{\delta}{1-\theta} + (\phi_0 + \phi_1 L)(1 + \theta L + \theta^2 L^2 + \cdots)x_t + (1 + \theta L + \theta^2 L^2 + \cdots)\varepsilon_t \\ &= \frac{\delta}{1-\theta} + \phi_0 x_t + (\phi_0\theta + \phi_1)x_{t-1} + \theta(\phi_0\theta + \phi_1)x_{t-2} + \theta^2(\phi_0\theta + \phi_1)x_{t-3} + \cdots \\ &\quad + \varepsilon_t + \theta\varepsilon_{t-1} + \theta^2\varepsilon_{t-2} + \cdots . \end{aligned}$$

The effect of x_t on y after h periods (or interim multiplier) is measured by

the coefficient on x_{t-h}:

$$\frac{\partial y_{t+h}}{\partial x} = \frac{\partial y_t}{\partial x_{t-1}} = \theta^{h-1}(\phi_0\theta + \phi_1)$$

for $h \geq 1$ 3. The effect on y of a permanent change in x (or long-run multiplier) is then given by the sum of the contemporaneous effect and all subsequent changes:

$$\frac{\partial y_t}{\partial x_t} + \sum_{i=1}^{\infty} \frac{\partial y_{t+h}}{\partial x_t} = \frac{\partial y_t}{\partial x_t} + \sum_{i=1}^{\infty} \frac{\partial y_t}{\partial x_{t-h}} = \phi_0 + (\phi_0\theta + \phi_1)\frac{1}{1-\theta}$$

$= \frac{\phi_0 + \phi_1}{1-\phi_1}$ (3) Note that the long-run multiplier in (3) can be obtained more directly by setting $L = 1$ in the polynomial attached to x_t in (1), which amounts to summing all the coefficients on x_{t-h} for $h = 0, 1, ...\infty$ in (2). The long-run multiplier is also interpretable as the effect on the (unconditional) mean of y

of a change in the (unconditional) mean of x. In fact, taking expected values we have:

$E(y_t) = \delta + \theta E(y_{t-1}) + \phi_0 E(x_t) + \phi_1 E(x_{t-1})$ from which, exploiting the stationarity of y

and x (i.e. $E(y_t) = E(y_{t-1})$ and $E(x_t) = E(x_{t-1})) : E(y_t) = \frac{\delta}{1-\theta} + \frac{\phi_0+\phi_1}{1-\theta}E(x_t)$

$$\Rightarrow \frac{\partial E(y_t)}{\partial E(x_t)} = \frac{\phi_0 + \phi_1}{1-\theta}$$

2

Notes

• the $ADL(1,1)$ model can be reparameterized (i.e. rewritten) in terms of an error-correction - or, perhaps more appropriately, equilibrium-correction - mechanism (ECM) as follows (subtracting y_{t-1} from both sides and adding and subtracting $\phi_0 x_{t-1}$ on the right-hand side):

$$y_t - y_{t-1} = \delta - (1-\theta)y_{t-1} + \phi_0(x_t - x_{t-1}) + (\phi_0 + \phi_1)x_{t-1} + \varepsilon_t$$
$$\Rightarrow \Delta y_t = \phi_0 \Delta x_t - (1-\theta)\left[y_{t-1} - \left(\frac{\delta}{1-\theta} + \frac{\phi_0+\phi_1}{1-\theta}x_{t-1}\right)\right] + \varepsilon_t$$

where the term in [.] captures the deviation of y_{t-1} from the long-run mean $E(y)$ for $E(x) = x_{t-1}$. The "correction" of such deviation in the following period t is measured by the negative coefficient $-(1-\theta) < 0$. In fact, when $[.] > 0$ it means that y_{t-1} was above its long-run mean (related to x_{-1}) and this deviation determines a negative adjustment of y_t (i.e. $\Delta y_t < 0$) towards

tits long-run mean; the opposite occurs ($\Delta y_t > 0$) when y_{t-1} was below its long-run mean. The size of the adjustment depends on the magnitude of θ: the larger θ, the smaller is the "correction" and therefore the slower the adjustment path of y towards its long-run mean.

• the $ADL(1,1)$ model can be generalized in two ways:

− more than one lag of y and x can appear in the autoregressive and dis- tributed lag components. The general $ADL(p,q)$ model then becomes $\theta(L)u_t = \delta + \phi(L)x_t + \varepsilon_t$ where $\theta(L) = 1 - \theta_1 L - \ldots - \theta_n L^p$ $\phi(L) = \phi_0 + \phi_1 L \ldots + \phi_q L^q$

and the roots of $\theta(L)$ satisfy the stationarity conditions for $\{y_t\}$, that include $\theta(1) < 1$. The dynamic properties of the simple case extend naturally to the general model; in particular, the long-run multiplier is $\frac{\partial E(y_t)}{\partial E(x_t)} = \frac{\phi_0+\phi_1+\ldots+\phi_q}{1-\theta_1-\ldots-\theta_t}$

− more than one explanatory variable x can enter the model; in this case the model becomes $\theta(L)y_t = \delta + \phi_1(L)x_{1t} + \phi_2(L)x_{2t} + \ldots + \phi_k(L)x_{kt} + \varepsilon_t$

1.1 Some economic fundamentals of ADL models In some instances, dynamic models of the ADL variety have a clear underlying economic interpretation.

We present here two examples: a model with "adaptive expectations" and a model featuring a "partial adjustment" mechanism.

Adaptive expectations 1.1.1 Consider the following economic model, which describes the relation between a (dependent) variable y and the expectation of an (independent) variable x:

$y_t = \alpha + \beta\, x^*_{t+1} + \eta_t$ where x^*_{t+1} is the expectation formed by agents at time t on the value of x

at $t+1$.

No dynamic element enters the relation (no lags of either y or x^* appear there). We assume that expectations are formed adaptively, according to the following recursive relation, with $0 < \gamma < 1$:

$x^*_{t+1} - x^*_t = \gamma(x_t - x^*_t)$ Expectations of x_{t+1} formed at t are revised on the basis of the deviation between the realized value of the variable at t and its expected value formed one period earlier, at $t-1$, denoted x^*_t (i.e. the "forecast error"). The parameter γ measures by how much agents change their expectations as a consequence of their forecast

errors. By rewriting the expectation formation mechanism as:

$x^*_{t+1} = \gamma x_t + (1-\gamma)x^*_t \Rightarrow x^*_{t+1} = \frac{\gamma}{1-(1-\gamma)L}x_t$ and substituting the latter equation into the original model for y we get:

$y_t = \alpha + \beta\frac{\gamma}{1-(1-\gamma)L}x_t + \eta_t$ from which:

$(1-(1-\gamma)L)y_t = \alpha\gamma + \beta\gamma x_t + (1-(1-\gamma)L)\eta_t$ that can be finally expressed as $y_t = \alpha\gamma + (1-\gamma)y_{t-1} + \beta\gamma x_t + \varepsilon_t$ with $\varepsilon_t = \eta_t - (1-\gamma)\eta_{t-1}$

A dynamic relation between y and x is then obtained from a static economic model with adaptive expectations (note that this transformation produces an $MA(1)$ error term, affecting the choice of the appropriate estimation method).

1.1.2 Partial adjustment Now consider an economic model describing the relation between a target (or "de- sired") value for y and the value of another variable, x:

$y^*_t = \alpha + \beta x_t + \eta_t$ where y^*_t is the target value for y for time t (e.g. a firm's stock of desired capital).

There are two similarities with the adaptive expectation model: (i) even in this case the model is static; (ii) the target value y^* is not directly observed, as the expected value of x in the preceding case (in fact, an explicit formulation of an expectation formation mechanism was needed). We adopt the following "partial adjustment" (or stock adjustment) hypothesis which links the actual (and observable) value of y with its target (unobservable) value, with $0 < \theta < 1$:

$\underbrace{y_t - y_{t-1}}_{\text{actual}} = (1-\theta)\underbrace{(y^*_t - y_{t-1})}_{\text{desired}}$ adjustment This relation says that in every period t the actual value of y is changed so as to fill a fraction of the gap between the target value for that period, and the actual existing

value y_{t-1}. The parameter $1-\theta$ measures the fraction of the desired adjustment that is actually carried out in each period t. This adjustment mechanism can be rewritten as:

$y_t = (1-\theta)y^*_t + \theta y_{t-1} \Rightarrow \quad y_t = \frac{1-\theta}{1-\theta I}y^*_t$ Substituting for y^*_t from the original model we get:

$$y_t = \frac{1-\theta}{1-\theta L}(\alpha + \beta x_t + \eta_t)$$
$$\Rightarrow \quad y_t = (1-\theta)\alpha + \theta y_{t-1} + (1-\theta)\beta x_t + \varepsilon_t$$

with $\varepsilon_t = (1-\theta)\eta_t$ which is a dynamic relation between x and the actual (and observable) value of y.

Exercise. Start from the following special case of a dynamic model with an infinite number of lags $y_t = \delta + \phi_0 x_t + \phi_1 x_{t-1} + \ldots + \phi_j x_{t-j} + \ldots + \eta_t$

where x is stationary, $\eta \sim WN(0, \sigma^2)$ and $E(x_t\eta_t) = 0$. Assume that the parameters ϕ_j are all of the same sign and decrease geometrically following the relation (for $j = 1, 2, ...$) $\phi_j = \phi_0\,\lambda^j$ with $0 < \lambda < 1$

1. Using the relation among the parameters, find an ADL specification of the model (expressing y_t as a function of y_{t-1} and x_t) and find the long-run mul- tiplier, discussing how it depends on the parameter λ;

2. discuss possible estimation problems if OLS is applied to the model in ADL form.

Multivariate time series models single-equation dynamic models for stationary variables dynamic models for non-stationary variables vector autoregressive (VAR) models even in the presence of non-stationary variables

Dynamic models for stationary series Class of single-equation models describing the dynamic response of y_t (stationary) to changes of x_t (stationary):

autoregressive distributed lag (ADL) models $ADL(1,1)$ model:

$v_t = \delta + \theta y_{t-1} + \phi_0 x_t + \phi_1 x_{t-1} + \varepsilon_t$ with $\varepsilon \sim WN(0, \sigma^2)$, $E(x_t \varepsilon_t) = 0$ and $|\theta| < 1$

PROPOSITION 1.10– ADL dynamic effects and long-run multiplier

1. The contemporaneous effect (or impact multiplier) on y of a change in x is simply measured by:

$$\frac{\partial y_t}{\partial x_t} = \phi_0$$

2. In order to find the effect on y of the change in x in subsequent periods (i.e. after t), we rewrite the model using the lag operator L:

$(1 - \theta L)v_t = \delta + (\phi_0 + \phi_1 L)x_t + \varepsilon_t$ getting $y_t = \frac{\delta}{1-\theta} + \frac{\phi_0 + \phi_1 L}{1-\theta L} x_t + \frac{1}{1-\theta L}\varepsilon_t$

Unfolding the polynomials:

$$\begin{aligned} y_t &= \frac{\delta}{1-\theta} + (\phi_0 + \phi_1 L)(1 + \theta L + \theta^2 L^2 + \cdots)x_t + (1 + \theta L + \theta^2 L^2 + \cdots)\varepsilon_t \\ &= \frac{\delta}{1-\theta} + \phi_0 x_t + (\phi_0\theta + \phi_1)x_{t-1} + \theta(\phi_0\theta + \phi_1)x_{t-2} + \theta^2(\phi_0\theta + \phi_1)x_{t-3} + \cdots \\ &\quad + \varepsilon_t + \theta\varepsilon_{t-1} + \theta^2\varepsilon_{t-2} + \cdots . \end{aligned}$$

The effect of x_t on y after h periods (or interim multiplier) is

measured by the coefficient on x_{t-h}:

$$\frac{\partial y_{t+h}}{\partial x_t} = \frac{\partial y_t}{\partial x_{t-h}} = \theta^{h-1}(\phi_0\theta + \phi_1) \quad \text{for } h \geq 1$$

3. The effect on y of a permanent change in x (or long-run multiplier) is then given by the sum of the contemporaneous effect and all subsequent changes:

$$\frac{\partial y_t}{\partial x_t} + \sum_{h=1}^{\infty} \frac{\partial y_{t+h}}{\partial x_t} = \frac{\partial y_t}{\partial x_t} + \sum_{h=1}^{\infty} \frac{\partial y_t}{\partial x_{t-h}} = \phi_0 + (\phi_0\theta + \phi_1)\frac{1}{1-\theta}$$

$= \frac{\phi_0 + \phi_1}{1-\rho}$ #

The long-run multiplier is also interpretable as the effect on the (unconditional) mean of

v of a change in the (unconditional) mean of x.

Taking expected values:

$E(v_t) = \delta + \theta E(v_{t-1}) + \phi_0 E(x_t) + \phi_1 E(x_{t-1})$ from which, exploiting the stationarity of y and x, $E(y_t) = \frac{\delta}{1-\theta} + \frac{\phi_0+\phi_1}{1-\theta} E(x_t)$

$$\Rightarrow \frac{\partial E(y_t)}{\partial E(x_t)} = \frac{\phi_0 + \phi_1}{1-\theta}$$

Notes

the $ADL(1,1)$ model can be reparameterized in terms of an error-correction mechanism (ECM):

$$y_t - y_{t-1} = \delta - (1-\theta) y_{t-1} + \phi_0 (x_t - x_{t-1}) + (\phi_0 + \phi_1) x_{t-1} + \varepsilon_t$$
$$\Rightarrow \Delta y_t = \phi_0 \Delta x_t - (1-\theta) \left[y_{t-1} - \left(\frac{\delta}{1-\theta} + \frac{\phi_0+\phi_1}{1-\theta} x_{t-1} \right) \right] + \varepsilon_t$$

deviation of y_{t-1} from long-run mean error-correction mechanism

the $ADL(1,1)$ model can be generalized in two ways:

more than one lag of y and x can appear in the autoregressive and distributed lag components: $ADL(p,q)$ $\theta(L) v_t = \delta + \phi(L) x_t + \varepsilon_t$ where $\theta(L) = 1 - \theta_1 L - \ldots - \theta_p L^p$ $\phi(L) = \phi_0 + \phi_1 L \ldots + \phi_q L^q$

and $\theta(1) < 1$ under stationarity of $\{y_t\}$. The long-run multiplier is $\frac{\partial E(y_t)}{\partial E(x_t)} = \frac{\phi_0+\phi_1+\ldots+\phi_q}{1-\theta_1-\ldots-\theta_n}$

- more than one explanatory variable x can enter the model; in this case the model becomes $\theta(L) y_t = \delta + \phi_1(L) x_{1t} + \phi_2(L) x_{2t} + \ldots + \phi_k(L) x_{kt} + \varepsilon_t$

Economic fundamentals of ADL models Adaptive expectations Relation (with no dynamics) between y and the expectation of x:

$y_t = \alpha + \beta x^*_{t+1} + \eta_t$ where x^*_{t+1} is the expectation formed by agents at time t on the value of x at $t+1$.

Expectations are formed adaptively, with $0 < \gamma < 1$:

$$x^*_{t+1} - x^*_t = \gamma (x_t - x^*_t)$$

Rewriting the expectation formation mechanism as:

$x^*_{t+1} = \gamma x_t + (1-\gamma) x^*_t \Rightarrow x^*_{t+1} = \frac{\gamma}{1-(1-\gamma)L} x_t$ and substituting into the original model for y:

$$y_t = \alpha + \beta \frac{\gamma}{1-(1-\gamma)L} x_t + \eta_t$$
$$(1-(1-\gamma)L) v_t = \alpha\gamma + \beta\gamma x_t + (1-(1-\gamma)L)\eta_t$$
$$\Rightarrow$$

expressed as dynamic model (with $MA(1)$ error):

$v_t = \alpha\gamma + (1-\gamma)v_{t-1} + \beta\gamma x_t + \varepsilon_t$ with $\varepsilon_t = \eta_t - (1-\gamma)\eta_{t-1}$

Partial adjustment Static relation between a target (or "desired") y* and the value of another variable, x:

$v_t^* = \alpha + \beta x_t + \eta_t$ "partial adjustment" (or stock adjustment) hypothesis linking the actual y with its target, with $0 < \theta < 1$:

$\underbrace{y_t - y_{t-1}}_{t-1} = (1-\theta) \qquad \underbrace{(y_t^* - y_{t-1})}_{t-1}$ actual desired adjustment

The adjustment mechanism can be rewritten as:

$y_t = (1-\theta)y_t^* + \theta y_{t-1} \Rightarrow y_t = \frac{1-\theta}{1-\theta L} y_t^*$ Substituting for y_t^* from the original model:

$$y_t = \frac{1-\theta}{1-\theta L}(\alpha + \beta x_t + \eta_t)$$
$$\Rightarrow y_t = (1-\theta)\alpha + \theta y_{t-1} + (1-\theta)\beta x_t + \varepsilon_t$$

with $\varepsilon_t = (1-\theta)\eta_t$

1.6 Set 6 – Dynamic Models for Non-stationary Multivariate Series

Dynamic models for non-stationary series When the series investigated are non-stationary, standard statistical results (which apply in regressions involving stationary variables) are not valid. In particular, regressions involving independent non-stationary series tend to yield statistically significant, but meaningless, results: this problem is discussed in section 2.1. How- ever, there are cases in which non-stationary variables are linked by some existing underlying economic mechanism: the conditions under which this case arises and the suitable econometric methodology to use are discussed in sections 2.2 and 2.3.

2.1 The problem of spurious regressions In a bivariate context, a regression is defined as spurious (or nonsense) if the coeffi- cient in the estimated regression between two independent variables is significantly different from zero.

The probability to obtain such result usually tends to zero as the sample dimension increases. Let us consider two cases:

• stationary and ergodic independent processes: the OLS regression delivers a consistent estimate which converges to the population parameter (zero) as the sample size increases;

• non-stationary ($I(1)$) independent processes: the results from an OLS re- gression are not valid and apparently statistically significant estimates can be obtained even if there is no relation between the variables, i.e. the regression is spurious.

As an example, take the following two independent, non-stationary processes of the random walk type:

$y_t = y_{t-1} + \varepsilon_t^y$ with $y_0 = 0$ $x_t = x_{t-1} + \varepsilon_t^x$ with $x_0 = 0$ where ε_t^y and ε_s^x are independently and identically distributed random variables

for all pairs (t, s), with mean zero and variances σ_y^2 and σ_x^2. Therefore, all signifi- cant relations between y and x are spurious by construction. The following linear regression can be estimated by OLS:

$y_t = \alpha + \beta x_t + e_t$ Here the true parameter values are: $\alpha = 0$ and $\beta = 0$; moreover, the error term of the estimated equation is non-stationary:

$$e_t = y_t = \sum_{i=1}^{t} \varepsilon_i^y \sim I(1)$$

The classical assumptions on the properties of the error term are then violated and the values $\hat{\alpha}$, $\hat{\beta}$, R^2 and the t and F statistics cannot be relied upon for correct inference, even if the sample size increases. In what follows, a set of results charac- terizing a spurious regression is presented, with illustrations obtained from a Monte Carlo simulation exercise.

Monte Carlo simulation. In this simulation exercise, two independent random walk processes $\{y_t\}$ and $\{x_t\}$ are generated, using independently distributed stan- dard normal distributions for $\{\varepsilon_t^y\}$ and $\{\varepsilon_t^x\}$. Two sample sizes are considered: a

relatively small sample ($T = 50$) and a relatively large sample ($T = 1000$), and the number of replications of the experiment is set at 2000. In each replication, and for either sample size, an OLS regression of y_t on a constant and x_t is run, yielding values for the estimated coefficient on $x_t(\beta)$ and the associated t-statistic, the R^2 and F-statistic of the regression, and the Durbin-Watson (DW) statistic for first-order residual autocorrelation. The empirical distributions of $\hat{\beta}$, t, R^2, F and

DW for $T = 50$ and $T = 1000$ are displayed in the figures below.

We highlight a series of important results from the Monte Carlo simulation:

- distribution of $\hat{\beta}$. Although $\hat{\beta}$ is an unbiased estimator of the true β, i.e.

$E(\hat{\beta}) = 0$, as the sample size T increases, the variance of $\hat{\beta}$ does not shrink:

$\hat{\beta}$ does not tend to a random variable with a degenerate distribution (the constant true value of zero), as shown in Figure 1(a);

- distribution of $t_{\hat{\beta}}$. As the sample size T increases, the variance of $t_{\hat{\beta}}$ increases, with critical values increasingly far away from those of the t distribution, as shown in Figure 1(b). Therefore, if the latter are used for inference, as the sample lengthens, the probability of rejecting the (true) null hypothesis $\beta = 0$ increases.

Intuition on the distortion in the estimated t-statistic on β. To test the null hypothesis of $\beta = 0$ in the linear bivariate regression above, the computed t-statistic is $t_{\beta=0} = \frac{\hat{\beta}}{\hat{\sigma}_{\hat{\alpha}}}$

where $\hat{\beta} = \frac{\sum_{t=1}^{T}(x_t-\bar{x})(y_t-\bar{y})}{\sum_{t=1}^{T}(x_t-\bar{x})^2}$, $\hat{\sigma}_{\hat{\beta}} = \frac{\hat{\sigma}_e^2}{\sqrt{\sum_{t=1}^{T}(x_t-\bar{x})^2}}$ and $\hat{\sigma}_e = \sqrt{\frac{\sum_{t=1}^{T}\hat{e}_t^2}{T}}$

(with $\hat{e}$ being the estimated residuals from the OLS regression). $\hat{\beta}$ is an unbiased estimator of β with a large variance, but its computed $\hat{\sigma}_{\hat{\beta}}$ seriously under-estimates the true value, determining very large values of the t-statistic.

This under-estimation of $\sigma_{\hat{\beta}}^2$ occurs for two reasons:

(i) when $\beta = 0$ the estimated residual variance $\hat{\sigma}_e^2 = \frac{\sum(y_t-\hat{y}_t)^2}{T}$ will be lower than $\sigma_e^2 = \frac{\sum(y_t-\bar{y})^2}{T}$ because the estimator $\hat{\beta}$ will usually be $\neq 0$ and widely so (given its large

(ii) $\sum(x_t - \bar{x})^2$ is not an appropriate measure of the variance of x_t when it is nonstationary. For a stationary variable, deviations from the sample mean are good measures of how much the variable changes; but if x_t is nonstationary, deviations from the previous value x_{t-1} would be more appropriate measures of the stochastic change in x_t. Then $\sum(x_t - \bar{x})^2 \geq$

$$\sum(x_t - x_{t-1})^2$$

Both effects contribute to a serious downward bias in the estimation of the standard error of β and tend to inflate the calculated value of the associated t- statistic. This leads to over-rejection of the null hypothesis of $\beta = 0$, therefore mistakenly signalling a significant relation between y_t and x_t.

• distribution of R^2. It does not concentrate towards zero as the sample size increases, as shown in Figure 2.

.10 T = 50 T = 1000 .7 .08 .6 .5 .06 Density .4 .04 .3 .2 .02 .1 .0 .00

-120

-160

-4

-3

-2

-1

-80

-40 (b) t-statistic for coefficient on x (a) Coefficient on x Figure 1. Empirical distributions across 2000 simulations of: (a) estimated coefficient on x_t ($\hat{\beta}$) and (b) t-statistic on $\hat{\beta}$ for $T = 50$ and $T = 1000$.

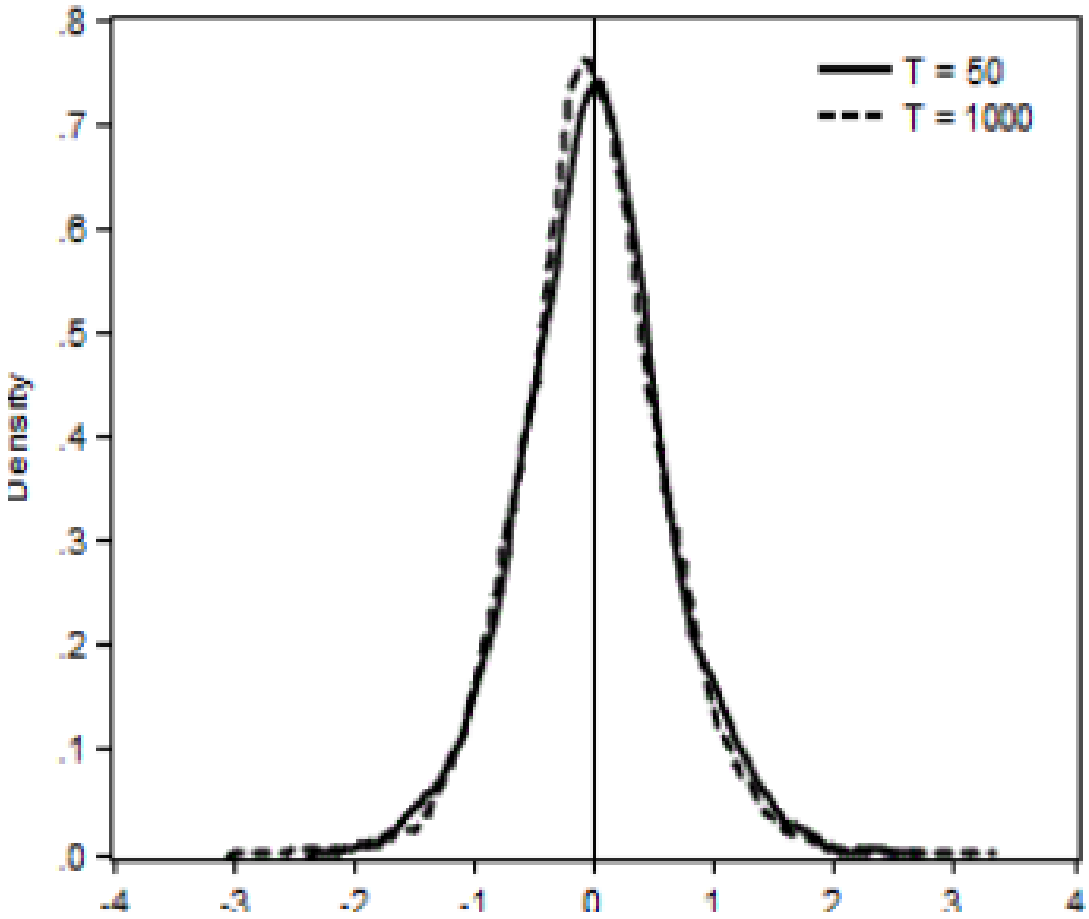


Figure 1.61: Course figure, source page 3, crop 1.

3.2 T = 50 (vertical line: median) T = 1000 (vertical line: median) 2.8 2.4 2.0 Density 1.6 1.2 0.8 0.4 0.0 0.2 0.3 0.4 0.5 R2 0.6 0.7 0.8 0.9 0.0 0.1 1.0 Figure 2. Empirical distributions of R^2 across 2000 simulations for $T = 50$ and $T = 1000$. Vertical lines denote the median values of the distributions.

• distribution of the regression F. If the critical values of the $F(1, T-2)$ distribution are used, the test tends to reject the null hypothesis $\beta = 0$ with a very large size (type-I error), see Figure 3.

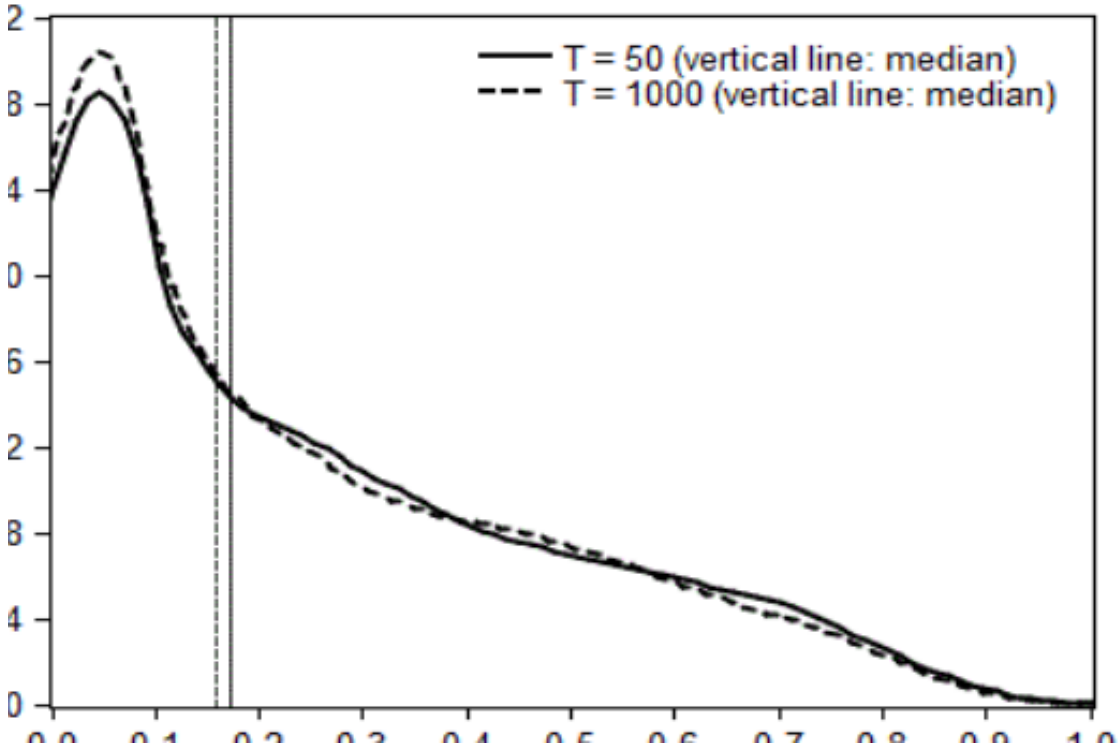


Figure 1.62: Course figure, source page 4, crop 1.

T = 1000 $T = 50$ .0020 .032 5% crit, value 5% crit. value of empirical distr.

of empirical distr .028 .0016 .024 .020 .0012 Density .016 .0008 .012 .008 .0004 .004 0000 .000 1,000 2,000 3,000 4,000 5,000 F of the regression Figure 3. Empirical distributions across 2000 simulations of the regression F-statistic for $T = 50$ and $T = 1000$.

• distribution of the DW (Durbin-Watson) statistic (for testing first order resid- ual autocorrelation). With $\alpha = \beta = 0$ the error term coincides with y and is a random walk; in particular, the coefficient of first-order autocorrelation ρ_1 tends to 1 as T increases. The DW statistics is a simple test for first- order residual autocorrelation computed from the estimated residuals (e_t) and closely related to the sample first-order residual autocorrelation coefficient $\hat{\rho}_1$.

In fact $DW = \frac{\sum_{t=2}^{T}(e_t - e_{t-1})^2}{\sum_{t=1}^{T} e_t^2} = \frac{\sum_{t=2}^{T} e_t^2 + \sum_{t=2}^{T} e_{t-1}^2 - 2\sum_{t=2}^{T} e_t e_{t-1}}{\sum_{t=1}^{T} e_t^2}$

If T is "sufficiently large", different ranges in the summations have negligible effects and therefore $\sum_{t=2}^{T} e_t^2 \approx \sum_{t=2}^{T} e_{t-1}^2 \approx \sum_{t=1}^{T} e_t^2$

leading to the following (approximate) relation:

$DW \approx 2 - 2\frac{\sum_{t=2}^{T} e_t e_{t-1}}{\sum_{t=1}^{T} e_t^2} = 2(1 - \hat{\rho}_1)$ Then, with random walk residuals, $\hat{\rho}_1 \to 1$ and the DW statistic tends to 0,

as shown in Figure 4.

Note

in general, the finding of a relatively high value for R^2 and a very low value of the DW statistic is a symptom of the possibility that the regression is spurious and the associated test statistics meaningless.

T = 50 T = 1000 2.5 2.0 1.5 Density 1.0 0.5 0.0 0.8 2.0 0.0 0.2 0.4 0.6 1.0 1.2 1.6 1.8 1.4 0.2 0.4 0.6 0.0 0.8 1.0 1.2 1.4 1.6 1.8 2.0 DW statistic Figure 4. Empirical distributions across 2000 simulations of the DW statistic for $T = 50$ and $T = 1000$.

Comparison with bivariate regression with stationary series. For the sake of comparison, we perform the same Monte Carlo simulation on series generated inde- pendently from stationary $AR(1)$ processes, with first-lag parameter 0.5 as follows:

$y_t = 0.5y_{t-1} + \varepsilon_t^y$ with $y_0 = 0$ $x_t = 0.5x_{t-1} + \varepsilon_t^x$ with $x_0 = 0$ with $\{\varepsilon_t^y\}$ and $\{\varepsilon_t^x\}$ generated by independently distributed standard normal dis-

tributions. Again, the sample sizes considered are $T = 50$ and $T = 1000$, and the number of replications of the experiment is 2000. The empirical distributions of β, t, R^2, and DW (for $T = 50$ and $T = 1000$) from a bivariate OLS regression of y_t on a constant and x_t are displayed in the figures below.

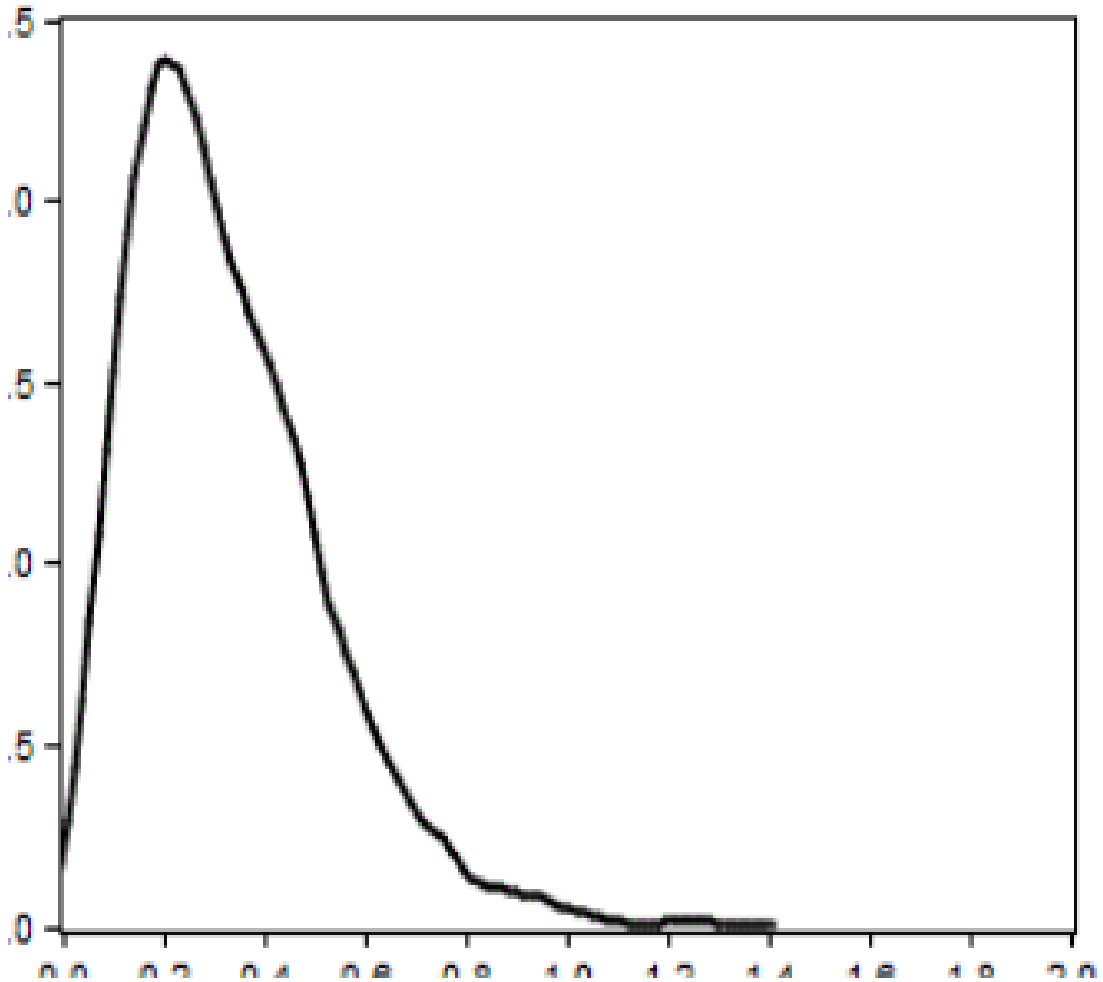


Figure 1.63: Course figure, source page 6, crop 1.

Independent stationary AR(1) processes (theta = 0.5) T = 50

- T = 1000 T = 1000 .30 .20 Density .15 .10 .05 .00 0.6 0.8

-1.0

-0.6

-0.4

-0.2 0.0 0.2 0.4 1.0

-0.8

-2

-1

-6

-5

-3 Coefficient on x t-statistic for coefficient on x Figure 5. Empirical distributions of $\hat{\beta}$ and t-statistic on $\hat{\beta}$ for $T = 50$ and $T = 1000$ with independent $AR(1)$ stationary processes.

As expected, the empirical distribution of the estimated coefficients $\hat{\beta}$ shrinks around the true

value (zero) as sample size increases, and the distribution of the as- sociated t-statistic displays a more standard shape even in the small sample (Figure 5). Figure 6 shows that the distribution of the R^2 statistic has an extremely low median value (0.0145) already in the small sample and concentrates further towards zero as sample size increases.

R2 T = 1000 T = 50 (vertical line: median) Density .00 .05 .10 .20 .25 .30 .35 .40 .00 .05 .10 .15 .20 .25 .30 .35 .40 Figure 6. Empirical distributions of R^2 for $T = 50$ and $T = 1000$ with independent stationary $AR(1)$ processes.

Finally, Figure 7 shows that the distribution of the DW statistic signals substantial positive first-order autocorrelation of the estimated residuals (in fact, given the true value $\beta = 0$, the residual series coincides with the dependent variable y_t and therefore is a $AR(1)$ process with first-lag parameter 0.5) and is centered around 1 (since $DW \approx 2(1 - \hat{\rho}_1)$ and the distribution of $\hat{\rho}_1$ is centered around 0.5) with

sharply decreasing variance as the sample lengthens.

DW statistic T = 50 T = 1000 Density 0.4 0.0 0.2 0.6 0.8 1.0 1.2 1.4 1.6 1.8 2.0 Figure 7. Empirical distributions of DW statistic for $T = 50$ and $T = 1000$ with independent stationary $AR(1)$ processes.

2.2 Cointegration: the bivariate case Basic idea. In some instances the regression between non-stationary variables is not spurious but captures the true relation between the series.

This happens when the error term ε_t in the regression of y_t on x_t (both $I(1)$ series) is stationary:

$\varepsilon_t \sim I(0)$ In this case the two variables $y_t \sim I(1)$ and $x_t \sim I(1)$ are cointegrated (Engle- Granger, 1987) The cointegrating relation between y_t and x_t captures the tendency of the series (drifting over time on a non-stationary path) to move together, i.e.

not in an independent way. The two variables share the stochastic trend which determines their non-stationarity (i.e. y_t and x_t share a common stochastic trend).

This concept can be easily formalized with the following example, where the relation between two variables $y_t \sim I(1)$ and $x_t \sim I(1)$ is described by the model

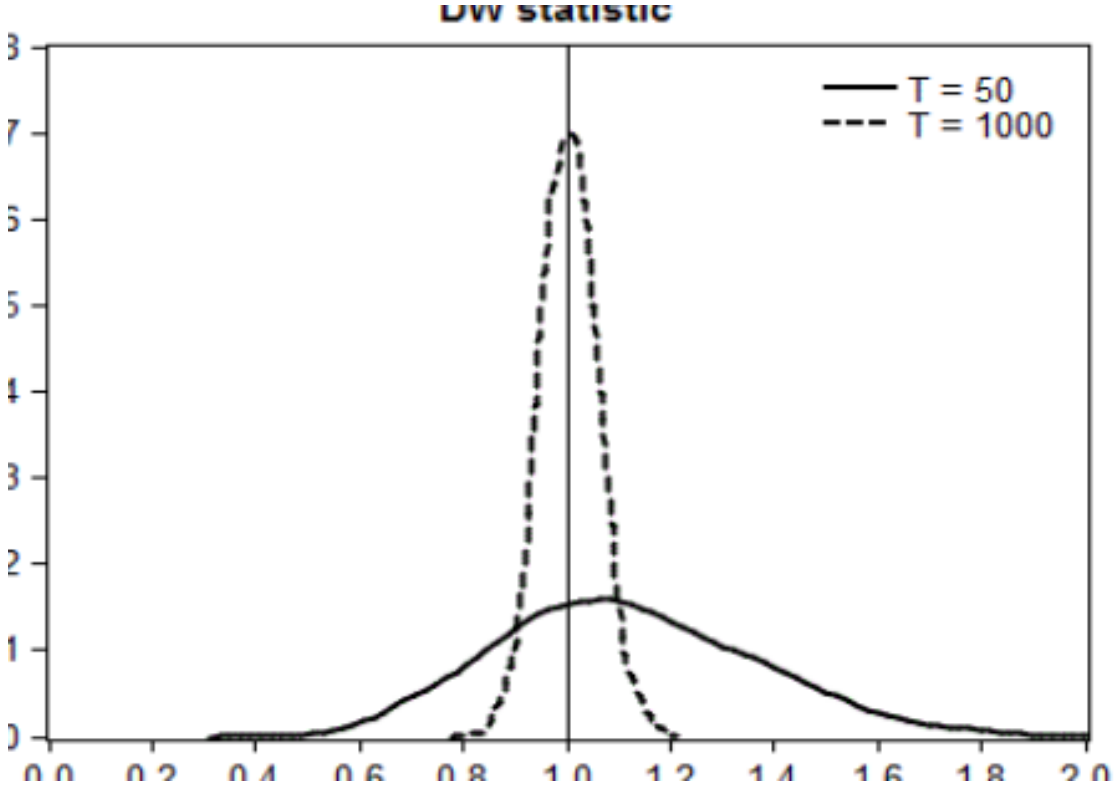


Figure 1.64: Course figure, source page 8, crop 1.

(with $y_0 = x_0 = 0$):

$y_t - \beta x_t = \varepsilon_t$ with $\varepsilon_t = \rho\varepsilon_{t-1} + \eta_t^y$ $x_t = x_{t-1} + \eta_t^x$ where $\{\eta^y\}$ and $\{\eta^x\}$ are independent white noise. The second equation defines x_t

as a random walk; the first equation defines ε_t as a linear combination of y_t and x_t with coefficients $(1, -\beta)$. If $\rho = 1$ the linear combination ε_t is non-stationary ($\varepsilon_t \sim I(1)$) and the series y_t and x_t are not cointegrated. If instead $|\rho| < 1$ the

linear combination defined by the coefficients $(1, -\beta)$ yields a stationary variable:

$\varepsilon_t \sim I(0)$ and the two series y_t and x_t are cointegrated.1 Cointegration and common stochastic trends. More generally, we can show the relationship between cointegration and the presence of a common stochastic trend in the case of two random walk plus noise stochastic processes y_t and x_t (obviously non stationary):

$y_t = \mu_t^y + \varepsilon_t^y$ with $\mu_t^y = \mu_{t-1}^y + \eta_t^y$ $x_t = \mu_t^x + \varepsilon_t^x$ with $\mu_t^x = \mu_{t-1}^x + \eta_t^x$

where μ_t^y and μ_t^x are the stochastic trends of the two variables, and ε_t^y and ε_t^x are independent $I(0)$ series. A generic linear combination of y_t and x_t with coefficients $\beta_1, \beta_2 \neq 0$ is:

$\beta_1 y_t + \beta_2 x_t = (\beta_1 \mu_t^y + \beta_2 \mu_t^x) + (\beta_1 \varepsilon_t^y + \beta_2 \varepsilon_t^x)$ Cointegration between y_t and x_t implies the existence of non-zero coefficients β_1

and β_2 that produce stationarity of the linear combination $(\beta_1 y_t + \beta_2 x_t) \sim I(0)$ For this to occur, the coefficients must make the combination of the stochastic trends in the series equal to zero, thereby eliminating the non-stationarity:

$\beta_1 \mu_t^y + \beta_2 \mu_t^x = 0$ This occurs only if for all t $\mu_t^y = -\frac{\beta_2}{\beta_1} \mu_t^x$ 1In a bivariate context, if a cointegrating relation exists between two $I(1)$ processes, it is unique.

Let $y_t - \beta_1 x_t = \varepsilon_t \sim I(0)$ be a valid cointegrating relation. All linear combinations of y_t and x_t can be expressed as $y_t - \beta_2 x_t = \eta_t$, where β_2 is a generic coefficient and rewritten equivalently as $\underbrace{y_t - \beta_1 x_t}_{\varepsilon_*} - (\beta_2 - \beta_1)x_t = \eta_t$

where ε_t is stationary and $(\beta_2 - \beta_1)x_t$ is $I(1)$. The only possibility for η_t to be stationary is when $\beta_2 = \beta_1$, proving the uniqueness of the assumed cointegrating relation with coefficient β_1.

The two $I(1)$ variables must therefore have the same stochastic trend, with a con- stant of proportionality given by $-\frac{\beta_2}{\beta_1}$. Using vectors, with $\boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$

we have $\boldsymbol{\beta}' \begin{pmatrix} y_t \\ x_t \end{pmatrix} = \boldsymbol{\beta}' \begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^x \end{pmatrix} \sim I(0)$

Vector $\boldsymbol{\beta}$ is the cointegrating vector and is usually "normalized" with respect to one of the variables for ease of economic interpretation (e.g., normalizing with respect to y_t, the vector becomes $(1, \frac{\beta_2}{\beta_1})'$).

Economic interpretation. Cointegrating relationships have the economic interpre- tation of equilibrium relations (toward which the variables tend in the long-run), often a direct implication of a theoretical model. For example, according to the efficient market hypothesis applied to the foreign currency market, the forward ex- change rate at t (f_t), is equal to the spot rate (rationally) expected for $t+1$ on the basis of the information set available to agents at time t: $E(s_{t+1}|I_t) = f_t$. Since, with rational expectations, the forecast error $\varepsilon_{t+1} \equiv s_{t+1} - E(s_{t+1}|I_t)$ has a zero

expected value at t, $E(\varepsilon_{t+1}|I_t) = 0$, we get the following relation between s_{t+1} and f_t:

$s_{t+1} = f_t + \varepsilon_{t+1}$ with $E(\varepsilon_{t+1}|I_t) = 0$ Since the observed $\{s_t\}$ and $\{f_t\}$ series are typically non-stationary $I(1)$, the ef- ficiency hypothesis implies a cointegrating relation between the variables, yield-

ing a stationary forecasting error: $\varepsilon_{t+1} \sim I(0)$. The cointegrating vector here is $\beta = (1, -1)'$. General definition (Engle-Granger, 1987). The definition of a cointegrating relation given above for two variables can be generalized to n series. Given a vector of n non-stationary series $\mathbf{y}_t' = (y_{1t}, y_{2t}, \ldots y_{nt})'$, the elements of $\mathbf{y}_t$ are cointegrated of

order d, b, denoted by $\mathbf{y}_t \sim CI(d, b)$, if:

- all series in $\mathbf{y}_t$ are integrated of order d, $I(d)$;
- there exists a vector $\beta' = (\beta_1, \beta_2, ..., \beta_n)$, the cointegrating vector, such that the linear combination $\boldsymbol{\beta}'\mathbf{y}_t = \beta_1 y_{1t} + \beta_2 y_{2t} + \ldots + \beta_n y_{nt}$

is integrated of order $d - b$, with $b > 0$.

Warning: when $n > 2$ variables are involved, more than one valid cointegrating relation can exist. The problem of the multiplicity of cointegrating relations in a multivariate context is addressed is subsequent sections of the following sections.

2.3 Estimation and testing procedures In this section we present two simple empirical strategies for testing the existence of a cointegrating relationship among two non-stationary variables and for estimating the cointegration coefficient. Both strategies are based on the estimation of a single equation. The issue of how to conduct inference on the cointegration parameter is also discussed.

2.3.1 The Engle-Granger procedure and the equilibrium-correction rep- resentation R. Engle and C. Granger (EG, 1987) designed a multi-step procedure to test the hypothesis of cointegration between $I(1)$ variables, and to estimate the coefficients of the cointegrating vector. In the bivariate

case with processes $\{y_t\}$ and $\{x_t\}$ the EG procedure entails the following steps:

1. Unit-root tests on the individual series. Test the order of integration of y_t and x_t to check their nature of $I(1)$ variables by means of the usual DF, ADF and/or $KPSS$ tests.

2. Estimation of static "cointegrating" regression. Estimate with OLS the following "static" regression (called cointegrating regression) $y_t = \alpha + \beta x_t + \varepsilon_t$ If y_t and x_t are cointegrated with cointegration parameter β, the OLS esti- mator β is superconsistent (Stock 1987):

plim $\hat{\beta} = \beta$ with speed of convergence measured by T (and not by $\sqrt{T}$ as in the case of OLS estimation applied to stationary se- ries). Intuitively, all linear combinations of y_t and x_t with coefficients different from $(1, -\beta)$ yield non-stationary residuals (therefore with infinite variance), whereas the linear combination defined by $(1, -\beta)$ is the only one to generate

stationary residuals (with finite variance). By construction, then, the OLS method estimates β by finding that coefficient β that minimizes the variance of the estimated residuals, and therefore yields an estimate of the cointegration parameter "quickly" converging to β as the sample size increases.

3. Testing for cointegration. After estimation of the static regression, the estimated residuals e_t are used to test the existence of a cointegrating relation between y_t and x_t. The test proposed by EG is similar to the DF and ADF tests; the difference is that in this case the test is implemented on residuals estimated from a previously run regression, and not on an original series. The

EG test is performed using the t statistic on the OLS estimate of γ_1 from the following equation:

$\Delta e_t = \gamma_0 + \gamma_1 e_{t-1} + \sum_{i=1}^{p} \delta_i \Delta e_{t-i} + u_t$ with $H_0 : \gamma_1 = 0 \Rightarrow e_t \sim I(1) \Rightarrow y_t$ and x_t not cointegrated

$H_1 : \gamma_1 < 0 \Rightarrow e_t \sim I(0) \Rightarrow y_t$ and x_t cointegrated The distribution of the $t_{\hat{\gamma}_1}$ statistic does not coincide with that of the DF and ADF statistics, since e_t is obtained from a regression in which the parame- ters α and β have been estimated (if those parameters were known, then the

distribution of the statistic would be the same as that of the DF and ADF statistics). For example, with only a constant γ_0 (and not a linear trend) in the deterministic part of the equation, the critical values of the EG test are the following, for different sample sizes:

4. Representation and estimation of adjustment dynamics as an "equi- librium correction mechanism" (ECM). If y_t and x_t are cointegrated (i.e.

the null hypothesis of the test under (3) can be rejected) the two variables have an equilibrium-correction representation. This result, known as the Granger representation theorem (1983), can be illustrated using the model intro- duced at the beginning of the present section:

$y_t = \beta x_t + \varepsilon_t$ with $\varepsilon_t = \rho \varepsilon_{t-1} + \eta_t^y$ $x_t = x_{t-1} + \eta_t^x$ where we assume cointegration, i.e. $|\rho| < 1$. Writing ε_t as

$\varepsilon_t = \frac{1}{1-\alpha I} \eta_t^y$ the process y_t can be expressed as:

$y_t = \beta x_t + \frac{1}{1-\rho L} \eta_t^y$ yielding $(1 - \rho L) y_t = \beta (1 - \rho L) x_t + \eta_t^y$

Subtracting y_{t-1} from both sides of the equation and adding and subtracting βx_{t-1} on the right-hand side, we get the error-correction representation:

$$\Delta y_t = \beta \Delta x_t - (1 - \rho) y_{t-1} + \beta (1 - \rho) x_{t-1} + \eta_t^y$$
$$\Rightarrow \Delta y_t = \beta \Delta x_t - (1 - \rho)[y_{t-1} - \beta x_{t-1}] + \eta_t^y$$

The term in [.] is the deviation of y_{t-1} from its long-run equilibrium value βx_{t-1}, given by the cointegrating relation with x. Since $-(1 - \rho) < 0$ (with $|\rho| < 1$), any deviation from equilibrium

in $t-1$ causes a "correction" of y in the following period t. This correction tends to bring y gradually back toward its equilibrium relation with x. As the cointegrating relation captures the long-run equilibrium between y and x, the ECM (Equilibrium-Correction

- or, less appropriately, error-correction -Mechanism) term describes the dynamic adjustment (i.e. the "short-run dynamics") towards equilibrium. In practice, having obtained the estimated residual series $\{e_t\}$ from the static cointegrating regression, the adjustment dynamics is estimated by OLS ap- plied to the ECM equation:

$\Delta y_t = \delta_0 + \delta_1 e_{t-1} + \xi_t$ delivering consistent estimates of the short-run parameters.

Example of cointegration and equilibrium-correction behaviour. To illustrate the cointegration property and the associated equilibrium-correction behaviour in a two-variable framework, consider the familiar model (where we set $\beta = 1$ and $\{\eta_t^y\}$ and $\{\eta_t^x\}$ are independent white noise, normally distributed processes with $\sigma_y^2 = 1$ and $\sigma_x^2 = 0.25$)

$y_t = x_t + \varepsilon_t$ with $\varepsilon_t = \rho_i \varepsilon_{t-1} + \eta_t^y \; i = 1, 2 \; x_t = x_{t-1} + \eta_t^x$ that can be rewritten in terms of first differences as follows:

$\Delta y_t = \Delta x_t - (1 - \rho_i)[y_{t-1} - x_{t-1}] + \eta_t^y \; \Delta x_t = \eta_t^x$ Here we assume two values for ρ_i (i.e. $\rho_1 = 0.99$ and $\rho_2 = 0.01$) that both

imply cointegration between y_t and x_t but with a very different magnitude of the equilibrium-correction dynamics for y_t. In fact, with $\rho_1 = 0.99$ any sto- chastic deviation from the long-run equilibrium (cointegrating) relation due to η_t^y persists for a long time, keeping y_t away from its long-run relation with x_t for a prolonged period (the "correction" parameter is in fact very small:

$-(1 - \rho_1) = -0.01$). Instead, with $\rho_2 = 0.01$ the corresponding correction pa- rameter is very large (-0.99) and any deviation from the long-run equilibrium

is quickly corrected by subsequent changes of y_t (note that in the present ex- ample only y_t adjusts towards the long-run equilibrium, given the independent random walk assumption on x_t, whereas in a broader bivariate framework also x_t could display a correction behaviour, with a positive correction parameter).

Figure 8 shows time series plots of the $\{x_t\}$ and $\{y_t\}$ processes generated by the above model using 2000 (independent) draws from the distributions of $\{\eta_t^x\}$, the innovation generating the random walk behaviour of $\{x_t\}$, and of

$\{\eta_t^y\}$, the innovation in $\{\varepsilon_t\}$, the AR(1) process for the deviation of y_t from its long-run equilibrium relation with x_t. In Figure 8(a) $\rho_1 = 0.99$ and therefore y_t shows a relatively slow adjustment towards its long-run cointegrating rela- tion with x_t; in Figure 8(b) $\rho_2 = 0.01$ and the adjustment is much quicker, so

that the two series are hardly distinguishable. Figure 9 shows the same dif- ference in the adjustment by plotting y_t against x_t over the simulation sample for the two values of ρ. In both cases the cointegrating relation is displayed by the regression line. When the adjustment is fast, as in Figure 9(b), deviations from the long-run equilibrium tend to disappear quickly and the position of the variables in the plot is always very close to the cointegration line (acting as an "attractor" for the two series), whereas a slower speed of correction implies long-lasting deviations.

Time series plots of cointegrated x and y (b) "fast" correction (rho = 0.01) 0.99

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-60 1750 2000 1000 1750 1000 1250 1500 1250 1500 2000 Time Figure 8. Time series plots of x_t and y_t for $\rho_1 = 0.99$ and $\rho_2 = 0.01$.

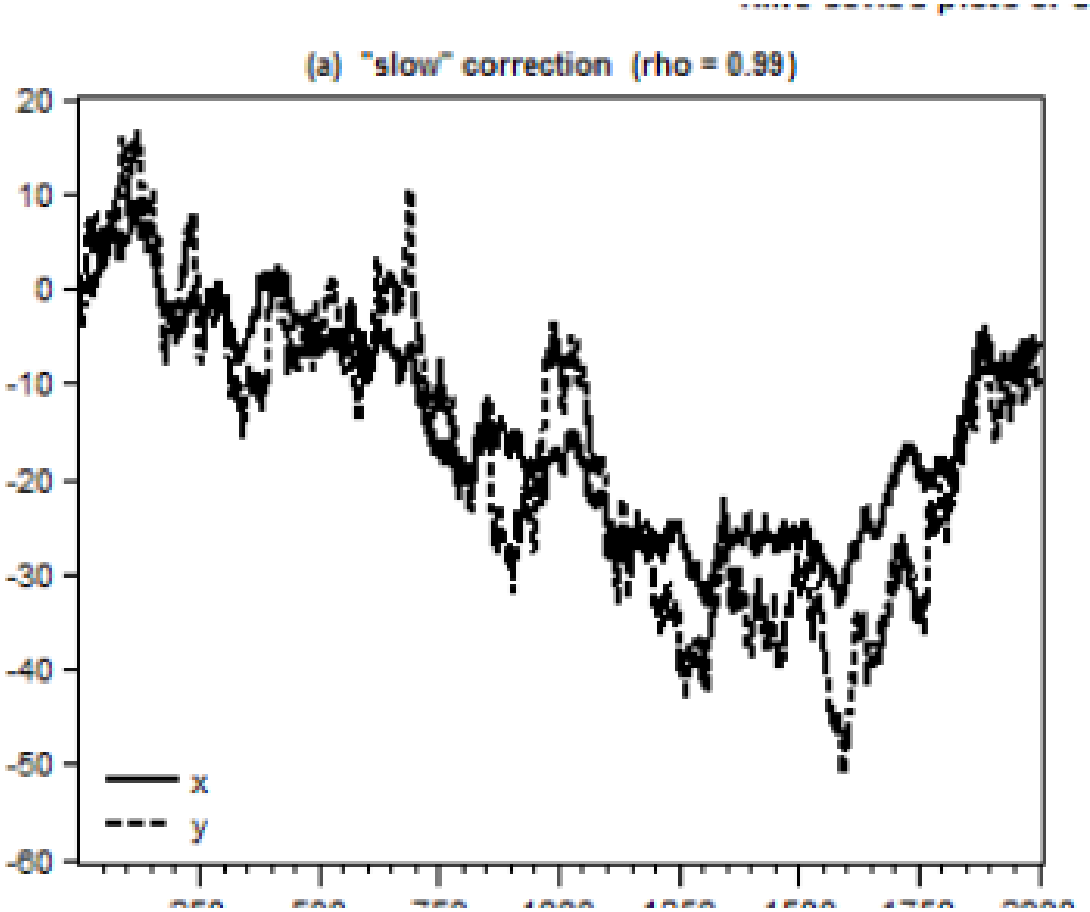


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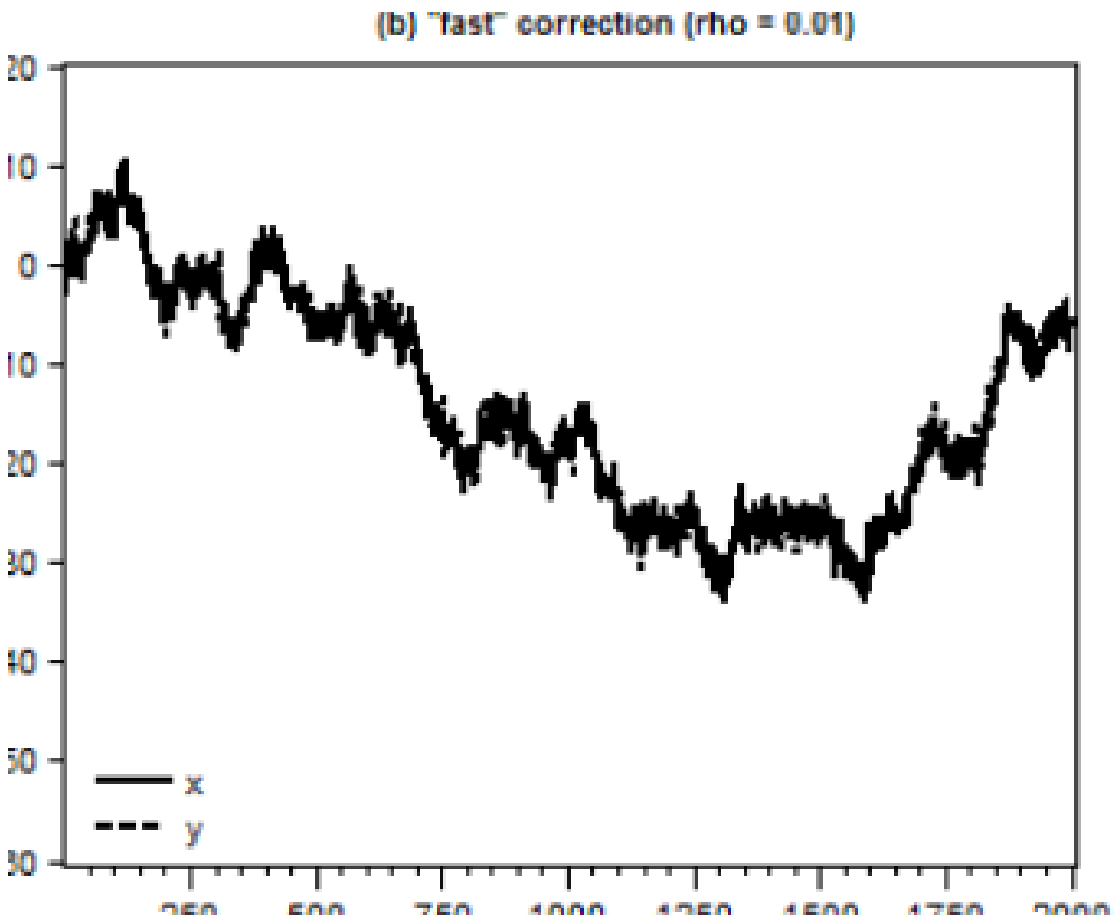


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XY line" plots of cointegrated x and y (with estimated cointegration line) (a) "slow" correction (rho (b) "fast" correction (rho = 0.01)

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-5 x Figure 9. Plots of y_t against x_t for $\rho_1 = 0.99$ and $\rho_2 = 0.01$ with estimated cointegration lines.

Exercise. Consider the following model for two $I(1)$ processes $\{x_t\}$ and $\{y_t\}$:

$y_t = \beta x_t + \varepsilon_t$ with $\varepsilon_t = \rho_1 \varepsilon_{t-1} + \rho_2 \varepsilon_{t-2} + \eta_t^y$ $x_t = x_{t-1} + \eta_t^x$

where we assume cointegration, i.e. $\{\varepsilon_t\}$ is a stationary $(I(0))$ process (a necessary condition for stationarity being $1 - \rho_1 - \rho_2 > 0$). Find the ECM representation of the system and the "equilibrium-correction" coefficient cap- turing the speed of adjustment of y_t to the long-run equilibrium relation with x_t.

5. Problems of inference on the cointegration parameter. The EG pro- cedure provides a (superconsistent) estimate of the cointegration parameter β and a test of the hypothesis of no cointegration using the estimated residuals from the static OLS regression. However, in general this procedure does not allow to use standard t statistics to test particular hypotheses on β. Consid- ering the usual model $y_t = \beta x_t + \varepsilon_t$ $x_t = x_{t-1} + \eta_t^x$

th inference problem is due to the potential presence of:

(i) autocorrelation in the process for ε_t;

(ii) correlation between ε_t and η_s^x, also when $t \neq s$, determining the endo- geneity of x_t Benchmark case. To illustrate the inference problem, let us start from a "benchmark" case (which rarely occurs in empirical applications) in which there is no autocorrelation in ε_t and $E(\varepsilon_t \eta_s^x) = 0$ for all pairs (t, s). Under

these assumptions, x_t is (strictly) exogenous with respect to the error term ε_t, and the classic hypotheses that allow for the use of OLS and the associated standard inference procedures apply. In particular:

- $\hat{\beta}_{OLS}$ is superconsistent

- $p \lim s^2 = \sigma_\varepsilon^2$ (the estimator of the error variance is consistent)

- the distribution of the t-statistic on $\hat{\beta}_{OLS}$ converges to the $N(0,1)$ dis- tribution; therefore inference on β can be carried out in a standard way, using the critical values of the $N(0,1)$ distribution, since under the null hypothesis $H_0 : \beta = \beta_0 : t_{\hat{\beta}_{OLS}} = \frac{\hat{\beta}_{OLS} - \beta_0}{s_{\hat{\beta}_{OLS}}} \xrightarrow{d} N(0,1)$

Autocorrelation in ε_t and correlation with η_s^x. In general, if ε_t is au- to correlated and $E(\varepsilon_t \eta_s^x) \neq 0$ for some pairs (t, s), then x_t is correlated with the error term of the cointegrating equation and, though OLS still yields a superconsistent estimate of β, it is not possible to use the associated t statistic

for inference. In fact, in this case:

- $\hat{\beta}_{OLS}$ is superconsistent but the distribution of $T(\hat{\beta}_{OLS} - \beta)$ converges to a non-normal random variable, whose distribution depends on unknown parameters:

- $t_{\hat{\beta}_{OLS}}$ converges to a random variable different from a standard normal $N(0,1)$ and dependent on unknown parameters, making inference im- possible.

To proceed further, an extension of the static EG cointegration regression approach is needed. To

this purpose, the dynamic ordinary least squares ($DOLS$) technique due to Stock and Watson (1993) can be employed. This procedure consists in estimating an "augmented" version of the cointegrating regression including as additional regressors, beside x_t, also Δx_t and lags (i.e.

past values) and leads (i.e. future values) of Δx_t:

$$y_t = \beta x_t + \sum_{j=-l}^{l} \phi_j \Delta x_{t+j} + v_t.$$

where now the error term v_t is uncorrelated with all regressors. In fact, the in- clusion of Δx_t removes the correlation between ε_t and η_t^x, and the additional lags and leads of Δx_t remove the correlation between ε_t and η_s^x for $t \neq s$,

yielding an error term v_t that is uncorrelated with contemporaneous, past and future values of η_t^x. OLS applied to this equation delivers the DOLS estimate of the cointegrating parameter (β_{DOLS}), on which standard hypoth- esis testing is allowed using the associated t-statistic, which converges to a standard normal distribution.

2.3.2 The cointegration estimation and testing procedure based on an ECM model Although the EG procedure delivers a superconsistent estimator of β under the null hypothesis of cointegration, the bias can be substantial in small samples. Al- ternative approaches to estimation and testing for cointegration have been pursued in order to correct the bias and increase the power of the tests (i.e. the ability to reject the null hypothesis of no cointegration when it is false).

In our bivariate framework, the problem with the EG testing procedure, imple- mented as a unit root test on the residuals estimated from a static regression, is that the formulation of the test implicitly imposes strong restrictions on the joint dy- namics of the two variables, resulting in a loss of power. Therefore, we present here an alternative procedure based on a general dynamic model of the autoregressive- distributed lag (ADL) class which more appropriately describes the autocovariance structure of the series (leaving white noise residuals). This procedure is designed to yield:

(i) a cointegration test with increased power with respect to the EG test;

(ii) an estimate of the cointegration parameter β (derived from the long-run so- lution of the model) on which hypothesis testing can be carried out.

To illustrate, assume that the following $ADL(1,1)$ model $y_t = \theta y_{t-1} + \phi_0 x_t + \phi_1 x_{t-1} + \varepsilon_t$ (with $0 \leq \theta \leq 1$) appropriately describes the dynamics of the two $I(1)$ processes $\{y_t\}$ and $\{x_t\}$: any autocorrelation in the data for y_t is captured by the lags y_{t-1} and

x_{t-1} and therefore $\{\varepsilon_t\}$ is a white noise process. $\{x_t\}$ is generated by a random walk with innovation process $\{\eta_t^x\}$ independent of $\{\varepsilon_t\}$. This model can be rewritten (by

subtracting y_{t-1} from both sides and adding and subtracting $\phi_0 x_{t-1}$ to the right- hand side) in unrestricted equilibrium-correction (ECM) form as:

$\Delta y_t = \phi_0 \Delta x_t - (1-\theta) y_{t-1} + (\phi_0 + \phi_1) x_{t-1} + \varepsilon_t$ (*)

with no restrictions imposed among the coefficients θ, ϕ_0 and ϕ_1. The long-run solution of the model, capturing the long-run effect on y_t of permanent changes in x_t, is obtained (setting $\Delta y_t = \Delta x_t = 0$, and $\varepsilon_t = 0$) as $\beta = \frac{\phi_0 + \phi_1}{1-\theta}$

that can be explicitly included in the equilibrium-correction model yielding $\Delta y_t = \phi_0 \, \Delta x_t - \gamma \, (y_{t-1} - \beta \, x_{t-1}) + \varepsilon_t$ with $\gamma = 1 - \theta$.

Now, when ε_t, Δy_t and Δx_t are $I(0)$ variables, two cases are possible concerning the term in

lagged levels:

(i) $\gamma = 0$ and $(y_{t-1} - \beta x_{t-1})$ is a non-stationary, $I(1)$ linear combination of the two variables:

(ii) $\gamma > 0$ and $(y_{t-1} - \beta x_{t-1})$ is a stationary, $I(0)$ variable.

In the first case y_t is non-stationary and is not cointegrated with x_t. In the second case, y_t is cointegrated with x_t and reacts to past deviations from its long-run relationship with x_t. This reaction contributes to the dynamics of the variable (Δy_t) in addition to changes in x_t and random elements. The stationary nature of the equilibrium-deviation term $(y_t - \beta x_t)$ can be shown by subtracting $\beta \Delta x_t$ from

both sides of $(*)$, obtaining:

$$\Delta y_t - \beta\,\Delta x_t = (\phi_0 - \beta)\,\Delta x_t - \gamma\,(y_{t-1} - \beta\,x_{t-1}) + \varepsilon_t$$
$$\Rightarrow (y_t - \beta x_t) = (1-\gamma)(y_{t-1} - \beta x_{t-1}) + (\phi_0 - \beta)\Delta x_t + \varepsilon_t$$

$(**)$ which defines a stationary autoregressive process since $1-\gamma = \theta$ and $\theta < 1$ under cointegration. This suggests a simple test for cointegration based on the unrestricted ECM mod $(*)$. The ECM cointegration test. Operationally, a test for cointegration can be obtained by estimating the unrestricted ECM equation (*) by OLS, regressing Δy_t on Δx_t, y_{t-1} and x_{t-1}, and testing the statistical significance of the estimated

coefficient on y_{t-1} by means of its t-ratio (given by the estimated κ_1 divided by its estimated standard error). Thus, the estimated equation is $\Delta y_t = \delta + \kappa_0\,\Delta x_t + \kappa_1\,y_{t-1} + \kappa_2\,x_{t-1} + u_t$

(where a constant coefficient δ has been added). Under the null hypothesis of no cointegration, $\kappa_1 = 0$, whereas cointegration implies $\kappa_1 < 0$. The t-ratio associated with the estimated coefficient $\hat{\kappa}_1$ $t_{\hat{\kappa}_1} = \frac{\hat{\kappa}_1}{S_{\hat{\kappa}_1}}$

(where $s_{\hat{\kappa}_1}$ is the estimated standard error of $\hat{\kappa}_1$) is compared with the appropriate critical values tabulated by N. Ericsson and J. MacKinnon (2002).2 For example, the asymptotic critical values for the model above with only a constant term in the deterministic component are -2.91, -3.21 and -3.80 at the 10%, 5% and 1% level

respectively.

Relation with the EG cointegration test. To see the relation between the ECM and the EG cointegration tests, let $e_t = y_t - \beta x_t$, and rearrange $(**)$ to obtain $\Delta e_t = -\gamma\,e_{t-1} + (\phi_0 - \beta)\,\Delta x_t$ $+ \varepsilon_t$ Imposing the restriction $\phi_0 = \beta$ yields the equation used to perform the EG cointe-

gration test, with an estimate of e_t obtained from a first-step static OLS regression in levels. The null hypothesis of no cointegration ($\gamma = 0$) is tested by means of the t-statistic on the estimated coefficient of e_{t-1}, $\hat{\gamma}$. Therefore, the EG cointegration test imposes equality between the contemporaneous (impact) effect of changes in x_t onto y_t, captured by ϕ_0, and the long-run effect measured by β, a strong restriction

on the dynamics of the relationship between the two series which is likely to be invalid in the data. On the other hand, the ECM specification allows for a more general form of dynamics, including relevant lags of both variables.3 Estimation and inference on the cointegration parameter. An estimate of the cointegration parameter β, capturing the long-run effect of x_t on y_t, is obtained from the estimated dynamic model in equilibrium-correction form as the ratio (with a changed sign) of the estimated coefficients on the lagged level variables (x_{t-1}) and y_{t-1}):

$\hat{\beta} = -\frac{\hat{\kappa}_2}{\hat{\kappa}_1}$ As shown in the simulation exercise below, the estimator of the cointegration parameter derived from a dynamic model is preferable to the EG estimator in terms of reduced small-sample bias and asymptotic properties. Moreover, given cointegra- tion, the t-statistic

constructed using the estimated standard error of β (involving

a non-linear function of elements of the variance-covariance matrix of estimated co- efficients) can be used to test hypotheses on β, such as $H_0 = \beta = \beta_0$, since $t_{\hat{\beta}} = \frac{\hat{\beta}-\beta_0}{s_{\hat{\beta}}}$

is asymptotically distributed as a standard normal random variable.

2N. Ericsson and J. MacKinnon, "Distributions of error correction tests for cointegration", Econometrics Journal, 2002.

3Also the ECM cointegration test is based on an implicit (but testable) assumption, namely the weak exogeneity of x_t for the cointegration parameter β, which amounts to assuming that x_t does not adjust to the lagged disequilibrium term $(y_{t-1} - \beta x_{t-1})$. This is the case, for example, if x_t is generated by an independent random walk, so that $\Delta x_t = \eta_t^x$, with no role for past deviations

from the long-run equilibrium.

Properties of the ECM cointegration procedure: a Monte Carlo simula- tion. To assess the major properties of the cointegration estimation and testing procedure based on a dynamic model compared with the EG procedure, we em- ploy a small-scale Monte Carlo simulation exercise, based on the following data generating process for the two $I(1)$ series $\{y_t\}$ and $\{x_t\}$:

$y_t = 0.5y_{t-1} + 0.3x_t + 0.2x_{t-1} + \varepsilon_t$ $x_t = x_{t-1} + \eta_t^x$ where $\{\varepsilon_t\}$ and $\{\eta_t^x\}$ are independent white noise processes with standard normal

distributions. Given the chosen parameterization, $\{y_t\}$ and $\{x_t\}$ are cointegrated with long-run cointegration parameter $\beta = \frac{0.3+0.2}{1-0.5} = 1$. In each replication, independent drawings from the distributions of $\{\varepsilon_t\}$ and $\{\eta_t^x\}$

are used to generate the series $\{y_t\}$ and $\{x_t\}$ using the above model over two different sample sizes: a relatively small sample ($T = 100$) and a larger sample ($T = 1000$).

For each replication, the following two equations are estimated by OLS:

$y_t = \alpha + \beta_{EG} x_t + v_t$ EG static regression:

ECM dynamic regression: $\Delta y_t = \delta + \kappa_0 \Delta x_t + \kappa_1 y_{t-1} + \kappa_2 x_{t-1} + u_t$ From the EG static cointegrating regression a direct estimate of the long-run coin- tegration parameter β is obtained as $\hat{\beta}_{EG}$, with associated t-ratio $t_{\hat{\beta}_{EG}}$. From the

ECM dynamic regression an estimate of β is obtained as $\hat{\beta}_{ECM} = -\frac{\hat{\kappa}_2}{\hat{\kappa}_1}$, and the t- ratios are denoted $t_{\hat{\beta}_{ECM}}$. We want to compare the distributions of $\hat{\beta}_{EG}$ and $\hat{\beta}_{ECM}$,

and the distributions of the associated t-ratio under the true null hypothesis that $\beta = 1$, in small and large samples.

• Distributions of estimated long-run cointegration parameters. Figure 10 shows the empirical distributions of β_{EG} and β_{ECM} for $T = 100$ and $T = 1000$;

in both plots the vertical line is set at the true value of the cointegration parameter $\beta = 1$. In the smaller sample, the bias of the EG estimator is substantial (the average estimate being 0.933), whereas the ECM estimator has a distribution centered around the true value. In the larger sample the two distributions are closer due to the superconsistency of the EG estimator (though its average is 0.993, still lower than the true value).

(a) T = 100 (b) T = 1000 EG static regression ECM dynamic regression Density 0.6 1.3 1.4 1.5 0.5 0.7 0.8 0.9 1.0 1.1 1.2 0.90 0.92 0.94 0.96 0.98 1.00 1.02 1.04 1.06 1.08 1.10 Estimated long-run cointegration parameter Figure 10. Empirical distributions of $\hat{\beta}_{EG}$ and $\hat{\beta}_{ECM}$ for small ($T = 100$) and large ($T = 1000$) samples.

• Test of the (true) null hypothesis $\beta = 1$. To test the (true) null hypothesis on the cointegration parameter $\beta = 1$, we compute the values of associated the t-ratios from the static and dynamic regressions as:

$$t_{\hat{\beta}_{EG}} = \frac{\hat{\beta}_{EG} - 1}{s_{\hat{\beta}_{EG}}} \quad \text{and} \quad t_{\hat{\beta}_{ECM}} = \frac{\hat{\beta}_{ECM} - 1}{s_{\hat{\beta}_{ECM}}}$$

where $s_{\hat{\beta}_{EG}}$ and $s_{\hat{\beta}_{ECM}}$ are the estimated standard errors of the respective coefficients in the two regressions. Figure 11 shows that standard inference on the value of β conducted using the t-ratios from the EG static regression would be highly misleading. In fact, the distibution of $t_{\hat{\beta}_{EG}}$ is not centered

around the true value (zero) in both small and large samples. On the contrary, the t-ratio obtained from the dynamic ECM regression is centered around zero and its distribution is close to a standard normal in both small and large samples.

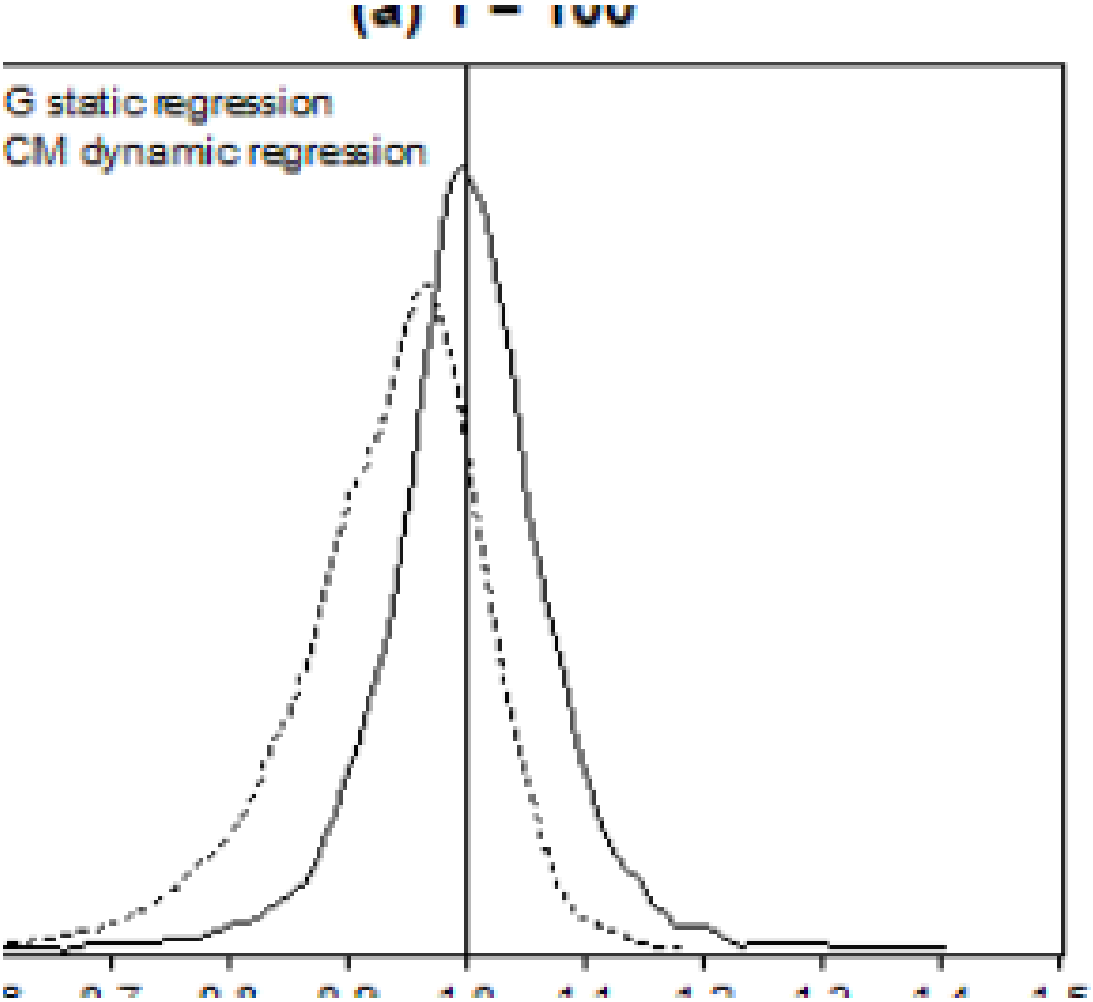


Figure 1.67: Course figure, source page 21, crop 1.

EG static regression ECM dynamic regression

—- T = 100 T = 1000

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-8 t - ratio Figure 11. Empirical distributions of t-ratios for testing $\beta = 1$ from EG and ECM regressions in small and large samples.

The results above point to the estimation of an ECM dynamic regression as a more appropriate empirical specification to carry out estimation of, and inference on, long-run cointegration parameters.

Finally, to assess the robustness of the ECM dynamic regression procedure to misspecification of the estimated ADL model we present the results obtained by replicating the above Monte Carlo exercise keeping the same data generating process for $\{y_t\}$ and $\{x_t\}$ (which implies cointegration with long-run parameter $\beta = 1$) but estimating an ECM dynamic regression obtained by reparameterizing an $ADL(2,2)$ model as follows:

$$y_t = \delta + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \phi_0 x_t + \phi_1 x_{t-1} + \phi_2 x_{t-2} + u_t$$
$$\Delta y_t = \delta + \kappa_0 \Delta x_t + \kappa_1 y_{t-1} + \kappa_2 x_{t-1} + \kappa_3 \Delta y_{t-1} + \kappa_4 \Delta x_{t-1} + u_t$$

$\Rightarrow$ where $\kappa_0 = \phi_0$, $\kappa_1 = -(1 - \theta_1 - \theta_2)$, $\kappa_2 = \phi_0 + \phi_1 + \phi_2$, $\kappa_3 = -\theta_2$ and $\kappa_4 = -\phi_2$.

Also in this case an estimate of the long-run cointegration parameter is obtained as $\hat{\beta}_{ECM} = -\frac{\hat{\kappa}_2}{\hat{\kappa}_1}$ and the (true) null hypothesis $\beta = 1$ can be tested using the associated t-ratio. For each replication of the experiment, two ECM dynamic regressions are

estimated: one based on the (true) $ADL(1,1)$ model and one based on the more general (over-parameterized) $ADL(2,2)$ model. The empirical distributions of the estimated long-run cointegration parameters $\hat{\beta}$ and the associated t-ratios $t_{\hat{\beta}} = \frac{\hat{\beta}-1}{s_{\hat{\beta}}}$

are compared for the two sample sizes $T = 100, 1000$. Figure 12 shows that the shape of the distribution of β in the case of an over-parameterized $ADL(2,2)$ specification is very close to that of the correctly specified $ADL(1,1)$ dynamic model for both sample sizes. Similarly, Figure 13 displays the distributions of the t-ratios for testing the true null hypothesis that $\beta = 1$; again, the over-parameterized model yields t-statistics with a distribution very close to that of the correctly specified

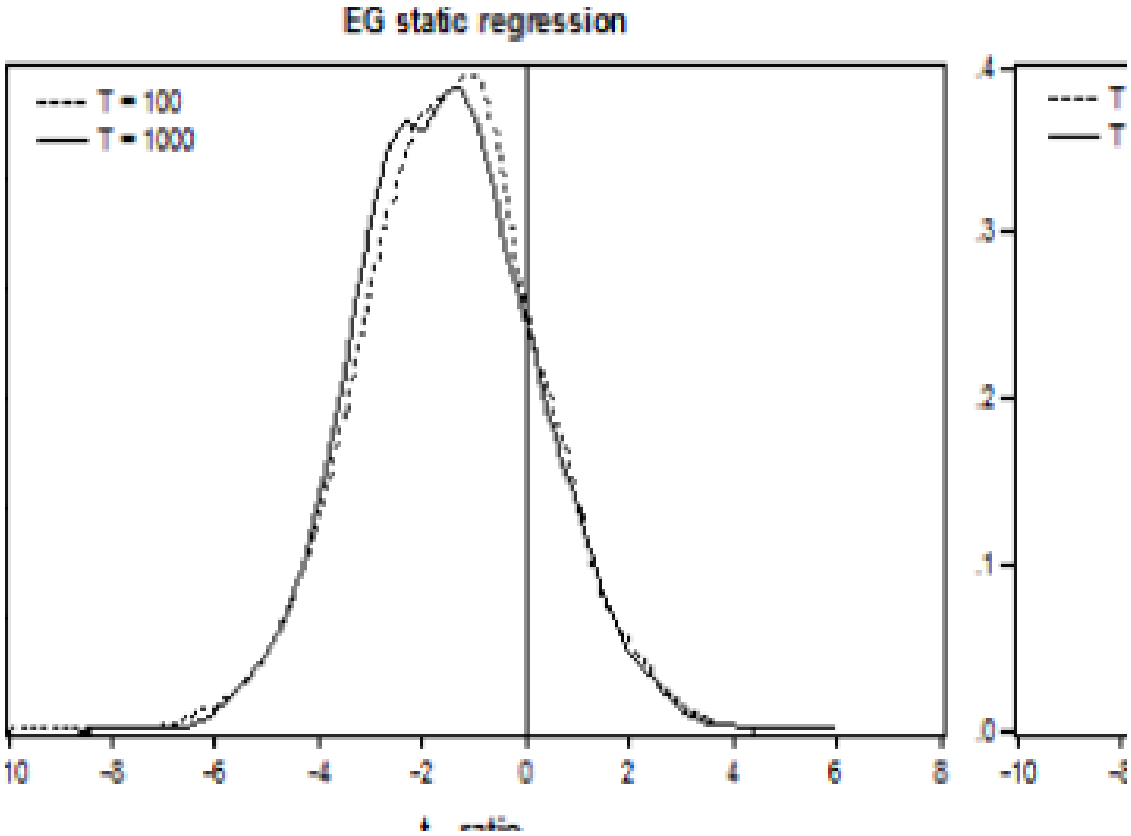


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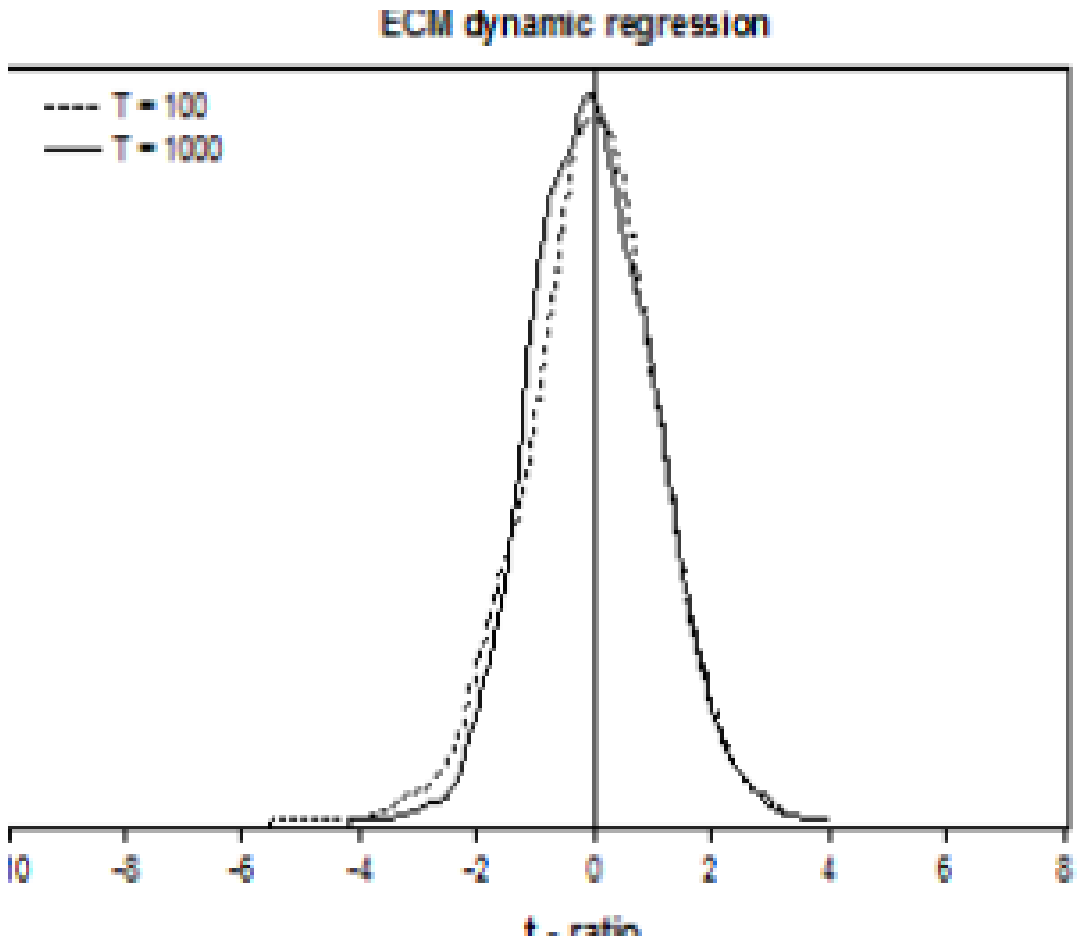


Figure 1.69: Course figure, source page 22, crop 2.

model. Overall, those results favour the estimation and testing procedure based on a dynamic ECM model as a more reliable approach to (bivariate) cointegration analysis, being also robust to the inclusion of redundant dynamic regressors (e.g.

y_{t-2} and x_{t-2} in our example above) in the estimated specification.

(a) T = 100 (b) T = 1000 Correctly specified dynamic model Over-parameterized dy mic model Over-parameterized dynam c model Density 1.04 1.05 0.96 0.97 0.98 0.99 1.00 1.01 1.02 1.03 1.1 0.5 0.6 0.7 0.8 0.9 1.0 1.2 1.3 1.4 1.5 Estimated long-run cointegration parameter Figure 12. Empirical distributions of $\hat{\beta}$ from $ADL(1,1)$ and $ADL(2,2)$ dynamic models for small ($T = 100$) and large ($T = 1000$) samples.

(a) T = 100 (b) T = 1000 Correctly specified dynamic model Over-parameterized dynamic Over-parameterized dvn cmodel ode Density .2 .1 .0

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-1 t-ratio Figure 13. Empirical distributions of t-ratios for testing $\beta = 1$ from $ADL(1,1)$ and $ADL(2,2)$ dynamic models in small and large samples.

ALGORITHM – Problem

Dynamic models for non-stationary variables **Problem.** spurious regressions A bivariate regression is defined as spurious if the coefficient in the estimated regression between two independent variables is significantly different from zero

ALGORITHM – Problem

Problem. spurious regressions A bivariate regression is defined as spurious if the coefficient in the estimated regression between two independent variables is significantly different from zero. The probability to obtain such result usually tends to zero as the sample dimension increases.

Two cases:

stationary and ergodic independent processes: OLS regression delivers a consistent estimate which converges to the population parameter (zero);

- non-stationary ($I(1)$) independent processes: results from OLS regression are not valid and statistically significant estimates can be obtained even if there is no relation between the variables, i.e.

the regression is spurious

EXAMPLE – Illustration

Example. two independent random walks $y_t = y_{t-1} + \varepsilon_t^y$ with $y_0 = 0$ $x_t = x_{t-1} + \varepsilon_t^x$ with $x_0 = 0$ where ε_t^y and ε_s^x are IID for all pairs (t, s), with mean zero and

variances σ_v^2 and σ_x^2.

$\Rightarrow$ all significant relation between y and x is spurious

Linear regression estimated by OLS:

$v_t = \alpha + \beta x_t + e_t$ True parameter values: $\alpha = 0$ and $\beta = 0$; the estimated error term is non-stationary:

$e_t = y_t = \sum_{i=1}^{t} \varepsilon_i^y \sim I(1)$ Classical assumptions on the properties of the error term are then violated and the values $\hat{\alpha}$, $\hat{\beta}$, R^2 and the t and F statistics cannot be relied upon for correct inference, even if the sample

size increases

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Sample: 1980M01 2005M12 - 312 obs Order: Name Sample: 1 1000 - 1000 obs Order: Name Dependent Variable: LUKRPA M britpus V ruk5y ruk6m V rus10y V rus1y V rus2y V rus3m alpha1 alpha2 beta1 beta2 c BC Method: Least Squares M ddluscpi M dexuk Date: 11/21/07 Time: 11:34 M dlukrpa distr_beta M dluscpi Sample (adjusted): 1980M01 2005M08 M dlusip distr_beta_t M druk10y distr_dw Included observations: 308 after adjustments druk3m distr_f druk5y distr_r2 drus10y distr_t M drus1y M dw1 M drus2y Variable Std. Error M dw2 Coefficient t-Statistic Prob.

M drus3m M epsilonx M drus5y M epsilony vsr1_infl_month_up M drus6m M f1 M dukinfl f1emp950 f2 −0.779158 0.110248

-7.067300 0.0000 M e lusip e_ruk # f2emp950 LUSIP 1.266464 0.024831 51.00424 0.0000 ecm_ruk graf_b_t Var_coint1 eq_lukrpa graf_r_f grafb1 grafb2 var_coint2 eq_r5y var_ruk3m6m1y graf_drate 4.836958 R-squared 0.894753 Mean dependent var Wdi76axdf grafdw' graf_dspur graf infl grafdw2 Adjusted R-squared 0.894409 S.D. dependent var 0.297306 graff1 graf_infl_ann Graff2 u graf infl ann m S.E. of regression 0.096609 Akaike info criterion −1.829815 graf_infl_mon_var grafrsq1 graf_infl_nonstat grafrsq2 graf_nonstat_trend graftb1 2.855996 −1.805594 Sum squared resid Schwarz criterion graf_rate Graftb2 graf_rates m Log likelihood 283.7915 0.010543 Hannan-Quinn criter.

graf_spur # r1emp500 graf_spurs # r2emp500 F-statistic 1.2E-151 M resid graf_uk_cpi graf_us_indprod M rsq1 M rsq2 graf_uscpi M lukip spurt1000 M lukip_hp spurt50 M lukrpa # t1 M luscpi # t2 M talpha1 M lusip Iusip_dt M talpha2 M lusip_hp # tb1emp025 Iusip st # tb1emp975 M lusipf # tb2emp025 # tb2emp975 M mean_infl_1 M mean_infl_2 M tbeta1 M resid M tbeta2 ruk10y ruk1y ruk2y ruk2y # thetax # thetay X Untitled / Untitled1 / Untitled2 / New Page RW / AR_stat / New Page 1.266464348219718 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 14:25 A ∼ e ×∃ Fz S P2)

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druk10y distr_f Autocorrelation Partial Correlation PAC Q-Stat Prob AC Spruk3m1y M druk3m distr_r2 spruk6m1y wkgap wkinfl M druk5y distr_t drus10y M dw1 drus1y M dw2 ukinfl_m wkinprodg M drus2y 0.972 0.972 293.65 0.000 M epsilonx M drus3m M epsilony N ∼ drus5y ukrpaxmif 569.89 0.000 r F 0.941 -0.059 M f1 drus6m uscp7500f f1emp950 M dukinfl usgap 828.62 0.000 0.909 −0.032 usinfl e_lusip e_ruk ecm_ruk # f2emp950 usinprodg 1073.3 0.000 0.883 0.079 graf_b_t Var1_infl_month_up graf_r_f eq_lukrpa Var_coint1 1306.3 0.000 0.860 0.047 eq_r5y Var_coint2 Ju grafb2 graf_drate var var_ruk3m6m1y grafdw' 1529.0 0.000 0.017 wdi76axdf 0.

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-0.062 2625.7 graf_uscpi spurt1000 M lukip spurt50 0.000 0.648 -0.041 2761.5 M lukip_hp # t1 M lukrpa # t2 M luscpi 0.000 0.616 -0.008 2884.8 M talpha1 M lusip M talpha2 Iusip_dt 0.585 -0.018 2996.4 0.000 # tb1emp025 M lusip_hp # tb1emp975 Iusip_st 0.000 0.560 0.062 3098.7 # tb2emp025 M lusipf # tb2emp975 M mean_infl_1 0.000 0.534 -0.024 3192.1 M tbeta1 M mean_infl_2 M tbeta2 M resid 0.507 -0.027 3276.7 0.000 # thetax ruk10y # thetay X 0.478 -0.024 3352.2 0.000 ruk2y ruk3m 0.450 0.010 3419.3 0.000 Contitled / Untitled1 / Untitled2 / New Page <» RW / AR_stat / New Page Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 14:34 .

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Statistical features of a spurious regression (Monte Carlo results):

distribution of $\hat{\beta}$. As the sample size T increases, the variance of $\hat{\beta}$ does not shrink: $\hat{\beta}$ does not tend to a random variable with a degenerate distribution

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-7.940761 0.0000 var1_infl_month_up graf_b_t graf_r_f grafb1 grafb2 eq_lukrpa Var_coint1 eq_r5y Var_coint2 war_ruk3m6m1y wdi76axdf 2.353000 graf_drate 0.567784 Mean dependent var R-squared grafdw' In graf dspur grafdw2 graf_infl 0.558780 S.D. dependent var 1.615182 Adjusted R-squared graff1 graf_infl_ann Graff2 graf_infl_ann_m S.E. of regression 1.072875

Akaike info criterion 3.017739 grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat 55.25094 3.094220 Sum squared resid Schwarz criterion Ju graftb1 graf_nonstat_trend Graftb2 graf_rate Log likelihood

-73.44348 3.046864 Hannan-Quinn criter.

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distr_t drus10y drus10y drus1y drus2y drus3m drus5y drus5y M dw1 Variable Coefficient Std. Error t-Statistic Prob.

M dw2 M epsilonx M epsilony M f1 uscp7500f f1emp950 f2 9.975579 0.483035 20.65188 0.0000 usgap M dukinfl e_lusip e_ruk ecm_ruk usinfl # f2emp950 0.457052 0.032548 14.04256 var1_infl_month_up 0.0000 graf_b_t graf_r_f grafb1 grafb2 eq_lukrpa Var_coint1 eq_r5y Var_coint2 war_ruk3m6m1y wdi76axdf graf_drate 15.02257 0.164989 R-squared Mean dependent var grafdw' In graf dspur grafdw2 graf_infl 11.16260 Adjusted R-squared 0.164152 S.D. dependent var graff1 graf_infl_ann Graff2 graf_infl_ann_m 10.20538 Akaike info criterion 7.485704 S.E. of regression grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat 7.495520 Sum squared

resid 103941.4 Schwarz criterion graftb1 graf_nonstat_trend Graftb2 graf_rate 7.489435 m Log likelihood

-3740.852 Hannan-Quinn criter.

graf_rates # r1emp500 graf_spur # r2emp500 F-statistic 197.1936 Durbin-Watson stat 0.011688 graf_spurs M resid graf_uk_cpi M rsq1 Prob(F-statistic) 0.000000 graf_us_indprod rsq2 spurt1000 graf_uscpi M lukip spurt50 Iukip_hp Iukipa Iukrpa Iuscpi # t1 # t2 M talpha1 M talpha2 Iusip_dt # tb1emp025 M lusip_hp # tb1emp975 Iusip_st # tb2emp025 M lusipf # tb2emp975 M mean_infl_1 M tbeta1 mean_infl_2 mean_infl_2 resid ruk10y ruk1y ruk2y M tbeta2 # thetax # thetay $\sum$ $\sum$ y ruk3m Contitled / Untitled1 / Untitled2 / New Page RW / AR_stat / New Page 0.4570522522858043 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR 16:02 14/03/2022 e D r S Fz S P2

- w G D PB I 3)

– X EViews - [Equation: SPURT1000 Workfile: SPUR_PLOTS::RW\] File Edit Object View Proc Quick Options Add-ins Window Help _ X View Proc Object Print Name Freeze Estimate Forecast Stats Resids

-10 Y Residuals

-20 1000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR 16:05 14/03/2022 .

S D G .

Statistical features of a spurious regression (Monte Carlo results):

distribution of $\hat{\beta}$. As the sample size T increases, the variance of $\hat{\beta}$ does not shrink: $\hat{\beta}$ does not tend to a random variable with a degenerate distribution

× EViews - [Graph: DISTR_BETA Workfile: SPUR_PLOTS::RW\]

- File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom .8

— T = 50

— T = 1000 .7 .6 .5 Density .4 - .3 .2 .1 - .0

-2

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-4 Coefficient on x Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR ^ 16:09 14/03/2022 O Sa e Fz r

- S w x∃ *** P a)

Statistical features of a spurious regression (Monte Carlo results):

distribution of $\hat{\beta}$. As the sample size T increases, the variance of $\hat{\beta}$ does not shrink: $\hat{\beta}$ does not tend to a random variable with a degenerate distribution distribution of $t_{\hat{\beta}}$. As the sample size T increases, the variance of $t_{\hat{B}}$ increases, with critical values increasingly far away from those

of the t distribution. If the latter are used for inference, as the sample lengthens the probability

of rejecting the null hypothesis $\beta = 0$ increases.

× EViews - [Graph: DISTR_T Workfile: SPUR_PLOTS::RW\]

-8 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom .10

— T = 50

— T = 1000 .08 .06 Density .04 .02 .00

-80

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-120

-160 t-statistic for coefficient on x Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR ⌢ 16:13 14/03/2022 .

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distribution of R^2. It does not concentrate towards zero as the sample size increases.

distribution of the regression F. If the critical values of the $F(1, T-2)$ distribution are used, the test tends to reject the null hypothesis $\beta = 0$ with a very large size (type-I error)

× EViews - [Graph: DISTR_R2 Workfile: SPUR_PLOTS::RW\]

-8 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom 3.2

— T = 50 (vertical line: median)

— T = 1000 (vertical line: median) 2.8 2.4 2.0 Density 1.6 1.2 0.8 0.4 0.0 0.3 0.2 0.5 0.8 0.6 0.9 0.4 0.7 0.1 0.0 1.0 R^2 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR V V 16:17 14/03/2022 O S e **N** D

- Fz r w x∃ S P 3)

× EViews - [Graph: DISTR_F Workfile: SPUR_PLOTS::RW\]

- File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom T = 1000 T = 50 .0020 .032 5% crit. value of empirical distr.

.028 .0016 .024 .020 .0012 Density .016 .0008 .012 .008 .0004 .004 .0000 .000 2,000 3,000 4,000 1,000 5,000 250 300 F of the regression Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR ⌢ 16:19 14/03/2022 e A O .

Fz r **k**

- w × S P (3)

distribution of the DW (Durbin-Watson) statistic (for testing first order residual autocorrelation). With $\alpha = \beta = 0$ the error term coincides with y and is a random walk; in particular, the coefficient of first-order autocorrelation ρ_1 tends to 1 as T increases. Recalling that the DW statistics is

linked to ρ_1 by the following relation $DW \approx 2(1-\rho_1)$ we see that DW tends to 0.

The DW statistics is used to test for first-order autocorrelation of the estimated residuals (e_t) and is closely related to the sample first-order residual autocorrelation coefficient $\hat{\rho}_1$.

$$DW = \frac{\sum_{t=2}^{T}(e_t - e_{t-1})^2}{\sum_{t=1}^{T} e_t^2} = \frac{\sum_{t=2}^{T} e_t^2 + \sum_{t=2}^{T} e_{t-1}^2 - 2\sum_{t=2}^{T} e_t e_{t-1}}{\sum_{t=2}^{T} e_t^2}$$

If T is "sufficiently large" we have $\sum_{t=1}^{T} e_t^2 \approx \sum_{t=1}^{T} e_{t-1}^2 \approx \sum_{t=1}^{T} e_t^2$ leading to the following (approximate) relation:

$$DW \approx 2 - 2\frac{\sum_{t=2}^{I} e_t e_{t-1}}{\sum_{t=1}^{T} e_t^2} = 2(1-\hat{\rho}_1)$$

× EViews - [Graph: DISTR_DW Workfile: SPUR_PLOTS::RW\]

-2 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom T = 50 T = 1000 2.5 2.0 1.5 Density 1.0 0.5 0.0 1.8 0.4 1.6 0.2 0.6 0.8 1.0 1.4 2.0 0.0 1.2 0.0 0.2 0.4 0.6 1.6 1.8 2.0 0.8 1.2 1.4 1.0 DW statistic 1.15 | 1.79 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR 16:24 14/03/2022 e S D P Fz **N**

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Note

in general, the finding of a relatively high value for R^2 and a very low value of the DW statistic is a symptom of the possibility that the regression is spurious and the associated test statistics meaningless

Comparison with bivariate regression with stationary series.

Monte Carlo simulation with independent stationary $AR(1)$ processes:

$y_t = 0.5y_{t-1} + \varepsilon_t^y$ with $y_0 = 0$ $x_t = 0.5x_{t-1} + \varepsilon_t^x$ with $x_0 = 0$ $\{\varepsilon_t^\gamma\}$ and $\{\varepsilon_t^x\}$ independent standard normal distributions

Sample sizes: $T = 50$ and $T = 1000$ Number of replications: 2000

× EViews - [Group: UNTITLED Workfile: SPUR_PLOTS::AR_stat\] G File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Default V Options Position Sample Sheet Stats Spec 1 - 0 -

-1 -

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-4 - 1000 1 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR 16:29 14/03/2022 D G

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× EViews - [Group: UNTITLED Workfile: SPUR_PLOTS::AR_stat\]

-3 G File Edit Object View Proc Quick Options Add-ins Window Help _ X View Proc Object Print Name Freeze Default V Options Position Sample Sheet Stats Spec

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-4 1 III III II 17 | -2.1 50 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR ⌢ 16:31 14/03/2022 ....

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-0 File Edit Object View Proc Quick Options Add-ins Window Help Workfile: SPUR_PLOTS - (c:\bagli...

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Equation: SPURT50 Workfile: SPUR_PLOTS::AR_stat\

- X View Proc Object Save Freeze Details+/- Sho View Proc Object Print Name Freeze Estimate Forecast Stats Resids View Proc Object Save Freeze Details+/- Show Fetch Sto Range: 1 2000 – 2000 obs Filter: * Range: 1980M01 2010M12 - 372 obs Filter: * Sample: 1 1000 - 1000 obs Order: Name Dependent Variable: Y Sample: 1980M01 2005M12 - 312 obs Order: Name alpha1 alpha2 beta1 beta2 c britpus ruk5y ruk6m rus10y rus10y rus1y Method: Least Squares M ddluscpi Date: 03/14/22 Time: 16:32 M dexuk M dlukrpa distr_beta Sample: 1 50 dluscpi distr_beta_t ∇ dlusip distr_dw Included observations: 50 rus6m druk10y distr_r2 druk3m rusom spruk3m1y spruk6m1y ukgap ukinfl ukinfl_m ukinprodg ukrpaxmif uscp7500f distr_t M druk5y

dw1 drus10y drus10y drus1y drus2y drus3m drus5y drus5y M dw2 Coefficient Std. Error Variable t-Statistic Prob.

M epsilonx M epsilony M f1 # f1emp950 0.9751 −0.005639 0.179552 −0.031405 M f2 usgap usinfl M dukinfl # f2emp950 e_lusip e_ruk ecm_ruk X 0.084568 0.153546 0.5843 0.550768 graf_b_t var1_infl_month_up graf_dw graf_r_f grafb1 grafb2 eq_lukrpa Var_coint1 eq_r5y Var_coint2 0.006280 −0.010739 R-squared Mean dependent var war_ruk3m6m1y wdi76axdf graf_drate grafdw' In graf dspur grafdw2 graf_infl Adjusted R-squared

-0.014422 S.D. dependent var 1.258891 graff1 graf_infl_ann Graff2 1.267937 S.E. of regression Akaike info criterion 3.351837 graf_infl_ann_m grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat 77.16786 3.428318 Sum squared resid Schwarz criterion graftb1 graf_nonstat_trend Graftb2 graf_rate 3.380961 Log likelihood

-81.79593 Hannan-Quinn criter.

m graf_rates # r1emp500 graf_spur F-statistic 0.303345 0.699065 Durbin-Watson stat # r2emp500 graf_spurs M resid graf_uk_cpi 0.584347 Prob(F-statistic) M rsq1 graf_us_indprod M rsq2 graf_uscpi spurt1000 M lukip spurt50 Iukip_hp Iukipa Iukrpa Iuscpi # t1 # t2 M talpha1 M talpha2 Iusip_dt # tb1emp025 M lusip_hp # tb1emp975 Iusip_st # tb2emp025 M lusipf # tb2emp975 M mean_infl_1 M tbeta1 mean_infl_2 mean_infl_2 resid ruk10y ruk1y ruk2y M tbeta2 # thetax # thetay X ruk3m Contitled / Untitled1 / Untitled2 / New Page \ RW \ AR_stat \ New Page Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur plots 16:32 e r ∼

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– X EViews - [Equation: SPURT50 Workfile: SPUR_PLOTS::AR_stat\] E File Edit Object View Proc Quick Options Add-ins Window Help _ X View Proc Object Print Name Freeze Estimate Forecast Stats Resids

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-3 Y Residuals

-4 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots eR 16:35 14/03/2022 ....

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- File Edit Object View Proc Quick Options Add-ins Window Help Equation: SPURT50 Workfile: SPUR_PLOTS::AR_stat\

- X Workfile: SPUR_PLOTS - (c:\bagli... - Workfile: USUK05 - (c:\bagliano\corsi\ects\b...

View Proc Object | Print Name Freeze | Estimate Forecast Stats Resids Correlogram of Residuals View Proc Object Save Freeze Details+/- Sho View Proc Object Save Freeze Details+/- Show Fetch St Range: 1 2000 - 2000 obs Filter: * Range: 1980M01 2010M12 - 372 obs Filter: * Date: 03/14/22 Time: 16:36 Sample: 1 1000 - 1000 obs Order: Name Sample: 1980M01 2005M12 - 312 obs Order: Name alpha1 Alpha2 beta1 beta2 c M britpus ruk5y ruk6m Sample: 1 50 BC M ddluscpi rus10y rus1y rus2y Included observations: 50 M dexuk dlukrpa distr_beta rus3m dluscpi distr_beta_t rus5y ∇ dlusip Autocorrelation Partial Correlation PAC AC Q-Stat Prob distr_dw rus6m ∼ druk10y

distr_r2 druk3m Spruk3m1y distr_t Spruk6m1y Spruk6m1y Wukgap Wukinfl Wukinfl_m Wukinprodg M druk5y M dw1 M drus10y M dw2 0.650 0.650 22.422 0.000 M drus1y M epsilonx drus2y drus3m M epsilony 30.042 0.000

-0.082 0.375 M f1 ∼ drus5y ukrpaxmif # f1emp950 drus6m uscp7500f 0.172 -0.067 31.673 0.000 M f2 M dukinfl usgap # f2emp950 usinfl e_lusip e_ruk ecm_ruk

-0.027 -0.158 31.716 0.000 graf_b_t usinprodg graf_dw var1_infl_month_up 36.142 0.000

-0.277 -0.295 graf_r_f grafb1 grafb2 eq_lukrpa Var_coint1 eq_r5y Var_coint2

-0.425 -0.156 46.834 0.000 graf_drate var var_ruk3m6m1y grafdw' wdi76axdf In graf dspur grafdw2 0.052 56.030 0.000 7 -0.390 graf_infl graff1 graf_infl_ann Graff2 −0.329 62.741 0.000

-0.038 graf_infl_ann_m grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat 0.056 65.548 0.000 −0.210 graftb1 graf_nonstat_trend J graftb2 graf_rate 10 -0.154 -0.160 67.084 0.000 m graf_rates # r1emp500 graf_spur 0.000 0.082 67.093 0.012 # r2emp500 graf_spurs M resid graf_uk_cpi 0.000 0.015 -0.243 67.108 M rsq1 graf_us_indprod M rsq2 graf_uscpi 0.000 0.052 67.331 0.056 spurt1000 M lukip spurt50 M lukip_hp 67.500 0.000 # t1

-0.090 0.048 V lukrpa # t2 M talpha1 0.000 67.786 M lusip 0.062 0.048 M talpha2 Iusip_dt # tb1emp025 68.008 0.000 M lusip_hp 0.054 -0.027 # tb1emp975 Iusip_st # tb2emp025 M lusipf 0.138 68.656 0.000 0.091 # tb2emp975 M mean_infl_1 M tbeta1 M mean_infl_2 70.256 0.000 0.140 -0.005 M tbeta2 M resid # thetax ruk10y 70.659 0.000 0.069 -0.105 # thetay X ruk2y 0.000 ~ 0.058 -0.033 70.951 ruk3m Untitled / Untitled1 / Untitled2 / New Page < \ RW \ AR_stat \ New Page Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots 16:36 **N** .

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-0 File Edit Object View Proc Quick Options Add-ins Window Help Workfile: SPUR_PLOTS - (c:\bagli... - Workfile: USUK05 - (c:\bagliano\corsi\ects\b...

Equation: SPURT1000 Workfile: SPUR_PLOTS::AR_stat\

- X View Proc Object Save Freeze Details+/- Sho View Proc Object Save Freeze Details+/- Show Fetch Sto View Proc Object Print Name Freeze Estimate Forecast Stats Resids Range: 1 2000 – 2000 obs Filter: * Range: 1980M01 2010M12 - 372 obs Filter: * Sample: 1 1000 - 1000 obs Order: Name Sample: 1980M01 2005M12 - 312 obs Order: Name Dependent Variable: Y alpha1 alpha2 beta1 beta2 c britpus ruk5y ruk6m rus10y rus1y rus2y rus2y Method: Least Squares M ddluscpi Date: 03/14/22 Time: 16:37 M dexuk M dlukrpa distr_beta dluscpi Sample: 1 1000 distr_beta_t rus5y ∇ dlusip distr_dw rus6m druk10y Included observations: 1000 distr_r2 druk3m Spruk3m1y distr_t Spruk6m1y Spruk6m1y Wukgap Wukinfl_m Wukinprodg Wukinprodg

M druk5y dw1 drus10y drus10y drus1y drus2y drus3m drus5y drus5y M dw2 Variable Coefficient Std. Error t-Statistic Prob.

M epsilonx epsilony M f1 f1emp950 uscp7500f C

-0.016646 0.037306 −0.446207 0.6555 usgap M dukinfl # f2emp950 e_lusip e_ruk ecm_ruk usinfl graf_b_t X 0.2707 0.034472 0.031279 1.102084 var1_infl_month_up graf_dw graf_r_f grafb1 grafb2 eq_lukrpa Var_coint1 eq_r5y Var_coint2 war_ruk3m6m1y wdi76axdf graf_drate

-0.017551 0.001216 R-squared Mean dependent var grafdw' In graf dspur grafdw2 graf_infl 1.179573 Adjusted R-squared 0.000215 S.D. dependent var graff1 graf_infl_ann Graff2 graf_infl_ann_m 1.179446 Akaike info criterion 3.169965 S.E. of regression grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat 3.179780 1388.311 Sum squared resid Schwarz criterion Ju graftb1 graf_nonstat_trend Graftb2 graf_rate 3.173695 m Log likelihood −1582.982 Hannan-Quinn criter.

graf_rates # r1emp500 graf_spur # r2emp500 F-statistic 1.214589 Durbin-Watson stat 1.018677 graf_spurs M resid graf_uk_cpi M rsq1 0.270691 Prob(F-statistic) graf_us_indprod rsq2 spurt1000 graf_uscpi M lukip spurt50 Iukip_hp Iukipa Iukrpa Iuscpi # t1 # t2 M talpha1 M talpha2 Iusip_dt # tb1emp025 M lusip_hp # tb1emp975 Iusip_st # tb2emp025 M lusipf # tb2emp975 M mean_infl_1 M tbeta1 mean_infl_2 mean_infl_2 resid ruk10y ruk1y ruk2y

M tbeta2 # thetax # thetay X ruk3m Contitled / Untitled1 / Untitled2 / New Page \ RW \ AR_stat \ New Page Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur plots 16:37 ∼

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-3 EViews - [Equation: SPURT1000 Workfile: SPUR_PLOTS::AR_stat\] E File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Estimate Forecast Stats Resids

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-3 Y Residuals

-4 1000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR ^ 16:39 14/03/2022 D S C .

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- File Edit Object View Proc Quick Options Add-ins Window Help Equation: SPURT1000 Workfile: SPUR_PLOTS::AR_stat\

- X Workfile: SPUR_PLOTS - (c:\bagli... - Workfile: USUK05 - (c:\bagliano\corsi\ects\b... = View Proc Object Print Name Freeze Estimate Forecast Stats Resids View Proc Object Save Freeze Details+/- Sho Correlogram of Residuals View Proc Object Save Freeze Details+/- Show Fetch St Range: 1 2000 - 2000 obs Filter: * Range: 1980M01 2010M12 - 372 obs Filter: * Sample: 1 1000 - 1000 obs Order: Name Date: 03/14/22 Time: 16:40 Sample: 1980M01 2005M12 - 312 obs Order: Name alpha1 Alpha2 beta1 beta2 c M britpus ruk5y ruk6m Sample: 1 1000 BC M ddluscpi rus10y rus1y rus2y Included observations: 1000 M dexuk dlukrpa distr_beta rus3m dluscpi distr_beta_t rus5y ∇

dlusip distr_dw Autocorrelation Partial Correlation AC PAC Q-Stat Prob rus6m N druk10y distr_r2 ∼ M druk3m spruk3m1y distr_t spruk6m1y wkgap wkinfl M druk5y dw1 M drus10y M dw2 drus1y 0.489 0.489 240.09 0.000 M epsilonx ukinfl_m wkinprodg M drus2y M epsilony drus3m 0.249 0.013 302.28 0.000 M f1 ∼ drus5y ukrpaxmif # f1emp950 drus6m uscp7500f 0.094 -0.043 311.08 0.000 M f2 M dukinfl usgap # f2emp950 usinfl e_lusip e_ruk ecm_ruk 0.048 0.021 313.42 0.000 graf_b_t usinprodg graf_dw var1_infl_month_up graf_r_f grafb1 grafb2 0.048 0.033 315.79 0.000 eq_lukrpa Var_coint1 eq_r5y Var_coint2 0.025 -0.017 316.39 0.000 graf_drate var var_ruk3m6m1y grafdw wdi76axdf In graf dspur grafdw2 graf_infl 0.011 316.86 0.000

0.021 graff1 graf_infl_ann Graff2 graf_infl_ann_m 317.88 0.000 0.032 0.026 grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat 0.066 323.87 0.000 0.077 J graftb1 graf_nonstat_trend J graftb2 graf_rate 0.086 0.024 331.42 0.000 m graf_rates # r1emp500 graf_spur 0.052 -0.019 334.21 0.000 # r2emp500 graf_spurs M resid graf_uk_cpi M rsq1 0.025 -0.003 334.87 0.000 graf_us_indprod rsq2 spurt1000 graf_uscpi 0.028 0.024 335.67 0.000 M lukip spurt50 M lukip_hp # t1 V lukrpa 0.015 -0.012 335.91 0.000 # t2 M talpha1 M lusip 0.023 0.016 336.45 0.000 M talpha2 Iusip_dt # tb1emp025 M lusip_hp 0.014 -0.002 336.66 0.000 # tb1emp975 Iusip_st # tb2emp025 M lusipf 0.020 0.013 337.06 0.000 # tb2emp975 M mean_infl_1 M tbeta1 M mean_infl_2 337.27 0.000 18 -0.014 -0.045 M tbeta2 M resid # thetax ruk10y thetay 0.022 0.048 337.77 0.000

ruk2y 0.046 0.030 339.94 0.000 ruk3m Contitled / Untitled1 / Untitled2 / New Page \ RW \

AR_stat \ New Page > < Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots 16:40 ....

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- D gR ^ 14/03/2022 w ×∃ C PE (a)

Results from estimation of $y_t = \alpha + \beta x_t + e_t$:

• empirical distributions of $\hat{\beta}$ and associated t-statistic • empirical distributions of R^2 empirical distributions of DW statistic ($\simeq 1$ with $\rho_1 = 0.5$)

× EViews - [Graph: DISTR_BETA Workfile: SPUR_PLOTS::AR_stat\]

-2 III File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Independent stationary AR(1) processes

— T = 50

— T = 1000 Density

-0.4

-1.0 -0.8 -0.6

-0.2 0.0 0.2 0.4 8.0 0.6 1.0 Coefficient on x Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR 14:11 e Fz r **N** O * ×∃ S w P (3)

× EViews - [Graph: DISTR_T Workfile: SPUR_PLOTS::AR_stat\]

-0 III File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom Independent stationary AR(1) processes .35

— T = 50

— T = 1000 .30 .25 .20 Density .15 .10 .05 .00

-5

-3

-2

-6

-4 t-statistic for coefficient on x Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR ^ 14:14 15/03/2022 e **N** O Fz = 1 S w I PB

× EViews - [Graph: DISTR_R2 Workfile: SPUR_PLOTS::AR_stat\]

-3 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom R^2 T = 1000 T = 50 (vertical line: median) 500 - Density 300 - 100 - 4 - 0 - .15 .20 .25 .30 .35 .00 .05 .10 .40 .10 .15 .20 .25 .05 .35 .30 .00 .40 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR 14:16 15/03/2022 e **V** Sa i Fz x = S • w × PB 3)

× EViews - [Graph: DISTR_DW Workfile: SPUR_PLOTS::AR_stat\]

-8 File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom DW statistic

—— T = 50

— T = 1000 Density 0.6 1.2 1.8 0.2 0.4 0.8 1.0 1.6 0.0 1.4 2.0 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = spur_plots xR ^ 14:18 15/03/2022 **N** e O N Fz r

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Cointegration: the bivariate case Basic idea. In some instances the regression between non-stationary variables captures the true relation between the series. This happens when the error term ε_t in the regression of y_t on x_t (both $I(1)$ series) is stationary:

$\varepsilon_t \sim I(0)$ In this case the two variables $y_t \sim I(1)$ and $x_t \sim I(1)$ are cointegrated (Engle-Granger, 1987) The stochastic trend driving the non-stationarity of y_t and x_t is common to the two variables: common stochastic trend

EXAMPLE – Illustration

Example. $y_t \sim I(1)$ and $x_t \sim I(1)$ (with $y_0 = x_0 = 0$):

$y_t - \beta x_t = \varepsilon_t$ with $\varepsilon_t = \rho\varepsilon_{t-1} + \eta_t^y$ $x_t = x_{t-1} + \eta_t^x$ $\{\eta^y\}$ and $\{\eta^x\}$ independent white noise

Two cases:

- if $\rho = 1$ the linear combination ε_t is non-stationary ($\varepsilon_t \sim I(1)$) and y_t and x_t are not cointegrated.
- if $|\rho| < 1$ the linear combination defined by the coefficients $(1, -\beta)$ yields a stationary variable: $\varepsilon_t \sim I(0)$ and the two series v_t and x_t are cointegrated

Cointegration and common stochastic trends. Case of two random walk plus noise processes v_t and x_t:

$v_t = \mu_t^y + \varepsilon_t^y$ with $\mu_t^y = \mu_{t-1}^y + \eta_t^y$ $x_t = \mu_t^x + \varepsilon_t^x$ with $\mu_t^x = \mu_{t-1}^x + \eta_t^x$

μ_t^y and μ_t^x are the stochastic trends, and ε_t^y and ε_t^x are independent $I(0)$ series. A generic linear combination of y_t and x_t with coefficients $\beta_1, \beta_2 \neq 0$ is:

$\beta_1 v_t + \beta_2 x_t = (\beta_1\mu_t^y + \beta_2\mu_t^x) + (\beta_1\varepsilon_t^y + \beta_2\varepsilon_t^x)$ Cointegration between y_t and x_t implies the existence of

non-zero coefficients β_1 and β_2 that produce stationarity of the linear combination $(\beta_1 v_t + \beta_2 x_t) \sim I(0)$

For this to occur, the coefficients must make the combination of the stochastic trends in the series equal to zero, thereby eliminating the non-stationarity:

$\beta_1\mu_t^y + \beta_2\mu_t^x = 0$ This occurs only if for all t $\mu_t^{\mathcal{Y}} = -\frac{\beta_2}{\beta_1}\mu_t^{\mathcal{X}}$

The two $I(1)$ variables must therefore have the same stochastic trend, with a constant of proportionality given by $\frac{\beta_2}{\beta_1}$

Using vectors, with $\boldsymbol{\beta} = \begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix}$ we have $\boldsymbol{\beta}' \begin{pmatrix}\mathcal{Y}_t\\ x_t\end{pmatrix} = \boldsymbol{\beta}' \begin{pmatrix}\varepsilon_t^{\mathcal{Y}}\\ \varepsilon_t^{\mathcal{X}}\end{pmatrix} \sim I(0)$

Vector β is the cointegrating vector and is usually "normalized" with respect to one of the variables

for ease of economic interpretation (e.g., normalizing with respect to y_t, the vector becomes $(1, \frac{\beta_2}{\beta_1})'$)

Economic interpretation Cointegrating relationships have the economic interpretation of equilibrium relations (toward which the variables tend in the long-run), often a direct implication of a theoretical model.

EXAMPLE – Illustration

Example.

according to the efficient market hypothesis, the one-period forward exchange rate at t (f_t) is equal to the spot rate (rationally) expected for $t+1$ on the basis of information at t:

$E(s_{t+1}|I_t) = f_t$ Since the forecast error $\varepsilon_{t+1} \equiv s_{t+1} - E(s_{t+1}|I_t)$ has zero expected value at t, $E(\varepsilon_{t+1}|I_t) = 0$:

$s_{t+1} = f_t + \varepsilon_{t+1}$ with $E(\varepsilon_{t+1}|I_t) = 0$ In the data $\{s_t\}, \{f_t\} \sim I(1)$, and the efficiency hypothesis implies a cointegrating relation such as $\varepsilon_{t+1} \sim I(0)$ with cointegrating

vector $\boldsymbol{\beta} = (1, -1)'$

General definition (Engle-Granger, 1987). Given a vector of n non-stationary series $\mathbf{y}_t' = (y_{1t}, y_{2t}, \ldots, y_{nt})'$, the elements of $\mathbf{y}_t$ are cointegrated of order d, b, denoted by $\mathbf{y}_t \sim CI(d, b)$, if:

all series in y_t are integrated of order d, $I(d)$;

• there exists a vector $\boldsymbol{\beta}' = (\beta_1, \beta_2, \ldots, \beta_n)$, the cointegrating vector, such that the linear combination $\boldsymbol{\beta}'\mathbf{y}_t = \beta_1 y_{1t} + \beta_2 y_{2t} + \ldots + \beta_n y_{nt}$

is integrated of order $d - b$, with $b > 0$

Estimation and testing procedure (1) Engle-Granger procedure (EG, 1987) Multi-step procedure (bivariate case with $\{y_t\}$ and $\{x_t\}$):

1. Test of the order of integration of y_t and x_t to check their $I(1)$ nature, by means of DF, ADF and/or KPSS tests

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- File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Options AddText Line/Shade Remove Template Zoom RUS5Y RUK5Y Corr. = 0.91 1980 2005 1985 1990 1995 2000 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 xR ⌒ 14:48 15/03/2022 • • .

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View Proc Object Properties Print Name Freeze Sample Genr Sheet Graph Stats Ident View Proc Object Save Freeze Details+/- Show Fetc View Proc Object Save Freeze Details+/- Show Fetch Store Delete Correlogram of RUK5Y Range: 1 2000 - 2000 obs Filter: 1 Range: 1980M01 2010M12 - 372 obs Filter: * Sample: 1 1000 - 1000 obs Sample: 1980M01 2005M12 - 312 obs Order: Name Date: 03/15/22 Time: 14:50 alpha1 Alpha2 beta1 beta2 c M britpus ruk3m ruk5y ruk6m rus10y BC Sample: 1980M01 2005M12 M ddluscpi M dexuk Included observations: 309 rus1y rus2y rus3m rus5y ∼ dlukrpa distr_beta M dluscpi distr_beta_t dlusip druk10y distr_dw rus6m distr_f M druk3m Autocorrelation Partial Correlation Q-Stat Prob AC PAC distr_r2 M druk5y Spruk3m1y Spruk6m1y distr_t drus10y M dw1 M drus1y M dw2 drus2y ukinfl

0.976 0.976 297.01 0.000 M epsilonx M drus3m ukinfl_m M epsilony M drus5y ukinprodg 581.56 0.953 0.032 0.000 M f1 M ukrpaxmif M drus6m f1emp950 uscp7500f M dukinfl 853.01 0.000 0.930 -0.041 usgap M e_lusip # f2emp950 M usinfl e_ruk graf_b_t usinprodg 1113.8 0.000 ecm_ruk 0.910 0.065 graf_r_f grafb1 grafb2 eq_lukrpa var1_infl_month_up eq_r5y var_coint1 0.000 1364.7 0.891 0.018 graf_drate Var_coint2 grafdw' graf_dspur var var_ruk3m6m1y 1607.5 0.000 0.875 0.047 grafdw2 wdi76axdf graf_infl graff1 graf_infl_ann 1844.3 0.000 0.863 0.080 Graff2 graf_infl_ann_m grafrsq1 graf_infl_mon_var 2075.2 0.000 0.851 0.003 grafrsq2 graf_infl_nonstat graftb1 graf_nonstat_trend 2302.0 0.000 0.841 0.064 J graftb2 graf_rate m graf_rates 2521.4 0.000 0.826 −0.114 # r1emp500 graf_spur # r2emp500 graf_spurs

M resid 2734.3 0.000 0.813 0.023 graf_uk_cpi M rsq1 graf_us_indprod M rsq2 graf_uscpi 2939.3 0.000 0.796 -0.041 spurt1000 M lukip spurt50 lukip_hp 3136.6 0.000 0.779 -0.024 # t1 # t2 Iuscpi 3328.7 0.000 0.768 0.110 M talpha1 M talpha2 0.758 0.017 3516.2 0.000 # tb1emp025 M lusip_hp # tb1emp975 Iusip_st 0.749 3700.4 0.000 0.022 # tb2emp025 M lusipf # tb2emp975 Mean_infl_1 3880.0 0.000 M tbeta1 0.739 -0.036 M mean_infl_2 M tbeta2 M resid # thetax ruk10y 4056.4 0.000 0.731 0.033 # thetay ruk1y X ruk2y 0.719 -0.051 4227.8 0.000 \ Untitled / Untitled1 / Untitled2 / New Page 4394.6 0.000 20 0.708 0.009 RW / AR_stat / New Page Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 14:51

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-0 File Edit Object View Proc Quick Options Add-ins Window Help Series: RUK5Y Workfile: USUK05::Untitled _ X Workfile: SPUR_PLOTS - (c:\bagliano\co... - Workfile: USUK05 - (c:\bagliano\corsi\ects\bbs\ects2\en...

View Proc Object Properties | Print Name Freeze | Sample Genr Sheet Graph Stats Ident View Proc Object Save Freeze Details+/- Show Fetc View Proc Object Save Freeze Details+/- Show Fetch Store Delete Augmented Dickey-Fuller Unit Root Test on RUK5Y Range: 1 2000 – 2000 obs Filter: 1 Range: 1980M01 2010M12 - 372 obs Filter: * Sample: 1 1000 - 1000 obs Order: Name Sample: 1980M01 2005M12 - 312 obs Order: Name Null Hypothesis: RUK5Y has a unit root alpha1 alpha2 beta1 beta2 ruk3m ruk5y ruk6m rus10y M britpus BC Exogenous: Constant M ddluscpi M dexuk Lag Length: 0 (Automatic - based on SIC, maxlag=15) B C rus1y rus2y rus3m rus5y M dlukrpa distr_beta M dluscpi distr_beta_t dlusip druk10y distr_dw rus6m Prob.* distr_f t-Statistic druk3m distr_r2 M druk5y Spruk3m1y spruk6m1y wkgap ukinfl distr_t drus10y M dw1 M drus1y M dw2 drus2y 0.3252 Augmented Dickey-Fuller test statistic

−1.914917 M epsilonx M drus3m ukinfl_m M epsilony M drus5y M ukinprodg Test critical values:

1% level −3.451421 M f1 M drus6m M ukrpaxmif f1emp950 f2 uscp7500f M dukinfl 5% level −2.870712 usgap M e_lusip # f2emp950 e_ruk M usinfl 10% level

-2.571728 graf_b_t ecm_ruk var1_infl_month_up graf_r_f grafb1 grafb2 eq_lukrpa eq_r5y graf_drate Var_coint2 grafdw' *MacKinnon (1996) one-sided p-values.

var var_ruk3m6m1y graf_dspur grafdw2 wdi76axdf graf_infl graff1 graf infl ann Graff2 graf_infl_ann_m grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat graftb1 graf_nonstat_trend Augmented Dickey-Fuller Test Equation Graftb2 graf_rate m graf_rates Dependent Variable: D(RUK5Y) # r1emp500 graf_spur # r2emp500 graf_spurs Method: Least Squares M resid graf_uk_cpi M rsq1 graf_us_indprod Date: 03/15/22 Time: 14:53 M rsq2 graf_uscpi spurt1000 M lukip Sample (adjusted): 1980M02 2005M09 spurt50 lukip_hp # t1 # t2 Iusip Iusip Included observations: 308 after adjustments M talpha1 M talpha2 # tb1emp025 M lusip_hp # tb1emp975 Iusip_st Variable Coefficient Std. Error t-Statistic Prob.

tb2emp025 M lusipf # tb2emp975 Mean_infl_1 M tbeta1 M mean_infl_2 M tbeta2 M resid RUK5Y(-1) −0.017785 0.009288 −1.914917 0.0564 # thetax ruk10y # thetay ruk1y 0.116544 1.385174 0.1670 0.084137 $\sum$ $\sum$ y ruk2y Untitled / Untitled1 / Untitled2 / New Page RW / AR_stat / New Page

-1.914916542978798 Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 xR 14:53 e **k** P Z S T

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2. Estimate with OLS the "static" regression (called cointegrating regression) $y_t = \alpha + \beta x_t + \varepsilon_t$ If y_t and x_t are cointegrated, the OLS estimator $\hat{\beta}$ is superconsistent (Stock 1987):

plim $\hat{\beta} = \beta$ with speed of convergence measured by T (and not by $\sqrt{T}$ as in the case of OLS estimation applied to stationary series)

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- X View Proc Object Save Freeze Details+/- Show Fetc View Proc Object Save Freeze Details+/- Show Fetch Store Delete View Proc Object Print Name Freeze Estimate Forecast Stats Resids Range: 1 2000 – 2000 obs Filter: 1 Range: 1980M01 2010M12 - 372 obs Filter: * Sample: 1 1000 - 1000 obs Order: Name Sample: 1980M01 2005M12 - 312 obs Order: Name Dependent Variable: RUK5Y alpha1 alpha2 beta1 beta2 c V ruk3m ruk5y ruk6m rus10y rus1y rus2y rus2y rus2y rus5y rus6m M britpus B c ddluscpi Method: Least Squares M dexuk Date: 03/15/22 Time: 14:58 dlukrpa dluscpi dlusip druk10y distr_beta Sample: 1980M01 2005M09 distr_beta_t distr_dw Included observations: 309 distr_f druk3m distr_r2 Spruk3m1y M druk5y distr_t drus10y spruk6m1y wkgap wkinfl M dw1 Variable Coefficient Std. Error t-Statistic Prob.

M dw2 M drus2y epsilonx ukinfl_m M drus3m M epsilony wkinprodg drus5y drus6m M f1 f1emp950 f2 uscp7500f C 1.908238 0.188222 10.13825 0.0000 M dukinfl usgap M e_lusip # f2emp950 e_ruk M usinfl RUS5Y 0.893899 0.023479 38.07254 0.0000 graf_b_t ecm_ruk var1_infl_month_up graf_r_f grafb1 grafb2 grafdw1 eq_lukrpa eq_r5y graf_drate var_coint2 var_ruk3m6m1y 8.528573 0.825222 Mean dependent var R-squared graf_dspur grafdw2 graff1 wdi76axdf graf_infl 3.024378 0.824653 S.D. dependent var Adjusted R-squared graf_infl_ann Graff2 graf_infl_ann_m 1.266442 3.316751 S.E. of regression Akaike info criterion grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat Sum squared resid 492.3894 3.340915 Schwarz criterion Ju graftb1 graf_nonstat_trend J graftb2 graf_rate Log likelihood

-510.4380 3.326412 Hannan-Quinn criter.

m graf_rates # r1emp500 graf_spur # r2emp500 0.168343 F-statistic 1449.518 Durbin-Watson

stat graf_spurs M resid graf_uk_cpi M rsq1 graf_us_indprod Prob(F-statistic) 0.000000 rsq2 spurt1000 graf_uscpi Iukip Iukip Iukip_hp Iukrpa Iuscpi Iusip Iusip Iusip_dt spurt50 # t1 # t2 M talpha1 M talpha2 V lusip_hp # tb1emp025 # tb1emp975 Iusip_st # tb2emp025 M lusipf # tb2emp975 Mean_infl_1 M tbeta1 M mean_infl_2 M tbeta2 M resid # thetax M ruk10y thetay x y ruk1y <» Untitled / Untitled1 / Untitled2 / New Page RW / AR_stat / New Page Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 14:58 e ∼ r S Fz S T x2 ⌢ 15/03/2022

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- X View Proc Object Print Name Freeze Estimate Forecast Stats Resids

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3. After estimation of the static regression, the estimated residuals e_t are used to test the existence of a cointegrating relation between y_t and x_t. The test is similar to the DF and ADF tests but it is implemented on residuals estimated from a previously run regression, and not on an original series.

The EG test is performed using the t statistic on the OLS estimate of γ_1 from the following equation:

$\Delta e_t = \gamma_0 + \gamma_1 e_{t-1} + \sum_{i=1}^{p} \delta_i \Delta e_{t-i} + u_t$ with H_0: $\gamma_1 = 0 \Rightarrow e_t \sim I(1) \Rightarrow y_t$ and x_t not cointegrated

H_1: $\gamma_1 < 0 \Rightarrow e_t \sim I(0) \Rightarrow y_t$ and x_t cointegrated

The distribution of the $t_{\hat{\gamma}_1}$ statistic does not coincide with that of the DF and ADF statistics (e_t is obtained from a previous regression in which α and β have been estimated). With only a constant γ_0 in the deterministic part, the critical values of the test are, for different sample sizes:

1% 5% 10% $T = 50$ -4.09 -3.44 -3.09 $T = 500$ -3.91 -3.34 -3.05 $T = \infty$ | -3.90 -3.34 -3.04

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- X View Proc Object Print Name Freeze Estimate Forecast Stats Resids Workfile: SPUR_PLOTS - (c:\bagliano\co... - Workfile: USUK05 - (c:\bagliano\corsi\ects\bbs\ects2\en...

Correlogram of Residuals View Proc Object Save Freeze Details+/- Show Fetc View Proc Object Save Freeze Details+/- Show Fetch Store Delete Range: 1 2000 - 2000 obs Filter: 1 Range: 1980M01 2010M12 - 372 obs Filter: * Date: 03/15/22 Time: 15:08 Sample: 1 1000 - 1000 obs Sample: 1980M01 2005M12 - 312 obs Order: Name Sample: 1980M01 2005M09 alpha1 Alpha2 beta1 beta2 c M britpus M ruk3m ruk5y ruk6m rus10y BC Included observations: 309 M ddluscpi M dexuk rus1y rus2y rus3m rus5y ∼ dlukrpa distr_beta M dluscpi distr_beta_t Autocorrelation Partial Correlation M dlusip PAC Q-Stat Prob AC distr_dw M druk10y rus6m distr_f M druk3m distr_r2 Spruk3m1y M druk5y Spruk6m1y distr_t drus10y 0.902 0.902 253.71 0.000 M dw1

drus1y M dw2 drus2y ukinfl 461.04 0.000 0.814 0.004 M epsilonx M drus3m

ukinfl_m M epsilony M drus5y M ukinprodg 628.75 0.000 0.731

-0.020 M f1 M ukrpaxmif M drus6m f1emp950 f2 uscp7500f M dukinfl .03 771.89 0.000 0.674 0.096 usgap M e_lusip # f2emp950 M usinfl e_ruk graf_b_t 897.35 0.000 usinprodg 0.630 0.047 ecm_ruk graf_r_f grafb1 grafb2 var1_infl_month_up eq_lukrpa eq_r5y var_coint1 0.000 0.590 1007.7 0.006 graf_drate Var_coint2 grafdw' graf_dspur var var_ruk3m6m1y 1106.5 0.000 0.557 0.039 grafdw2 wdi76axdf graf_infl graff1 graf_infl_ann 1195.1 0.000 0.527 0.017 Graff2 graf_infl_ann_m grafrsq1 graf_infl_mon_var 0.522 1282.3 0.000 0.132 grafrsq2 graf_infl_nonstat graftb1 graf_nonstat_trend 1365.2 0.000 0.508

-0.019 J graftb2 graf_rate m graf_rates 0.497 0.028 1444.9 0.000 # r1emp500 graf_spur # r2emp500 graf_spurs M resid 1515.9 0.000 I 0.468 -0.061 graf_uk_cpi M rsq1 graf_us_indprod M rsq2 1574.4 0.000 graf_uscpi 0.424 -0.090 spurt1000 M lukip spurt50 M lukip_hp 0.393 1624.6 0.000 0.045 # t1 M lukrpa # t2 V luscpi 0.369 1669.2 0.000 0.023 M talpha1 M talpha2 Iusip_dt 0.372 0.112 1714.6 0.000 # tb1emp025 M lusip_hp # tb1emp975 Iusip_st 1758.5 0.000 0.365 −0.019 # tb2emp025 M lusipf # tb2emp975 Mean_infl_1 1802.8 0.000 0.366 0.046 M tbeta1 mean_infl_2 M tbeta2 M resid 1843.0 0.000 0.348 -0.068 # thetax ruk10y # thetay ruk1y X ruk2y 0.354 0.110 1884.5 0.000 Contitled / Untitled1 / Untitled2 / New Page > RW / AR_stat / New Page Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05

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View Proc Object Properties Print Name Freeze Sample Genr Sheet Graph Stats Ident View Proc Object Save Freeze Details+/- Show Fetc View Proc Object Save Freeze Details+/- Show Fetch Store Delete Augmented Dickey-Fuller Unit Root Test on E RUK Range: 1 2000 – 2000 obs Filter: 1 Range: 1980M01 2010M12 - 372 obs Filter: * Sample: 1 1000 - 1000 obs Sample: 1980M01 2005M12 - 312 obs Order: Name Null Hypothesis: E_RUK has a unit root alpha1 alpha2 beta1 beta2 ruk3m ruk5y ruk6m rus10y M britpus BC Exogenous: Constant M ddluscpi M dexuk Lag Length: 0 (Automatic - based on SIC, maxlag=15) B C rus1y rus2y rus3m rus5y M dlukrpa distr_beta M dluscpi distr_beta_t dlusip druk10y distr_dw rus6m distr_f druk3m Prob.* t-Statistic distr_r2 M druk5y Spruk3m1y spruk6m1y wkgap ukinfl distr_t drus10y M dw1 M drus1y M dw2 drus2y 0.0010 Augmented Dickey-Fuller test statistic

−4.119901 M epsilonx M drus3m ukinfl_m M epsilony M drus5y M ukinprodg Test critical values:

1% level −3.451421 M f1 M drus6m M ukrpaxmif f1emp950 f2 M dukinfl uscp7500f 5% level −2.870712 usgap M e_lusip # f2emp950 e_ruk M usinfl 10% level −2.571728 graf_b_t ecm_ruk usinprodg graf_r_f grafb1 grafb2 var1_infl_month_up eq_lukrpa eq_r5y Var_coint1 graf_drate Var_coint2 grafdw' *MacKinnon (1996) one-sided p-values.

graf_dspur var var_ruk3m6m1y grafdw2 wdi76axdf graf_infl graff1 graf_infl_ann Graff2 graf_infl_ann_m grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat graftb1 graf_nonstat_trend Augmented Dickey-Fuller Test Equation J graftb2 graf_rate m graf_rates Dependent Variable: D(E_RUK) # r1emp500 graf_spur # r2emp500 graf_spurs Method: Least

Squares M resid graf_uk_cpi M rsq1 graf_us_indprod Date: 03/15/22 Time: 15:16 M rsq2 graf_uscpi spurt1000 M lukip spurt50 lukip_hp Sample (adjusted): 1980M02 2005M09 # t1 # t2 Iusip Iusip Included observations: 308 after adjustments M talpha1 M talpha2 # tb1emp025 M lusip_hp # tb1emp975 Iusip_st Variable Coefficient Std. Error t-Statistic Prob.

tb2emp025 M lusipf # tb2emp975 Mean_infl_1 M tbeta1 M mean_infl_2 M tbeta2 M resid E_RUK(-1) −0.094223 0.022870 −4.119901 0.0000 # thetax ruk10y # thetay ruk1y $\sum \sum$ y −0.015436 0.028853 −0.534986 0.5930 ruk2y Untitled / Untitled1 / Untitled2 / New Page RW / AR_stat / New Page Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 xR V 15:16 15/03/2022 e **k** P × Fz T

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4. If y_t and x_t are cointegrated (i.e. the null hypothesis of the test can be rejected) the two variables have an equilibrium-correction representation (Granger representation theorem, 1983), that can be illustrated using the model introduced at the beginning of the present section:

$y_t = \beta x_t + \varepsilon_t$ with $\varepsilon_t = \rho\varepsilon_{t-1} + \eta_t^y$ $x_t = x_{t-1} + \eta_t^x$ where we assume cointegration, i.e. $|\rho| < 1$

Writing ε_t as $\varepsilon_t = \frac{1}{1-\rho L}\eta_t^y$ the process y_t can be expressed as:

$y_t = \beta x_t + \frac{1}{1-\rho L}\eta_t^y$ yielding $(1-\rho L)y_t = \beta(1-\rho L)x_t + n_t^y$ Subtracting y_{t-1} to both sides of the equation and adding and subtracting βx_{t-1} on the right-hand side, we get the

equilibrium-correction representation:

$$\Delta v_t = \beta\Delta x_t - (1-\rho)v_{t-1} + \beta(1-\rho)x_{t-1} + \eta_t^y$$
$$\Rightarrow \Delta v_t = \beta\Delta x_t - (1-\rho)[v_{t-1} - \beta x_{t-1}] + \eta_t^y$$

ECM representation:

$$\Delta y_t = \beta\Delta x_t \underbrace{-(1-\rho)\left[y_{t-1} - \beta x_{t-1}\right]}_{+\eta_t^y} + \eta_t^y$$

equilibrium correction mechanism Since $-(1-\rho) < 0$ (with $|\rho| < 1$), any deviation from equilibrium in $t-1$ causes a "correction" of y in the following period t. This correction tends to bring y gradually back toward its equilibrium relation with x.

As the cointegrating relation captures the long-run equilibrium relation, the ECM term describes the dynamic adjustment ("short-run dynamics") towards equilibrium

In practice, having obtained the estimated residual series $\{e_t\}$ from the static cointegrating regression, the adjustment dynamics is estimated by OLS applied to the ECM equation:

$\Delta y_t = \delta_0 + \delta_1 e_{t-1} + \xi_t$ delivering consistent estimates of the short-run parameters

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D(RUK5Y) alpha1 alpha2 beta1 beta2 c ruk3m ruk5y ruk6m rus10y rus1y M britpus c ddluscpi Method: Least Squares M dexuk Date: 05/13/12 Time: 18:59 dlukrpa dluscpi dlusip druk10y distr_beta Sample (adjusted): 1980M02 2005M09 distr_beta_t distr_dw Included observations: 308 after adjustments distr_f rus6m druk3m distr_r2 Spruk3m1y M druk5y distr_t drus10y spruk6m1y wkgap wkinfl M dw1 M drus1y Coefficient Variable Std. Error t-Statistic Prob.

M dw2 M drus2y epsilonx M drus3m ukinfl_m M epsilony M ukinprodg drus5y drus6m M f1 M ukrpaxmif f1emp950 f2 −0.026517 0.025847 −1.025930 0.3057 uscp7500f M dukinfl usgap M e_lusip # f2emp950 e_ruk ecm_ruk M usinfl −0.083653 0.020498 −4.080963 0.0001 E_RUK(-1) graf_b_t usinprodg graf_r_f grafb1 grafb2 grafdw1 eq_lukrpa var1_infl_month_up D(RUS5Y) 0.388330 0.057530 6.750083 0.0000 eq_r5y graf_drate Var_coint2 var var_ruk3m6m1y graf_dspur grafdw2 wdi76axdf graf_infl −0.035390 0.162523 Mean dependent var R-squared graff1 graf_infl_ann Graff2 graf_infl_ann_m 0.493467 Adjusted R-squared 0.157031 S.D. dependent var grafrsq1 graf_infl_mon_var grafrsq2 graf_infl_nonstat 0.453069 S.E. of regression

Akaike info criterion 1.264146 Ju graftb1 graf_nonstat_trend J graftb2 graf_rate 62.60771 1.300478 Sum squared resid Schwarz criterion m graf_rates # r1emp500 graf_spur # r2emp500 −191.6785 1.278673 Log likelihood Hannan-Quinn criter.

graf_spurs M resid graf_uk_cpi M rsq1 graf_us_indprod F-statistic 29.59445 2.124103 Durbin-Watson stat rsq2 spurt1000 graf_uscpi Iukip Iukip Iukip_hp Iukrpa Iuscpi Iusip Iusip Iusip_dt Prob(F-statistic) 0.000000 spurt50 # t1 # t2 M talpha1 M talpha2 M lusip_hp # tb1emp025 # tb1emp975 Iusip_st # tb2emp025 M lusipf # tb2emp975 Mean_infl_1 M tbeta1 M mean_infl_2 M tbeta2 M resid # thetax M ruk10y thetay x y ruk1y Untitled / Untitled1 / Untitled2 / New Page RW / AR_stat / New Page E_RUK(-1) FileZilla Client Path = c:\bagliano\corsi\ects\bbs\ects2\english\eviews 2016 DB = none WF = usuk05 x9 15:26 15/03/2022 ∼ e r Fz S T w S D

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Example of cointegration and ECM behaviour Consider the familiar model (with $\beta = 1$ and $\{\eta_t^\gamma\}$ and $\{\eta_t^x\}$ independent white noise) $y_t = x_t + \varepsilon_t$ with $\varepsilon_t = \rho_i \varepsilon_{t-1} + \eta_t^y \; i = 1, 2$

$x_t = x_{t-1} + \eta_t^x$ rewritten in first differences:

$\Delta y_t = \Delta x_t - (1 - \rho_i)[y_{t-1} - x_{t-1}] + \eta_t^y \; \Delta x_t = \eta_t^x$ Assume $\rho_1 = 0.99$ and $\rho_2 = 0.01$ Both imply cointegration but with a very different magnitude of

the equilibrium-correction dynamics for y_t

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ALGORITHM – Problem

5. Inference on cointegration parameter **Problem.** EG procedure in general does not allow to use standard t-statistics to test hypotheses on β Model:

$y_t = \beta x_t + \varepsilon_t \; x_t = x_{t-1} + \eta_t^x$ Inference problems due to the potential presence of:

(i) autocorrelation in the process for ε_t;

(ii) correlation between ε_t and η_s^x, also when $t \neq s \Rightarrow x_t$ endogenous

Benchmark case:

no autocorrelation in ε_t and no correlation with η_s^x $E(\varepsilon_t \varepsilon_{t-s}) = 0$ and $E(\varepsilon_t \eta_s^x) = 0$ for all pairs $(t, s) \Rightarrow x_t$ is (strictly) exogenous with respect to the error term ε_t,

and OLS and associated inference procedures apply:

• $\hat{\beta}_{OLS}$ is superconsistent • $p \lim s^2 = \sigma_\varepsilon^2$ • standard inference on β is possible, with $N(0, 1)$ reference distribution. Under $H_0 : \beta = \beta_0$:

$$t_{\hat{\beta}_{OLS}} = \frac{\beta_{OLS} - \beta_0}{S_{\hat{\beta}_{OLS}}} \xrightarrow{J} N(0, 1)$$

General case:

autocorelation in ε_t and correlation with η_s^x $E(\varepsilon_t \varepsilon_{t-s}) \neq 0$ and $E(\varepsilon_t \eta_s^x) \neq 0$ for some pairs (t, s)

$\Rightarrow x_t$ is correlated with the error term of the cointegrating

equation and, though OLS yield a superconsistent estimate of β.

it is not possible to use t-statistics for inference In this case:

• $\hat{\beta}_{OLS}$ is superconsistent but the distribution of $T(\hat{\beta}_{OLS} - \beta)$ converges to a non-normal random variable;

also $t_{\hat{\beta}_{OIS}}$ converges to a non-standard normal random variable, making inference impossible

To proceed further:

dynamic ordinary least squares (DOLS) technique (Stock-Watson 1993)

Standard asymptotic distributions are obtained (allowing for the use of $t_{\hat{\beta}_{OIS}}$ to test hypotheses on β) adding lags and leads of Δx_t:

$$y_t = \beta x_t + \phi_0 \Delta x_t + \phi_{-1} \Delta x_{t-1} + \phi_{-2} \Delta x_{t-2} + \cdots + \phi_{-l} \Delta x_{t-l} + \phi_{-l} \Delta x_{t-l}$$

$+\phi_1 \Delta x_{t+1} + \phi_2 \Delta x_{t+2} + \ldots + \phi_l \Delta x_{t+l} + v_t$ where v_t is uncorrelated with all regressors.

Inclusion of leads and lags of Δx_t yields v_t uncorrelated with contemporaneous, past and future values of η_t^x.

OLS applied to this equation produce the DOLS estimate of the cointegrating relation and standard inference is allowed on β

Estimation and testing procedure (2) Equilibrium-correction (ECM) procedure Problems with EG procedure:

bias in estimating cointegration parameter may be substantial in small samples low power of cointegration test due to implicit strong restrictions on the joint dynamics of $\{y_t\}$ and $\{x_t\}$

$\Rightarrow$ alternative procedure based on general dynamic ADL model yielding:

cointegration test with increased power estimate of cointegration parameter on which inference can be carried out

Consider an $ADL(1,1)$ dynamic model:

$y_t = \theta y_{t-1} + \phi_0 x_t + \phi_1 x_{t-1} + \varepsilon_t \qquad 0 \leq \theta \leq 1$ $\{\varepsilon_t\}$ white noise (not autocorrelated) process $\{x_t\}$ generated by random walk with innovations $\{\eta_t^x\}$

independent of $\{\varepsilon_t\}$

$0 \leq \theta \leq 1$ $y_t = \theta y_{t-1} + \phi_0 x_t + \phi_1 x_{t-1} + v_t$ rewritten in equilibrium-correction form:

$\Delta y_t = \phi_0 \Delta x_t - (1-\theta) y_{t-1} + (\phi_0 + \phi_1) x_{t-1} + \varepsilon_t$ with no restrictions on the coefficients θ, ϕ_0, ϕ_1

Long-run solution of the model obtained as $\beta \equiv \frac{\phi_0 + \phi_1}{1-\theta}$ long-run multiplier that can be explicitly included in the equilibrium-correction model yielding $\Delta y_t = \phi_0 \Delta x_t - \gamma (y_{t-1} - \beta x_{t-1}) + \varepsilon_t$

with $\gamma = 1 - \theta$.

$\Delta y_t = \phi_0 \Delta x_t - \gamma(y_{t-1} - \beta x_{t-1}) + \varepsilon_t$ When v_t, Δy_t and Δx_t are $I(0)$, two cases are possible:

(i) $\gamma = 0$ and $(y_{t-1} - \beta x_{t-1})$ is a non-stationary, $I(1)$ linear combination of the two variables $\Rightarrow$ yt non-stationary and not cointegrated with xt;

$\Delta y_t = \phi_0 \Delta x_t - \gamma(y_{t-1} - \beta x_{t-1}) + \varepsilon_t$ When v_t, Δy_t and Δx_t are $I(0)$, two cases are possible:

(ii) $\gamma > 0$ and $(y_{t-1} - \beta x_{t-1})$ is a stationary, $I(0)$ variable $\Rightarrow$ yt reacts to past deviations from

long-run relationship with xt

Subtracting $\beta\Delta x_t$ from both sides:

$$\Delta y_t - \beta\Delta x_t = (\phi_0 - \beta)\Delta x_t - \gamma(y_{t-1} - \beta x_{t-1}) + v_t$$
$$\Rightarrow (y_t - \beta x_t) = (1-\gamma)(y_{t-1} - \beta x_{t-1}) + (\phi_0 - \beta)\Delta x_t + v_t$$

which is stationary since $1 - \gamma = \theta < 1$ under cointegration

ECM cointegration test Estimate by OLS $\Delta y_t = \delta + \kappa_0\Delta x_t + \kappa_1 y_{t-1} + \kappa_2 x_{t-1} + u_t$ with H_0: (no cointegration) $\kappa_1 = 0$ H_A: (cointegration) $\kappa_1 < 0$

t-ratio associated with $\hat{\kappa}_1$ $t_{\hat{\kappa}_1} = \frac{\hat{\kappa}_1}{S_{\hat{\kappa}_1}}$ is compared with the appropriate critical values

Relation with EG cointegration test Letting $e_t = y_t - \beta x_t$ and $\Delta e_t = \Delta y_t - \beta\Delta x_t$ we can rewrite the equilibrium-correction form of the dynamic model as $\Delta e_t = -\gamma e_{t-1} + (\phi_0 - \beta)\Delta x_t + \varepsilon_t$

Relation with EG cointegration test Letting $e_t = y_t - \beta x_t$ and $\Delta e_t = \Delta y_t - \beta\Delta x_t$ we can rewrite the equilibrium-correction form of the dynamic model as $\Delta e_t = -\gamma e_{t-1} + (\phi_0 - \beta)\Delta x_t + \varepsilon_t$

The EG cointegration test imposes $\phi_0 = \beta$ and tests the null hypothesis of no cointegration ($\gamma = 0$) by means of the t-statistic on the estimated coefficient of e_{t-1}

the EG cointegration test imposes a strong restriction on the dynamics of the relationship between x and y:

equality between $\begin{cases} \phi_0: \text{ the impact effect of } x_t \text{ on } y_t \\ \beta: \text{ the long-run effect of } x_t \text{ on } y_t \end{cases}$

which is absent from the formulation of the ECM cointegration test

Estimation and inference on the cointegration parameter From estimation of $\Delta y_t = \delta + \kappa_0\Delta x_t + \kappa_1 y_{t-1} + \kappa_2 x_{t-1} + u_t$ the cointegration parameter is estimated as $\hat{\beta} = -\frac{\hat{\kappa}_2}{\hat{\kappa}_1}$

which is preferable to the EG estimator (reduced small-sample bias and better asymptotic properties)

Moreover, given cointegration $t_{\hat{\beta}} = \frac{\hat{\beta} - \beta_0}{s_{\hat{\beta}}}$ can be used to test $H_0 = \beta = \beta_0$ since it is asymptotically

distributed as $N(0,1)$

Properties of the ECM cointegration procedure:

Monte Carlo simulation Data generating process for $\{y_t\}$ and $\{x_t\}$:

$y_t = 0.5y_{t-1} + 0.3x_t + 0.2x_{t-1} + \varepsilon_t$ $x_t = x_{t-1} + \eta_t^x$ where $\{\varepsilon_t\}$ and $\{\eta_t^x\}$ independent white noise with standard

normal distributions Long-run cointegration parameter $\beta = \frac{0.3+0.2}{1-0.5} = 1$ Sample sizes: $T = 100, T = 1000$

Two equations estimated by OLS in each replication:

EG static regression: $y_t = \alpha + \beta_{EG} x_t + v_t$ ECM dynamic regression: $\Delta y_t = \delta + \kappa_0\Delta x_t + \kappa_1 y_{t-1} + \kappa_2 x_{t-1} + u_t$ From EG regression:

$\hat{\beta}_{EG}$ with t-ratio $t_{\hat{\beta}_{EG}} = \frac{\beta_{EG} - 1}{S_{\hat{\beta}_{EG}}}$

From ECM regression:

$\hat{\beta}_{ECM} = -\frac{\hat{\kappa}_2}{\hat{\kappa}_1}$ with t-ratio $t_{\hat{\beta}_{ECM}} = \frac{\hat{\beta}_{ECM}-1}{S_{\hat{\beta}_{ECM}}}$

Robustness to dynamic misspecification Same data generating process; two ECM dynamic equations estimated:

ECM correctly specified: $\Delta y_t = \delta + \kappa_0 \Delta x_t + \kappa_1 y_{t-1} + \kappa_2 x_{t-1} + u_t$ ECM over-parameterized: Δy_t $= \delta + \kappa_0 \Delta x_t + \kappa_1 y_{t-1} + \kappa_2 x_{t-1} + \kappa_2 x_{t-1}$

$+\kappa_3 \Delta y_{t-1} + \kappa_4 \Delta x_{t-1} + u_t$ From each ECM specification:

$\hat{\beta}_{ECM} = -\frac{\hat{\kappa}_2}{\hat{\kappa}_1}$ with t-ratio $t_{\hat{\beta}_{ECM}} = \frac{\hat{\beta}_{ECM}-1}{S_{\hat{\beta}_{ECM}}}$

1.7 Set 7 – Vector Autoregressive Models

Vector Autoregressive (VAR) models Univariate autoregressive (AR) models can be extended to the multivariate case to study dynamic interrelationships among several variables, all viewed as endogenous.

The resulting vector autoregressive (VAR) models describe the evolution over time of a vector of n variables $\mathbf{y}_t = (y_{1t}\ y_{2t} \dots y_{nt})'$ as a function of its past realizations $\mathbf{y}_{t-1}, \mathbf{y}_{t-2}, \dots$ and a vector of stochastic terms $\mathbf{u}_t = (u_{1t}\ u_{2t} \dots u_{nt})'$. A VAR model

with p lags of the endogenous variables in $\mathbf{y}_t$ is called $VAR(p)$ and has the following general form:

$$\mathbf{y}_t = \delta + \mathbf{\Theta}_1 \mathbf{y}_{t-1} + \mathbf{\Theta}_2 \mathbf{y}_{t-2} + ... + \mathbf{\Theta}_p \mathbf{y}_{t-p} + \mathbf{u}_t$$

where $\boldsymbol{\delta} = (\delta_1\ \delta_2 \dots \delta_n)'$ is a vector of constant terms, $\mathbf{\Theta}_1, \mathbf{\Theta}_2 \dots \mathbf{\Theta}_p$ are $n \times n$ matrices

and $\mathbf{u}_t$ is a vector of white noise processes with the following variance-covariance structure:

$$E(\mathbf{u}_t \mathbf{u}_s') = \begin{cases} \mathbf{\Sigma} & \text{if } t = s \\ \mathbf{0} & \text{if } t \neq s \end{cases}$$

Matrix Σ is not assumed to be diagonal: therefore the error terms of the individual equations can be (contemporaneously) correlated.

To understand the nature of the contemporaneous correlation among the el- ements of $\mathbf{u}_t$ it is useful to view the VAR model as the "reduced form" of a "structural" model capturing behavioral relationships among the endogenous vari- ables. As an example, in the case of two variables with dynamics limited to one lag only, the structural form of the model is given by $y_{1t} = \gamma_{10} + a_{12} y_{2t}$ $+ \gamma_{11} y_{1t-1} + \gamma_{12} y_{2t-1} + v_{1t}$

$y_{2t} = \gamma_{20} + a_{21} y_{1t} + \gamma_{21} y_{1t-1} + \gamma_{22} y_{2t-1} + v_{2t}$ with $E(\mathbf{v}_t \mathbf{v}_s') = \begin{cases} \mathbf{D} \text{ if } t = s \text{ with } \mathbf{D} \text{ diagonal} \\ \mathbf{0} \text{ if } t \neq s \end{cases}$

The elements of $\mathbf{v}_t$ are the structural shocks (or disturbances) of the system. The fact that they are uncorrelated (since D is diagonal, implying $E(v_{1t} v_{2t}) = 0$), allows for a precise economic interpretation of the structural disturbances in terms, for example, of demand shocks, supply shocks, innovations to monetary policy, etc.

Indeed, one of the main purposes of VAR modelling is the estimation of the dynamic response of all endogenous variables to structural disturbances hitting the system.

In matrix notation, the structural form of the system can be expressed as:

$$\underbrace{\begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \underbrace{\begin{pmatrix} \gamma_{10} \\ \gamma_{20} \end{pmatrix}}_{\mathbf{A}} + \underbrace{\begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \end{pmatrix} + \underbrace{\begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix}}_{\mathbf{A}}$$

$$\mathbf{A}\,\mathbf{y}_t \;=\; \boldsymbol{\gamma} + \Gamma\,\mathbf{y}_{t-1} + \mathbf{v}_t$$

Matrix A captures the contemporaneous relations between the two endogenous variables. Inverting A we get the reduced form of the model as a $VAR(1)$:

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix}^{-1} \begin{pmatrix} \gamma_{10} \\ \gamma_{20} \end{pmatrix} + \begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix}^{-1} \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix} \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \end{pmatrix}$$

$$+ \begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix}^{-1} \begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix}$$

$$\mathbf{v}_t = \mathbf{A}^{-1}\boldsymbol{\gamma} + \mathbf{A}^{-1}\Gamma\mathbf{v}_{t-1} + \mathbf{A}^{-1}\mathbf{v}_t$$

$\Rightarrow \mathbf{v}_t = \boldsymbol{\delta} + \boldsymbol{\Theta}\mathbf{v}_{t-1} + \mathbf{u}_t$ The VAR error terms (elements of $\mathbf{u}_t$), interpreted as one-step forecast errors or "innovations", are linear combinations of the structural disturbances v_{1t} and v_{2t}:

$$\mathbf{u}_t = \mathbf{A}^{-1}\mathbf{v}_t = \frac{1}{1 - a_{12}a_{21}} \begin{pmatrix} 1 & a_{12} \\ a_{21} & 1 \end{pmatrix} \begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix} = \frac{1}{1 - a_{12}a_{21}} \begin{pmatrix} v_{1t} + a_{12}v_{2t} \\ a_{21}v_{1t} + v_{2t} \end{pmatrix}$$

with, in general, non-zero covariance. In fact, the variance-covariance matrix of the error terms Σ is given by $\boldsymbol{\Sigma} = \frac{1}{(1-a_{12}a_{21})^2} \begin{pmatrix} \sigma^2_{v_1} + a^2_{12}\sigma^2_{v_2} & a_{21}\sigma^2_{v_1} + a_{12}\sigma^2_{v_2} \\ a_{21}\sigma^2_{v_1} + a_{12}\sigma^2_{v_2} & a^2_{21}\sigma^2_{v_1} + \sigma^2_{v_2} \end{pmatrix}$

3.1 Stationarity Using the VAR representation, it is possible to study the dynamic response of the variables in $\mathbf{y}_t$ to each element of vector $\mathbf{u}_t$.

If the VAR is stationary (that is, it has finite and time-invariant first and second moments), the response to innovations gradually dies down, tending to zero in the long-run. It is therefore important to establish under what conditions the VAR system is stationary. Such conditions are a multivariate extension of those valid for the unvariate AR process.

In the $VAR(p)$ case $\mathbf{y}_t = \boldsymbol{\delta} + \boldsymbol{\Theta}_1\,\mathbf{y}_{t-1} + \boldsymbol{\Theta}_2\,\mathbf{y}_{t-2} + ... + \boldsymbol{\Theta}_p\,\mathbf{y}_{t-p} + \mathbf{u}_t$

$$(\mathbf{I} - \boldsymbol{\Theta}_1 L - \boldsymbol{\Theta}_2 L^2 - \cdots - \boldsymbol{\Theta}_n L^p)\mathbf{v}_t = \boldsymbol{\delta} + \mathbf{u}_t$$

$\Theta(L)\mathbf{v}_t = \boldsymbol{\delta} + \mathbf{u}_t$ for $\mathbf{y}_t$ to be stationary, the matrix polynomial in the lag operator $\Theta(L)$ must be invertible. Invertibility requires that the roots of the following equation $\det\,(\mathbf{I} - \boldsymbol{\Theta}_1 z - \boldsymbol{\Theta}_2 z^2 - \ldots - \boldsymbol{\Theta}_n z^p) = 0$

be (in modulus) outside the unit circle.

EXAMPLE – Illustration

Example. for a $VAR(1)$ bivariate system $\mathbf{y}_t = \boldsymbol{\delta} + \boldsymbol{\Theta}_1 \mathbf{y}_{t-1} + \mathbf{u}_t \Rightarrow (\mathbf{I} - \boldsymbol{\Theta}_1 L)\mathbf{y}_t = \boldsymbol{\delta} + \mathbf{u}_t$

$\begin{pmatrix} 1-\theta_{11}L & -\theta_{12}L \\ -\theta_{21}L & 1-\theta_{22}L \end{pmatrix} \mathbf{y}_t = \boldsymbol{\delta} + \mathbf{u}_t$ the stationarity condition is that the roots (z_1, z_2) of the equation

$\det(\mathbf{I} - \boldsymbol{\Theta}_1 z) = 0 \Rightarrow 1 - (\theta_{11} + \theta_{22})z + (\theta_{11}\theta_{22} - \theta_{21}\theta_{12})z^2 = 0$

be outside the unit circle $(|z_1|, |z_2| > 1)$.

Under stationarity, the VAR can be expressed as a vector moving average of infinite order, $VMA(\infty)$:

$$\begin{aligned} \mathbf{y}_t &= \boldsymbol{\Theta}(L)^{-1}(\boldsymbol{\delta} + \mathbf{u}_t) \\ &= \Theta(1)^{-1}\delta + \Theta(L)^{-1}\mathbf{u}_t \end{aligned}$$

from which we get the expected value (unconditional mean) of $\mathbf{y}_t$:

$$E(\mathbf{v}_t) = \boldsymbol{\Theta}(1)^{-1}\boldsymbol{\delta} = (\mathbf{I} - \boldsymbol{\Theta}_1 - \boldsymbol{\Theta}_2 - \cdots - \boldsymbol{\Theta}_n)^{-1}\boldsymbol{\delta}$$

3.2 Estimation

The VAR can be estimated by means of OLS applied equation by equation. This procedure yields consistent estimates of the parameters in δ, Θ_1, Θ_2, ... Θ_p, and of the variances and covariances of the innovations in matrix Σ.

The "identification" problem 3.3 If we are interested in analyzing the dynamic response of the endogenous variables in $\mathbf{y}_t$ to each element of the innovation vector $\mathbf{u}_t$, the $VMA(\infty)$ form of the VAR model can be used:

$$\begin{aligned} \mathbf{y}_t &= \boldsymbol{\Theta}(1)^{-1}\boldsymbol{\delta} + \boldsymbol{\Theta}(L)^{-1}\mathbf{u}_t \\ &= \Theta(1)^{-1}\boldsymbol{\delta} + (\mathbf{I} + \boldsymbol{\Psi}_1 L + \boldsymbol{\Psi}_2 L^2 + ...)\mathbf{u}_t \end{aligned}$$

The $n \times n$ matrices Ψ_s (for $s = 0, ...\infty$ and with $\Psi_0 = \mathbf{I}$) describe the response over time of each endogenous variable y_i $(i = 1, ...n)$ to each shock u_j $(j = 1, ...n)$.

However, the economic interpretability of such responses is often difficult since the VAR innovations (elements of $\mathbf{u}_t$) are linear combinations of the system's struc- tural disturbances and display correlation. Indeed, what we would like to get from

the dynamic analysis of the VAR model is the response of $\mathbf{y}_t$ to the structural dis- turbances, to which it is generally possible to attach an economic interpretation.

Extending the above example (with $n = 2$ and $p = 1$) to $n > 2$ endogenous variables and to $p > 1$ lags, we can write the relationship between the structural and reduced forms of the system (and the associated disturbances) as:

$$\mathbf{A}\mathbf{y}_t = \gamma + \boldsymbol{\Gamma}_1 \mathbf{y}_{t-1} + \boldsymbol{\Gamma}_2 \mathbf{y}_{t-2} + ... + \boldsymbol{\Gamma}_p \mathbf{y}_{t-p} + \mathbf{v}_t$$

$$\Rightarrow \mathbf{y}_t = \mathbf{A}^{-1}\boldsymbol{\gamma} + \mathbf{A}^{-1}\boldsymbol{\Gamma}_1 \mathbf{y}_{t-1} + \mathbf{A}^{-1}\boldsymbol{\Gamma}_2 \mathbf{y}_{t-2} + \cdots + \mathbf{A}^{-1}\boldsymbol{\Gamma}_p \mathbf{y}_{t-n} + \mathbf{A}^{-1}\mathbf{v}_t$$

$= \delta + \Theta_1 \mathbf{y}_{t-1} + \Theta_2 \mathbf{y}_{t-2} + ... + \Theta_n \mathbf{y}_{t-n} + \mathbf{u}_t$ from which we get the relationship between the VAR residuals ($\mathbf{u}_t$) and the struc-

tural disturbances ($\mathbf{v}_t$):

$\mathbf{u}_t = \mathbf{A}^{-1}\mathbf{v}_t \Rightarrow \mathbf{A}\mathbf{u}_t = \mathbf{v}_t$ To obtain the structural disturbances $\mathbf{v}_t$ from estimation of the VAR's innovations

$\mathbf{u}_t$, it is necessary to identify the elements of matrix $\mathbf{A}$ (containing the contem- poraneous relationships among the endogenous variables). Only after solving this identification problem it is possible to construct (estimates of) the structural disturbances $\mathbf{v}_t$ (from $\mathbf{A}\mathbf{u}_t = \mathbf{v}_t$) and to proceed to the analysis of the dynamic response

of $\mathbf{y}_t$ to each shock in $\mathbf{v}_t$.

To obtain the elements of A we can use the relation between the variance-covariance matrices of $\mathbf{u}_t$ ($\mathbf{\Sigma}$) and $\mathbf{v}_t$ ($\mathbf{D}$), the latter being a diagonal matrix:

$$E(\mathbf{A}\mathbf{u}_t(\mathbf{A}\mathbf{u}_t)') = \mathbf{A}\underbrace{E(\mathbf{u}_t\mathbf{u}_t')}_{\Sigma}\mathbf{A}' = \underbrace{E(\mathbf{v}_t\mathbf{v}_t')}_{\mathbf{D}}$$

$\Rightarrow$ A Σ A' = D This is a system of n^2 non-linear equations in the unknown parameters in A and D. The parameters to be identified are the $n(n-1)$ off-diagonal elements of A (by construction the elements on the main diagonal being set equal to 1), and the n elements on the diagonal of D (the variances of the structural shocks): overall, n^2 parameters. Estimation of the VAR yields estimates of n variances and $\frac{n(n-1)}{2}$

covariances among the innovations: overall, $\frac{n(n+1)}{2}$ estimates, corresponding to the distinct elements of the symmetric matrix Σ. Since the number of available es- timates is less than the number of parameters to be identified, it is necessary to impose a set of $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$ restrictions on the elements of A. Such restrictions tions, concerning the simultaneous relationships among the endogenous variables,

can be suggested by economic theory.

3.3.1 A "recursive" identification strategy As originally proposed by C. Sims ("Macroeconomics and reality", Econometrica, 1980), a possible procedure to achieve identification of the contemporaneous para- meters imposes a recursive structure on matrix A by means of "zero" restrictions on the elements above its main diagonal.

The number of such restrictions is $\frac{n(n-1)}{2}$, sufficient to exactly identify the (remaining) elements of A and D. Imposing this set of restrictions, matrix A takes the following form:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ -a_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & 1 \end{pmatrix}.$$

From the economic point of view, adopting the above recursive identification scheme amounts to imposing a specific "causal chain" pattern of relationships among the n variables: from the "most exogenous" variable (corresponding to the first element of vector $\mathbf{y}_t$) to the "most endogenous"

one (corresponding to the last element of $\mathbf{y}_t$). Such causal chain must be justified on economic theory grounds.

Example (for $n = 2$). In the bivariate case, the relationship between the VAR residuals and the structural disturbances ($\mathbf{A}\mathbf{u}_t = \mathbf{v}_t$) is:

$$\begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix} \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix} = \begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix}$$

There are 4 parameters to be identified: two off-diagonal elements of A ($-a_{12}$ and $-a_{21}$), and the two variances of the structural shocks $\sigma^2_{v_1}$ and $\sigma^2_{v_2}$. From VAR estimation we get estimates s_{ij} (with $i, j = 1, 2$) of the elements of Σ (with $s_{12} =$

s_{21}). The relationship between Σ and $\mathbf{D}(\mathbf{A}\Sigma\mathbf{A}' = \mathbf{D})$ is then:

$$\begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix} \begin{pmatrix} s_{11} & s_{12} \\ s_{12} & s_{22} \end{pmatrix} \begin{pmatrix} 1 & -a_{21} \\ -a_{12} & 1 \end{pmatrix} = \begin{pmatrix} \sigma^2_{v_1} & 0 \\ 0 & \sigma^2_{v_2} \end{pmatrix}$$

from which we get the following system of 3 distinct equations:

$$\begin{aligned} s_{11} - 2a_{12}s_{12} + a^2_{12}s_{22} &= \sigma^2_{v_1} \\ a^2_{21}s_{11} - 2a_{21}s_{12} + s_{22} &= \sigma^2_{v_2} \end{aligned}$$

$-a_{21}s_{11} + (1 + a_{21}a_{12})s_{12} - a_{12}s_{22} = 0$ It is necessary to impose one restriction to achieve identification. For example, if the chosen restriction is $a_{12} = 0$, the system becomes:

$$\begin{aligned} s_{11} &= \sigma^2_{v_1} \\ a^2_{21}s_{11} - 2a_{21}s_{12} + s_{22} &= \sigma^2_{v_2} \\ -a_{21}s_{11} + s_{12} &= 0 \end{aligned}$$

from which we get the 3 remaining parameters:

$$\begin{aligned} \sigma^2_{v_1} &= s_{11} \\ a_{21} &= \frac{s_{12}}{s_{11}} \\ \sigma^2_{v_2} &= s_{22} - \frac{s^2_{12}}{s_{11}} \end{aligned}$$

This is the simplest example of "triangular" (or "Choleski") identification, which imposes a recursive structure to the system of contemporaneous relations among the variables. In this case the structural shock v_{1t} is immediately identified as the first element of the vector of VAR residuals (u_{1t}), and the shock v_{2t} is identified as that portion of u_{2t} not correlated with (i.e. "orthogonal" to) u_{1t}:

$u_{1t} = v_{1t}$ $u_{2t} = a_{21}u_{1t} + v_{2t}$ The chosen restriction for identification ($a_{12} = 0$) corresponds to a specific "or- dering" of the variables in the system (with y_1 coming "before" y_2): y_1 affects y_2 within period t whereas y_2 does affect y_1 only in subsequent periods (through its

lags $y_{2t-1,\ldots}$ in the VAR equation for y_{1t}). The alternative restriction $a_{21} = 0$ leads to a different identification, based on the opposite ordering of the variables, with y_2 before y_1; in this case we have $u_{2t} = v_{2t}$ $u_{1t} = a_{12}u_{2t} + v_{1t}$

In general, simple recursive identification schemes of this kind make use of the Choleski decomposition of a symmetric, positive definite matrix such as Σ.4 General point: within a recursive identification approach, all possible orderings of the n variables in $\mathbf{y}_t$ yield the necessary number of identification restrictions leading to "just-(or exact-)identification" of the system. All orderings describe the data equally well and it is not possible to apply statistical tests to determine the most appropriate recursive identification scheme. Once a specific ordering of the variables has been chosen, the economic plausibility of the following dynamic analysis is an (indirect) way to assess its validity for the data set at hand.

4 Choleski decomposition: any symmetric positive definite matrix Σ can be decomposed as $\Sigma = TT'$ with $\mathbf{T}$ a lower-triangular matrix (with all zeros above the main diagonal).

3.4 Dynamic analysis

After achieving identification of the parameters in A and D, the (orthogonal) struc- tural disturbances can be constructed from the relation $\mathbf{A}\mathbf{u}_t = \mathbf{v}_t$. It is then possi- ble to proceed to the analysis of the dynamic response of $\mathbf{y}_{t+i}$ to $\mathbf{v}_t$. This analysis, called innovation accounting, is carried out by constructing the (orthogonal-

ized) impulse response functions (IRF) and forecast errors variance decompositions ($FEVD$), both derived from the $VMA(\infty)$ representation of the VAR system:

$$\begin{aligned}\mathbf{y}_t &= \boldsymbol{\Theta}(1)^{-1}\,\boldsymbol{\delta} + (\mathbf{I} + \boldsymbol{\Psi}_1 L + \boldsymbol{\Psi}_2 L^2 + \dots)\,\mathbf{u}_t \\ &= \boldsymbol{\Theta}(1)^{-1}\,\boldsymbol{\delta} + (\underbrace{\mathbf{A}^{-1}}_{\boldsymbol{\Phi}_0} + \underbrace{\boldsymbol{\Psi}_1\mathbf{A}^{-1}}_{\boldsymbol{\Phi}_1} L + \underbrace{\boldsymbol{\Psi}_2\mathbf{A}^{-1}}_{\boldsymbol{\Phi}_2} L^2 + \dots)\,\mathbf{v}_t\end{aligned}$$

$= \Theta(1)^{-1}\boldsymbol{\delta} + \Phi_0\mathbf{v}_t + \Phi_1\mathbf{v}_{t-1} + \Phi_2\mathbf{v}_{t-2} + \dots$ IRF: the elements of the matrices Φ_s trace out the effects over time (impulse re-

sponse functions) of each structural disturbance in $\mathbf{v}$ keeping all the other disturbances at zero, under the set of identifying assumptions used in the preceding step (note that $\Phi_0 = \mathbf{A}^{-1}$). Each matrix Φ_s captures the effect of the structural shocks at time t on the endogenous variables at time $t + s$; the typical element $\phi_{s,ij} \equiv \frac{\partial y_{i,t}}{\partial v_{i,t-s}} = \frac{\partial y_{i,t+s}}{\partial v_{i,t}}$

captures the response of the ith element of $\mathbf{y}_{t+s}$ to an "impulse" due to the jth element of $\mathbf{v}_t$.

FEVD: from the $VMA(\infty)$ representation of the VAR it is possible to obtain the fore- cast of future y's over an h-period horizon ($\hat{\mathbf{y}}_{t+h}$) on the basis of information in current (time t) and past values of the variables in the system $\mathbf{y}_t, \mathbf{y}_{t-1}, \dots$ The associated forecast error is:

$$\mathbf{y}_{t+h} - E(\mathbf{y}_{t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, ...) = \boldsymbol{\Phi}_0\mathbf{v}_{t+h} + \boldsymbol{\Phi}_1\mathbf{v}_{t+h-1} + ... + \boldsymbol{\Phi}_{h-1}\mathbf{v}_{t+1}$$

The forecast error variance is given by the following symmetric matrix $var(\mathbf{y}_{t+h} - E(\mathbf{y}_{t+h}|\mathbf{y}_t,$ $\mathbf{y}_{t-1}, ...)) = \boldsymbol{\Phi}_0\mathbf{D}\boldsymbol{\Phi}_0' + \boldsymbol{\Phi}_1\mathbf{D}\boldsymbol{\Phi}_1' + ... + \boldsymbol{\Phi}_{h-1}\mathbf{D}\boldsymbol{\Phi}_{h-1}'$

The elements on the main diagonal of this matrix capture the forecast er- ror variances of each variable in y: $var(y_{i,t+h} - E(y_{i,t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, \dots))$. This variance can be expressed as the sum of the contributions of each structural disturbance in $\mathbf{v}$ to the total variance of each endogenous variable over the relevant horizon (forecast error variance decomposition). For example, the

fraction of the forecast error variance of the i-th endogenous variable at hori- zon h attributable to the j-th structural disturbance is given by $\frac{\sigma_{v_j}^2 \sum_{s=0}^{h-1} \phi_{s,ij}^2}{\sigma_{v_1}^2 \sum_{s=0}^{h-1} \phi_{s,i1}^2 + \sigma_{v_2}^2 \sum_{s=0}^{h-1} \phi_{s,i2}^2 + ... + \sigma_{v_n}^2 \sum_{s=0}^{h-1} \phi_{s,in}^2}$

Again (as for the $IRFs$), this step is economically meaningful only if there are no covariance terms, which is warranted by the chosen identification procedure (D is diagonal). (A fully worked-out bivariate case of innovation accounting is presented in Appendix A in section 3.6).

3.4.1 A small-scale application: the effects of monetary policy shocks

We present here a small-scale (three-variable) application of VAR modelling to evaluate the effects of monetary policy actions on real activity, closely following J. Stock and M. Watson ("Vector Autoregressions", Journal of Economic Perspec- tives, 2001, available on the Moodle page of the course, together with the EViews data file). The empirical analysis of monetary policy by means of VAR models requires a careful handling of the identification problem due to the main feature of monetary policy conduct: in setting their policy instruments, monetary authorities react to current and foreseen developments in the economy. In order to capture the dynamic effects of a monetary policy action on the economy (the "monetary policy transmission mechanism"), the endogenous response of monetary policy to

macroeconomic and financial variables (the "systematic" policy component) must be separated from the changes in the policy instruments due to "deviations" from such systematic policy (the monetary policy "shock"). Only when the latter is identified, the VAR analysis can yield reliable information on the transmission of policy impulses to the economy. Possible rationalizations of such shocks include, for example, changes in the "preferences" of the monetary authorities (e.g. as to the relative weight of inflation control and output stabilization goals in the central bank's objective function).

Using quarterly data for the US over the $1960(1) - 2000(4)$ period (as in the Stock- Watson paper) we study a small-scale macroeconomic system including the inflation rate (π), the unemployment rate (U) and the Federal Funds rate (R), the very short- term policy rate directly influenced by the monetary policy decisions of the Federal Reserve (Fed). The structural form of the system is then:

$$\mathbf{A}\begin{pmatrix}\pi_t\\U_t\\R_t\end{pmatrix} = \mathbf{\Gamma}(L)\begin{pmatrix}\pi_{t-1}\\U_{t-1}\\R_{t-1}\end{pmatrix} + \begin{pmatrix}v_t^1\\v_t^2\\v_t^P\end{pmatrix}$$

where A is a 3×3 matrix capturing the contemporaneous (i.e. within quarter) re- lations among the variables, and $\Gamma(L)$ is a 3×3 matrix of finite-order polynomials in the lag operator. The vector $\mathbf{v}_t$ collects three structural disturbances. Since the focus of our exercise is the analysis of the monetary policy transmission mech- anism, we are mostly interested in identifying a "monetary policy shock", named v_t^P, and leave the other two disturbances, named generically v_t^1 and v_t^2, without

a detailed economic (structural) interpretation. The reduced form (VAR) of the

system (obtained by pre-multiplying by A^{-1}) is the following:

$$\begin{pmatrix}\pi_t\\U_t\\R_t\end{pmatrix} = \mathbf{A}^{-1}\mathbf{\Gamma}(L)\begin{pmatrix}\pi_{t-1}\\U_{t-1}\\R_{t-1}\end{pmatrix} + \mathbf{A}^{-1}\begin{pmatrix}v_t^1\\v_t^2\\v_t^P\end{pmatrix}$$

$$= \Theta(L)\begin{pmatrix}\pi_{t-1}\\U_{t-1}\\R_{t-1}\end{pmatrix} + \begin{pmatrix}u_t^\pi\\u_t^U\\u_t^R\end{pmatrix}$$

where u_t^{π}, u_t^{U}, and u_t^{R} are the reduced form innovations corresponding to the infla- tion, unemployment and interest rate equations, respectively, being linear combina- tions of the structural disturbances in $\mathbf{v}_t$. Estimation of the VAR system (performed by OLS applied to each equation with four lags) yields consistent estimates of the parameters in $\Theta(L)$ and of the variance-covariance innovation matrix Σ. From the

elements of Σ we can obtain the correlation coefficients among the estimated VAR innovations $\widehat{u}_t^{\pi}$, $\widehat{u}_t^{U}$ and $\widehat{u}_t^{R}$:

A sizable negative correlation exists between $\hat{u}_t^{U}$ and $\hat{u}_t^{R}$ and a smaller but not negligible correlation is also detected between $\hat{u}_t^{\pi}$ and $\hat{u}_t^{R}$, whereas $\hat{u}_t^{\pi}$ and $\hat{u}_t^{U}$ do

not appear to be correlated.

A recursive identification scheme is adopted, ordering the variables as shown above.

The innovation in the inflation rate u_t^{π} is not affected contemporaneously (i.e.

whithin the same quarter) by any other innovation in the system (since π_t is or- dered first in $\mathbf{y}_t$; the unemployment innovation u_t^{U} is contemporaneously affected by u_t^{π} but not by u_t^{R} (U_t being ordered second) and the interest rate innovation u_t^{R} is affected contemporaneously by both u_t^{π} and u_t^{U} (R_t being ordered last). This

set of identification restrictions imposes a lower-triangular structure on A and the relation between the structural disturbances can be represented as:

$$\begin{pmatrix} 1 & 0 & 0 \\ a_{21} & 1 & 0 \\ a_{21} & a_{22} & 1 \end{pmatrix} \begin{pmatrix} u_t^{\pi} \\ u_t^{U} \\ u_t^{R} \end{pmatrix} = \begin{pmatrix} v_t^{1} \\ v_t^{2} \\ v_t^{P} \end{pmatrix}$$

$$\begin{aligned} & u_t^{\pi} = v_t^{1} \\ \Rightarrow \quad & u_t^{U} = -a_{21} u_t^{\pi} + v_t^{2} \\ & u_t^{R} = -a_{21} u_t^{\pi} - a_{32} u_t^{U} + v_t^{P} \end{aligned}$$

(*) where, from the third equation, the structural "monetary policy shock" v_t^{P} is inter- preted as an unanticipated movement of the interest rate (u_t^{R}) due to a monetary

policy action that is not a response of the Fed to (unanticipated) movements in π_t and U_t (those responses being captured by the coefficients $-a_{31}$ and $-a_{32}$ on u_t^{π} and u_t^{U}, respectively).

Inverting A yields the VAR innovations as linear combinations of the structural disturbances:

$$\begin{aligned} u_t^{\pi} &= v_t^{1} \\ u_t^{U} &= -a_{21} v_t^{1} + v_t^{2} \\ u_t^{R} &= -(a_{31} - a_{32} a_{21}) v_t^{1} - a_{32} v_t^{2} + v_t^{P} \end{aligned}$$

Estimating each equation in $(*)$ by OLS we get the estimates of the elements of A (standard errors in parentheses):

$$\begin{aligned} u_t^{\pi} &= v_t^{1} \\ u_t^{U} &= -0.013 u_t^{\pi} + v_t^{2} \qquad\qquad R^2 = 0.207 \\ u_t^{R} &= 0.093 u_t^{\pi} - 1.619 u_t^{U} + v_t^{P} \end{aligned}$$

and we also obtain estimates of the standard errors of the structural disturbances:

$\sigma_{v^1} = 0.956$ $\sigma_{v^2} = 0.228$ $\sigma_{v^P} = 0.758$ (**) The interest rate shows a negative reaction to the unemployment rate (i.e. when U_t is unexpectedly high, R_t is reduced by the Fed), and a positive (but small and not strongly statistically significant) reaction to the inflation rate (i.e. when π_t is

unexpectedly high, R_t is raised by the Fed). Any deviation from those responses, that define the "systematic" component of monetary policy conduct, are captured by the shock v_t^P, whose effect on π_t and U_t is investigated by analysing the impulse response functions (IRF) and the forecast error variance decompositions ($FEVD$).

Relying on the above recursive identification scheme, it is possible to implement the innovation accounting analysis, by estimating IRF of all endogenous variables to the structural disturbances and the $FEVD$ of the series at various horizons.

Results are presented in the "slides" pdf file by means of plots of:

• IRF of π, U and R to "Choleski one standard deviation innovations" esti- mated as in (**) with a 95% confidence interval (± 2 standard deviations of the estimated IRF). The shocks are named $INFL$ (corresponding to v^1), $UNEMP(v^2)$ and $R(v^P)$.

• FEVD of π, U and R at various horizons (up to 120 quarters) into the fractions attributable to the structural disturbances v^1, v^2 and v^P.

3.5 An alternative identification strategy based on long-run restrictions

A different way of identifying structural disturbances, in order to proceed to the impulse response and forecast error variance decomposition analysis, has been pro- posed by O. Blanchard and D. Quah ("The dynamic effects of aggregate demand and supply disturbances" American Economic Review, 1989; BQ) and relies on re- strictions on the long-run responses of endogenous variables to structural shocks rather than on their contemporaneous reactions.

This methodology has been implemented on small sets of macroeconomic variables to separate the effects of permanent from temporary disturbances. Here, we cast the argument in terms of a bivariate model including one non-stationary ($I(1)$) variable y and one stationary ($I(0)$) variable z. Since, in order to apply the BQ identification procedure, only stationary variables must enter the VAR, we specify the vector of endogenous variables as $(\Delta y\ z)'$. Note that if the two variables of interest are both non-stationary, the BQ methodology can be applied to the vector of first differences of both variables provided that they are not cointegrated (this

point will be clarified later, when we will deal with cointegrated VAR systems).

Let us start from the reduced (VAR) form of the bivariate system:

$$\begin{pmatrix} \Delta y_t \\ z_t \end{pmatrix} = \Theta(L) \begin{pmatrix} \Delta y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} u_t^Y \\ u_t^Z \end{pmatrix}$$

(4) where the innovations are defined as u^Y and u^Z. The relation between the VAR inovations and the structural disturbances (named v_t^1 and v_t^2) is now specified as $\begin{pmatrix} u_t^Y \\ u_t^Z \\ u_t^Z \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} v_t^1 \\ v_t^2 \end{pmatrix}$

(5) where the two structural disturbances are given the economic interpretation of a shock having a permanent effect on the non-stationary variable y (the "permanent" disturbance) and a shock having only a transitory effect on y (the "transitory" disturbance). Both shocks have only temporary effects on the stationary variable z.

The two structural disturbances are orthogonal and, for simplicity, their variance is normalized to

unity, so that $E(\mathbf{v}_t\mathbf{v}_t') = \mathbf{I}$. In (5) the VAR innovations for Δy and z are linear combinations of permanent and transitory disturbances. In order to retrieve the structural disturbances v^1 and v^2 from estimation of the VAR innovations u^Y and u^Z (and of their covariance matrix), identification of the four

elements b_{ij} $(i, j = 1, 2)$ is necessary. The BQ methodology achieves identification by imposing a long-run restriction on the impulse response function of Δy with respect to the transitory shock.

From (5) we can write the covariance matrix of VAR innovations in terms of the

coefficients b_{ii}:

$$E(\mathbf{u}_t\,\mathbf{u}_t') = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} E(\mathbf{v}_t\,\mathbf{v}_t') \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}'$$

$$\Rightarrow \begin{pmatrix} s_Y^2 & s_{YZ} \\ s_{YZ} & s_Z^2 \end{pmatrix} = \begin{pmatrix} b_{11}^2 + b_{12}^2 & b_{11}b_{21} + b_{12}b_{22} \\ b_{11}b_{21} + b_{12}b_{22} & b_{21}^2 + b_{22}^2 \end{pmatrix}$$

since $E(\mathbf{v}_t\mathbf{v}_t') = \mathbf{I}$. Given the estimated values for the variances and covariance of VAR innovations (denoted by s_Y^2, s_Z^2 and s_{YZ}), the above relation delivers the following three equations to obtain the b_{ij} coefficients:

$b_{11}^2 + b_{12}^2 = s_V^2$ (6) $b_{21}^2 + b_{22}^2 = s_Z^2$ (7) $b_{11}b_{21} + b_{12}b_{22} = s_{VZ}$

(8) One additional restriction is needed for (just-)identification. This restriction is derived by obtaining the bivariate vector moving average (VMA) form from the VAR representation of the system in (4):

$$\begin{pmatrix} \Delta y_t \\ z_t \end{pmatrix} = [\mathbf{I} - \mathbf{\Theta}(L)L]^{-1} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} v_t^1 \\ v_t^2 \end{pmatrix}$$

$$= \begin{pmatrix} \pi_{11}(L) & \pi_{12}(L) \\ \pi_{21}(L) & \pi_{22}(L) \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} v_t^1 \\ v_t^2 \end{pmatrix} = \begin{pmatrix} d_{11}(L) & d_{12}(L) \\ d_{21}(L) & d_{22}(L) \end{pmatrix} \begin{pmatrix} v_t^1 \\ v_t^2 \end{pmatrix}$$

(9) In (9) the polynomials $d_{ii}(L)$ trace out the responses of Δy and z to the permanent and transitory structural disturbances. The restriction that the transitory shock, say v^2, does not affect the level of y in the long run amounts to imposing:5 $d_{12}(1) = 0$ In fact, note that by definition $y_{t+h} \equiv y_t + \Delta y_{t+1} + \Delta y_{t+2} + \ldots + \Delta y_{t+h}$. If y is

restricted not to be affected in the long run (i.e. for $h \to \infty$) by a disturbance occurring at time t, then the sum of all changes Δy_{t+j} must be zero, which is the constraint imposed by $d_{12}(1) = 0$. This restriction can be expressed in terms of the b_{ij}'s and the $\pi_{ij}(1)$'s as6 $\pi_{11}(1)b_{12} + \pi_{12}(1)b_{22} = 0$

(10) 5Recall that setting $L = 1$ in a polynomial $a(L) = 1 + a_1L + a_2L^2 + \ldots + a_iL^i + \ldots$ yields the sum of all coefficients a_i: $a(1) = 1 + a_1 + a_2 + ... + a_i + ... = \sum_{i=0}^{n} a_i$.

6In fact, from the from the VMA representation of Δy in (9):

$$\Delta y_t = \underbrace{[\pi_{11}(L)\,b_{11} + \pi_{12}(L)\,b_{21}]}_{d_{12}(L)}\, v_t^1 + \underbrace{[\pi_{11}(L)\,b_{12} + \pi_{12}(L)\,b_{22}]}_{d_{12}(L)}\, v_t^2$$

From estimation of the reduced form of the system the values of $\pi_{11}(1)$ and $\pi_{12}(1)$ are obtained, so that (10) can be added to the three equations in (6)-(8) to form the set of four just-identifying

restrictions needed for retrieving the structural dis- turbances from VAR innovations. Once the structural shocks are obtained, the impulse response functions of Δy and z to the permanent (v^1) and the transitory

(v^2) shocks can be computed and the forecast error variance of the endogenous variables can be decomposed into the fractions attributable to the two structural disturbances.

3.6 Appendix A: Innovation accounting in a VAR: a bivariate example

Consider a stationary bivariate VAR system in $VMA(\infty)$ representation (omitting constant terms for simplicity):

$$\mathbf{y}_t = \mathbf{\Phi}_0\,\mathbf{v}_t + \mathbf{\Phi}_1\,\mathbf{v}_{t-1} + \mathbf{\Phi}_2\,\mathbf{v}_{t-2} + \ldots + \mathbf{\Phi}_s\,\mathbf{v}_{t-s} + \ldots$$

written in full form as:

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \phi_{0,11} & \phi_{0,12} \\ \phi_{0,21} & \phi_{0,22} \end{pmatrix} \begin{pmatrix} v_{1,t} \\ v_{2,t} \end{pmatrix} + \begin{pmatrix} \phi_{1,11} & \phi_{1,12} \\ \phi_{1,21} & \phi_{1,22} \end{pmatrix} \begin{pmatrix} v_{1,t-1} \\ v_{2,t-1} \end{pmatrix} +$$

$$+ \begin{pmatrix} \phi_{2,11} & \phi_{2,12} \\ \phi_{2,21} & \phi_{2,22} \end{pmatrix} \begin{pmatrix} v_{1,t-2} \\ v_{2,t-2} \end{pmatrix} + \ldots + \begin{pmatrix} \phi_{s,11} & \phi_{s,12} \\ \phi_{s,21} & \phi_{s,22} \end{pmatrix} \begin{pmatrix} v_{1,t-s} \\ v_{2,t-s} \end{pmatrix} + \ldots$$

Impulse response functions The elements of the matrices Φ_i trace out the effects over time (impulse response) functions) of each structural disturbance in $\mathbf{v}$ keeping all the other disturbances at zero, under the set of identifying assumptions used ($\Phi_0 = \mathbf{A}^{-1}$). Each matrix Φ_s captures the effect of the structural shocks at time t on the endogenous variables

at time $t + s$; the typical element $\phi_{s,ij} = \frac{\partial y_{i,t+s}}{\partial v_{i,t}}$ captures the response of the ith element of $\mathbf{y}_{t+s}$ to an "impulse" due to the jth element of $\mathbf{v}_t$. Noting that $\frac{\partial y_{i,t+s}}{\partial v_{j,t}} = \frac{\partial y_{i,t}}{\partial v_{j,t-s}}$, from the VMA representation above

we get:

• IRF of y_1 to v_1: $\phi_{0,11}, \phi_{1,11}, \phi_{2,11}, ...$ (elements in position 1, 1 in $\Phi_0, \Phi_1, \Phi_2, ...$) • IRF of y_1 to v_2: $\phi_{0,12}, \phi_{1,12}, \phi_{2,12}, ...$ (elements in position 1, 2 in $\Phi_0, \Phi_1, \Phi_2, ...$)

• IRF of y_2 to v_1: $\phi_{0,21}, \phi_{1,21}, \phi_{2,21}, \ldots$ (elements in position 2, 1 in $\Phi_0, \Phi_1, \Phi_2, \ldots$) • IRF of y_2 to v_2: $\phi_{0,22}, \phi_{1,22}, \phi_{2,22}, \ldots$ (elements in position 2, 2 in $\Phi_0, \Phi_1, \Phi_2, \ldots$)

Forecast error variance decomposition From the $VMA(\infty)$ representation of the VAR it is possible to obtain the forecast of future $\mathbf{y}$'s over an h-period horizon on the basis of information in current (time t) and past values of the variables in the system $\mathbf{y}_t, \mathbf{y}_{t-1}, \ldots$ The associated forecast error is:

$$\mathbf{y}_{t+h} - E(\mathbf{y}_{t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, ...) = \mathbf{\Phi}_0\mathbf{v}_{t+h} + \mathbf{\Phi}_1\mathbf{v}_{t+h-1} + ... + \mathbf{\Phi}_{h-1}\mathbf{v}_{t+1}$$

written in full form as $\begin{pmatrix} y_{1,t+h} - E(y_{1,t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, \ldots) \\ y_{2,t+h} - E(y_{2,t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, \ldots) \end{pmatrix} = \begin{pmatrix} \phi_{0,11} & \phi_{0,12} \\ \phi_{0,21} & \phi_{0,22} \end{pmatrix} \begin{pmatrix} v_{1,t+h} \\ v_{2,t+h} \end{pmatrix} +$

$$+ \begin{pmatrix} \phi_{1,11} & \phi_{1,12} \\ \phi_{1,21} & \phi_{1,22} \end{pmatrix} \begin{pmatrix} v_{1,t+h-1} \\ v_{2,t+h-1} \end{pmatrix} + \cdots + \begin{pmatrix} \phi_{h-1,11} & \phi_{h-1,12} \\ \phi_{h-1,21} & \phi_{h-1,22} \end{pmatrix} \begin{pmatrix} v_{1,t+1} \\ v_{2,t+1} \end{pmatrix}$$

The forecast error variance is given by the following symmetric matrix var $(\mathbf{y}_{t+h} - E(\mathbf{y}_{t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, ...)) = \mathbf{\Phi}_0\mathbf{D}\mathbf{\Phi}_0' + \mathbf{\Phi}_1\mathbf{D}\mathbf{\Phi}_1' + ... + \mathbf{\Phi}_{h-1}\mathbf{D}\mathbf{\Phi}_{h-1}'$

The elements on the main diagonal capture the forecast error variances of each variable in y. Writing the matrix in full form we get:

$$\text{Var}(\mathbf{y}_{t+h} - E(\mathbf{y}_{t+h} \mid \mathcal{I}_t)) = \sum_{s=0}^{h-1} \mathbf{\Phi}_s\mathbf{D}\mathbf{\Phi}_s'.$$

Writing the two-variable case explicitly gives

$$\begin{pmatrix} \sigma_{v_1}^2 \sum_{s=0}^{h-1} \phi_{s,11}^2 + \sigma_{v_2}^2 \sum_{s=0}^{h-1} \phi_{s,12}^2 & \cdots \\ \cdots & \sigma_{v_1}^2 \sum_{s=0}^{h-1} \phi_{s,21}^2 + \sigma_{v_2}^2 \sum_{s=0}^{h-1} \phi_{s,22}^2 \end{pmatrix}.$$

At any horizon h, the forecast error variance for each variable i (with $i = 1, 2$) is the sum of two components: (i) $\sigma_{v_1}^2 \sum_{s=0}^{h-1} \phi_{s,i1}^2$, that captures the variance due to the first structural disturbance v_1, and (ii) $\sigma_{v_2}^2 \sum_{s=0}^{h-1} \phi_{s,i2}^2$, capturing the variance

due to v_2. The FEVD exercise consists in computing the fractions of the total variance attributable to each of the two structural shocks. Letting $FEVD_{h,ij}$ denote the portion of the forecast error variance of variable i attributable to shock j at forecasting horizon h, we have:

- for a one-period forecasting horizon, $h = 1$:

$$FEVD_{1,ij} = \frac{\sigma_{v_j}^2 \phi_{0,ij}^2}{\sigma_{v_1}^2 \phi_{0,i1}^2 + \sigma_{v_2}^2 \phi_{0,i2}^2}, \qquad i, j = 1, 2.$$

- for a two-period forecasting horizon, $h = 2$:

$$FEVD_{2,ij} = \frac{\sigma_{v_j}^2 \sum_{s=0}^{1} \phi_{s,ij}^2}{\sigma_{v_1}^2 \sum_{s=0}^{1} \phi_{s,i1}^2 + \sigma_{v_2}^2 \sum_{s=0}^{1} \phi_{s,i2}^2}, \qquad i, j = 1, 2.$$

and so on for more extended forecasting horizons. In general, for horizon $\boldsymbol{h}$, variable i and shock j, we have:

$$FEVD_{h,ij} = \frac{\sigma_{v_j}^2 \sum_{s=0}^{h-1} \phi_{s,ij}^2}{\sigma_{v_1}^2 \sum_{s=0}^{h-1} \phi_{s,i1}^2 + \sigma_{v_2}^2 \sum_{s=0}^{h-1} \phi_{s,i2}^2}$$

Vector Autoregression (VAR) models Univariate autoregressive (AR) models can be extended to the multivariate case to study dynamic interrelationships among several "endogenous" variables.

Vector autoregression (VAR) models describe the evolution over time of a vector of n variables $\mathbf{y}_t = (y_{1t} y_{2t} \dots y_{nt})'$ as a function of its past realizations $y_{t-1}, y_{t-2}, \dots$ and a vector of stochastic terms $\mathbf{u}_t = (u_{1t}\ u_{2t} \dots u_{nt})'$

A VAR model with p lags of $\mathbf{y}_t$ is called $VAR(p)$ and has the following form:

$$\mathbf{y}_t = \delta + \mathbf{\Theta}_1\mathbf{y}_{t-1} + \mathbf{\Theta}_2\mathbf{y}_{t-2} + \dots + \mathbf{\Theta}_p\mathbf{y}_{t-p} + \mathbf{u}_t$$

where $\boldsymbol{\delta} = (\delta_1\ \delta_2 \dots \delta_n)'$ is a vector of constant terms, $\mathbf{\Theta}_1, \mathbf{\Theta}_2 \dots \mathbf{\Theta}_p$

are $n \times n$ matrices and $\mathbf{u}_t$ is a vector of white noise processes with the variance-covariance structure:

$$E(\mathbf{u}_t\mathbf{u}_s') = \begin{cases} \mathbf{\Sigma} & \text{if } t = s \\ \mathbf{0} & \text{if } t \neq s \end{cases}$$

Matrix Σ is not assumed to be diagonal: therefore the error terms of the individual equations can be (contemporaneously) correlated.

The VAR is the reduced form of a structural model capturing the relationships among the endogenous variables.

With only 2 variables and 1 lag, the structural form of the model is:

$$y_{1t} = \gamma_{10} + a_{12}y_{2t} + \gamma_{11}y_{1t-1} + \gamma_{12}y_{2t-1} + v_{1t}$$
$$y_{2t} = \gamma_{20} + a_{21}y_{1t} + \gamma_{21}y_{1t-1} + \gamma_{22}y_{2t-1} + v_{2t}$$

with $E(\mathbf{v}_t\mathbf{v}_s') = \begin{cases} \mathbf{D} & \text{if } t = s \text{ with } \mathbf{D} \text{ diagonal} \\ \mathbf{0} & \text{if } t \neq s \end{cases}$

The elements of $\mathbf{v}_t$ are the structural shocks (or disturbances) of the system. They are uncorrelated and therefore can bear a precise economic interpretation.

One of the main purposes of VAR modelling is the estimation of the dynamic response of all endogenous variables to the structural disturbances in v.

In matrix notation, the structural form is:

$$\begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix} \begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \gamma_{10} \\ \gamma_{20} \end{pmatrix} + \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix} \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \end{pmatrix}$$

$$+ \begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix} \text{ V } \mathbf{A}\mathbf{y}_t = \gamma + \mathbf{\Gamma}\mathbf{y}_{t-1} + \mathbf{v}_t$$

$\mathbf{A}\mathbf{y}_t = \gamma + \mathbf{\Gamma}\mathbf{y}_{t-1} + \mathbf{v}_t$ A captures the contemporaneous relations between the two endogenous variables. Inverting A we get the reduced form of the model as a $VAR(1)$

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix} \begin{pmatrix} \gamma_{10} \\ \gamma_{20} \end{pmatrix} +$$

$$+ \begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix}^{-1} \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix} \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \end{pmatrix}$$

$$+ \begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix}^{-1} \begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix}$$

$$\mathbf{y}_t = \mathbf{A}^{-1}\boldsymbol{\gamma} + \mathbf{A}^{-1}\mathbf{\Gamma}\mathbf{y}_{t-1} + \mathbf{A}^{-1}\mathbf{v}_t$$

$$\Rightarrow$$

$$\mathbf{y}_t = \delta + \mathbf{\Theta}\mathbf{y}_{t-1} + \mathbf{u}_t$$

$$\mathbf{y}_t = \mathbf{A}^{-1}\boldsymbol{\gamma} + \mathbf{A}^{-1}\mathbf{\Gamma}\mathbf{y}_{t-1} + \mathbf{A}^{-1}\mathbf{v}_t$$

$\Rightarrow$ yt = δ + Θ yt-1 + ut The elements of $\mathbf{u}_t$ are linear combinations of the structural disturbances v_{1t} and v_{2t}:

$$\mathbf{u}_t = \mathbf{A}^{-1}\mathbf{v}_t = \frac{1}{1 - a_{12}a_{21}} \begin{pmatrix} 1 & a_{12} \\ a_{21} & 1 \end{pmatrix} \begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix}$$

$$= \frac{1}{1 - a_{12}a_{21}} \begin{pmatrix} v_{1t} + a_{12}v_{2t} \\ a_{21}v_{1t} + v_{2t} \end{pmatrix}$$

and the variance-covariance matrix of the error terms Σ is given by $\mathbf{\Sigma} = \frac{1}{(1-a_{12}a_{21})^2} \begin{pmatrix} \sigma_{v_1}^2 + a_{12}^2\sigma_{v_2}^2 & a_{21}\sigma_{v_1}^2 + a_{12}\sigma_{v_2}^2 \\ a_{21}\sigma_{v_1}^2 + a_{12}\sigma_{v_2}^2 & a_{21}^2\sigma_{v_1}^2 + \sigma_{v_2}^2 \end{pmatrix}$

The VAR error terms (elements of $\mathbf{u}_t$) in general have non-zero covariance.

Stationarity.

The VAR representation allows to study the dynamic response of the variables in $\mathbf{y}_t$ to each element of vector $\mathbf{u}_t$.

If the VAR is stationary the response to shocks gradually declines, tending to zero in the long run Stationarity conditions for the VAR are a multivariate extension of those valid for univariate AR process.

In the $VAR(p)$ case $\mathbf{y}_t = \delta + \mathbf{\Theta}_1\mathbf{y}_{t-1} + \mathbf{\Theta}_2\mathbf{y}_{t-2} + \ldots + \mathbf{\Theta}_p\mathbf{y}_{t-p} + \mathbf{u}_t$

$$(\mathbf{I} - \mathbf{\Theta}_1 L - \mathbf{\Theta}_2 L^2 - \cdots - \mathbf{\Theta}_p L^p)\mathbf{y}_t = \delta + \mathbf{u}_t$$

$\Theta(L)\mathbf{y}_t = \delta + \mathbf{u}_t$ for $\mathbf{y}_t$ to be stationary, the polynomial in the lag operator $\Theta(L)$ must be invertible.

Invertibility requires that the roots of the following equation det $(\mathbf{I} - \mathbf{\Theta}_1 z - \mathbf{\Theta}_2 z^2 - \ldots - \mathbf{\Theta}_p z^p) = 0$ be (in modulus) outside the unit circle.

EXAMPLE – Illustration

Example. for a $VAR(1)$ $\mathbf{y}_t = \delta + \mathbf{\Theta}_1\mathbf{y}_{t-1} + \mathbf{u}_t \Rightarrow (\mathbf{I} - \mathbf{\Theta}_1 L)\mathbf{y}_t = \delta + \mathbf{u}_t \begin{pmatrix} 1 - \theta_{11}L & -\theta_{12}L \\ -\theta_{21}L & 1 - \theta_{22}L \end{pmatrix} \mathbf{y}_t = \delta + \mathbf{u}_t$

the stationarity condition is that the roots of the equation $\det(\mathbf{I} - \mathbf{\Theta}_1 z) = 0 \Rightarrow 1 - (\theta_{11} + \theta_{22})z + (\theta_{11}\theta_{22} - \theta_{21}\theta_{12})z^2 = 0$

be outside the unit circle.

Under stationarity, the VAR can be expressed as a vector moving average of infinite order, $VMA(\infty)$:

$$\mathbf{y}_t = \mathbf{\Theta}(L)^{-1}(\delta + \mathbf{u}_t)$$
$$= \Theta(1)^{-1}\delta + \Theta(L)^{-1}\mathbf{u}_t$$

from which we get the expected value (unconditional mean) of $\mathbf{y}_t$ $E(\mathbf{y}_t) = \mathbf{\Theta}(1)^{-1}\delta = (\mathbf{I} - \mathbf{\Theta}_1 - \mathbf{\Theta}_2 - \ldots - \mathbf{\Theta}_p)^{-1}\delta$

Estimation.

The VAR can be estimated by OLS applied equation by equation.

This procedure yields consistent estimates of the parameters in δ, Θ_1, Θ_2, ... Θ_p, and of the variances and covariances of the innovations in matrix Σ.

The "identification" problem To analyze the dynamic response of the endogenous variables in $\mathbf{y}_t$ to each element of the innovation vector $\mathbf{u}_t$, the $VMA(\infty)$ form of the VAR model can be used:

$$\begin{aligned}\mathbf{y}_t &= \boldsymbol{\Theta}(1)^{-1}\,\delta + \boldsymbol{\Theta}(L)^{-1}\,\mathbf{u}_t \\ &= \Theta(1)^{-1}\boldsymbol{\delta} + (\mathbf{I} + \boldsymbol{\Psi}_1 L + \boldsymbol{\Psi}_2 L^2 + \dots)\mathbf{u}_t\end{aligned}$$

The $n \times n$ matrices Ψ_i (for $j = 0, ...\infty$ and with $\Psi_0 = I$) describe the response over time of each endogenous variable y_i $(i = 1, \dots n)$ to each shock u_i $(j = 1, \dots n)$.

ALGORITHM – Problem

Problem.

The economic interpretability of such responses is difficult since the VAR innovations are linear combinations of structural shocks and are therefore correlated.

We would like to get the response of y_t to the structural (unobserved) disturbances, to which it is generally possible to attach an economic interpretation.

For $n > 2$ endogenous variables and $p > 1$ lags, the relationship between the structural and reduced forms of the system (and the associated disturbances) is:

$$\mathbf{A}\mathbf{y}_t = \gamma + \boldsymbol{\Gamma}_1\mathbf{y}_{t-1} + \boldsymbol{\Gamma}_2\mathbf{y}_{t-2} + \dots + \boldsymbol{\Gamma}_p\mathbf{y}_{t-p} + \mathbf{v}_t$$

$$\Rightarrow \mathbf{y}_t = \mathbf{A}^{-1}\boldsymbol{\gamma} + \mathbf{A}^{-1}\boldsymbol{\Gamma}_1\mathbf{y}_{t-1} + \mathbf{A}^{-1}\boldsymbol{\Gamma}_2\mathbf{y}_{t-2} + \dots + \mathbf{A}^{-1}\boldsymbol{\Gamma}_p\mathbf{y}_{t-p} + \mathbf{A}^{-1}\mathbf{v}_t$$

$$= \delta + \boldsymbol{\Theta}_1\mathbf{y}_{t-1} + \boldsymbol{\Theta}_2\mathbf{y}_{t-2} + \dots + \boldsymbol{\Theta}_p\mathbf{y}_{t-p} + \mathbf{u}_t$$

from which we get the relationship between the VAR residuals ($\mathbf{u}_t$) and the structural disturbances ($\mathbf{v}_t$):

$$\begin{aligned}\mathbf{u}_t &= \mathbf{A}^{-1}\mathbf{v}_t \\ \Rightarrow \mathbf{A}\mathbf{u}_t &= \mathbf{v}_t\end{aligned}$$

To obtain the structural disturbances $\mathbf{v}_t$ from the estimated $\mathbf{u}_t$ it is necessary to identify the elements of A.

Only after identification it is possible to proceed to the analysis of the dynamic response of $\mathbf{y}_t$ to each shock in $\mathbf{v}_t$ To obtain the elements of A we can use the relation between the variance-covariance matrices of $\mathbf{u}_t$ ($\boldsymbol{\Sigma}$) and $\mathbf{v}_t$ ($\mathbf{D}$):

$$E(\mathbf{A}\mathbf{u}_t(\mathbf{A}\mathbf{u}_t)') = \mathbf{A}\underbrace{E(\mathbf{u}_t\mathbf{u}_t')}_{\boldsymbol{\Sigma}}\mathbf{A}' = \underbrace{E(\mathbf{v}_t\mathbf{v}_t')}_{\mathbf{D}}$$

D $\Rightarrow$ AΣA' = D

Parameters to be identified:

$$\left\{ \begin{array}{l} n(n-1) \text{ off-diagonal elements of } \mathbf{A} \\ n \text{ elements on the diagonal of } \mathbf{D} \end{array} \right\} \Rightarrow n^2 \text{ parameters}$$

Estimates (from VAR) of the elements of Σ:

$$\left\{ \begin{array}{l} n \text{ residual variances} \\ \frac{n(n-1)}{2} \text{ residual covariances} \end{array} \right\} \Rightarrow \frac{n(n+1)}{2} \text{ estimates } < n^2$$

Restrictions to impose on the elements of A:

$$n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$$

A "recursive" identification strategy This strategy (C. Sims 1980) imposes a recursive structure on A by means of "zero" restrictions on the $\frac{n(n-1)}{2}$ elements above its main diagonal $\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ a_{21} & 1 & 0 & \dots & 0 & 0 \\ a_{31} & a_{32} & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{(n-1)1} & a_{(n-1)2} & a_{(n-1)3} & \dots & 1 & 0 \\ a_{n1} & a_{n2} & a_{n3} \dots & a_{n(n-1)} & 1 & \end{pmatrix}$

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ a_{21} & 1 & 0 & \dots & 0 & 0 \\ a_{31} & a_{32} & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{(n-1)1} & a_{(n-1)2} & a_{(n-1)3} & \dots & 1 & 0 \\ a_{n1} & a_{n2} & a_{n3} \dots & a_{n(n-1)} & 1 & \end{pmatrix}$$

$\Rightarrow$ specific "causal chain" pattern of relationships among the n variables: from the "most exogenous" (first element of y_t) to the "most endogenous" (last element of y_t)

EXAMPLE – Illustration

Example. $n = 2$. Relationship between VAR residuals and structural disturbances:

$$\begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix} \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix} = \begin{pmatrix} v_{1t} \\ v_{2t} \end{pmatrix}$$

Parameters to identify: $-a_{12}$, $-a_{21}$ and $\sigma^2_{\nu_1}$, $\sigma^2_{\nu_2}$.

The VAR yields estimates s_{ij} $(i, j = 1, 2)$ of the elements of Σ (with $s_{12} = s_{21}$) The relationship between Σ and D is:

$$\begin{pmatrix} 1 & -a_{12} \\ -a_{21} & 1 \end{pmatrix} \begin{pmatrix} s_{11} & s_{12} \\ s_{12} & s_{22} \end{pmatrix} \begin{pmatrix} 1 & -a_{21} \\ -a_{12} & 1 \end{pmatrix} = \begin{pmatrix} \sigma^2_{v_1} & 0 \\ 0 & \sigma^2_{v_2} \end{pmatrix}$$

This relation yields a 3-equation system:

$$s_{11} - 2a_{12}s_{12} + a_{12}^2 s_{22} = \sigma_{v_1}^2$$
$$a_{21}^2 s_{11} - 2a_{21}s_{12} + s_{22} = \sigma_{\nu_2}^2$$

$-a_{21}s_{11} + (1 + a_{21}a_{12})s_{12} - a_{12}s_{22} = 0$ To achieve identification of the 4 parameters it is necessary to impose 1 additional restriction.

Imposing $a_{12} = 0$ the system becomes:

$$s_{11} = \sigma_{v_1}^2$$
$$a_{21}^2 s_{11} - 2a_{21}s_{12} + s_{22} = \sigma_{\nu_2}^2$$
$$-a_{21}s_{11} + s_{12} = 0$$

from which we get:

$$\sigma_{v_1}^2 = s_{11}$$
$$a_{21} = \frac{S_{12}}{S_{11}}$$
$$\sigma_{v_2}^2 = s_{22} - \frac{s_{12}^2}{s_{11}}$$

$\Rightarrow$ "triangular" or "Choleski" identification scheme (imposes a recursive structure on the system of simultaneous relations among the endogenous variables).

v_{1t} is immediately identified as the first element of the VAR residual vector u_{1t} v_{2t} is identified as the portion of u_{2t} uncorrelated with ("orthogonal" to) u_{1t}:

$u_{1t} = v_{1t}$ $u_{2t} = a_{21}u_{1t} + v_{2t}$ The restriction $a_{12} = 0$ corresponds to a specific "ordering" of the variables:

y_1 affects y_2 contemporaneously (within period t), whereas v_2 affects v_1 only in subsequent periods

The alternative restriction $a_{21} = 0$ leads to a different identification, based on a reversed ordering of the variables, with y_2 placed "before" y_1 $u_{2t} = v_{2t}$ $u_{1t} = a_{12}u_{2t} + v_{1t}$ Important:

• All orderings of the n variables in $\mathbf{y}_t$ yield the necessary number of restrictions leading to "just-(or exact-)identification" • All orderings describe the data equally well and it is not possible to use statistical tests to determine the most appropriate recursive scheme

Dynamic analysis After achieving identification of the parameters in A and D, the (orthogonal) structural disturbances can be constructed from the relation $A\mathbf{u}_t = \mathbf{v}_t$.

It is then possible to proceed to the analysis of the dynamic response of $\mathbf{y}_{t+i}$ to $\mathbf{v}_t$.

This analysis, called innovation accounting, is carried out by constructing:

the (orthogonalized) impulse response functions (IRF) the forecast errors variance decompositions (FEVD)

Both the IRF and FEVD are derived from the $VMA(\infty)$ representation of the VAR system:

$$\mathbf{y}_\star = \mathbf{\Theta}(1)^{-1}\,\delta + (\mathbf{I} + \mathbf{\Psi}_1 L + \mathbf{\Psi}_2 L^2 + \dots)\,\mathbf{u}_t$$

$$= \Theta(1)^{-1}\delta + (\underbrace{\mathbf{A}^{-1}}_{t} + \underbrace{\mathbf{\Psi}_1}_{t}\mathbf{A}^{-1}L + \underbrace{\mathbf{\Psi}_2}_{t}\mathbf{A}^{-1}L^2 + \dots)\mathbf{v}_t$$

$$\begin{matrix}\mathbf{\Phi}_0 \\ \mathbf{\Phi}_1 \\ \Phi_\gamma\end{matrix}$$

$$= \mathbf{\Theta}(1)^{-1}\boldsymbol{\delta} + \mathbf{\Phi}_0\mathbf{v}_t + \mathbf{\Phi}_1\mathbf{v}_{t-1} + \mathbf{\Phi}_2\mathbf{v}_{t-2} + \dots$$

$$\mathbf{y}_t = \mathbf{\Theta}(1)^{-1}\delta + \mathbf{\Phi}_0\mathbf{v}_t + \mathbf{\Phi}_1\mathbf{v}_{t-1} + \mathbf{\Phi}_2\mathbf{v}_{t-2} + \dots$$

IRF: (impulse response functions) the elements of matrices Φ_s trace out the effects over time of each structural disturbance in v keeping all other disturbances at zero (note: $\Phi_0 = A^{-1}$).

Each matrix Φ_s captures the effect of the structural shocks at t on the endogenous variables at $t + s$; the typical element $\phi_{s,ij} \equiv \frac{\partial y_{i,t}}{\partial v_{i,t-s}} = \frac{\partial y_{i,t+s}}{\partial v_{i,t}}$

captures the response of the i-th element of y_{t+s} to an "impulse" due to the j-th element of $\mathbf{v}_t$

FEVD: (forecast error variance decomposition) from the $VMA(\infty)$ representation it is possible to obtain the forecast of future y's over an h-period horizon ($\hat{y}_{t+h}$) on the basis of information in current and past values of the variables $\mathbf{y}_t, \mathbf{y}_{t-1}, \dots$ The associated forecast error is:

$$\mathbf{y}_{t+h} - E(\mathbf{y}_{t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, \dots) = \mathbf{\Phi}_0\mathbf{v}_{t+h} + \mathbf{\Phi}_1\mathbf{v}_{t+h-1} + \dots + \mathbf{\Phi}_{h-1}\mathbf{v}_{t+1}$$

The forecast error variance is given by the following symmetric matrix $var(\mathbf{y}_{t+h} - E(\mathbf{y}_{t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, \dots)) = \mathbf{\Phi}_0\mathbf{D}\mathbf{\Phi}_0' + \mathbf{\Phi}_1\mathbf{D}\mathbf{\Phi}_1' + \dots + \mathbf{\Phi}_{h-1}\mathbf{D}\mathbf{\Phi}_{h-1}'$

The elements on the main diagonal capture the forecast error variances of each variable in y:

$var(y_{i,t+h} - E(y_{i,t+h}|\mathbf{y}_t, \mathbf{y}_{t-1}, ...))$. This variance can be expressed as the sum of the contributions of each structural disturbance in v to the total variance of each endogenous variable over the relevant horizon.

For example, the fraction of the forecast error variance of the i-th endogenous variable at horizon h attributable to the j-th structural disturbance is given by $\frac{\sigma_{v_j}^2 \sum_{s=0}^{h-1}\phi_{s,ij}^2}{\sigma_{v_1}^2 \sum_{s=0}^{h-1}\phi_{s,i1}^2 + \sigma_{v_2}^2 \sum_{s=0}^{h-1}\phi_{s,i2}^2 + \dots + \sigma_{v_n}^2 \sum_{s=0}^{h-1}\phi_{s,in}^2}$

Again (as for the IRFs), this step is economically meaningful only if there are no covariance terms, which is warranted by the chosen identification procedure (D is diagonal).

Application: effects of monetary policy shocks (Stock-Watson, JEP 2001) Structural form:

$$\mathbf{A}\begin{pmatrix}\pi_t \\ U_t \\ R_t\end{pmatrix} = \mathbf{\Gamma}(L)\begin{pmatrix}\pi_{t-1} \\ U_{t-1} \\ R_{t-1}\end{pmatrix} + \begin{pmatrix}v_t^1 \\ v_t^2 \\ v_t^P\end{pmatrix}$$

v^1 and v^2 = "macroeconomic shocks" v^P = "monetary policy shock"

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Reduced form (VAR):

$$\begin{pmatrix} \pi_t \\ U_t \\ R_t \end{pmatrix} = \mathbf{A}^{-1}\mathbf{\Gamma}(L) \begin{pmatrix} \pi_{t-1} \\ U_{t-1} \\ R_{t-1} \end{pmatrix} + \mathbf{A}^{-1} \begin{pmatrix} v_t^1 \\ v_t^2 \\ v_t^P \end{pmatrix}$$

$$= \mathbf{\Theta}(L) \begin{pmatrix} \pi_{t-1} \\ U_{t-1} \\ R_{t-1} \end{pmatrix} + \begin{pmatrix} u_t^\pi \\ u_t^U \\ u_t^R \end{pmatrix}$$

u_t^π, u_t^U, $u_t^R = VAR$ (reduced form) innovations

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-01 File Edit Object View Proc Quick Options Add-ins Window Help Var: UNTITLED Workfile: SW_JEP01::VAR_analysis\

- X View Proc Object Print Name Freeze Estimate Stats Impulse Resids Vector Autoregression Estimates Workfile: SW_JEP01 - (c:\bagliano\corsi\monet2\var\sw_jep01.wf1)

- X Date: 09/13/22 Time: 14:35 View Proc Object Save Freeze Details+/- Show Fetch Store Delete Genr Sample Range: 1955Q1 2000Q4 – 184 obs Filter: * Sample: 1960Q1 2000Q4 Sample: 1955Q1 2000Q4 – 184 obs Order: Name Included observations: 164 BC torr inn Standard errors in () & t-statistics in [] L endog Le fevd 12 fevd 120 M infl R INFL UNEMP L infl r I if 40 Mr r M resid INFL(-1) 0.546602 0.030733 0.086682 u infl u r (0.07866) (0.01870) (0.06935) u unemp unemp [6.94890] [1.64377] [1.24984] unemp r v2 var var4 INFL(-2) 0.075629

-0.030821 0.193629 Vp (0.02125) (0.08941) (0.07884) StockWatson_data VAR_analysis New Page [0.84583] [-1.45026] [2.45610] INFL(-3) 0.111728 0.029789 −0.078252 (0.09006) (0.02141) (0.07941) [1.24052] [1.39157] [-0.98541] INFL(-4) 0.266606

-0.024084 −0.027216 (0.08228) (0.01956) (0.07255) [3.24007] [-1.23145] [-0.37514] UNEMP(-1)

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-0.003358 R(-1) fevd 120 M infl (0.10046) (0.02388) (0.08858) L infl r I inf 40 [2.26651] [-0.14061] [10.7084] Mr r M resid u_infl u_r −0.228725 0.050389 R(-2) −0.381030 u unemp M unemp (0.03199) (0.13458) (0.11866) unemp_r v2 [1.57529] [-1.69956] [-3.21117] war var4 M vp

-0.020360 R(-3) 0.062624 0.332095 \ StockWatson_data \ VAR_analysis \ New Page (0.03183) (0.11806) (0.13390) [0.46768] [-0.63971] [2.81287] 0.005464 R(-4)

-0.069266 0.029857 (0.02434) (0.10242) (0.09030) [-0.67630] [0.22444] [0.33064] C 1.027154 0.101297 0.505753 (0.37358) (0.08880) (0.32939) [2.74946] [1.14079] [1.53544] Path = c:\users\fabio bagliano\dropbox (esomas)\econometrics - historical papers\di fenizio DB = none WF = sw_jep01 xR 14:36 13/09/2022 e S

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-

OLS estimation delivers $\hat{\Sigma}$, from which we get correlations among estimated innovations $\widehat{u}_t^{\pi}$ $\widehat{u}_t^{U}$ $\widehat{u}_t^{R}$ correl. coeff.

$$\hat{u}_t^{\pi}$$
$$\widehat{u}_t^{U}$$

-0.056 1 0.129 -0.442 1 $\widehat{u}_t^{R}$

Relation between VAR innovations and structural disturbances based on a Choleski (recursive) identification scheme:

$$\begin{pmatrix} 1 & 0 & 0 \\ a_{21} & 1 & 0 \\ a_{31} & a_{32} & 1 \end{pmatrix} \begin{pmatrix} u_t^{\pi} \\ u_t^{U} \\ u_t^{R} \end{pmatrix} = \begin{pmatrix} v_t^{1} \\ v_t^{2} \\ v_t^{P} \end{pmatrix}$$

$$\begin{aligned} u_t^{\pi} &= v_t^{1} \\ \Rightarrow u_t^{U} &= -a_{21} u_t^{\pi} + v_t^{2} \\ u_t^{R} &= -a_{31} u_t^{\pi} - a_{32} u_t^{U} + v_t^{P} \end{aligned}$$

$$\begin{aligned} u_t^{\pi} &= v_t^{1} \\ \Rightarrow u_t^{U} &= -a_{21} u_t^{\pi} + v_t^{2} \\ u_t^{R} &= -a_{31} u_t^{\pi} - a_{32} u_t^{U} + v_t^{P} \end{aligned}$$

Inverting A:

$$u_t^{\pi} = v_t^1$$
$$u_t^{U} = -a_{21}v_t^1 + v_t^2$$
$$u_t^{R} = -(a_{31} - a_{32}a_{21})v_t^1 - a_{32}v_t^2 + v_t^P$$

OLS estimation yields:

$u_t^{\pi} = v_t^1$ $u_t^{U} = -0.013\ u_t^{\pi} + v_t^2$ $R^2 = 0.003$ (0.019) $u_t^{R} = 0.093\ u_t^{\pi} - 1.619\ u_t^{U} + v_t^P$ $R^2 = 0.207$

(0.062) (0.261) and estimates of the standard errors of the structural disturbances:

$$\sigma_{v^1} = 0.956$$
$$\sigma_{v^2} = 0.228$$
$$\sigma_{v^P} = 0.758$$

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-2 I File Edit Object View Proc Quick Options Add-ins Window Help

- X View Proc Object Print Name Freeze Estimate Stats Impulse Resids Response to Cholesky One S.D. Innovations 2 S.E.

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Identification with long-run restrictions (Blanchard-Quah, AER 1989) separate the effects of permanent and transitory disturbances Aim:

on endogenous variables Bivariate model:

non-stationary, $I(1)$, variable y entering the model in differenced form: Δy stationary, $I(0)$, variable z

Reduced (VAR) form (all included variables stationary) $\begin{pmatrix} \Delta y_t \\ z_t \end{pmatrix} = \Theta(L) \begin{pmatrix} \Delta y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} u_t^Y \\ u_t^Z \end{pmatrix}$

innovations: u^Y and u^Z.

Relation between VAR residuals u and structural disturbances (orthogonal) $\begin{pmatrix} u_t^Y \\ u_t^Z \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} v_t^1 \\ v_t^2 \end{pmatrix}$

where $E(\mathbf{v}_t \mathbf{v}_t') = \mathbf{I}$ Identification of structural v^1 and v^2 from VAR innovations u^Y and u^Z (and their covariance matrix), requires identification of the four elements b_{ij} $(i, j = 1, 2)$

BQ methodology:

impose long-run restriction on the impulse response function of Δy with respect to the transitory shock

The covariance matrix of VAR innovations in terms of the b_{ij}'s is $E(\mathbf{u}_t \, \mathbf{u}_t') = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} E(\mathbf{v}_t \, \mathbf{v}_t') \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}'$

$$\Rightarrow \begin{pmatrix} s_Y^2 & s_{YZ} \\ s_{YZ} & s_Z^2 \end{pmatrix} = \begin{pmatrix} b_{11}^2 + b_{12}^2 & b_{11}b_{21} + b_{12}b_{22} \\ b_{11}b_{21} + b_{12}b_{22} & b_{21}^2 + b_{22}^2 \end{pmatrix}$$

Given the estimated s_Y^2, s_Z^2, s_{YZ}, 3 equations are obtained:

$$b_{11}^2 + b_{12}^2 = s_v^2$$
$$b_{21}^2 + b_{22}^2 = s_7^2$$
$$b_{11}b_{21} + b_{12}b_{22} = s_{YZ}$$

one additional restriction is needed for (just)identification.

This restriction is derived from the bivariate VMA form of the VAR:

$$\begin{pmatrix}\Delta y_t \\ z_t\end{pmatrix} = \left[\mathbf{I} - \mathbf{\Theta}(L)L\right]^{-1} \begin{pmatrix}b_{11} & b_{12} \\ b_{21} & b_{22}\end{pmatrix} \begin{pmatrix}v_t^1 \\ v_t^2\end{pmatrix}$$

$$= \begin{pmatrix}\pi_{11}(L) & \pi_{12}(L) \\ \pi_{21}(L) & \pi_{22}(L)\end{pmatrix} \begin{pmatrix}b_{11} & b_{12} \\ b_{21} & b_{22}\end{pmatrix} \begin{pmatrix}v_t^1 \\ v_t^2\end{pmatrix}$$

$$= \begin{pmatrix}d_{11}(L) & d_{12}(L) \\ d_{21}(L) & d_{22}(L)\end{pmatrix} \begin{pmatrix}v_t^1 \\ v_t^2\end{pmatrix}$$

The polynomials $d_{ij}(L)$ trace out the responses of Δy and z to the permanent and transitory structural disturbances.

Focusing on the VMA representation of Δy:

$$\Delta y_t = \left[\pi_{11}(L)b_{11} + \pi_{12}(L)b_{21}\right] v_t^1 + \left[\pi_{11}(L)b_{12} + \pi_{12}(L)b_{22}\right] v_t^2$$

$d_{11}(L)$ $d_{12}(L)$ The restriction that the transitory shock, say v^2, does not affect y in the long run amounts to imposing:

$d_{12}(1) = 0$ which can be expressed in terms of the b_{ij}'s and the $\pi_{ij}(1)$'s as $\pi_{11}(1)b_{12} + \pi_{12}(1)b_{22} = 0$

1.8 Set 8 – Cointegration in VAR Models

Cointegration in VAR models VAR models are analyzed in the previous section under the assumption of stationarity (section 3.1). When all the series involved are stationary, we can include in the VAR, as elements of $\mathbf{y}_t$, the levels of all variables. If some of the series are non-stationary ($I(1)$) we can obtain stationarity by taking (first) differences of the original series and include those differences in the vector of endogenous variables, together with $I(0)$ variables in levels.

However, by mechanically differencing all non-stationary series, we run the risk of neglecting the potential stationary combinations that may exists among those variables, with an economically important interpretation in terms of long-run equi- librium relations (i.e. cointegration, defined in section 2.2). To address this issue, we introduce cointegration into a VAR specification when non-stationary variables are present, starting with a bivariate case (that extends the example used in section 2.2) and then presenting an outline of the general multivariate case.

4.1 The bivariate case

To introduce cointegration in a multivariate framework and the related vector equi- librium correction (VECM) form, we start from a simple bivariate example (fol- lowing Engle and Granger, Econometrica, 1987). Consider two $I(1)$ variables, y and x, generated by the following model:

$$\begin{array}{rcll} y_t - \gamma\, x_t & = & v_{1t} & v_{1t} = \rho\, v_{1t-1} + \varepsilon_{1t} \\ \delta\, y_t + x_t & = & \theta_1\, y_{t-1} + \theta_2\, x_{t-1} + v_{2t} & v_{2t} = v_{2t-1} + \varepsilon_{2t} \end{array}$$

(11) (12) where ε_{1t} and ε_{2t} are white noise processes. Imposing $|\rho| < 1$ makes the two variables cointegrated of order $(1, 1)$ since equation (11) describes a stationary linear combination of y

and x with cointegrating vector $(1, -\gamma)$. By differencing and substituting for v_{1t} into (11)[7] the following reduced form is obtained, where a term

in the levels of the variables captures the tendency of the system to "correct" any 7Differencing (11) we obtain on the right-hand side $\Delta v_{1t} = (\rho - 1)v_{1t-1} + \varepsilon_{1t}$ and, lagging (11) by one period and substituting for v_{1t-1}:

$$\Delta v_{1t} = (\rho - 1)\,(y_{t-1} - \gamma\, x_{t-1}) + \varepsilon_{1t}$$

deviation from the long-run, cointegrating relationship:

$$\Delta y_t = -\frac{1-\rho}{1+\gamma\delta}(y_{t-1} - \gamma x_{t-1}) + \frac{\gamma\theta_1}{1+\gamma\delta}\Delta y_{t-1} + \frac{\gamma\theta_2}{1+\gamma\delta}\Delta x_{t-1}$$

$$+\frac{1}{1+\gamma\delta}(\varepsilon_{1t} + \gamma\varepsilon_{2t})$$

$$\Delta x_t = \delta\frac{1-\rho}{1+\gamma\delta}(y_{t-1} - \gamma x_{t-1}) + \frac{\theta_1}{1+\gamma\delta}\Delta y_{t-1} + \frac{\theta_2}{1+\gamma\delta}\Delta x_{t-1}$$

$+\frac{1}{1+\sqrt{\delta}}(\varepsilon_{2t} - \delta\varepsilon_{1t})$ The existence of this equilibrium-correction representation of the system in (11) and (12) crucially depends on the magnitude of ρ: if $\rho = 1$, y and x are not cointegrated and the equilibrium-correction terms vanish (the reduced form is simply a $VAR(1)$)

for the variables expressed in differenced form). In matrix notation we have:

$$\begin{pmatrix}\Delta y_t \\ \Delta x_t\end{pmatrix} = \Pi\begin{pmatrix}y_{t-1} \\ x_{t-1}\end{pmatrix} + \Gamma\begin{pmatrix}\Delta y_{t-1} \\ \Delta x_{t-1}\end{pmatrix} + \begin{pmatrix}u_{1t} \\ u_{2t}\end{pmatrix}$$

The correspondence between the existence of a cointegrating vectors and an equilibrium-correction representation of the bivariate system (one proposition of the "Granger representation theorem") is the basis of the simple two-step Engle-Granger estima- tion procedure, whereby the long-run equilibrium and the short-run dynamics are modelled sequentially. In the first step, after pre-testing the variables entering the cointegrating relation in order to ensure that they are of the same order of integration, an estimate of the cointegrating vector is obtained by means of a static OLS regression. Under the hypothesis of cointegration, such regression with the variables in levels yields superconsistent estimates of the cointegrating vector, with the parameters rapidly converging to their true values (Stock, Econometrica, 1987).

The intuition behind this result is that, since in general a linear combination of $I(1)$ variables is also $I(1)$, almost all vectors obtained in static levels regressions will yield a residual series with asymptotically infinite variance. The exception will be any cointegrating vector. Since OLS estimation minimizes the residual vari- ance, the estimated vector derived from a static OLS regression should yield a very good approximation to a true cointegrating vector, if it exists. In the second step of the procedure, the residuals from the cointegrating regression are used as an equilibrium-correction term in a dynamic model for the differenced variables in order to model the short-run adjustment dynamics.

However, when more than two variables are involved in the analysis, multiple coin- tegrating vectors may exist; in this case, the first-step Engle-Granger static equation yields an (obviously stationary) linear combination of the cointegrating vectors with no means to separate them. Appropriate tools to address the issue of multiple coin- tegrating vectors are needed. Appendix B in section 4.5 presents this issue in the general context of multivariate modelling of economic relationship.

4.2 The multivariate case

To illustrate the case of multiple cointegrating vectors (Johansen, Journal of Eco- nomics Dynamics and Control 1988; Econometrica 1991), consider the VAR process for $\mathbf{y}_t$, a vector of n non-stationary ($I(1)$) variables (with only two lags for simplic- ity):

$\mathbf{y}_t = \boldsymbol{\delta} + \mathbf{A}_1\mathbf{y}_{t-1} + \mathbf{A}_2\mathbf{y}_{t-2} + \mathbf{u}_t$ (where δ is a vector of constant terms) rewritten as a reduced-form equilibrium-

correction model as follows $\Delta\mathbf{y}_t = \boldsymbol{\delta} + (\mathbf{A}_1 + \mathbf{A}_2 - \mathbf{I})\mathbf{y}_{t-1} - \mathbf{A}_2\Delta\mathbf{y}_{t-1} + \mathbf{u}_t$

$$\Rightarrow \Delta\mathbf{y}_t = \boldsymbol{\delta} + \boldsymbol{\Pi}\,\mathbf{y}_{t-1} + \boldsymbol{\Gamma}\Delta\,\mathbf{y}_{t-1} + \mathbf{u}_t$$

(13) This system is a multivariate generalization of the equation used to test for the existence of a unit root with the (augmented) Dickey-Fuller method (section 7.3.2) of the notes on "Univariate time series models"). There, the coefficient π on y_{t-1} captured relevant information about non-stationarity of the variable. Here, the elements of matrix Π contain the long-run relations among the levels of the variables in y (obtained by setting $\Delta y_t = \Delta y_{t-1} = 0$ and $u_t = 0$). Each row of Π defines

a linear combination of the elements in y that, if stationary, represents a valid long-run relation (i.e. a cointegrating vector). With $n > 2$ variables in y there may exist more than one cointegrating vector capturing the long-run behaviour of the data. The number of linearly independent rows of Π (the rank of Π) yields the number of valid cointegrating vectors, that is the number of distinct stationary linear combinations of the n non-stationary variables.

Various cases can arise:

• if rank(Π) = 0 $\Rightarrow \Pi = 0$ and no cointegrating vector exist; there are n stochastic trends in the system, each driving one of the $I(1)$ variables in y.

The stationary form of the system is in first differences of all variables, with no term in levels.

• if rank(Π) = n (full rank) $\Rightarrow$ there is a set of n independent restrictions on the long-run values of the elements in y. To derive the long-run solution, setting $\Delta\mathbf{y}_t = \Delta\mathbf{y}_{t-1} = \mathbf{0}$ and $\mathbf{u}_t = \mathbf{0}$, we get

$$\pi_{11}y_{1t-1} + \pi_{12}y_{2t-1} + \cdots + \pi_{1n}y_{nt-1} = -\delta_1$$
$$\pi_{21}y_{1t-1} + \pi_{22}y_{2t-1} + \ldots + \pi_{2n}y_{nt-1} = -\delta_2$$

... ...

$\pi_{n1}y_{1t-1} + \pi_{n2}y_{2t-1} + \ldots + \pi_{nn}y_{nt-1} = -\delta_n$ If Π is of full rank this system admits a solution for the long-run values of y_1, $y_2, \ldots y_n$, that we call $\bar{y}_1, \bar{y}_2, \ldots \bar{y}_n$. In matrix terms:

$$\bar{\mathbf{v}} = -\boldsymbol{\Pi}^{-1}\boldsymbol{\delta}$$

Vactor $\bar{\mathbf{y}}$ captures the unconditional mean of the n-dimensional stochastic process $\{\mathbf{y}_t\}$: $E(\mathbf{y}_t) = \bar{\mathbf{y}}$. Then, all variables are stationary and no stochastic trend is present in the system.

• if 0 < rank(Π) = $r < n$ $\Rightarrow$ there are r cointegrating vectors (capturing the only r linear combinations of the variables in $\mathbf{y}$ that are stationary) and the number of common stochastic trends in the system is $n - r$.8 In this case the correct representation of the system is the vector equilibrium-correction mechanism ($VECM$) in (13), including the term in the lagged levels of the variables $\mathbf{y}_{t-1}$. In this case, if the $n \times n$ matrix $\boldsymbol{\Pi}$ has reduced rank ($r < n$),

it can be written as the product of two $n \times r$ matrices, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, as follows:

$$\Pi = \boldsymbol{\alpha}\boldsymbol{\beta}' = (\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_r) \begin{pmatrix} \boldsymbol{\beta}_1' \\ \vdots \\ \boldsymbol{\beta}_r' \end{pmatrix}$$

where α_i and β_i $(i = 1, ..., r)$ denote the columns of α and β respectively.

Matrix β contains the cointegrating vectors (one in each column), whereas matrix α contains the weights (loadings) with which each cointegrating vector enters the n equations in the VAR. Since $\beta'\mathbf{y}_{t-1}$ represents the vector of deviations of the variables from the set of r long-run equilibrium relations, the coefficients in α measure the adjustment of Δy_t to the system's long-run

equilibrium.

8(To clarify this point) Example for $n = 3$ and $r = 2$. Consider 3 non-stationary variables generated by random walk plus noise processes:

$y_t^i = \mu_t^i + \varepsilon_t^i$ with $i = 1, 2, 3$ where μ_t^i $(i = 1, 2, 3)$ are the stochastic trends and ε_t^i $(i = 1, 2, 3)$ are independent $I(0)$ processes.

If there exist two cointegrating vectors among the three variables $(r = 2)$, then the coefficients of each vector eliminate the linear combination of the stochastic trends in the y variables. Denoting with β_{i1} and β_{i2} $(i = 1, 2, 3)$ the coefficients of the two cointegrating vectors, we have:

$$\begin{aligned} \beta_{11}\mu_t^1 + \beta_{21}\mu_t^2 + \beta_{31}\mu_t^3 &= 0 \\ \beta_{12}\mu_t^1 + \beta_{22}\mu_t^2 + \beta_{32}\mu_t^3 &= 0 \end{aligned}$$

It is possible to express two stochastic trends in terms of the third, for example as:

$$\mu_t^1 = \frac{\beta_{21}\beta_{32} - \beta_{31}\beta_{22}}{\beta_{11}\beta_{22} - \beta_{21}\beta_{12}}\mu_t^3$$

$$\mu_t^2 = \frac{\beta_{31}\beta_{12} - \beta_{11}\beta_{32}}{\beta_{11}\beta_{22} - \beta_{21}\beta_{12}}\mu_t^3$$

Therefore, in the 3-variable system, there exist only 1 stochastic trend (for example, μ_t^3) which is common to all 3 non-stationary variables (μ_t^1 and μ_t^2 being simply proportional to μ_t^3).

EXAMPLE – Illustration

Example. consider a system of $n = 4$ non-stationary variables with $r = 2$ cointegrating vectors. In the $VECM$ representation of the system the term in levels, $\Pi\mathbf{y}_{t-1}$ is expressed as $\mathbf{\Pi y}_{t-1} = \begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \\ \alpha_{31} & \alpha_{32} \\ \alpha_{41} & \alpha_{42} \end{pmatrix} \begin{pmatrix} \beta_{11} & \beta_{21} & \beta_{31} & \beta_{41} \\ \beta_{12} & \beta_{22} & \beta_{32} & \beta_{42} \end{pmatrix} \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \\ y_{3t-1} \\ y_{4t-1} \end{pmatrix}$

whereby the previous period deviation of the variables from the first (second) long-run equilibrium relation, given by the first (second) cointegrating vector, enters the equations of the $VECM$ with weights given by the elements of the first (second) column of α.

4.3 A note on cointegration tests

When more than two variables are involved in the analysis, the possibility arises of multiple cointegrating vectors. In this case, the Engle-Granger estimation proce- dure is not appropriate, since a static cointegrating regression would yield a linear combination of all the valid cointegrating vectors. To test for the existence and number of cointegrating vectors, the Johansen procedure (based on maximum like- lihood estimation) is usually employed.

This approach tests various hypotheses on the rank of Π, which corresponds to the number of cointegrating vectors in the system. The "Johansen tests" exploit the fact that, if there is cointegration and therefore Π has not full rank, then $\det(\mathbf{\Pi}) = \lambda_1 \lambda_2 \dots \lambda_n = 0$ where the λ_i are the n eigenvalues (characteristic roots) of Π, and the number of

valid cointegrating vectors is equal to the number of non-zero eigenvalues.9 The 9The eigenvalues (characteristic roots) of Π are scalars λ such that $\mathbf{\Pi}\,\mathbf{x} = \lambda\,\mathbf{x}$ (*) where $\mathbf{x}$ is a non-null vector. (*) can be rewritten as $(\mathbf{\Pi} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$

where I is the identity matrix. Since $\mathbf{x} \neq \mathbf{0}$, the matrix $\mathbf{\Pi} - \lambda\mathbf{I}$ must not be of full rank (its rows must be linearly dependent), implying that $\det(\mathbf{\Pi} - \lambda\mathbf{I}) = 0$

The characteristic roots of Π are the values λ that satisfy this (polynomial) equation.

statistics proposed by Johansen (so called λ_{TRACE} and λ_{MAX}) test for the number of eigenvalues λ that are significantly different from zero (with slightly different null and alternative hypotheses):

$$\lambda_{TRACE} = -T \sum_{i=k+1}^{n} \ln\left(1 - \hat{\lambda}_i\right) \quad \text{with } H_0 : \ r \leq k \text{ and } H_A : \ r > k$$

$\lambda_{MAX} = -T \ln(1 - \hat{\lambda}_{k+1})$ with $H_0 : r = k$ and $H_A : r = k + 1$ λ_{TRACE} tests the null hypothesis that there are at most k cointegrating vectors (against the alternative that there are more than k), whereas λ_{MAX} tests the hy-

pothesis that there are k vectors against $k + 1$. The tests are performed in sequence, starting with $k = 0$ and continued until for the first time the null hypothesis can- not be rejected (the appropriate critical values of their non-standard distributions are obtained by simulation and, in general, automatically provided by econometric software packages). The corresponding value of k gives the rank of Π and then the number of valid cointegrating vectors.

The Johansen's maximum-likelihood procedure (not described here) leads also to estimation of the elements of Π with the reduced rank restriction imposed (i.e.

imposing that Π has a rank equal to the number of valid cointegrating vectors detected by the λ_{TRACE} and λ_{MAX} tests); note that, since the reduced rank of Π implies cross-equation restrictions on the VAR system, OLS estimation (equation by equation) is not implementable here, and a system procedure is needed. The estimated $(n \times n)$ matrix Π is then expressed as Π $= \alpha\,\beta'$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are $n \times r$ matrices.

However, the estimated coefficients of β and α delivered by the Johansen's proce- dure suffer from a fundamental identification problem: they are obtained from the estimated reduced rank Π by imposing an arbitrary normalization which rules out an immediate economic interpretation for the estimated cointegrating vectors. To see this point note that for any non-singular $r \times r$ matrix Θ we can express the same long-run matrix Π as $\Pi = (\boldsymbol{\alpha\Theta})\,(\boldsymbol{\Theta}^{-1}\boldsymbol{\beta}')$

The cointegrating vectors yielded by the Johansen procedure might then be linear combinations

(obviously stationary) of the r valid cointegrating vectors of economic interest. In order to identify the long-run economic relationships of interest, struc- tural long-run hypotheses must be imposed and tested on the elements of β. The idea of the test is that only r linear combinations of the variables are station- ary (i.e. the valid cointegrating vectors), whereas all different combinations are

non-stationary. If the imposed restrictions on the elements of β define linear com- binations of the variables that are "very far" from the stationary ones, then the number of estimated cointegrating vectors (i.e. non-zero eigenvalues) of the re- stricted system should be reduced. In this case, a test comparing the eigenvalues of the unrestricted and restricted systems should reject the long-run restrictions. The opposite occurs when the imposed restrictions are not binding.

An example of cointegrated VAR: output and consump- 4.4 tion J. Cochrane (Quarterly Journal of Economics, 1994) studies the dynamic effects and relative importance of permanent and transitory components in the behaviour of important macroeconomic (GNP and aggregate consumption) and financial (stock prices and dividends) series using a bivariate cointegrated VAR approach. In what follows, we present the main steps of the analysis (with results from estimation and related plots in the slides file).

Focusing on the GNP-consumption case, the starting point is a bivariate VAR system with two lags of the rates of change of consumption and GNP (Δc_t and Δy_t) specified as (constant terms omitted):

$$\begin{pmatrix}\Delta c_t \\ \Delta y_t\end{pmatrix} = \begin{pmatrix}\pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22}\end{pmatrix}\begin{pmatrix}c_{t-1} \\ y_{t-1}\end{pmatrix} + \Gamma_1\begin{pmatrix}\Delta c_{t-1} \\ \Delta y_{t-1}\end{pmatrix} + \Gamma_2\begin{pmatrix}\Delta c_{t-2} \\ \Delta y_{t-2}\end{pmatrix} + \begin{pmatrix}u_t^c \\ u_t^y\end{pmatrix}$$

where c and y are the logs of consumption and GNP, both $I(1)$ series, and u_t^c and u_t^y are the (reduced form) VAR innovations in the consumption and output equation, respectively. Cointegration between c and y with cointegrating vector $(1, -1)$ (given the strong evidence of stationarity of the consumption/GNP ratio) is imposed on the system, yielding the following $VECM$ model to be estimated:

$$\begin{pmatrix}\Delta c_t \\ \Delta y_t\end{pmatrix} = \begin{pmatrix}\alpha_c \\ \alpha_y\end{pmatrix}(c_{t-1} - y_{t-1}) + \Gamma_1\begin{pmatrix}\Delta c_{t-1} \\ \Delta y_{t-1}\end{pmatrix} + \Gamma_2\begin{pmatrix}\Delta c_{t-2} \\ \Delta y_{t-2}\end{pmatrix} + \begin{pmatrix}u_t^c \\ u_t^y\end{pmatrix}$$

Estimation of this system (Table I in the paper) yields $\hat{\alpha}_y = 0.08$ (t-stat. $= 3.45$) pointing to the importance of lagged consumption/GDP ratio in predicting future GNP movements, whereas $\hat{\alpha}_c$ is not statistically significantly different from zero.

Moreover, consumption is not far away from a random walk.

Impulse response functions (Figure I in the paper) are derived after imposing the following recursive (Choleski) identification scheme:

$$\begin{aligned} u_t^c &= v_t^c \\ u_t^y &= a\, u_t^c + v_t^y \end{aligned}$$ where v^c and v^y are “structural disturbances”. Within this framework, a distur-

bance v_t^y that affects y_t without changing c_t contemporaneously has no long-run

effect on c (since, for a random walk, the contemporaneous response to a shock coincides with the long-run response); and, given cointegration between c and y, it has no long-run effect on y. Therefore, the economic interpretation of v_t^y is that of a transitory shock, with no long-run effect on GNP and consumption. On the other hand, v_t^c has contemporaneous effects on c and y

and also long-run effects on both variables; its interpretation is then that of a permanent shock, driving GNP and consumption in the long-run.

This identification scheme finds a rationale within the theoretical framework of the "permanent income model" of consumption (with rational expectations), according to which consumption should follow a random walk and should be cointegrated with income. A simple formalization of this idea that generates a version (with no lags) of the VAR system estimated by Cochrane is given by the following three equations:

$y_t = y_t^P + \eta_t$ $\eta_t = \rho\eta_{t-1} + v_t$ $(0 < \rho < 1)$ (14) $y_t^P = y_{t-1}^P + \mu + \varepsilon_t$ (15) $c_t = y_t^P$

(16) Equation (14) defines observed GNP as the sum of two components: the unobserved "permanent income" component (y_t^P) and a stationary $AR(1)$ process (η_t), inter- preted as "transitory income"; v_t is a white noise innovation to transitory income.

Permanent income in (15) is generated by a random walk (with drift μ) plus noise stochastic process, where ε_t is a white noise uncorrelated with η_t. Finally, accord- ing to (16), consumption is equal to permanent income and is not affected by the transitory component of GNP.10 First, note that in this model consumption and income are cointegrated with coin- tegrating vector (1, -1) since $c_t - y_t = -\eta_t$ with $\eta_t \sim I(0)$. Therefore, a VECM

representation exists and may be derived by taking first differences of (14)-(16) and using the fact that $\Delta\eta_{t-1} = (1-\rho)(c_{t-1} - y_{t-1}) + v_t$. The following cointegrated VAR system is obtained

$$\begin{pmatrix}\Delta c_t \\ \Delta y_t\end{pmatrix} = \begin{pmatrix}\mu \\ \mu\end{pmatrix} + \begin{pmatrix}0 \\ 1-\rho\end{pmatrix}(c_{t-1} - y_{t-1}) + \begin{pmatrix}u_t^c \\ u_t^y\end{pmatrix}$$

10It can be shown that a standard version of the “permanent income model” implies that consumption is set in each period t at the level given by income expected in the long-run (less deterministic growth), that is $c_t = \lim_{h\to\infty}(E_t\, y_{t+h} - h\,\mu)$ (*) Using (14) and (15) above we have $E_t y_{t+h} = E_t y_{t+h}^P + \underbrace{E_t v_{t+h}}_{\circ} = y_t^P + h\mu$

which, substituted into $(*)$, yields $c_t = y_t^P$ as in (16).

where the innovations are given by:

$u_t^c = \varepsilon_t$ (17) $u_t^y = \varepsilon_t + v_t$ (18) Therefore, the model above generates a recursive structure in the relation between the VAR innovations and the structural disturbances ε_t and v_t: the innovation in consumption growth captures only disturbances to permanent income, that also affect contemporaneously the GNP growth rate, whereas transitory income shocks have a contemporaneous impact only on Δy_t.

After this identification is imposed, innovation accounting is carried out with the following notable result:

• the long-run responses of c and y to the two structural disturbances are the same. Intuitively, since the consumption/GNP ratio is stationary, in the long-run the two variables have to show the same response to any shock to restore the ratio. This point can be proved formally using a simplified version of Cochrane's $VECM$ (no constant terms and no lags), in which c follows a random walk exactly, i.e.

$\Delta c_t = u_t^c$ $\Delta y_t = \alpha_y\,(c_{t-1} - y_{t-1}) + u_t^y$ Subtracting the second equation from the first we get $\Delta c_t - \Delta y_t = -\alpha_y(c_{t-1} - y_{t-1}) + (u_t^c - u_t^y)$

$\Rightarrow c_t - y_t = (1-\alpha_u)(c_{t-1} - y_{t-1}) + (u_t^c - u_t^y)$ which is a (stationary) $AR(1)$ process for $c - y$. To derive the VMA (vector moving average) representation for Δc_t and Δy_t we first express $c - y$ as $[1 - (1-\alpha_y)L](c_t - y_t) = u_t^c - u_t^y$

$\Rightarrow c_t - y_t = \frac{1}{1-(1-\alpha_u)L}\left(u_t^c - u_t^y\right)$ and then substitute it into the equation for Δy_t, obtaining Δy_t $= \alpha_y \frac{1}{1-(1-\alpha_x)L} L(u_t^c - u_t^y) + u_t^y$

$= \frac{\alpha_y}{1-(1-\alpha_y)L} L u_t^c + \left(1 - \frac{\alpha_y}{1-(1-\alpha_y)L} L\right) u_t^y$ The $VMA(\infty)$ representation of the bivariate system is then $\begin{pmatrix} \Delta c_t \\ \Delta y_t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{\alpha_y}{1-(1-\alpha_v)L} L & 1 - \frac{\alpha_y}{1-(1-\alpha_v)L} L \end{pmatrix} \begin{pmatrix} u_t^c \\ u_t^y \end{pmatrix}$

The long-run response of c and y to the two innovations is found simply by taking $L = 1$:

$$\begin{pmatrix} \Delta c_t \\ \Delta u_t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_t^c \\ u_t^y \end{pmatrix}$$

Using the recursive identification scheme in (17) and (18) we obtain the re- sponses to the permanent and transitory structural disturbances as $\begin{pmatrix} \Delta c_t \\ \Delta y_t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_t \\ v_t \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \varepsilon_t \\ v_t \end{pmatrix}$ showing that in the long run c and y change by the same amount in response to the permanent shock (ε_t) and to the transitory disturbance (v_t).

Exercise.

Consider the following alternative "permanent income" model in which:

(i) permanent income y^P is directly observable (by assumption);

(ii) consumption adjusts gradually to permanent income.

Formally, the model is:

$c_t = y_t^P + \zeta_t$ with $\zeta_t = \rho\zeta_{t-1} + \eta_t$ $(0 < \rho < 1)$ $y_t^P = y_{t-1}^P + \mu + \varepsilon_t$ where η_t and ε_t are uncorrelated.

1. Find the cointegrated VECM representation of the system formed by Δc_t and Δy_t^P;

2. find the (recursive) identification scheme appropriate to recover the impulse response functions of c and y^P to permanent and transitory disturbances.

Appendix B: The problem of multiple long-run rela- 4.5 tions The modelling of many economic relationships involves specification and hypothesis testing on both their long-run links (interpreted as "equilibrium" relations) and the specification of the short-run dynamics (interpreted as the "adjustment" process towards the long-run equilibrium). Often, the empirical analysis is carried out using either a single-equation approach, focusing on the explanation of a variable y in terms of a set of determinants $\mathbf{x}$, or a multivariate system approach, in which y and the variables in $\mathbf{x}$ are jointly modelled.

Taking the first route, the (so called "general-to-specific") modelling strategy starts with an unrestricted dynamic ADL model of y in terms of its own lags and current and lagged values of the variables in $\mathbf{x}$ (see section 1) Such general model is then reduced by testing a series of restrictions on coefficients (e.g. exclusion restrictions or difference restrictions). From the resulting final form of the dynamic model (usually including variables in first and higher-order differences and in levels), a long-run solution can be computed and interpreted as the equilibrium relationship among the variables. Deviations from this equilibrium can enter the equation as an important element of the short-run adjustment dynamics (the equilibrium-correction

term).

This approach may run into various problems due to:

• the presence of simultaneity between the dependent variable y and (some of) the regressors in $\mathbf{x}$; this problem may be solved by means of instrumental variable estimation techniques, applying formal tests for checking the validity of the instrument set;

• existence of multiple long-run relations among the variables under analy- sis (y and the $x's$). If this is the case, the long-run solution derived from the single-equation estimate cannot bear the interpretation of an equilibrium relation, being a combination of the multiple long-run relations linking the variables.

A multivariate approach, treating all relevant variables as potentially en- dogenous and directly addressing the issue of the existence of multiple long-run equilibrium relations, may be a more effective modelling strategy. The following example, focused on the modelling of a money demand relation, illustrates this point.

4.5.1 An example: modelling money demand

When an economic behavioural function (e.g. money demand) is part of a larger system of equations involving the variables of interest (e.g. money balances, income, interest rates, inflation, etc., all assumed to be non-stationary, $I(1)$, series in the sample available for estimation), the possibility arises of the existence multiple long-run relations, empirically interpreted as cointegrating vectors and capturing the equilibrium links among the whole set of variables.

In the context of money demand estimation, consider the following pair of long-run relations:

$m - p = \theta y$ $R^m = \gamma R^b$ The first describes a long-run money demand for transaction purposes, positing a long-run equilibrium relation between real money balances (including bank de- posits) and income (a scale variable), whereas the second captures a long-run equi- librium relation between the return on an alternative (short-run) asset (R^b) and the own-return on money ($\mathbb{R}^m$), the latter set by banks as a "mark-down" (i.e. $\gamma < 1$)

on the former. Now let the short-run dynamics of all the variables involved be determined by the following set of equations (all coefficients positive):

$$\Delta(m-p)_t = a_1\Delta(m-p)_{t-1} - a_2\left[(m-p) - \theta y\right]_{t-1} + a_3\left(R^m - \gamma R^b\right)_{t-1} + u_{1t}$$
$$\Delta y_t = b_1\Delta y_{t-1} + b_2[(m-p) - \theta y]_{t-1} + u_{2t}$$

$$\Delta R^m_t = c_1\Delta R^m_{t-1} - c_2\left(R^m - \gamma R^b\right)_{t-1} + u_{3t}$$

$\Delta R^b_t = d_1\Delta R^b_{t-1} + u_{4t}$ For simplicity, we assume $E(u_{it}u_{jt}) = 0$ for $i \neq j$, which rules out the simultaneity issue. Estimation of a single equation for money balances (interpreted as a money demand function) yields the following result, observationally equivalent to the first equation of the system above:

$$\Delta(m-p)_t = \delta_1\Delta(m-p)_{t-1} - \delta_2(m-p)_{t-1} + \delta_3 y_{t-1} + \delta_4 R^m_{t-1} - \delta_5 R^b_{t-1} + \varepsilon_t$$

from which the following long-run solution is derived (setting $\Delta(m-p)_t = \Delta(m-p)_t$ $p_{t-1} = 0$ and ignoring the disturbance ε_t):

$m - p = \frac{\delta_3}{\delta_2}y + \frac{\delta_4}{\delta_2}R^m - \frac{\delta_5}{\delta_2}R^b$ This relationship is potentially misinterpreted as a long-run money demand function

involving all variables. Apparently, in the long-run money demand reacts to its own rate of return R^m (with a positive elasticity measured by $\frac{\delta_4}{\delta_2}$) and to the return on an alternative asset R^b (with

a negative elasticity measured by $-\frac{\delta_5}{\delta_2}$), in addition to

showing a positive reaction to income y (with elasticity $\frac{\delta_3}{\delta_2}$). The reactions to $\mathbb{R}^m$ and R^b are spurious results, since the true long-run money demand depends only on income.

Carrying out cointegration tests within the four-variable system would allow for detection and identification of the multiple cointegrating vectors, to be included in the specification of the short-run dynamic adjustment of all variables in the system to the estimated long-run equilibrium relationships.

Cointegration in a VAR: bivariate case Two $I(1)$ variables, y and x, generated by:

$$v_t - \gamma x_t = v_{1t}$$
$$v_{1t} = \rho v_{1t-1} + \varepsilon_{1t}$$
$$\delta y_t + x_t = \theta_1 y_{t-1} + \theta_2 x_{t-1} + v_{2t}$$
$$v_{2t} = v_{2t-1} + \varepsilon_{2t}$$

If $|\rho| < 1$ the two variables are cointegrated of order (1,1) since the first equation describes a stationary linear combination of y and x with cointegrating vector $(1, -\gamma)$.

Then an equilibrium-correction form representation exists:

$$\Delta y_t = -\frac{1-\rho}{1+\gamma\delta}(y_{t-1} - \gamma x_{t-1}) + \frac{\gamma\theta_1}{1+\gamma\delta}\Delta y_{t-1} + \frac{\gamma\theta_2}{1+\gamma\delta}\Delta x_{t-1} + \frac{\varepsilon_{1t} + \gamma\varepsilon_{2t}}{1+\gamma\delta}$$

$$\Delta x_t = \delta\frac{1-\rho}{1+\gamma\delta}(y_{t-1} - \gamma x_{t-1}) + \frac{\theta_1}{1+\gamma\delta}\Delta y_{t-1} + \frac{\theta_2}{1+\gamma\delta}\Delta x_{t-1} + \frac{\varepsilon_{2t} - \delta\varepsilon_{1t}}{1+\gamma\delta}$$

The existence of this representation crucially depends on the magnitude of ρ: if $\rho = 1$, y and x are not cointegrated and the equilibrium-correction terms vanish.

Then an equilibrium-correction form representation exists:

$$\Delta y_t = -\frac{1-\rho}{1+\gamma\delta}(y_{t-1} - \gamma x_{t-1}) + \frac{\gamma\theta_1}{1+\gamma\delta}\Delta y_{t-1} + \frac{\gamma\theta_2}{1+\gamma\delta}\Delta x_{t-1} + \frac{\varepsilon_{1t} + \gamma\varepsilon_{2t}}{1+\gamma\delta}$$

$$\Delta x_t = \delta\frac{1-\rho}{1+\gamma\delta}(y_{t-1} - \gamma x_{t-1}) + \frac{\theta_1}{1+\gamma\delta}\Delta y_{t-1} + \frac{\theta_2}{1+\gamma\delta}\Delta x_{t-1} + \frac{\varepsilon_{2t} - \delta\varepsilon_{1t}}{1+\gamma\delta}$$

The existence of this representation crucially depends on the magnitude of ρ: if $\rho = 1$, y and x are not cointegrated and the equilibrium-correction terms vanish.

In matrix form:

$$\begin{pmatrix}\Delta y_t \\ \Delta x_t\end{pmatrix} = \Pi\begin{pmatrix}y_{t-1} \\ x_{t-1}\end{pmatrix} + \Gamma\begin{pmatrix}\Delta y_{t-1} \\ \Delta x_{t-1}\end{pmatrix} + \begin{pmatrix}u_{1t} \\ u_{2t}\end{pmatrix}$$

Simpler example of data generating process in section 2.3.1:

$y_t - \gamma x_t = u_{1t}$ $u_{1t} = \rho u_{1t-1} + \varepsilon_{1t}$ $x_t = x_{t-1} + \mathcal{E}_{2t}$ with ε_{1t} and ε_{2t} uncorrelated white noise.

In this case the reduced form (reparameterized in first differences) is:

$$\Delta v_t = -(1-\rho)(v_{t-1} - \gamma x_{t-1}) + (\varepsilon_{1t} + \gamma \varepsilon_{2t})$$
$$\Delta x_t = \varepsilon_{2t}$$

Modelling economic relationships in a multivariate framework Strategies to model empirically long-run equilibrium relationships short-run adjustment dynamics among several economic and variables:

single-equation approach (explanation of a variable y in terms of a set of determinants x), following a "general-to-specific" modelling procedure:

⇒ estimation of a long-run relation and an “equilibrium-correction” form for the short-run dynamics;

- multivariate system approach (y and the variables in x are jointly) modelled)

Problems of the single-equation approach:

simultaneity between y and (some of) the regressors in x; this problem may be solved by means of IV estimation techniques, with formal tests for checking the validity of the instrument set;

multiple long-run relations among y and the x's. If this is the case, the long-run solution derived from the single-equation estimate cannot bear the interpretation of an equilibrium relation, being a combination of the underlying multiple long-run relations.

The problem of multiple long-run relations (money demand example) Variables under study: $m - p$, y, R^m and R^b "True" long-run equilibrium relations:

$$m - p = \theta y$$
$$R^m = \gamma R^b$$

Short-run dynamics:

$$\Delta(m-p)_t = a_1 \Delta(m-p)_{t-1} - a_2[(m-p) - \theta y]_{t-1} + a_3(R^m - \gamma R^b)_{t-1} + u_{1t}$$
$$\Delta y_t = b_1 \Delta y_{t-1} + b_2[(m-p) - \theta y]_{t-1} + u_{2t}$$

$$\Delta R^m_t = c_1 \Delta R^m_{t-1} - c_2(R^m - \gamma R^b)_{t-1} + u_{3t}$$
$$\Delta R^b_t = d_1 \Delta R^b_{t-1} + u_{4t}$$

Assumption of $E(u_{it}u_{jt}) = 0$ for $i \neq j$ rules out simultaneity.

Estimated single-equation for money demand:

$$\Delta(m-p)_t = \delta_1 \Delta(m-p)_{t-1} - \delta_2(m-p)_{t-1} + \delta_3 y_{t-1} + \delta_4 R^m_{t-1} - \delta_5 R^b_{t-1} + \varepsilon_t$$

from which the following long-run solution is derived:

$$m - p = \frac{\delta_3}{\delta_2} y + \frac{\delta_4}{\delta_2} R^m - \frac{\delta_5}{\delta_2} R^b$$

Cointegration in a VAR: multivariate case n-dimensional VAR process for y_t:

$$\mathbf{y}_t = \delta + \mathbf{A}_1 \mathbf{y}_{t-1} + \mathbf{A}_2 \mathbf{y}_{t-2} + \mathbf{u}_t$$

reduced-form equilibrium-correction:

$$\Delta \mathbf{y}_t = \delta + (\mathbf{A}_1 + \mathbf{A}_2 - \mathbf{I})\mathbf{y}_{t-1} - \mathbf{A}_2 \Delta \mathbf{y}_{t-1} + \mathbf{u}_t$$

$$\Rightarrow \Delta \mathbf{y}_t = \delta + \mathbf{\Pi} \mathbf{y}_{t-1} + \mathbf{\Gamma} \Delta \mathbf{y}_{t-1} + \mathbf{u}_t$$

The rows of Π are not linearly independent if the variables in y are cointegrated The rank of Π provides the number of the cointegrating vectors:

(a) if rank(Π) = 0 ⇒ Π = 0, no cointegrating vector exist (n) stochastic trends in the system)

(b) if rank(Π) = n (full rank) ⇒ there is a set of n independent restrictions on the long-run values of the elements in y. To derive the long-run solution, setting $\Delta y_t = \Delta y_{t-1} = 0$ and $\mathbf{u}_t = \mathbf{0}$ we get $\pi_{11} y_{1t-1} + \pi_{12} y_{2t-1} + \ldots + \pi_{1n} y_{nt-1} = -\delta_1$

$$\pi_{21} y_{1t-1} + \pi_{22} y_{2t-1} + \ldots + \pi_{2n} y_{nt-1} = -\delta_2$$
$$\pi_{n1} y_{1t-1} + \pi_{n2} y_{2t-1} + \ldots + \pi_{nn} y_{nt-1} = -\delta_n$$

If Π is of full rank this system admits a solution for the long-run values of $\mathbf{y}_t$: $\bar{\mathbf{y}}_t$, with $E(\mathbf{y}_t) = \bar{\mathbf{y}}_t$ $\bar{\mathbf{v}} = -\mathbf{\Pi}^{-1}\delta$

All variables are stationary and no stochastic trend is present in the system.

(b) if rank(Π) = $r < n$ ⇒ there are r cointegrating vectors and the number of common stochastic trends in the system is $n - r$.

(To clarify this point) Example for $n = 3$ and $r = 2$.

Consider 3 non-stationary variables generated by random walk plus noise processes:

$v_t^i = \mu_t^i + \varepsilon_t^i$ with $i = 1, 2, 3$ where $\mu_t^i = \mu_{t-1}^i + \eta_t^i$ with $i = 1, 2, 3$

are the stochastic trends and ε_t^i $(i = 1, 2, 3)$ are independent $I(0)$ processes.

If $r = 2$, then the coefficients of each vector (β_{i1} and β_{i2} for $i = 1, 2, 3$) eliminate the linear combination of the stochastic trends in the y variables:

$$\beta_{11}\mu_t^1 + \beta_{21}\mu_t^2 + \beta_{31}\mu_t^3 = 0$$
$$\beta_{12}\mu_t^1 + \beta_{22}\mu_t^2 + \beta_{32}\mu_t^3 = 0$$

It is then possible to express two stochastic trends in terms of the third, for example as:

$$\mu_t^1 = \frac{\beta_{21}\beta_{32} - \beta_{31}\beta_{22}}{\beta_{11}\beta_{22} - \beta_{21}\beta_{12}} \mu_t^3$$

$$\mu_t^2 = \frac{\beta_{31}\beta_{12} - \beta_{11}\beta_{32}}{\beta_{11}\beta_{22} - \beta_{21}\beta_{12}} \mu_t^3$$

(c) if rank(Π) = $r < n$ ⇒ there are r cointegrating vectors and the number of common stochastic trends in the system is $n - r$.

⇒ vector equilibrium correction mechanism (VECM):

$$\Delta \mathbf{y}_t = \delta + \alpha \beta' \mathbf{y}_{t-1} + \mathbf{\Gamma} \Delta \mathbf{y}_{t-1} + \mathbf{u}_t$$

$$\Pi = \underline{\alpha} \quad \underline{\beta'} \text{ n×r r×n} = \underbrace{(\boldsymbol{\alpha}_1 \ \dots \ \boldsymbol{\alpha}_r)}_{\text{loadings}} \begin{pmatrix} \boldsymbol{\beta}_1' \\ \\ \boldsymbol{\beta}_r' \end{pmatrix}$$

cointegrating vectors

Example for $n = 4$ and $r = 2$ $\mathbf{\Pi} \mathbf{y}_{t-1} = \begin{pmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \\ \alpha_{31} & \alpha_{32} \\ \alpha_{41} & \alpha_{42} \end{pmatrix} \begin{pmatrix} \beta_{11} & \beta_{21} & \beta_{31} & \beta_{41} \\ \beta_{12} & \beta_{22} & \beta_{32} & \beta_{42} \end{pmatrix} \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \\ y_{3t-1} \\ y_{4t-1} \end{pmatrix}$

$$= \begin{pmatrix} \alpha_{11} \\ \alpha_{21} \\ \alpha_{31} \\ \alpha_{41} \end{pmatrix} \boldsymbol{\beta}_1' \mathbf{y}_{t-1} + \begin{pmatrix} \alpha_{12} \\ \alpha_{22} \\ \alpha_{32} \\ \alpha_{42} \end{pmatrix} \boldsymbol{\beta}_2' \mathbf{y}_{t-1}$$

In the bivariate example above ($n = 2, r = 1$):

$$\begin{pmatrix} -\frac{1-\rho}{1+\gamma\delta} & \frac{1-\rho}{1+\gamma\delta}\gamma \\ \delta\frac{1-\rho}{1+\gamma\delta} & -\frac{1-\rho}{1+\gamma\delta}\delta\gamma \end{pmatrix} \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix} =$$

In the bivariate example above ($n = 2, r = 1$):

$$\begin{pmatrix} -\frac{1-\rho}{1+\gamma\delta} & \frac{1-\rho}{1+\gamma\delta}\gamma \\ \delta\frac{1-\rho}{1+\gamma\delta} & -\frac{1-\rho}{1+\gamma\delta}\delta\gamma \end{pmatrix} \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix} = \begin{pmatrix} -\frac{1-\rho}{1+\gamma\delta} \\ \delta\frac{1-\rho}{1+\gamma\delta} \end{pmatrix} \begin{pmatrix} 1 & -\gamma \end{pmatrix} \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix}$$

Testing for cointegration (Johansen) The Johansen test exploits the fact that $\det(\mathbf{\Pi}) = \lambda_1\lambda_2 \dots \lambda_n = 0$ where the λ_i are the n eigenvalues (characteristic roots) of Π.

$\Rightarrow$ the number of valid cointegrating vectors is equal to the number of non-zero eigenvalues.

The procedure proposed by Johansen provides estimates of the eigenvalues λ and test statistics for the number of eigenavalues that are significantly different from zero (λ_{TRACE} and λ_{MAX})

The Johansen λ_{TRACE} and λ_{MAX} test statistics are:

$\lambda_{TRACE} = -T \sum \ln(1 - \hat{\lambda}_i)$ with H_0: $r \leq k$ and H_A: $r > k$ $i = k + 1$ $\lambda_{MAX} = -T \ln(1 - \hat{\lambda}_{k+1})$ with H_0: $r = k$ and H_A: $r = k + 1$

Johansen's ML procedure leads also to estimation of the elements of Π with the reduced rank restriction imposed. The estimated $(n \times n)$ matrix Π is then expressed as $\Pi = \alpha\beta'$

(Long-run) identification problem:

$\Rightarrow$ the estimated coefficients of β and α are obtained by imposing an arbitrary normalization which rules out an immediate economic interpretation for the estimated cointegrating vectors. Infact, for any non-singular $r \times r$ matrix Θ we can express the same long-run matrix Π as $\Pi = (\alpha\mathbf{\Theta})(\mathbf{\Theta}^{-1}\beta')$

The cointegrating vectors yielded by the Johansen procedure might then be linear combinations (obviously stationary) of the r valid cointegrating vectors of economic interest.

$\Rightarrow$ structural long-run hypotheses must be imposed and tested on the elements of β.

A cointegrated VAR: Cochrane (1994) Aim: study dynamic effects and relative importance of permanent and transitory components in the behaviour of some macroeconomic (GNP and

consumption) and financial (stock prices and dividends) series using a bivariate cointegrated VAR.

GNP-consumption case. Estimated bivariate VAR system:

$$\begin{pmatrix}\Delta c_t\\ \Delta y_t\end{pmatrix} = \begin{pmatrix}\pi_{11} & \pi_{12}\\ \pi_{21} & \pi_{22}\end{pmatrix}\begin{pmatrix}c_{t-1}\\ v_{t-1}\end{pmatrix} +$$

$$+\Gamma_1\begin{pmatrix}\Delta C_{t-1}\\ \Delta V_{t-1}\end{pmatrix} + \Gamma_2\begin{pmatrix}\Delta C_{t-2}\\ \Delta V_{t-2}\end{pmatrix} + \begin{pmatrix}u_t^c\\ u_t^y\end{pmatrix}$$

where c and y are $I(1)$ series.

Cointegration between c and y with cointegrating vector $(1, -1)$ is imposed, yielding the VECM model:

$$\begin{pmatrix}\Delta c_t\\ \Delta y_t\end{pmatrix} = \begin{pmatrix}\alpha_c\\ \alpha_v\end{pmatrix}(c_{t-1} - y_{t-1}) +$$
$$+\Gamma_1\begin{pmatrix}\Delta c_{t-1}\\ \Delta v_{t-1}\end{pmatrix} + \Gamma_2\begin{pmatrix}\Delta c_{t-2}\\ \Delta v_{t-2}\end{pmatrix} + \begin{pmatrix}u_t^c\\ u_t^y\end{pmatrix}$$

Estimation of this system yields $\hat{\alpha}_v = 0.08$ (t-stat. $= 3.45$) pointing to the importance of lagged consumption/GDP ratio in predicting future GNP movements.

Moreover, consumption is nearly a random walk.

TABLE I CONSUMPTION AND GNP REGRESSIONS v_i denotes log real GNP. c_i denotes log nondurable + services consumption.

Δ denotes first differences, $\Delta y_t = y_t - y_{t-1}$. Data sample 1947:1–1989:3.

1. Vector autoregression Right-hand variable R^2 Left-hand $c_{t-1} - y_{t-1}$ Δc_{t-1} Δc_{t-2} Δy_{t-1} Δy_{t-2} p-valueb

variable const.

Δc_t coeff.

0.06 −0.43 −0.02 0.07 −0.02 0.09 −0.02 −0.19 0.7 −0.49 −1.23 0.90 1.91 −0.40 t-stat.

0.29 p-valuea (percent) 5.19 0.08 0.52 0.16 0.22 0.27 0.14 Δv_ℓ coeff.

3.49 3.45 3.81 1 12 2.74 1.89 0.0 t-stat.

p-valuea (percent) 1.8

TABLE I
CONSUMPTION AND GNP REGRESSIONS

y_t denotes log real GNP. c_t denotes log nondurable + services consumption.
Δ denotes first differences, $\Delta y_t = y_t - y_{t-1}$. Data sample 1947:1–1989:3.

1. Vector autoregression

Left-hand variable	Right-hand variable						R^2
	const.	$c_{t-1} - y_{t-1}$	Δc_{t-1}	Δc_{t-2}	Δy_{t-1}	Δy_{t-2}	p-value[b]
Δc_t coeff.	−0.43	−0.02	0.07	−0.02	0.09	−0.02	0.06
t-stat.	−0.49	−1.23	0.90	−0.19	1.91	−0.40	0.7

Figure 1.70: Course figure, source page 25, crop 1.

"Permanent" c shock	97	30	100	15
"Temporary" y shock	3	70	0	85

3. Univariate autoregression

Left-hand variable	Right-hand variable					R^2
Δy_t	const.	Δy_{t-1}	Δy_{t-2}	Δy_{t-3}	Δy_{t-4}	p-value[b]
Coefficient	0.56	0.33	0.19	−0.11	−0.11	0.18
t-statistic	4.82	4.17	2.39	−1.37	−1.36	0.0

a. Percent of replications with a coefficient farther from 0, under the null that the coefficient = 0 and c/y has a unit root (bootstrap).
b. Percent probability value of an F-test for the joint significance of the right-hand variables.

Figure 1.71: Course figure, source page 25, crop 2.

Impulse response functions are derived after imposing the following recursive (Choleski) identification scheme:

$u_t^c = v_t^c$ $u_t^y = au_t^c + v_t^y$ where v^c and v^y are "structural disturbances".

Theoretical motivation:

"permanent income model" of consumption (with rational expectations), according to which

- consumption should follow a random walk

- consumption should be cointegrated with income.

$u_t^c = v_t^c$ $u_t^y = au_t^c + v_t^y$ 1. a disturbance v_t^y that affects y without changing c contemporaneously has no long-run effect on c; and, given cointegration between c and y, it has no long-run effect on y.

$\Rightarrow$ the economic interpretation of v_t^y is that of a transitory shock, with no long-run effect on GNP and consumption.

2. v_t^c has contemporaneous effects on c and y and also long-run effects on both variables $\Rightarrow$ its interpretation is then that of a permanent shock, driving GNP and consumption in the long-run.

Example of "permanent income model" $v_t = v_t^P + \eta_t$ $v_t^P = v_{t-1}^P + \mu + \varepsilon_t$ $c_t = v_t^P$ with η_t and ε_t white noise uncorrelated processes Implications:

• c_t is a random walk (with drift) $\Delta c_t = \mu + \varepsilon_t$ • c_t and y_t are cointegrated with cointegrating vector $(1, -1)$ $c_t - v_t = -\eta_t \sim I(0)$

The cointegrated VECM is $\begin{pmatrix} \Delta c_t \\ \Delta v_t \end{pmatrix} = \begin{pmatrix} \mu \\ \mu \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} (c_{t-1} - y_{t-1}) + \begin{pmatrix} u_t^c \\ u_t^y \end{pmatrix}$

where innovations are given by:

$$u_t^c = \varepsilon_t$$
$$u_t^y = \varepsilon_t + \eta_t$$

Notable result from innovation accounting:

the long-run responses of c and y to the two structural disturbances are the same

A. Multivariate 1.2 r Consumption response GNP response 1.0 Response to 1 a shack 9.0 a shack 7.0 a shack Consumption shock 0.2 GNP shock 0.0 Quarters after shock

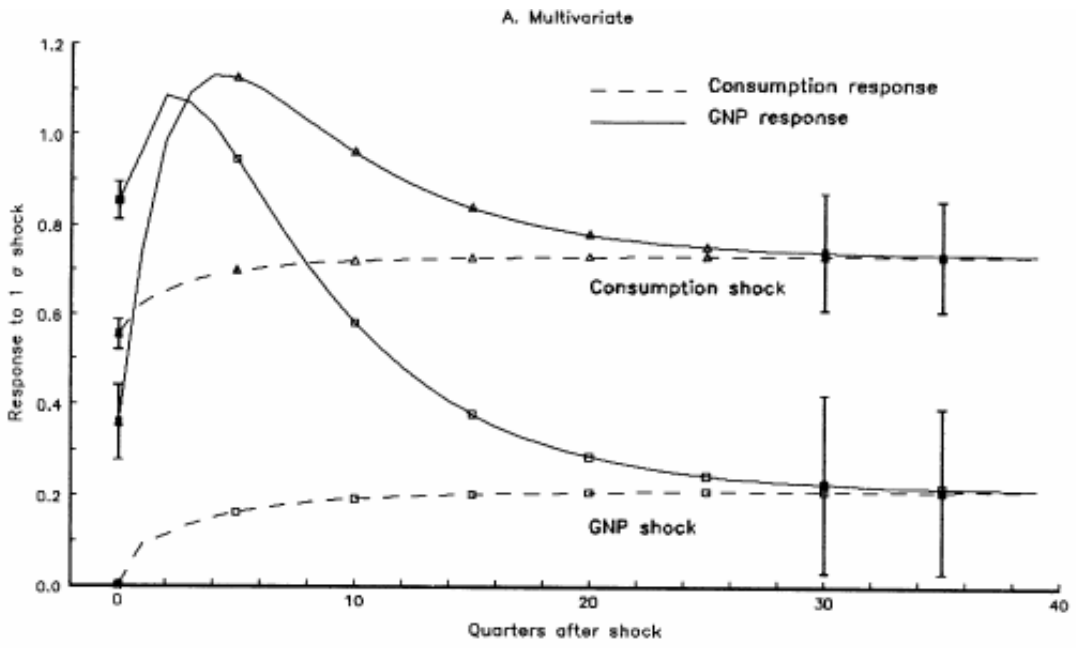


Figure 1.72: Course figure, source page 32, crop 1.

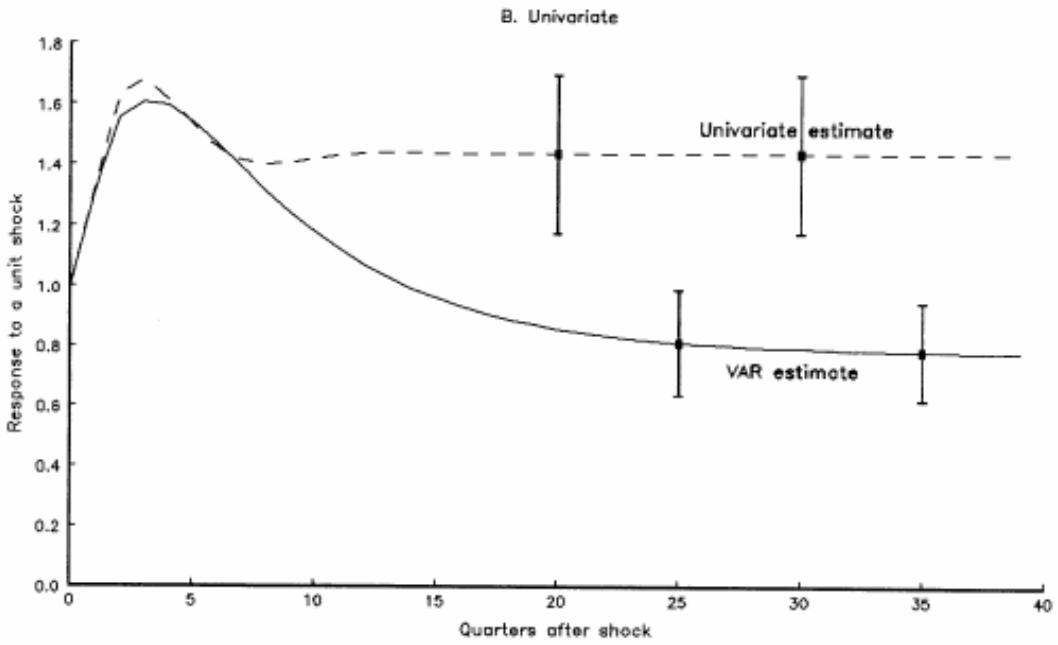


FIGURE I
Impulse-Response Functions for Consumption and GNP

Top panel: Response of consumption (c) and GNP (y) to one-standard-deviation shocks in the consumption-GNP VAR (Table I).

Bottom panel: Response of GNP to a unit univariate GNP shock. The "univariate estimate" is based on the regression of GNP growth on past GNP growth, panel 3 of Table I; the "VAR estimate" is based on the consumption-GNP VAR, panel 1 of Table I.

Bars show bootstrap one standard error bands.

Figure 1.73: Course figure, source page 32, crop 2.

Notable results from innovation accounting:

the long-run responses of c and y to the two structural disturbances are the same.

Simplified version of Cochrane's VECM, in which c is an exact random walk and lags are limited to the first order, i.e.

$\Delta c_t = u_t^c$ $\Delta v_t = \alpha_v(c_{t-1} - v_{t-1}) + u_t^y$ Subtracting the second equation from the first we get $\Delta c_t - \Delta y_t = -\alpha_v(c_{t-1} - v_{t-1}) + (u_t^c - u_t^y)$

$\Rightarrow c_t - y_t = (1 - \alpha_v)(c_{t-1} - v_{t-1}) + (u_t^c - u_t^y)$ which is a (stationary) $AR(1)$ process for $c - y$.

To derive the VMA (vector moving average) representation for Δc_t and Δy_t we express $c - y$ as $[1 - (1 - \alpha_v)L](c_t - v_t) = u_t^c - u_t^y \Rightarrow c_t - y_t = \frac{1}{1-(1-\alpha_v)L}(u_t^c - u_t^y)$

and then substitute it into the equation for Δy_t, obtaining $\Delta y_t = \alpha_y \frac{1}{1-(1-\alpha_y)L} L(u_t^c - u_t^y) + u_t^y$

$$= \frac{\alpha_y}{1 - (1 - \alpha_y)L} L u_t^c + \left(1 - \frac{\alpha_y}{1 - (1 - \alpha_y)L} L\right) u_t^y$$

The $VMA(\infty)$ representation of the bivariate system is then $\begin{pmatrix} \Delta c_t \\ \Delta y_t \end{pmatrix} =$

$$\begin{pmatrix} 1 & 0 \\ \frac{\alpha_y}{1-(1-\alpha_y)L} & 1-\frac{\alpha_y}{1-(1-\alpha_y)L}L \end{pmatrix} \begin{pmatrix} u_t^c \\ u_t^y \end{pmatrix}$$

The long-run response of c and y to the two innovations is found simply by taking $L = 1$:

$$\begin{pmatrix} \Delta c_t \\ \Lambda v_t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_t^c \\ u_t^y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_t \\ v_t \end{pmatrix}$$

$= \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \varepsilon_t \\ v_t \end{pmatrix}$ showing the same long-run responses

Chapter 2

Exam and Exercise Materials

2.0.1 Time series Solutions to exercises 1

Time series econometrics Solutions to Exercises (1) Exercise (Univariate Time Series models, p. 17) Suppose that the stochastic process $\{y_t\}$ is generated by two unobserved components $\{x_t\}$ and $\{z_t\}$, defined as:

$x_t = v_t + v_{t-1}$ with $v_t \sim WN(0,1)$ $z_t \sim WN(0,1)$ independent of v_t $y_t = x_t + z_t$ (*) 1. Prove that the variance and the autocovariances of $\{y_t\}$ in (*) have the features typical of an $MA(1)$ process.

2. Now, suppose that you want to represent $\{y_t\}$ by means of a specific $MA(1)$ process of the following form:

$y_t = \varepsilon_t + \alpha\,\varepsilon_{t-1}$ with $\varepsilon_t \sim WN(0,\sigma_\varepsilon^2)$ (**) Find α and σ_ε^2 such that the variance and the autocovariances of $\{y_t\}$

obtained from $(*)$ are equal to those obtained from $(**)$.

Answer

Answers.

1. For the process in $(*)$ we have:

$E(y_t) = 0$ $var(y_t) = var(v_t) + var(v_{t-1}) + var(z_t) = 3$ $cov(y_t, y_{t-1}) = var(v_{t-1}) = 1$ $cov(y_t, y_{t-k}) = 0$ for $k > 1$ Only the autocovariance at the first lag is different from zero, which is a typical feature of $MA(1)$ processes.

2. From the $MA(1)$ representation in (**) we get:

$$E(y_t) = 0$$
$$var(y_t) = var(\varepsilon_t) + \alpha^2 var(\varepsilon_{t-1}) = (1+\alpha^2)\sigma_\varepsilon^2$$
$$cov(y_t, y_{t-1}) = \alpha var(\varepsilon_{t-1}) = \alpha\sigma_\varepsilon^2$$

$cov(y_t, y_{t-k}) = 0$ for $k > 1$

Equating $var(y_t)$ and $cov(y_t, y_{t-1})$ obtained from (*) and from (**) yields the following system of two equations:

$3 = (1+\alpha^2)\sigma_\varepsilon^2$ $1 = \alpha\sigma_\epsilon^2$ that can be solved for the parameters α and σ_ε^2. In particular, substituting for σ_ε^2 into the first equation we get

$\alpha^2 - 3\alpha + 1 = 0$ that yields the following two solutions (with approximation):

$\alpha = \begin{cases} 2.62 \\ 0.38 \end{cases}$ with the corresponding values for σ_ε^2 (again with approximation):

$\sigma_\varepsilon^2 = \begin{cases} 0.38 \\ 2.62 \end{cases}$ Therefore, the process $\{y_t\}$ has the following two $MA(1)$ representa- tions, with the same variance and autocovariance structure:

$$y_t = \varepsilon_t + 2.62\varepsilon_{t-1} \quad \text{with } \varepsilon_t \sim WN(0, 0.38)$$
$$y_t = \varepsilon_t + 0.38\varepsilon_{t-1} \quad \text{with } \varepsilon_t \sim WN(0, 2.62)$$

Note

(**Note.** only the second $MA(1)$ representation is invertible).

2

Exercise (Univariate Time Series models, p. 17) $AR(1)$ plus noise process. Let the process $\{y_t\}$ be the sum of two components, $\{x_t\}$ and $\{v_t\}$:

$(*)$ $y_t = x_t + v_t$ with $\{v_t\}$ being a white noise process, $v_t \sim WN(0, \sigma_v^2)$, and $\{x_t\}$ being a stationary $AR(1)$ process $(**)$ $x_t = \theta x_{t-1} + u_t$

$|\theta| < 1$ where $\{u_t\}$ is a white noise process, $u_t \sim WN(0, \sigma_u^2)$. The two white noise processes $\{v_t\}$ and $\{u_t\}$ are uncorrelated for all t.

1. Check that the process $\{y_t\}$ is (weakly) stationary.

2. Find the autocorrelation function (ACF) of $\{y_t\}$ and evaluate the effect of $\{v_t\}$ on the autocorrelation pattern of the process.

Answer

Answers.

1. From $(**)$, recalling that under stationarity of $\{x_t\}, E(x_t) = E(x_{t-1})$ and that $E(u_t) = 0$:

$E(x_t) = \theta E(x_t) + E(u_t) \implies E(x_t) = 0$ Then, since $E(v_t) = 0$, $E(y_t) = E(x_t) + E(v_t) = 0$ so the unconditional mean is finite and time-invariant.

To compute the variance (γ_0) and the autocovariances (γ_k) of $\{y_t\}$ recall that, under stationarity of $\{x_t\}$, $var(x_t) = var(x_{t-1})$, so we have $var(x_t) = \theta^2 var(x_t) + \sigma_u^2 \implies var(x_t) = \frac{\sigma_u^2}{1-\theta^2}$

(note that $cov(x_{t-1}, u_t) = 0$, since x_{t-1} depends on past u_{t-i}, all uncor- related with u_t), and $cov(x_t, x_{t-1}) = cov(\theta x_{t-1} + u_t, x_{t-1}) = \theta var(x_t)$ Then, $\gamma_0 \equiv var(y_t) = var(x_t) + var(v_t) = \frac{\sigma_u^2}{1-\rho^2} + \sigma_v^2$

finite and time-invariant (note that $cov(x_t, v_t) = 0$, since x_t depends on current and past u_{t-i}, all uncorrelated with v_t). The autocovariances of $\{y_t\}$ are:

$$\gamma_1 \equiv cov(y_t, y_{t-1}) = cov(x_t + v_t, x_{t-1} + v_{t-1}) = cov(x_t, x_{t-1}) = \theta var(x_t)$$
$$= \frac{\theta}{1-\theta^2}\sigma_u^2$$

$$\gamma_2 \equiv cov(y_t, y_{t-2}) = cov(x_t + v_t, x_{t-2} + v_{t-2})$$
$$= cov\left(\theta^2 x_{t-2} + \theta u_{t-1} + u_t, x_{t-2} + v_{t-2}\right)$$

$$= \theta^2 var(x_t) = \theta\gamma_1$$

$\gamma_k \equiv cov(y_t, y_{t-k}) = \theta\gamma_{k-1}$ Also the autocovariances (obtained recursively from $k \geq 2$) are time-invariant, confirming the (weak) stationarity of $\{y_t\}$.

2. The autocorrelation function of $\{y_t\}$ is obtained from the variance and the autocovariances as (with $\rho_0 = 1$):

$$\rho_1 \equiv \frac{\gamma_1}{\gamma_0} = \frac{\frac{\theta}{1-\theta^2}\sigma_u^2}{\frac{\sigma_u^2}{\sigma_u^2} + \sigma_u^2} = \theta\frac{\sigma_u^2}{\sigma_u^2 + (1-\theta^2)\sigma_u^2}$$

$$= \frac{\theta}{1 + (1-\theta^2)\frac{\sigma_v^2}{\sigma^2}}$$
$$\rho_2 \equiv \frac{\gamma_2}{\gamma_0} = \theta\frac{\gamma_1}{\gamma_0} = \theta\,\rho_1$$

$\rho_k \equiv \frac{\gamma_k}{\gamma_c} = \theta\frac{\gamma_{k-1}}{\gamma_c} = \theta\rho_{k-1}$ For $k \geq 2$ the autocorrelations can be obtained recursively and decrease in magnitude (since $|\theta|0$ or with alternate signs if $\theta < 0$. The value of ρ_1 (and therefore of all the autocorrelations ρ_k) is affected by the variability of $\{v_t\}$ (the purely transitory component of $\{y_t\}$) through the variance ratio $\frac{\sigma_v^2}{\sigma_u^2}$. In fact, the larger is σ_v^2 (relative

to σ_u^2), the smaller is $|\rho_1|$ (and all the $|\rho_k|$): a larger white noise (i.e.

not autocorrelated) component reduces the overall autocorrelation of $\{y_t\}$ due to the $AR(1)$ process $\{x_t\}$.

EXAMPLE – Exercise

Exercise (Univariate Time Series models, p. 26) Suppose that we want to estimate the following dynamic model $y_t = \theta y_{t-1} + u_t$ with $u_t = \varepsilon_t + \alpha\varepsilon_{t-1}$, $|\theta| < 1$, $\varepsilon \sim WN(0, \sigma^2)$ being interested only in estimating θ.

1. What are the properties of the OLS estimator of θ?

2. Is it possible to use y_{t-2} as an instrument to get a consistent estimator of θ by means of instrumental variable (IV) estimation?

Answer

Answers.

1. The OLS estimator of θ is $\hat{\theta}_{OLS} = \frac{\sum_{t=2}^{T} y_{t-1}y_t}{\sum_{t=2}^{T} y_{t-1}^2}$

with $p\lim \hat{\theta}_{OLS} = \frac{E(y_{t-1}y_t)}{E(y_t^2)} = \frac{\theta E(y_{t-1}^2)+\alpha E(y_{t-1}\varepsilon_{t-1})}{E(y_t^2)}$

$= \theta + \frac{\alpha E(y_{t-1}\varepsilon_{t-1})}{E(u^2)} \neq \theta$ since $E(y_{t-1}u_t) = E(y_{t-1}(\varepsilon_t + \alpha\varepsilon_{t-1})) = \alpha E(y_{t-1}\varepsilon_{t-1}) = \alpha\sigma^2 \neq 0$

In particular, $E(y_t^2) = Var(y_t) = \theta^2 Var(y_{t-1}) + (1+\alpha^2)\sigma^2 + 2\alpha\theta \underbrace{E(y_{t-1}\varepsilon_{t-1})}_{\sigma^2}$

$\Rightarrow E(y_t^2) = \frac{(1+\alpha^2+2\alpha\theta)\sigma^2}{1-\alpha^2}$ from which we get $p\lim \hat{\theta}_{OLS} = \theta + \frac{\alpha(1-\theta^2)}{1\pm\alpha^2\pm2\alpha^2}$

Note

(**Note.** $\hat{\theta}_{OLS}$ is a consistent estimator of ρ_1 - the coefficient of first-order autocorrelation, which is different from θ - for an $ARMA(1,1)$ process)

2. Applying IV estimation with y_{t-2} as an instrument, the estimator of θ becomes $\hat{\theta}_{IV} = \frac{\sum_{t=3}^{T} y_{t-2}y_t}{\sum_{t=3}^{T} y_{t-2}y_{t-1}}$

with $p\lim \hat{\theta}_{IV} = \frac{E(y_{t-2}y_t)}{E(y_{t-2}y_{t-1})} = \frac{E(y_{t-2}(\theta y_{t-1}+u_t))}{E(y_{t-2}y_{t-1})}$

$$= \frac{\theta E(y_{t-2}y_{t-1}) + E(y_{t-2}u_t)}{E(y_{t-2}y_{t-1})}$$
$$= \theta + \frac{E(y_{t-2}u_t)}{E(y_{t-2}y_{t-1})} = \theta$$

since $E(y_{t-2}u_t) = E(y_{t-2}(\varepsilon_t + \alpha\varepsilon_{t-1})) = 0$ $\hat{\theta}_{IV}$ is a consistent estimator of θ.

6

2.0.2 Time series Solutions to exercises 2

Time series econometrics Solutions to Exercises (2) Exercise (Univariate Time Series models, p. 39).

Consider the following random walk plus noise stochastic process $y_t = y_0 + \sum_{i=1}^{t} \varepsilon_i + \eta_t$ where $\varepsilon \sim WN(0, \sigma_\varepsilon^2)$ and $\eta \sim WN(0, \sigma_\eta^2)$ are not correlated: $E(\varepsilon_t \eta_{t-s}) = 0$

for all t and s.

(a) Show that y_t follows an $ARIMA(p,d,q)$ process and find the values of p, d and q. (b) Derive (conditional onto the initial observation y_0) the following prop- erties of y: $E(.), Var(.), \gamma_k$ and ρ_k. Compare the ACF of this process with that of a random walk (without noise).

(c) Check the stationarity of Δy_t, find its ACF and compare it with that of an $MA(1)$ process.

Solution

Solution.

(a) From $y_{t-1} = y_0 + \sum_{i=1}^{t-1} \varepsilon_i + \eta_{t-1}$ we get $\Delta y_t = y_t - y_{t-1} = \varepsilon_t + \eta_t - \eta_{t-1} \Rightarrow$ $y \sim ARIMA(0,1,1)$

(b) Conditional on y_0:

$$E(y_t) = y_0$$
$$Var(y_t) = t\sigma_\varepsilon^2 + \sigma_\tau^2$$
$$\gamma_{kt} = E\left[(\varepsilon_t + \varepsilon_{t-1} + \cdots + \varepsilon_1 + \eta_t)(\varepsilon_{t-k} + \cdots + \varepsilon_1 + \eta_{t-k})\right]$$
$$= (t-k)\sigma_\varepsilon^2$$
$$\rho_{kt} = \frac{\gamma_{kt}}{\sqrt{Var(y_t)Var(y_{t-k})}}$$
$$= \frac{(t-k)\sigma_\varepsilon^2}{\sqrt{\left[(t-k)\sigma_\varepsilon^2 + \sigma_\eta^2\right]\left[t\sigma_\varepsilon^2 + \sigma_\eta^2\right]}}$$

As k increases (for given t) the ACF decreases more quickly than in the case of a pure random walk, because of the presence of the noise component (which is not autocorrelated) (c) Δy_t has the following properties:

$$E(\Delta y_t) = 0$$
$$Var(\Delta y_t) = \sigma_\varepsilon^2 + 2\sigma_n^2$$
$$\gamma_k = \begin{cases} -\sigma_\eta^2 & \text{for } k = 1 \\ 0 & \text{for } k > 1 \end{cases}$$

$$\rho_k = \begin{cases} -\frac{1}{2} \leq -\frac{\sigma_\eta^2}{\sigma_\varepsilon^2 + 2\sigma_\eta^2} \leq 0 & \text{for } k = 1 \\ 0 & \text{for } k > 1 \end{cases}$$

The process is stationary (expected value and autocovariances are finite and independent of t) and has an $MA(1)$ nature, with ACF different from zero only at the first lag.

2

EXAMPLE – Exercise

Exercise (Multivariate Time Series models, p. 6).

Start from the following special case of a dynamic model with an infinite number of lags $y_t = \delta + \phi_0 x_t + \phi_1 x_{t-1} + \ldots + \phi_i x_{t-i} + \ldots + \eta_t$ where x is stationary, $\eta \sim WN(0, \sigma^2)$ and $E(x_t \eta_t) = 0$. Assume that the

parameters ϕ_i are all of the same sign and decrease geometrically following the relation (for $j = 1, 2, ...$) $\phi_i = \phi_0 \lambda^j$ with $0 < \lambda < 1$ 1. Using the relation among the parameters, find an ADL specification of the model (expressing y_t as a function of y_{t-1} and x_t) and

find the long-run multiplier, discussing how it depends on the parameter λ;

2. discuss possible estimation problems if OLS is applied to the model in ADL form.

Solution

Solution.

1. Given the assumption on the parameters ϕ_i the model can be expressed as $y_t = \delta + \frac{\phi_0}{1-\lambda T} x_t + \eta_t \Rightarrow (1-\lambda L) y_t = (1-\lambda)\delta + \phi_0 x_t + (1-\lambda L)\eta_t$

$\Rightarrow y_t = (1-\lambda)\,\delta + \lambda\, y_{t-1} + \phi_0\, x_t + \varepsilon_t$ with $\varepsilon_t = \eta_t - \lambda \eta_{t-1}$. The long-run multiplier is $\sum_{j=1}^{\infty} \phi_j = \frac{\phi_0}{1-\lambda}$

The larger is λ (determining a slower decline of the ϕ_i's), the larger is the multiplier.

2. The error term η_t in the autoregressive model follows an $MA(1)$ process.

Therefore $Cov(y_{t-1}, \varepsilon_t) = Cov(y_{t-1}, \eta_t - \lambda\eta_{t-1}) = -\lambda\sigma^2 \neq 0$ and the OLS estimator is biased and inconsistent. IV estimation is needed (with, for example, y_{t-2} as an instrument for y_{t-1}).

EXAMPLE – Exercise

Exercise (Multivariate Time Series models, p. 21).

Consider the following model for two $I(1)$ processes $\{x_t\}$ and $\{y_t\}$:

$y_t = \beta x_t + \varepsilon_t$ with $\varepsilon_t = \rho_1 \varepsilon_{t-1} + \rho_2 \varepsilon_{t-2} + \eta_t^y$ $x_t = x_{t-1} + \eta_t^x$

where we assume cointegration, i.e. $\{\varepsilon_t\}$ is a stationary $(I(0))$ process (a necessary condition for stationarity being $1 - \rho_1 - \rho_2 > 0$). Find the ECM representation of the system and the "error-correction" coefficient capturing the speed of adjustment of y_t to the long-run equilibrium relation with x_t.

Solution

Solution.

Noting that $\varepsilon_t = \frac{1}{1-\rho_1 L - \rho_2 L}\eta_t^y$ the process y_t can be rewritten as:

$y_t = \beta\, x_t + \frac{1}{1-\rho_1 L - \rho_2 L}\eta_t^y$ yielding $(1-\rho_1 L - \rho_2 L) y_t = \beta(1-\rho_1 L - \rho_2 L) x_t + \eta_t^y$

that can be rearranged in an $ADL(2,2)$ form (with common factor restric- tions imposed):

$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \beta x_t - \rho_1 \beta x_{t-1} - \rho_2 \beta x_{t-2} + \eta_t^y$ Subtracting y_{t-1} from both sides of the equation, adding and subtracting

$\rho_2 y_{t-1}$ and $\rho_2 \beta x_{t-1}$ from the right-hand side, and rearranging yields the error-correction representation:

$$\Delta y_t = \beta\, \Delta x_t - \rho_2\, \Delta y_{t-1} + \rho_2 \beta\, \Delta x_{t-1} - (1-\rho_1-\rho_2)\,(y_{t-1} - \beta\, x_{t-1}) + \eta_t^y$$

(using $\Delta x_t = \eta_t^x$) where the last term captures the "correcting" reaction of y_t to any deviation of y_{t-1} from its long-run relation with x_{t-1}. The speed of adjustment of y_t depends inversely on the magnitude of the sum of the autoregressive parameters $(\rho_1 + \rho_2)$ in the process for ε_t.

EXAMPLE – Exercise

Exercise (Multivariate Time Series models, p. 56) Consider the following alternative "permanent income" model in which:

(i) permanent income y^P is directly observable (by assumption);

(ii) consumption adjusts gradually to permanent income.

Formally, the model is:

$$c_t = y_t^P + \zeta_t \quad \text{with} \quad \zeta_t = \rho\zeta_{t-1} + \eta_t \quad (0 < \rho < 1)$$
$$y_t^P = y_{t-1}^P + \mu + \varepsilon_t$$

where η_t and ε_t are uncorrelated.

1. Find the cointegrated $VECM$ representation of the system formed by Δc_t and Δy_t^P;

2. find the (recursive) identification scheme appropriate to recover the impulse response functions of c and y^P to permanent and transitory disturbances.

Solution

Solution.

1. First differencing and using the process for ζ_t we get:

$$\begin{aligned}\Delta c_t &= \Delta u_t^P + \Delta\zeta_t \\ &= \Delta y_t^P + (\rho - 1)\zeta_{t-1} + \eta_t \\ &= \Delta y_t^P + (\rho - 1)(c_{t-1} - y_{t-1}^P) + \eta_t\end{aligned}$$

and the $VECM$ form is $\begin{pmatrix}\Delta c_t \\ \Delta y_t^P\end{pmatrix} = \begin{pmatrix}\mu \\ \mu\end{pmatrix} + \begin{pmatrix}-(1-\rho) \\ 0\end{pmatrix}(c_{t-1} - y_{t-1}^P) + \begin{pmatrix}u_t^c \\ u_t^{y^P}\end{pmatrix}$

with $u_t^c = \eta_t + \varepsilon_t$ and $u_t^{y^P} = \varepsilon_t$.

2. The appropriate recursive structure to achieve identification of the per- manent and transitory shocks is:

$$u_t^{y^P} = \varepsilon_t$$
$$u_t^c = \eta_t + \varepsilon_t$$

implying an ordering of the variables in the VAR with Δy_t^P first and Δc_t second. Now, the shock to permanent income ε_t affects consump- tion contemporaneously, whereas η_t, affecting only consumption and not y_t^P, is due to deviations from the long-run (cointegrating) relation with permanent income and has a purely transitory nature. In this al- ternative model, consumption is the series which contains a transitory

component and therefore adjusts to the long-run equilibrium relation with permanent income (in fact, the cointegrating vector enters the equation for Δc_t only).

6

2.0.3 Ectx2 Time series Exam sample 1

Econometrics II /Second Part (Time Series) 1. Univariate time series. Consider a time series $\{y_t\}$ generated by the following $ARMA(1,2)$ process:

$y_t = \beta\, y_{t-1} + \varepsilon_t + \alpha_1\, \varepsilon_{t-1} + \alpha_2\, \varepsilon_{t-2}$ where ε_t is a white noise process with variance σ^2, and β, α_1, $\alpha_2 \neq 0$.

(a) Under what conditions is $\{y_t\}$ stationary? (explain). Under stationarity, find $E(y_t)$. (b) Is it possible that the above process displays the same autocorrelation function (ACF) of the following:

$y_t = \varepsilon_t + \delta\varepsilon_{t-1}$ where $\delta \neq 0$? If so, under what conditions?

(c) Under stationarity, given a sample of T observations, find the optimal predictors (forecast) for $\{y_t\}$ at horizons $h = 1, h = 2$, and $h = 3$. Show what happens to the optimal predictor $\hat{y}_{T+h|T}$ when $h \to \infty$.

2. Multivariate time series. Consider the following model:

$y_t^* = \alpha + \beta x_t + \gamma x_{t-1} + \delta y_{t-1} + \varepsilon_t$ where y_t^* is the "desired" (target) value of y for time t, $0 < \delta < 1$, and the process $\{x_t\}$ is stationary with finite and time-invariant unconditional mean $E(x_t) = \mu$; moreover,

$\varepsilon_t \sim WN(0, \sigma^2)$. The relation between the observable value of y and its target value is:

$y_t - y_{t-1} = \lambda\,(y_t^* - y_{t-1})$ where $0 < \lambda < 1$.

(a) Derive an autoregressive distributed lag (ADL) model for y_t.

(b) Find the effect of a change in x_t on y_t (i.e. the "impact multiplier"); is it affected by the parameter λ, and, if so, why?

(c) Find $E(y_t)$ and the effect on y of a permanent change in x (i.e. the "long-run multiplier"). Explain why the magnitude of the parameter δ affects the long-run multiplier.

2.0.4 Ectx2 Time series Exam sample 2

Econometrics II / Second Part (Time Series) 1. Univariate time series. Consider the following autoregressive stochastic process for y_t:

$y_t = \kappa + \theta y_{t-1} + \varepsilon_t$ where κ is a constant and ε_t is generated by the following stochastic process:

$\varepsilon_t = v_t\sqrt{\alpha + \beta\,\varepsilon_{t-1}^2}$ where v_t is white noise with $\sigma_v^2 = 1$, independent of ε_{t-1}.

(a) What assumptions on the parameters θ, α and β are necessary to ensure that the process for y_t is weakly (covariance) stationary and why?

(b) Find the mean and variance of $\{y_t\}$ conditional on $I_{t-1} = \{y_{t-1}, y_{t-2}, ...\}$ and unconditional.

(c) Given a sample of length T, find the forecasts for the horizons $h = 1, 2$ and 3 for the process $\{y_t\}$ and for the conditional variance $\sigma_t^2 = E(\varepsilon_t^2 \mid I_{t-1})$. What happens to those forecasts when the forecast horizon $h \to \infty$? (explain).

2. Multivariate time series. Consider the following model:

$y_t = \delta + \beta_1 x^*_{t+1} + \beta_2 x_t + \varepsilon_t$ where x^*_{t+1} is the expectation of x_{t+1} formed on the basis of the agents' information set at time t, δ is a constant and $\varepsilon_t \sim WN(0, \sigma^2)$. The expectation-formation mechanism

is of the following (adaptive) form $x^*_{t+1} - x^*_t = \lambda\,(x_t - x^*_t)$ where $0 < \lambda < 1$ and $\{x_t\}$ is generated by a stationary process.

(a) Derive an autoregressive distributed lag (ADL) model for y_t.

(b) Find the effect of a change in x_t on y_t (i.e. the "impact multiplier") and on y_{t+1} (i.e. the "interim multiplier" after one period). Compute the long-run multiplier of a permanent change in x_t on y_t, and comment on the effect of a change in the parameter λ on its magnitude (explain).

(c) Construct the equilibrium-correction representation for Δy_t. What does the mag- nitude of the equilibrium-correction coefficient depend on? (explain)

2.0.5 Ectx2 Time series Exam sample 3

Econometrics II /Second Part (Time Series) 20th June, 2022 Univariate time series. Consider a time series $\{\Delta y_t\}$ generated by the following 1.

$ARMA(1, 2)$ process:

$\Delta y_t = c + \beta\,\Delta y_{t-1} + \varepsilon_t + \gamma_1\,\varepsilon_{t-1} + \gamma_2\,\varepsilon_{t-2}$ where c is a constant, ε_t is a white noise process with variance σ^2, and $\beta, \gamma_1, \gamma_2 \neq 0$.

(a) Assume $0 < \beta < 1$; under this assumption, is the process for $\{\Delta y_t\}$ stationary (or additional conditions are needed)? What are the conditions for invertibility of $\{\Delta y_t\}$? Is it possible that $\{\Delta y_t\}$ has the same autocorrelation function (ACF)

of an $MA(1)$ process (and, if so, under what conditions)?

(b) Assume $\beta = 1$. What are the orders of integration of $\{y_t\}$ and $\{\Delta y_t\}$? (explain).

(c) Assume $0 < \beta < 1$ and $\gamma_2 = 0$. Find the unconditional variance of $\{\Delta y_t\}$, i.e.

$var(\Delta y_t)$. Now, given a sample of T observations, find the optimal predictors (forecasts) for $\{y_t\}$ (in levels) at horizons $h = 1$ and $h = 2$, and the associated forecast error variances.

2. Multivariate time series. Consider an autoregressive distributed lag (ADL) model for $\{y_t\}$ of the following form:

$y_t = c + \alpha y_{t-1} + \delta x_t + \gamma x_{t-1} + \varepsilon_t$ (*) where c is a constant and $\{x_t\}$ is generated by (**) $x_t = k + \beta x_{t-1} + u_t + \rho u_{t-1}$ (where k is a constant), with $\{\varepsilon_t\}$ and $\{u_t\}$ being two independent white noise processes.

(a) Under what conditions on the parameters of the above processes ($\alpha, \beta, \gamma, \delta,$ and ρ), does the unconditional mean of y_t, i.e. $E(y_t)$, exist? Impose those conditions and find the (linear) relationship between $E(y_t)$ and $E(x_t)$, giving an interpretation of your result. Is this relationship affected, or not affected, by the parameter ρ?

(explain) (b) Under the same conditions imposed in (a), write the ADL model in terms of an error-correction mechanism (ECM) for Δy_t. Comment on the determinants of the magnitude of the "correction" effect. (Hint: do not substitute for x_t with ($**$) above) (c) Now suppose $\alpha = \gamma = 0$ and $\beta = 1$ in the processes (*) and (**) generating $\{y_t\}$ and $\{x_t\}$. What are the orders of integration of $\{y_t\}$ and $\{x_t\}$? Are the two

processes $\{y_t\}$ and $\{x_t\}$ cointegrated? (explain why, or why not).

2.0.6 Ectx2 Time series Exam sample 4

Econometrics II /Second Part (Time Series) Univariate time series. Consider a time series $\{\Delta y_t\}$ generated by the following 1.

$ARMA(2,1)$ process:

$\Delta y_t = k + \alpha_1 \Delta y_{t-1} + \alpha_2 \Delta y_{t-2} + \varepsilon_t + \delta_1 \varepsilon_{t-1}$ (*) where k is a constant, ε_t is a white noise process with variance σ^2, and $\alpha_1, \alpha_2, \delta_1 \neq 0$.

(a) What are the conditions for the process $\{\Delta y_t\}$ to be stationary? Under those conditions, find $E(\Delta y_t)$.

(b) Is it possible for $\{\Delta y_t\}$ to display the same autocorrelation function (ACF) of the following:

$\Delta y_t = c + \rho \Delta y_{t-1} + \varepsilon_t$ where $\rho \neq 0$? If so, under what conditions?

(c) With reference to the process in $(*)$, given a sample of T observations, find the optimal predictors (forecasts) for $\{y_t\}$ (in levels) at horizons $h = 1$ and $h = 2$, and the associated forecast error variances.

2. Multivariate time series. Consider an autoregressive distributed lag (ADL) model for $\{y_t\}$ of the following form:

$y_t = k + \rho y_{t-1} + \alpha x_t + \beta x_{t-1} + \eta_t$ $(*)$ where k is a constant and $\{x_t\}$ is generated by $x_t = c + \varepsilon_t + \gamma \varepsilon_{t-1}$ (**)

where c is a constant and $\{\eta_t\}$ and $\{\varepsilon_t\}$ are two independent white noise processes.

(a) Under what conditions on the parameters of the above processes ($\rho, \alpha, \beta,$ and γ), do the unconditional mean and variance of y_t, $E(y_t)$ and $var(y_t)$, exist? Impose those conditions and find $E(y_t)$ and $var(y_t)$.

(b) Under the above conditions, find the effect of a change in x_t on y_t (the "impact" multiplier) and on y_{t+1} (the "interim" multiplier after one period). Which of those multipliers is affected by the parameter ρ, and why?

(c) Find the effect of a permanent change in x on y (the "long-run" multiplier).

Write the ADL model in terms of an error-correction mechanism (ECM) for Δy_t. (Hint: do not substitute for x_t with (**) above)

2.0.7 Ectx2 Time series Exam sample 5

Econometrics II /Second Part (Time Series) 1. Univariate time series. Consider the following process for $\{\Delta y_t\}$ (with $\Delta y_t \equiv y_t - y_{t-1}$):

$\Delta y_t = c + \varepsilon_t + \gamma_1 \varepsilon_{t-1} + \gamma_2 \varepsilon_{t-2}$ where c is a constant, $\{\varepsilon_t\}$ is a white noise process with variance σ^2, and $\gamma_1, \gamma_2 \neq 0$.

(a) Under what conditions is $\{\Delta y_t\}$ stationary? (explain). Is the process for the level $\{y_t\}$ stationary? (explain). Under what conditions is $\{\Delta y_t\}$ invertible?

(b) Now suppose that the process is invertible and you estimate the following equation by OLS:

$\Delta y_t = \beta_0 + \beta_1 \Delta y_{t-1} + \beta_2 \Delta y_{t-2} + \beta_3 \Delta y_{t-3} + u_t$ Can we interpret any of the estimated coefficients $\hat{\beta}_{1OLS}$, $\hat{\beta}_{2OLS}$ and $\hat{\beta}_{3OLS}$ in

terms of autocorrelations and partial autocorrelations of $\{\Delta y_t\}$? (explain). How does your answer change if $\gamma_2 = 0$ in the process for $\{\Delta y_t\}$? (explain) (c) Given a sample of T observations, find

the optimal predictors (forecast) for $\{\Delta y_t\}$

at horizons $h = 1, 2$ and 3, and the associated forecast error variances. Show what happens to the optimal predictors for Δy_t and the forecast error variances when $h \to \infty$ (explain).

2. Multivariate time series. Consider the following model:

$y_t = k + \alpha_1 x^*_{t+1} + \alpha_2 x_t + \varepsilon_t$ where x^*_{t+1} is the expectation of x_{t+1} formed on the basis of the agents' information set at time t, k is a constant and $\varepsilon_t \sim WN(0, \sigma^2)$. The expectation-formation mechanism

is of the following (adaptive) form $x^*_{t+1} - x^*_t = \theta\,(x_t - x^*_t)$ where $0 < \theta < 1$ and $\{x_t\}$ is generated by a stationary process.

(a) Derive an autoregressive distributed lag (ADL) model for y_t.

(b) Find the effect of a change in x_t on y_t (i.e. the "impact multiplier") and on y_{t+1} (i.e. the "interim multiplier" after one period). Compute the long-run multiplier of a permanent change in x_t on y_t, and comment on the effect of a change in the parameter θ on its magnitude, if any (explain).

(c) Construct the equilibrium-correction representation for Δy_t. What does the mag- nitude of the equilibrium-correction coefficient depend on? (explain)

2.0.8 Ectx2 Time series Exam sample 6

Econometrics II /Second Part (Time Series) Consider a time series $\{\Delta y_t\}$ generated by the following Univariate time series.

1.

$ARMA(1, 1)$ process:

$\Delta y_t = c + \beta\,\Delta y_{t-1} + \varepsilon_t + \gamma\,\varepsilon_{t-1}$ where c is a constant, ε_t is a white noise process with variance σ^2, and $\beta, \gamma \neq 0$.

(a) Under what conditions on the parameters β and γ, do the unconditional mean and variance of $\{\Delta y_t\}$ exist? Under those conditions, what is the order of integration of $\{\Delta y_t\}$ and of the series in levels $\{y_t\}$? Now assume that $\beta = 1$; what is the order

of integration of $\{\Delta y_t\}$ and $\{y_t\}$ in this case? (Explain clearly, adding parameter conditions if needed).

(b) Assuming stationarity of $\{\Delta y_t\}$, find the unconditional mean of the process $\{\Delta y_t\}$, that is $E(\Delta y_t)$.

(c) Under stationarity, given a sample of T observations, find the optimal predictors (forecasts) for $\{y_t\}$ at horizons $h = 1$ and $h = 2$, and the associated forecast error variances.

2. Multivariate time series. Consider an autoregressive distributed lag (ADL) model for $\{y_t\}$ of the following form:

$y_t = \theta\,y_{t-1} + \beta\,x_t + \varepsilon_t$ (*) where $\varepsilon_t = u_t + \varphi u_{t-1}$ with u_t being a white noise, and $\{x_t\}$ is generated by (**) $x_t = \gamma x_{t-1} + v_t$

where $\{v_t\}$ is a white noise process independent of u_t.

(a) Under what conditions on the parameters θ, β, γ and φ does the unconditional mean of y_t ($E(y_t)$) exist? Impose those conditions and find the (linear) relation- ship between $E(y_t)$ and $E(x_t)$. Give an interpretation of your result.

(b) Under the above conditions, find the effect of a change in x_t on y_t (i.e. the "impact multiplier") and on y_{t+1} and y_{t+2} (i.e. the "interim multipliers" after one and two periods).

(c) Now assume $\theta = 0$ and $\gamma = 1$. Are $\{y_t\}$ and $\{x_t\}$ cointegrated? Explain clearly why, introducing additional conditions if needed.

Chapter 3

Optional Readings

3.0.1 Optional reading – BlanchardQuah AER89

The Dynamic Effects of Aggregate Demand and Supply Disturbances By OLIVIER JEAN BLANCHARD AND DANNY QUAH* We interpret fluctuations in GNP and unemployment as due to two types of disturbances: disturbances that have a permanent effect on output and disturbances that do not. We interpret the first as supply disturbances, the second as demand disturbances. Demand disturbances have a hump-shaped mirror-image effect on output and unemployment. The effect of supply disturbances on output increases steadily over time, peaking after two years and reaching a plateau after five years.

It is now widely accepted that GNP is However, if GNP is affected by more than reasonably characterized as a unit root pro- one type of disturbance, as is likely, the cess: a positive innovation in GNP should interpretation becomes more difficult. In that lead one to revise upward one's forecast on case, the univariate-moving average repre- GNP for all horizons. Following the influ- sentation of output is some combination of ential work of Charles Nelson and Charles the dynamic response of output to each of Plosser (1982), this statistical characteriza- the disturbances. The work in Stephen tion has been recorded and refined by nu- Beveridge and Nelson (1981), Andrew Har- merous authors including John Campbell and vey (1985), and Watson (1986) can be viewed N. Gregory Mankiw (1987a), Peter Clark as early attempts to get at this issue.1 (1987, 1988), John Cochrane (1988), Francis

To proceed, given the possibility that out- Diebold and Glenn Rudebusch (1988), put may be affected by more than one type George Evans (1987), and Mark Watson of disturbance, one can impose a priori re- strictions on the response of output to each (1986).

of the disturbances, or one can exploit infor- How should this finding affect one's views mation from macroeconomic variables other about macroeconomic fluctuations? Were than GNP. In addition to the work named there only one type of disturbance in the above, Clark (1987) has also used the first economy, then the implications of these findings would be straightforward. That dis- approach. This paper adopts the second, and turbance would affect the economy in a way considers the joint behavior of output and characterized by estimated univariate-mov- unemployment. Campbell and Mankiw ing average representations, such as those (1987b), Clark (1988), and Evans (1987) have given by Campbell and Mankiw. The prob- also taken this approach. Our analysis differs lem would simply be to find out what this mainly in its choice of identifying restric- disturbance was, and why its dynamic effects

had the shape that they did. The way to proceed would be clear.

1As will become clear, our work differs from these in that we wish to examine the dynamic effects of distur- *Both authors are with the Economics Department, bances that have permanent effects; such issues cannot MIT, Cambridge MA 02139, and the NBER. We thank be addressed by studies that restrict the permanent Stanley Fischer, Julio Rotemberg, Mark Watson for helpful discussions, and the NSF for financial assis- component to be a random walk. In other work, one of us has characterized the effects of different parametric tance. We are also grateful for the

comments of two specifications (such as lag length restrictions, a rational anonymous referees and of participants at an NBER form for the lag distribution) for the question of the Economic Fluctuations meeting, and for the hospitality relative importance of permanent and transitory com-

of the MIT Statistics Center.

ponents. See Quah (1988).

THE AMERICAN ECONOMIC REVIEW SEPTEMBER 1989 variance decompositions of output at various tions; as we shall argue, we find our restric- horizons, we find that the respective contri- tions more appealing than theirs.

Our approach is conceptually straightfor- butions of supply and demand disturbances ward. We assume that there are two kinds of are not precisely estimated. For instance, at a forecast horizon of four quarters, we find disturbances, each uncorrelated with the other, and that neither has a long-run effect that, under alternative assumptions, the con- tribution of demand disturbances ranges on unemployment. We assume however that from 40 percent to over 95 percent.

the first has a long-run effect on output while The rest of the paper is organized as fol- the second does not. These assumptions are lows. Section I analyzes identification, and sufficient to just identify the two types of Section II discusses our economic interpreta- disturbances, and their dynamic effects on tion of the disturbances. Section III dis- output and unemployment.

cusses estimation, and Section IV charac- While the disturbances are defined by the terizes the dynamic effects of demand and identification restrictions, we believe that supply disturbances on output and unem- they can be given a simple economic inter- ployment. Section V characterizes the rela- pretation. Namely, we interpret the distur- tive contributions of demand and supply bances that have a temporary effect on out- put as being mostly demand disturbances, disturbances to fluctuations in output and and those that have a permanent effect on unemployment.

output as mostly supply disturbances. We I. Identification present a simple model in which this inter- pretation is warranted and use it to discuss the justification for, as well as the limitations In this section, we show how our assump- tions characterize the process followed by of, this interpretation.

output and unemployment, and how this Under these identification restrictions and process can be recovered from the data.

this economic interpretation, we obtain the We make the following assumptions. There following characterization of fluctuations:

are two types of disturbances affecting un- demand disturbances have a hump-shaped employment and output. The first has no effect on both output and unemployment;

long-run effect on either unemployment or the effect peaks after a year and vanishes output. The second has no long-run effect on after two to three years. Up to a scale factor, unemployment, but may have a long-run the dynamic effect on unemployment of de- effect on output. Finally, these two distur- mand disturbances is a mirror image of that bances are uncorrelated at all leads and lags.

on output. The effect of supply disturbances These restrictions in effect define the two on output increases steadily over time, to disturbances. As indicated in the introduc- reach a peak after two years and a plateau after five years. "Favorable" supply distur- tion, and discussed at length in the next section, we will refer to the first as demand bances may initially increase unemployment.

disturbances, and to the second as supply This is followed by a decline in unemploy- disturbances. How we name the disturbances ment, with a slow return over time to its however is irrelevant for

the argument of original value.

this section.

While this dynamic characterization is The demand and supply components de- fairly sharp, the data are not as specific as to scribed above are permitted to be serially the relative contributions of demand and correlated. Under regularity conditions, each supply disturbances to output fluctuations.

of these components can always be uniquely On the one hand, we find that the time-series represented as an invertible distributed lag of demand-determined output fluctuations, of serially uncorrelated disturbances. Thus, that is the time-series of output constructed we can refer to the associated serially uncor- by putting all supply disturbance realiza- related disturbances as the demand and sup- tions equal to zero, has peaks and troughs ply disturbances themselves: this is without which coincide with most of the NBER ambiguity or loss of generality. We will then troughs and peaks. But, when we turn to

VOL. 79 NO. 4 BLANCHARD AND QUAH: DEMAND AND SUPPLY DISTURBANCES also require a further technical condition:

tation:

the innovations in the bivariate Wold de- $X(t) = v(t) + C(1)v(t-1) + \cdots$ (2) composition of output growth and unem- ployment are linear combinations of these underlying demand and supply disturbances.

$= \sum_{j=0}^{\infty} C(j)\nu(t-j)$, We now derive the joint process followed by output and unemployment implied by our assumptions. Let Y and U denote the $\mathrm{Var}(\nu) = \Omega$.

logarithm of GNP and the level of the unem- ployment rate, respectively, and let e_d and e_s be the two disturbances. Let X be the This moving average representation is unique vector $(\Delta Y, U)'$ and e be the vector of dis- and can be obtained by first estimating and turbances $(e_d e_s)'$. The assumptions above then inverting the vector autoregressive rep- imply that X follows a stationary process resentation of X in the usual way.

given by:

Comparing equations (1) and (2) we see that ν, the vector of innovations, and e, the vector of original disturbances, are related $X(t) = A(0)e(t) + A(1)e(t-1) + \cdots$ (1) by $\nu = A(0)e$, and that $A(j) = C(j)A(0)$, for all j. Thus knowledge of $A(0)$ allows one to recover e from ν, and similarly to obtain $= \sum_{j=0}^{\infty} A(j)e(t-j)$,

$A(j)$ from $C(j)$.

Is $A(0)$ identified? An informal argument suggests that it is. Equations (1) and (2) $Var(e) = I$.

imply that $A(0)$ satisfies: $A(0)A(0)' = \Omega$, and that the upper left-hand entry in $\sum_{i=0}^{\infty} A(j)$ where the sequence of matrices A is such $= (\sum_{i=0}^{\infty} C(j))A(0)$ is 0. Given Ω, the first that its upper left-hand entry, $a_{11}(j)$, $j =$

relation imposes three restrictions on the $1, 2, \ldots$, sums to zero.

four elements of $A(0)$; given $\sum_{j=0}^{\infty} C(j)$, the Equation (1) gives Y and U as distributed other implication imposes a fourth restric- lags of the two disturbances, e_d and e_s.

tion. This informal argument is indeed cor- Since these two disturbances are assumed to rect. A rigorous and constructive proof, be uncorrelated, their variance covariance which we actually use to obtain $A(0)$ is as matrix is diagonal; the assumption that the follows: Let S denote the unique lower tri- covariance matrix is the identity is then sim- angular Choleski factor of Ω. Any

matrix ply a convenient normalization. The contem- $A(0)$ such that $A(0)A(0)' = \Omega$ is an orthonor- poraneous effect of e on X is given by $A(0)$;

mal transformation of S. The restriction that subsequent lag effects are given by $A(j)$, the upper left-hand entry in $(\sum_{j=0}^{\infty} C(j))A(0)$ $j \geq 1$. As X has been assumed to be station- be equal to 0 is an orthogonality restriction ary, neither disturbance has a long-run effect that then uniquely determines this orthonor- on either unemployment, U, or the rate of mal transformation.2 change in output, ΔY. The restriction $\sum_{j=0}^{\infty} a_{11}(j) = 0$ implies that e_d also has no

effect on the level of Y itself. To see why 2Notice that identification is achieved by a long-run this is, notice that $a_{11}(j)$ is the effect of e_d restriction. This raises a knotty technical issue. Without on ΔY after j periods, and therefore, precise prior knowledge of lag lengths, inference and $\sum_{j=0}^{k} a_{11}(j)$ is the effect of e_d on Y itself after k periods. For e_d to have no effect on

restrictions on the kind of long-run behavior we are interested in here is delicate. See for instance Y in the long run, we must have then that Christopher Sims (1972); we are extrapolating here from Sims's results which assume strictly exogenous regres- $\sum_{j=0}^{\infty} a_{11}(j) = 0$. sors. Similar problems may arise in the VAR case, We now show how to recover this repre- although the results of Kenneth Berk (1974) suggest sentation from the data. Since X is station- otherwise. Nevertheless, we can generalize our long-run ary, it has a Wold-moving average represen-

restriction to one that applies to some neighborhood of

THE AMERICAN ECONOMIC REVIEW SEPTEMBER 1989 In summary, our procedure is as follows.

ity. Notice that productivity is allowed to We first estimate a vector autoregressive rep- affect aggregate demand directly; it can do resentation for X, and invert it to obtain so through investment demand for example, (2). We then construct the matrix $A(0)$; and in which case $a > 0$. Equation (4) is the use this to obtain $A(j) = C(j)A(0)$, $j = 0, 1, 2, ...$, and $e_t = A(0)^{-1}v_t$. This gives production function: it relates output, em- ployment, and productivity, and assumes a output and unemployment as functions of constant returns-to-scale technology. Equa- current and past demand and supply distur-

tion (5) describes price-setting behavior, and bances.

gives the price level as a function of the nominal wage and of productivity. Finally II. Interpre- tation the last equation, (6), characterizes wage-set- ting behavior in the economy: the wage is Interpreting residuals in small dimen- chosen one period in advance, and is set so sional systems as "structural" disturbances as to achieve (expected) full employment.

is always perilous, and our interpretation of To close the model, we need to specify disturbances as supply and demand distur- how M and θ evolve. We assume that they bances is no exception. We discuss various follow:

issues in turn.

$M(t) = M(t-1) + e_d(t)$, (7) Our interpretation of disturbances with permanent effects as supply disturbances, $\theta(t) = \theta(t-1) + e_\epsilon(t)$, (8) and of disturbances with transitory effects as demand disturbances is motivated by a tra- ditional Keynesian view of fluctuations. For where e_d and e_s are the serially uncorrelated illustrative purposes, as well as to focus the and pairwise orthogonal demand and supply discussion below, we now provide a simple disturbances. Define unemployment U to be model which delivers those implications. The

$N - N$; solving for unemployment and out- model is a variant of that in Stanley Fischer put growth then gives:

(1977):

$\Delta Y = e_d(t) - e_d(t-1)$ $Y(t) = M(t) - P(t) + a \cdot \theta(t)$, (3) $+a \cdot (e_s(t) - e_s(t-1)) + e_s(t)$, $Y(t) = N(t) + \theta(t)$, (4) $P(t) = W(t) - \theta(t)$. $U = -e_d(t) - a \cdot e_s(t)$. (5) $W(t) = W|\{E_{t-1}N(t) = \overline{N}\}$.

(6) These two equations clearly satisfy the re- strictions in equation (1) of the previous The variables Y, N, and θ denote the log of section. Due to nominal rigidities, demand output, employment, and productivity, re- disturbances have short-run effects on out- spectively. Full employment is represented put and unemployment, but these effects dis- by $\overline{N}$; and P, W, and M are the log of the appear over time. In the long run, only sup- price level, the nominal wage, and the money ply, that is, productivity disturbances here, supply.

affect output. Neither of the disturbances Equation (3) states that aggregate demand have a long-run impact on unemployment.

is a function of real balances and productiv- This model is clearly only illustrative.

More complex wage and price dynamics, such as in John Taylor (1980), will also satisfy the long-run properties embodied in equation (1). This model is nevertheless a frequency zero, instead of just a restriction at the point useful vehicle to discuss the limitations of zero. Under appropriate regularity conditions, we can our interpretation of permanent and transi- show that our results are the limit of those from that tory disturbances.

kind of restriction, as the neighborhood shrinks to zero.

VOL. 79 NO. 4 BLANCHARD AND QUAH: DEMAND AND SUPPLY DISTURBANCES Granting our interpretation of these dis- with many demand disturbances, some with turbances as demand and supply distur- permanent and others with transitory effects, bances, one may nevertheless question the and if they all play an equally important role assumption that the two disturbances are in aggregate fluctuations, our decomposition uncorrelated at all leads and lags. We think is likely to be meaningless. A more interest- of this as a nonissue. The model makes clear ing case is that where all the supply distur- that this orthogonality assumption does not bances have permanent output effects, and eliminate for example the possibility that where all the demand disturbances have only supply disturbances directly affect aggregate transitory output effects. One may then hope

demand. Put another way, the assumption that, in this case, what we present as "the" that the two disturbances are uncorrelated demand shock represents an average of the does not restrict the channels through which dynamic effects of the different shocks (in demand and supply disturbances affect out- the sense of Clive Granger and M. J. Morris.

put and unemployment.

1976, for example), and similarly for supply Again granting our interpretation of these shocks. This however is not true in general: a disturbances as demand and supply distur- simple counterexample that illustrates this is bances, one may argue that even demand provided in the technical appendix. How- disturbances have a long-run impact on out- ever, we also present in the appendix neces- put: changes in the subjective discount rate, sary and sufficient conditions such that an or changes in fiscal policy may well affect the aggregation proposition does hold. Those savings rate, and subsequently the long-run conditions will be satisfied if for instance, capital stock and output. The presence of the economy is subject to only one supply increasing returns, and of learning by doing, disturbance but many demand disturbances, also raise the possibility that demand distur-

where each of the demand disturbances has bances may have some long-run effects. Even different dynamic effects on output, but all if not, their effects through capital accumula- the demand disturbances leave unaffected tion may be sufficiently long lasting as to be the dynamic relation between output and indistinguishable from truly permanent ef- unemployment. That demand disturbances fects in a finite data sample. We agree that should leave the relation between output

and demand disturbances may well have such unemployment nearly unaffected is highly long-run effects on output. However, we also plausible. That the economy is subject to believe that if so, those long-run effects are only one, or at least to one dominant, source small compared to those of supply distur- of supply disturbances is more questionable.

bances. To the extent that this is true then, If there are many supply disturbances of our decomposition is "nearly correct" in the roughly equal importance, and if, as is likely, following sense: in a sequence of economies each of them affects the dynamic relation where the size of the long-run effect of de- between unemployment and output, our de- mand disturbances becomes arbitrarily small composition is likely to be meaningless.

relative to that of supply, the correct identi- In summary, our interpretation of the dis- fying scheme approaches that which we ac- turbances is subject to various caveats. Nev- tually use. This result is proven in the techni- ertheless we believe that interpretation to be cal appendix.

reasonable and useful in understanding the This raises a final set of issues, one inher- results below. We now briefly discuss the ent in the estimation and interpretation of relation of our paper to others on the same any low-dimensional dynamic system. It is topic. We first examine how our approach likely that there are in fact many sources of relates to the business-cycle-versus-trend dis- disturbances, each with different dynamic tinction.

effects on output and on unemployment, Following estimation, we can construct rather than only two as we assume here.

two output series, a series reflecting only the Certainly if there are many supply distur- effects of supply disturbances, obtained by bances, some with permanent and others setting all realizations of the demand distur- with transitory effects on output, together bances to zero, and a series reflecting only

THE AMERICAN ECONOMIC REVIEW SEPTEMBER 1989 the effects of demand disturbances, obtained "trend" and "cycle" disturbances, which are by setting supply realizations to zero. By assumed to be uncorrelated. Their identify- construction, the first series, the supply com- ing restriction is then that trend disturbances ponent of output, will be nonstationary while do not affect unemployment. The discussion the second, the demand component, is sta- above suggests that this assumption of zero tionary.3 correlation between cycle and trend compo- A standard distinction in describing out- nents is unattractive; if their two distur- put movements is the "business cycle versus bances are instead reinterpreted as supply trend" distinction. While there is no stan- and demand disturbances, respectively, the dard definition of these components, the identifying restriction that supply distur-

trend is usually taken to be that part of bances do not affect unemployment is equally output that would realize, were all prices unattractive.

perfectly flexible; business cycles are then Clark (1988) also assumes the existence of taken to be the dynamics of actual output "trend" and "cycle" disturbances, and also around its trend.4 assumes that "trend" disturbances do not It is tempting to associate the first series affect unemployment but allows for contem- we construct with the "trend" component of poraneous correlation between trend and cy- output and the second series with the "busi- cle disturbances. While this may be seen as ness cycle" component. In our view, that an improvement over Campbell and Mankiw.

association is unwarranted. If prices are in it still severely constrains the dynamic effects fact imperfectly flexible, deviations from of disturbances on output and unemploy- trend will arise not only from demand dis- ment in ways that are difficult to interpret.

turbances, but also from supply distur- The paper closest to ours is that of bances: business cycles will occur due to Evans (1987). Evans assumes two distur- both supply and demand

disturbances. Put bances, "unemployment" and "output" dis- another way, supply disturbances will affect turbances, which can be reinterpreted as both the business cycle and the trend com- supply and demand disturbances, respec- ponent. Identifying separately business cy- tively. By assuming the existence of a re- cles and trend is likely to be difficult, as the duced form identical to equation (2) above, two will be correlated through their joint he also assumes that neither supply nor de- dependence on current and past supply dis- mand disturbances have a long-run effect on turbances.

unemployment, but that both may have a With this discussion in mind, we now re- long-run effect on the level of output. How- view the approaches to identification used by ever, instead of using the long-run restriction others.

that we use here, he assumes that supply Campbell and Mankiw (1987b) assume the disturbances have no contemporaneous ef- existence of two types of disturbances, fect on output. We find this restriction less appealing as a way of achieving identifica- tion; it should be clear however that our 3There is a technical subtlety here: strictly speaking, paper builds on Evans' work.

the fact that the sum of coefficients approaches zero is a necessary but not sufficient condition for the demand III. Estimation component to be stationary. However it turns out to be sufficient when unemployment and output growth are individually ARMA processes. This is proven in Quah We need to confront one final problem (1988).

before estimation. The representation we use 4A precise definition would obviously be tricky but is in Section I assumes that both the level of not needed for our argument. In models with imperfect unemployment and the first difference of the information, this would be the path of output, absent imperfect information. In models with nominal rigidi- logarithm of GNP are stationary around ties, this would be the path of output, absent nominal given levels. Postwar-U.S. data however sug- rigidities. In models that assume market clearing and gest instead both a small but steady increase perfect information, such as in Edward Prescott (1987), in the average unemployment rate over the the distinction between business cycles and trend is not sample, as well as a decline in the average a useful one.

VOL. 79 NO. 4 BLANCHARD AND QUAH: DEMAND AND SUPPLY DISTURBANCES growth rate of GNP since the mid-1970s.5 for eight lags is estimated using observations This raises two issues.

from 1950:2 through 1987:4.7 The GNP data The first is that our basic assumptions are quarterly; the monthly unemployment may be wrong in fundamental ways. For data are averaged to provide quarterly obser- instance, unemployment might in fact be vations. Evans (1987) has estimated essen- nonstationary, and affected even in the long tially the same bivariate VAR representa- run by demand and supply disturbances.

tion, although he uses instead the aggregate This is predicted by models with a "hyster- civilian unemployment rate. He has also esis" effect, as developed in Blanchard and tested the stationarity assumptions that we Lawrence Summers (1986), and used by them use here. The properties of the VAR repre- to explain European unemployment. This sentation and of the moving average repre- property also obtains in some recent growth sentation found by direct inversion do not models with increasing returns to scale, have any meaning within our framework, so where changes in the savings rate may affect we do not discuss those further here.

not only the level but also the growth rate of The mean growth rates for output are 3.62 output. While we cannot claim that such percent and 2.43 percent, at an annual rate, effects are not present here, we are willing to over 1948:2 through 1973:4, and 1974:1 assume that their importance is minimal, for through 1987:4, respectively. This break the period and the economy at hand.

point is chosen to coincide with the first Next, there is the issue of how to handle OPEC oil

shock. The fitted-time-trend re- the apparent time trend in unemployment, gression coefficient for the unemployment and the apparent slowdown in growth since rate series is 0.019, which implies a secular the mid-1970s. There is no clean solution for increase of 2.97 percentage points over the this, and we take an eclectic approach.6 To sample period. When we allow for a change focus the discussion, we present as a base in the output growth, we simply remove the case the results from estimation allowing for different sample means before estimating the a change in the growth rate of output, and vector autoregression; similarly when we al- for a secular increase in the unemployment low for a secular change in the unemploy- rate, as captured by a fitted-linear time-trend

ment rate, the fitted-trend line is removed regression line. There are three other cases of before VAR analysis.

interest: (a) there is no change in the growth It turns out that the results for cases $(a) - (c)$ rate of output, but there is a secular change are qualitatively similar to those for the base in the unemployment rate; (b) there is no case. More precisely, the moving average secular trend in the unemployment rate, but responses to demand and supply distur- there is a break in the average growth rate of bances are sufficiently close to those of the output; and finally, (c) there is neither a base case in their main features; the princi- change in the growth rate of output nor a pal differences lie in the magnitudes of the secular change in the unemployment rate.

responses. These differences are notable only A VAR system in real GNP growth (ΔY) in forecast error variance decompositions;

and the unemployment rate (U), allowing we will therefore present four such decom- position tables for the different cases below.

Because of the similarity in the other quali- 5The increase in the unemployment rate, sometimes tative features however, and to conserve attributed to demographic changes, is evident even in space, we will present results for the impulse the relatively homogenous labor group on which we focus our attention. We use the seasonally adjusted unemployment rate for Males, age 20 and over. This is from the U.S. Department of Labor, Bureau of Labor Statistics (BLS), 1982, and BLS Table A-39, February 7Estimation with twelve lags produced little differ- issues.

oSee for example Pierre Perron (1987) and Lawrence ence in the results. We also experimented with omitting Christiano (1988) on the statistical evidence for and the first five years, as the Korean War experience seemed against a break in average growth over the postwar anomalous. Again, the empirical results remain practi- period.

cally unchanged.

THE AMERICAN ECONOMIC REVIEW SEPTEMBER 1989 responses and historical decom- positions only 1.40 for the base case.8 1.20 We turn next to the dynamic effects of 1.00 demand and supply disturbances.

0.80 IV. Dynamic Effects of Demand and Supply 0.60 Disturbances 0.40 0.20

The dynamic effects of demand and sup- ply disturbances are reported in Figures 1 0.00 and 2. The vertical axes in Figures 1 and 2 10- -0.20 denote simultaneously the log of output and the rate of unemployment; the horizontal

-0.40 axis denotes time in quarters. Figures 3-6

-0.60 provide the same information, but now with one standard deviation bands around the FIGURE 1. RESPONSE TO DEMAND, $--$ = OUTPUT,

—- = UNEMPLOYMENT point estimates.9 Demand disturbances have a hump-shaped effect

on output and unemployment. Their effects peak after two to four quarters. The 1.00 effects of demand then decline to vanish 0.80 after about three to five years. The responses in output and unemployment are mirror im- 0.60 ages of each other; we return to this aspect of the results below after discussing the ef- 0.40 fects of supply disturbances.

0.20 The output response is smallest when the raw data are used, without allowing for a 0.00 ++++ break or a secular change in unemployment .10' (case c, not shown); it also decays the most

-0.20 rapidly in this case. Once a change in the

-0.40 average growth rate of output is allowed, the treatment of possible secular changes in un- −0.60 employment seems to be relatively unimpor- tant for the responses to demand distur- FIGURE 2. RESPONSE TO SUPPLY, −− = OUTPUT, − = $UNEMPLOYMENT$ bances.

These dynamic effects are consistent with a traditional view of the dynamic effects of aggregate demand on output and unemploy- and wages leads the economy back to equi- ment, in which movements in aggregate de- librium.

mand build up until the adjustment of prices Supply disturbances have an effect on the level of output which cumulates steadily over time. In the base case, the peak response is about eight times the initial effect and takes 8The other graphs are available from the authors place after eight quarters. The effect de- upon request.

creases to stabilize eventually. For good sta- 9More precisely, these boundaries are separated from tistical reasons, the long-run impact is im- the point estimate by the square root of mean squared precisely estimated. The dynamic response deviations in each direction, over 1000 bootstrap replications. Thus the bands need not be and indeed are not in unemployment is quite different: a posi- symmetric. By construction, they will of course neces- tive supply disturbance (that is, a supply sarily include the point estimate. In each case, pseudo- disturbance that has a positive long-run ef- histories are created by drawing with replacement from fect on output) initially increases unemploy- the empirical distribution of the VAR innovations.

VOL. 79 NO. 4 BLANCHARD AND QUAH: DEMAND AND SUPPLY DISTURBANCES 1.40 0.10 1.20 0.00 1.00

-0.10 0.80

-0.20 0.60

-0.30 0.40

-0.40 0.20

-0.50 0.00

-0.20

-0.60 FIGURE 3. OUTPUT RESPONSE TO DEMAND FIGURE 5. UNEMPLOYMENT RESPONSE TO Demand 1.40 0.50 1.20 1.00 0.40 0.80 0.30 0.60 0.20 0.40 0.10 0.20 0.00

-0.10

-0.20 FIGURE 4. OUTPUT RESPONSE TO SUPPLY FIGURE 6. UNEMPLOYMENT RESPONSE TO SUPPLY ment slightly. Following this increase, the effect is reversed after a few quarters, and output needed to maintain constant unem- unemployment slowly returns to its original ployment; real wage rigidities can explain steady-state value. The dynamic effects of a why increases in productivity can lead to a supply disturbance on unemployment are decline in unemployment after a few quar- largely over by about five years.

ters which persists until real wages have The qualitative results are similar across caught up with the new higher level of pro- all alternative treatments of breaks and time ductivity.

trends. The only significant difference ap- Figures 1 and 2 also shed interesting light pears in the initial unemployment response on the relation between changes in unem- to demand disturbances: in the case when ployment and output known as Okun's law.

neither break nor time trend is permitted, The textbook value of Okun's coefficient is the response is initially negative rather than about 2.5. Under our interpretation, this co- positive as in the base case. The one stan- efficient is a mongrel coefficient, as the joint dard deviation band does however include behavior of output and unemployment de- positive values.

pends on the type of disturbance affecting The response of unemployment and out- the economy. In the case of demand distur- put are suggestive of the presence of rigidi- bances, Figure 1 suggests that there is indeed ties, both nominal and real. Nominal rigidi- a tight relation between output and unem- ties can explain why in response to a positive ployment. At the peak responses, the graph supply shock, say an increase in productiv- suggests an implied coefficient between out- ity, aggregate demand does not initially put and unemployment that is slightly greater increase enough to match the increase in than 2. In the case of supply disturbances,

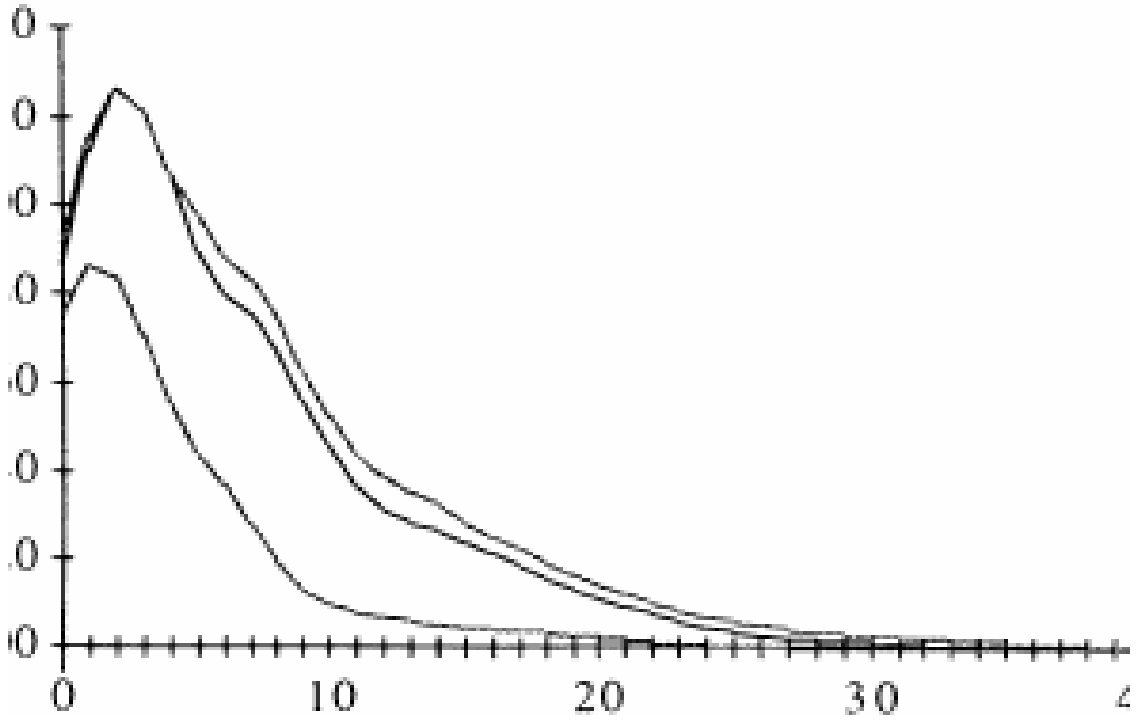


Figure 3.1: Course figure, source page 9, crop 1.

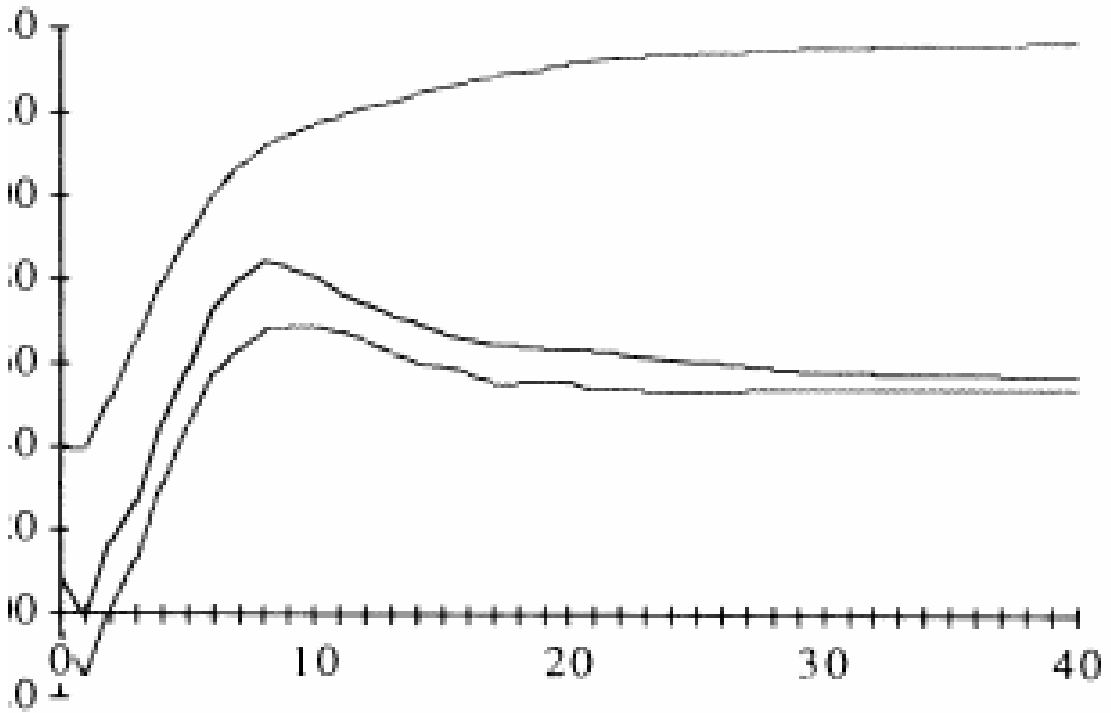


Figure 3.2: Course figure, source page 9, crop 2.

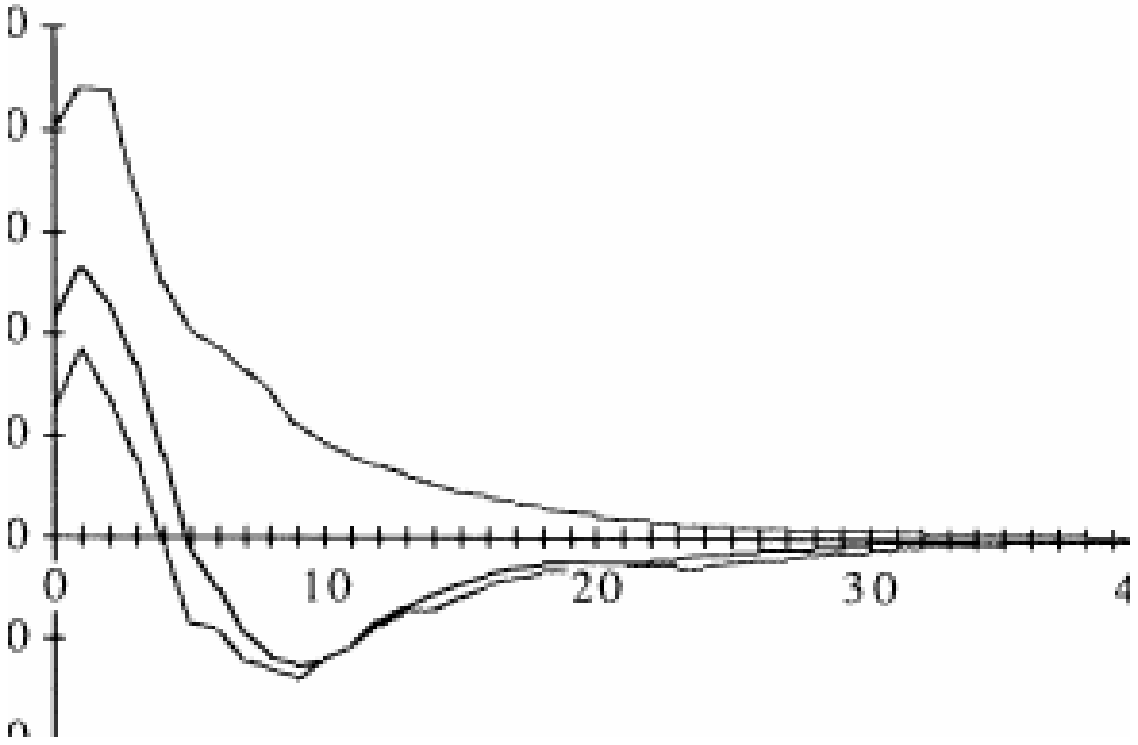


Figure 3.3: Course figure, source page 9, crop 3.

SEPTEMBER 1989 THE AMERICAN ECONOMIC REVIEW 8.40 there is no such close relation between out- put and unemployment. In the short run, 8.20 output increases, unemployment may

rise or 8.00 fall; in the long run, output remains higher whereas-by assumption-unemployment 7.80 returns to its initial value. In the intervening 7.60 period, unemployment and output devia- 7.40 tions are of opposite sign. At the peak re- sponses, Figure 2 suggests an implied coef- 7.20 ficient slightly exceeding four, higher in ab- 7.00 solute value than Okun's coefficient. That 1970 1980 1960 1950 the absolute value of the coefficient is higher FIGURE 7. OUTPUT FLUCTUATIONS ABSENT for supply disturbances than for demand Demand disturbances is exactly what we expect. Sup- ply disturbances are likely to affect the rela- tion between output and employment, and 0.10 to increase output with little or no change in

0.08 employment.

0.06 0.04 V. Relative Contributions of Demand and 0.02 Supply Disturbances.

0.00

-0.02 Having shown the dynamic effects of each

-0.04 type of disturbance, the next step is to assess

-0.06 their relative contribution to fluctuations in

-0.08 output and unemployment. We do this in

-0.10 1970 1980 1950 1960 two ways. The first is informal, and entails a comparison of the historical time-series of FIGURE 8. OUTPUT FLUCTUATIONS DUE TO the demand component of output to the DEMAND NBER chronology of business cycles. The second examines variance decompositions of components of unemployment are station- output and unemployment in demand and supply disturbances at various horizons.

ary.

The time-series for these components are presented in Figures 7 through 10. Superim- A. Demand Disturbances and NBER posed on these time series are the NBER Business Cycles peaks and troughs. Peaks are drawn as verti- cal lines above the horizontal axis, troughs From estimation of the joint process for as vertical lines below the axis.

output and unemployment, and our identify- The peaks and troughs of the demand ing restrictions, we can form the "demand component in output match closely the components" of output and unemployment.

NBER peaks and troughs. The two reces- These are the time paths of output and un- sions of 1974-1975 and 1979-1980 deserve employment that would have obtained in the special mention. Our decomposition at- absence of supply disturbances. Similarly, by tributes them in about equal proportions to setting demand innovations to zero, we can adverse supply and demand disturbances.

generate the time-series of "supply compo- This is best shown by giving the estimated nents" in output and unemployment. From values of the supply and demand innova- the identifying restriction that demand dis- tions over these periods. These are collected turbances have no long-run effect on output, in Table 1. The recession of 1974-75 is the resulting series of the demand compo- therefore explained by an initial string of nent in the level of output is stationary. By negative supply disturbances, and then of the same token, both the demand and supply

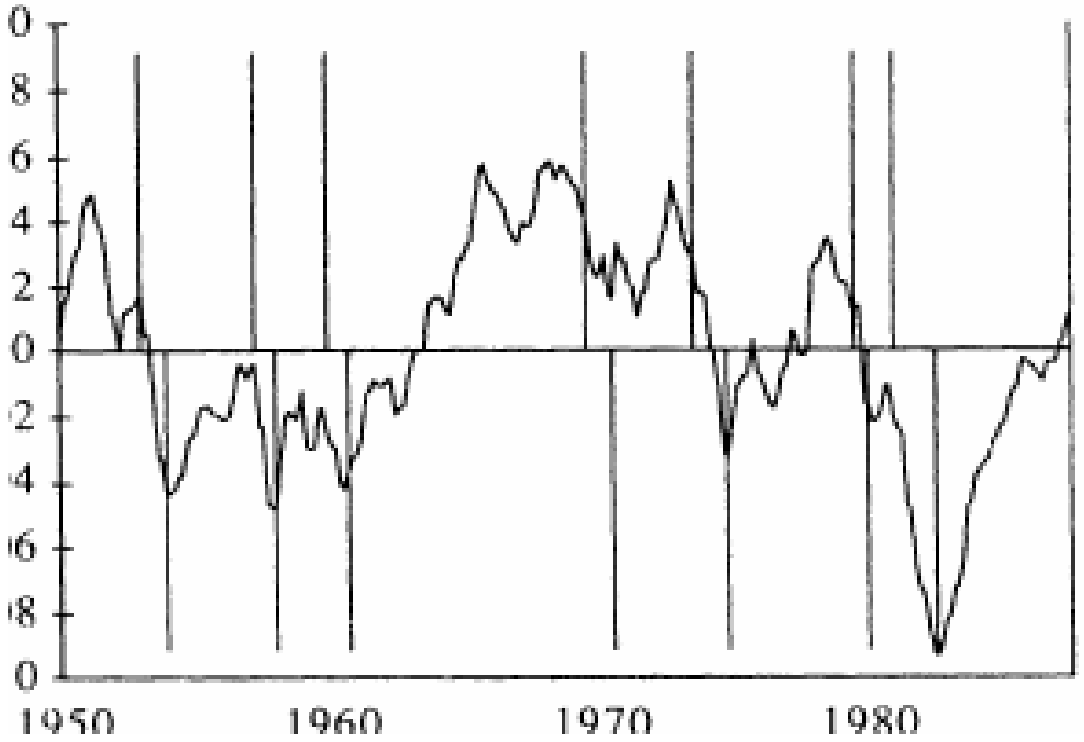


Figure 3.4: Course figure, source page 10, crop 1.

VOL. 79 NO. 4 BLANCHARD AND QUAH: DEMAND AND SUPPLY DISTURBANCES 7.00 TABLE 1 – DEMAND AND SUPPLY INNOVATIONSa BASE CASEb 6.00 mm Quarter Demand (Percent) Supply (Percent) 5.00 1973 − 3 −0.8 −1.9 4.00 1973 − 4 0.3 −0.4 3.00 1974 − 1 −0.7 −0.0 1974 − 2 0.5 −1.5 2.00 1974 − 3 −1.8 −1.0 1974 − 4 −0.7 1.1 1.00 1975 − 1 −1.5 2.8 0.00 1950 1960 1970 1980 1979 − 1 −0.5

−0.3 1979 − 2 −0.4 −1.7 FIGURE 9. UNEMPLOYMENT FLUCTUATIONS 1979-3 0.7 0.6 ABSENT DEMAND 1979-4 −0.8 0.2 1980 − 1 0.2 2.0 1980 − 2 −3.2 1.9 4.00 Notes:

3.00 a The identified innovations are obtained by applying 2.00 the transformation of Section I to the fitted-VAR resid- uals. By construction, the standard deviations of these 1.00 innovations are equal to 1 percent.

0.00 bThe estimated innovations for the other cases follow the same pattern as above.

-1.00

-2.00

-3.00 Notice that the supply component in out- put, presented in Figure 7, is clearly not a

-4.00 1950 1960 1970 1980 deterministic trend. It exhibits slower growth in the late 1950s, as well as in the 1970s.

FIGURE 10. UNEMPLOYMENT FLUCTUATIONS DUE Figures 9 and 10 give the supply and TO DEMAND demand components in unemployment. Un- employment fluctuations due to demand cor- negative demand disturbances. Similarly, the respond closely to those in the demand com- 1979-80 recession is first dominated by a ponent of GNP. This is consistent with our large negative supply disturbance in the sec- earlier finding on the mirror image moving ond quarter of 1979, and then a large average responses of unemployment and negative demand disturbance a year later.

output growth to demand disturbances. The Without appearing to interpret every single model attributes substantial fluctuation in residual, we find these estimated sequences unemployment to supply disturbances, again of demand and supply disturbances consis- with increases in the late 1950s and around the time of the oil disturbances of the 1970s.11 tent with less formal descriptions of these episodes.10 out 1973-1 to 1976-4. The estimated dynamic effects of 10Formal evidence of a slightly different nature is both demand and supply disturbances were nearly iden- also available. In Blanchard and Watson (1986), evi- tical to those described above. 11By construction, the supply component of unem- dence from four time-series is used to decompose fluc- tuations into supply and demand disturbances. There ployment is close to actual unemployment for the first

the recession of 1975 is attributed in roughly equal few observations in the sample. Thus, the large decrease proportions to adverse demand and supply distur- from 1950 to 1952 in the supply component simply bances, that of 1980 mostly to demand disturbances. To reflects the actual movement in unemployment in this see how much our characterization of the dynamic period. In light of this, we reestimated the model from effects of demand and supply disturbances depend on 1955-2 through the end of our sample. We found little the 1973-76 episode, we reestimated the model, leaving change in the empirical results.

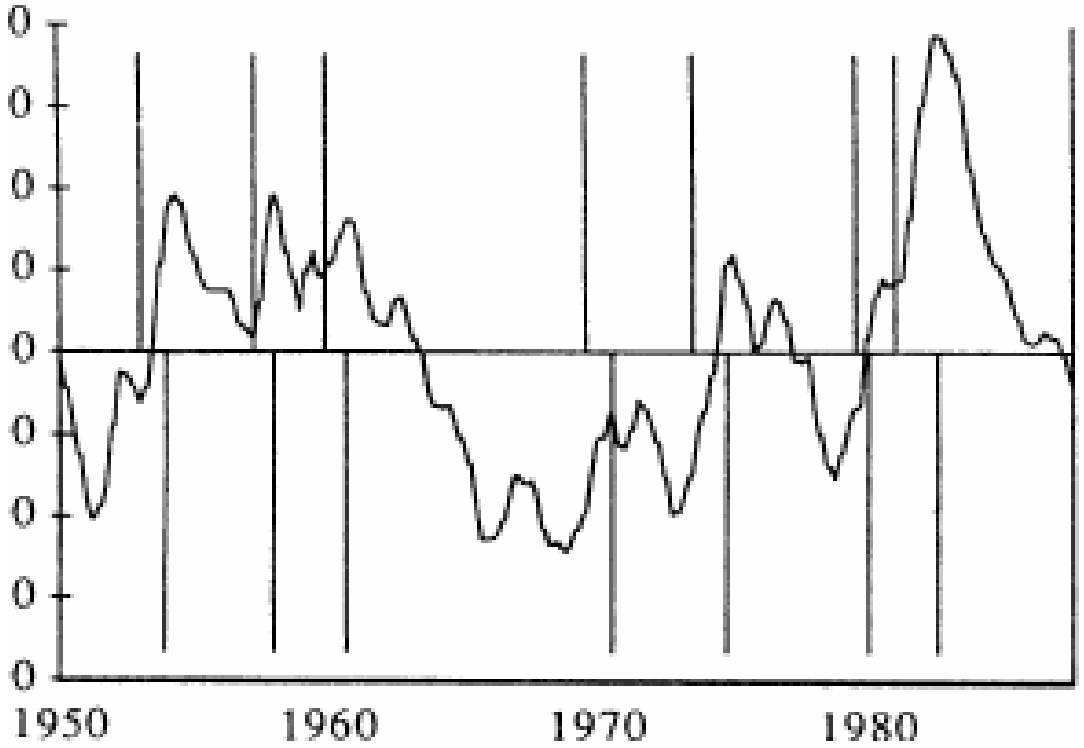


Figure 3.5: Course figure, source page 11, crop 1.

THE AMERICAN ECONOMIC REVIEW SEPTEMBER 1989 TABLE 2—VARIANCE DECOMPOSITION OF OUTPUT AND UNEMPLOYMENT (CHANGE IN OUTPUT GROWTH AT 1973/1974; UNEMPLOYMENT DETRENDED) Percentage of Variance Due to Demand:

Horizon (Quarters) Output Unemployment 99.0 51.9 (76.9, 99.7) (35.8, 77.6) 99.6 63.9 (78.4, 99.9) (41.8, 80.3) 99.0 73.8 (76.0, 99.6) (46.2, 85.6)

Δ 97.9 80.2 (71.0, 98.9) (49.7, 89.5) 81.7 87.3 (53.6, 92.9) (46.3, 87.0) 67.6 86.2 (30.9, 73.9) (52.9, 92.1) 39.3 85.6 (7.5, 39.3) (52.6, 91.6) TABLE 2A-VARIANCE DECOMPOSITION OF OUTPUT AND UNEMPLOYMENT (NO DUMMY BREAK, TIME TREND IN UNEMPLOYMENT)

Percentage of Variance Due to Demand:

Horizon (Quarters) Output Unemployment 83.8 79.7 (59.4, 93.9) (55, 3, 92.0) 87.5 88.2 (62.8, 95.4) (58.9, 95.2) 83.4 93.5 (58.8, 93.3) (61.3, 97.5) 78 9 95.7 (53.5, 90.0) (63.9, 98.2) 52.5 88.9 (31.4,68.6) (63.5, 94.5) 37.8 79.7 (21.3, 51.4) (58.8, 90.3) 18.7 75.9 (7.4, 23.5) (56.9, 88.6) B. Variance Decompositions ply disturbances in the last k quarters. The number for output at horizon $k, k = 1, ..., 40$ While the above empirical evidence is sug- gives the percentage of variance of the gestive, a more formal statistical assessment k-quarter ahead forecast error due to de- can be given by computing variance decom- mand. The contribution of supply, not re- positions for output and unemployment at ported, is given by 100 minus that number.

various horizons.

A similar interpretation holds for the num- Tables 2, and 2A-C give this variance bers for unemployment. The numbers in decomposition for the different cases. The parentheses are one standard deviation bands, surrounding the point estimate.12 table has the following interpretation. Define the k quarter-ahead forecast error in output as the difference between the actual value of output and its forecast from equation (2) as of k quarters earlier. This forecast error is 12Again, these bands are asymmetric, and obtained due to both unanticipated demand and sup- as described above.

VOL. 79 NO. 4 BLANCHARD AND QUAH: DEMAND AND SUPPLY DISTURBANCES TABLE 2B—VARIANCE DECOMPOSITION OF OUTPUT AND UNEMPLOYMENT (CHANGE IN OUTPUT GROWTH AT 1973/1974; NO TREND IN UNEMPLOYMENT) Percentage of Variance Due to Demand:

Horizon (Quarters) Output Unemployment 99.3 50.7 (75.0, 99.8) (32.0, 79.9) 99.7 63.2 (77.6, 99.9) (36.6, 83.3) 99.4 73.4 (76.1, 99.7) (40.8, 88.3) 98.6 80.0 (72.9, 99.2) (44.3, 91.1) 86.3 88.4 (53.2, 91.5) (50.0, 94.6) 75.5 88.9 (40.9, 83.0) (49.9, 94.6) 50.4 90.0 (12.5, 54.8) (49.7, 95.0) TABLE 2C-VARIANCE DECOMPOSITION OF OUTPUT AND UNEMPLOYMENT (NO DUMMY BREAK, NO TREND IN UNEMPLOYMENT) Percentage of Variance Due to Demand:

Horizon (Ouarters) Output Unemployment 45.2 99.8 (20.1, 77.6) (76.6, 100.0) 50.2 98.3 (23.4, 79.9) (72.8, 99.3) 44.2 92.7 (20.4, 77.0) (67.1, 97.8) 38.9 85.9 (17.1, 72.7) (62.7, 95.8) 19.6 60.5 (8.8, 54.4) (44.3, 89.6) 12.9 47.6 (6.5, 43.5) (35.2, 87.8) 5.2 40.5 (2.4, 17.7) (31.4, 87.1) Our identifying restrictions impose only case, the relative contribution of demand one restriction on the variance decomposi- disturbances to output fluctuations, at a four tions, namely that the contribution of supply quarters horizon, in 98 percent. This contri- disturbances to the variance of output tends bution falls to 79 percent when no break is allowed but there is a time trend in unem- to unity as the horizon increases. All other aspects are unconstrained.

ployment, remains about the same when a Two principal conclusions emerge from break is allowed in output growth but there these tables.

is no trend in the unemployment rate. When First, the data do not give a precise an- neither a break nor a trend is permitted, it is swer as to the relative contribution of de- only 39 percent. Next, the standard error mand and supply disturbances to move- bands are quite large in each case, ranging ments in output at short and medium-term from 71 to 99 percent in the base case, 54 to 90 percent in case A, 73 to 99 percent in case horizons. The results vary across alternative treatments of break and trend. In the base B, and 17 to 73 percent in case C. Evidently

THE AMERICAN ECONOMIC REVIEW SEPTEMBER 1989 when a break is permitted in output growth, the demand and supply components of GNP the treatment of the trend in unemployment with a larger set of macroeconomic vari- appears to be quite unimportant. These cases ables. Preliminary results appear to confirm are also when the demand contribution is our interpretation of shocks. We find in par- more precisely estimated. Despite the dif- ticular the

supply component of GNP to be ferences across estimates, and the uncer- positively correlated with real wages at high tainty associated with each set, we view the to medium frequencies, while no such corre- results as suggesting an important role for lation emerges for the demand component.13 demand disturbances in the short run.

The second extension is to enlarge the Second, estimates of the relative contribu- system to one in four variables, unemploy- tion of the different disturbances to unem- ment, output, prices, and wages. This would ployment do not appear to vary a great deal also allow examination of different ques- across alternative treatments of break and tions from an alternative perspective, as in trend. The contribution of demand distur- Blanchard (1989). As one might expect, wage bances, four quarters ahead, to unemploy- and price data will help identify more explic- ment fluctuations varies from 80 to 96 per- itly supply and demand disturbances. Re- cent. In the base case, the one standard error search by Jordi Gali (1988), Sung-in Jun band ranges from 50 to 90 percent with a (1988), and Matthew Shapiro and Watson point estimate of 80 percent. In all cases, the

(1988) has already extended our work in that demand disturbance appears to be quite im- particular direction.

portant for unemployment fluctuations at all horizons.

Technical Appendix VL Conclusion and Extensions This technical appendix discusses further and establishes the claims made in the sec- We have assumed the existence of two tion on interpretation.

types of disturbances generating unemploy- First, we asserted in the text that our ment and output dynamics, the first type identification scheme is approximately cor- having permanent effects on output, the sec- rect even when both disturbances have per- ond having only transitory effects. We have manent effects on the level of output, pro- argued that these two types of disturbances vided that the long-run effect of demand on could usefully be interpreted as supply and output is small. We now prove this.

demand shocks. Under that interpretation, The first element of the model, output we have concluded that demand distur- growth, has the moving average representa- tion in demand and supply disturbances:

bances have a hump-shaped effect on output and unemployment which disappears after $\Delta Y_t = a_{11}(L)e_{dt} + a_{12}(L)e_{st}$, approximately two to three years, and that supply disturbances have an effect on output where $a_{11}(1)$ is the cumulative effect on the which cumulates over time to reach a plateau level of output Y of the disturbance e_d. The after five years. We have also concluded that demand disturbances make a substantial

moving average representation $C(L)$, to- gether with the innovation covariance matrix contribution to output fluctuations at short- $\hat{\Omega}$, is related to our desired interpretable and medium-term horizons; however, the data do not allow us to quantify this contri- representation through some identifying ma- bution with great precision.

While we find this simple exercise to have been worthwhile, we also believe that further work is needed, especially to validate and 13The methodology and results will be described in a future paper. The statement in the text refers to the sum refine our identification of shocks as supply of correlations from lags -5 to $+5$ between the supply and demand shocks. We have in mind two innovation derived in this paper and the innovations specific extensions. The first is to examine in real wages obtained from univariate ARIMA es- the co-movements of what we have labeled timation.

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S, such that:

ply (under our identifying assumption that the long-run effect of demand is zero) multi- $SS' = \Omega$, and $A(L) = C(L)S$.

plied by the $(2, 2)$ element of the orthogonal matrix $V(\delta)$. As δ tends to zero, the (2,2) The model is identified by choosing a unique element of $V(\delta)$ tends continuously to zero identifying matrix S. In the paper, we as well. But, up to a column sign change, the selected the unique matrix S such that unique $V(\delta)$ with (2,2) element equal to zero $a_{11}(1) = 0$. is the identity matrix. This establishes that Let the long-run effect of the demand $S(\delta) \to S(0)$, element by element. Hence, we

disturbance be δ instead, where $\delta > 0$ with- have shown that $|S(\delta) - S(0)| \to 0$ as $\delta \to 0$.

out loss of generality. For each δ, this im- plies a different identifying matrix $S(\delta)$. Let $|S(\delta)$ $- S(0)| = \max_{j,k}(S_{jk}(\delta) - S_{jk}(0))^2$; this Next, we turn to the effects of multiple measures the deviation in the implied identi- demand and supply disturbances: Suppose fying matrix from that which we use. Since that there is a $p_d \times 1$ vector of demand dis-

the approximation is thus seen to be a turbances f_{dt}, and a $p_s \times 1$ vector of supply disturbances f_{st}, so that:

finite-dimensional problem, any matrix norm will induce the same topology, which is all that is needed to study the continuity prop- $$\begin{pmatrix} \Delta Y_t \\ U_t \end{pmatrix} = \begin{pmatrix} B_{11}(L)' & B_{12}(L)' \\ B_{21}(L)' & B_{22}(L)' \end{pmatrix} \begin{pmatrix} f_{dt} \\ f_{st} \end{pmatrix},$$

erties of our identification scheme. All of the empirical results vary continuously in S rel- ative to this topology. Thus, it is sufficient to show that where B_{jk} are column vectors of analytic functions; B_{j1} has the same dimension as f_d, B_{j2} has the same dimension as f_s, and $B_{11}(z) = (1 - z)\beta_{11}(z)$, for some vector of analytic functions β_{11}. Each disturbance has

$|S(\delta) - S(0)| \to 0$ as $\delta \to 0$.

In words, if an economy has long-run effects in demand that are small but different from a different distributed lag effect on output zero, our identifying scheme which incor- and unemployment.

rectly assumes the long-run effects to be zero Since our VAR method allows identifica- nevertheless recovers approximately the cor- tion of only as many disturbances as ob- rect point estimates.

served variables, it is immediate that we will not be able to recover the individual compo- PROOF:

nents of $f = (f_d' f_s')'$. To clarify the issues involved, we provide We prove this as follows. Since both $S(0)$ and $S(\delta)$ are matrix square roots of Ω, there an explicit example where our procedure exists an orthogonal matrix $V(\delta)$ such that:

produces misleading results. Suppose that there is only one supply disturbance and $S(\delta) = S(0)V(\delta)$, where $V(\delta)V(\delta)' = I$.

two demand disturbances: $f_{dt} = (f_{d1,t}, f_{d2,t})'$.

Suppose further that the first demand distur- Then the long-run effect of demand is the bance affects only output, while the second $(1, 1)$ element in the matrix:

demand disturbance affects only unemploy- ment. The supply disturbance affects both $A(1; \delta) = C(1)S(\delta) = C(1)S(0)V(\delta)$. output and unemployment. Formally, as- sume that the true model is:

But recall that the elements of the first row of $C(1)S(0)$ are respectively, the long-run $X_t = \begin{pmatrix} \Delta Y_t \\ U_t \end{pmatrix} = \begin{pmatrix} 1 - L & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} J_{d1,t} \\ f_{d2,t} \\ f_t \end{pmatrix}.$

effects of demand and of supply on the level of output, when the long-run effect of de- mand is restricted to be zero. Thus for any $V(\delta)$, the new implied long-run effect of demand is simply the long-run effect of sup- An unrestricted VAR representation corre-

THE AMERICAN ECONOMIC REVIEW SEPTEMBER 1989 sponding to this data generating process is number of actual and explicitly modeled dis- found by applying the calculations in Yu turbances be benign?

Rozanov (1967), Theorem 10.1, (pp. 44–48).

We state the necessary and sufficient con- The implied moving average representation ditions for this as a theorem which is proved is:

below.

THEOREM 3.1– Bivariate moving-average representation theorem

Let X be a bivariate stochastic $X_t = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}(2-L) \\ \sqrt{2} & 0 \end{pmatrix} \eta_t, \quad E\eta_t\eta_t' = I.$

sequence generated by (i) $X_t = B(L)f_t$; (i) $f_t = (f_{dt}'f_{st}')'$, with $f_d p_d \times 1$, $f_s p_s \times 1$; (ii) $Ef_tf_{t-k} = I$ if $k = 0$, and 0 otherwise; (iv) $B(z) = \begin{pmatrix} B_{11}(z)' & B_{12}(z)' \\ B_{21}(z)' & B_{22}(z)' \end{pmatrix}$; (v) $B_{11}(z) = (1-z)\beta_{11}($

It is straightforward to verify that the matrix covariogram implied by this moving aver- age matches that of the true underlying model.

(vi) β_{11}, B_{21}, B_{12}, B_{22} are column vectors of analytic functions; β_{11} and $B_{21} p_d \times 1$, B_{12} Further, the unique zero of the determinant is 2, and consequently lies outside the unit

circle. Therefore this moving average repre- and $B_{22} p_s \times 1$;

(vii) BB^* is full rank on $|z| = 1$, where * sentation is, as asserted, obtained from the denotes complex conjugation followed by vector autoregressive representation of the transposition.

true model.

Then there exists a bivariate moving aver- However, this moving average does not age representation for X, $X_t = A(L)e_t$, such satisfy our identifying assumption that the that:

"demand" disturbance has only transitory (viii) $A(z) = \begin{pmatrix} a_{11}(z) & a_{12}(z) \\ a_{21}(z) & a_{22}(z) \end{pmatrix}$, with a_{11}, a_{12}, a_{21}, a_{22} scalar functions, det $A \neq 0$ for all $|z|$

effects on the level of output. We therefore apply our identifying transformation to ob- tain:

≤ 1; (ix) $a_{11}(z) = (1-z)\alpha_{11}(z)$, with α_{11} analytic on $|z| \leq 1$; and (x) $e_t = \begin{pmatrix} e_{dt} \\ e_{st} \end{pmatrix}$, $Ee_te_{t-k} = I$ if $k = 0$, and is 0

$$X_t = \begin{pmatrix} \frac{1}{2}(1-L) & \frac{1}{2}(3-L) \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e_{dt} \\ e_{st} \end{pmatrix}.$$

otherwise.

This moving average representation is what In the bivariate representation, ed is or- we would recover if in fact the data are thogonal to f_s, and e_s is orthogonal to f_d, at all leads and lags if and only if there exists a generated by the three disturbances (f_{d1}, f_{d2}, f_s). Notice that while the supply disturbance pair of scalar functions γ_1, γ_2 such that:

f_s affects both output growth and unemploy- $B_{21} = \gamma_1 \cdot \beta_{11}$ ment equally and only contemporane- ously, we would identify e_s to have a larger effect $B_{22} = \gamma_2 \cdot B_{12}$. on output than on unemployment, together with a distributed lag effect on output. Fur- ther a positive demand disturbance, re-

Conditions (i)–(vii) describe the true data stricted to have only a transitory effect on generating process for the observed data in output, is seen to have a contemporaneous output growth and unemployment. There are p_d demand and p_s supply disturbances; (v) negative impact on unemployment. In the expresses the requirement that demand dis- true model however, no demand disturbances affect output and unemployment to- turbances have only transitory effects on the gether, either contemporaneously or at any level of output. Condition (vii) is a regularity lag. In conclusion, a researcher following our condition that allows the existence of a VAR bivariate procedure is likely to be seriously mean square approximation. The moving av- erage recovered by our VAR procedure is

misled when in fact the true underlying described by $(viii)$-(x): the theorem guaran- model is driven by more than two disturbances. Having seen this, we ask under what tees that there always exists such a represen- circumstances will this mismatch in the tation.

VOL. 79 NO. 4 BLANCHARD AND QUAH: DEMAND AND SUPPLY DISTURBANCES The second part of the theorem establishes rem, notice that A has been constructed so necessary and sufficient conditions on the that on $|z| = 1$:

underlying model such that the bivariate $|1 - z|^2|\alpha_{11}|^2 + |\alpha_{12}|^2$ identification procedure does not inappro- priately confuse demand and supply distur- $|1 - z|^2\beta'_{11}(\beta'_{11})^*$ bances. In words, correct identification is possible if and only if the individual dis- $+B'_{12}(B'_{12})^*$;

tributed lag responses in output growth and unemployment are sufficiently similar across $(1 - z)\alpha_{11}a^*_{21} + a_{12}a^*_{22}$ the different demand disturbances, and across the different supply disturbances. This $= (1 - z)\beta'_{11}(B'_{21})^*$ does not mean that the dynamic responses in output growth and unemployment across de-

$+B'_{12}(B'_{22})^*$; mand disturbances must be identical or pro- portional, simply that they differ up to a $|a_{21}|^2 + |a_{22}|^2 = B'_{21}(B_{21})^*$ scalar lag distribution.

Thus even though in general a bivariate $+B'_{22}(B_{22})^*$.

procedure is misleading, there are important and reasonable sets of circumstances under For e_d to be orthogonal to f_s, and e_s to be which our technique provides the "correct" orthogonal to f_d at all leads and lags, it is answers. For instance, suppose that there is necessary and sufficient that on $|z| = 1$:

only one supply disturbance but multiple (a) $|\alpha_{11}|^2 = \beta'_{11}(\tilde{\beta}'_{11})^*$; demand disturbances. Suppose further that (b) $|a_{12}|^2 = B'_{12}(B'_{12})^*$; each of the demand components in the level

(b) $|a_{12}| = B_{12}(B_{12})^*$; (c) $a_{11}a^*_{21} = \beta'_{11}(B'_{21})^*$; (d) $a_{12}a^*_{22} = B'_{12}(B'_{22})^*$; (e) $|a_{21}|^2 = B'_{21}(B'_{21})^*$; (f) $|a_{22}|^2 = B'_{22}(B'_{22})^*$.

of output has the same distributed lag rela- tion with the corresponding demand compo- nent in unemployment. This assumption is consistent with our "production func- Consider relations (a), $(c), and (e)$. Denot- tion"-based interpretations below. Then our procedure correctly distinguishes

the dy- ing complex conjugation of B by B, the trian- namic effects of demand and supply compogle inequality implies that:

nents in output and unemployment.

$$|\beta_{11} \cdot \overline{B}_{21}| = \left| \sum_{j=1}^{p_d} \beta_{11j} B^*_{21j} \right| \leq \sum_{j=1}^{p_d} |\beta_{11j} B^*_{21j}|,$$

PROOF OF THEOREM: By (i)-(vi), the matrix spectral density of X is given by $S_X(\omega) = B(z)B(z)^*|_{|z|=1}$. By reasoning analogous to that in pp. 44-48 in Rozanov (1967), there where the inequality is strict unless B_{21} is a complex scalar multiple of β_{11} for each z on exists a 2×2 matrix function C, each of $|z| = 1$. Next, by the Cauchy-Schwarz inequal-

whose elements are analytic, with det $C \neq 0$ for $|z| \leq 1$, and $S_x = CC^*$ on $|z| = 1$. This ity, represents X as a moving average in unit $\sum_i |\beta_{11j} B^*_{21j}| \leq \left(\sum_i |\beta_{11j}|^2\right)^{1/2} \cdot \left(\sum_i |B_{21j}|^2\right)^{1/2}$,

variance orthogonal white noise, obtainable from its VAR mean square approximation.

However, such a moving average C need not again with strict inequality except when B_{21} is satisfy the condition that the first (demand) a complex scalar multiple of β_{11}, for each z on disturbance have only transitory impact on the $|z| = 1$. Therefore:

level of output (condition (ix)). Form the 2×2 orthogonal matrix M whose second col- $|\alpha_{11} a^*_{21}|^2 \leq |\alpha_{11}|^2 \cdot |a_{22}|^2$, on $|z| = 1$, umn is the transpose of the first row of C evaluated at $z = 1$, normalized to have length

where the inequality is strict except when B_{21} 1, as a vector. Then $A \equiv CM$ provides the moving average representation satisfying is a complex scalar multiple of β_{11} on $|z| = 1$.

$(viii) - (x)$. For the second part of the theo- But the strict inequality is a contradiction as

THE AMERICAN ECONOMIC REVIEW SEPTEMBER 1989 and Unemployment," Stanford Univer- α_{11} and α_{22} are just scalar functions. Thus sity, mimeo., 1988.

$(a), (c), and (e)$ can be simultaneously satis- fied if and only if there exists some complex Cochrane, John, "How Big Is the Random Walk in GNP?," Journal of Political Econ- scalar function $\gamma_1(z)$ such that $B_{21} = \gamma_1 \cdot \beta_{11}$.

A similar argument applied to (b), (d), and omy, October 1988, 96, no. 5, 893-920.

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3.0.2 Optional reading – Cochrane QJE94

PERMANENT AND TRANSITORY COMPONENTS OF GNP AND STOCK PRICES* JOHN H. COCHRANE This paper uses two-variable autoregressions to characterize transitory components in GNP and stock prices. Shocks to GNP holding consumption constant are almost entirely transitory, and account for large fractions of the variance of GNP growth. If consumption does not change, consumers must think that any GNP change is transitory. The facts that the consumption/GNP ratio forecasts GNP growth and that consumption is nearly a random walk drive this result. An implication is that consumption provides a good estimate of the "trend" in GNP.

Prices and dividends behave similarly: shocks to prices holding dividends constant are almost entirely transitory.

I. INTRODUCTION A recent voluminous literature has examined the long-run properties of GNP and stock prices, with surprising results. One would expect that GNP reverts to "potential GNP" or some other trend following a shock. Yet many studies have found no mean- reversion, especially in postwar U. S. GNP.1 This view obviously challenges a broad spectrum of macroeconomic theories designed to produce and understand transitory fluctuations.

Conventional wisdom once held that stock prices are random walks (martingales) that display no mean-reversion. Yet a large number of recent studies have instead found mean-reversion or transitory components in stock prices.2 Depending on the author's tastes, these findings are interpreted as evidence for "fads"- irrational investor behavior-or as evidence for as-yet unmodeled time-variation in real investment opportunities.

To examine long-run properties of GNP and stock prices, I focus on simple two-variable autoregressions: GNP and consump- *I thank Olivier Blanchard, John Campbell, Frank Diebold, Lars Hansen, John Huizinga, Robert Lucas, Guillermo Mondino, Mark Watson, James Stock, Michael Woodford, referees, and especially Eugene Fama for helpful comments. Two drafts of this paper were completed while I was a Visiting Scholar at the Hoover Institution, whose hospitality I gratefully acknowledge. This resear School of Business at the University of Chicago.

1. This view has a long history, starting at least with Fisher [1925] and McCulloch [1975]. More recently, among others, see Nelson and Plosser [1982], Campbell and Mankiw [1987], and Cogley [1990]. Clark [1987] and Cochra exception.

2. Fama [1991] reviews predictability of stock returns.

1994 by the President and Fellows of Harvard College and the Massachusetts Institute of Technology.

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QUARTERLY JOURNAL OF ECONOMICS tion growth are regressed on their lags and the lagged consumption/ GNP ratio: stock returns and dividend growth are regressed on their lags

and the lagged dividend/price ratio.

The results of these VARs confirm and characterize statisti- cally and economically important transitory components in GNP and stock prices. For example, transitory shocks account for an estimated 70-80 percent of the variance of GNP growth and 57 percent of the variance of annual stock returns.

Why do I find such strong transitory variation in postwar U.S.

GNP data, where others have not? Transitory variation requires long-horizon forecastability. If GNP growth is unforecastable, GNP follows a random walk. If a negative shock signals many quarters of above-average growth, then GNP contains a transitory component that reverts to its mean. Most of the GNP literature is based on univariate forecasts. It documents that lagged GNP growth forecasts future GNP growth poorly. The consumption/ GNP ratio is a much more potent forecaster of long-horizon GNP growth. Thus, it can imply much larger transitory variation.

The forecastability of GNP growth can be documented by regressing it on a variety of variables. The consumption/GNP ratio is special, because it is stable over long time periods (consumption and GNP are cointegrated), while consumption is nearly a random walk. As a result, if GNP is more than its customary ratio to consumption. GNP must be forecast to decline until the ratio is reestablished. In this way, consumption defines the "trend" in GNP.

It is natural to interpret these features of the data via the simple permanent income model. The model predicts that consump- tion is a random walk and that consumption and total income are cointegrated. If consumption does not change, consumers must think any fluctuation in GNP is transitory. Thus, by observing consumption, we separate GNP into permanent and transitory components, as viewed by consumers. Of course, the permanent income model can be statistically rejected, since consumption is not exactly a random walk. Nonetheless, it is a good approximation that helps to describe and interpret the VAR results.

Similar points hold for dividends and stock prices. Dividend/ price ratios forecast returns better than lagged returns, while dividends are nearly a random walk. Therefore, price shocks with constant dividends are almost entirely transitory. As detailed below, the present value model plus the hypothesis that managers smooth dividends predicts and interprets these features of the data. In addition, this model suggests that we interpret the price

COMPONENTS OF GNP AND STOCK PRICES shocks as "discount rate" shocks and the dividends shocks as "earnings" shocks.

II. MEASURING THE PERSISTENCE OF GNP SHOCKS IN POSTWAR U. S. DATA A. VAR and Impulse-Response Functions Table I presents a vector autoregression of log GNP and log nondurable + services consumption growth on their lags and the lagged log consumption/GNP ratio. It also presents a univariate TABLE I CONSUMPTION AND GNP REGRESSIONS y_t denotes log real GNP. c_t denotes log nondurable + services consumption.

Δ denotes first differences, $\Delta y_t = y_t - y_{t-1}$. Data sample 1947:1–1989:3.

1. Vector autoregression Right-hand variable $\mathbb{R}^2$ Left-hand variable p-valueb $c_{t-1} - y_{t-1}$ Δc_{t-1} Δc_{t-2} Δy_{t-1} Δy_{t-2}

const.

Δc_t coeff.

-0.43 -0.02 0.07 -0.02 0.09 -0.02 0.06 t-stat.

-0.49 -1.23 0.90 -0.19 1.91 -0.40 0.7 p-valuea (percent) 0.29 5.19 0.08 0.52 Δv_t coeff.

0.16 0.22 0.14 0.27 t-stat.

3.49 3.45 3.81 1.12 2.74 1.89 0.0 p-valuea (percent) 1.8 2. Variance decomposition Variance of Due to $\Delta c_t - E_{t-1}\Delta c_t$ Δc_t Δy_t $\Delta y_t - E_{t-1}\Delta y_t$ "Permanent" c shock "Temporary" y shock 3. Univariate autoregression Left-hand variable Right-hand variable R^2

const.

p-valueb Δy_t Δy_{t-4} Δy_{t-1} Δy_{t-2} Δy_{t-3} Coefficient 0.56 0.33 0.19 −0.11 0.18 t-statistic −1.37 −1.36 4.82 4.17 2.39 0.0 a. Percent of replications with a coefficient farther from 0, under the null that the coefficient = 0 and c/y has

a unit root (bootstrap).

b. Percent probability value of an F-test for the joint significance of the right-hand variables.

QUARTERLY JOURNAL OF ECONOMICS autoregression of GNP growth on lagged GNP growth. Both regressions are generous; for example, the last lag is statistically insignificant.

The regressions verify that GNP growth is predictable. In particular, the t-statistic of GNP growth on the lagged consumption/ GNP ratio is 3.45, verifying the hunch that the ratio helps to forecast GNP.3 Consumption growth is slightly predictable in the VAR, but with a much lower R^2 than GNP growth. Most of this slight predictability comes from the first-order serial correlation of quarterly consumption growth and the first lag of GNP, as noted by the permanent income literature [Flavin 1981; Campbell and Mankiw 1989].

Table I also presents probability values for t-statistics on the lagged consumption/GNP ratio obtained from a bootstrap in which that coefficient is zero. The other coefficients are reestimated imposing this null. Since the consumption/GNP ratio has a unit root under this null, the distribution of the t-statistic is nonstand- ard. Nonetheless, the coefficient is again significant at normal $|evels|^4$ The top panel of Figure I presents impulse-response functions of the consumption-GNP VAR. The VAR errors are orthogonalized so that consumption does not respond contemporaneously to a GNP shock. Equivalently, current consumption growth is included in the GNP growth regression. I discuss this assumption below.

Note three features of the impulse-response functions. First, the eventual response of the two series to each shock is the same. If 3. Private GNP—GNP less government purchases—paints an even stronger picture. The consumption/private GNP ratio ignores long-term fluctuations in government purchases, and responds more strongly to business-cycle related f produces similar results.

4. Under this null, the estimated coefficients are biased to positive values. Thus, the negative consumption coefficient is "more significant" than the larger, positive GNP coefficient. Again, private GNP gives stronger results: the bootstrap p-value is less than 1/2000 (the number of replications).

Augmented Dickey-Fuller tests also reject the unit root null for the consumption/ GNP ratio at better than 1 percent levels. For private GNP, the statistic is three times larger than the 1 percent critical value.

I include these tests to verify that the consumption/GNP ratio forecasts GNP growth. Stationarity of the consumption/GNP ratio is best regarded as an a priori assumption based on common sense and basic economic theory. Blough [1992] and Cochrane [1991] warn of the dangers of pretesting for unit roots or cointegration and then imposing that form in subsequent analysis. Their poin forecast slow mean reversion in GNP.

COMPONENTS OF GNP AND STOCK PRICES A. Multivariate 1.2 Consumption response GNP response 1.0 Response to 1 shock 0.8 Consumption shock 0.6 0.4 0.2 GNP shock 0.0 Quarters after shock B. Univariate 1.8 1.6 Univariate estimate 1.4 Response to a unit shock

1.2 1.0 0.8 VAR estimate 0.6 0.4 0.2 0.0 L Quarters after shock FIGURE I Impulse-Response Functions for Consumption and GNP Top panel: Response of consumption (c) and GNP (y) to one-standard-deviation shocks in the consumption-GNP VAR (Table I).

Bottom panel: Response of GNP to a unit univariate GNP shock. The "univariate estimate" is based on the regression of GNP growth on past GNP growth, panel 3 of Table I; the "VAR estimate" is based on the consumption-GNP VA Bars show bootstrap one standard error bands.

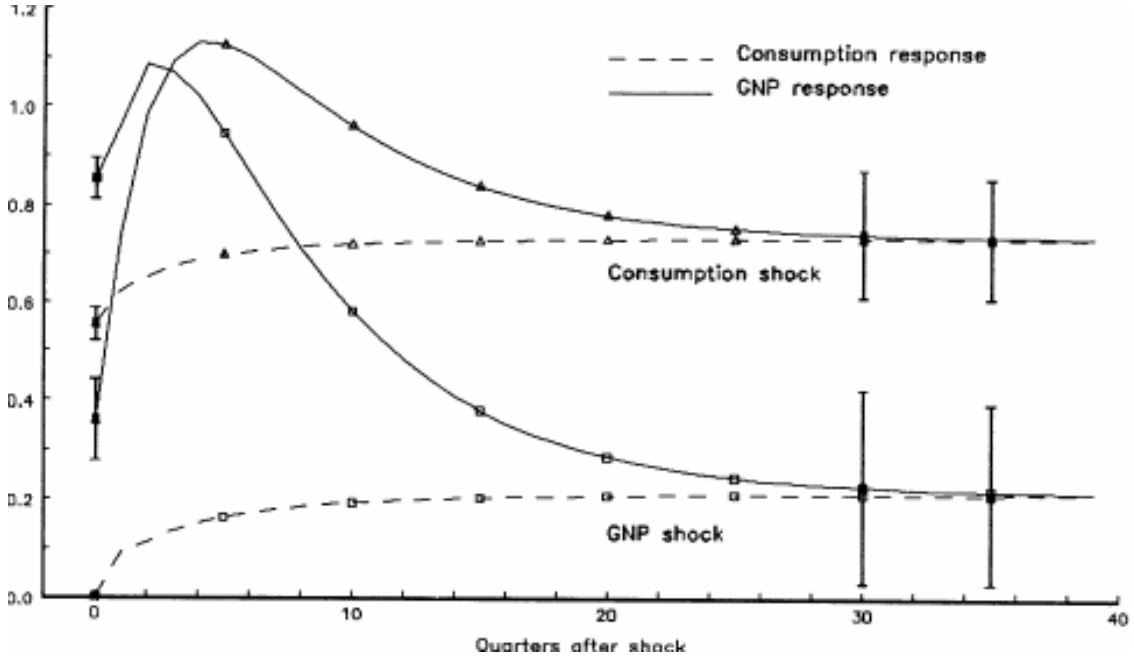


Figure 3.6: Course figure, source page 5, crop 1.

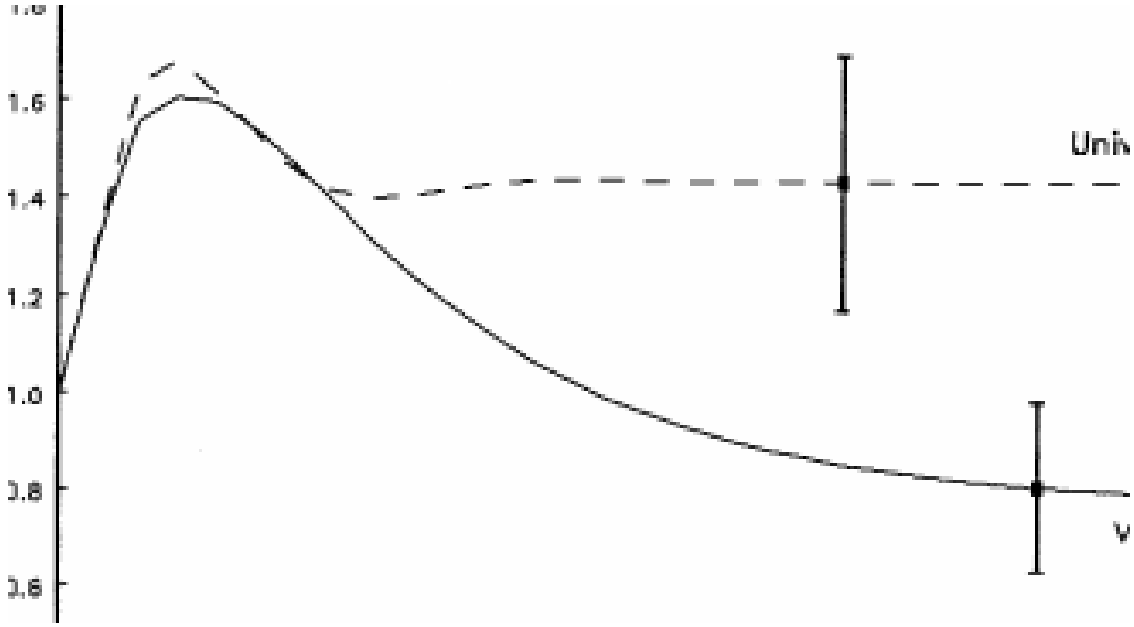


Figure 3.7: Course figure, source page 5, crop 2.

QUARTERLY JOURNAL OF ECONOMICS the two responses did not end up at the same value, the consump- tion/GNP ratio would not be restored following a shock. This feature results from

the inclusion of the consumption/GNP ratio on the right-hand side, and the fact that it enters with nonzero coefficients.

Second, look at the responses of consumption and GNP to a consumption shock. Consumption is almost a random walk: its impulse response function is almost flat. The slight rise in the long-run consumption impulse-response is statistically insignifi- cant. Also, it disappears with slight changes in specification, such as changes in the sample period, annual data, or with private GNP.

GNP has a hump-shaped response to the consumption shock.

Fama [1993] interprets this pattern as the lagged response of investment to a wealth shock.

Third, and most important, look at the responses to a GNP shock. This is a shock to GNP that does not contemporaneously shock consumption (by the orthogonalization assumption). This shock has a small impact on consumption at any horizon, but the response of GNP to this shock is almost completely transitory.

How important are the transitory components of GNP? Figure I gives some indication. The figure plots the responses to one- standard-deviation shocks. Thus, the relative size of the two GNP impulse-response functions gives some measure of the relative importance of the two shocks. More formally, Table I includes a variance decomposition of GNP and consumption growth. Almost all of the variance of consumption growth is due to the permanent.

or consumption shock. But 70 percent of the variance of GNP growth and 85 percent of the variance of one-step-ahead GNP prediction errors are due to the transitory, or GNP shock.5 And this calculation does not include the transitory variation in GNP seen in the hump-shaped response to the consumption shock. Thus, not only does there exist a transitory component to GNP, but also it is economically important.

The pattern of the impulse-response functions has a natural permanent income interpretation. (The Appendix derives the relevant predictions of the PIH model more formally.) The model predicts that consumption should be a random walk, which is pretty much what we find. It also predicts that if consumption does not change, then consumers must regard a movement in GNP as transitory. Thus, it predicts that a GNP shock with no contempora- 5. With private GNP, the fractions rise to 80 percent and 89 percent.

COMPONENTS OF GNP AND STOCK PRICES neous consumption change (i.e., orthogonalized as above) should be completely transitory. Again, this is pretty much what we find.

B. Comparison with a Univariate Estimate The bottom panel of Figure I presents univariate impulse- response functions for GNP—the response of GNP to a change in GNP growth not forecast by past GNP growth. The figure includes the response implied by the univariate autoregression presented in Table I. and the univariate impulse-response function implied by the VAR. (See the Appendix for construction.) The latter are two consistent estimates of the same object. However, they may differ in finite samples, and because the univariate process implied by a given order VAR is a different order ARMA than the directly estimated univariate process.

The impulse-response function estimated from the univariate autoregression displays a good deal of persistence: in response to a unit shock, GNP climbs to about 1.6 after a year, and then declines to only about 1.4. This univariate behavior of postwar GNP is found by many authors using a wide variety of univariate tech- niques (for example, Campbell and Mankiw [1987]). It results from the positive serial correlation of GNP growth at short horizons, captured in the autoregression in Table I.

The univariate impulse-response estimated from the VAR has quite similar short-run dynamics, but it displays much more mean reversion at long horizons, ending up at about half its peak

value.6 Why does the question "how persistent are shocks to GNP?" lead to such widely different answers? There are three reasons.

First, the consumption/GNP ratio forecasts long-term GNP growth better than lagged GNP growth. Second, one can isolate two shocks in a bivariate system, and they are different objects than univariate shocks. A univariate GNP shock is a movement in GNP not forecast by lagged GNP. A bivariate GNP shock is a movement in GNP not forecast by lagged GNP and consumption. One of the bivariate shocks has permanent effects, and one has transitory effects. The single univariate shock mixes the permanent and transitory effects. Third, cointegration means that the long-run properties of consumption and GNP are the same. Since consump- tion is nearly a random walk, its long-run properties are almost the 6. Using private GNP, the long-run response estimated from the VAR is otherwise similar, but declines to 0.4 rather than about 0.8, and is much more

precisely estimated. The basic pattern of the graph is robust to this and other variations in specification.

QUARTERLY JOURNAL OF ECONOMICS same as its short-run properties, and thus easy to estimate. The cointegrated VAR measures long-run properties of GNP by looking at consumption, which is why it can reveal more univariate persistence than the GNP autoregression.7 C. Orthogonalization and Interpretation of the Shocks To calculate the impulse-response functions in Figure I. I identified a shock to GNP with no contemporaneous change in consumption. The purpose of this assumption is to create a permanent and a transitory shock. As explained above, the permanent income interpretation suggests this orthogonalization, since it predicts that changes in GNP with consumption (contemporane- ously) held constant will be transitory.

An obvious alternative is to force the long-run response to one shock to zero, following Blanchard and Quah [1989]. Since the long-run response to the GNP shock is close to zero (Figure I), this orthogonalization produces very similar results.

In fact, if consumption were a pure random walk, then the conventional c first and Blanchard-Quah orthogonalizations would produce exactly the same result (see the Appendix). The intuition is simple: if consumption is a random walk, and if a shock does not change consumption contemporaneously, then that shock does not affect long-run consumption. Since consumption and GNP are cointegrated, the shock does not affect long-run GNP, which is the Blanchard-Quah assumption. The conventional orthogonalization has the slight advantage in that it does not require an estimate of the spectral density at frequency zero.

The contemporaneous correlation of the nonorthogonalized shocks is about 0.3, so other orthogonalizations (y first, for example) do influence the results. Responses to such shocks are linear combinations of the responses in Figure I. They typically do not change the shape of impulse-response functions much, but they have mixtures of permanent and transitory responses. Again, the aim is to find an interpretable orthogonalization that does produce permanent and transitory shocks, so there is not much reason to pursue these alternative orthogonalizations.

7. In the frequency domain, cointegration means that the spectral density of 7. In the frequency domain, contegration means that the spectral density of consumption growth and GNP growth must be the same at frequency zero. The spectral density of GNP growth implied by the univariate autoregressio tion growth, implying lower univariate persistence measures. See Watson [1990] and Cochrane [1990] for plots.

COMPONENTS OF GNP AND STOCK PRICES I do not attach labels to the shocks beyond "permanent" or "consumption" shock and "transitory" or "GNP" shock. VAR shocks are linear forecast errors. For these VAR shocks to recover the underlying shocks to the economy, we must impose an eco- nomic model in which there are exactly two true shocks: one shock has no

permanent effects: and the model's responses to the two shocks is linear and invertible.8 Within the context of simple models that satisfy the above conditions. Blanchard and Quah [1989] and Shapiro and Watson [1988] can label their shocks "supply" and "demand."9 But without such a model, the VAR says nothing about the number, persistence, relative importance, or identity ("supply" versus "demand," preferences versus technol- ogy, money versus fiscal policy, etc.) of the underlying shocks to the

economy. For example, suppose that there is a single underlying technology shock, as in real business cycle models. The VAR may decompose this single shock into two components: a "consumption" or "permanent" component, and a "GNP" or "transitory" compo- nent. Conversely, if there are many underlying shocks with various persistences, the VAR will still find two shocks, corresponding to permanent and transitory movements in GNP.

D. VAR Specification A variety of related VARs have been run. Blanchard and Quah [1989] use a detrended unemployment rate to help forecast GNP.

Like the consumption/GNP ratio, the unemployment rate is a "level" variable that can indicate higher or lower than average GNP growth for many subsequent quarters. The consumption/ GNP VAR improves on this idea:10 the consumption/GNP ratio does not need to be detrended; it is directly linked to GNP growth;

it provides almost the same result with conventional versus 8. Lippi and Reichlin [1993] examine non-invertible representations of the Blanchard-Quah VAR and note the sensitivity of the impulse response function and variance decomposition to the choice of representation.

Cassou and Mittnik [1991] criticize Blanchard and Quah's economic model. They argue that in alternative models, "demand" variables such as money growth can have permanent effects, while "supply" variables such have very little permanent effect.

10. Evans [1989] also uses unemployment to forecast GNP. Fama [1993] uses ratios to uncover transitory movement in GNP—he essentially studies the consump- tion shock of the VAR in detail. Cochrane and Sbordone [1988] use the consumption/ GNP and dividend/price ratios, but they characterize persistence with variance ratios rather than impulse-response functions. Harvey and Stock [1988] and King, Plosser, Stock, and Watson [1991] present the most similar coi 1990-1991 recession.

QUARTERLY JOURNAL OF ECONOMICS long-run orthogonalization; and its results are robust to extra variables, as explained next.

Variables beyond the consumption/GNP ratio are useful for forecasting short-term GNP growth, and the forecasting literature therefore typically uses larger systems.11 However, further variables or longer lags just add wiggles to the short end of the impulse-response functions, with little impact on the long-run behavior of GNP.

There is a good reason. If consumption were a pure random walk-i.e., if consumption growth were unpredictable by any variable-then including any other variable would leave the long-run properties of consumption and GNP completely unaffected. In particular, the response of GNP to a shock with constant consump- tion would still be completely transitory. In the permanent income interpretation, consumption summarizes all consumers' informa- tion about long-run GNP, so other variables are superfluous.12 Since consumption growth is not very predictable, even when many other variables are included, this result holds as an approximation in the data.

I use the bivariate systems to focus on long-run behavior in this paper. A model designed to produce good short- and long-term forecasts should include short-term forecasting variables as well as lagged consumption/GNP.

III. "DETRENDING" OR "CYCLICALLY ADJUSTING" GNP Since the above results indicate

a large transitory component in GNP, they suggest that we can meaningfully decompose GNP into "trend" and "stationary" or "cyclical" components. Bever- idge and Nelson [1981] suggest an attractive way to do this. They define the trend in GNP as the level GNP will reach after all transitory dynamics work themselves out. Equivalently, the trend in GNP is GNP plus all expected future GNP growth: if GNP growth is expected to be higher than average in the future, GNP is below trend; if it is expected to be below average, GNP is above trend. Formally, the trend z_t is defined by $z_t = \lim_{k\to\infty} E_t(y_{t+k} - k\mu) = y_t + \sum_{i=1}^{\infty} E_t(\Delta y_{t+i} - \mu)$,

11. For example Sims [1980] and Litterman and Weiss [1985]. Shapiro and Watson [1988] add short-run forecasting variables to the Blanchard-Quah VAR. 12. Hansen, Roberds, and Sargent [1992] show how consumption summarizes information about permanent income.

COMPONENTS OF GNP AND STOCK PRICES where $\mu \equiv E(\Delta y)$. Figure II presents log GNP and the Beveridge- Nelson stochastic trend. constructed from the estimated consump- tion-GNP VAR of Table I (see the Appendix for construction). This stochastic trend responds to long-run movements in GNP growth A. Multivariate Dividend shock Response to 1 a shack Dividend response Price response o Price shock

—— Years after shock B. Univariate 1.25 1.00 VAR estimate Response to a unit shock 0.75 Univariate estimate 0.50 0.25 0.00 Years after shock FIGURE II Beveridge-Nelson GNP Trend and Consumption The “VAR trend” is the Beveridge-Nelson [1981] trend calculated from the consumption-GNP VAR (Table I) as GNP plus all expected future above-average growth in GNP. “CN&S + Mean Ratio” gives log nondurable + services consum plus the mean log GNP/consumption ratio. The graph is limited to 1963–1990 for clarity.

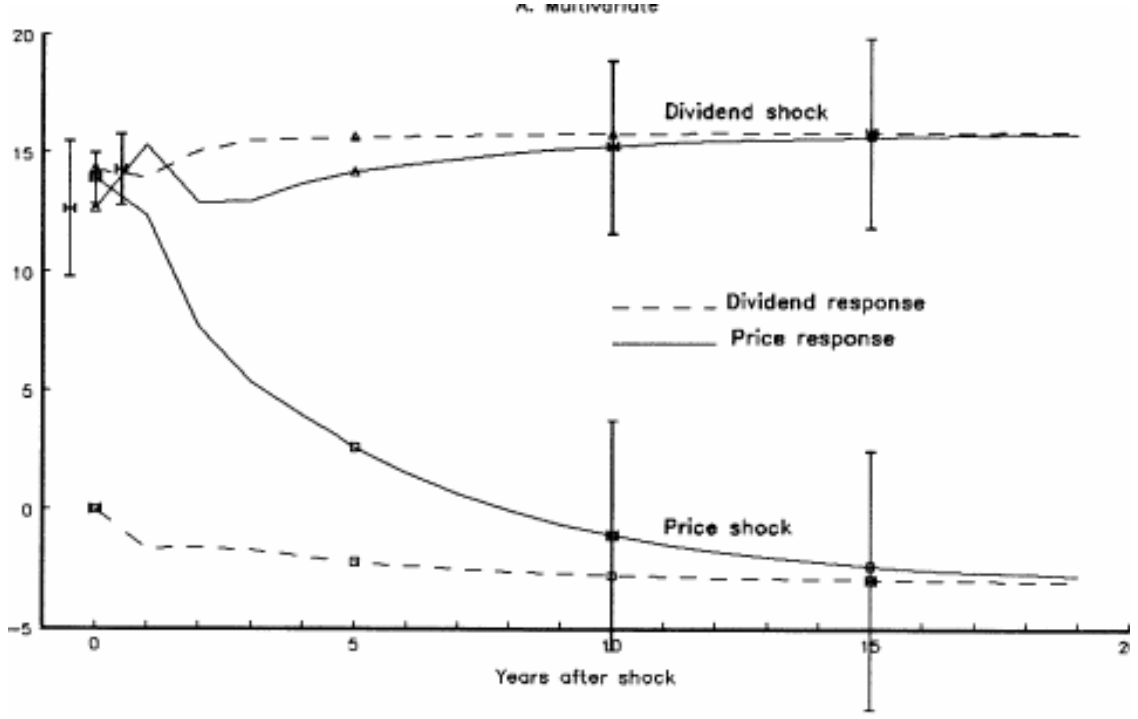


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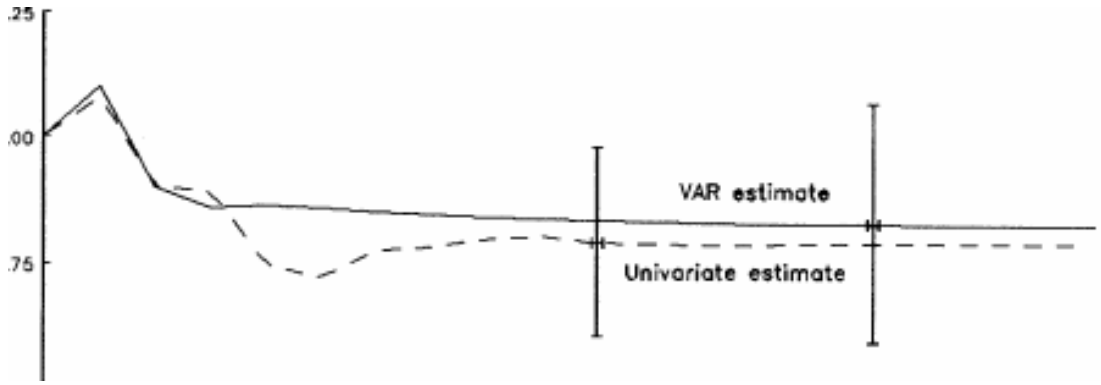


Figure 3.9: Course figure, source page 11, crop 2.

QUARTERLY JOURNAL OF ECONOMICS during the seventies and eighties, yet shows the traditional NBER business cycles as transitory variations about that trend.

If consumption were a pure random walk, the Beveridge- Nelson trend would be exactly consumption less the mean log GNP/consumption ratio (see the Appendix). Figure II also plots this quantity. Since consumption growth is poorly forecastable, this quantity is almost the same as the Beveridge-Nelson trend.

Thus, consumption provides a good measure of the trend in GNP, since it measures consumers' expectations of long-run GNP.

In place of the traditional potential GNP calculation, "what would GNP be if the unemployment rate were x percent," this measure is "what would GNP be if the consumption/GNP ratio was at its historical mean." This measure could easily be used in place of Hodrick-Prescott filtering, or other decompositions based on uni- variate representations. Since univariate representations of GNP typically show little transitory variation, this procedure will isolate much larger cyclical components (see Cochrane [1990] for plots of one example).

IV. STOCK PRICES AND DIVIDENDS A. Results Table II presents a VAR of log dividends and log prices.13 and a univariate regression of log returns on lagged log returns. The data are from the CRSP value-weighted NYSE portfolio. They are annual to avoid the seasonal in dividends. Again, more lags and other right-hand variables (term premium, default premium, interest rate) add more wiggles to the short-run impulse-response functions and raise the one-step-ahead forecast R^2, but do not alter the pattern of the long-run impulse-response functions.

The results in Table II are similar to the previous results for GNP and consumption, with dividends in place of consumption and prices in place of GNP. The dividend/price ratio forecasts returns much more strongly than it forecasts dividend growth, so prices rather than dividends adjust to bring the ratio back to its mean.14 13. I define the "price" variable to be cumulated returns, so that price growth regressions are return forecasting regressions, and so that price growth "price" response can be interpreted as variation in expected returns. Since dividends are poorly forecastable, results in which the price variable i distribution theory. However, the bootstrap only rejects a zero coefficient at a 9.2 percent level. Similarly, an augmented Dickey-Fuller test rejects at the 2.5 percent level with one lag but at 10 percent with two lags. The dividend/price ratio is more

slow-moving than the consumption/GNP ratio, so the greater difficulty of rejecting a unit root is expected.

COMPONENTS OF GNP AND STOCK PRICES TABLE II DIVIDEND AND PRICE REGRESSIONS d_t denotes log dividends and p_t denotes log price (cumulated return) on the value-

weighted NYSE portfolio. Δ denotes first difference; Δp is the log return. Data sample 1927-1988.

1. Vector autoregression Right-hand variable Left-hand R^2 variable const. $d_{t-1} - p_{t-1} \Delta d_{t-1} \Delta d_{t-2}$ Δp_{t-2} p-valueb Δp_{t-1} 20.01 0.062 − 0.082 Δd_t coeff.

0.038 0.046 0.040 0.038 0.78 0.47 0.25 t-stat.

0.34 −0.65 0.32 32.0 p-valuea (percent) 20.7 Δp_t coeff.

78.65 0.225 0.060 − 0.086 0.114 − 0.090 0.140 t-stat.

2.34 2.11 0.25 − 0.36 0.68 − 0.55 1.2 p-valuea (percent) 9.9 2. Variance decomposition Variance of $\Delta d_t - E_{t-1} \Delta d_t$ Due to Δd_f $\Delta p_t - E_{t-1} \Delta p_t$ Δp_t "Permanent" d shock "Temporary" p shock Univariate autoregression Left-hand

variable Right-hand variable R^2 const.

p-valueb Δp_t Δp_{t-1} Δp_{t-2} Δp_{t-3} Δp_{t-4} Coefficient 11.91 0.075 −0.179 0.015 −0.18 .061 t-statistic 0.57 −1.37 0.12 −1.37 3.46 49.7 a. Percent of replications with a coefficient farther from 0, under the null that the coefficient = 0 and d/p

has a unit root (bootstrap).

b. Percent probability value of an F-test for the joint significance of the right-hand variables.

The R^2 of the dividend growth forecasting regression is lower than the R^2 of the return forecasting regression, and the bivariate return regression has a higher R^2 than the univariate return In postwar data the dividend/price ratio forecasts both returns and dividend growth more strongly. The t-statistics rise from 2.11 to 4.00, and 0.78 to 2.71 respectively. Fama and French [1988b] and Hodrick [1992] also find more statisti- cally significant forecasts of returns from dividend/price ratios in postwar data.

Given this evidence, the strong a priori reasons to believe the dividend/price ratio is stationary, and the dangers noted above of pretesting for unit roots. I assume the dividend/price ratio is stationary and proceed to study the implications of this assumption.

QUARTERLY JOURNAL OF ECONOMICS 8.4 8.2 8.0 log GNP 7.8 7.6 GNP VAR trend CN&S + mean ratio 7.4 Date FIGURE III Impulse-Response Functions for Dividends and Prices Top panel: Response of dividends (d) and prices (p) to one-standard-deviation shocks in the dividend-price VAR (Table II).

Bottom panel: Response of price to a unit univariate price shock. The "univariate estimate" is based on the regression of returns on past returns, panel 3 of Table II; the "VAR estimate" is based on the dividend-price VAR, panel 1 of Table Bars show bootstrap one-standard-error bands.

regression. Dividends look a lot like a random walk, as do returns when regressed only on lagged returns.15 The top panel of Figure III presents VAR impulse-response functions. The pattern is similar to that of the consumption-GNP impulse-response in Figure I. In response to a dividend shock, prices and dividends move immediately to their long-run values.

On the other hand, a price shock with no movement in dividends 15. The return forecasting literature has focused on long-horizon returns. One can infer long-horizon properties from a VAR. (See Hodrick [1992].) The long- horizon return R^2 implied by the VAR in Table II rises to a peak of 0.24 at a seven-year horizon and then gradually declines back to 0.15 at a twenty-year horizon. Again, greater predictability occurs in the postwar sample. The variance decomposition also varies with horizon. Since the dividend shock is permanent, it gradually accounts for more of the variance of longer-horizon dividend growth and returns.

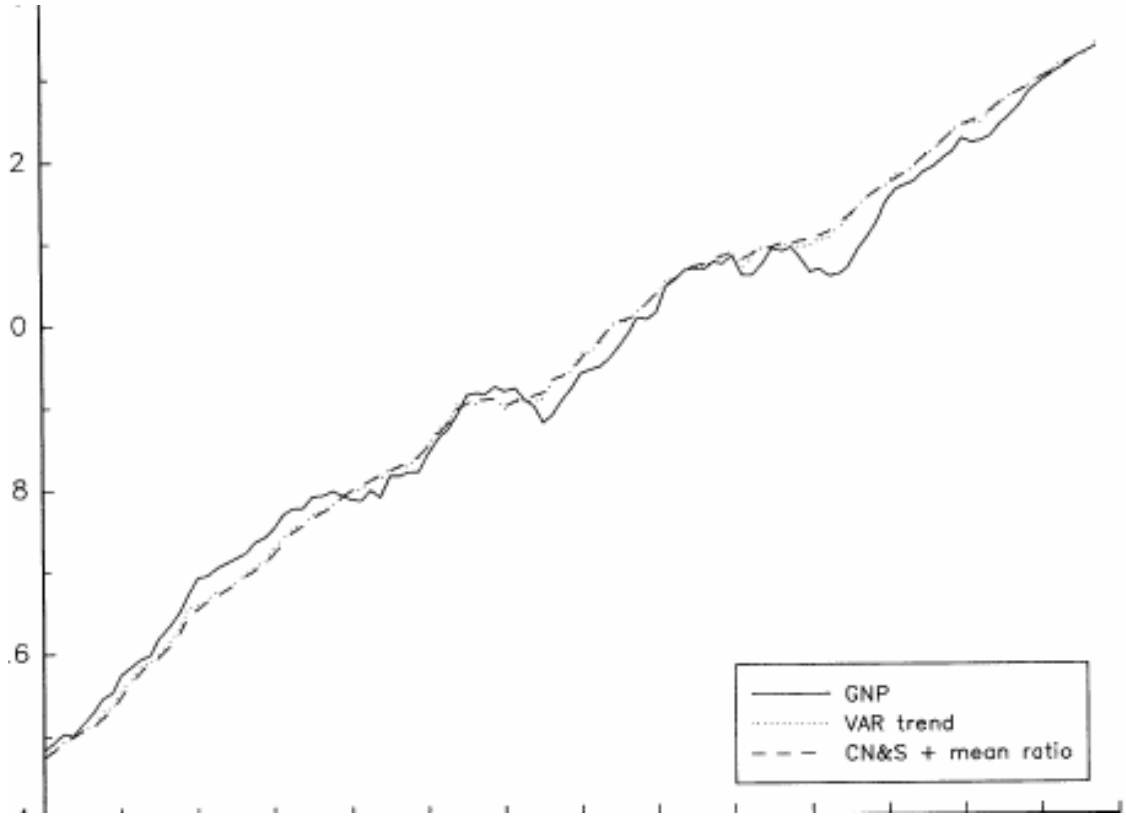


Figure 3.10: Course figure, source page 14, crop 1.

COMPONENTS OF GNP AND STOCK PRICES has a completely transitory effect on prices and no effect on dividends. As the swings in the dividend/price ratio are longer, the transitory movement in prices occurs over a longer horizon (half- life of about five years) than was the case for GNP (half-life of about one-two years).16 The variance decomposition finds that 57 percent of the variance of returns and 55 percent of the variance of one-step- ahead return forecast errors are attributed to the price shock.

Thus, the transitory component of price is economically as well as statistically significant.

The bottom panel of Figure III presents univariate impulse- response functions for stock prices. Here, the VAR and univariate estimates look the same. There is little evidence for univariate mean reversion no matter how estimated; evidence for mean reversion in prices (or predictability in returns) comes when one isolates a transitory multivariate shock. (The point estimates that suggest univariate predictability come from long-horizon regres- sions as in Fama and French [1988a] or variance ratios as in Poterba and Summers [1988]. Both estimates are equivalent to much longer AR's with restrictions on the coefficients.) B. Interpretation: Present Value-Dividend Smoothing Model The present value model together with the hypothesis that managers smooth dividends can be used to interpret the price/ dividend VAR. This section presents the ideas verbally, while the

Appendix contains a more formal treatment.

The present value model states that the price/dividend ratio depends on expectations of future discount rates and dividend growth rates.17 (Prices rise on news of high future long-term 16. In postwar data the long-run impulse-response functions are quite similar.

Dividend growth is more forecastable in postwar data, so its impulse-response function is not flat. However, it does not end up far from where it starts, so the price shock is still almost completely

permanent.

17. With time-varying discount rates, the present value relation is $P_t = E_t\left[\sum_{i=1}^{\infty}\left(\prod_{k=1}^{J} m_{t+k}\right) D_{t+j}\right]$, where P = price, D = dividend, and m = stochastic discount factor ($m = 1/$

(1 + discount rate). Dividing by D_t and letting $\Delta D_t = D_t/D_{t-1}$, $\frac{P_t}{D_t} = E_t\left[\sum_{i=1}^{\infty}\prod_{k=1}^{j} m_{t+k}\Delta D_{t+k}\right]$.

Thus, price/dividend ratios are related to future discount factors m and dividend growth rates ΔD. "Time-varying discount rates" or "time-varying expected returns" do not require time-varying interest rates, since risk premiums may vary over time.

QUARTERLY JOURNAL OF ECONOMICS dividends, or on news of lower long-term discount rates or expected returns.) Therefore, if dividend growth and discount rates (ex- pected returns) are stationary, the present value model implies that the price/dividend ratio is stationary.

If managers smooth dividends by setting dividends equal to the discounted value of earnings (discounted at the risk-free rate), then dividends follow a random walk just as consumption does in the PIH model. In turn, a stationary price/dividend ratio and random walk in dividends imply that the price shock is completely transitory. Thus, the model predicts the impulse-response pattern of Figure III.

In addition, the present value-dividend smoothing model suggests that we label the shocks as an "earnings" shock and a "discount rate" or "expected return" shock. In the model an increase in dividends signals an increase in long-term earnings, as perceived by managers. Since discount rates or expected returns have not changed, prices rise simultaneously. Dividends, following a random walk, are expected to remain at the new level, so prices are too. Thus, the earnings shock gives rise to simultaneous, permanent shocks to prices and dividends, just like the "dividend" shock in the VAR.

On the other hand, if there is a decrease in expected returns (risk premiums, not risk-free rates), prices rise with no concurrent change in dividends. As expected, returns revert to their mean, and prices revert too. Thus, a discount rate or expected return shock gives rise to a transitory price shock with no change in dividends, just like the "price" shock in the VAR.

Of course, the dividend smoothing model can be statistically rejected, just like the PIH model. Dividend growth is statistically predictable. However, the result that price shocks with constant dividends are transitory requires only that the instantaneous response of dividends to any shock equals their limiting response;

the impulse-response functions can wiggle on their way, and dividend growth may be predictable (see the Appendix). Much of dividend predictability is of this type, and so has little effect on the long-run results. Also, as found extensively in the variance bound literature.18 almost all variation in price/dividend ratios is in fact due to changing forecasts of expected returns rather than changing forecasts of dividend growth. In this sense, the economic impor- 18. See Campbell and Shiller [1988] and Cochrane [1992] for evidence beyond that presented here.

COMPONENTS OF GNP AND STOCK PRICES tance of dividend forecastability is small, even though it is statisti- cally significant. Furthermore, much evidence against the simple dividend smoothing model comes from individual firm data. These price and dividend movements largely define the idiosyncratic or diversifiable component of firm returns, so index dividends, as used here, are much less forecastable. In any case, the dividend smooth- ing model, like the PIH model, is a useful approximation that helps us to interpret the dividend/price VAR.

Comparing the price/dividend and consumption/GNP VARs, one might think that prices should take the place of consumption, since prices are a forecast of future dividends as consumption is a forecast of future income. If expected returns were constant, this is in fact what we would see: price/dividend ratios would forecast long-term dividend growth and not returns, and prices would be a random walk. However, expected returns are not constant, and aggregate long-run dividends happen to be nearly unpredictable.

Hence, price, like GNP, is the series with the interesting temporary component; dividends, like consumption, is the near-random walk that defines the "trend;" and the PIH analogy is that dividends equal "permanent earnings," not that prices equal "permanent dividends " APPENDIX Subsection A formalizes the relations between time-series representations discussed in the text. It shows that the following three statements are equivalent: (1) c and y are cointegrated and c is a random walk, (2) c measures the trend in y, and (3) conven- tional (Sims) orthogonalization with c first is equivalent to Blan- chard-Quah orthogonalization. Subsections B and C discuss the predictions of the permanent income and present value-dividend

smoothing models. Subsection D details the VAR estimation and construction of impulse-response functions.

A. Relation Between Statistical Assumptions Let γ denote log GNP or log stock prices, and c denote log consumption or log dividends. Let $I(t)$ denote an information set available to consumers at time t, including at least all lags of c_t and y_t. $E_t(\cdot)$ denotes $E(\cdot|I(t))$.

I assume that $(1-L)y_t$ is stationary with zero mean. (It is easy to generalize to nonzero means.) Stationarity implies that there exists a moving average representation $(1-L)y_t = A(L)w_t$, where

QUARTERLY JOURNAL OF ECONOMICS w_t is a vector of unforecastable random variables that generates the information set $I(t)$. Thus, I maintain ASSUMPTION A1. (Stationarity) $(1-L)y_t$ is stationary, or $(1-L)\mathbf{v}_t = A(L)w_t$. Next, we formalize three possible characterizations of the (c, y) process. First, y and c may be cointegrated. Cointegration requires that $(1-L)c_t$ is stationary, and thus has a moving average representation; and it implies that c and y have the same long-run

response to any shock. Thus, cointegration means ASSUMPTION A2. (Cointegration) A1 and $y_t - c_t$ stationary, or $(1-L)c_t = B(L)w_t$, with $B(1) = A(1)$.

Second, the long-run response of c to shocks might equal its immediate response.

ASSUMPTION A3.

$c_t - E_{t-1}(c_t) = \lim_{j\to\infty} E_t c_{t+j} - \lim_{j\to\infty} E_{t-1}c_{t+j}$ or $B_0 = B(1)$. A3 is weaker than a random walk. The impulse-response of a

process that obeys A3 can wiggle on the way to infinity, and Δc can be predictable.

Third, innovations in c might measure innovations in long-run y:

ASSUMPTION A4.

$c_t - E_{t-1}(c_t) = \lim_{j\to\infty} E_t y_{t+j} - \lim_{j\to\infty} E_{t-1}y_{t+j}$, or $B_0 = A(1)$. These assumptions are related by

PROPOSITION 1. Given stationarity (A1) and cointegration (A2), innovations in c measure innovations in long-run y (A4) if and only if the immediate and long-run responses of c to a shock are the same ($A3$).

Proof of Proposition 1. Given $B(1) = A(1)$, $B_0 = B(1)$ iff $B_0 = $ A(1).

A pure random walk in c is sufficient for A3 and hence Proposition 1. It also implies that c equals

long-run y. Formally, consider ASSUMPTION A5. (Random walk) $c_t = E_t(c_{t+1})$, or $B(L) = B_0$.

COMPONENTS OF GNP AND STOCK PRICES ASSUMPTION A6. (c equals long-run y) $c_t = \lim_{i\to\infty} E_t(y_{t+j})$, or $B(L) = A(1)$. PROPOSITION 2. Given stationarity (A1) and cointegration (A2), c is a random walk (A5) if and only if c equals the long-term forecast of v (A6).

Proof of Proposition 2. First, A2, A5 $\Rightarrow$ A6. $B(1) = A(1)$, $B(L) = B_0$ imply that $B(L) = A(1)$. Second, A6 $\Rightarrow$ A2, A5. $B(L) = A(1)$ implies that $B(L) = B_0$ and $B(1) = A(1)$.

The text claimed that Blanchard-Quah and conventional orthogonalization might be the same. The conditions are specified in PROPOSITION 3. Cointegration (A2) and property (A3) imply that Blanchard-Quah [1989] and Sims [1980] orthogonalization with c first are equivalent.

Proof of Proposition 3. Sims orthogonalization with c first states that the first element of B_0 is zero; i.e., c does not respond to the y shock. Blanchard-Quah orthogonalization states that the first shock has no long-run impact on γ (or c); i.e., the first element of $A(1)$ (or $B(1)$) is zero. Cointegration (A2) $A(1) = B(1)$ and A3 $B_0 = B(1)$ imply that the first element of B_0 is zero if and only if the first elements of $A(1)$ or $B(1)$ are zero.

A pure random walk or c = expected long-run v (A5 and A6) are sufficient for Proposition 3, but the weaker conditions A3 or A4 will do. Proposition 3 has an important corollary.

COROLLARY. Cointegration (A2) and A3 imply that shocks to v with c held constant are completely transitory.

This is just an interpretation of the Blanchard-Quah identifica- tion condition. Keep in mind that the information set is not limited to past c and y in any of these representations. A3 with respect to a large information set implies that y shocks with constant c are transitory even if many other variables are added to the VAR.

B. Permanent Income Model PROPOSITION 4. If labor income is stationary in levels or first differences, the permanent income model predicts that con- sumption is a random walk and that consumption and total income are cointegrated.

QUARTERLY JOURNAL OF ECONOMICS Proof of Proposition 4. The random walk implication is well- known; see Hall [1978] or Sargent [1987]. To show cointegration, start with the permanent income decision rule:

$c_t = rk_t + r\beta \sum_{i=0}^{\infty} \beta^j E_t e_{t+j}$, where r = interest and discount rate, $\beta = 1/(1+r)$, e_t = endowment or labor income, k_t = wealth. Total income is $y_t = e_t + rk_t$.

Hence, $c_t - y_t = rk_t + r\beta \sum_{i=0}^{\infty} \beta^j E_t e_{t+j} - e_t - rk_t = r\beta \sum_{i=0}^{\infty} \beta^j (E_t e_{t+j} - e_t)$.

If e_t is either level- or difference-stationary, then $E_t e_{t+j} - e_t$, and hence $c_t - y_t$, are stationary.

Propositions 1-3 then show that the PIH model implies that $c_t = \lim_{i\to\infty} E_t(y_{t+i})$ or consumption is the Beveridge-Nelson trend in GNP, that Sims orthogonalization with c first should be the same as Blanchard- Quah orthogonalization; i.e., that GNP shocks with constant consumption should be transitory.

Proposition 4 is not true for consumption and labor income.

The PIH does not predict cointegration of consumption and labor income, and it predicts that $c_t = rk_t + r\beta \sum_{i=0}^{\infty} \beta^j E_t e_{t+j}$ not that $c_t = \lim_{j\to\infty} E_t e_{t+j}$.

Proposition 4 is true for labor income only in the limit $\beta \to 1$.

Instead, Proposition 4 describes a permanent income-like (limit as $\beta \to 1$) relation between consumption and total income. This relation could be used to test the PIH avoiding the difficulties inherent in separating labor from total income.

The permanent income model predicts that the (per capita) level of consumption is a random walk, and that the consumption- GNP difference is stationary. I follow Campbell and Mankiw [1989] in applying this model as an approximation to log consumption and GNP.

Stochastic growth models (for example, King, Plosser, Stock, and Watson [1991]) also typically imply stationarity of the consump- tion/GNP ratio, as well as other "great ratios." They do not predict

COMPONENTS OF GNP AND STOCK PRICES a random walk in consumption, since they predict a small amount of interest rate variation. However, they do predict less transitory variation in consumption than in investment or GNP. Therefore, consumption still helps VAR estimation of permanent versus transitory components in GNP, but the above representation results only hold approximately.

C. Present Value and Dividend Smoothing Models If dividend growth and discount rates are strongly stationary, then the dividend/price ratio is strongly stationary. This proposi- tion is discussed in detail in Cochrane [1992] and Craine [1993].

The basic idea comes from staring at the present value relation:

$$\ln\left(\frac{D_t}{P_t}\right) = -\ln\left(E_t \sum_{i=0}^{\infty} \prod_{k=1}^{j} m_{t+k} \Delta D_{t+k}\right),$$

where P_t = price, D_t = dividend, $\Delta D_t = D_t/D_{t-1}$, and m_t = discount rate. Assume that m_t, ΔD_t, and the variables generating E_t are strongly stationary. The right-hand side is then a time-invariant function of stationary variables, so the log dividend-price ratio is strongly stationary.

To motivate a random walk in dividends, we can map the standard dividend-smoothing model into the permanent income framework given above. Suppose that managers can borrow and lend at constant interest rates. (Expected risky returns may still vary over time.) Let k_t = accumulated earnings. Then, if managers set dividends to "permanent" earnings e_t, $D_t = rk_t + r\beta E_t \sum_{i=0}^{\infty} \beta^j e_{t+j}$,

dividends will follow a random walk, just like consumption. (The right-hand side of the last equation is not the present value of the firm, since risk-adjusted or stochastic discount factors must be used to find the latter. For this reason, a separate proof of dividend/price stationarity, given above, is required.) Propositions 2 and 3 then imply that dividends define permanent and transitory components of prices with respect to managers' presumably large information set.

D. VAR Transformations The VARs in Tables I and II can be represented as $A(L)\Delta x_t = \beta_0 + \gamma\alpha' x_{t-1} + \epsilon_t \qquad E(\epsilon_t\epsilon_t') = \Sigma$, (A.1)

 where $\Delta x_t = \begin{bmatrix} \Delta c_t \\ \Lambda v_t \end{bmatrix}$, $\epsilon_t = \begin{bmatrix} \epsilon_t^c \\ \epsilon_y^c \end{bmatrix}$, $\gamma = \begin{bmatrix} \beta^c \\ \beta^y \end{bmatrix}$,

$\alpha = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$,

and $A(L)$ is a matrix of lag polynomials, $A(0) = I$.

One can also run $[\Delta cc - y]'$, $[\Delta yc - y]'$, or $[cy]$ on their lags. If $c - y$ is really stationary, these procedures are equivalent up to lag length restrictions. I use the error-correction representation (A.1) to focus on growth rate forecasts, and because it naturally gives rise to an interesting orthogonalization.

I orthogonalize the error terms by Choleski decomposing the variance-covariance matrix of the innovations in the order c, then v. Precisely, I find a triangular matrix R such that $RR' = E(\epsilon_t\epsilon_t')$

$= \Sigma$. Then I define new errors $v_t = R^{-1}\epsilon_t$ $E(v_t v_t') = R^{-1}RR'R^{-1'} = I$.

Rewriting the VAR (A.1) in terms of these new errors, $A(L)\Delta x_t = \beta_0 + \gamma\alpha' x_{t-1} + R\nu_t$ $E(\nu_t\nu_t')$ $= I$. To find the impulse-response function, I simulate the response of the VAR (without constants) to the v_t shocks starting with all variables set to zero. This calculation is most easily performed by writing the VAR in AR(1) notation. For example, a two-lag VAR

(without means) can be written compactly as

$$\mathbf{x}_t = \mathbf{B}\mathbf{x}_{t-1} + \mathbf{R}\boldsymbol{\nu}_t,$$

where β_c and β_y denote vectors of OLS regression coefficients. This simulation leads to the Wold moving average representation, $\Delta x_t = \mu + D(L)\nu_t$. (A.3) Cumulative sums of the elements of $D(L)$ are plotted in Figures I and III. For subsequent calculations, I retained 50 terms of the

moving average (A.3).

COMPONENTS OF GNP AND STOCK PRICES I calculated variance decompositions as follows. Express (A.3) as $\begin{bmatrix}\Delta c_t \\ \Delta y_t\end{bmatrix} = \mu + \begin{bmatrix}D_{cc}(L)D_{cy}(L) \\ D_{cy}(L)D_{cy}(L)\end{bmatrix}\begin{bmatrix}v_t^c \\ v_t^y\end{bmatrix}$.

Since the shocks are all orthogonal with unit variance, $\text{var}(\Delta y_t - E_{t-1}\Delta y_t) = D_{vc0}^2 + D_{vv0}^2$ $\text{var}(\Delta y_t)$ $= \sum_{i=1}^{\infty} D_{yc,j}^2 + \sum_{i=1}^{\infty} D_{yy,j}^2$.

The first terms on the right-hand side give the variance attributed to the c shock; the second terms give the variance attributed to the γ shock.

I constructed spectral densities from (A.3) from the definition.

$S_{\Lambda\pi}(e^{-i\omega}) = D(e^{-i\omega})D(e^{i\omega})'$. (A.4) To find the univariate impulse response function for GNP implied by the VAR, I factored the spectral density of Δy_t. The second row of (A.4) is $S_{\Lambda\nu}(z) = D_{11}(z)D_{11}(z^{-1}) + D_{12}(z)D_{12}(z^{-1})$.

To find the Wold representation of Δy_t and hence its univariate impulse response, one must find a polynomial $a(z)$ whose roots are all on or outside the unit circle, and such that $a(z)a(z^{-1}) = D_{11}(z)D_{11}(z^{-1}) + D_{12}(z)D_{12}(z^{-1})$. To do this, I found the roots of the right-hand side numerically,

selected the roots outside the unit circle, and then constructed the polynomial $a(z)$ with those roots. Since the Wold moving average is unique, $a(L)$ gives the univariate impulse-response function.

To obtain the Beveridge-Nelson decomposition, I use the AR(1) form of the VAR, as in (A.2). Then the trend is given by $z_t = y_t + \sum_{i=1}^{\infty} E_t(\Delta y_{t+i})$ or $z_t = y_t + [0\ 1\ 0\ 0]\sum_{i=1}^{\infty} B^j x_t = [0\ 1\ 0\ 0]B(I-B)^{-1}x_t$.

DEPARTMENT OF ECONOMICS, UNIVERSITY OF CHICAGO, N.B.E.R., AND N.O.R.C.

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3.0.3 Optional reading – Engle AER04

Risk and Volatility: Econometric Models and Financial Practice By Robert Engle* The advantage of knowing about risks is that tion between expected returns and variance.

we can change our behavior to avoid them. Of These contributions were recognized by Nobel course, it is easily observed that to avoid all prizes in 1981 and 1990.

risks would be impossible; it might entail no Fisher Black and Myron Scholes (1972) and flying, no driving, no walking, eating and drink- Robert C. Merton (1973) developed a model to ing only healthy foods, and never being touched evaluate the pricing of options. While the theory by sunshine. Even a bath could be dangerous. I is based on option replication arguments could not receive this prize if I sought to avoid through dynamic trading strategies, it is also all risks. There are some risks we choose to take consistent with the CAPM. Put options give the because the benefits from taking them exceed owner the right to sell an asset at a particular the possible costs. Optimal behavior takes risks price at a time in the future. Thus these options can be thought of as insurance. By purchasing that are worthwhile. This is the central para- digm of finance; we must take risks to achieve

such put options, the risk of the portfolio can be rewards but not all risks are equally rewarded.

completely eliminated. But what does this in- Both the risks and the rewards are in the future, surance cost? The price of protection depends so it is the expectation of loss that is balanced upon the risks and these risks are measured by the variance of the asset returns. This contribu- against the expectation of reward. Thus we op- timize our behavior, and in particular our portion was recognized by a 1997 Nobel prize.

folio, to maximize rewards and minimize risks.

When practitioners implemented these finan- This simple concept has a long history in cial strategies, they required estimates of the economics and in Nobel citations. Harry M.

variances. Typically the square root of the vari- Markowitz (1952) and James Tobin (1958) as- ance, called the volatility, was reported. They sociated risk with the variance in the value of a immediately recognized that the volatilities portfolio. From the avoidance of risk they de- were

changing over time. They found different answers for different time periods. A simple rived optimizing portfolio and banking behav- ior. William Sharpe (1964) developed the approach, sometimes called historical volatility, implications when all investors follow the same was, and remains, widely used. In this method, objectives with the same information. This the- the volatility is estimated by the sample stan- dard deviation of returns over a short period.

ory is called the Capital Asset Pricing Model or But, what is the right period to use? If it is too CAPM, and shows that there is a natural rela- long, then it will not be so relevant for today and if it is too short, it will be very noisy.

Furthermore, it is really the volatility over a This article is a revised version of the lecture Robert future period that should be considered the risk, Engle delivered in Stockholm, Sweden, on December 8, 2003, when he received the Bank of Sweden Prize in Eco- hence a forecast of volatility is needed as well nomic Sciences in Memory of Alfred Nobel. The article is as a measure for today. This raises the possibil- copyright The Nobel Foundation 2003 and is published ity that the forecast of the average volatility here with the permission of the Nobel Foundation.

over the next week might be different from the * Stern School of Business, New York University, 44 forecast over a year or a decade. Historical West Fourth Street, New York, NY 10012 (e-mail:

volatility had no solution for these problems.

rengle@stern.nyu.edu). This paper is the result of more than On a more fundamental level, it is logically two decades of research and collaboration with many, many people. I would particularly like to thank the audiences in inconsistent to assume, for example, that the B.I.S., Stockholm, Uppsala, Cornell University, and the variance is constant for a period such as one Universit de Savoie for listening as this talk developed.

year ending today and also that it is constant for David Hendry, Tim Bollerslev, Andrew Patton, and Robert the year ending on the previous day but with a Ferstenberg provided detailed suggestions. Nevertheless, different value. A theory of dynamic volatilities any lacunas remain my responsibility.

THE AMERICAN ECONOMIC REVIEW JUNE 2004 is needed; this is the role that is filled by the information and less to the distant past. Clearly ARCH models and their many extensions that the ARCH model was a simple generalization of we discuss today.

the sample variance.

In the next section, I will describe the genesis The big advance was that the weights could of the ARCH model, and then discuss some of be estimated from historical data even though the true volatility was never observed. Here is its many generalizations and widespread empir- how this works. Forecasts can be calculated ical support. In subsequent sections, I will show how this dynamic model can be used to forecast every day or every period. By examining these volatility and risk over a long horizon and how forecasts for different weights, the set of it can be used to value options.

weights can be found that make the forecasts closest to the variance of the next return. This procedure, based on Maximum Likelihood, I. The Birth of the ARCH Model gives a systematic approach to the estimation of The ARCH model was invented while I was the optimal weights. Once the weights are de- on sabbatical at the London School of Econom- termined, this dynamic model of time-varying ics in 1979. Lunch in the Senior Common volatility can be used to measure the volatility at Room with David Hendry, Dennis Sargan, Jim any time and to forecast it into the near and distant future. Granger's test for bilinearity Durbin, and many leading econometricians pro- vided a stimulating environment. I was looking turned out to be the optimal, or Lagrange Mul- for a model that could assess the validity of a tiplier test for ARCH, and is widely used

today.

conjecture of Milton Friedman (1977) that the There are many benefits to formulating an unpredictability of inflation was a primary explicit dynamic model of volatility. As men- cause of business cycles. He hypothesized that tioned above, the optimal parameters can be the level of inflation was not a problem; it was estimated by Maximum Likelihood. Tests of the the uncertainty about future costs and prices that adequacy and accuracy of a volatility model can would prevent entrepreneurs from investing and be used to verify the procedure. One-step and lead to a recession. This could only be plausible multi-step forecasts can be constructed using these parameters. The unconditional distribu- if the uncertainty were changing over time so this was my goal. Econometricians call this tions can be established mathematically and are heteroskedasticity. I had recently worked exten-

generally realistic. Inserting the relevant vari- sively with the Kalman Filter and knew that a ables into the model can test economic models likelihood function could be decomposed into that seek to determine the causes of volatility.

the sum of its predictive or conditional densi- Incorporating additional endogenous variables ties. Finally, my colleague, Clive Granger, with and equations can similarly test economic mod- whom I share this prize, had recently developed els about the consequences of volatility. Several a test for bilinear time series models based on applications will be mentioned below.

the dependence over time of squared residuals.

David Hendry's associate, Frank Srba, wrote That is, squared residuals often were autocorre- the first ARCH program. The application that appeared in Engle (1982) was to inflation in the lated even though the residuals themselves were not. This test was frequently significant in eco- United Kingdom since this was Friedman's nomic data; I suspected that it was detecting conjecture. While there was plenty of evidence something besides bilinearity but I did not know that the uncertainty in inflation forecasts was time varying, it did not correspond to the U.K.

what.

business cycle. Similar tests for U.S. inflation The solution was autoregressive conditional heteroskedasticity or ARCH, a name invented data, reported in Engle (1983), confirmed the by David Hendry. The ARCH model described finding of ARCH but found no business-cycle the forecast variance in terms of current observ- effect. While the trade-off between risk and ables. Instead of using short or long sample return is an important part of macroeconomic standard deviations, the ARCH model proposed theory, the empirical implications are often dif- taking weighted averages of past squared fore- ficult to detect as they are disguised by other dominating effects, and obscured by the reli- cast errors, a type of weighted variance. These weights could give more influence to recent ance on relatively low-frequency data. In fi-

VOL. 94 NO. 3 ENGLE: RISK AND VOLATILITY nance, the risk/return effects are of primary soup of ARCH models that include: AARCH, importance and data on daily or even intradaily APARCH, FIGARCH, FIEGARCH, STARCH, frequencies are readily available to form accu- SWARCH, GJR-GARCH, TARCH, MARCH, rate volatility forecasts. Thus finance is the field NARCH, SNPARCH, SPARCH, SQGARCH, in which the great richness and variety of CESGARCH, Component ARCH, Asymmetric Component ARCH, Taylor-Schwert, Student-t- ARCH models developed.

ARCH, GED-ARCH, and many others that I II. Generalizing the ARCH Model have regrettably overlooked. Many of these models were surveyed in Bollerslev et al.

(1992), Bollerslev et al. (1994), Engle (2002b), Generalizations to different weighting schemes can be estimated and tested. The very and Engle and Isao Ishida (2002). These models

recognize that there may be important nonlin- important development by my outstanding student Tim Bollerslev (1986), called Generalized earity, asymmetry, and long memory properties Autoregressive Conditional Heteroskedasticity, of volatility and that returns can be nonnormal or GARCH, is today the most widely used with a variety of parametric and nonparametric model. This essentially generalizes the purely distributions.

autoregressive ARCH model to an autoregres- A closely related but econometrically distinct sive moving average model. The weights on class of volatility models called Stochastic Vol- atility, or SV models, have also seen dramatic past squared residuals are assumed to decline geometrically at a rate to be estimated from the development. See, for example, Peter K. Clark (1973), Stephen Taylor (1986), Andrew C. Har- data. An intuitively appealing interpretation of the $GARCH(1,$ 1) model is easy to understand.

vey et al. (1994), and Taylor (1994). These The GARCH forecast variance is a weighted models have a different data-generating process average of three different variance forecasts.

which makes them more convenient for some purposes but more difficult to estimate. In a One is a constant variance that corresponds to the long-run average. The second is the forecast linear framework, these models would simply that was made in the previous period. The third be different representations of the same process;

is the new information that was not available but in this nonlinear setting, the alternative when the previous forecast was made. This specifications are not equivalent, although they could be viewed as a variance forecast based on are close approximations.

one period of information. The weights on these three forecasts determine how fast the variance III. Modeling Financial Returns changes with new information and how fast it reverts to its long-run mean.

The success of the ARCH family of models is A second enormously important generaliza- attributable in large measure to the applications tion was the Exponential GARCH or EGARCH in finance. While the models have applicability model of Daniel B. Nelson (1991), who prema- for many statistical problems with time series turely passed away in 1995 to the great loss of data, they find particular value for financial time our profession as eulogized by Bollerslev and series. This is partly because of the importance Peter E. Rossi (1995). In his short academic of the previously discussed trade-off between career, his contributions were extremely influ- risk and return for financial markets, and partly because of three ubiquitous characteristics of ential. He recognized that volatility could re- spond asymmetrically to past forecast errors. In financial returns from holding a risky asset.

a financial context, negative returns seemed to Returns are almost unpredictable, they have sur- be more important predictors of volatility than prisingly large numbers of extreme values, and positive returns. Large price declines forecast both the extremes and quiet periods are clus- greater volatility than similarly large price in- tered in time. These features are often described creases. This is an economically interesting ef- as unpredictability, fat tails, and volatility clus- fect that has wide-ranging implications to be tering. These are precisely the characteristics discussed below.

for which an ARCH model is designed. When volatility is high, it is likely to remain high, and Further generalizations have been proposed by many researchers. There is now an alphabet when it is low, it is likely to remain low. However,

THE AMERICAN ECONOMIC REVIEW JUNE 2004 these periods are time limited so that the fore- small. If the economy has low interest rates and cast is sure to eventually revert to less extreme surplus labor, it may be easy to develop this volatilities. An ARCH process produces dy- new product. With everything else equal, the namic, mean-reverting patterns in volatility that

response will be greater in a recession than in a can be forecast. It also produces a greater numboom period. Hence we are not surprised to find ber of extremes than would be expected from a higher volatility in economic downturns even if standard normal distribution, since the extreme the arrival rate of new inventions is constant.

values during the high volatility period are This is a slow moving type of volatility cluster- ing that can give cycles of several years or greater than could be anticipated from a con- stant volatility process.

longer.

The GARCH(1,1) specification is the work- The same invention will also give rise to a horse of financial applications. It is remarkable high-frequency volatility clustering. When the that one model can be used to describe the vola- invention is announced, the market will not tility dynamics of almost any financial return se- immediately be able to estimate its value on the ries. This applies not only to U.S. stocks but also stock price. Agents may disagree but be suffi- to stocks traded in most developed markets, to ciently unsure of their valuations that they pay most stocks traded in emerging markets, and to attention to how others value the firm. If an investor buys until the price reaches his estimate most indices of equity returns. It applies to ex- change rates, bond returns, and commodity re- of the new value, he may revise his estimate

turns. In many cases, a slightly better model can after he sees others continue to buy at succesbe found in the list of models above, but GARCH sively higher prices. He may suspect they have better information or models and consequently is generally a very good starting point.

The widespread success of GARCH(1,1) raise his valuation. Of course, if the others are begs to be understood. What theory can explain selling, then he may revise his price downward.

why volatility dynamics are similar in so many This process is generally called price discovery different financial markets? In developing such and has been modeled theoretically and empirically in market microstructure. It leads to vol- a theory, we must first understand why asset prices change. Financial assets are purchased atility clustering at a much higher frequency and owned because of the future payments that than we have seen before. This process could last a few days or a few minutes.

can be expected. Because these payments are uncertain and depend upon unknowable future But to understand volatility we must think of developments, the fair price of the asset will more than one invention. While the arrival rate of require forecasts of the distribution of these inventions may not have clear patterns, other types payments based on our best information today.

of news surely do. The news intensity is generally As time goes by, we get more information on high during wars and economic distress. During these future events and revalue the asset. So at important global summits, congressional or regu- a basic level, financial price volatility is due to latory hearings, elections, or central bank board the arrival of new information. Volatility clusmeetings, there are likely to be many news events.

tering is simply clustering of information arriv- These episodes are likely to be of medium duraals. The fact that this is common to so many tion, lasting weeks or months.

assets is simply a statement that news is typi- The empirical volatility patterns we observe cally clustered in time.

are composed of all three of these types of To see why it is natural for news to be clus- events. Thus we expect to see rather elaborate tered in time, we must be more specific about volatility dynamics and often rely on long time the information flow. Consider an event such as series to give accurate models of the different an invention that will increase the value of a time constants.

firm because it will improve future earnings and dividends. The effect on stock prices of this IV. Modeling the Causes and Consequences of event will depend on economic conditions in Financial Volatility the economy and in the firm. If the firm is near bankruptcy, the effect can be very large and if it Once a model has been developed to measure is already operating at full capacity, it may be volatility, it is natural to attempt to explain the

VOL. 94 NO. 3 ENGLE: RISK AND VOLATILITY causes of volatility and the effects of volatility in forecasting volatility rather than as causes. If the true causes were included in the specifica- on the economy. There is now a large literature examining aspects of these questions. I will tion, then the lags would not be needed.

only give a discussion of some of the more A small collection of papers has followed this route. Torben G. Andersen and Bollerslev limited findings for financial markets.

(1998b) examined the effects of announcements In financial markets, the consequences of vol- atility are easy to describe although perhaps on exchange rate volatility. The difficulty in difficult to measure. In an economy with one finding important explanatory power is clear risky asset, a rise in volatility should lead in- even if these announcements are responsible in vestors to sell some of the asset. If there is a important ways. Another approach is to use the fixed supply, the price must fall sufficiently so volatility measured in other markets. Engle et that buyers take the other side. At this new al. (1990b) find evidence that stock volatility lower price, the expected return is higher by just causes bond volatility in the future. Engle et al.

enough to compensate investors for the in- (1990a) model the influence of volatility in mar- creased risk. In equilibrium, high volatility kets with earlier closing on markets with later should correspond to high expected returns.

closing. For example, they examine the influ- Merton (1980) formulated this theoretical ence of currency volatilities in European, Asian model in continuous time, and Engle et al.

markets, and the prior day U.S. market on to- (1987) proposed a discrete time model. If the day's U.S. currency volatility. Yasushi Hamao price of risk were constant over time, then rising et al. (1990), Pat Burns et al. (1998), and others conditional variances would translate linearly have applied similar techniques to global equity into rising expected returns. Thus the mean of markets.

the return equation would no longer be esti- mated as zero, it would depend upon the past V. An Example squared returns exactly in the same way that the conditional variance depends on past squared To illustrate the use of ARCH models for returns. This very strong coefficient restriction financial applications, I will give a rather ex- can be tested and used to estimate the price of tended analysis of the Standard & Poors 500 risk. It can also be used to measure the coeffi- Composite index. This index represents the bulk cient of relative risk aversion of the representa- of the value in the U.S. equity market. I will tive agent under the same assumptions.

look at daily levels of this index from 1963 Empirical evidence on this measurement has through late November 2003. This gives a been mixed. While Engle et al. (1987) find a sweep of U.S. financial history that provides an positive and significant effect, Ray-Yeutien ideal setting to discuss how ARCH models are Chou et al. (1992) and Lawrence R. Glosten et used for risk management and option pricing.

al. (1993) find a relationship that varies over All the statistics and graphs are computed in EViews 4.1.

time and may be negative because of omitted variables. Kenneth R. French et al. (1987) The raw data are presented in Figure 1 where showed that a positive volatility surprise should prices are shown on the left axis. The rather and does have a negative effect on asset prices.

smooth lower curve shows what has happened There is not simply one risky asset in the econ- to this index over the last 40 years. It is easy to omy and the price of risk is not likely to be see the great growth of equity prices over the constant; hence the instability is not surprising period and the subsequent decline after the new and does not disprove the existence of the risk millennium. At the beginning of 1963 the index return trade-off, but it is a challenge to better was priced at $63 and at the end it was $1,035.

modeling of this trade-off.

That means that one dollar invested in 1963 The causes of volatility are more directly would have become $16 by November 21, 2003 modeled. Since the basic ARCH model and its (plus the stream of dividends that would have been received, as this index does not take ac- many variants describe the conditional variance as a function of lagged squared returns, these count of dividends on a daily basis). If this are perhaps the proximate causes of volatility. It investor were clever enough to sell his position on March 24, 2000, it would have been worth is best to interpret these as observables that help

THE AMERICAN ECONOMIC REVIEW JUNE 2004 0.06 0.04 0.0 0.02 1,600 0.00

-0.1 0.02 0.04

-0.2 0.06

-0.3 1965 1970 1975 1980 1985 SP500 SPRETURNS SP500 SPRETURNS FIGURE 2. S&P 500 DAILY BEFORE 1987 FIGURE 1. S&P 500 DAILY PRICES AND RETURNS FROM JANUARY 1963 TO NOVEMBER 2003 0.08 0.04 $24. Hopefully he was not so unlucky as to have 0.00 purchased on that day. Although we often see pictures of the level of these indices, it is obvi-

-0.04 1,600 ously the relative price from the purchase point 0.08 to the sale point that matters. Thus economists focus attention on returns as shown at the top of the figure. This shows the daily price change on the right axis (computed as the logarithm of the price today divided by the price yesterday). This return series is centered around zero throughout 1988 1990 1992 1994 1996 1998 2000 the sample period even though prices are some- SPRETURNS SP500 times increasing and sometimes decreasing.

Now the most dramatic event is the crash of FIGURE 3. S&P 500 1988 TO 2000 October 1987 which dwarfs all other returns in the size of the decline and subsequent partial tility period of the middle 1990's. This was recovery.

Other important features of this data series accompanied by a slow and steady growth of can be seen best by looking at portions of the equity prices. It was frequently discussed whole history. For example, Figure 2 shows the whether we had moved permanently to a new same graph before 1987. It is very apparent that era of low volatility. History shows that we did the amplitude of the returns is changing. The not. The volatility began to rise as stock prices magnitude of the changes is sometimes large got higher and higher, reaching very high levels from 1998 on. Clearly, the stock market was and sometimes small. This is the effect that risky from this perspective but investors were ARCH is designed to measure and that we have called volatility clustering. There is, however, willing to take this risk because the returns were another interesting feature in this graph. It is

so good. Looking at the last period since 1998 clear that the volatility is higher when prices are in Figure 4, we see the high volatility continue falling. Volatility tends to be higher in bear as the market turned down. Only at the end of markets. This is the asymmetric volatility effect the sample, since the official conclusion of the that Nelson described with his EGARCH Iraq war, do we see substantial declines in vol- atility. This has apparently encouraged inves- model.

tors to come back into the market which has Looking at the next subperiod after the 1987 crash in Figure 3, we see the record low vola- experienced substantial price increases.

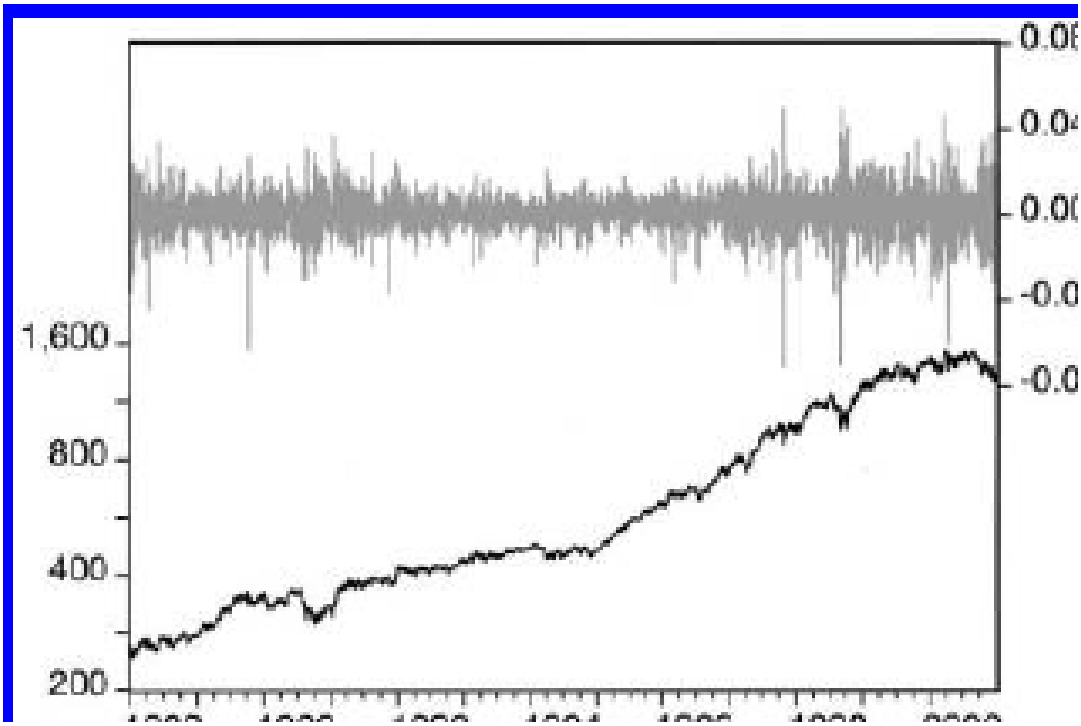


Figure 3.11: Course figure, source page 6, crop 1.

VOL. 94 NO. 3 ENGLE: RISK AND VOLATILITY 0.08 0.04 0.00

-0.04 1.600 1,400

-0.08 Normal Quantile 1,200 1.000

-1

-2 2003 1998 1999 2000 2001 2002

-3 SP500 SPRETURNS

-4 FIGURE 4. S&P 500 1998 TO 2003

-0.04 0.00 0.04

-0.08 0.08 SPRETURNS TABLE 1—S&P 500 RETURNS FIGURE 5. QUANTILE PLOT OF S&P 500 RETURNS POST 1990 Sample Full Since 1990 0.0003 Mean 0.0094 Standard deviation 0.0104 tering of volatility can be concisely shown by Skewness −1.44 −0.10 looking at autocorrelations. Autocorrelations 6.78 Kurtosis 41.45 are correlations calculated between the value of a random variable today and its value some days in the past. Predictability may show up as sig- We now show some statistics that illustrate nificant autocorrelations in returns, and volatil- the three stylized facts mentioned above: almost ity clustering will show up as significant unpredictable returns, fat tails, and volatility autocorrelations in squared or absolute returns.

clustering. Some features of returns are shown Figure 6 shows both of these plots for the post- in Table 1. The mean is close to zero relative to 1990 data. Under conventional criteria,1 auto- the standard deviation for both periods. It is correlations bigger than 0.033 in absolute value would

be significant at a 5-percent level.

0.03 percent per trading day or about 7.8 per- cent per year.

The standard deviation is slightly Clearly, the return autocorrelations are almost higher in the 1990's. These standard deviations all insignificant while the square returns have all autocorrelations significant. Furthermore, the correspond to annualized volatilities of 15 per- squared return autocorrelations are all positive, cent and 17 percent. The skewness is small which is highly unlikely to occur by chance.

throughout.

This figure gives powerful evidence for both the The most interesting feature is the kurtosis which measures the magnitude of the extremes.

unpredictability of returns and the clustering of If returns are normally distributed, then the kur- volatility.

tosis should be three. The kurtosis of the nine- Now we turn to the problem of estimating ties is substantial at 6.8, while for the full volatility. The estimates called historical vola- sample it is a dramatic 41. This is strong evi- tility are based on rolling standard deviations of dence that extremes are more substantial than returns. In Figure 7 these are constructed for would be expected from a normal random vari- five-day, one-year, and five-year windows. While able. Similar evidence is seen graphically in each of these approaches may seem reasonable, Figure 5, which is a quantile plot for the post- 1990 data. This is designed to be a straight line if returns are normally distributed and will have 1 The actual critical values will be somewhat greater as an s-shape if there are more extremes.

the series clearly are heteroskedastic. This makes the case The unpredictability of returns and the clus- for unpredictability in returns even stronger.

THE AMERICAN ECONOMIC REVIEW JUNE 2004 0.25 0.2 0.15 0.1 0.05

-0.05

-0.1 SP Returns SQ Returns FIGURE 6. RETURN AND SQUARED RETURN AUTOCORRE- LATIONS eters based on the historical data, we have fore- casts for each day in the sample period and for 1.6 any period after the sample. The natural first model to estimate is the $GARCH(1,1)$. This 1.2 model gives weights to the unconditional vari- ance, the previous forecast, and the news mea- 0.8 sured as the square of yesterday's return. The weights are estimated to be (0.004, 0.941, 0.4 0.055), respectively.2 Clearly the bulk of the information comes from the previous day fore- 0.0 cast. The new information changes this a little and the long-run average variance has a very V260 V 1,300 small effect. It appears that the long-run vari- ance effect is so tiny that it might not be im- FIGURE 7. HISTORICAL VOLATILITIES WITH VARIOUS

WINDOWS portant. This is incorrect. When forecasting many steps ahead, the long-run variance even- tually dominates as the importance of news and other recent information fades away. It is natu- the answers are clearly very different. The five- rally small because of the use of daily data.

day estimate is extremely variable while the other two are much smoother. The five-year In this example, we will use an asymmetric estimate smooths over peaks and troughs that volatility model that is sometimes called GJR- GARCH for Glosten et al. (1993) or TARCH the other two see. It is particularly slow to recover after the 1987 crash and particularly for Threshold ARCH (Jean Michael Zakoian, slow to reveal the rise in volatility in 1998- 1994). The statistical results are given in Table 2000. In just the same way, the annual estimate 2. In this case there are two types of news.

fails to show all the details revealed by the There is a squared return and there is a variable five-day volatility. However, some of these de- that is the squared return when returns are negtails may be just noise. Without any true mea- ative, and zero otherwise. On average, this is sure of volatility, it is difficult to pick from these candidates.

The ARCH model provides a solution to this 2 For a conventional GARCH model defined as h_{t+1} = dilemma. From estimating the unknown param- $\omega + \alpha r_t^2 + \beta h_t$, the weights are $((1-\alpha-\beta), \beta, \alpha)$.

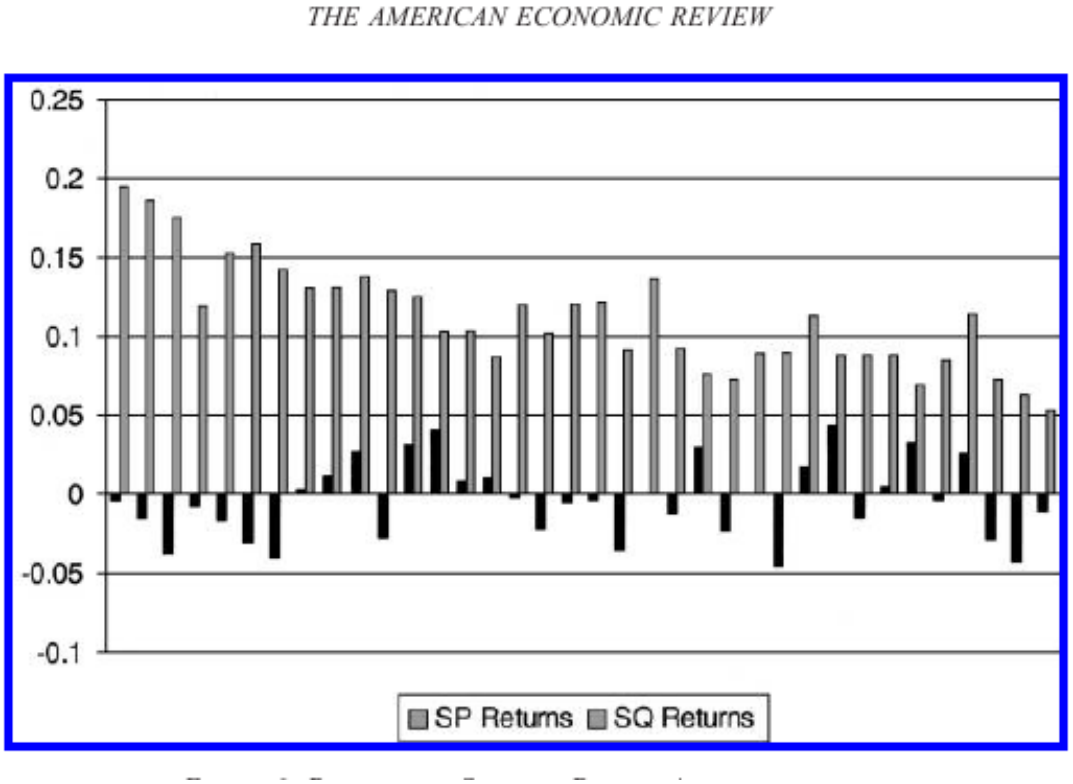


Figure 3.12: Course figure, source page 8, crop 1.

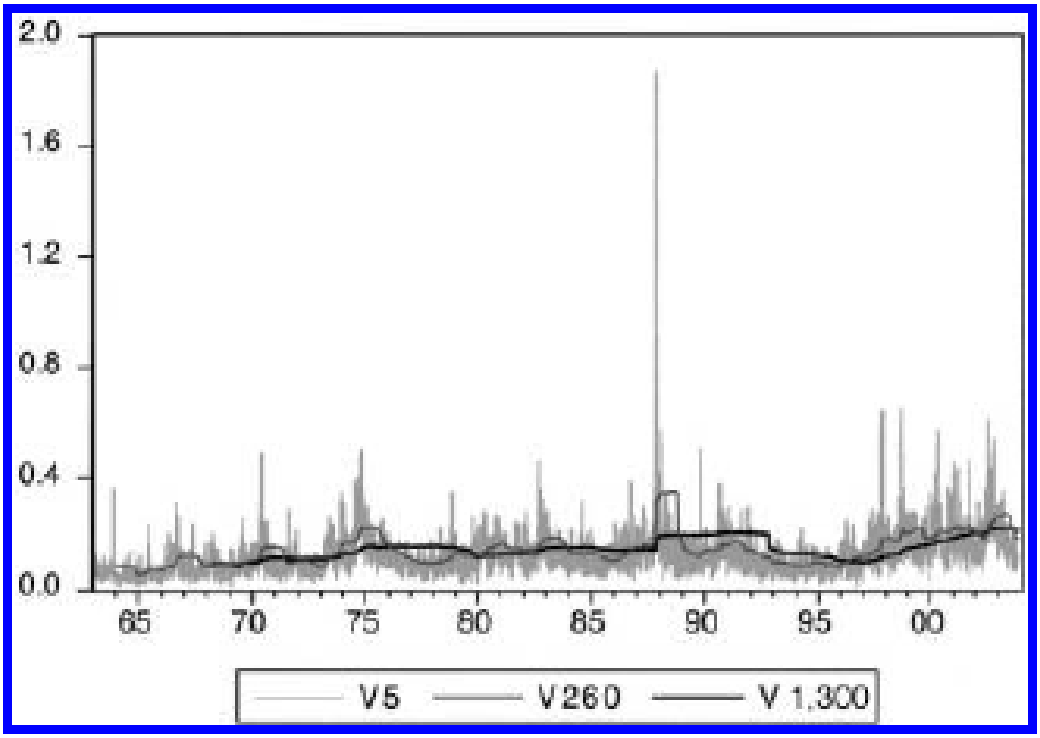


Figure 3.13: Course figure, source page 8, crop 2.

VOL. 94 NO. 3 ENGLE: RISK AND VOLATILITY TABLE 2-TARCH ESTIMATES OF S&P 500 RETURN DATA Dependent variable: NEWRET SP Method: ML-ARCH (Marquardt) Date: 11/24/03 Time: 09:27 Sample (adjusted): 1/03/1963-11/21/2003 Included observations: 10,667 after adjusting endpoints Convergence achieved after 22 iterations Variance backcast: ON Coefficient Standard error z-statistic Probability 0.000301 6.67E-05 4.512504 0.0000 Variance equation 4.55E-07 8.980473 0.0000 C 5.06E-08 ARCH(1) 0.028575 0.003322 8.602582 0.0000 (RESID $<$ 0) $*$ ARCH(1) 19.93374 0.076169 0.003821 0.0000 0.930752 0.002246 414.4693 0.0000 GARCH(1) 1.2 0.05 1.0 0.8 0.0 0.6

-0.05 0.4 0.2 1990 1992 1994 1996 1998 2000 2002 0.0 3*GARCHSTD SPRETURNS

-3*GARCHSTD o GARCHVOL FIGURE 9. GARCH CONFIDENCE INTERVALS: THREE STANDARD DEVIATIONS FIGURE 8. GARCH VOLATILITIES than the annual or five-year historical volatili- half as big as the variance, so it must be doubled implying that the weights are half as big. The ties, but is less variable than the five-day vola- tilities. Since it is designed to measure the weights are now computed on the long-run av- erage, the previous forecast, the symmetric volatility of returns on the next day, it is natural news, and the negative news. These weights are to form confidence intervals for returns. In Fig- estimated to be (0.002, 0.931, 0.029, 0.038) respectively.3 Clearly the asymmetry is impor- ure 9 returns are plotted against plus and minus three TARCH standard deviations. Clearly the tant since the last term would be zero otherwise.

confidence intervals are changing in a very be- In fact, negative returns in this model have more lievable fashion. A constant band would be too than three times the effect of positive returns on wide in some periods and too narrow in others.

future variances. From a statistical point of The TARCH intervals should have 99.7-percent view,

the asymmetry term has a t-statistic of probability of including the next observation if almost 20 and is very significant.

the data are really normally distributed. The expected number of times that the next return is The volatility series generated by this model is given in Figure 8. The series is more jagged outside the interval should then be only 29 out of the more than 10,000 days. In fact, there are 75 indicating that there are more outliers than would be expected from normality.

3 If the model is defined as $h_t = \omega + \beta h_{t-1} + \alpha r_{t-1}^2 + \beta h_{t-1}$ Additional information about volatility is $\gamma r_{t-1}^2 J_{r_{t-1}<0}$, then the weights are $(1 - \alpha - \beta - \gamma/2, \beta, \alpha, \gamma/2)$.

available from the options market. The value of

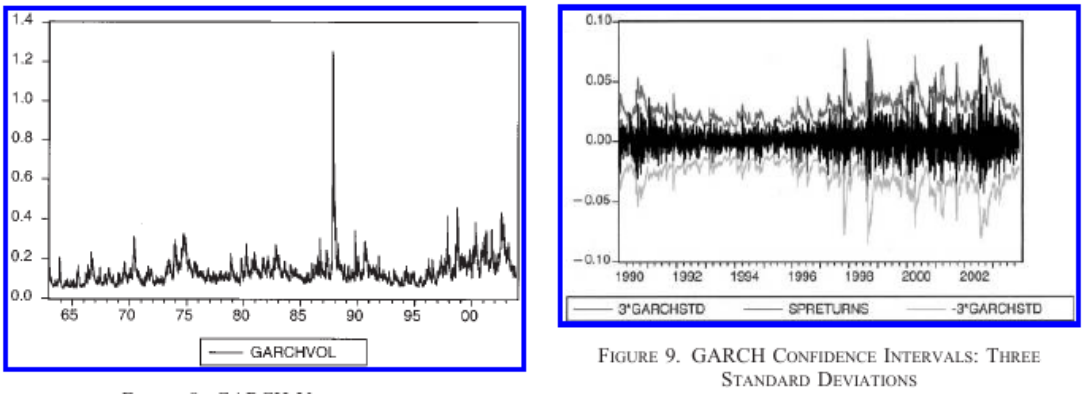


FIGURE 9. GARCH CONFIDENCE INTERVALS: THREE STANDARD DEVIATIONS

Figure 3.14: Course figure, source page 9, crop 1.

THE AMERICAN ECONOMIC REVIEW JUNE 2004 two widely used applications. In the presenta- tion, some novel implications of asymmetric volatility will be illustrated.

VI. Financial Practice—Value at Risk Every morning in thousands of banks and financial services institutions around the world, the Chief Executive Officer is presented with a risk profile by his Risk Management Officer. He 1994 1996 1998 2000 2002 1990 1992 is given an estimate of the risk of the entire GARCHM portfolio and the risk of many of its compo- VIX nents. He would typically learn the risk faced by FIGURE 10. IMPLIED VOLATILITIES AND GARCH the firm's European Equity Division, its U.S.

VOI ATILITIES Treasury Bond Division, its Currency Trading Unit, its Equity Derivative Unit, and so forth.

These risks may even be detailed for particular trading desks or traders. An overall figure is traded options depends directly on the volatility of the underlying asset. A carefully constructed then reported to a regulatory body although it portfolio of options with different strikes will may not be the same number used for internal have a value that measures the option market purposes. The risk of the company as a whole is estimate of future volatility under rather weak less than the sum of its parts since different assumptions. This calculation is now performed portions of the risk will not be perfectly by the CBOE for S&P 500 options and is re- correlated.

The typical measure of each of these risks is ported as the VIX. Two assumptions that un- derlie this index are worth mentioning. The Value at Risk, often abbreviated as VaR. The price process should be continuous and there VaR is a way of measuring the probability of should be no risk premia on volatility shocks. If losses that could occur to the portfolio. The these assumptions are good approximations, 99-percent one-day VaR is a number of dollars then implied volatilities can be compared with that the manager is 99 percent certain will be worse than whatever loss occurs

on the next ARCH volatilities. Because the VIX represents the volatility of one-month options, the TARCH day. If the one-day VaR for the currency desk is volatilities must be forecast out to one month.

$50,000, then the risk officer asserts that only The results are plotted in Figure 10. The on one day out of 100 will losses on this port- general pattern is quite similar, although the folio be greater than $50,000. Of course this TARCH is a little lower than the VIX. These means that on about 2.5 days a year, the losses differences can be attributed to two sources.

will exceed the VaR. The VaR is a measure of First, the option pricing relation is not quite risk that is easy to understand without knowing correct for this situation and does not allow for any statistics. It is, however, just one quantile volatility risk premia or nonnormal returns.

of the predictive distribution and therefore it These adjustments would lead to higher options has limited information on the probabilities of prices and consequently implied volatilities that loss.

were too high. Second, the basic ARCH models Sometimes the VaR is defined on a multi-day have very limited information sets. They do not basis. A 99-percent ten-day VaR is a number of dollars that is greater than the realized loss over use information on earnings, wars, elections, etc. Hence the volatility forecasts by traders ten days on the portfolio with probability 0.99.

should be generally superior; differences could This is a more common regulatory standard but be due to long-lasting information events.

is typically computed by simply adjusting the one-day VaR as will be discussed below. The This extended example illustrates many of loss figures assume that the portfolio is un- the features of ARCH/GARCH models and how they can be used to study volatility processes.

changed over the ten-day period which may be We turn now to financial practice and describe counterfactual.

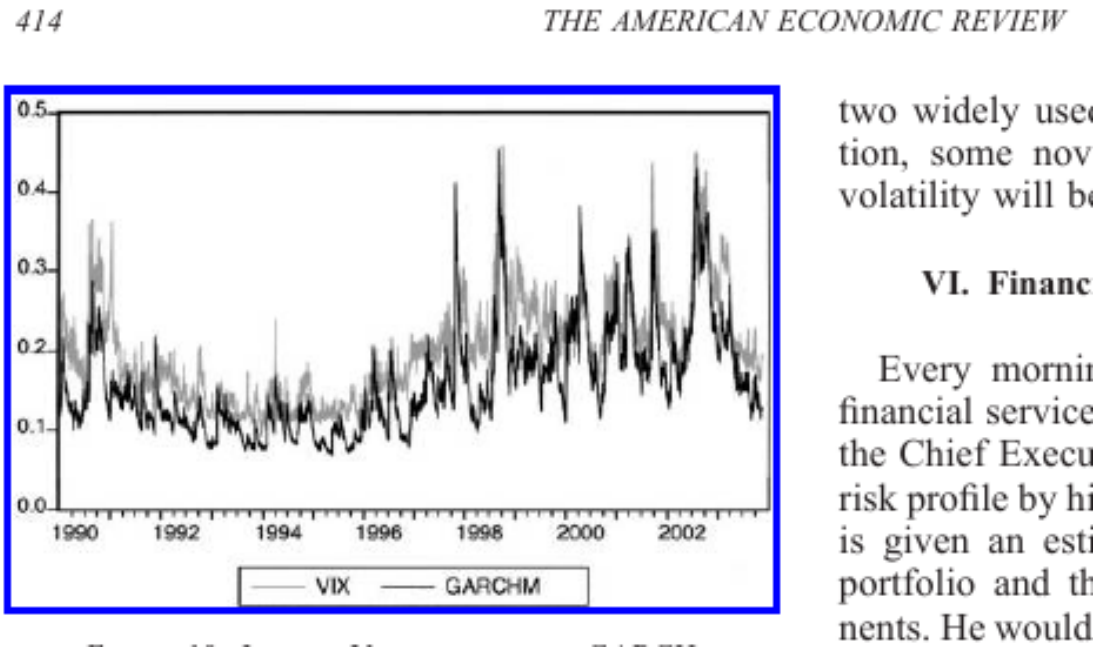


Figure 3.15: Course figure, source page 10, crop 1.

ENGLE: RISK AND VOLATILITY VOL. 94 NO. 3 To calculate the VaR of a trading unit or a portfolio can lose in ten days is a lot greater firm as a whole, it is necessary to have variances

than it can lose in one day. But how much and covariances, or equivalently volatilities and greater is it? If volatilities were constant, then correlations, among all assets held in the port- the answer would be simple; it would be the folio. Typically, the assets are viewed as re- square root of ten times as great. Since the sponding primarily to one or more risk factors ten-day variance is ten times the one-day vari- that are modeled directly. Riskmetrics , for ance, the ten-day volatility multiplier would example, uses about 400 global risk factors.

be the square root of ten. We would take the BARRA uses industry risk factors as well as one-day standard deviation and multiply it by risk factors based on firm characteristics and 3.16 and then with normality we would mul- other factors. A diversified U.S. equity portfolio tiply this by 2.33, giving 7.36 times the stan- would have risks determined primarily by the dard deviation. This is the conventional solution in industry practice. For November aggregate market index such as the S&P 500.

We will carry forward the example of the pre- 24, the ten-day 99-percent VaR is 5.6 percent vious section to calculate the VaR of a portfolio of portfolio value.

that mimics the S&P.

However, this result misses two important The one-day 99-percent VaR of the S&P can features of dynamic volatility models. First, it be estimated using ARCH. From historical data, makes a difference whether the current volatil- the best model is estimated, and then the stan- ities are low or high relative to the long-run dard deviation is calculated for the following average, so that they are forecast to increase or day. In the case of S&P on November 24, this decrease over the next ten days. Since the vol- forecast standard deviation is 0.0076. To con- atility is relatively low in November, the TARCH model will forecast an increase over vert this into VaR we must make an assumption about the distribution of returns. If normality is the next ten days. In this case, this effect is not assumed, the 1 percent point is -2.33 standard

very big as the standard deviation is forecast to deviations from zero. Thus the value at risk is increase to 0.0077 from 0.0076 over the ten-day 2.33 times the standard deviation or in the case period.

of November 24, it is 1.77 percent. We can be More interesting is the effect of asymmetry in 99 percent sure that we will not lose more than variance for multi-period returns. Even though each period has a symmetric distribution, the 1.77 percent of portfolio value on November 24.

In fact the market went up on the 24th so there multi-period return distribution will be asym- metric. This effect is simple to understand but were no losses.

has not been widely recognized. It is easily The assumption of normality is highly ques- tionable. We observed that financial returns illustrated with a two-step binomial tree, Figure have a surprising number of large returns. If we 11, as used in elementary option pricing models.

divide the returns by the TARCH standard de- In the first period, the asset price can either increase or decrease and each outcome is viations, the result will have a constant volatil- ity of one but will have a nonnormal equally likely. In the second period, the vari- distribution. The kurtosis of these "devolatized ance will depend upon whether the price went returns," or "standardized residuals," is 6.5, up or down. If it went up, then the variance will which is much less than the unconditional kur- be lower so that the binomial branches will be tosis, but is still well above normal. From these relatively close together. If the price went down, devolatized returns, we can find the 1-percent the variance will be higher so that the outcomes quantile and use this to give a better idea of the will be further apart. After two periods, there VaR. It turns out to be 2.65 standard deviations

are four outcomes that are equally likely. The below the mean. Thus our portfolio is riskier

distribution is quite skewed, since the bad out- come is far worse than if the variance had been than we thought using the normal approxima- tion. The one-day 99-percent VaR is now esti- constant.

mated to be 2 percent.

To calculate the VaR in this setting, a simu- A ten-day value at risk is often required by lation is needed. The TARCH model is simu- regulatory agencies and is frequently used lated for ten days using normal random variables and starting from the values of internally as well. Of course, the amount a

THE AMERICAN ECONOMIC REVIEW JUNE 2004 such as options. These are typically valued the- oretically assuming some particular process for the underlying asset and then market prices of the derivatives are used to infer the parameters Low of the underlying process. This strategy is often variance called "arbitrage free pricing." It is inadequate for some of the tasks of financial analysis. It cannot determine the risk of a derivative posi- tion since each new market price may corre- spond to a different set of parameters and it is the size and frequency of these parameter changes that signify risk. For the same reason, it is difficult to find optimal hedging strategies. Fi- nally, there is no way to determine the price of a new issue or to determine whether some de- rivatives are trading at discounts or premiums.

High A companion analysis that is frequently car- ried out by derivatives traders is to develop variance fundamental pricing models that determine the appropriate price for a derivative based on the observed characteristics of the underlying asset.

FIGURE 11. TWO-PERIOD BINOMIAL TREE WITH These models could include measures of trading ASYMMETRIC VOLATILITY cost, hedging cost, and risk in managing the options portfolio.

November 21.4 This was done 10,000 times and In this section, a simple simulation-based op- then the worst outcomes were sorted to find the tion pricing model will be employed to illustrate Value at Risk corresponding to the 1-percent the use of ARCH models in this type of funda- quantile. The answer was 7.89 times the stan- mental analysis. The example will be the pric- dard deviation. This VaR is substantially larger ing of put options on the S&P 500 that have ten than the value assuming constant volatility.

trading days left to maturity.

To avoid the normality assumption, the simu- A put option gives the owner the right to sell lation can also be done using the empirical dis- an asset at a particular price, called the strike tribution of the standardized residuals. This price, at maturity. Thus if the asset price is simulation is often called a bootstrap; each draw below the strike, he can make money by selling of the random variables is equally likely to be any at the strike and buying at the market price. The observation of the standardized residuals. The Oc- profit is the difference between these prices. If, tober 1987 crash observation could be drawn once however, the market price is above the strike, or even twice in some simulations but not in then there is no value in the option. If the others. The result is a standard deviation multiplier investor holds the underlying asset in a portfolio

of 8.52 that should be used to calculate VaR. For and buys a put option, he is guaranteed to have our case, the November 24 ten-day 99-percent at least the strike price at the maturity date. This VaR is 6.5 percent of portfolio value.

is why these options can be thought of as insur- ance contracts.

VII. Financial Practice—Valuing Options The simulation works just as in the previous section. The TARCH model is simulated from Another important area of financial practice the end of the sample period, 10,000 times. The is valuation and management of derivatives bootstrap approach

is taken so that nonnormal- ity is already incorporated in the simulation.

This simulation should be of the "risk-neutral" 4 In the example here, the simulation was started at the distribution, i.e., the distribution in which assets unconditional variance so that the time aggregation effect are priced at their discounted expected values.

could be examined alone. In addition, the mean was taken to be The risk-neutral distribution differs from the zero but this makes little difference over such short horizons.

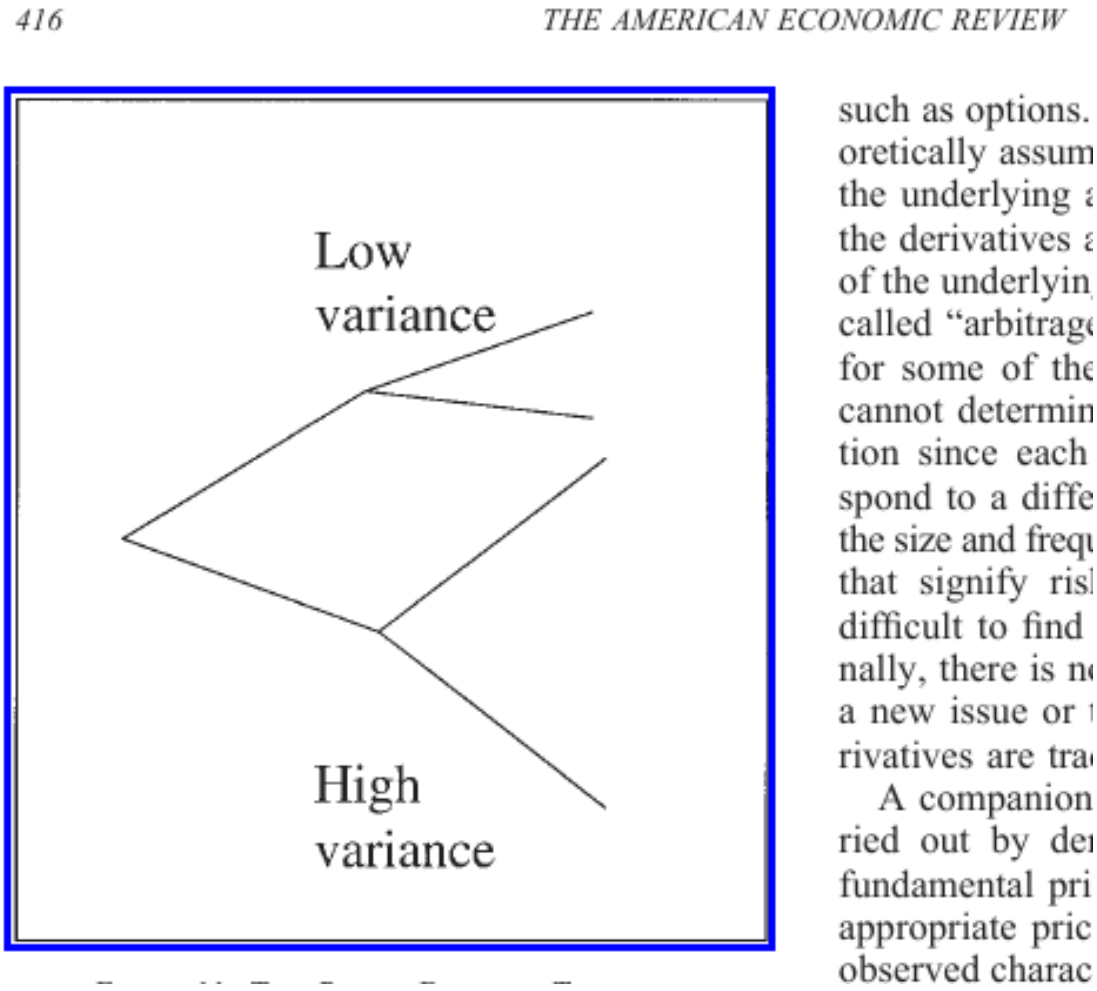


Figure 3.16: Course figure, source page 12, crop 1.

ENGLE: RISK AND VOLATILITY VOL. 94 NO. 3 0.168 0.164 0.160 PUTIMP PUT 0.156 0.152 0.148 0.144 1.000 1,040 1,080 1,000 1,040 1,080 FIGURE 12. PUT PRICES FROM GARCH SIMULATION FIGURE 13. IMPLIED VOLATILITIES FROM GARCH SIMULATION empirical distribution in subtle ways so that there is an explicit risk premium in the empiri- cal distribution which is not needed in the risk ent from those generated by the Black-Scholes neutral. In some models such as the Black- model. This is easily seen by calculating the Scholes, it is sufficient to adjust the mean to be implied volatility for each of these put options.

the risk-free rate. In the example, we take this The result is shown in Figure 13. The implied route. The distribution is simulated with a mean volatilities are higher for the out-of-the-money of zero, which is taken to be the risk-free rate.

puts than they are for the at-the-money puts, and As will be discussed below, this may not be a

the in-the-money put volatilities are even lower.

If the put prices were generated by the Black- sufficient adjustment to risk-neutralize the distribution.

Scholes model, these implied volatilities would From the simulation, we have 10,000 equally all be the same. This plot of implied volatilities likely outcomes for ten days in the future. For against strike is a familiar feature for options each of these outcomes we can compute the traders. The downward slope is called a "vola- value of a particular put option. Since these are tility skew" and corresponds to a skewed distri- equally likely and since the riskless rate is taken bution of the underlying assets. This feature is to be zero, the fair value of the put option is the very pronounced for index options, less so for average of these values. This can be done for individual equity options, and virtually nonex- put options with different strikes. The result is istent for currencies, where it is called a plotted in Figure 12. The S&P is assumed to

"smile." It is apparent that this is a consequence begin at 1,000 so a put option with a strike of of the asymmetric volatility model and corre- 990 protects this value for ten days. This put spondingly, the asymmetry is not found for cur- option should sell for $11. To protect the port- rencies and is weaker for individual equity folio at its current value would cost $15 and to options than for indices.

be certain that it was at least worth 1,010 would This feature of options prices is strong con- cost $21. The VaR calculated in the previous firmation of asymmetric volatility models. Un- section was $65 for the ten-day horizon. To fortunately, the story is more complicated than protect the portfolio at this point would cost this. The actual options skew is generally some- around $2. These put prices have the expected what steeper than that generated by asymmetric shape; they are monotonically increasing and ARCH models. This calls into question the risk neutralization adopted in the simulation. There convex.

However, these put prices are clearly differ- is now increasing evidence that investors are

THE AMERICAN ECONOMIC REVIEW JUNE 2004 particularly worried about big losses and are at which trades arrive is random, the formula- willing to pay extra premiums to avoid them.

tion of ultra high frequency volatility models This makes the skew even steeper. The required requires a model of the arrival process of trades.

Engle and Jeffrey R. Russell (1998) propose the risk neutralization has been studied by several authors such as Jens C. Jackwerth (2000), Autoregressive Conditional Duration or ACD Joshua V. Rosenberg and Engle (2002), and model for this task. It is a close relative of David S. Bates (2003).

ARCH models designed to detect clustering of trades or other economic events; it uses this information to forecast the arrival probability of VIII. New Frontiers the next event.

Many investigators in empirical market mi- crostructure are now studying aspects of finan- It has now been more than 20 years since the cial markets that are relevant to this problem. It ARCH paper appeared. The developments and turns out that when trades are clustered, the applications have been fantastic and well be- yond anyone's most optimistic forecasts. But volatility is higher. Trades themselves carry in- what can we expect in the future? What are the formation that will move prices. A large or next frontiers?

medium-size buyer will raise prices, at least partly because market participants believe he There appear to be two important frontiers of could have important information that the stock research that are receiving a great deal of atten- tion and have important promise for applica- is undervalued. This effect is called price im- tions. These are high-frequency volatility pact and is a central component of liquidity risk, models and high-dimension multivariate mod- and a key feature of

volatility for ultra high els. I will give a short description of some of the frequency data. It is also a central concern for promising developments in these areas.

traders who do not want to trade when they will Merton was perhaps the first to point out the have a big impact on prices, particularly if this is just a temporary impact. As financial markets benefits of high-frequency data for volatility measurement. By examining the behavior of become ever more computer driven, the speed stock prices on a finer and finer time scale, and frequency of trading will increase. Methods better and better measures of volatility can be to use this information to better understand the volatility and stability of these markets will be achieved. This is particularly convenient if vol- atility is only slowly changing so that dynamic ever more important.

considerations can be ignored. Andersen and The other frontier that I believe will see sub- Bollerslev (1998a) pointed out that intra-daily stantial development and application is high- data could be used to measure the performance dimension systems. In this presentation, I have of daily volatility models. Andersen et al.

focused on the volatility of a single asset. For most financial applications, there are thousands (2003) and Engle (2002b) suggest how intra- daily data can be used to form better daily of assets. Not only do we need models of their volatility forecasts.

volatilities but also of their correlations. Ever However, the most interesting question is since the original ARCH model was published how to use high-frequency data to form high- there have been many approaches proposed for frequency volatility forecasts. As higher and multivariate systems. However, the best method higher frequency observations are used, there is to do this has not yet been discovered. As the apparently a limit where every transaction is number of assets increase, the models become observed and used. Engle (2000) calls such data extremely difficult to accurately specify and ultra high frequency data. These transactions estimate. Essentially there are too many possi- occur at irregular intervals rather than equally bilities. There are few published examples of models with more than five assets. The most spaced times. In principle, one can design a

volatility estimator that would update the vola- successful model for these cases is the constant conditional correlation model, CCC, of Boller- tility every time a trade was recorded. However, even the absence of a trade could be information slev (1990). This estimator achieves its perfor- useful for updating the volatility so even more mance by assuming that the conditional correlations are constant. This allows the vari- frequent updating could be done. Since the time

VOL. 94 NO. 3 ENGLE: RISK AND VOLATILITY ances and covariances to change but not the . "Modelling the Coherence in Short- Run Nominal Exchange Rates: A Multivari- correlations.

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3.0.4 Optional reading – Engle JEP01

Journal of Economic Perspectives—Volume 15, Number 4—Fall 2001—Pages 157–168 GARCH 101: The Use of ARCH/GARCH Models in Applied Econometrics Robert Engle he great workhorse of applied econometrics is the least squares model.

This is a natural choice, because applied econometricians are typically called upon to determine how much one variable will change in response to a change in some other variable. Increasingly however, econometricians are being asked to forecast and analyze the size of the errors of the model. In this case, the questions are about volatility, and the standard tools have become the ARCH/ GARCH models.

The basic version of the least squares model assumes that the expected value of all error terms, when squared, is the same at any given point. This assumption is called homoskedasticity, and it is this assumption that is the focus of ARCH/ GARCH models. Data in which the variances of the error terms are not equal, in which the error terms may reasonably be expected to be larger for some points or ranges of the data than for others, are said to suffer from heteroskedasticity. The standard warning is that in the presence of heteroskedasticity, the regression coefficients for an ordinary least squares regression are still unbiased, but the standard errors and confidence intervals estimated by conventional procedures will be too narrow, giving a false sense of precision. Instead of considering this as a problem to be corrected, ARCH and GARCH models treat heteroskedasticity as a

variance to be modeled. As a result, not only are the deficiencies of least squares corrected, but a prediction is computed for the variance of each error term. This prediction turns out often to be of interest, particularly in applications in finance.

The warnings about heteroskedasticity have usually been applied only to cross-section models, not to time series models. For example, if one looked at the Robert Engle is the Michael Armellino Professor of Finance, Stern School of Business, New York University, New York, New York, and Chancellor's Associates Professor of Economics, University of California at San Diego, La Jolla, California.

158 Journal of Economic Perspectives cross-section relationship between income and consumption in household data, one might expect to find that the consumption of low-income households is more closely tied to income than that of high-income households, because the dollars of savings or deficit by poor households are likely to be much smaller in absolute value than high income households. In a cross-section regression of household consump- tion on income, the error terms seem likely to be systematically larger in absolute value for high-income than for low-income households, and the assumption of homoskedasticity seems implausible. In contrast, if one looked at an aggregate time series consumption function, comparing national income to consumption, it

seems more plausible to assume that the variance of the error terms doesn't change much over time.

A recent development in estimation of standard errors, known as "robust standard errors," has also reduced the concern over heteroskedasticity. If the sample size is large, then robust standard errors give quite a good estimate of standard errors even with heteroskedasticity. If the sample is small, the need for a heteroskedasticity correction that does not affect the coefficients, and only asymptotically corrects the standard errors, can be debated.

However, sometimes the natural question facing the applied econometrician is the accuracy of the predictions of the model. In this case, the key issue is the variance of the error terms and what makes them large. This question often arises in financial applications where the dependent variable is the return on an asset or portfolio and the variance of the return represents the risk level of those returns.

These are time series applications, but it is nonetheless likely that heteroskedasticity is an issue. Even a cursory look at financial data suggests that some time periods are riskier than others; that is, the expected value of the magnitude of error terms at some times is greater than at others. Moreover, these risky times are not scattered randomly across quarterly or annual data. Instead, there is a degree of autocorre- lation in the riskiness of financial returns. Financial analysts, looking at plots of daily returns such as in Figure 1, notice that the amplitude of the returns varies over time and describe this as "volatility clustering." The ARCH and GARCH models, which stand for autoregressive conditional heteroskedasticity and generalized autoregressive conditional heteroskedasticity, are designed to deal with just this set of issues. They have become widespread tools for dealing with time series heteroske-

dastic models. The goal of such models is to provide a volatility measure—like a standard deviation-that can be used in financial decisions concerning risk analy- sis, portfolio selection and derivative pricing.

ARCH/GARCH Models Because this paper will focus on financial applications, we will use financial notation. Let the dependent variable be labeled r_t, which could be the return on an asset or portfolio. The mean value m and the variance h will be defined relative to a past information set. Then, the return r in the present will be equal to the mean

Robert Engle Figure 1 Nasdaq, Dow Jones and Bond Returns RATE DOW JONES .03 .06 .02 .04 .01 .02 .00 −.01 −.02

-.04 -.06 −.03 −.04

-.08

-.05 94 95 .08 .06 PORTFOLIO NASDAQ .04 .02 .00 −.02 −.04 −.08 −.06 −.12 −.08 96 97 value of r (that is, the expected value of r based on past information) plus the standard deviation of r (that is, the square root of the variance) times the error term for the present period.

The econometric challenge is to specify how the information is used to forecast the mean and variance of the return, conditional on the past information. While many specifications have been considered for the mean return and have been used in efforts to forecast future returns, virtually no methods were available for the variance before the introduction of ARCH models. The primary descriptive tool was the rolling standard deviation. This is the standard deviation calculated using a fixed number of the most recent observations. For example, this could be calcu- lated every day using the most recent month (22 business days) of data. It is convenient to think of this formulation as the first ARCH model; it assumes that the variance of tomorrow's return is an equally weighted average of the squared residuals from the last 22 days. The assumption of equal weights seems unattractive,

as one would think that the more recent events would be more relevant and therefore should have higher weights. Furthermore the assumption of zero weights for observations more than one month old is also unattractive. The ARCH model proposed by Engle (1982) let these weights be parameters to be estimated. Thus, the model allowed the data to determine the best weights to use in forecasting the variance.

A useful generalization of this model is the GARCH parameterization intro- duced by Bollerslev (1986). This model is also a weighted average of past squared residuals, but it has declining weights that never go completely to zero. It gives parsimonious models that are easy to estimate and, even in its simplest form, has proven surprisingly successful in predicting conditional variances. The most widely used GARCH specification asserts that the best predictor of the variance in the next period is a weighted average of the long-run average variance, the variance

160 Journal of Economic Perspectives predicted for this period, and the new information in this period that is captured by the most recent squared residual. Such an updating rule is a simple description of adaptive or learning behavior and can be thought of as Bayesian updating.

Consider the trader who knows that the long-run average daily standard deviation of the Standard and Poor's 500 is 1 percent, that the forecast he made yesterday was 2 percent and the unexpected return observed today is 3 percent.

Obviously, this is a high volatility period, and today is especially volatile, which suggests that the forecast for tomorrow could be even higher. However, the fact that the long-term average is only 1 percent might lead the forecaster to lower the forecast. The best strategy depends upon the dependence between days. If these three numbers are each squared and weighted equally, then the new forecast would be $2.16 = \sqrt{(1+4+9)/3}$. However, rather than weighting these equally, it is generally found for daily data that weights such as those in the em- pirical example of (.02, .9, .08) are much more accurate. Hence the forecast is $2.08 = \sqrt{.02*1 + .9*4 + .08*9}$.

To be precise, we can use h_t to define the variance of the residuals of a regression $r_t = m_t + \sqrt{h_t \varepsilon_t}$. In this definition, the variance of ε is one. The GARCH model for variance looks like this:

$h_{t+1} = \omega + \alpha(r_t - m_t)^2 + \beta h_t = \omega + \alpha h_t \varepsilon_t^2 + \beta h_t$. The econometrician must estimate the constants ω, α, β; updating simply requires knowing the previous forecast h and residual. The weights are $(1-\alpha-\beta, \beta, \alpha)$,

and the long-run average variance is $\sqrt{\omega}/(1-\alpha-\beta)$. It should be noted that this only works if $\alpha + \beta < 1$, and it only really makes sense if the weights are positive, requiring $\alpha > 0, \beta > 0, \omega > 0$.

The GARCH model that has been described is typically called the GARCH(1,1) model. The $(1,1)$ in parentheses is a standard notation in which the first number refers to how many autoregressive lags, or ARCH terms, appear in the equation, while the second number refers to how many moving average lags are specified, which here is often called the number of GARCH terms. Sometimes models with more than one lag are needed to find good variance forecasts.

Although this model is directly set up to forecast for just one period, it turns out that based on the one-period forecast, a two-period forecast can be made.

Ultimately, by repeating this step, long-horizon forecasts can be constructed. For the GARCH(1, 1), the two-step forecast is a little closer to the long-run average variance than is the one-step forecast, and, ultimately, the distant-horizon forecast is the same for all time periods as long as $\alpha + \beta < 1$. This is just the unconditional variance. Thus, the GARCH models are mean reverting and conditionally het- eroskedastic, but have a constant unconditional variance.

I turn now to the question of how the econometrician can possibly estimate an equation like the GARCH(1,1) when the only variable on which there are data is r_t.

The simple answer is to use maximum likelihood by substituting h_t for σ^2 in the normal likelihood and then maximizing with respect to the parameters. An even

GARCH 101: The Use of ARCH/GARCH models in Applied Econometrics simpler answer is to use software such as EViews, SAS, GAUSS, TSP, Matlab, RATS and many others where there exist already packaged programs to do this.

But the process is not really mysterious. For any set of parameters ω, α, β and a starting estimate for the variance of the first observation, which is often taken to be the observed variance of the residuals, it is easy to calculate the variance forecast for the second observation. The GARCH updating formula takes the weighted average of the unconditional variance, the squared residual for the first observation and the starting variance and estimates the variance of the second observation. This is input into the forecast of the third variance, and so forth. Eventually, an entire time series of variance forecasts is constructed. Ideally, this series is large when the

residuals are large and small when they are small. The likelihood function provides a systematic way to adjust the parameters ω, α, β to give the best fit.

Of course, it is entirely possible that the true variance process is different from the one specified by the econometrician. In order to detect this, a variety of diagnostic tests are available. The simplest is to construct the series of $\{\varepsilon_i\}$, which are supposed to have constant mean and variance if the model is correctly specified.

Various tests such as tests for autocorrelation in the squares are able to detect model failures. Often a "Ljung box test" with 15 lagged autocorrelations is used.

A Value-at-Risk Example Applications of the ARCH/GARCH approach are widespread in situations where the volatility of returns is a central issue. Many banks and other financial institutions use the concept of "value at risk" as a way to measure the risks faced by their portfolios. The 1 percent value at risk is defined as the number of dollars that one can be 99 percent certain exceeds any losses for the next day. Statisticians call this a 1 percent quantile, because 1 percent of the outcomes are worse and 99 percent are better. Let's use the $GARCH(1, 1)$ tools to estimate the 1 percent value at risk of a $1,000,000 portfolio on March 23, 2000. This portfolio consists of 50 percent Nasdaq, 30 percent Dow Jones and 20 percent long bonds. The long bond is a ten-year constant maturity Treasury bond.1 This date is chosen to be just

before the big market slide at the end of March and April. It is a time of high volatility and great anxiety.

First, we construct the hypothetical historical portfolio. (All calculations in this example were done with the EViews software program.) Figure 1 shows the pattern of returns of the Nasdaq, Dow Jones, bonds and the composite portfolio leading up to the terminal date. Each of these series appears to show the signs of ARCH effects in that the amplitude of the returns varies over time. In the case of the equities, it is clear that this has increased substantially in the latter part of the sample period.

Visually, Nasdaq is even more extreme. In Table 1, we present some illustrative 1 The portfolio has constant proportions of wealth in each asset that would entail some rebalancing over time.

Journal of Economic Perspectives Table 1 Portfolio Data NASDAQ Dow Jones Rate Portfolio Mean 0.0009 0.0005 0.0001 0.0007 Std. Dev.

0.0115 0.0090 0.0073 0.0083 Skewness −0.5310 −0.2031 −0.4738 −0.3593 7.4936 7.0026 Kurtosis 8.3288 4.9579 Sample: March 23, 1990 to March 23, 2000.

statistics for each of these three investments separately and for the portfolio as a whole in the final column. From the daily standard deviation, we see that the Nasdaq is the most volatile and

interest rates the least volatile of the assets. The portfolio is less volatile than either of the equity series even though it is 80 percent equity-yet another illustration of the benefits of diversification. All the assets show evidence of fat tails, since the kurtosis exceeds 3, which is the normal value, and evidence of negative skewness, which means that the left tail is particularly extreme.

The portfolio shows substantial evidence of ARCH effects as judged by the autocorrelations of the squared residuals in Table 2. The first order autocorrelation is .210, and they gradually decline to .083 after 15 lags. These autocorrelations are not large, but they are very significant. They are also all positive, which is uncom- mon in most economic time series and yet is an implication of the $GARCH(1,1)$ model. Standard software allows a test of the hypothesis that there is no autocor- relation (and hence no ARCH). The test p-values shown in the last column are all zero to four places, resoundingly rejecting the "no ARCH" hypothesis.

Then we forecast the standard deviation of the portfolio and its 1 percent quantile. We carry out this calculation over several different time frames: the entire ten years of the sample up to March 23, 2000; the year before March 23, 2000; and from January 1, 2000, to March 23, 2000.

Consider first the quantiles of the historical portfolio at these three different time horizons. To do this calculation, one simply sorts the returns and finds the 1 percent worst case. Over the full ten-year sample, the 1 percent quantile times $1,000,000 produces a value at risk of $22,477. Over the last year, the calculation produces a value at risk of $24,653—somewhat higher, but not enormously so.

However, if the 1 percent quantile is calculated based on the data from January 1, 2000, to March 23, 2000, the value at risk is $35,159. Thus, the level of risk apparently has increased dramatically over the last quarter of the sample. Each of these numbers is the appropriate value at risk if the next day is equally likely to be the same as the days in the given sample period. This assumption is more likely to be true for the shorter period than for the long one.

The basic $GARCH(1,1)$ results are given in Table 3. Under this table it lists the dependent variable, PORT, and the sample period, indicates that it took the algorithm 16 iterations to maximize the likelihood function and computed stan-

Robert Engle Table 2 Autocorrelations of Squared Portfolio Returns AC Q-Stat Prob 0.210 115.07 0.000 0.183 202.64 0.000 237.59 0.116 0.000 0.082 255.13 0.000 0.122 294.11 0.000 0.163 363.85 0.000 0.090 384.95 0.000 410.77 0.099 0.000 427.88 0.081 0.000 0.081 445.03 0.000 457.68 0.069 0.000 0.080 474.29 0.000 0.076 489.42 0.000 503.99 0.074 0.000 0.083 521.98 0.000 Sample: March 23, 1990 to March 23, 2000.

Notes

Table 3 GARCH(1,1) Variance Equation Variable Coef St. Err Z-Stat P-Value 1.40E-06 4.48E-07 3.1210 0.0018 ARCH(1) 0.0772 0.0179 4.3046 0.0000 GARCH(1) 0.9046 0.0196 46.1474 0.0000 **Notes.** Dependent Variable: PORT.

Sample (adjusted): March 23, 1990 to March 23, 2000.

Convergence achieved after 16 iterations.

Bollerslev-Woodridge robust standard errors and covariance.

dard errors using the robust method of Bollerslev-Wooldridge. The three coeffi- cients in the variance equation are listed as C, the intercept; $ARCH(1)$, the first lag of the squared return; and GARCH(1), the first lag of the conditional variance.

Notice that the coefficients sum up to a number less than one, which is required to have

a mean reverting variance process. Since the sum is very close to one, this process only mean reverts slowly. Standard errors, Z-statistics (which are the ratio of coefficients and standard errors) and p-values complete the table.

The standardized residuals are examined for autocorrelation in Table 4.

Clearly, the autocorrelation is dramatically reduced from that observed in the portfolio returns themselves. Applying the same test for autocorrelation, we now

164 Journal of Economic Perspectives Table 4 Autocorrelations of Squared Standardized Residuals AC Q-Stat Prob 0.005 0.0589 0.808 0.039 4.0240 0.134 4.3367 0.997 −0.011 −0.017 5.0981 0.277 0.002 5.1046 0.403 0.009 5.3228 0.503 −0.015 5.8836 0.553 −0.013 6.3272 0.611 −0.024 7.8169 0.553 −0.006 7.9043 0.638 −0.023 9.3163 0.593 −0.013 9.7897 0.634 −0.003 9.8110 0.709 10.038 0.009 0.759 −0.012 0.791 10.444 find the p-values are about 0.5 or more, indicating that we can accept the hypothesis of "no residual ARCH." The forecast standard deviation for the next day is 0.0146, which is almost double the average standard deviation of 0.0083 presented in the last column of

Table 1. If the residuals were normally distributed, then this would be multiplied by 2.327, because 1 percent of a normal random variable lies 2.327 standard deviations below the mean. The estimated normal value at risk = $33,977. As it turns out, the standardized residuals, which are the estimated values of $\{\varepsilon_i\}$, are not very close to a normal distribution. They have a 1 percent quantile of 2.844, which reflects the fat tails of the asset price distribution. Based on the actual distribution, the estimated 1 percent value at risk is $39,996. Notice how much this value at risk has risen to reflect the increased risk in 2000.

Finally, the value at risk can be computed based solely on estimation of the quantile of the forecast distribution. This has recently been proposed by Engle and Manganelli (2001), adapting the quantile regression methods of Koenker and Basset (1978) and Koenker and Hallock in this symposium. Application of their method to this data set delivers a value at risk = $38,228.

What actually did happen on March 24, 2000, and subsequently? The portfolio lost more than $1000 on March 24 and more than $3000 on March 27.

The biggest hit was $67,000 on April 14. We all know that Nasdaq declined substantially over the next year. The Dow Jones average was much less affected, and bond prices increased as the Federal Reserve lowered interest rates. Fig- ure 2 plots the value at risk estimated each day using this methodology within the sample period and the losses that occurred the next day. There are about 1 percent of times the value at risk is exceeded, as is expected, since this is in-sample. Figure 3 plots the same graph for the next year and a quarter, during

GARCH 101: The Use of ARCH/GARCH models in Applied Econometrics Figure 2 Value at Risk and Portfolio Losses In-Sample 80000 LOSS 60000

– VAR 40000 20000 O −20000 −40000 −60000 Figure 3 Value at Risk and Portfolio Losses Out of Sample LOSS 80000

– VAR 60000 40000 20000 −20000 −40000 −60000 −80000 2000:10 2001:01 2001:04 2000:04 2000:07 which the equity market tanks and the bond yields fall. The parameters are not reestimated, but the formula is simply updated each day. The computed value at risk rises substantially from the $40,000 initial figure as the volatility rises in April 2000. Then the losses decline, so that the value at risk is well above the realized losses. Toward the end of the period, the losses approach the value at risk again, but at a lower level. In this year and a quarter, the value at risk is exceeded only once; thus, this is actually a slightly conservative estimate of the risk. It is not easy to determine whether a particular value-at-risk number is correct, although statistical tests can be

formulated for this in the same way they

are formulated for volatilities. For example, Engle and Manganelli (2001) present a "dynamic quantile test."

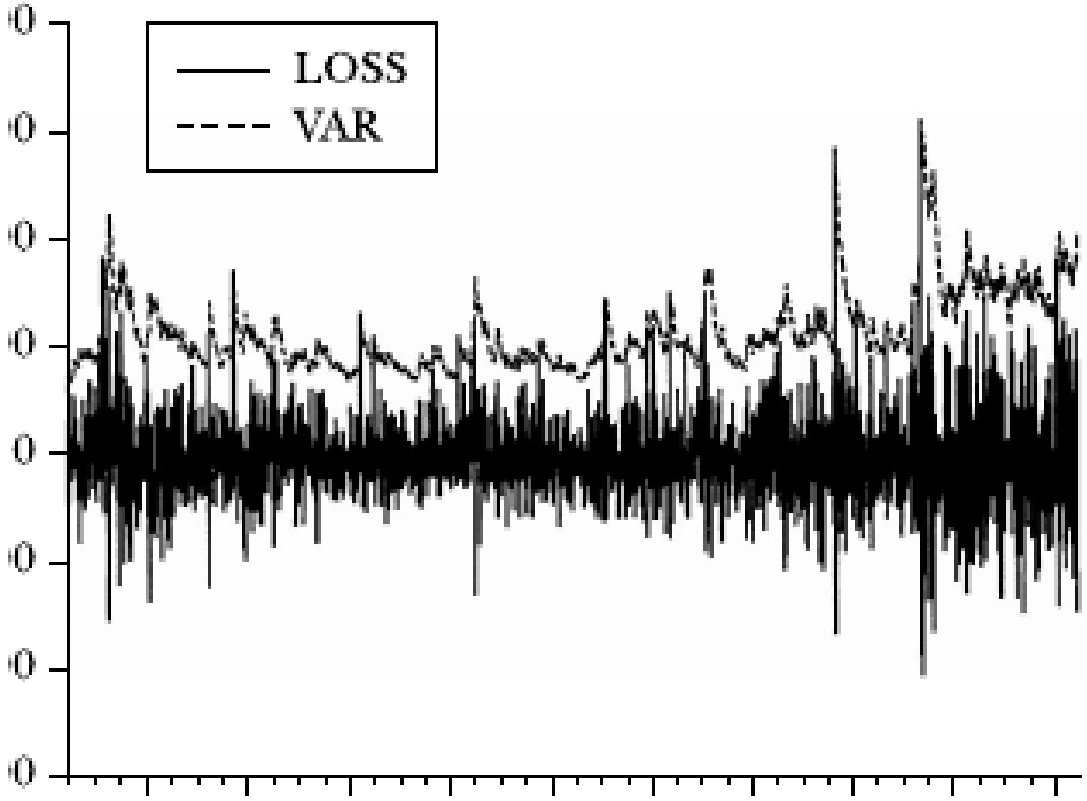


Figure 3.17: Course figure, source page 9, crop 1.

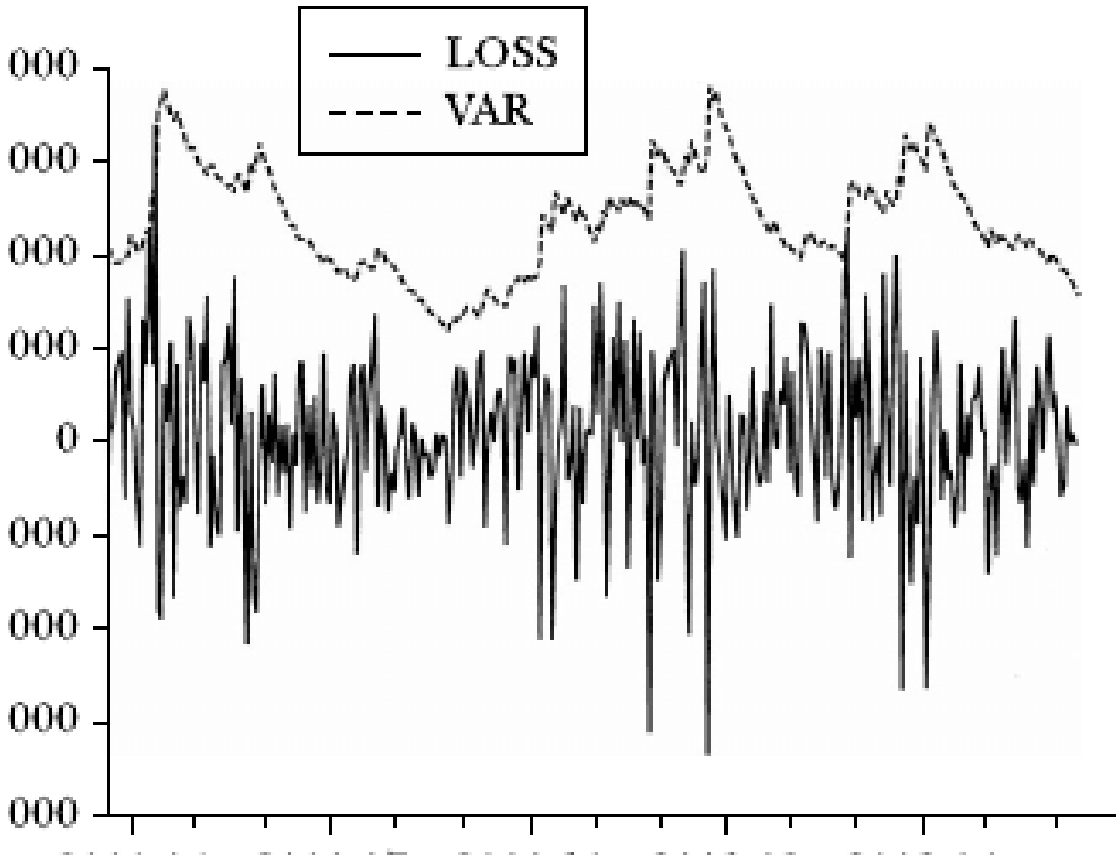


Figure 3.18: Course figure, source page 9, crop 2.

166 Journal of Economic Perspectives Extensions and Modifications of GARCH The $GARCH(1, 1)$ is the simplest and most robust of the family of volatility models. However, the model can be extended and modified in many ways. I will briefly mention three modifications, although the number of volatility models that can be found in the literature is now quite extraordinary.

The GARCH(1,1) model can be generalized to a GARCH(p, q) model—that is, a model with additional lag terms. Such higher-order models are often useful when a long span of data is used, like several decades of daily data or a year of hourly data. With additional lags, such models allow both fast and slow decay of information. A particular specification of the GARCH(2,2) by Engle and Lee (1999), sometimes called the "component model," is a useful starting point to this approach.

ARCH/GARCH models thus far have ignored information on the direction of returns; only the magnitude matters. However, there is very convincing evidence that the direction does affect volatility. Particularly for broad-based equity indices and bond market indices, it appears that market declines forecast higher volatility than comparable market increases do. There is now a variety of asymmetric GARCH models, including the EGARCH model of Nelson (1991), the TARCH model- threshold ARCH—attributed to Rabemananjara and Zakoian (1993) and Glosten, Jaganathan and Runkle (1993), and a collection and comparison by Engle and Ng (1993).

The goal of volatility analysis must ultimately be to explain the causes of volatility. While time series structure is valuable for forecasting, it does not satisfy our need to explain volatility. The estimation strategy introduced for ARCH/GARCH models can be directly applied if there are predetermined or exogenous variables. Thus, we can think of the estimation problem for the

variance just as we do for the mean. We can carry out specification searches and hypothesis tests to find the best formulation. Thus far, attempts to find the ultimate cause of volatility are not very satisfactory. Obviously, volatility is a response to news, which must be a surprise. However, the timing of the news may not be a surprise and gives rise to predictable components of volatility, such as economic announcements. It is also possible to see how the amplitude of

news events is influenced by other news events. For example, the amplitude of return movements on the United States stock market may respond to the volatility observed earlier in the day in Asian markets as well as to the volatility observed in the United States on the previous day. Engle, Ito and Lin (1990) call these "heat wave" and "meteor shower" effects.

A similar issue arises when examining several assets in the same market. Does the volatility of one influence the volatility of another? In particular, the volatility of an individual stock is clearly influenced by the volatility of the market as a whole.

This is a natural implication of the capital asset pricing model. It also appears that there is time variation in idiosyncratic volatility (for example, Engle, Ng and Rothschild, 1992).

Robert Engle This discussion opens the door to multivariate modeling where not only the volatilities but also the correlations are to be investigated. There are now a large number of multivariate ARCH models to choose from. These turn out often to be difficult to estimate and to have large numbers of parameters. Research is continu- ing to examine new classes of multivariate models that are more convenient for fitting large covariance matrices. This is relevant for systems of equations such as vector autoregressions and for portfolio problems where possibly thousands of assets are to be analyzed.

Conclusion ARCH and GARCH models have been applied to a wide range of time series analyses, but applications in finance have been particularly successful and have been the focus of this introduction. Financial decisions are generally based upon the tradeoff between risk and return; the econometric analysis of risk is therefore an integral part of asset pricing, portfolio optimization, option pric- ing and risk management. This paper has presented an example of risk mea- surement that could be the input to a variety of economic decisions. The analysis of ARCH and GARCH models and their many extensions provides a statistical stage on which many theories of asset pricing and portfolio analysis can be exhibited and tested.

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3.0.5 Optional reading – StockWatson JEP01

Journal of Economic Perspectives—Volume 15, Number 4—Fall 2001—Pages 101–115 Vector Autoregressions James H. Stock and Mark W. Watson acroeconometricians do four things: describe and summarize macro- economic data, make macroeconomic forecasts, quantify what we do or do not know about the true structure of the macroeconomy, and advise (and sometimes become) macroeconomic policymakers. In the 1970s, these four tasks—data description, forecasting, structural inference and policy analysis—were performed using a variety of techniques. These ranged from large models with hundreds of equations to single-equation models that focused on interactions of a few variables to simple univariate time series models involving only a single variable.

But after the macroeconomic chaos of the 1970s, none of these approaches appeared especially trustworthy.

Two decades ago, Christopher Sims (1980) provided a new macroeconometric framework that held great promise: vector autoregressions (VARs). A univariate autoregression is a single-equation, single-variable linear model in which the cur- rent value of a variable is explained by its own lagged values. A VAR is an n-equation, n-variable linear model in which each variable is in turn explained by its own lagged values, plus current and past values of the remaining $n - 1$ variables.

This simple framework provides a systematic way to capture rich dynamics in multiple time series, and the statistical toolkit that came with VARs was easy to use and to interpret. As Sims (1980) and others argued in a series of influential early papers, VARs held out the promise of providing a coherent and credible approach to data description, forecasting, structural inference and policy analysis.

In this article, we assess how well VARs have addressed these four macroecono- James H. Stock is the Roy E. Larsen Professor of Political Economy, John F. Kennedy School of Government, Harvard University, Cambridge, Massachusetts. Mark W. Watson is Profes- sor of Economics and Public Affairs, Department of Economics and Woodrow Wilson School of Public and International Affairs, Princeton University, Princeton, New Jersey. Both authors are Research Associates, National Bureau of Economic Research, Cambridge, Massachusetts.

102 Journal of Economic Perspectives metric tasks.1 Our answer is "it depends." In data description and forecasting, VARs have proven to be powerful and reliable tools that are now, rightly, in everyday use.

Structural inference and policy analysis are, however, inherently more difficult because they require differentiating between correlation and causation; this is the "identification problem," in the jargon of econometrics. This problem cannot be solved by a purely statistical tool, even a powerful one like a VAR. Rather, economic theory or institutional knowledge is required to solve the identification (causation versus correlation) problem.

A Peek Inside the VAR Toolkit What, precisely, is the effect of a 100-basis-point hike in the federal funds interest rate on the rate of inflation one year hence? How big an interest rate cut is needed to offset an expected half percentage point rise in the unemployment rate? How well does the Phillips curve predict inflation? What fraction of the variation in inflation in the past 40 years is due to monetary policy as opposed to external shocks?

Many macroeconomists like to think they know the answers to these and similar questions, perhaps with a modest range of uncertainty. In the next two sections, we take a quantitative look at these and related questions using several three-variable VARs estimated using quarterly U.S. data on the rate of price inflation (π_t), the unemployment rate (u_t) and the interest rate (R_t), specifically, the federal funds rate) from 1960:I-2000:IV.2 First, we construct and examine these models as a way to display the VAR toolkit; criticisms are reserved for the next section.

VARs come in three varieties: reduced form, recursive and structural.

A reduced form VAR expresses each variable as a linear function of its own past values, the past values of all other variables being considered and a serially uncor- related error term. Thus, in our example, the VAR involves three equations:

current unemployment as a function of past values of unemployment, inflation and the interest rate; inflation as a function of past values of inflation, unemployment and the interest rate; and similarly for the interest rate equation. Each equation is estimated by ordinary least squares regression. The number of lagged values to include in each equation can be determined by a number of different methods, and we will use four lags in our examples.3 The error terms in these regressions are the "surprise" movements in the variables after taking its past values into account.

If the different variables are correlated with each other—as they typically are in 1 Readers interested in more detail than provided in this brief tutorial should see Hamilton's (1994) textbook or Watson's (1994) survey article.

2 The inflation data are computed as $\pi_t = 400 \ln(P_t/P_{t-1})$, where P_t is the chain-weighted GDP price index and u_t is the civilian unemployment rate. Quarterly data on u_t and R_t are formed by taking quarterly averages of their monthly values.

3 Frequently, the Akaike (AIC) or Bayes (BIC) information criteria are used; for a discussion, see L tkepohl (1993, chapter 4).

James H. Stock and Mark W. Watson macroeconomic applications-then the error terms in the reduced form model will also be correlated across equations.

A recursive VAR constructs the error terms in each regression equation to be uncorrelated with the error in the preceding equations. This is done by judiciously including some contemporaneous values as regressors. Consider a three-variable VAR, ordered as 1) inflation, 2) the unemployment rate, and 3) the interest rate.

In the first equation of the corresponding recursive VAR, inflation is the dependent variable, and the regressors are lagged values of all three variables. In the second equation, the unemployment rate is the dependent variable, and the regressors are lags of all three variables plus the current value of the inflation rate. The interest rate is the dependent variable in the third equation, and the regressors are lags of all three variables, the current value of the inflation rate plus the current value of the unemployment rate. Estimation of each equation by ordinary least squares produces residuals that are uncorrelated across equations.4 Evidently, the results depend on the order of the variables: changing the order changes the VAR equations, coefficients, and residuals, and there are n! recursive VARs representing all possible orderings.

A structural VAR uses economic theory to sort out the contemporaneous links among the variables (Bernanke, 1986; Blanchard and Watson, 1986; Sims, 1986).

Structural VARs require "identifying assumptions" that allow correlations to be interpreted causally. These identifying assumptions can involve the entire VAR, so that all of the causal links in the model are spelled out, or just a single equation, so that only a specific causal link is identified. This produces instrumental variables that permit the contemporaneous links to be estimated using instrumental vari- ables regression. The number of structural VARs is limited only by the inventiveness of the researcher.

In our three-variable example, we consider two related structural VARs. Each incorporates a different assumption that identifies the causal influence of monetary policy on unemployment, inflation and interest rates. The first relies on a version of the "Taylor rule," in which the Federal Reserve is modeled as setting the interest rate based on past rates of inflation and unemployment.5 In this system, the Fed sets the federal funds rate R according to the rule $R_t = r^* + 1.5(\bar{\pi}_t - \pi^*) - 1.25(\bar{u}_t - u^*) + laggedvaluesofR, \pi, u + \varepsilon_t$

where r^* is the desired real rate of interest, $\bar{\pi}_i$ and $\bar{u}_i$ are the average values of inflation and unemployment rate over the past four quarters, π^* and u^* are the target values of inflation and unemployment, and ε_t is the error in the equation.

This relationship becomes the interest rate equation in the structural VAR.

4 In the jargon of VARs, this algorithm for estimating the recursive VAR coefficients is equivalent to estimating the reduced form, then computing the Cholesky factorization of the reduced form VAR covariance matrix; see L tkepohl (1993, chapter 2).

5 Taylor's (1993) original rule used the output gap instead of the unemployment rate. Our version uses Okun's Law (with a coefficient of 2.5) to replace the output gap with unemployment rate.

104 Journal of Economic Perspectives The equation error, ε_t, can be thought of as a monetary policy "shock," since it represents the extent to which actual interest rates deviate from this Taylor rule.

This shock can be estimated by a regression with $R_t - 1.5\ \bar{\pi}_t + 1.25\ \bar{u}_t$ as the dependent variable, and a constant and lags of interest rates, unemployment and inflation on the right-hand side.

The Taylor rule is "backward looking" in the sense that the Fed reacts to past information ($\bar{\pi}_t$ and $\bar{u}_t$ are averages of the past four quarters of inflation and unemployment), and several researchers have argued that Fed behavior is more appropriately described by forward-looking behavior. Because of this, we consider another variant of the model in which the Fed reacts to forecasts of inflation and unemployment four quarters in the future. This Taylor rule has the same form as the rule above, but with $\bar{\pi}_t$ and $\bar{u}_t$ replaced by four-quarter ahead forecasts com-

puted from the reduced form VAR.

Putting the Three-Variable VAR Through Its Paces The different versions of the inflation-unemployment-interest rate VAR are put through their paces by applying them to the four macroeconometric tasks.

First, the reduced form VAR and a recursive VAR are used to summarize the comovements of these three series. Second, the reduced form VAR is used to forecast the variables, and its performance is assessed against some alternative benchmark models. Third, the two different structural VARs are used to estimate the effect of a policy-induced surprise move in the federal funds interest rate on future rates of inflation and unemployment. Finally, we discuss how the structural VAR could be used for policy analysis.

Data Description Standard practice in VAR analysis is to report results from Granger-causality tests, impulse responses and forecast error variance decompositions. These statistics are computed automatically (or nearly so) by many econometrics packages (RATS, Eviews, TSP and others). Because of the complicated dynamics in the VAR, these statistics are more informative than are the estimated VAR regression coefficients or R^2 statistics, which typically go unreported.

Granger-causality statistics examine whether lagged values of one variable help to predict another variable. For example, if the unemployment rate does not help predict inflation, then the coefficients on the lags of unemployment will all be zero in the reduced-form inflation equation. Panel A of Table 1 summarizes the Granger-causality results for the three-variable VAR. It shows the p-values associated with the F statistics for testing whether the relevant sets of coefficients are zero. The unemployment rate helps to predict inflation at the 5 percent significance level (the p-value is 0.02, or 2 percent), but the federal funds interest rate does not (the p-value is 0.27). Inflation does not help to predict the unemployment rate, but the federal funds rate does. Both inflation and the unemployment rates help predict the federal funds interest rate.

Vector Autoregressions Table 1 VAR Descriptive Statistics for (π, u, R) A. Granger-Causality Tests Dependent Variable in Regression R Regressor π u 0.00 0.31 0.00 π 0.02 0.00 u R 0.27 0.01 0.00 B. Variance Decompositions from the Recursive VAR Ordered as π, u, R B.i. Variance Decomposition of π Variance Decomposition (Percentage Points) Forecast Standard Error R Horizon π u 0.96 1.34 1.75 1.97 B.ii. Variance Decomposition of u Variance Decomposition (Percentage Points) Forecast Standard Error R Horizon π u 0.23 0.64 0.79 0.92 B.iii. Variance Decomposition of R

Notes

Variance Decomposition (Percentage Points) Forecast Standard Error R Horizon π u 0.85 1.84 2.44 2.63 **Notes.** π denotes the rate of price inflation, u denotes the unemploy- ment rate and R denotes the Federal Funds interest rate. The entries in Panel A show the p-values for F-tests that lags of the variable in the row labeled Regressor do not enter the reduced form equation for the column variable labeled Dependent Variable. The results were com- puted from a VAR with four lags and a constant term over the 1960:I-2000:IV sample period.

106 Journal of Economic Perspectives Impulse responses trace out the response of current and future values of each of the variables to a one-unit increase in the current value of one of the VAR errors, assuming that this error returns to zero in subsequent periods and that all other errors are equal to zero. The implied thought experiment of changing one error while holding the others constant makes most sense when the errors are uncorre- lated across equations, so impulse responses are typically calculated for recursive and structural VARs.

The impulse responses for the recursive VAR, ordered π_ν, u_ν, R_ν are plotted in Figure 1. The first row shows the effect of an unexpected 1 percentage point increase in inflation on all three variables, as it works through the recursive VAR system with the coefficients estimated from actual data. The second row shows the effect of an unexpected increase of 1 percentage point in the unemployment rate, and the third row shows the corresponding effect for the interest rate. Also plotted

are ± 1 standard error bands, which yield an approximate 66 percent confidence interval for each of the impulse responses. These estimated impulse responses show patterns of persistent common variation. For example, an unexpected rise in inflation slowly fades away over 24 quarters and is associated with a persistent increase in unemployment and interest rates.

The forecast error decomposition is the percentage of the variance of the error made in forecasting a variable (say, inflation) due to a specific shock (say, the error term in the unemployment equation) at a given horizon (like two years). Thus, the forecast error decomposition is like a partial R^2 for the forecast error, by forecast horizon. These are shown in Panel B of Table 1 for the recursive VAR. They suggest considerable interaction among the variables. For example, at the 12-quarter horizon, 75 percent of the error in the forecast of the federal funds interest rate is attributed to the inflation and unemployment shocks in the recursive VAR.

Forecasting Multistep-ahead forecasts, computed by iterating forward the reduced form VAR, are assessed in Table 2. Because the ultimate test of a forecasting model is its out-of-sample performance, Table 2 focuses on pseudo out-of-sample forecasts over the period from 1985:I to 2000:IV. It examines forecast horizons of two quarters, four quarters and eight quarters. The forecast h steps ahead is computed by estimating the VAR through a given quarter, making the forecast h steps ahead, reestimating the VAR through the next quarter, making the next forecast and so on through the forecast period.6 As a comparison, pseudo out-of-sample forecasts were also computed for a univariate autoregression with four lags-that is, a regression of the variable on lags 6 Forecasts like these are often referred to as pseudo or "simulated" out-of-sample forecasts to emphasize

that they simulate how these forecasts would have been computed in real time, although, of course, this exercise is conducted retrospectively, not in real time. Our experiment deviates slightly from what would have been computed in real time because we use the current data, which includes later revisions made to the inflation and unemployment data by statistical agencies, rather than the data available in real time.

James H. Stock and Mark W. Watson Figure 1 Impulse Responses in the Inflation-Unemployment-Interest Rate Recursive VAR Inflation Shock to Inflation Unemployment Interest Rate 1.25 0.75 50 0 7F Percent 00 0 95 70 70 Lag Unemployment Shock to

Inflation Unemployment nterest Rate 1.0 0 6 Percent Lag Interest Rate Shock to Inflation Unemployment Interest Rate 0.50 Percent 0.0.01 Percent 0.50 0.0 L To **24** Lag of its own past values-and for a random walk (or "no change") forecast. Inflation rate forecasts were made for the average value of inflation over the forecast period, while forecasts for the unemployment rate and interest rate were made for the final quarter of the forecast period. Table 2 shows the root mean square forecast error for each of the forecasting methods. (The mean squared forecast error is computed

as the average squared value of the forecast error over the 1985–2000 out-of-sample period, and the resulting square root is the root mean squared forecast error reported in the table.) Table 2 indicates that the VAR either does no worse than or improves upon the univariate autoregression and that both improve upon the random walk forecast.

Structural Inference What is the effect on the rates of inflation and unemployment of a surprise 100 basis-point increase in the federal funds interest rate? Translated into VAR jargon,

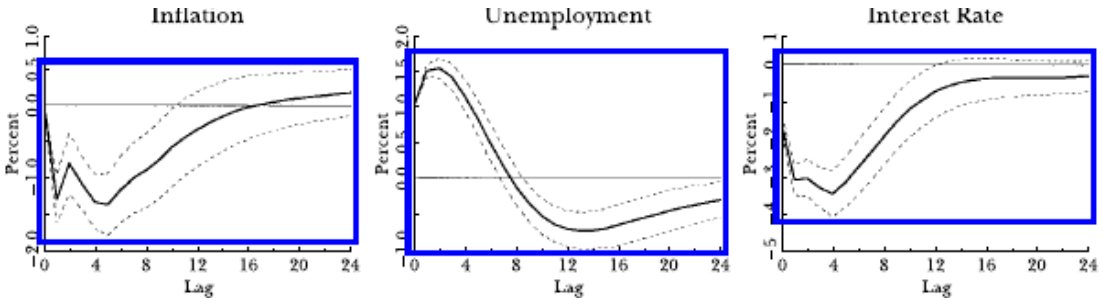


Figure 3.19: Course figure, source page 7, crop 1.

108 Journal of Economic Perspectives Table 2 Root Mean Squared Errors of Simulated Out-Of-Sample Forecasts, 1985:1-2000:IV Inflation Rate Unemployment Rate Interest Rate Forecast RW Horizon AR VAR AR VAR AR VAR 2 quarters 0.82 0.70 0.68 0.34 0.28 0.29 0.79 0.77 0.68 0.73 0.65 0.63 0.62 0.52 0.53 1.36 1.25 1.07 4 quarters 0.75 1.12 0.95 0.78 2.18 1.92 1.70 8 quarters Notes: Entries are the root mean squared error of forecasts computed recursively for univariate and vector autoregressions (each with four lags) and a random walk ("no change") model. Results for the random walk and univariate autoregressions are shown in columns labeled RW and AR, respectively.

Each model was estimated using data from 1960:I through the beginning of the forecast period.

Forecasts for the inflation rate are for the average value of inflation over the period. Forecasts for the unemployment rate and interest rate are for the final quarter of the forecast period.

this question becomes: What are the impulse responses of the rates of inflation and unemployment to the monetary policy shock in a structural VAR?

The solid line in Figure 2 plots the impulse responses computed from our model with the backward-looking Taylor rule. It shows the inflation, unemploy- ment and real interest rate (R_t $- \pi_t$) responses to a 1 percentage point shock in the nominal federal funds rate. The initial rate hike results in the real interest rate exceeding 50 basis points for six quarters. Although inflation is eventually reduced by approximately 0.3 percentage points, the lags are long, and most of the action occurs in the third year after the contraction. Similarly, the rate of unemployment rises by approximately 0.2 percentage points, but most of the economic slowdown is in the third year

after the rate hike.

How sensitive are these results to the specific identifying assumption used in this structural VAR—that the Fed follows the backward-looking Taylor rule? As it happens, very sensitive. The dashed line in Figure 2 plots the impulse responses computed from the structural VAR with the forward-looking Taylor rule. The impulse responses in real interest rates are broadly similar under either rule.

However, in the forward-looking model the monetary shock produces a 0.5 per- centage point increase in the unemployment rate within a year, and the rate of inflation drops sharply at first, fluctuates, then leaves a net decline of 0.5 percent- age points after six years. Under the backward-looking rule, this 100 basis-point rate hike produces a mild economic slowdown and a modest decline in inflation several years hence; under the forward-looking rule, by this same action the Fed wins a major victory against inflation at the cost of a swift and sharp recession.

Policy Analysis In principle, our small structural VAR can be used to analyze two types of policies: surprise monetary policy interventions and changing the policy rule, like shifting from a Taylor rule (with weight on both unemployment and inflation) to an explicit inflation targeting rule.

Vector Autoregressions Figure 2 Impulse Responses of Monetary Policy Shocks for Different Taylor Rule Identifying Assumptions Response of Inflation Response of Unemployment 1.0 0.5 0.0 0.5 −0.5 Percent 0.0 −0.5 −2.5 **24** Lag Response of Real Interest Rates 3.5 2.5 1.5 Percent 0.5 −0.5

Notes

—- 0_2.5 Lag **Notes.** The solid line is computed with the backward-looking Taylor rule; the dashed line, with the forward-looking Taylor rule.

If the intervention is an unexpected movement in the federal funds interest rate, then the estimated effect of this policy on future rates of inflation and unemployment is summarized by the impulse response functions plotted in Figure 2. This might seem a somewhat odd policy, but the same mechanics can be used to evaluate a more realistic intervention, such as raising the federal funds rate by 50 basis points and sustaining this increase for one year. This policy can be engineered in a VAR by using the right sequence of monetary policy innovations to hold the federal funds interest rate at this sustained level for four quarters, taking into account that in the VAR, actions on interest rates in earlier quarters affect those in later quarters (Sims, 1982; Waggoner and Zha, 1999).

Analysis of the second type of policy—a shift in the monetary rule itself—is more complicated. One way to evaluate a new policy rule candidate is to ask what would be the effect of monetary and nonmonetary shocks on the economy under the new rule. Since this question involves all the structural disturbances, answering

110 Journal of Economic Perspectives it requires a complete macroeconomic model of the simultaneous determination of all the variables, and this means that all of the causal links in the structural VAR must be specified. In this case, policy analysis is carried out as follows: a structural VAR is estimated in which all the equations are identified, then a new model is formed by replacing the monetary policy rule. Comparing the impulse responses in the two models shows how the change in policy has altered the effects of monetary and nonmonetary shocks on the variables in the model.

How Well Do VARs Perform the Four Tasks?

We now turn to an assessment of VARs in performing the four macroecono- metric tasks, highlighting both successes and shortcomings.

Data Description Because VARs involve current and lagged values of multiple time series, they capture comovements that cannot be detected in univariate or bivariate models.

Standard VAR summary statistics like Granger-causality tests, impulse response functions and variance decompositions are well-accepted and widely used methods for portraying these comovements. These summary statistics are useful because they provide targets for theoretical macroeconomic models. For example, a theoretical model that implied that interest rates should Granger-cause inflation but unem- ployment should not would be inconsistent with the evidence in Table 1.

Of course, the VAR methods outlined here have some limitations. One is that the standard methods of statistical inference (such as computing standard errors for impulse responses) may give misleading results if some of the variables are highly persistent.7 Another limitation is that, without modification, standard VARs miss nonlinearities, conditional heteroskedasticity and drifts or breaks in parameters.

Forecasting Small VARs like our three-variable system have become a benchmark against which new forecasting systems are judged. But while useful as a benchmark, small VARs of two or three variables are often unstable and thus poor predictors of the future (Stock and Watson, 1996).

State-of-the-art VAR forecasting systems contain more than three variables and allow for time-varying parameters to capture important drifts in coefficients (Sims, 1993). However, adding variables to the VAR creates complications, because the number of VAR parameters increases as the square of the number of variables: a nine-variable, four-lag VAR has 333 unknown coefficients (including the inter- 7 Bootstrap methods provide some improvements (Kilian, 1999) for inference about impulse responses, but treatments of this problem that are fully satisfactory theoretically are elusive (Stock, 1997; Wright, 2000).

James H. Stock and Mark W. Watson cepts). Unfortunately, macroeconomic time series data cannot provide reliable estimates of all these coefficients without further restrictions.

One way to control the number of parameters in large VAR models is to impose a common structure on the coefficients, for example using Bayesian meth- ods, an approach pioneered by Litterman (1986) (six variables) and Sims (1993) (nine variables). These efforts have paid off, and these forecasting systems have solid real-time track records (McNees, 1990; Zarnowitz and Braun, 1993).

Structural Inference In our three-variable VAR in the previous section, the estimated effects of a monetary policy shock on the rates of inflation and unemployment (summarized by the impulse responses in Figure 2) depend on the details of the presumed mone- tary policy rule followed by the Federal Reserve. Even modest changes in the assumed rule resulted in substantial changes in these impulse responses. In other words, the estimates of the structural impulse responses hinge on detailed institu- tional knowledge of how the Fed sets interest rates.8 Of course, the observation that results depend on assumptions is hardly new.

The operative question is whether the assumptions made in VAR models are any more compelling than in other econometric models. This is a matter of heated debate and is thoughtfully discussed by Leeper, Sims and Zha (1996), Christiano, Eichenbaum and Evans (1999), Cochrane (1998), Rudebusch (1998) and Sims (1998). Below are three important criticisms of structural VAR modeling.9 First, what really makes up the VAR "shocks?" In large part, these shocks, like those in conventional regression, reflect factors omitted from the model. If these factors are correlated

with the included variables, then the VAR estimates will contain omitted variable bias. For example, officials at the Federal Reserve might scoff at the idea that they mechanically followed a Taylor rule, or any other fixed-coefficient mechanical rule involving only a few variables; rather, they suggest

that their decisions are based on a subtle analysis of very many macroeconomic factors, both quantitative and qualitative. These considerations, when omitted from the VAR, end up in the error term and (incorrectly) become part of the estimated historical "shock" used to estimate an impulse response. A concrete example of this in the VAR literature involves the "price puzzle." Early VARs showed an odd result:

inflation tended to increase following monetary policy tightening. One explanation for this (Sims, 1992) was that the Fed was looking forward when it set interest rates and that simple VARs omitted variables that could be used to predict future inflation. When these omitted variables intimated an increase in inflation, the Fed tended to increase interest rates. Thus, these VAR interest rate shocks presaged 8 In addition, the institutional knowledge embodied in our three-variable VAR is rather na ve; for example, the Taylor rule was designed to summarize policy in the Greenspan era, not the full sample in our paper.

9 This list hits only the highlights; other issues include the problem of "weak instruments" discussed in Pagan and Robertson (1998) and the problem of noninvertible representations discussed in Hansen and Sargent (1991) and Lippi and Reichlin (1993).

112 Journal of Economic Perspectives increases in inflation. Because of omitted variables, the VAR mistakenly labeled these increases in interest rates as monetary shocks, which led to biased impulse responses. Indeed, Sims's explanation of the price puzzle has led to the practice of including commodity prices in VARs to attempt to control for predicted future inflation.

Second, policy rules change over time, and formal statistical tests reveal widespread instability in low-dimensional VARs (Stock and Watson, 1996). Con- stant parameter structural VARs that miss this instability are improperly identified.

For example, several researchers have documented instability in monetary policy rules (for example, Bernanke and Blinder, 1992; Bernanke and Mihov, 1998;

Clarida, Gali and Gertler, 2000; Boivin, 2000), and this suggests misspecification in constant coefficient VAR models (like our three-variable example) that are esti- mated over long sample periods.

Third, the timing conventions in VARs do not necessarily reflect real-time data availability, and this undercuts the common method of identifying restrictions based on timing assumptions. For example, a common assumption made in struc- tural VARs is that variables like output and inflation are sticky and do not respond "within the period" to monetary policy shocks. This seems plausible over the period of a single day, but becomes less plausible over a month or quarter.

In this discussion, we have carefully distinguished between recursive and structural VARs: recursive VARs use an arbitrary mechanical method to model contemporaneous correlation in the variables, while structural VARs use economic theory to associate these correlations with causal relationships. Unfortunately, in the empirical literature the distinction is often murky. It is tempting to develop economic "theories" that, conveniently, lead to a particular recursive ordering of the variables, so that their "structural" VAR simplifies to a recursive VAR, a structure called a "Wold causal chain." We think researchers yield to this temptation far too often. Such cobbled-together theories, even if superficially plausible, often fall apart on deeper inspection. Rarely does it add value to repackage a recursive VAR and sell it as structural.

Despite these criticisms, we think it is possible to have credible identifying assumptions in a VAR.

One approach is to exploit detailed institutional knowledge.

An example of this is the study by Blanchard and Perotti (1999) of the macroeco- nomic effects of fiscal policy. They argue that the tax code and spending rules impose tight constraints on the way that taxes and spending vary within the quarter, and they use these constraints to identify the exogenous changes in taxes and spending necessary for causal analysis. Another example is Bernanke and Mihov (1998), who use a model of the reserves market to identify monetary policy shocks.

A different approach to identification is to use long-run restrictions to identify shocks; for example, King, Plosser, Stock and Watson (1991) use the long-run neutrality of money to identify monetary shocks. However, assumptions based on the infinite future raise questions of their own (Faust and Leeper, 1997).

A constructive approach is to recognize explicitly the uncertainty in the assumptions that underlie structural VAR analysis and see what inferences, or range of inferences, still can be made. For example, Faust (1998) and Uhlig (1999)

Vector Autoregressions discuss inference methods that can be applied using only inequality restrictions on the theoretical impulse responses (for example, monetary contractions do not cause booms).

Policy Analysis Two types of policies can be analyzed using a VAR: one-off innovations, in which the same rule is maintained; and changes in the policy rule. The estimated effect of one-off innovations is a function of the impulse responses to a policy innovation, and potential pitfalls associated with these have already been discussed.

Things are more difficult if one wants to estimate the effect of changing policy rules. If the true structural equations involve expectations (say, an expectational Phillips curve), then the expectations will depend on the policy rule; thus, in general, all the VAR coefficients will depend on the rule. This is just a version of the Lucas (1976) critique. The practical importance of the Lucas critique for this type of VAR policy analysis is a matter of debate.

After Twenty Years of VARs VARs are powerful tools for describing data and for generating reliable mul- tivariate benchmark forecasts. Technical work remains, most notably extending VARs to higher dimensions and richer nonlinear structures. Even without these important extensions, however, VARs have made lasting contributions to the mac- roeconometrician's toolkit for tackling these two tasks.

Whether 20 years of VARs have produced lasting contributions to structural inference and policy analysis is more debatable. Structural VARs can capture rich dynamic properties of multiple time series, but their structural implications are only as sound as their identification schemes. While there are some examples of thoughtful treatments of identification in VARs, far too often in the VAR literature the central issue of identification is handled by ignoring it. In some fields of economics, such as labor economics and public finance, identification can be obtained credibly using natural experiments that permit some exogenous variation to be teased out of a relationship otherwise fraught with endogeneity and omitted variables bias. Unfortunately, these kinds of natural experiments are rare in macroeconomics.

Although VARs have limitations when it comes to structural inference and policy analysis, so do the alternatives. Calibrated dynamic stochastic general equi- librium macroeconomic models are explicit about causal links and expectations and provide an intellectually coherent framework for policy analysis. But the current generation of these models do not fit the data well. At the other extreme, simple single-equation models, for example, regressions of inflation against lagged interest rates, are easy to estimate and sometimes can produce good forecasts. But if it is difficult to distinguish correlation and causality in a VAR, it is even more so in single-equation models,

which can, in any event, be viewed as one equation pulled from a larger VAR. Used wisely and based on economic reasoning and

114 Journal of Economic Perspectives institutional detail, VARs both can fit the data and, at their best, can provide sensible estimates of some causal connections. Developing and melding good theory and institutional detail with flexible statistical methods like VARs should keep macroeconomists busy well into the new century.

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